# Pyrexia of unknown origin (PUO)



This chapter gives guidance on hundreds of pyrexia-causing infections; but what should you do if your patient has a fever that you cannot explain? Pyrexia of unknown origin (PUO)[24] has a differential of >200 diseases. 15–30% of these patients will eventually be given an infective diagnosis (depending on your corner of the globe). ~20% will remain undiagnosed, but in most of these the fever will resolve within 4wks.

## Diagnostic criteria

*Pyrexia >3wks with no identified cause after evaluation in hospital for 3d or ≥3 out-patient visits.*

Fever may also be undiagnosed in specific subgroups despite appropriate evaluation for 3d, including negative cultures at ≥2 days:

- *Nosocomial PUO:* Patient hospitalized for >48h with no infection at admission.
- *Immunodeficient PUO:* Pyrexia in patient with <500 neutrophils/microlitre.
- *HIV PUO:* Pyrexia in HIV infection lasting 3d as an in-patient or >4wks as an outpatient.

## History

Most PUO are due to common diseases with atypical presentation. Consider all details as potentially relevant. Include: travel (p414), diet, animal contact, changes in medication, recreational drug use, obstetric/sexual history, family history (table 9.30).

## Examination

Confirm fever. Pattern of fever is rarely helpful (contrary to the textbooks, most malaria has no specific pattern). Do not forget: mouth, genitals, skin, thyroid, lymphatic system, eyes including retina, temporal arteries (table 9.30).

## Investigation

►Extent of investigation depends on immune status and how well the patient is.

- *Blood tests:* FBC, U&E, LFT, CRP, ESR, electrophoresis, LDH, CK, ANA, ANCA, rheumatoid factor, HIV test, malaria smear, interferon-gamma release assay for TB (p394).
- *Microscopy and culture:* Blood ×3, urine, sputum (including AFB), stool, CSF.
- *Imaging:* CXR, abdominal/pelvic USS, venous Doppler. Consider: CT(PA), MRI, echo (TOE). Fluorodeoxyglucose-PET (FDG-PET) highlights areas of ↑glucose uptake including tumour and inflammation. It may aid/direct diagnosis in up to 50% of PUO.
- *Other:* Hepatitis serology, CMV, EBV, autoimmune screen, cryoglobulins, toxoplasmosis, brucellosis, *Coxiella*, lymph node biopsy, endoscopy, temporal artery biopsy.

Table 9.30 PUO differential according to history and examination findings

| History | Differential |
|---|---|
| Animal contact | Brucellosis, toxoplasmosis, *Bartonella*, leptospirosis, Q fever, psittacosis |
| Cough | TB, PE, Q fever, enteric fever, sarcoidosis, legionnaire's disease |
| Nasal symptoms | Sinusitis, GPA, relapsing fever, psittacosis |
| Confusion | TB, *Cryptococcus*, sarcoid, carcinomatosis, brucellosis, enteric fever |
| Arthralgia | SLE, infective endocarditis, Lyme disease, brucellosis, TB, IBD |
| Weight loss | Malignancy, vasculitis, TB, HIV, IBD, thyrotoxicosis |
| Family history | Familial Mediterranean fever |
| Drug history | Drug-induced fever (~7–10d after new drug) |
| **Examination** | **Differential** |
| Conjunctivitis | Leptospirosis, relapsing fever, spotted fever, trichinosis |
| Uveitis | TB, sarcoid, adult Still's disease, SLE, Behçet's disease |
| Mouth | Dental abscess, Behçet's disease, CMV, IBD |
| Lymphadenopathy | Lymphoma, TB, EBV, CMV, HIV, toxoplasmosis, brucellosis, *Bartonella* |
| Rash | HIV, EBV, SLE, vasculitis, Still's disease, endocarditis |
| Hepatomegaly | TB, EBV, malignancy, malaria, enteric fever, granulomatous hepatitis, Q fever, visceral leishmaniasis |
| Splenomegaly | Leukaemia, lymphoma, TB, brucellosis, infective endocarditis, CMV, EBV, rheumatoid arthritis, sarcoid, enteric fever, relapsing fever |
| Renal | Chronic pyelonephritis, perinephric abscess, renal tumour |
| Epididymo-orchitis | TB, lymphoma, EBV, brucellosis, leptospirosis |

## Eliciting the weird and the wonderful

**Listen to your patient** You have two cultures to master: the host and the pathogen. Prolonged immersion in both may be needed.

**Ask** Do not expect to find apposite questions such as these in any other textbook:

1. *'Have you delivered any septic babies in the last year?'* Impress and cure your obstetrician friends who tell you that 'I'm so depressed about not being able to shake off this flu', and who have forgotten about transfer of brucellosis from baby to obstetrician.
2. *'Are your carp well at present?' Mycobacterium marinum* skin infection.
3. *'Could that be a Hyalomma tick bite from when you rode your ostrich in last week's race?'* Crimean-Congo haemorrhagic fever.
4. *'Who has been licking your face recently?' Pasteurella multocida.*
5. *'Has your dog been on holiday this year?'* Monkeypox from prairie dogs.
6. *'Has your pet hedgehog lost weight?' Salmonella.*
7. *'Did you develop your headache after you adopted your pet magpie?'* Zoonotic transmission of *Cryptococcus neoformans* causing meningitis.
8. *'Did you have a stray pig living under your house when the monsoon started?'* Pigs + standing water + mosquitoes = Japanese encephalitis.
9. *'Did you sample the local frog paella?'* Angiostrongyliasis causing eosinophilic meningitis.
10. *'Did your goat miscarry last year?' Coxiella burnetii.*
11. *'Did your depressed pig develop a purple snout?' Erysipelothrix rhusiopathiae* endocarditis via swine erysipelas.
12. *'Can I see your pet lobster: he may be the cause of your bad hand?'* Lobsterman's hand, an erysipeloid infection from *Erysipelothrix*. Pet lobsters have a grand pedigree. Gérard de Nerval used to take his pet lobster for walks, on a blue silk lead, beside the Seine (fig 9.60). A lobster, he said, is, 'serious-minded and quiet, doesn't scratch or bark like a dog, and knows all of the secrets of the deep'. His lobster's mission was to combat the Philistinism chaining us all to mediocrity.

Fig 9.60 'Is your pet lobster well?' (Gérard de Nerval).

Ask also, 'Where have you been?' Though you will not be absolved of thought, even when the answer is, 'Southend'. It may be a question of amnesic stopovers. Or perhaps your patient is an airport baggage-handler, bitten by a hitch-hiking mosquito. And even when there has been no travel to the tropics, global warming is ensuring that the tropics are travelling to us. To the first writers of medical books, Paradise was just beyond the Far East, and the world was a disc surrounded by oceans of blue water. But the world moves on, tarnished, tawdry, and trashed; and Paradise appears to be evolving with ever more serpents in the garden, beguiling us with ambiguous answers to our great questions.

**Don't give up** If the culture is negative, tests may need repeating. Perhaps the organism is 'fastidious' in its nutritional requirement or requires a longer incubation? Even if culture is achieved, it may be that the organism grown is flora not pathogen. If culture fails, look for antibodies or antigen. PCR is increasingly used for identification, but it is far from infallible; beware of inhibitors, contamination, and a primer that is not as unique as your patient.

Remember Sherlock: *'My mind,' he said, 'rebels at stagnation. Give me problems, give me work, give me the most abstruse cryptogram or the most intricate analysis, and I am in my own proper atmosphere. I can dispense then with artificial stimulants. But I abhor the dull routine of existence. I crave for mental exaltation. That is why I have chosen my own particular profession, - or rather created it, for I am the only one in the world.'*

*The Sign of the Four*, Arthur Conan Doyle, 1890.

No, Sherlock, you are not; you have the infectious disease physicians for company.

# 10 Neurology

## Contents



Fig 10.1 Sir Roger Bannister CBE ran the first ever sub-4 minute mile on 6 May 1954 at the age of 25. At that time he was a medical student at Oxford University, and graduated the following year. He went on to become a prominent academic neurologist, conducting research into the autonomic nervous system, which he regarded as his greatest achievement. Autonomic activity and running are, of course, intimately linked, as Bannister reflected in a more poetic manner than the usual 'fight or flight' cliché: 'As a child I ran barefoot along damp, fresh sand by the seashore. The air there had a special quality.... The sound of breakers shut out all others, and I was startled, almost frightened, by the tremendous excitement a few steps could create. It was an intense moment of discovery of a source of power and beauty that one previously hardly dreamt existed.' In 2014, Bannister spoke publicly of his sense of irony at being diagnosed with Parkinson's disease.

Bettmann / Bettmann collection / Getty Images.

## I think, therefore I am

It is the brain, more than any other organ, that marks *Homo sapiens* apart from other animals. Our ability to be self-aware, to think, and to reason has formed the basis of scientific inquiry and philosophical speculation for millennia, as we attempt to rationalize and define this cognitive capability. What is the mind? Less tangible than other aspects of our being, throughout time humans have looked to explanations from philosophy, folklore, religion, and now science. The concept of the sense of self and being that defines us all—whether it be called the ego, nous, or the soul—remains intriguing yet elusive to explain. Aristotle held that the psyche (Greek: ψυχή=*soul*) was not separate to its housing body, as one could not exist without the other. This view was directly contradicted by the 17th-century French philosopher René Descartes. He proposed the theory of 'mind-body dualism' in which the mind, located in the pineal gland, is an entirely separate entity to the material being, and controls the avatar of the physical body like a puppeteer.

These conflicting viewpoints well illustrate the spectrum of neurological disease and its blurred boundaries with psychiatry and psychology. Jean Martin Charcot (1825–1893), often credited as the father of neurology, appeared to be more interested in this crossover than neuronal dysfunction: he spent two decades of his career studying hysteria in Paris, initially attributing symptoms of crying, fainting, and temporary blindness to an organic, inherited cause, before revising his view in later life to conclude that this was a psychological disease. His controversial work in this field inspired his student Sigmund Freud's psychoanalytical theories, and illustrated the very real power of the conscious, or sub-conscious, to produce physical symptoms, a phenomenon familiar to all physicians. Perhaps reflecting our frustration with this challenging area as much as his limited success, we remember Charcot for his more tangible outputs, for example, in giving the inaugural description of, among other things, multiple sclerosis, Parkinson's disease, amyotrophic lateral sclerosis (ALS), Charcot-Marie-Tooth disease, and of course his eponymous misshapen joint resulting from proprioceptive loss.

However, both Aristotle and Charcot would have surely agreed with Descartes' proposition of 'cogito ergo sum': I think, therefore I am. Whether mind and brain are dual or one, they are inextricably linked and the very nature of awareness proves its existence.

# Where is the lesion?

This is an important first question to ask and depends on recognizing characteristic patterns of cognitive, cranial nerve, motor, and sensory deficits. Locating a focal lesion can be aided by features such as asymmetry (eg one pupil dilated, one upgoing plantar response) or a spinal level (effects may be symmetrical below the lesion). ►Note that sometimes there is no single lesion, rather, a *general insult* causing a falsely localizing sign, eg abducens nerve palsy in ↑ICP. Other generalized causes of specific local effects are: trauma, encephalitis, anoxia, poisoning, or post-ictal states.

**Patterns of loss** Are crucial in locating the lesion (see BOX 'From findings to neuroanatomy'):

*Upper motor neuron (UMN):* Patterns of weakness are caused by damage anywhere along the corticospinal (=pyramidal) tracts: pathways that carry motor information from the precentral gyrus of the frontal cortex up to the synapse with anterior horn cells in the spinal cord (via the internal capsule, brainstem, and cord).[1] •UMN weakness affects groups rather than individual muscles, typically in a '*pyramidal*' pattern: in the arm, extensors are predominantly affected; in the leg the opposite is true and the flexors are the weaker muscle group. •*Spasticity* develops in the stronger muscle groups (arm flexors and leg extensors). It manifests as ↑*tone* that is *velocity-dependent*, ie the faster you move the patient's muscle, the greater the resistance, until it finally gives way (like a 'clasp-knife').[2] •Muscle wasting is less prominent. •There is *hyperreflexia:* reflexes are brisk. •*Plantars are upgoing* (+ve Babinski sign) ± *clonus* (elicited by rapidly dorsiflexing the foot; ≤3 rhythmic, downward beats of the foot are normal). •*Loss of skilled fine finger movements* may be greater than expected from the overall grade of weakness (see BOX 'Muscle weakness grading'). ►UMN lesions can mimic LMN lesions in the first few hours before spasticity and hyperreflexia develop.

*Lower motor neuron (LMN):* Lesions are caused by damage anywhere from the anterior horn cells distally, including the nerve roots, plexuses, and peripheral nerves. The pattern of weakness corresponds to the muscles supplied by the involved neurons. •Affected muscles show *wasting* ± *fasciculation* (spontaneous involuntary twitching). •There is *hypotonia/flaccidity:* the limb feels soft and floppy, providing little resistance to passive stretch. •*Reflexes are reduced* or absent; the *plantars remain flexor*. ►The chief differential for LMN weakness is a primary muscle disease (p510)—but here there is symmetrical loss, reflexes are normal or lost late, and there is no sensory component.

*Mixed LMN and UMN signs:* Can occur, eg in MND, ↓$B_{12}$, taboparesis (see p466).

*Sensory deficits:* It is important to test individual modalities and remember the quirks of our normal wiring: correctly interpreted, the distribution of sensory loss and the modality involved (pain, T°, touch, vibration, joint-position sense) will help refine and increase your confidence in localizing the lesion (see BOX 'From findings to neuroanatomy'). Pain and T° sensations travel along small fibres in peripheral nerves and the *anterolateral (spinothalamic) tracts* in the cord and brainstem (p516), whereas joint-position and vibration sense travel in large fibres in peripheral nerves and the large *dorsal columns* of the cord.

### Muscle weakness grading (MRC classification)

| | |
|---|---|
| • Grade 0 No muscle contraction | • Grade 3 Active movement against gravity |
| • Grade 1 Flicker of contraction | • Grade 4 Active movement against resistance |
| • Grade 2 Some active movement | • Grade 5 Normal power (allowing for age) |

Grade 4 covers a big range: 4−, 4, and 4+ denote movement against slight, moderate, and stronger resistance; avoid fudging descriptions—'strength 4/5 throughout' suggests a mild quadriparesis or myopathy. It is better to document 'poor effort' and the maximum grade for each muscle tested. ►Distribution of weakness tells us more than grade of weakness (grade does help document improvement).

1 Most of the fibres of the corticospinal (=pyramidal) tract decussate at the medullary pyramids, hence the name and the contralateral nature of symptoms. *Extrapyramidal* denotes damage to the basal ganglia and presents as parkinsonism (see p468, 494).
2 Whereas with *rigidity*, ↑ tone is not velocity-dependent but constant throughout passive movement.

## From findings to neuroanatomy: localizing the lesions

*Cortical lesions* may cause a particularly localized problem with hand or foot movements, with normal or even ↓tone—but ↑reflexes more proximally in the arm or leg will point to this being an UMN rather than LMN lesion. Sensory loss may be confined to discriminative functions, eg stereognosis and two-point discrimination (p499). *Internal capsule* (p470) and *corticospinal tract* lesions cause contralateral hemiparesis (UMN signs). There is generalized contralateral sensory loss. A cranial nerve palsy (III–XII) contralateral to a hemiplegia implicates the *brainstem* on the side of the cranial nerve palsy. *Lateral brainstem* lesions show both dissociated and crossed sensory loss with pain and T° loss on the side of the face ipsilateral to the lesion, and contralateral arm and leg sensory loss. *Cord lesions* causing paraparesis (both legs) or quadriparesis/tetraplegia (all limbs) are suggested by finding a motor and reflex level (power is unaffected above the lesion, with LMN signs *at* the level of the lesion, and UMN signs *below* the lesion). A *sensory level* is the hallmark (albeit a rather unreliable one)—ie decreased sensation below the level of the lesion with normal sensation above. Hemi-cord lesions cause a Brown-Séquard picture (p696): dorsal column loss on the side of the lesion and contralateral spinothalamic loss. *Dissociated sensory loss* may occur, eg in *cervical cord lesions*—loss of fine touch and proprioception without loss of pain and temperature (or vice versa, eg syringomyelia, p516; or cord tumours). *Peripheral neuropathies:* (p502.) Most cause distal weakness, eg foot-drop; weak hand (note: although Guillain-Barré syndrome typically presents as distal weakness that ascends over time, some atypical forms of Guillain-Barré syndrome may present with proximal weakness due to nerve root involvement). Sensory loss is typically worse distally (may involve all sensory modalities or be selective, depending on nerve fibre size involved). Involvement of a single nerve (mononeuropathy) occurs with trauma or entrapment (carpal tunnel, p503); involvement of several nerves (mononeuritis multiplex) is seen, eg in DM or vasculitis. Sensory loss from individual nerve lesions will follow anatomical territories (dermatomes, p454), which are usually more sharply defined than those of root lesions.

## Also ask

**What is the lesion?** Are the cells diseased, dysfunctional, disconnected (after a stroke), or under- or overexcited (migraine; epilepsy)? Is there loss of a specific type of nerve cell, as in MND or subacute combined degeneration of the cord (↓$B_{12}$, p334). Arteriopaths get strokes, tropical travellers get wormy lesions.

**Why?** Is there a systemic disease causing the neurology? Eg atrial fibrillation allowing an embolus to form, which then lodges in the patient's dominant hemisphere, causing an infarct that presents with dysphasia. Do a full systems examination and always beware the irregularly irregular pulse.

# Drugs and the nervous system

The brain is a gland that secretes both thoughts and molecules: both products are modulated by neurotransmitter systems. Some target sites for drugs:

1 Precursor of the transmitter (eg levodopa).
2 Interference with the storage of transmitter in vesicles within the pre-synaptic neuron (eg tetrabenazine).
3 Binding to the post-synaptic receptor site (eg bromocriptine).
4 Binding to the receptor-modulating site (eg benzodiazepines).
5 Interference with the breakdown of neurotransmitter within the synaptic cleft (eg acetylcholinesterase inhibitors; monoamine oxidase inhibitors—MAOIs).
6 Reduce reuptake of transmitter from synaptic cleft into pre-synaptic cell (eg selective serotonin reuptake inhibitors—SSRIs, eg fluoxetine; or serotonin and noradrenaline reuptake inhibitors—SNRIs, eg mirtazapine).
7 Binding to pre-synaptic autoreceptors (eg pindolol, a β-blocker with partial 5HT autoreceptor antagonist effects, can be used to augment antidepressant therapy).

Important neurotransmitters (and some associated drugs) are listed in table 10.1.



## Storms on the sea of neurotransmission

The complex and subtle mixture of chemicals that bathes our hundreds of trillions of synapses has been likened to a 'sea' of neurotransmitters. If so, it is surely a seascape of exquisite beauty, no matter how disturbed cognition may become by the storms that whip the waves on the surface. A well-chosen prescription may offer a lifeboat from such storms, but before prescribing any drug that modulates neurotransmission, consider that you are about to release a blunt and poorly understood force into a delicate environment. • The drug (or a metabolite) must be able to pass through the blood-brain barrier to have an effect. • The consequences of any sedative effects may be severe. • There will be short- and long-term SEs (eg tardive dyskinesia with neuroleptic drugs). • Most drugs affect many neurotransmitters, increasing therapeutic scope (and uncertainty) eg risperidone (blocks $D_2$, $5HT_2$, $\alpha_1$ and $\alpha_2$ receptors). • Metabolites of drugs may have equal or more important pharmacological effects resulting in clinically important interactions with drugs affecting eg hepatic metabolism. • One neurotransmitter may have many effects, eg dopaminergic neurons go awry in Parkinson's disease, schizophrenia, and addiction to drugs and gambling, by affecting motor control, motivation, effort, reward, analgesia, stress, learning, attention, and cognition.

Table 10.1 Major neurotransmitters and associated drugs

| Drugs increasing activity (≈agonists) | Drugs decreasing activity (≈antagonists) |
|---|---|
| **Dopamine** Acts on receptors $D_{1-5}$; affects mood and reward-seeking behaviour. | |
| Pramipexole, ropinirole, levodopa, apomorphine (Parkinson's*)<br>Cabergoline (hyperprolactinaemia; acromegaly). | Chlorpromazine (schizophrenia, *OHCS* p340)<br>Metoclopramide (nausea)<br>*Inhibition of dopamine signalling may lead to drug induced parkinsonism.* |
| **Serotonin (5-hydroxytryptamine; 5HT)** Many receptor types $5HT_{1-7}$; multiple effects. | |
| Lithium (mood stabilizer)<br>Sumatriptan (migraine)<br>Buspirone (partial agonist; anxiety)<br>Fluoxetine, sertraline (reuptake inhibitors; depression). | Ondansetron (nausea)<br>Mirtazapine (depression)<br>Olanzapine, clozapine (schizophrenia) |
| **Amino acids** Glutamate and aspartate act as excitatory transmitters on NMDA and non-NMDA receptors—relevant in epilepsy and CNS ischaemia. γ-aminobutyric acid (GABA) is mostly inhibitory. | |
| Gabapentin, valproate (GABA agonists; epilepsy and neuropathic pain)<br>Benzodiazepines (GABA agonists; sedation)<br>Baclofen (GABA agonists; spasticity)<br>Alcohol (GABA agonist)† | Memantine (glutamate antagonist; dementia) |
| **Acetylcholine** Multiple receptors classed into muscarinic and nicotinic types. Peripheral agonists used in glaucoma (pilocarpine); myasthenia (anticholinesterases). Peripheral antagonists used in asthma (ipratropium); incontinence; to dry secretions pre-op; to dilate pupils; to ↑ heart rate (atropine). Centrally acting drugs include: | |
| Donepezil, galantamine, rivastigmine (acetylcholinesterase inhibitors; dementia) | Procyclidine, trihexyphenidyl (drug-induced parkinsonism) |
| **Histamine and purines** (eg ATP) | |
| | Cyclizine (antihistamine; nausea)<br>Purinergic receptor blockers (emerging role in chronic pain). |
| **Neuropeptides** Multiple and growing list; includes opioids and substance P | |
| Exogenous opioids (wide-ranging analgesic and mood-related effects). | Aprepitant (↓chemotherapy-related nausea by blocking substance P receptors). |
| **Noradrenaline, adrenaline (=norepinephrine, epinephrine)** 4 receptor types: $\alpha_{1-2}$, $\beta_{1-2}$. Noradrenaline is more specific for α-receptors but both transmitters affect all receptors. In the periphery, α-receptors drive arteriolar vasoconstriction and pupillary dilation; $\beta_1$ stimulation leads to ↑ pulse and myocardial contractility; $\beta_2$ stimulation leads to bronchodilatation, uterine relaxation, and arteriolar vasodilation. Centrally acting drugs include: | |
| Clonidine (refractory hypertension)<br>Tricyclic antidepressants and venlafaxine (5HT and noradrenaline reuptake inhibitors; depression)<br>MAOIs | |

*Agonism at $D_3$ receptor agonists may cause pathological behavioural patterns, eg hypersexuality, pathological gambling or hobbying, and disorders of impulse control in people having no history of these.
†In chronic alcohol use, GABA receptors are downregulated; acamprosate, used in alcoholism, may help to maintain GABA signalling after alcohol withdrawal.

# Cerebral blood supply



Knowledge of the anatomy of the blood supply of the brain helps diagnosis and management of cerebrovascular disease (pp470–8). Always try to identify the area of brain that correlates with a patient's symptoms and identify the affected artery.

**Internal carotid arteries** Supply the majority of blood to the anterior two-thirds of the cerebral hemispheres and the basal ganglia (via the lenticulostriate arteries). At worst, internal carotid artery occlusion causes fatal total infarction of these areas. More often, the picture is like middle cerebral artery occlusion (see later in topic).

**The circle of Willis** (fig 10.2) An anastomotic ring at the base of the brain fed by the three arteries that supply the brain with blood: the internal carotids (anteriorly) and the basilar artery (posteriorly, formed by the joining of the vertebral arteries, which supply the brainstem). This arrangement may compensate for the effects of occlusion of a feeder vessel by allowing supply from unaffected vessels; however, the anatomy of the circle of Willis is variable and in many people it does not provide much protection.

**Cerebral arteries** Three pairs of arteries leave the circle of Willis to supply the cerebral hemispheres: the anterior, middle, and posterior cerebral arteries (figs 10.2, 10.3). The anterior and middle cerebrals are branches of the internal carotid arteries; in 80%, the basilar artery divides into the two posterior cerebral arteries. Ischaemia from occlusion of any one of them may be lessened by retrograde supply from leptomeningeal vessels.

*Anterior cerebral artery:* (fig 10.2) Supplies the frontal and medial part of the cerebrum. Occlusion may cause a weak, numb contralateral leg ± similar, if milder, arm symptoms. The face is spared. Bilateral infarction is a rare cause of paraplegia and an even rarer cause of akinetic mutism.

*Middle cerebral artery:* (fig 10.2) Supplies the lateral part of each hemisphere. Occlusion may cause contralateral hemiparesis, hemisensory loss (esp. face and arm), contralateral homonymous hemianopia due to involvement of the optic radiation, cognitive change including dysphasia with dominant hemisphere lesions, and visuospatial disturbance (eg cannot dress; gets lost) with non-dominant lesions.

*Posterior cerebral artery:* (figs 10.2, 10.4) Supplies the occipital lobe. Occlusion gives contralateral homonymous hemianopia (often with macula sparing).

**Vertebrobasilar circulation** Supplies the cerebellum, brainstem, occipital lobes; occlusion causes signs relating to any or all three: hemianopia; cortical blindness; diplopia; vertigo; nystagmus; ataxia; dysarthria; dysphasia; hemi- or quadriplegia; unilateral or bilateral sensory symptoms; hiccups; coma. Infarctions of the brainstem can produce various syndromes, eg *lateral medullary syndrome*, in which occlusion of one vertebral artery or the posterior inferior cerebellar artery causes infarction of the lateral medulla and the inferior cerebellar surface (→vertigo, vomiting, dysphagia, nystagmus, ipsilateral ataxia, soft palate paralysis, ipsilateral Horner's syndrome, and a crossed pattern sensory loss—analgesia to pin-prick on ipsilateral face and contralateral trunk and limbs). *Locked-in syndrome* is caused by damage to the ventral pons due to pontine artery occlusion. Patients are unable to move, but retain full cognition and awareness, communicating by blinking, electronic boards, or special computers. Right-to-die legislation may be invoked...as one sufferer blinked: 'My life is dull, miserable, demeaning, undignified, and intolerable.' Locked-in syndrome is different from other right-to-die conditions because patients need someone to do the act for them.

*Subclavian steal syndrome:* Subclavian artery stenosis proximal to the origin of the vertebral artery may cause blood to be *stolen* by retrograde flow down this vertebral artery down into the arm, causing brainstem ischaemia typically after use of the arm. Suspect if the BP in each arm differs by >20mmHg.

### 'Dizzy-plus' syndromes and arterial events

- SCA→dizzy
- AICA→dizzy and deaf
- PICA→dizzy and dysphagic and dysphonic.

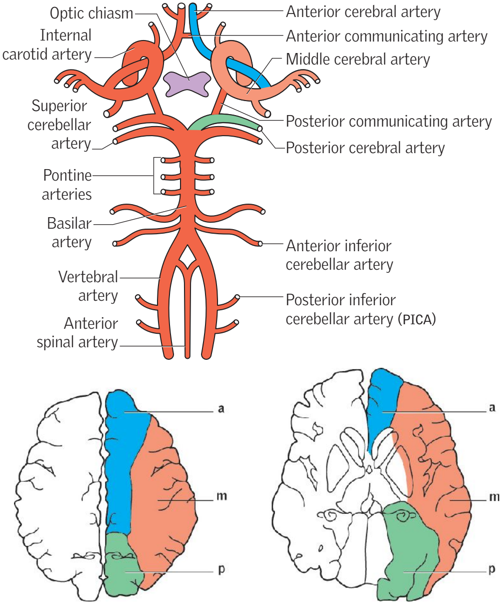



Fig 10.2 The circle of Willis at the base of the brain. See also figs 10.17, 10.18.

Fig 10.3 Berry aneurysm at junction of posterior communicating artery with internal carotid (p478). ©Dr D Hamoundi.

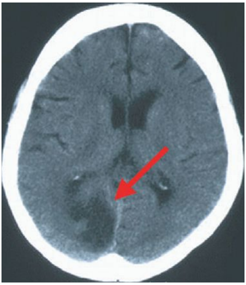


Fig 10.4 CT of stroke in posterior cerebral artery territory. ©J Trobe.

## Thomas Willis

Thomas Willis (1621–1675) is one of those happy Oxford heroes who hold a bogus DM degree, awarded in 1646 for his Royalist sympathies while at Christ Church, the most loyally royal college in the University. He had a busy life inventing terms such as 'neurology' and 'reflex'. Not only has his name been given to his famous circle, but he was the first to describe myasthenia gravis, whooping cough, and the sweet taste of diabetic urine. He was the first person (few have followed him) to know the course of the spinal accessory nerve. He is unusual among Oxford neurologists in that he developed the practice of giving his lunch away to the poor. He also espoused iatrochemistry: a theory of medicine according to which all morbid conditions of the body can be explained by disturbances in the fermentations and effervescences of its humours.

# Testing peripheral nerves

While there is some anatomical variation between individuals in ascribing particular nerve roots to muscles, tables 10.2–10.4 represent a reasonable compromise. Dermatomes and sensory nerve roots are shown in figs 10.5–10.9, pp454–5.

Remember to test proximal muscle power: ask the patient to sit from lying, to pull you towards him/herself, and to rise from squatting (if reasonably fit).

►Observe walking—easy to forget, even if the complaint is of difficulty walking!

►Don't be caught out by weakness secondary to musculoskeletal pathology—the traditional neurological examination relies on the musculoskeletal system being intact. Ruptured tendons and fractures may mimic focal neurological lesions (especially in patients who can't give a clear history).

Table 10.2 Assessment of peripheral nerve function in the lower limb

| Nerve root | Muscle | Test by asking the patient to: |
|---|---|---|
| **Femoral nerve** | | |
| L1, 2, 3 | Iliopsoas (also supplied via L1, 2, & 3 spinal nerves) | Flex hip against resistance with knee flexed and lower leg supported: patient lies on back |
| L2, 3, 4 | Quadriceps femoris | Extend at knee against resistance. Start with knee flexed |
| **Obturator nerve** | | |
| L2, 3, 4 | Hip adductors | Adduct leg against resistance |
| **Inferior gluteal nerve** | | |
| L5, S1, S2 | Gluteus maximus | Hip extension ('bury heel into the couch')—with knee in extension |
| **Superior gluteal nerve** | | |
| L4, 5, S1 | Gluteus medius and minimus | Abduction and internal hip rotation with leg flexed at hip and knee |
| **Sciatic and common peroneal* nerves; sciatic and tibial** nerves** | | |
| *L4, 5 | Tibialis anterior | Dorsiflex ankle |
| *L5, S1 | Extensor digitorum longus | Dorsiflex toes against resistance |
| *L5, S1 | Extensor hallucis longus | Dorsiflex hallux against resistance |
| *L5, S1 | Peroneus longus and brevis | Evert foot against resistance |
| *L5, S1 | Extensor digitorum brevis | Dorsiflex proximal phalanges of toes |
| (*)L5, S1, 2 | Hamstrings (short head of biceps femoris is from the common peroneal nerve) | Flex knee against resistance |
| **L4, 5 | Tibialis posterior | Invert plantarflexed foot |
| **S1, 2 | Gastrocnemius | Plantarflex ankle or stand on tiptoe |
| **L5, S1, 2 | Flexor digitorum longus | Flex terminal joints of toes |
| **S1, 2 | Small muscles of foot | Make the sole of the foot into a cup |

Table 10.3 Rapid screening tests for peripheral nerve roots

| | | | | | |
|---|---|---|---|---|---|
| **Shoulder** | Abduction | C5 | **Hip** | Flexion | L1–L2 |
| | Adduction | C5–C7 | | Adduction | L2–3 |
| **Elbow** | Flexion | C5–C6 | | Extension | L5–S1 |
| | Extension | C7 | **Knee** | Flexion | L5–S1 |
| **Wrist** | Flexion | C7–8 | | Extension | L3–L4 |
| | Extension | C7 | **Ankle** | Dorsiflexion | L4 |
| **Fingers** | Flexion | C8 | | Eversion | L5–S1 |
| | Extension | C7 | | Plantarflexion | S1–S2 |
| | Abduction | T1 | **Toe** | Big toe extension | L5 |

Table 10.4 Assessment of peripheral nerve function in the upper limb

| Nerve root | Muscle | Test by asking the patient to: |
|---|---|---|
| C3, 4 | Trapezius | Shrug shoulder (via accessory nerve) |
| C5, 6, 7 | Serratus anterior | Push arm forward against resistance; look for scapula winging (p511) if weak |
| C5, 6 | Pectoralis major (P major) clavicular head | Adduct arm from above horizontal, and push it forward |
| C6, 7, 8 | P major sternocostal head | Adduct arm below horizontal |
| C5, 6 | Supraspinatus | Abduct arm the first 15° |
| C5, 6 | Infraspinatus | Externally rotate semi-flexed arm, elbow at side |
| C6, 7, 8 | Latissimus dorsi | Adduct arm from horizontal position |
| C5, 6 | Biceps | Flex supinated forearm |
| C5, 6 | Deltoid | Abduct arm between 15° and 90° |
| **Radial nerve (p502)** | | |
| C6, 7, 8 | Triceps | Extend elbow against resistance |
| C5, 6 | Brachioradialis | Flex elbow with forearm half way between pronation and supination |
| C5, 6 | Extensor carpi radialis longus | Extend wrist to radial side |
| C6, 7 | Supinator | Arm by side, resist hand pronation |
| C7, 8 | Extensor digitorum | Keep fingers extended at MCP joint |
| C7, 8 | Extensor carpi ulnaris | Extend wrist to ulnar side |
| C7, 8 | Abductor pollicis longus | Abduct thumb at 90° to palm |
| C7, 8 | Extensor pollicis brevis | Extend thumb at MCP joint |
| C7, 8 | Extensor pollicis longus | Resist thumb flexion at IP joint |
| **Median nerve (p502)** | | |
| C6, 7 | Pronator teres | Keep arm pronated against resistance |
| C6, 7 | Flexor carpi radialis | Flex wrist towards radial side |
| C7, 8, T1 | Flexor digitorum superficialis | Resist extension at PIP joint (with proximal phalanx fixed by the examiner) |
| C7, 8 | Flexor digitorum profundus I & II | Resist extension at index DIP joint of index finger |
| C7, 8, T1 | Flexor pollicis longus | Resist thumb extension at interphalangeal joint (fix proximal phalanx) |
| C8, T1 | Abductor pollicis brevis | Abduct thumb (nail at 90° to palm) |
| C8, T1 | Opponens pollicis | Thumb touches base of 5th fingertip (nail parallel to palm) |
| C8, T1 | 1st lumbrical/interosseus (median and ulnar nerves) | Extend PIP joint against resistance with MCP joint held hyperextended |
| **Ulnar nerve (p502)** | | |
| C7, 8, T1 | Flexor carpi ulnaris | Flex wrist to ulnar side; observe tendon |
| C7, C8 | Flexor digitorum profundus III & IV | Resist extension of distal phalanx of 5th finger while you fix its middle phalanx |
| C8, T1 | Dorsal interossei | Finger abduction: cannot cross the middle over the index finger (tests index finger adduction too) |
| C8, T1 | Palmar interossei | Finger adduction: pull apart a sheet of paper held between middle and ring finger DIP joints of both hands; the paper moves on the weaker side* |
| C8, T1 | Adductor pollicis | Adduct thumb (nail at 90° to palm) |
| C8, T1 | Abductor digiti minimi | Abduct little finger |
| C8, T1 | Flexor digiti minimi | Flex little finger at MCP joint |

*Also, metacarpophalangeal joint flexion may be more on the affected side as flexor tendons are recruited—the basis of Froment's paper sign. Wartenberg's sign is persistent little finger abduction.

# Dermatomes and peripheral nerves



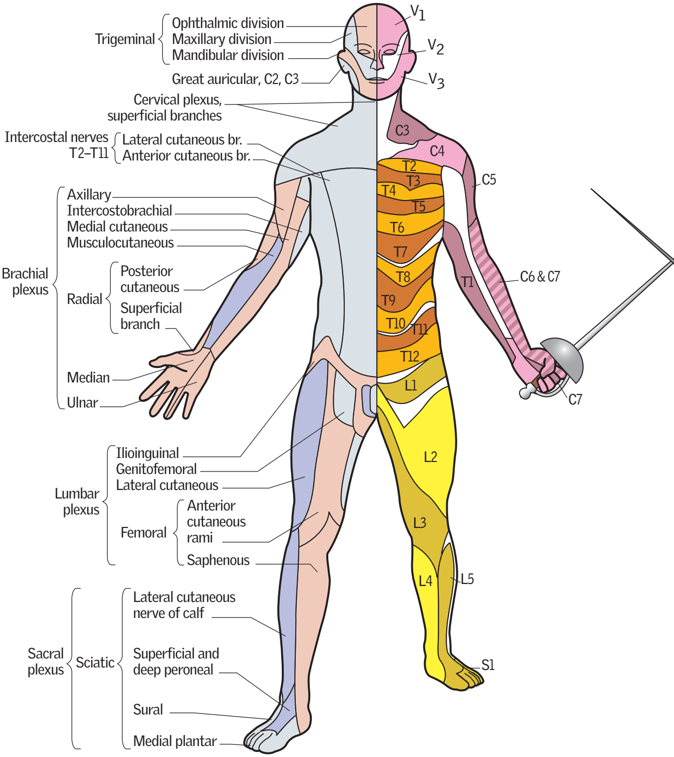



Fig 10.5 The white areas denote *terra incognita*: considerable inter-individual variation exists, and no single best option can be given.

Fig 10.6 Pain in a dermatomal distribution suggests a problem with a cranial nerve or dorsal root ganglion (radiculopathy)—where the cell bodies of sensory fibres live. What is the dermatome? What is the lesion? See p404 for the answer.

**Aim to keep a few key dermatomes up your sleeve (C5–T2)**

| | |
|---|---|
| C3–4 | Clavicles |
| C6–7 | Lateral arm/forearm |
| T1 | Medial side of arm |
| C6 | Thumb |
| C7 | Middle finger |
| C8 | Little finger |
| T4 | Nipples |
| T10 | Umbilicus |
| L1 | Inguinal ligament |
| L2–3 | Anterior and inner leg |
| L5 | Medial side of big toe |
| L5, S1–2 | Posterior and outer leg |
| S1 | Lateral margin of foot and little toe |
| S2–4 | Perineum |

Rough approximations!

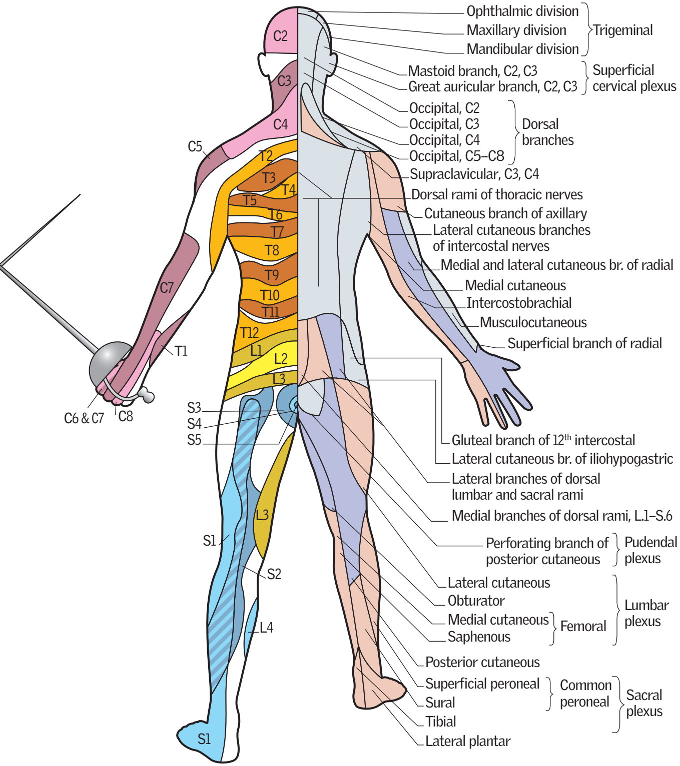



**Fig 10.7** Posterior view.

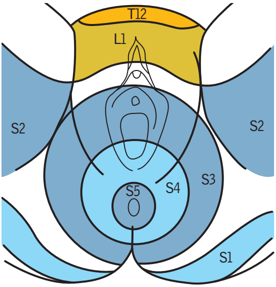



**Fig 10.8** The anterior ⅓ of the scrotum is L1; the posterior ⅔ is S3. The penis is S2/3 (L1 at its root).

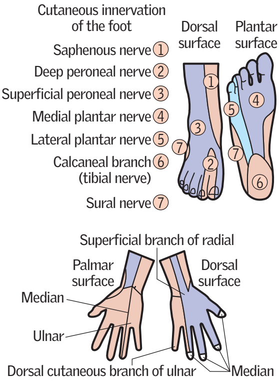



**Fig 10.9** Feet and hands.

# Headache



Every day, *thousands* of people visit the doctor complaining of headache. Tension headaches are the most common, but beware the disabling and treatable (migraine, cluster headache), and the sinister (space-occupying lesions, meningitis, subarachnoid haemorrhage). A good history is the key. Ask about:

**Onset** *Rapid onset* headaches are concerning; the key diagnosis to rule out here is •*subarachnoid haemorrhage* (SAH, p478), *sudden*-onset, 'worst ever' headache, often occipital, stiff neck, focal signs, ↓consciousness. Other differentials include: •*Meningitis* (p822): fever, photophobia, stiff neck, purpuric rash, coma. May be associated with neck stiffness (≈meningeal irritation). Do an LP, start antibiotics. •*Encephalitis* (p824): fever, odd behaviour, fits, or reduced consciousness. Do an urgent CT head and LP to look for signs of infection. •*Post-coital headache*. *Subacute/gradual onset headaches:* •*Venous sinus thrombosis* (p480): subacute headache, papilloedema. •*Sinusitis:* dull, constant ache over frontal or maxillary sinuses, with tenderness ± postnasal drip. Pain is worse on bending over. Ethmoid or sphenoid sinus pain is felt deep in the midline at the root of the nose. Common with coryza (p406). The pain lasts ~1–2 wks. CT can confirm diagnosis but is rarely needed. •*Tropical illness:* eg malaria: travel history, flu-like illness (p416); typhus (p415). •*Intracranial hypotension:* CSF leakage, eg iatrogenic after LP or epidural anaesthesia. Suspect if headaches worse on standing; treat with epidural blood patch over leak, if conservative management with IV fluids and caffeine fails.

**Character** Tight band? Think *tension headache* (the usual cause of bilateral, non-pulsatile headache ± scalp muscle tenderness). Throbbing/pulsatile/lateralizing? Think *migraine* (p458).

**Frequency** Headaches that recur tend to be benign: • *Migraine:* p458. • *Cluster headache:* (see BOX 'Cluster headache') • *Trigeminal neuralgia:* (see BOX 'Trigeminal neuralgia') • *Recurrent (Mollaret's) meningitis:* suspect if fever/meningism with each headache. Send CSF for herpes simplex PCR (HSV2). Is there access to subarachnoid spaces via a skull fracture, or a recurring cause of aseptic meningitis (SLE, eg abducens nerve palsy, Behçet's, sarcoid)?

**Duration** Chronic, progressive headaches can indicate ↑ICP. Typically worse on waking, lying, bending forward, or coughing. Also: vomiting, papilloedema, seizures, false localizing signs, or odd behaviour. Do imaging to exclude a space-occupying lesion, and consider idiopathic intracranial hypertension. ►LP is contraindicated until after imaging.

**Associated features** *Eye pain ± reduced vision:* Think *acute glaucoma*. Typically elderly, long-sighted people. Constant pain develops rapidly around one eye, radiating to the forehead with markedly reduced vision, visual haloes, and a red, congested eye (p561). ►Seek expert help at once. If delay in treatment of >1h is likely, give eye drops (eg 0.5% timolol maleate ± 2% pilocarpine) and acetazolamide 500mg PO. *Jaw claudication tender with thickened, pulseless temporal arteries: Giant cell arteritis:* (p556) Subacute-onset headache with ESR >40mm/h. ►Exclude in all >50yrs old with a headache that has lasted a few weeks: prompt diagnosis and steroids avoid blindness.

**Precipitating causes** *Head trauma:* Commonly causes localized pain but can be more generalized. It lasts ~2wks; often resistant to analgesia. Do CT to exclude subdural or extradural haemorrhage if drowsiness ± lucid interval, or focal signs (p482). *Also ask about:* Analgesia, sex, food (eg chocolate, cheese, coffee).

**Red flags** See p780.

**Drug history** *Exclude medication overuse (analgesic rebound) headache:* Culprits are mixed analgesics (paracetamol+codeine/opiates), ergotamine, and triptans. This is a common reason for episodic headache becoming chronic daily headache. Analgesia must be withdrawn—aspirin or naproxen may mollify the rebound headache. A preventive may help once off other drugs (eg tricyclics, valproate, gabapentin; p504). Limit use of over-the-counter analgesia (no more than 6d per month).

**Social history** Ask about stress or recent life events; may not explain the pathology, but will help you appreciate the context in which symptoms are experienced.

## Cluster headache

Cluster headache may be the most disabling of the primary headache disorders. The cause (unknown ♂:♀ ≥5:1; onset at any age; commoner in smokers).

**Symptoms** Rapid-onset of excruciating pain around one eye that may become watery and bloodshot with lid swelling, lacrimation, facial flushing, rhinorrhoea, miosis ± ptosis (20% of attacks). Pain is strictly unilateral and almost always affects the same side. It lasts 15–180min, occurs once or twice a day, and is often nocturnal. Clusters last 4–12wks and are followed by pain-free periods of months or even 1–2yrs before the next cluster. Sometimes it is chronic, not episodic.

**Treatment** *Acute attack:* 'Keep calm … carry oxygen': give 100% $O_2$ for ~15min via non-rebreathable mask (not if COPD); sumatriptan SC 6mg at onset (or zolmitriptan nasal spray 5mg).

**Preventives** *Avoid triggers:* Eg alcohol. *Medication:* Consider: corticosteroids (short term only; many SE); verapamil 360mg, lithium 900mg (monitor carefully).

## Trigeminal neuralgia

*Symptoms:* Paroxysms of intense, stabbing pain, lasting seconds, in the trigeminal nerve distribution. It is unilateral, typically affecting mandibular or maxillary divisions. The face screws up with pain (*hence tic douloureux*). *Triggers:* Washing affected area, shaving, eating, talking, dental prostheses. *Typical patient:* ♂ >50yrs old; in Asians ♀:♂ ≈ 2:1. *Secondary causes:* Compression of the trigeminal root by anomalous or aneurysmal intracranial vessels or a tumour, chronic meningeal inflammation, MS, zoster, skull base malformation (eg Chiari). *MRI:* Is necessary to exclude secondary causes (~14% of cases). *℞:* Carbamazepine (start at 100mg/12h PO; max 400mg/6h; lamotrigine; phenytoin 200–400mg/24h PO; or gabapentin (p504). If drugs fail, surgery may be necessary. This may be directed at the peripheral nerve, the trigeminal ganglion, or the nerve root. *Microvascular decompression:* Anomalous vessels are separated from the trigeminal root. Stereotactic gamma knife surgery can work, but length of pain relief and the time to treatment response are limiting factors. *Facial pain ΔΔ:* p65.

# Migraine

15% of us suffer from migraines, (♀:♂ ≈3:1); the economic costs extend to £billions/yr.

**Symptoms** *Classically:* •Visual or other aura (see below) lasting 15–30min followed within 1h by unilateral, throbbing headache. Or: •Isolated aura with no headache; •Episodic severe headaches without aura, often premenstrual, usually unilateral, with nausea, vomiting ± photophobia/phonophobia ('*common migraine*'). There may be allodynia—all stimuli produce pain: 'I can't brush my hair, wear earrings or glasses, or shave, it's so painful.' *Prodrome:* Precedes headache by hours/days: yawning, cravings, mood/sleep change. *Aura:* • *Visual:* chaotic distorting, 'melting' and jumbling of lines, dots, or zigzags, scotomata or hemianopia; • *Somatosensory:* paraesthesiae spreading from fingers to face; • *Motor:* dysarthria and ataxia (basilar migraine), ophthalmoplegia, or hemiparesis; • *Speech:* (8% of auras) dysphasia or paraphasia.

**Partial triggers** Seen in 50%: **CHOCOLATE** or: **c**hocolate, **h**angovers, **o**rgasms, **c**heese/caffeine, **o**ral contraceptives, **l**ie-ins, **a**lcohol, **t**ravel, or **e**xercise.

**Associations** Obesity, family history.

**Diagnosis** Clinical, based on the history. *Diagnostic criteria if no aura:* ≥5 headaches lasting 4–72h + nausea/vomiting (or photo/phonophobia) + any 2 of: • Unilateral • Pulsating • Impairs (or worsened by) routine activity.

**Differentials** Cluster or tension headache, cervical spondylosis, ↑BP, intracranial pathology, sinusitis/otitis media, dental caries. TIAs may mimic migraine aura.

**Management** ►Avoid identified triggers and ensure analgesic rebound headache is not complicating matters (p456). *Prophylactic treatment:* Can achieve ~50%↓ in attack frequency in most patients; consider after risks and benefits discussion. 1st line: Propranolol 40–120mg/12h or topiramate 25–50mg/12h (teratogenic, can interfere with Pill efficacy). Amitriptyline 10–75mg nocte can be used, though this is off-licence. ►Patients may be on previously-recommended prophylactic agents (eg valproate, pizotifen, pregabalin or ACE-i): if achieving good control then continue as required. 12-weekly botulinum toxin type A injections are a last resort in chronic migraine. *Treatment during an attack:* NICE recommends an oral triptan (or nasal in 12–17y) combined with either an NSAID or paracetamol.[1] Monotherapy with any of the above (or aspirin 900mg) can also be considered. Anti-emetics may help even in the absence of nausea and vomiting. Triptans are CI if IHD, coronary spasm, uncontrolled ↑BP, recent lithium, SSRIs, or ergot use. Rare SE: arrhythmias or angina ± MI, even if no pre-existing risk. *Non-pharmacological therapies:* Warm or cold packs to the head, or rebreathing into paper bag (↑$P_aCO_2$) may help abort attacks. Butterbur extracts or riboflavin supplementation may have a role. NICE recommend 10 sessions of acupuncture over 5–8 weeks if both topiramate and propranolol are unsuitable or ineffective. Transcutaneous nerve stimulation may help.

**Considerations in females** Incidence of migraine (especially with aura) + ischaemic stroke is increased by use of a combined OCP. Use progesterone-only or non-hormonal contraception in migraine + aura, though a low dose combined OCP can be used in those *without* aura. Further ↑risk: • Smoking • Age >35yrs • BP↑ • Obesity (body mass index >30) • Diabetes mellitus • Hyperlipidaemia • Family history of arteriopathy <45yrs. ►Warn patients to stop OCP at once if they develop aura or worsening migraine; see *OHCS* p301. *Perimenstrual migraine:* If uncontrolled with standard treatment and the onset of headache is predictable then consider frovatriptan 2.5mg BD or zolmitriptan 2.5mg BD/TDS on the days migraine is expected. *Pregnancy:* Migraine often improves; if not, get help—worsening headaches in pregnancy are associated with a greater risk of pre-eclampsia and cardiovascular complications. Offer paracetamol 1st line. Triptans and NSAIDs can be used but discuss risks and benefits with patients first. Don't use aspirin if breastfeeding. Anti-emetic: cyclizine or promethazine. *Prophylaxis:* seek specialist advice.

### What is going on in migraine?

Despite the high prevalence of migraines, the underlying pathophysiology is poorly understood. The previously favoured theory of dilatation of cerebral and meningeal arteries has been largely disproven, so what is the cause? • MRI during attacks shows episodic cerebral oedema, dilatation of intracerebral vessels, and ↓water diffusion not respecting vascular territories, so the primary event may be neurological. • PET suggests migraine is a subcortical disorder affecting the modulation of sensory processing • Magneto-encephalographic (MEG) studies have shown resting (interictal) hyperexcitability at least in the visual cortex, suggesting a failure of inhibitory circuits. • Hormones play a role: the incidence of migraine in both pre-pubertal and post-menopausal women is equal to men, yet increases to 3:1 during reproductive years, with 50% of females reporting synchrony of migraines with the menstrual cycle. • Elevated levels of 5-HT metabolites in the urine of patients during migraine attacks was first reported in 1972, and while its exact significance is controversial, the efficacy of triptans ($5HT_{1B/1D}$ agonists) support its role in migraine. • Triptans also inhibit release of substance P and pro-inflammatory neuropeptides, blocking transmission from the trigeminal nerve and implicating trigeminal nerve dysfunction.

### The bigger picture

Vincent Van Gogh suffered from 'sick headaches', widely believed to have been migraines. Could his swirling, cascading starry night (fig 10.10) be a visual aura? ►Just as with the fragile mental health of Van Gogh, migraine often co-exists with other chronic conditions—and the combined negative impact on physical and mental health is immense. Don't treat each disease in isolation. Rather, attempt to restore a good relationship with the self—and the recovery of the purpose of life through dialogue. This is the hardest but the most rewarding task, and may save some ears.

Fig 10.10 'The Starry Night' Vincent Van Gogh 1889
World History Archive/Ann Ronan Collection / Age Fotostock

# Blackouts

Causes of collapse ± loss of consciousness (LOC) are many; take a careful history (BOX).

**Vasovagal (neurocardiogenic) syncope** Occurs due to reflex bradycardia ± peripheral vasodilation provoked by emotion, pain, or standing too long (it cannot occur when lying down). Onset is over seconds (*not* instantaneous), and is often preceded by pre-syncopal symptoms, eg nausea, pallor, sweating, and narrowing of visual fields. Brief clonic jerking of the limbs may occur due to cerebral hypoperfusion, but there is no tonic/clonic sequence. Urinary incontinence is uncommon, and there is no tongue-biting. Unconsciousness usually lasts for ~2min and recovery is rapid.

**Situation syncope** Symptoms as for vasovagal syncope but with a clear precipitant: *cough syncope* occurs after a paroxysm of coughing; *effort syncope* is brought on by exercise; there is usually a cardiac cause, eg aortic stenosis, HCM; *micturition syncope* happens during or after urination: mostly men, at night.

**Carotid sinus syncope** Hypersensitive baroreceptors cause excessive reflex bradycardia ± vasodilation on minimal stimulation (eg head-turning, shaving).

**Epilepsy** (p490) Features suggestive of this diagnosis include: attacks when asleep or lying down; aura; identifiable triggers (eg TV); altered breathing; cyanosis; typical tonic-clonic movements; incontinence of urine; tongue-biting; prolonged post-ictal drowsiness, confusion, amnesia, and transient focal paralysis (Todd's palsy).

**Stokes-Adams attacks** Transient arrhythmias (eg bradycardia due to complete heart block) cause ↓cardiac output and LOC. The patient falls to the ground (often with *no* warning except palpitations; injuries are common), and is pale, with a slow or absent pulse. Recovery is in seconds: the patient flushes, the pulse speeds up, and consciousness is regained. As with vasovagal syncope, anoxic clonic jerks may occur in prolonged LOC. Attacks may happen several times a day and in any posture.

**Other causes** *Hypoglycaemia:* (p214) Tremor, hunger, and perspiration herald light-headedness or LOC; rare in non-diabetics. *Orthostatic hypotension:* Unsteadiness or LOC on standing from lying in those with inadequate vasomotor reflexes: the elderly; autonomic neuropathy (p505); antihypertensive medication; overdiuresis; multisystem atrophy (MSA; p494). *Anxiety:* Hyperventilation, tremor, sweating, tachycardia, paraesthesiae, light-headedness, and no LOC suggest a panic attack. *Drop attacks:* Sudden fall to the ground *without* LOC. Mostly benign and due to leg weakness but may also be caused by hydrocephalus, cataplexy, or narcolepsy. *Factitious blackouts:* Pseudoseizures, Münchausen's (p706).

**Examination** Cardiovascular, neurological. Measure BP lying and standing.

**Investigation** All with recurrent syncope (or falls) need cardiac assessment—urgently if associated with palpitations, arrhythmias, 3rd-degree AV block, or prolonged QT interval (p711). ECG ± 24h ECG (arrhythmia, long QT, eg Romano-Ward, p96); U&E, FBC, $Mg^{2+}$, $Ca^{2+}$, glucose; tilt-table test;[3] EEG, sleep EEG; echocardiogram; CT/MRI brain; ABG if practical (↓$P_aCO_2$ in attacks suggests hyperventilation as the cause).

►While the cause is being elucidated, advise against driving (see p158).

3 Patient is subject to continuous ECG and BP monitoring while strapped to a table and moved rapidly from resting horizontal position to vertical. Induction of symptoms with inappropriate BP drop >30mmHg or bradycardia suggests neurally mediated syncope. Consider pacing.

## Blackout history

It is vital to establish exactly what patients mean by 'blackout': loss of consciousness?—a fall to the ground without loss of consciousness?—vertigo or visual disturbance? Talk to the patient *and* witnesses and let them tell you as much as possible without prompting or leading. Ask:

- Does the patient lose awareness?
- Does the patient injure themselves?
- Does the patient move? Are they stiff or floppy? (A tonic phase preceding clonic jerking points towards epilepsy.)
- Is there incontinence? (More common in epilepsy, but can occur with syncope.)
- Does their complexion change? (Pale/cyanosis suggests epilepsy; very pale/white suggests syncope or arrhythmia.)
- Does the patient bite the side of their tongue? (Suggests epilepsy.)
- Are there associated symptoms eg palpitations, sweats, pallor, chest pain, dyspnoea (see fig 10.11)?
- How long does the attack last?

*Before the attack:*

- Is there any warning?—Eg typical epileptic aura or cardiac pre-syncope.
- In what circumstances do attacks occur? (If watching TV, consider epilepsy).
- Can the patient prevent attacks?

*After the attack:*

- How much does the patient remember about the attack?
- Is there muscle ache? (Suggests a tonic-clonic seizure.)
- Is the patient confused or sleepy? (Suggests epilepsy.)

*Background to attacks:*

- When did they start?
- Are they getting more frequent?
- Is anyone else in the family getting them? Sudden arrhythmic death may leave no evidence at postmortem, or there may be hereditary cardiomyopathy (refer those with a relative who has had a sudden unexplained death <40yrs old).

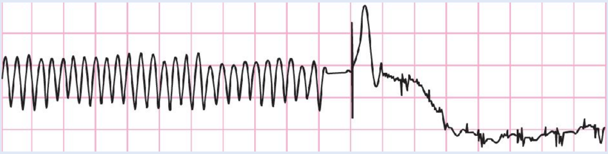


**Fig 10.11** VT causing blackout in Brugada syndrome (p695). This patient had been treated with an implantable defibrillator (see p132).

# Vertigo and dizziness



**Is this vertigo?** Complaints of 'dizzy spells' are very common and are used by patients to describe many different sensations. True vertigo is a *hallucination of movement*, often rotatory, of the patient or their surroundings. In practice, simple 'spinning' is rare—the floor may tilt, sink, or rise. The key to diagnosis is to find out exactly what the patient means by 'dizzy:' if this is not vertigo or if atypical symptoms are present consider other causes, eg if there is loss of awareness, think of epilepsy or syncope; if there is faintness, lightheadedness, or palpitations, think of anaemia, dysrhythmia, anxiety, or hypotension.

*Associated symptoms:* Difficulty walking or standing (may fall suddenly to the ground), relief on lying or sitting still (vertigo is almost always worsened by movement); nausea, vomiting, pallor, sweating. Associated hearing loss or tinnitus implies labyrinth or VIIIth nerve involvement.

## Causes

*Benign positional vertigo:* Occurs on head movement due to disruption of debris in the semicircular canal of the ears (canalolithiasis). Fatiguable nystagmus on performing the Hallpike manoeuvre is diagnostic; Epley manoeuvres clear the debris (*OHCS* p555).

*Acute labyrinthitis (vestibular neuronitis):* Abrupt onset of severe vertigo, nausea, vomiting ± prostration. No deafness or tinnitus. Causes: virus; vascular lesion. Severe vertigo subsides in days, complete recovery takes 3–4wks. ℞: reassure. Sedate.

*Ménière's disease:* Increased pressure in the endolymphatic system of the inner ear causes recurrent attacks of vertigo lasting >20min, fluctuating (or permanent) sensorineural hearing loss, and tinnitus (with a sense of aural fullness ± falling to one side). ℞: bed rest and reassurance in acute attacks. An antihistamine (eg cinnarizine) is useful if prolonged, or buccal prochlorperazine if severe, for up to 7d.

*Ototoxicity:* Aminoglycosides, loop diuretics, or cisplatin can cause deafness±vertigo.

*Acoustic neuroma:* (figs 10.12, 10.13) Doubly misnamed: it is a Schwannoma (not neuroma) arising from the vestibular (not auditory) nerve. They account for 80% of cerebellopontine angle tumours and often present with unilateral hearing loss, with vertigo occurring later. Growth rate is slow (usually 1–2 mm/year) and can be predicted by serial MRIs. With progression, ipsilateral Vth, VIth, IXth, and Xth nerves may be affected (also ipsilateral cerebellar signs). Signs of ↑ICP occur late, indicating a large tumour. Commoner in ♀ and neurofibromatosis (esp. NF2, p514).

*Traumatic damage:* If trauma affects the petrous temporal bone or the cerebellopontine angle then the auditory nerve may be damaged, causing vertigo, deafness, and/or tinnitus.

*Herpes zoster:* Herpetic eruption of the external auditory meatus; facial palsy ± deafness, tinnitus, and vertigo (Ramsay Hunt syndrome, see p501).

*Others:* Vertiginous epilepsy; MS; stroke/TIA; migraine; motion sickness; alcohol intoxication.

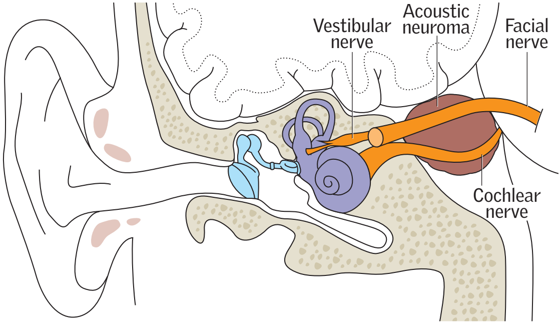



Fig 10.12 An acoustic neuroma (vestibular schwannoma) growing dangerously near the facial nerve.

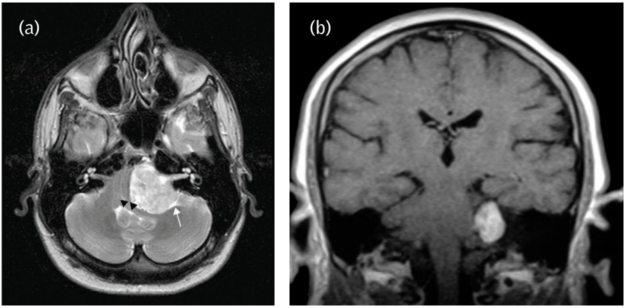



Fig 10.13 Large vestibular schwannoma: axial T2W (a) and contrast-enhanced coronal MRI (b).

## Hearing loss

**Whisper test** A simple but effective crude assessment of hearing: whisper numbers in one ear while blocking the other. Ask your patient to repeat the number. Make sure that failure is not from misunderstanding.

**Tuning fork tests** *Rinne:* Hold a vibrating tuning fork (512Hz or 256Hz) on the mastoid to test bone conduction (BC). When the sound is no longer audible move it in front of the ear with the prongs perpendicular to the auditory canal to test air conduction (AC). If there is no conductive deficit (ie in normal hearing or sensorineural hearing loss), AC is better than BC and the patient will be able to hear the note again. This is a 'Rinne positive' result. If BC is better than AC (Rinne negative), this indicates conductive deafness >20dB. A false-negative may occur in severe sensorineural hearing loss (SNHL) as the contralateral cochlea picks up the sound by bone conduction. *Weber:* With the vibrating tuning fork on the vertex or forehead, ask the patient which ear the sound is louder in. Sound localizes to the affected ear with conductive loss (>10dB loss), to the contralateral ear in SNHL, and to the midline if both ears are normal (or if bilateral SNHL).



**Conductive deafness** *Causes:* Wax (remove, eg by syringing with warm water after softening with olive oil drops), otosclerosis, otitis media, or glue ear (*OHCS* p546).

**Chronic sensorineural deafness** Often due to accumulated environmental noise toxicity, presbyacusis, or inherited disorders. *Presbyacusis:* Loss of acuity for high-frequency sounds starts before 30yrs old. We do not usually notice it until hearing of speech is affected. Hearing is most affected in the presence of background noise. Hearing aids are the usual treatment.

**Sudden sensorineural deafness** ►Get an ENT opinion *today* (steroids may cure)! *Causes:* Noise exposure; gentamicin/other toxin; mumps; acoustic neuroma; MS; stroke; vasculitis; TB. *Tests:* ESR, FBC, LFT, pANCA, viral titres and TB (see BOX 'Diagnostic tests for TB', p394); evoked response audiometry; CXR; MRI; lymph node and nasopharyngeal biopsy for culture.

## Tinnitus

This ringing or buzzing in the ears is common, and may cause depression or insomnia. ►Investigate unilateral tinnitus fully to exclude an acoustic neuroma (p462).

**Causes** Inner ear damage and hearing loss (leading to auditory cortex hyper-excitability), wax, excess noise, head injury, otitis media, post-stapedectomy, Ménière's, anaemia (if pulsatile then think of carotid artery stenosis or dissection, AV fistulae, and glomus jugulare tumours). *Drugs:* Aspirin (reversible), loop diuretics, aminoglycosides. *Mean age at onset:* 40–50yrs. ♂:♀≈1:1.

**Management** Exclude serious causes. ►Psychological support is very important: reassure that tinnitus does not mean madness or serious disease and that it often improves in time. *Cognitive therapy* helps, as do 'tinnitus coping training' and patient support groups. *Drugs* are disappointing: anticonvulsants (eg carbamazepine) are not of benefit; misoprostol appears to help (small-scale trials only); hypnotics at night may be of some benefit. Avoid tranquillizers, particularly if depressed (use tricyclic antidepressants here). If Ménière's disease is the cause, betahistine helps only a few. *Masking* may give relief: white noise (like an off-tuned radio) is given via a noise generator worn like a post-aural hearing aid. *Hearing aids* may help by amplifying desirable sounds. *Cochlear nerve section* is a drastic option that can relieve disabling tinnitus in 25% but at the expense of deafness.

**Bittersweet symphony**

German composer Ludwig van Beethoven (1770–1827) began to lose his hearing from his early 30s—first high-frequency sounds were lost, associated with debilitating tinnitus: 'My ears sing and buzz continually, day and night. I can truly say that I am living a wretched life...in my profession it is a frightful state.' However, despite becoming profoundly deaf by the age of 44, he continued to compose and perform throughout his auditory decline. As deafness crept into the ears of the French composer, Gabriel Fauré (1845–1924), he composed less, but only after losing his hearing did he manage to overcome his previous fear of writing a string quartert, telling his wife 'This is a genre which Beethoven in particular made famous, and causes all those who are not Beethoven to be terrified of it'. They are not the only masters of their field to overcome auditory impairment; so did cardiologist Helen Taussig (see p92).

# Acute bilateral leg weakness

It is crucial to establish a diagnosis quickly to avoid permanent disability. Look for specific patterns (see later in topic) and ask these questions to help elicit the diagnosis:

1. *Where is the lesion?* • Are the legs flaccid or spastic? (ie LMN or UMN?) • Is there sensory loss? A sensory *level* usually means spinal cord disease. • Is there loss of bowel or bladder control? (Lesion more likely to be in the conus medullaris or cauda equina.)
2. *What is the lesion?* • Was onset sudden or rapidly progressive? ►This is an emergency; it suggests cord compression so get urgent help (see next paragraph). • Are there any signs of infection (eg tender spine, ↑T°, ↑WCC, ↑ESR, ↑CRP: extradural abscess)?

**Cord compression** (See also p528.) *Symptoms:* Bilateral leg weakness (arm weakness—often less severe—suggests a cervical cord lesion, see p508) a sensory level ± preceding back pain (see p542). Bladder (and anal) sphincter involvement is late and manifests as hesitancy, frequency, and, later, as painless retention. *Signs:* Look for a motor, reflex, and sensory level, with normal findings above the level of the lesion, LMN signs at the level (especially in cervical lesions), and UMN signs *below* the level (but remember tone and reflexes are usually reduced in *acute* cord compression; *OHCS* p756). *Causes:* Secondary malignancy (breast, lung, prostate, thyroid, kidney) in the spine is commonest. Rarer: infection (epidural abscess), cervical disc prolapse, haematoma (warfarin), intrinsic cord tumour, atlanto-axial subluxation, myeloma. *ΔΔ:* Transverse myelitis, MS, carcinomatous meningitis, cord vasculitis (PAN, syphilis), spinal artery thrombosis or aneurysm, trauma, Guillain–Barré syndrome (p702). *Investigations:* ►Do not delay imaging at any cost. Spinal x-rays are unreliable; MRI is the definitive modality. Biopsy or surgical exploration may be needed to identify the nature of any mass. Do a CXR (primary lung malignancy, lung secondaries, TB). Bloods: FBC, ESR, $B_{12}$, syphilis serology, U&E, LFT, PSA, serum electrophoresis. *Treatment:* Give urgent dexamethasone in malignancy (p528) while considering more specific therapy, eg radiotherapy or chemotherapy ± decompressive laminectomy; which is most appropriate depends on tumour type, quality of life, and likely prognosis. Epidural abscesses must be surgically decompressed and antibiotics given.

**Cauda equina and conus medullaris lesions** The big difference between these lesions and those high up in the cord is that leg weakness is flaccid and areflexic, not spastic and hyperreflexic. *Causes:* As above, plus congenital lumbar disc disease and lumbosacral nerve lesions. *Signs: Conus medullaris lesions* feature mixed UMN/LMN signs, leg weakness, early urinary retention and constipation, back pain, sacral sensory disturbance and erectile dysfunction. *Cauda equina lesions* feature back pain and radicular pain down the legs; asymmetrical, atrophic, areflexic paralysis of the legs; sensory loss in a root distribution; and ↓sphincter tone; do PR.

## Other patterns of leg weakness

*Unilateral foot drop:* DM, common peroneal nerve palsy, stroke, prolapsed disc, MS.

*Weak legs with no sensory loss:* MND, polio, parasagittal meningioma (an exception to the rule that weak legs mean cord or distal lesion).

*Chronic spastic paraparesis:* MS, cord primary malignancy/metastasis, MND, syringomyelia, subacute combined degeneration of the cord (p334), hereditary spastic paraparesis, taboparesis (tertiary syphilis, see p412), histiocytosis X, parasites (eg schistosomiasis).

*Chronic flaccid paraparesis:* Peripheral neuropathy, myopathy.

*Absent knee jerks and extensor plantars:* (Ie combined LMN or UMN signs.) Combined cervical and lumbar disc disease, conus medullaris lesions, MND, myeloradiculitis, Friedreich's ataxia, subacute combined degeneration of the cord, taboparesis.[4]

4 Tertiary syphilis (p412): in tabes dorsalis the afferent pathways from muscle spindles are lost, with reduced tone and tendon reflexes (without weakness). Later, additional involvement of the pyramidal tracts causes taboparesis—a spastic paraparesis with the peculiar combination of extensor plantars (from the taboparesis) and absent tendon reflexes (from the tabes dorsalis).

## Gait disorders

*Spastic:* Stiff, circumduction of legs ± scuffing of the toe of the shoes: UMN lesions.

*Extrapyramidal:* Flexed posture, shuffling feet, slow to start, postural instability, eg Parkinson's disease.

*Apraxic:* Pathognomonic 'gluing-to-the-floor' on attempting walking or a wide-based unsteady gait with a tendency to fall, like a novice on an ice-rink. Seen in normal pressure hydrocephalus and multi-infarct states.

*Ataxic:* Wide-based; falls; cannot walk heel-to-toe. Caused by cerebellar lesions (eg MS, posterior fossa tumours, alcohol, phenytoin toxicity); proprioceptive sensory loss (eg sensory neuropathy, ↓$B_{12}$). Often worse in the dark or with eyes closed.

*Myopathic:* Waddling gait, cannot climb steps or stand from sitting due to hip girdle weakness.

*Psychogenic:* Suspect if there is a bizarre gait not conforming to any pattern of organic gait disturbance and without any signs when examined on the couch.

**Tests** Spinal X-rays; MRI; FBC; ESR; syphilis serology; serum $B_{12}$; U&E; LFT; PSA; serum electrophoresis; CXR; LP; EMG; muscle ± sural nerve biopsy.

## Non-neurological considerations in paralysed patients

Avoid pressure sores by turning and review weight-bearing areas often. Use appropriate pressure-relieving mattresses/cushions. Prevent thrombosis in paralysed limbs by frequent passive movement, pressure stockings, and LMWH (p350). Bladder care is vital; catheterization is only one option (do not control incontinence by decreasing fluid intake). Bowel evacuation may be manual or aided by suppositories; increasing dietary fibre intake may help. Exercise of unaffected or partially paralysed limbs is important to avoid unnecessary loss of function.

# Abnormal involuntary movements (dyskinesia)



These are characterized by impairment of the planning, control, or execution of movement. They can have multiple manifestations:

**Tremor** Note frequency, amplitude, and exacerbating factors (stress; fatigue).

- *Rest tremor:* Abolished on voluntary movement. *Cause:* parkinsonism (p494).
- *Intention tremor:* Irregular, large-amplitude, worse at the end of purposeful acts, eg finger-pointing or using a remote control. *Cause:* cerebellar damage (eg MS, stroke).
- *Postural tremor:* Absent at rest, present on maintained posture (arms outstretched) and may persist (but is not worse) on movement. *Causes:* benign essential tremor (autosomal dominant; improves with alcohol), thyrotoxicosis, anxiety, β-agonists.
- *Re-emergent tremor:* Postural tremor developing after a delay of ~10s. *Causes:* Parkinson's disease (don't mistake for essential tremor).

**Chorea** Non-rhythmic, jerky, purposeless movements flitting from one place to another—eg facial grimacing, raising the shoulders, flexing/extending the fingers. *Causes:* Huntington's disease, Sydenham's chorea (rare complication of group A streptococcal infection). Worsened by levodopa.

**Hemiballismus** Large-amplitude, flinging hemichorea (affects proximal muscles) contralateral to a vascular lesion of the subthalamic nucleus (often elderly diabetics). Recovers spontaneously over months.

**Athetosis** Slow, sinuous, confluent, purposeless movements (especially digits, hands, face, tongue), often difficult to distinguish from chorea. *Causes:* Commonest is cerebral palsy (*OHCS* p214). Most other 'athetoid' patterns may now be better classed as dystonias. *Pseudoathetosis:* Caused by severe proprioceptive loss.

**Tics** Brief, repeated, stereotyped movements which patients may suppress for a while. Tics are common in children (and usually resolve). In Tourette's syndrome (p700), motor and vocal tics occur. Consider psychological support, clonazepam or clonidine if tics are severe (haloperidol may help but risks tardive dyskinesia).

**Myoclonus** Sudden involuntary focal or general jerks arising from cord, brainstem, or cerebral cortex, seen in metabolic problems, neurodegenerative disease (eg lysosomal storage enzyme defects), CJD (p696), and myoclonic epilepsies (infantile spasms). *Benign essential myoclonus:* Childhood onset with frequent generalized myoclonus, without progression. Often autosomal dominant. It may respond to valproate, clonazepam, or piracetam. *Asterixis ('metabolic flap'):* Jerking (~1–2 jerks/sec) of outstretched hands, worse with wrists extended, from loss of extensor tone—ie incoordination between flexors and extensors (='negative myoclonus'). *Causes:* Liver or kidney failure, ↓$Na^+$, ↑$CO_2$, gabapentin, thalamic stroke (consider if unilateral).

**Tardive syndromes** Delayed onset yet potentially irreversible symptoms occuring after chronic exposure to dopamine antagonists (eg antipsychotics, antiemetics). *Classification:* •*Tardive dyskinesia:* orobuccolingual, truncal, or choreiform movements, eg vacuous chewing and grimacing movements. •*Tardive dystonia:* sustained, stereotyped muscle spasms of a twisting or turning character, eg retrocollis and back arching/opisthotonic posturing. •*Tardive akathisia:* sense of restlessness or unease ± repetitive, purposeless movements (stereotypies, eg pacing). •*Tardive myoclonus.* •*Tardive tourettism* (p700). •*Tardive tremor.* *Treating tardive dyskinesia:* Gradually withdraw neuroleptics and wait 3–6 months. Tetrabenazine may help. Quetiapine, olanzapine, and clozapine are examples of atypical antipsychotics that are less likely to cause tardive syndromes.

## Dystonia

Dystonia describes prolonged muscle contractions causing abnormal posture or repetitive movements.

*Idiopathic generalized dystonia:* Childhood-onset dystonia often starting in one leg with ipsilateral progression over 5–10yrs. Autosomal dominant inheritance is common (DYT1 deletion). Exclude Wilson's disease and dopa-responsive dystonia (needs an L-dopa trial). Anticholinergics and muscle relaxants may help. Deep brain stimulation for refractory, disabling symptoms.

*Focal dystonias:* Confined to one part of the body, eg *spasmodic torticollis* (head pulled to one side), *blepharospasm* (involuntary contraction of orbicularis oculi, *OHCS* p417), *writer's cramp*. Focal dystonias in adults are typically idiopathic, and rarely generalize. They are worsened by stress. Patients may develop a *geste antagoniste* to try to resist the dystonic posturing (eg a touch of the finger to the jaw in spasmodic torticollis). Injection of botulinum toxin into the overactive muscles is usually effective.

*Acute dystonia:* (fig 10.14) May occur on starting many drugs, including neuroleptics and some anti-emetics (eg metoclopramide, cyclizine). There is torticollis (head pulled back), trismus (oromandibular spasm), and/or oculogyric crisis (eyes drawn up). You may mistake this for tetanus or meningitis, but such reactions rapidly disappear after a dose of an anticholinergic, see p843.

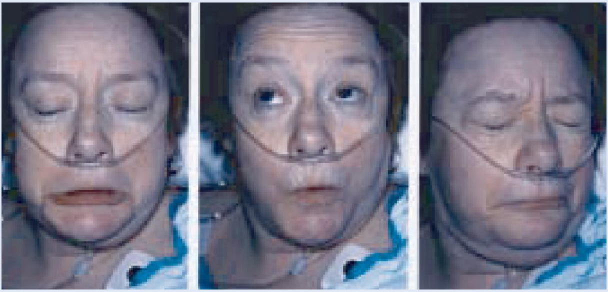


Fig 10.14 Oromandibular/oculogyric crisis in acute dystonia.

Reprinted from *Mayo Clinic Proceedings*, 78(9), Ritter *et al.*, 'Ondansetron-induced multifocal encephalopathy', 1150–2, 2003, with permission from Elsevier.

## St Vitus's dance

Throughout the middle ages, Europe was plagued by epidemics of 'dancing mania', in which afflicted individuals were described to have danced wildly, displaying strange contortions and convulsions until they collapsed from exhaustion. If the afflicted touched a relic of St Vitus they were miraculously cured: observing this, Paracelsus, 16th-century Swiss-German physician and philosopher, described the phenomenon of *chorea Sancti Viti* ('St Vitus's dance'). There may have been an infectious component, although mass hysteria induced by religious cults that swept across medieval Europe seems a more likely cause. Chorea was subsequently used as a general term for large-amplitude involuntary movements before being further refined by physicians such as Sydenham (though he did not connect his eponymous chorea seen in rheumatic fever with an infectious trigger) and Charcot. Nowaways, a more frequent cause of involuntary movements with behavioural disturbance is NMDA-receptor antibody encephalitis, the impact of which was documented in Susannah Cahalan's excellent 2012 autobiography *Brain on Fire: My Month of Madness*.

## Stroke: clinical features and acute management



Infarction or bleeding into the brain manifests with sudden-onset focal CNS signs. Someone in the UK has a stroke every 3.5 minutes; 1 in 4 of those will die within a year and half of survivors will have a permanent disability.

**Causes** • Small vessel occlusion/cerebral microangiopathy or thrombosis *in situ*. • Cardiac emboli (AF; endocarditis; MI—see BOX 'Cardiac causes of stroke', p473). • Atherothromboembolism (eg from carotids). • CNS bleeds (↑BP, trauma, aneurysm rupture, anticoagulation, thrombolysis). *Other causes:* ►Consider in younger patients: sudden BP drop by ≥40mmHg (most likely to affect the boundary zone between vascular beds), carotid artery dissection (spontaneous, or from neck trauma or fibromuscular dysplasia), vasculitis, subarachnoid haemorrhage (p478), venous sinus thrombosis (p480), antiphospholipid syndrome, thrombophilia (p374), Fabry disease (p698), CADASIL.[5]

**Differentials** Head injury, hypo/hyperglycaemia, subdural haemorrhage, intracranial tumours, hemiplegic migraine, post-ictal (Todd's palsy), CNS lymphoma, Wernicke's encephalopathy, hepatic encephalopathy, encephalitis, toxoplasmosis, cerebral abscesses, mycotic aneurysm, drug overdose (if comatose).

**Modifiable risk factors** ↑BP, smoking, DM, heart disease (valvular, ischaemic, AF), peripheral vascular disease, ↑PCV, carotid bruit, combined OCP, ↑lipids, ↑alcohol use, ↑clotting (eg ↑plasma fibrinogen, ↓antithrombin III, p374), ↑homocysteine, syphilis.

**Signs** Worst at onset. *Pointers to bleeding (unreliable!):* meningism, severe headache, coma. *Pointers to ischaemia:* carotid bruit, AF, past TIA, IHD. *Cerebral infarcts:* (50%.) Depending on site there may be contralateral sensory loss or hemiplegia—initially flaccid (floppy limb, falls like a dead weight when lifted), becoming spastic (UMN); dysphasia; homonymous hemianopia; visuo-spatial deficit. *Brainstem infarcts:* (25%.) Varied; include quadriplegia, disturbances of gaze and vision, locked-in syndrome (aware, but unable to respond). *Lacunar infarcts:* (25%). Basal ganglia, internal capsule, thalamus, and pons.) Five syndromes: ataxic hemiparesis, pure motor, pure sensory, sensorimotor, and dysarthria/clumsy hand. Cognition/consciousness are intact except in thalamic strokes.

**Acute management** ► *Protect the airway:* This avoids hypoxia/aspiration.

- *Maintain homeostasis:* Blood glucose: keep between 4–11 mmol/L. Blood pressure: only treat if there is a hypertensive emergency (eg encephalopathy or aortic dissection) or thrombolysis is considered (ideally aim for ≲185/110) as treating even very high BPs may impair cerebral perfusion.
- *Screen swallow:* 'Nil by mouth' until this is done (but keep hydrated).
- *CT/MRI within 1h:* Essential if: thrombolysis considered, high risk of haemorrhage (↓GCS, signs of ↑ICP, severe headache, meningism, progressive symptoms, bleeding tendency or anticoagulated), or unusual presentation (eg fluctuating consciousness, fever). Otherwise imaging less urgent (aim <24h). Diffusion-weighted MRI is most sensitive for an acute infarct, but CT helps rule out primary haemorrhage (fig 10.15).
- *Antiplatelet agents:* Once haemorrhagic stroke is excluded, give aspirin 300mg (continue for 2 weeks, then switch to long-term antithrombotic treatment, p472).
- *Thrombolysis:* Consider this as soon as haemorrhage has been excluded, provided the onset of symptoms was ≤4.5h ago.[2] The benefits of thrombolysis outweigh the risks within this window, though best results are within 90min. Alteplase is the agent of choice and must be given by trained staff, ideally within an expert acute stroke team. ► Always do CT 24h post-lysis to identify bleeds.[6] CI to thrombolysis: • Major infarct or haemorrhage on CT. • Mild/non-disabling deficit. • Recent surgery, trauma, or artery or vein puncture at uncompressible site. • Previous CNS bleed. • AVM/aneurysm. • Severe liver disease, varices, or portal hypertension. • Seizures at presentation. • Blood glucose (<3 or >22). • Stroke or serious head injury in last 3 months. • GI or urinary tract haemorrhage in the last 21 days. • Known clotting disorder. • Anticoagulants or INR >1.7. • Platelets $<100 \times 10^9$/L. • History of intracranial neoplasm. • Rapidly improving symptoms. • BP >180/105.
- *Thrombectomy:* Intra-arterial mechanical thrombectomy provides additional benefit for those with large artery occlusion in the proximal anterior circulation.

► Admit to an acute stroke unit: multidisciplinary care improves outcomes (p474).

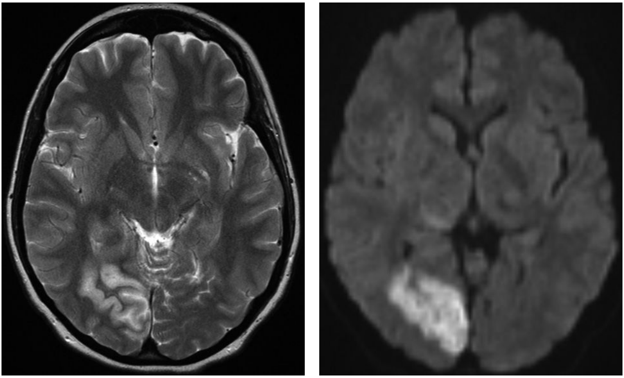


Fig 10.15 The T2-weighted (p746) image on the left shows oedema in the right occipital lobe. Differentials: infarct (right PCA), inflammation, or tumour. The diffusion-weighted image on the right shows limited diffusion in the region, indicating this is an infarct. ©Prof Peter Scally.

### Act FAST

Several public health measures have aimed to increase awareness of stroke and the seriousness of the condition: the relabelling of stroke as a 'brain attack,' and via the graphic mass media **FAST** campaign = **F**acial asymmetry, **A**rm/leg weakness, **S**peech difficulty, **T**ime to call 999. The publicity surrounding this acronym has increased recognition of the symptoms of stroke and emphasized the urgency of seeking medical help; following the introduction of the campaign in 2011 the NHS in England saw a 24% rise in stroke-related 999 calls.

5 Cerebral Autosomal Dominant Arteriopathy with Subcortical Infarcts & Leucoencephalopathy: the main genetic cause of stroke (there is also an autosomal recessive form).
6 If +ve, register at SITS, www.sitsinternational.org

**Primary prevention** (Ie before any stroke.)
*Control risk factors* (p470): look for and treat hypertension, DM, ↑lipids (p690), cardiac disease (see BOX 'Cardiac causes of stroke') and help quit smoking (see p93). Exercise helps (↑HDL, ↑glucose tolerance). Use *lifelong anticoagulation* in AF (see BOX 'Cardiac causes of stroke') and prosthetic heart valves. • For prevention post-TIA see p476.

**Secondary prevention** (Ie preventing further strokes.)
Control risk factors (as *Primary prevention* mentioned above): there is a considerable advantage from lowering blood pressure and cholesterol (even if not particularly raised). *Antiplatelet agents after stroke:* (See BOX 'Antiplatelets'.) If no primary haemorrhage on CT, give 2 weeks of aspirin 300mg, then switch to long-term clopidogrel monotherapy. If this is CI or not tolerated then give low dose aspirin plus slow-release dipyridamole. *Anticoagulation after stroke from AF:* See BOX 'Cardiac causes of stroke'.

**Tests** (See p470 for imaging.) Investigate promptly to identify risk factors for further strokes, but consider whether results will affect management. Look for:
- *Hypertension.* Look for retinopathy (p560), nephropathy, or cardiomegaly on CXR.
- *Cardiac source of emboli.* (See BOX 'Cardiac causes of stroke'.) 24h ECG to look for AF (p130). CXR may show an enlarged left atrium. Echocardiogram may reveal mural thrombus due to AF or a hypokinetic segment of cardiac muscle post-MI. It may also show valvular lesions in infective endocarditis or rheumatic heart disease. Transoesophageal echo is more sensitive than transthoracic.
- *Carotid artery stenosis.* Do carotid Doppler US ± CT/MRI angiography. Benefits and risks of revascularization should be individualized by an expert but generally most with ≥70% stenosis and life expectancy ≥5yrs will benefit while some (especially ♂) will benefit with 50–69% stenosis[7] (see p476). Carotid endarterectomy is the procedure of choice; endovascular carotid artery angioplasty with stenting is an alternative for those unfit for surgery and achieves similar long-term outcomes but has higher peri-procedure stroke and mortality rates.
- *Hypoglycaemia*, *hyperglycaemia, dyslipidaemia,* and *hyperhomocysteinaemia.*
- *Vasculitis.* ↑ESR, ANCA (p556). VDRL to look for active, untreated syphilis (p412).
- *Prothrombotic states,* eg thrombophilia (p374), antiphospholipid syndrome (p554).
- *Hyperviscosity,* eg polycythaemia (p366), sickle-cell disease (p340).
- *Thrombocytopenia* and other bleeding disorders.
- *Genetic tests.* CADASIL (p470); Fabry disease (p698).

**Prognosis** *Overall mortality:* 60 000/yr; UK 20% at 1 month, then ≲10%/yr. *Full recovery:* ≤40%. Drowsiness≈poor prognosis. Avoid pressure ulcers (fig 10.16).



**7** Interventions for 50–69% stenoses can be justifiable; individualize risk and check local guidelines. In particular, check which criteria used to estimate degree of stenosis since NASCET (North American Symptomatic Carotid Endarterectomy Trial) criteria tend to include some more severe lesions in 50–69% range as compared to the ECST (European Carotid Surgery Trialists' Collaborative Group) criteria.

## Cardiac causes of stroke

Cardioembolic causes are the source of stroke in >30% of patients, and may be hinted at if there are bilateral infarcts on imaging.

*Non-valvular atrial fibrillation:* (p130) Associated with an overall risk of stroke of 4.5%/yr, and ischaemic strokes in AF carry a worse prognosis.

- *$CHA_2DS_2VASc$ score* (p131) can be used to calculate risk of stroke in patients with AF. Offer anticoagulation in patients with a score of 2 or above. ►Take bleeding risk into account: calculate the risk of major bleeding using the HAS-BLED score. Caution and regular review of oral anticoagulants are required if the HAS-BLED score >3. ► Do not offer stroke prevention therapy in patients with AF if <65y and $CHA_2DS_2VASc$ score is 0 for men or 1 for women.
- *Anticoagulation* (see p350) can be commenced 2wks after a stroke (or from 7–10d if clinically and radiologically small). Offer a direct oral anticoagulant (DOAC) or warfarin (p350), following a discussion of risks and benefits.

*Other cardiac sources of emboli:* • Cardioversion. • Prosthetic valves. • Acute myocardial infarct with large left ventricular wall motion abnormalities on echocardiography. • Patent foramen ovale/septal defects. • Cardiac surgery. • Infective endocarditis (gives rise to septic emboli; 20% of those with endocarditis present with CNS signs).

## Antiplatelets: mechanism of action

*Aspirin:* Inhibits COX-1, suppressing prostaglandin and thromboxane synthesis.

*Clopidogrel:* A thienopyridine that inhibits platelet aggregation by modifying platelet ADP receptors, preventing further strokes and MIs.

*Dipyridamole:* ↑cAMP and ↓thromboxane A2.

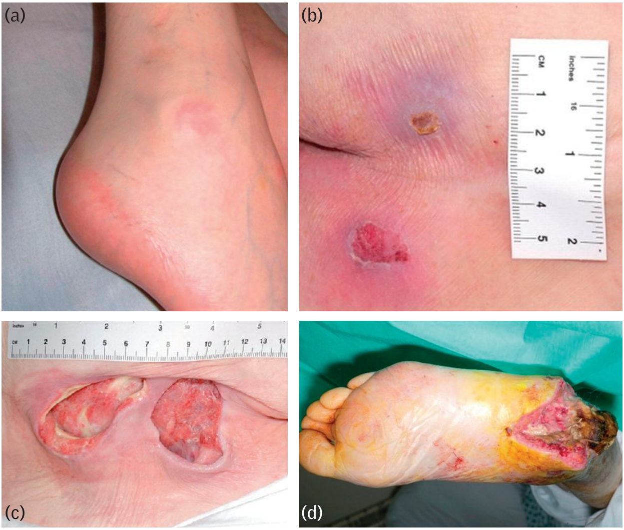



**Fig 10.16** Categorization of pressure ulcers. (a) Stage 1: non-blanchable redness of intact skin, typically over a bony prominence. (b) Stage 2: partial thickness loss of dermis presenting as a shallow open ulcer with a red/pink wound bed, without slough. May also present as an intact or open/ruptured sero-sanguinous blister. (c) Stage 3: full thickness skin loss with visible subcutaneous fat. Bone, tendon, or muscle are not exposed. (d) Stage 4: full thickness tissue loss wth exposed bone, tendon, or muscle.

Images (a) to (d) reproduced from Gosney *et al.*, *Oxford Desk Reference: Geriatric Medicine*, 2012, with permission from Oxford University Press.

# Re-enablement after stroke



Coordinated multidisciplinary care on a specialized stroke unit is essential, and leads to better patient outcomes. Rehabilitation must be started early post-stroke in order to maximize improvement and prevent complications related to immobility such as pressure sores, aspiration pneumonia, constipation, and contractures. Ongoing input after discharge consolidates inpatient gains and helps align the individual with their previous capability. It also helps with depression—both in the patient and their carer. ►Setting achievable goals and acknowledging the patient's own agenda is key.

## Imperatives for re-enablement

- Watch the patient swallow a small volume of water; if signs of aspiration (a cough or voice change) make *nil by mouth* until formal assessment by a speech therapist. Use IV fluids, then semi-solids (eg jelly; avoid soups and crumbly food). Avoid early NG tube feeds; these may be needed to safeguard nutrition in those with swallowing problems that persist beyond the first 2–3d. If swallowing fails to recover, consider benefits of enteral feeding tube placement (p759). Speech therapists skilled in assessing swallowing difficulties are invaluable here.
- Avoid further injury: minimize falls risk and take care when lifting the patient not to damage their shoulders.
- Ensure good bladder and bowel care through frequent toileting. Avoid early catheterization which may prevent return to continence.
- Position to minimize spasticity (occurs in ~40%). Get prompt physiotherapy. Splints and botulinum toxin injections are helpful for focal spasticity.
- Monitor progress: eg measure time taken to sit up and transfer to chair.
- Monitor mood: in pseudo-emotionalism/emotional lability (sobbing unprovoked by sorrow, from failure of cortical inhibition of the limbic system), tricyclics or fluoxetine may help.
- Engage the patient in their own recovery by making physiotherapy fun. Swimming (a hemiplegic arm may be supported on a special float), music, and video games are all enjoyable and ↑recovery through promoting cerebral reorganization. Constraint of the good arm may be helpful.

►Involve the carer/spouse with all aspects of care-giving. Good rehab saves lives.

**Tests** Asking to point to a named part of the body tests perceptual function. Copying matchstick patterns tests spatial ability. Dressing or copying a clock face tests for *apraxia* (p86). Picking out and naming easy objects from a pile tests for agnosia (acuity OK, but cannot mime use; guesses are way-out, semantically, and phonetically). Screen for depression (low mood; inability to feel pleasure or to concentrate).

**End-of-life decisions** ►See p13.

## Assessing dependence in daily life

*Handicap* entails inability to carry out social functions. 'A disadvantage for a given individual, resulting from an impairment or disability, that limits or prevents the fulfilment of a role.' Two people with the same *impairment* (eg paralysed arm) may have different *disabilities* (table 10.5, eg one may be able to dress but the other cannot). Disabilities are likely to determine quality of future life. Treatment is often best aimed at reducing disability, not curing disease. For example, Velcro® fasteners in place of buttons may enable a person to dress.

Table 10.5 Barthel's index of activities of daily living

| | | |
|---|---|---|
| Bowels | 0 | Incontinent (or needs to be given enemas) |
| | 1 | Occasional accidents (once a week) |
| | 2 | Continent |
| Bladder | 0 | Incontinent, or catheter inserted but unable to manage it |
| | 1 | Occasional accidents (up to once per 24h) |
| | 2 | Continent (for more than 7 days) |
| Grooming | 0 | Needs help with personal care: face, hair, teeth, shaving |
| | 1 | Independent (implements provided) |
| Toilet use | 0 | Dependent |
| | 1 | Needs some help but can do some things alone |
| | 2 | Independent (on and off, wiping, dressing) |
| Feeding | 0 | Unable |
| | 1 | Needs help in cutting, spreading butter, etc. |
| | 2 | Independent (food provided within reach) |
| Transfer | 0 | Unable to get from bed to commode: the vital transfer to prevent the need for 24-hour nursing care |
| | 1 | Major help needed (physical, 1–2 people), can sit |
| | 2 | Minor help needed (verbal or physical) |
| | 3 | Independent |
| Mobility | 0 | Immobile |
| | 1 | Wheelchair-independent, including corners, etc. |
| | 2 | Walks with help of one person (verbal or physical) |
| | 3 | Independent |
| Dressing | 0 | Dependent |
| | 1 | Needs help but can do about half unaided |
| | 2 | Independent (including buttons, zips, laces, etc.) |
| Stairs | 0 | Unable |
| | 1 | Needs help (verbal, physical, carrying aid) |
| | 2 | Independent up and down |
| Bath/shower | 0 | Dependent |
| | 1 | Independent (must get in and out unaided and wash self) |

Mahoney FI, Barthel DW: Functional evaluation: the Barthel Index. *Maryland State Medical Journal*. 1965; 14:61–65.

## Barthel's paradox

The more we contemplate Barthel's eulogy of independence, the more we see it as a mirage reflecting a greater truth about human affairs: ►there is no such thing as independence—only *interdependence*—and in fostering this interdependence lies our true vocation:

> *No man is an Island, intire of it selfe; every man is a peece of the Continent, a part of the maine; if a Clod bee washed away by the Sea, Europe is the lesse, as well as if a promontorie were, as well as if a Mannor of thy friends or of thine owne were. Any man's death diminishes me, because I am involved in mankinde; And therefore never send to know for whom the bell tolls: It tolls for thee.*

John Donne 1572–1631; *Meditation* XVII.

# Transient ischaemic attack (TIA)



This is an ischaemic (usually embolic) neurological event with symptoms lasting <24h (often much shorter). ► Without intervention, more than 1 in 12 patients will go on to have a stroke within a week, so prompt management is imperative.

**Signs** Specific to the arterial territory involved (p450). *Amaurosis fugax* occurs when the retinal artery is occluded, causing unilateral progressive vision loss 'like a curtain descending'. *Global* events (eg syncope, dizziness) are *not* typical of TIAs. Attacks may be single or many; multiple highly stereotyped attacks ('crescendo' TIAs) suggest a critical intracranial stenosis (commonly the superior division of the MCA).

**Causes** (See p470.) • *Atherothromboembolism* from the carotid is the chief cause: listen for bruits (though not a sensitive test). • *Cardioembolism:* mural thrombus post-MI or in AF, valve disease, prosthetic valve (p473). • *Hyperviscosity:* eg polycythaemia, sickle-cell anaemia, myeloma. • *Vasculitis* is a rare, non-embolic cause of TIA symptoms (eg cranial arteritis, PAN, SLE, syphilis, etc.).

**Differentials** Hypoglycaemia, migraine aura (p458), focal epilepsy (symptoms spread over seconds and often include twitching and jerking), hyperventilation, retinal bleeds. *Rare mimics of TIA:* Malignant hypertension, MS (paroxysmal dysarthria), intracranial tumours, peripheral neuropathy, phaeochromocytoma, somatization.

**Tests** FBC, ESR, U&Es, glucose, lipids, CXR, ECG, carotid Doppler ± angiography, CT or diffusion-weighted MRI, echocardiogram.

**Treatment**
- *Control cardiovascular risk factors:* Optimize: BP (cautiously lower; aim for <140/85mmHg, p140); hyperlipidaemia (p690); DM (p206); help to stop smoking (p93).
- *Antiplatelet drugs:* As with stroke, give aspirin 300mg OD for 2wks, then switch to clopidogrel 75mg OD. If this is contraindicated or not tolerated, give aspirin 75mg OD combined with slow-release dipyridamole.
- *Anticoagulation indications:* Cardiac source of emboli (see p473).
- *Carotid endarterectomy:* Perform within 2wks of first presentation if 70–99% stenosis[8] and operative risk is acceptable (higher risk in: ♀, >75y, ↑systolic BP, contralateral artery occluded; ipsilateral carotid syphon/external carotid stenosed). Do not stop aspirin preoperatively. Surgery is preferred to endovascular carotid artery angioplasty with stenting in those fit enough to tolerate due to higher peri-procedure stroke and mortality rates with stenting.

**Driving** Prohibited for at least 1 month, see p158.

**Prognosis** Long-term risks of stroke or cardiovascular events following TIAs are dependent on underlying vascular risk factors: calculate using the $ABCD^2$ score (see BOX and table 10.6).

8 Interventions for 50-69% stenoses can be justifiable; individualize risk and check local guidelines. In particular, check which criteria used to estimate degree of stenosis since NASCET (North American Symptomatic Carotid Endarterectomy Trial) criteria tend to include some more severe lesions in 50-69% range as compared to the ECST (European Carotid Surgery Trialists' Collaborative Group) criteria.

## When should TIA lead to emergency referral?

The ABCD$^2$ score is a helpful tool to stratify which patients are at higher risk of having a stroke following a suspected TIA (table 10.6).

Table 10.6 The ABCD$^2$ score

| | |
|---|---|
| **A**ge ≥60 yrs old | 1 point |
| **B**lood pressure ≥140/90 | 1 point |
| **C**linical features | |
| Unilateral weakness | 2 points |
| Speech disturbance without weakness | 1 point |
| **D**uration of symptoms | |
| Symptoms lasting ≥1h | 2 points |
| Symptoms lasting 10–59min | 1 point |
| **D**iabetes | 1 point |

A score of ≥4 indicates that the patient is at high risk of an early stroke, and must be assessed by a specialist within 24h. A score of ≥6 strongly predicts a stroke (8.1% within 2 days, 35.5% in the next week). Other factors that suggest increased risk are: • AF • >1 TIA in a week • TIA while anticoagulated. Crucially, risk is lowest if the patient is treated in a specialized stroke unit (p474).

## Subarachnoid haemorrhage (SAH)

Spontaneous bleeding into the subarachnoid space, often catastrophic (table 10.7).

**Incidence** 9/100 000/yr; typical age: 35–65.

**Symptoms** Sudden-onset excruciating headache, typically occipital—like a 'thunderclap'. Vomiting, collapse, seizures, and coma often follow. Coma/drowsiness may last for days. Some patients report a preceding, 'sentinel' headache, perhaps due to a small warning leak from the offending aneurysm (~6%).

**Signs** Neck stiffness; Kernig's sign (takes 6h to develop); retinal, subhyaloid and vitreous bleeds (=Terson's syndrome; ↑mortality × 5). Focal neurology at *presentation* may suggest site of aneurysm (eg pupil changes indicating a IIIrd nerve palsy with a posterior communicating artery aneurysm) or intracerebral haematoma. Later deficits suggest complications (see later in topic).

**Causes** • *Berry aneurysm rupture (80%).* Common sites: junctions of posterior communicating with the internal carotid (see fig 10.3, p451) or of the anterior communicating with the anterior cerebral artery, or bifurcation of the middle cerebral artery (fig 10.17). 15% are multiple. •*Arterio-venous malformations (15%).*• *Other causes;* encephalitis, vasculitis, tumour (invading blood vessels), idiopathic.

**Risk factors** Previous aneurysmal SAH (new aneurysms form, old ones get bigger), smoking, alcohol misuse, ↑BP, bleeding disorders, SBE (mycotic aneurysm), family history (3–5x ↑risk of SAH in close relatives). Polycystic kidneys, aortic coarctation, and Ehlers–Danlos syndrome (p149) are all associated with berry aneurysms.

**Differentials** Meningitis (p822), migraine (p458), intracerebral bleed, cortical vein thrombosis (p480), dissection of a carotid or vertebral artery, benign thunderclap headache (triggered by Valsalva manoeuvre, eg cough, coitus).

**Tests** • *Urgent CT:* Detects >95% of SAH within the 1st 24h (fig 10.18). • *Consider LP:* If CT −ve but the history is very suggestive of SAH (and no CI: p768). This needs to be done >12h after headache onset to allow breakdown of RBCs so that a positive sample is xanthrochromic (yellow, due to bilirubin: differentiaties between old blood from SAH vs a 'bloody tap').

**Management** ►Refer all proven SAH to neurosurgery immediately.
- *Re-examine CNS* often; chart BP, pupils, and GCS (p788). Repeat CT if deteriorating.
- *Maintain cerebral perfusion* by keeping well hydrated, but aim for SBP <160mmHg.
- *Nimodipine* (60mg/4h PO for 3wks, or 1mg/h IVI) is a $Ca^{2+}$ antagonist that reduces vasospasm and consequent morbidity from cerebral ischaemia.
- *Surgery:* endovascular coiling vs surgical clipping (requiring craniotomy): the decision depends on the accessibility and size of the aneurysm, though coiling is preferred where possible (fewer complications, better outcomes). Do catheter or CT angiography to identify single *vs* multiple aneurysms *before* intervening. Newer techniques such as balloon remodelling and flow diversion can be helpful in anatomically challenging aneurysms.

**Complications** *Rebleeding* is the commonest cause of death, and occurs in 20%, often in the 1st few days. *Cerebral ischaemia* due to vasospasm may cause a permanent CNS deficit, and is the commonest cause of morbidity. If this happens, surgery is not helpful at the time but may be so later. *Hydrocephalus,* due to blockage of arachnoid granulations, requires a ventricular or lumbar drain. *Hyponatraemia* is common but should not be managed with fluid restriction. Seek expert help.

Table 10.7 Mortality in subarachnoid haemorrhage

| Grade | Signs | Mortality: % |
|---|---|---|
| I | None | 0 |
| II | Neck stiffness and cranial nerve palsies | 11 |
| III | Drowsiness | 37 |
| IV | Drowsy with hemiplegia | 71 |
| V | Prolonged coma | 100 |

Most mortality occurs in 1st month. 90% of survivors of the 1st month, survive >1 year.

## Unruptured aneurysms: 'the time-bomb in my head'

Bear in mind the old adage: 'if it ain't broke, don't fix it'—usually, risks of *preventive* intervention outweigh any benefits, except perhaps in •young patients (more years at risk, and surgery is twice as hazardous if >45yrs old) who have •aneurysms >7mm in diameter, especially if located at the •junction of the internal carotid and the posterior communicating cerebral artery, or at the •rostral basilar artery bifurcation, and especially if there is •uncontrolled hypertension or a •past history of bleeds. Data from the 2003 International Study of Unruptured Intracranial Aneurysms (ISUIA) show that relative risk of rupture for an aneurysm 7–12mm across is 3.3 compared with aneurysms <7mm across; if the diameter is >12mm, the relative risk is 17.

Fig 10.17 CT images can be manipulated to show only high-density structures such as bones and arteries containing contrast. Here is a middle cerebral artery aneurysm.

We thank Prof. Peter Scally for these CT images and the commentaries on them.

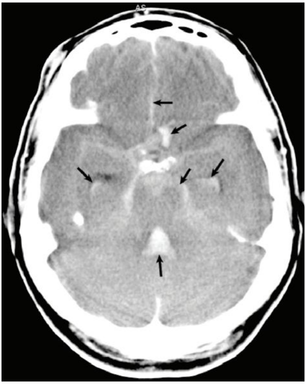


Fig 10.18 Blood from a ruptured aneurysm occupies the interhemispheric fissure (top arrow), a crescentic intracerebral area presumably near the aneurysm (2nd arrow), the basal cisterns, the lateral ventricles (temporal horns), and the 4th ventricle (bottom arrow).

We thank Prof. Peter Scally for these CT images and the commentaries on them.

# Intracranial venous thrombosis (IVT)

Thrombosis of the cerebral sinuses or veins causes cerebral infarction, though much less commonly than arterial disease. Seizures are common and focal; they can complicate diagnosis and post-ictal drowsiness may impair GCS assessment. Although ~80% will make a good functional recovery, death is mainly due to transtentorial herniation from mass effect or oedema.[9]

**Dural venous sinus thrombosis** Most commonly sagittal sinus thrombosis (figs 10.19, 10.20; 47% of all IVT) or transverse sinus thrombosis (35%). Sagittal sinus thrombosis often coexists if other sinuses are thrombosed. Symptom onset is gradual (over days or weeks). Features are dependent on the sinus affected:

- *Sagittal sinus:* Headache, vomiting, seizures, ↓vision, papilloedema.
- *Transverse sinus:* Headache ± mastoid pain, focal CNS signs, seizures, papilloedema.
- *Sigmoid sinus:* Cerebellar signs, lower cranial nerve palsies.
- *Inferior petrosal sinus:* Vth and VIth cranial nerve palsies, with temporal and retro-orbital pain (Gradenigo's syndrome, suggesting otitis media is the cause).
- *Cavernous sinus:* Often due to spread from facial pustules or folliculitis, causing headache, chemosis, oedematous eyelids, proptosis, painful ophthalmoplegia, fever.

**Cortical vein thrombosis (CVT)** Usually occurs with a sinus thrombus as it extends into the cortical veins, causing infarction in a venous territory (fig 10.21). These infarcts give rise to stroke-like focal symptoms that develop over days. There are often seizures, and an associated headache which may come on suddenly (thunderclap headache).

**Causes** Numerous, including anything that promotes a hypercoagulable state (p374). *Common causes:* Pregnancy/puerperium, combined OCP, head injury, dehydration, blood dyscrasias, tumours (local invasion/pressure), extracranial malignancy (hypercoagulability), recent LP. *Other causes:* Infection (meningitis, abscesses, otitis media, cerebral malaria, TB), Drugs (eg antifibrinolytics, androgens), SLE, vasculitis, Crohn's or UC.

**Differential diagnosis** Subarachnoid haemorrhage, meningitis, encephalitis, intracranial abscess, arterial infarction.

**Investigations** Exclude subarachnoid haemorrhage (if thunderclap headache, p478) and meningitis (p822). *Bloods:* Thrombophilia screen. *Imaging:* CT/MRI venography may show the absence of a sinus (fig 10.19), though an absent transverse sinus can be a normal variant. MRI T2-weighted gradient echo sequences can visualize thrombus directly (fig 10.20), and also identify haemorrhagic infarction. CT may be normal early, but show a filling defect at ~1wk (delta sign). *LP* (if no CI): raised opening pressure. CSF may be normal, or show RBCs and xanthochromia.

**Management** Seek expert help. Anticoagulation with heparin or LMWH and then warfarin (INR 2–3) may benefit even if there is secondary cerebral haemorrhage (unless otherwise CI). If there is deterioration despite adequate anticoagulation, endovascular thrombolysis or mechanical thrombectomy may provide limited benefit (but not in those with large infarcts and impending herniation). ↑ICP requires prompt attention (p830); decompressive hemicraniectomy may prevent impending herniation.

9 Predictors of poor prognosis include: GCS score on admission <9, deep CVT location, CNS infection, malignancy, intracranial haemorrhage, mental status abnormality, age >37 years, and ♂.

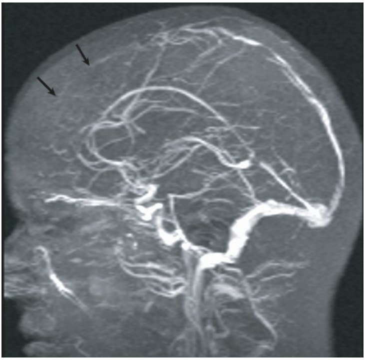


Fig 10.19 This magnetic resonance venogram (MRV) could look normal at first glance: the hardest thing to see in imaging is often that which is not there. Much of the superior sagittal sinus is not seen because it is filled with clot—a superior sagittal sinus thrombosis. The arrows point to where it should be seen. Posteriorly, the irregularity of the vessel indicates non-occlusive clot.

Image and commentary courtesy of Prof. P. Scally.

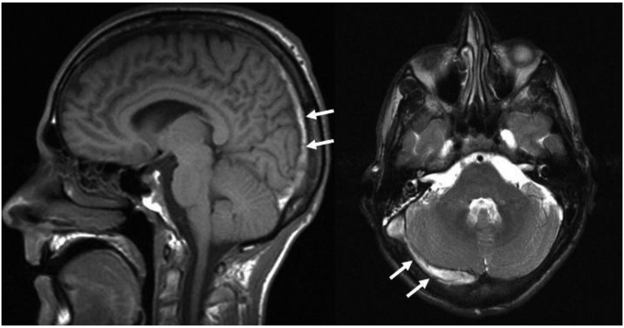


Fig 10.20 MRI showing thrombus (arrows) in the sagittal sinus (sagittal T1-weighted image, LEFT), and in the right transverse sinus (axial T2-weighted image, RIGHT). Often more than one sinus is involved.

Image courtesy of Dr David Werring.

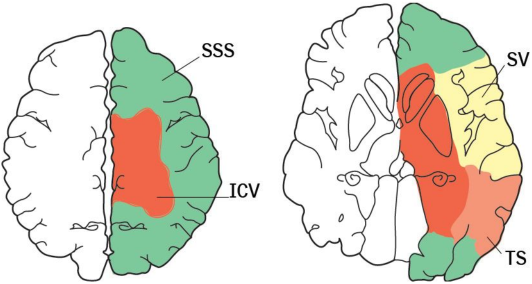



Fig 10.21 Venous territories (compare with arterial territories on p451). SSS—superior sagittal sinus; TS—transverse sinus; SV—Sylvian veins; ICV—internal cortical veins.

There is much greater variation in venous anatomy between individuals than there is in arterial anatomy, so this diagram is only a rough guide. The key point is to realize that infarction that crosses boundaries between arterial territories may be venous in origin.

## Subdural haematoma

►Consider this very treatable condition in all whose conscious level fluctuates, and also in those having an 'evolving stroke', especially if on anticoagulants. Bleeding is from bridging veins between cortex and venous sinuses (vulnerable to deceleration injury), resulting in accumulating haematoma between dura and arachnoid. This gradually raises ICP, shifting midline structures away from the side of the clot and, if untreated, eventual tentorial herniation and coning. Most subdurals are from trauma but the trauma is *often forgotten as it was so minor or so long ago* (up to 9 months). It can also occur without trauma (eg ↓ICP; dural metastases). The elderly are most susceptible, as brain atrophy makes bridging veins vulnerable. *Other risk factors:* falls (epileptics, alcoholics); anticoagulation.

**Symptoms** Fluctuating level of consciousness (seen in 35%) ± insidious physical or intellectual slowing, sleepiness, headache, personality change, and unsteadiness.

**Signs** ↑ICP (p830), seizures. Localizing neurological symptoms (eg unequal pupils, hemiparesis) occur late, often >1 month after the injury.

**Differentials** Stroke, dementia, CNS masses (eg tumours, abscesses).

**Imaging** (fig 10.22) CT/MRI shows clot ± midline shift (but beware bilateral isodense clots). Look for crescent-shaped collection of blood over 1 hemisphere. The sickle-shape differentiates subdural blood from extradural haemorrhage.

**Management** Reverse clotting abnormalities urgently. Surgical management depends on the size of the clot, its chronicity, and the clinical picture: generally those >10mm or with midline shift >5mm need evacuating (via craniotomy or burr hole washout). Address the cause of the trauma (eg falls, abuse).

## Extradural (epidural) haematoma

► Beware deteriorating consciousness after any head injury that initially produced no loss of consciousness or after initial drowsiness post injury seems to have resolved. This *lucid interval* pattern is typical of extradural bleeds.

**Cause** ►Suspect after any traumatic skull fracure. Often due to a fractured temporal or parietal bone causing laceration of the middle meningeal artery and vein, typically after trauma to a temple just lateral to the eye. Any tear in a dural venous sinus will also result in an extradural bleed. Blood accumulates between bone and dura.

**Clinical features** The lucid interval may last a few hours to a few days before a bleed declares itself by ↓GCS from rising ICP. Increasingly severe headache, vomiting, confusion, and seizures follow, ± hemiparesis with brisk reflexes and an upgoing plantar. If bleeding continues, the ipsilateral pupil dilates, coma deepens, bilateral limb weakness develops, and breathing becomes deep and irregular (brainstem compression). Death follows a period of coma and is due to respiratory arrest. Bradycardia and ↑BP are late signs.

**Differentials** Epilepsy, carotid dissection, carbon monoxide poisoning.

**Tests** CT (fig 10.23) shows a haematoma (often biconvex/lens-shaped; the blood forms a more rounded shape compared with the sickle-shaped subdural haematoma as the tough dural attachments to the skull keep it more localized). Skull X-ray may be normal or show fracture lines crossing the course of the middle meningeal vessels. ►*Lumbar puncture is contraindicated.*

**Management** Stabilize and transfer urgently (with skilled medical and nursing escorts) to a neurosurgical unit for clot evacuation ± ligation of the bleeding vessel. Care of the airway in an unconscious patient and measures to ↓ICP often require intubation and ventilation (+ mannitol IVI, p831).

**Prognosis** Excellent if diagnosis and operation early. Poor if coma, pupil abnormalities, or decerebrate rigidity are present pre-op.


**Fig 10.22** This image explains the cause as well as the pathology. On the patient's left, cerebral sulci are prominent and prior to this adverse event would have been even larger. The brain had shrunk within the skull as a result of atherosclerosis, and poor perfusion, leaving large subarachnoid spaces. A simple, quick rotation of the head is enough to tear a bridging vein, causing this *acute subdural haematoma*.

We thank Prof. Peter Scally
for these CT images and commentary.


**Fig 10.23** The blood (high attenuation, fusiform or biconvex collection) on the right side is limited anteriorly by the coronal suture and posteriorly by the lambdoid suture. This is therefore an *extradural haematoma*. The low-attenuation CSF density collection on the left is causing scalloping of the overlying bone. It is in the typical location of an arachnoid cyst; an incidental finding of a congenital abnormality.

We thank Prof. Peter Scally
for these CT images and commentary.

# Delirium (acute confusional state)

Delirium[10] affects up to 50% of inpatients >65y, and is associated with a longer admission, more complications, and higher mortality. ►Look for an underlying cause in *any* acute fluctuating, baffling behaviour change; it may be an early indication of treatable pathology (eg UTI).

**Clinical features** Globally impaired cognition, perception, and consciousness which develops over hours/days, characterized by a marked memory deficit, disordered or disorientated thinking, and reversal of the sleep-wake cycle. Some patients experience tactile or visual hallucinations. Delirium can be: *•hyperactive,* with restlessness, mood lability, agitation, or aggression *•hypoactive* in which the patient becomes slow and withdrawn or *•mixed.* ►Hypoactive and mixed delirium are much harder to recognize: it is crucial to compare current behaviour to the patient's baseline (see BOX).

**Risk factors** >65y, dementia/previous cognitive impairment, hip fracture, acute illness, psychological agitation (eg pain).

**Causes**
- Surgery/post-GA.
- Systemic infection: pneumonia, UTI, malaria, wounds, IV lines.
- Intracranial infection or head injury.
- Drugs/drug withdrawal: opiates, levodopa, sedatives, recreational.
- Alcohol withdrawal (2–5d post-admission; ↑LFTs, ↑MCV; history of alcohol abuse).
- Metabolic: uraemia, liver failure, $Na^+$ or ↑↓glucose, ↓Hb, malnutrition (beriberi, p268).
- Hypoxia: respiratory or cardiac failure.
- Vascular: stroke, myocardial infarction.
- Nutritional: thiamine, nicotinic acid, or $B_{12}$ deficiency.

**Differentials** Dementia (see BOX), anxiety, epilepsy: ►non-convulsive status epilepticus is an underdiagnosed cause of impaired cognition and odd behaviour: consider an EEG. Primary mental illness (eg schizophrenia) can also mimic delirium, but this is rare on the wards (especially if no past history).

**Tests** Look for the cause (eg UTI, pneumonia, MI): do FBC, U&E, LFT, blood glucose, ABG, septic screen (urine dipstick, CXR, blood cultures); also consider ECG, malaria films, LP, EEG, CT.

**Management** As well as identifying and treating the underlying cause, aim to:[3]
- Reorientate the patient: explain where they are and who you are at each encounter. Hunt down hearing aids/glasses. Visible clocks/calendars may help.
- Encourage visits from friends and family.
- Monitor fluid balance and encourage oral intake . Be vigilant for constipation.
- Mobilize and encourage physical activity.
- Practise sleep hygeine: restrict daytime napping, minimize night-time disturbance.
- Avoid or remove catheters, IV cannulae, monitoring leads and other devices (they increase infection risk and may get pulled out).
- Watch out for infection and physical discomfort/distress.
- Review medication and discontinue any unnecessary agents. Only use sedation if the patient is a risk to their own/other patients' safety (never use physical restraints). Consider haloperidol 0.5–2mg, or chlorpromazine 50–100mg, PO if they will take it, IM if not (p15). Wait 20min to judge effect—further doses can be given if needed. NB: avoid chlorpromazine in the elderly and in alcohol withdrawal (p280); avoid antipsychotics in those with Parkinson's disease or Lewy body dementia.

►Be aware that delirium may persist beyond the duration of the original illness by several weeks in the elderly. Do not assume this must be dementia—provide support and reassess 1–2 months later.

10 Delirium, from the Latin *de* (from) and *lira* (ridge between furrows), meaning 'out of one's furrow'.

### Delirium vs dementia

One is often mistaken for the other, yet perhaps the interconnectedness of these two conditions is greater than we realize: not only is dementia the leading risk factor for delirium, but delirium itself confers a greater risk of subsequently developing dementia.[4] It is likely that this is due to a number of factors: delirium is a marker of vulnerability of the brain, and may also emphasize previously unrecognized dementia symptoms. Furthermore, there may be direct causation through the noxious insults incurred during an episode of delirium, which can lead to permanent neuronal damage.

In distinguishing the two conditions (not always an easy task) the presence of inattention, distractibility, and disorganized thinking will all point you towards delirium. But the fundamental question is 'Has there been an *acute* change from the patient's cognitive baseline?' Family or carer collateral reports are invaluable, but may not always be available. Document cognition in all patients >65y admitted to hospital (eg AMTS, p64, many admission proformas allow for this). This will then allow you to compare their admission score with subsequent assessments and track any improvements, deteriorations, or fluctuations in cognition throughout the admission.

# Dementia

A neurodegenerative syndrome with progressive decline in several cognitive domains. The initial presentation is usually of memory loss over months or years (►look for other causes if over days/weeks). Prevalence increases with age: 20% of people >80yrs are known to have dementia, yet probably only half of cases are diagnosed.

**Diagnosis** Is made by: *History* from the patient with a thorough collateral narrative—ask about the timeline of decline and the domains affected. Non-cognitive symptoms such as agitation, aggression, or apathy indicate late disease.[5] *Cognitive testing:* Use a validated dementia screen such as the AMTS (p64) or similar, plus short tests of executive function and language. Carry out a mental state examination to identify anxiety, depression, or hallucinations. *Examination* may identify a physical cause, risk factors (eg for vascular dementia), or parkinsonism. *Medication review* is important to exclude drug-induced cognitive impairment.

**Investigations** Look for reversible/organic causes: ↑TSH/↓$B_{12}$/↓folate (treat low-normals, p334), ↓thiamine (eg alcohol), ↓$Ca^{2+}$. Check MSU, FBC, ESR, U&E, LFT, and glucose. An MRI (preferred to CT) can identify other reversible pathologies (eg subdural haematoma, p482; normal-pressure hydrocephalus[11]), as well as underlying vascular damage or structural pathology. Functional imaging (FDG, PET, SPECT) may help delineate subtypes where diagnosis is not clear. Consider EEG in: suspected delirium, frontotemporal dementia, CJD, or a seizure disorder. If clinically indicated then check autoantibodies, syphilis, HIV, CJD, or other rare causes (see later in topic).

**Subtypes** • *Alzheimer's disease (AD):* See p488. • *Vascular dementia:* (~25%.) Cumulative effect of many small strokes: sudden onset and stepwise deterioration is characteristic (but often hard to recognize). Look for evidence of arteriopathy (↑BP, past strokes, focal CNS signs). ►Do not use acetylcholinesterase inhibitors or memantine in these patients. • *Lewy body dementia:* (15–25%.) Fluctuating cognitive impairment, detailed visual hallucinations, and later, parkinsonism (p494). Histology is characterized by Lewy bodies (eosinophilic intracytoplasmic inclusion bodies) in brainstem and neocortex. ►Avoid using antipsychotics in Lewy body dementia (↑↑risk of SE, see p489). • *Fronto-temporal dementia:* Frontal and temporal atrophy with loss of >70% of spindle neurons. Patients display executive impairment; behavioural/personality change; disinhibition; hyperorality, stereotyped behaviour, and emotional unconcern. Episodic memory and spatial orientation are preserved until later stages. *Pick's disease* refers to the few fronto-temporal dementia patients who have Pick inclusion bodies on histology (spherical clusters of tau-laden neurons).

**Other causes** Alcohol/drug abuse; repeated head trauma; pellagra (p268), Whipple's disease (p716); Huntington's (p702); CJD (p696); Parkinson's (p494); HIV; cryptococcosis (p408); familial autosomal dominant Alzheimer's; CADASIL (p470).

**Management** ►Refer suspected or diagnosed dementia to integrated memory services for further assessment and management. *Medication:* (p489). Avoid drugs that impair cognition (eg neuroleptics, sedatives, tricyclics). *Non-pharmacological interventions:* Non-cognitive symptoms (eg agitation) may respond to measures such as aromatherapy, multisensory stimulation, massage, music, and animal-assisted therapy.

**Other considerations** • *Depression:* Common. Try an SSRI (eg citalopram 10–20mg OD) or, if severe, mirtazapine (15–45mg at night if eGFR >40). Cognitive behavioural therapy can help with social withdrawal and catastrophic thinking. • *Capacity:* Can the patient make decisions regarding medical or financial affairs? Wherever possible, allow them to. Suggest making an advanced directive or appointing a Lasting Power of Attorney in the early stages of the disease.

11 Dilated ventricles *without* enlarged sulci. *Signs:* gait apraxia, incontinence, dementia; CSF shunts help.

### Who will care for the carers?

Our ageing population and improvements in medicine mean that we not only have an increasing number of people with dementia in the UK, but that those people are living longer with the disease in more advanced stages. Their needs become more complex and they become increasingly dependent. Currently, informal (mostly family) carers of people with dementia save the UK £11 billion a year. Yet this is not an easy task: most dementia sufferers display behavioural or psychological symptoms, which can be particularly distressing for the carer. Carer stress is inevitable and causes ↑morbidity and mortality. Ameliorate this with:

- *A care coordinator* (via Social Services or the local Old Age Community Mental Healthcare Team); vital to coordinate the various teams and services available:
  - Laundry services for soiled linen
  - Car badge giving priority parking
  - Help from occupational therapist, district nurses, and community psychiatric nurses
  - Attendance allowance
  - Respite care in hospital
  - Council tax rebate (forms from local council office).
- *Day services* can be invaluable for stimulating patients and providing regular, much-needed breaks for carers.
- *Moral support* Support groups, telephone helplines, and charities can all ease the burden, eg UK Alzheimer's Disease Society.
- *Combatting challenging behaviour:* First rule out pain, infection, and depression. Then consider trazodone (50–300mg at night) or lorazepam (0.5–1mg/12–24h PO). Haloperidol (0.5–4mg) can be useful in the short term.

# Alzheimer's disease (AD)

This leading cause of dementia is *the* big neuropsychiatric disorder of our times, dominating the care of the elderly and the lives of their families who give up work, friends, and ways of life to support relatives through the long final years as they exit into their '*worlds of preoccupied emptiness*'. Suspect AD in adults >40yrs with persistent,[12] progressive, and *global* cognitive impairment: visuo-spatial skill, memory, verbal abilities, and executive function (planning) are all affected, unlike other dementias which may affect certain domains but not others (identify which with neuropsychometric tests). There is also anosognosia—a lack of insight into the problems engendered by the disease, eg missed appointments, misunderstood conversations or plots of films, and mishandling of money. Later there may be irritability; mood disturbance (depression or euphoria); behavioural change (eg aggression, wandering, disinhibition); psychosis (hallucinations or delusions); agnosia (may not recognize self in the mirror). There is no standard natural history. Cognitive impairment is progressive, but non-cognitive symptoms may come and go over months. Eventually many patients become sedentary, taking little interest in anything.



**Cause** Environmental and genetic factors both play a role. Accumulation of β-amyloid peptide, a degradation product of amyloid precursor protein, results in progressive neuronal damage, neurofibrillary tangles, ↑numbers of amyloid plaques, and loss of the neurotransmitter acetylcholine (fig 10.24). Neuronal loss is selective—the hippocampus, amygdala, temporal neocortex, and subcortical nuclei are most vulnerable. Vascular effects are also important—95% of AD patients show evidence of vascular dementia.

**Risk factors** 1st-degree relative with AD; Down's syndrome (in which AD is inevitable, often <40yrs); homozygosity for apolipoprotein E (ApoE) E4 allele (see BOX 'Genetics and the future'); PICALM, CL1 & CLU variants; vascular risk factors (↑BP, diabetes, dyslipidaemia, ↑homocysteine, AF); ↓physical/cognitive activity; depression; loneliness (↑risk × 2; simply living alone is not a risk factor); smoking.

**Management** ►See p486 for a general approach to management in dementia. • Refer to a specialist memory service. • Acetylcholinesterase inhibitors (see BOX 'Pharmacological treatment'). • BP control (in heart failure there is a 2x ↑risk of AD; extra risk halves with BP control).

*Prevention in the context of AD's time-course:* Changes in CSF β-amyloid are seen ~25yrs before onset of unequivocal symptoms (USy) and its deposition is detected 15yrs before USy. CSF tau protein and brain atrophy are also detected 15yrs before USy. Cerebral hypometabolism and impaired episodic memory occur 10yrs before USy. Global cognitive impairment occurs 5yrs before USy. Prevention will probably be most effective before any of this starts—though there is currently insufficient evidence to recommend any specific interventions (BOX 'Genetics and the future'). Ultimately, there is no simple relationship between brain structure, neurofibrillary tangles, and function.

**Prognosis** Mean survival = 7yrs from USy.

12 'Enduring' doesn't mean unfluctuating: cognition comes and goes, allowing poetic insights, as in Iris Murdoch's poignant self-diagnosis: 'I am sailing into the dark'.

## Pharmacological treatment of cognitive decline

There is overlap between Lewy body dementia, AD, and Parkinson's disease (PD), complicating treatment decisions: L-dopa (p495) can precipitate delusions, and antipsychotic drugs worsen PD. Rivastigmine may help all three.

*Acetylcholinesterase (AChE) inhibitors:* Donepezil, rivastigmine, and galantamine are all modestly effective in treating AD and are recommended by NICE.[6] There is also some evidence for their efficacy in the dementia of Parkinson's disease, and rivastigmine may improve behavioural symptoms in Lewy body dementia—►though none should be used in mild disease and they should be discontinued if there is no worthwile effect on symptoms. Doses:

- Donepezil: initially 5mg PO, eg doubled after 1 month.
- Rivastigmine: 1.5mg/12h initially, ↑ to 3–6mg/12h. Patches are also available.
- Galantamine: initially 4mg/12h, ↑ to 8–12mg/12h PO.

►The cholinergic effects of acetylcholinesterase inhibitors may exacerbate peptic ulcer disease and heart block. Ask about symptoms and do an ECG first.

*Antiglutamatergic treatment:* Memantine (an NMDA antagonist, p449) is reasonably effective in late-stage AD, and is recommended in patients with severe disease or those with moderate disease in which AChE inhibitors are not tolerated/CI. *Dose:* 5mg/24h initially, ↑ by 5mg/d weekly to 10mg/12h. *SE:* hallucinations, confusion, hypertonia, hypersexuality.

*Antipsychotics*: Consider in severe, non-cognitive symptoms only (eg psychosis or extreme agitation). ►Possible increased risk of stroke/TIA so discuss risks and assess cerebrovascular risk factors. Avoid in mild-to-moderate: Lewy body dementia (risk of neuroleptic sensitivity reactions), AD, and vascular dementia.

*Vitamin supplementation:* Trials of dietary and vitamin supplements have been mixed and disappointing. Perhaps the best evidence exists for vitamin E (2000IU OD) which may confer a modest benefit in delaying functional progression in mild to moderate AD, but with no effect on cognitive performance.

## Genetics and the future

Possession of the APOE4 allele on chromosome 19 is the leading genetic cause of AD; homozygosity increases risk of developing the disease 12x. Yet while identifying this risk could enable a person to make lifestyle changes, there is little else to be done but anticipate one's impending cognitive decline (just one of the dilemmas raised by genetic testing). While the proteins that make up the plaques and neurofibrillary tangles seen in AD were identified in the early 1980s, progress has since been slow. So much so that in 2013, world leaders pledged to have a drug that would halt dementia within the next 10 years. Now a second-generation tau aggregation inhibitor called LMTX just might do the job, having shown success in phase III clinical trials in patients with mild/moderate AD, with clinical improvement and a slowing of atrophy on MRI. Perhaps in the imminent future, identifying those at greater risk of developing AD through genetic testing will have a role.

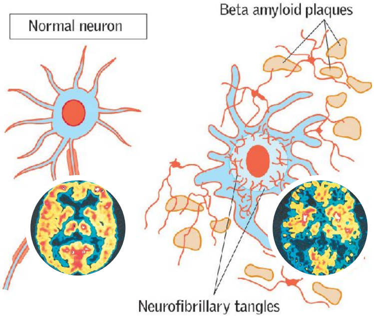



**Fig 10.24** Normal neuron (left) and one exhibiting senile plaques and neurofibrillary tangles (right). The corresponding changes on functional neuroimaging are also shown.

# Epilepsy: diagnosis

*Epilepsy* is a recurrent tendency to spontaneous, intermittent, abnormal electrical activity in part of the brain, manifesting as *seizures*. *Convulsions* are the motor signs of electrical discharges.

**Elements of a seizure** Some patients may experience a preceding *prodrome* lasting hours or days in which there may be a change in mood or behaviour. An *aura* implies a focal seizure, often, but not necessarily, from the temporal lobe. It may be a strange feeling in the gut, an experience such as *déjà vu* or strange smells or flashing lights. *Post-ictally* there may be headache, confusion, and myalgia; or temporary weakness after a focal seizure in the motor cortex (Todd's palsy, p712), or dysphasia following a focal seizure in the temporal lobe.

**Causes** ⅔ are idiopathic. *Structural:* Cortical scarring (eg head injury years before onset), developmental (eg dysembryoplastic neuroepithelial tumour or cortical dysgenesis), space-occupying lesion, stroke, hippocampal sclerosis (eg after a febrile convulsion), vascular malformations. *Others:* Tuberous sclerosis, sarcoidosis, SLE, PAN, antibodies to voltage-gated potassium channels.

**Diagnosis** Can be difficult due to the heterogenous nature of the disease (there are >40 different types of epilepsy). NICE estimate 5–30% of people with 'epilepsy' have been wrongly diagnosed. ► All patients with a seizure must be referred for specialist assessment and investigation in <2 wks.

*Take a thorough history:* Including a detailed description from a witness. Ask specifically about tongue-biting and a slow recovery. If this is a first seizure, enquire about past funny turns/odd behaviour. Déjà vu and odd episodic feelings of fear may well be relevant. Are there any triggers (eg alcohol, stress, flickering lights/TV)? Triggered attacks tend to recur.

*Establish the type of seizure:* See BOX 'Seizure classification'.[7] NB: if a seizure begins with focal features, it is a partial seizure, however rapidly it then generalizes. ►Don't forget non-epileptic attack disorder (= 'pseudo' seizures—psychogenic): this is not uncommon. (Suspect if seizures have a gradual onset, prolonged duration, and abrupt termination and are accompanied by closed eyes ± resistance to eye opening, rapid breathing, fluctuating motor activity, and episodes of motionless unresponsiveness. CNS exam, CT, MRI, and EEG are normal. It may coexist with true epilepsy.)

*Rule out provoking causes:* Most people would have a seizure given sufficient provocation (eg reflex anoxic seizures in faints) but would not be classed as epileptic: only 3–10% of provoked seizures recur; generally when the provocation is irreversible. *Causes:* trauma; stroke; haemorrhage; ↑ICP; alcohol or benzodiazepine withdrawal; metabolic disturbance (hypoxia, ↑↓$Na^+$, ↓$Ca^{2+}$, ↑↓glucose, uraemia, liver disease); infection (eg meningitis, encephalitis); ↑T°; drugs (tricyclics, cocaine). ►Unprovoked seizures have a recurrence rate of 30–50%.

**Investigations** Look for provoking causes. *Consider* an EEG: it cannot exclude epilepsy and can be falsely +ve, so don't do one if simple syncope is the likely diagnosis. Only do *emergency* EEGs if non-convulsive status is the problem. Other tests: MRI (structural lesions); drug levels (if on anti-epileptics: is the patient compliant?); drugs screen; LP (eg if infection suspected).

**Counselling** After any 'fit', advise about dangers, eg swimming, driving, heights until the diagnosis is known; then give *individualized* counselling on employment, sport, insurance, and conception (*OHCS* p28). The patient must contact DVLA and avoid driving until seizure-free for >1yr (p159).

Neurology

## Seizure classification

**Focal seizures** Originating within networks linked to one hemisphere and often seen with underlying structural disease. Various subclasses include:

- *Without impairment of consciousness:* (Previously described as 'simple'.) Awareness is unimpaired, with focal motor, sensory (olfactory, visual, etc.), autonomic, or psychic symptoms. No post-ictal symptoms.
- *With impairment of consciousness:* (Previously described as 'complex'.) Awareness is impaired—either at seizure onset or following a simple partial aura. Most commonly arise from the temporal lobe, in which post-ictal confusion is a feature.
- *Evolving to a bilateral, convulsive seizure:* (Previously described as 'secondary generalized'.) In ⅔ of patients with partial seizures, the electrical disturbance, which starts focally, spreads widely, causing a generalized seizure, which is typically convulsive.

**Generalized seizures** Originating at some point within, and rapidly engaging bilaterally distributed networks leading to simultaneous onset of widespread electrical discharge with no localizing features referable to a single hemisphere. Important subtypes include:

- *Absence seizures:* Brief (≤10s) pauses, eg suddenly stops talking in mid-sentence, then carries on where left off. Presents in childhood.
- *Tonic-clonic seizures:* Loss of consciousness. Limbs stiffen (tonic), then jerk (clonic). May have one without the other. Post-ictal confusion and drowsiness.
- *Myoclonic seizures:* Sudden jerk of a limb, face, or trunk. The patient may be thrown suddenly to the ground, or have a violently disobedient limb: one patient described it as '*my flying-saucer epilepsy*', as crockery which happened to be in the hand would take off.
- *Atonic (akinetic) seizures:* Sudden loss of muscle tone causing a fall, no LOC.
- *Infantile spasms:* (*OHCS* p206) Commonly associated with tuberous sclerosis.

NB: the classification of epileptic syndromes is separate to the classification of seizures, and is based on seizure type, age of onset, EEG findings, and other features such as family history.

## Localizing features of focal seizures

**Temporal lobe** •Automatisms—complex motor phenomena with impaired awareness, varying from primitive oral (lip smacking, chewing, swallowing) or manual movements (fumbling, fiddling, grabbing) , to complex actions. •Dysphasia. •*Déjà vu* (when everything seems strangely familiar), or *jamais vu* (everything seems strangely unfamiliar). •Emotional disturbance, eg sudden terror, panic, anger, or elation, and derealization (out-of-body experiences).• Hallucinations of smell, taste, or sound. •Delusional behaviour. • Bizarre associations—eg 'Canned music at Tesco always makes me cry and then pass out'.

**Frontal lobe** •Motor features such as posturing or peddling movements of the legs. •Jacksonian march (a spreading focal motor seizure with retained awareness, often starting with the face or a thumb). •Motor arrest. •Subtle behavioural disturbances (often diagnosed as psychogenic). •Dysphasia or speech arrest. •Post-ictal Todd's palsy (p712).

**Parietal lobe** •Sensory disturbances—tingling, numbness, pain (rare). •Motor symptoms (due to spread to the pre-central gyrus).

**Occipital lobe** •Visual phenomena such as spots, lines, flashes.

# Epilepsy: management

Living with epilepsy creates many problems: inability to drive and drug side-effects to name a few. Good management of the condition by an integrated specialized team is therefore of utmost importance.

**Anti-epileptic drugs (AEDs)** Should only be commenced by a *specialist*, after confirmed epilepsy diagnosis, ≥2 seizures (unless risk of recurrence is high, eg structural brain lesion, focal CNS deficit, or unequivocal epileptiform EEG), and following a detailed discussion of treatment options with the patient. AED choice depends on seizure type and epilepsy syndrome, comorbidities, lifestyle, and patient preference:

- *Focal (partial) seizures:* 1st line: carbamazepine or lamotrigine. 2nd line: levetiracetam, oxcarbazepine, or sodium valproate.[13]
- *Generalized tonic-clonic seizures:* 1st line: sodium valproate[13] or lamotrigine. 2nd line: carbamazepine, clobazam, levetiracetam, or topiramate.
- *Absence seizures:* 1st line: sodium valproate[13] or ethosuximide. 2nd line: lamotrigine.
- *Myoclonic seizures:* 1st line: sodium valproate.[13] 2nd line: levetiracetam, or topiramate (but ↑SE). Avoid carbamazepine and oxcarbazepine—may worsen seizures.
- *Tonic or atonic seizures:* Sodium valproate[13] or lamotrigine.

Treat with *one* drug and with *one* doctor in charge only. Slowly build up doses over 2–3 months (see BOX 'Anti-epileptic drugs (AEDs)') until seizures are controlled or maximum dosage is reached. If ineffective or not tolerated, switch to the next most appropriate drug. To switch drugs, introduce the new drug slowly, and only withdraw the 1st drug once established on the 2nd. Dual (adjunct) therapy is necessary in <10% of patients—consider if all appropriate drugs have been tried singly at the optimum dose.

*Stopping AEDs:* May be done *under specialist supervision* if the patient has been seizure-free for >2yrs and after assessing risks and benefits for the individual (eg the need to drive). The dose must be decreased slowly: over at least 2–3 months, or >6 months for benzodiazepines and barbiturates.

**Other interventions** *Psychological therapies:* Eg relaxation, CBT. May benefit some, but do not improve seizure frequency so only use as an adjunct to medication. *Surgical intervention:* Can be considered if a single epileptogenic focus can be identified (such as hippocampal sclerosis or a small low-grade tumour). Neurosurgical resection offers up to 70% chance of seizure resolution, but carries the risk of causing focal neurological deficits. Alternatives: vagal nerve stimulation, deep brain stimulation (DBS).

**Sudden unexpected death in epilepsy (SUDEP)** More common in uncontrolled epilepsy, and may be related to nocturnal seizure-associated apnoea or asystole. Those with epilepsy have 3x ↑mortality. >700 epilepsy-related deaths are recorded/yr in the UK; up to 17% are SUDEPs. The charity SUDEP Action may be of some help to families.

13 Sodium valproate is associated with significantly ↑risk of birth and developmental defects in children born to exposed mothers. Use in women of childbearing potential with caution and only after counselling.

## Anti-epileptic drugs (AEDs): typical adult doses and side-effects

*Carbamazepine:* (As slow-release.) Initially 100mg/12h, increase by 200mg/d every 2wks up to max 1000mg/12h. *SE:* leucopenia, diplopia, blurred vision, impaired balance, drowsiness, mild generalized erythematous rash, SIADH (rare; see p673).

*Lamotrigine:* As monotherapy, initially 25mg/d, ↑ by 50mg/d every 2wks up to 100mg/12h (max 250mg/12h). ►Halve monotherapy dose if on valproate; double if on carbamazepine or phenytoin (max 350mg/12h). *SE:* maculopapular rash—occurs in 10% (but 1/1000 develop *Stevens-Johnson syndrome* or *toxic epidermal necrolysis*) typically in 1st 8wks, especially if on valproate; warn patients to see a doctor at once if rash or flu symptoms develop; Other SEs: diplopia, blurred vision, photosensitivity, tremor, agitation, vomiting, aplastic anaemia.

*Levetiracetam:* Initially 250mg/24h, increase by 250mg/12h every 2wks up to max 1.5g/12h (if eGFR >80). *SE:* psychiatric side-effects are common, eg depression, agitation. Other SEs: D&V, dyspepsia, drowsiness, diplopia, blood dyscrasias.

*Sodium valproate:* Initially 300mg/12h, increase by 100mg/12h every 3d up to max 30mg/kg (or 2.5g) daily. *SE:* teratogenic. Nausea is very common (take with food). Other SEs: liver failure (watch LFT especially during 1st 6 months), pancreatitis, hair loss (grows back curly), oedema, ataxia, tremor, thrombocytopenia, encephalopathy (hyperammonaemia).

*Phenytoin:* No longer 1st line due to toxicity (nystagmus, diplopia, tremor, dysarthria, ataxia) and *SE:* ↓intellect, depression, coarse facial features, acne, gum hypertrophy, polyneuropathy, blood dyscrasias. Blood levels required for dosage.

►Carbamazepine, phenytoin, and barbiturates are liver enzyme inducing.

## Epilepsy and pregnancy

Epilepsy carries a 5% risk of fetal abnormalities, so good seizure control prior to conception and during pregnancy is vital. Yet some anti-epileptics are teratogenic: the patient must be given accurate information and counselling about contraception, conception, pregnancy, and breastfeeding in order to make informed decisions. In particular:

- Advise women of child-bearing age to take folic acid 5mg/d.
- Strictly avoid sodium valproate and polytherapy prior to conception and during pregnancy (lamotrigine is preferred but transition needs to be planned).
- Advise that most AEDs except carbamazepine and valproate are present in breast milk. Lamotrigine is not thought to be harmful to infants.
- Discuss contraceptive methods, bearing in mind that: enzyme-inducing AEDs make progesterone-only contraception unreliable, and oestrogen-containing contraceptives lower lamotrigine levels—an increased dose may be needed to achieve seizure control.

# Parkinsonism

This is the extrapyramidal triad of:

1 *Tremor.* Worse at rest; often 'pill-rolling' of thumb over fingers (see p468).
2 *Hypertonia.* Rigidity + tremor gives 'cogwheel rigidity', felt by the examiner during rapid pronation/supination.
3 *Bradykinesia.* Slow to initiate movement; actions slow and decrease in amplitude with repetition, eg ↓blink rate, micrographia. Gait is festinant (shuffling, pitched forward, fig 10.25) with ↓arm-swing and freezing at obstacles or doors (due to poor *simultaneous* motor and cognitive function). Expressionless face.

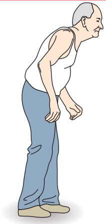


Fig 10.25 *'Marche à petit pas.'*

**Causes**

*Parkinson's disease (PD):* Loss of dopaminergic neurons in the substantia nigra, associated with Lewy bodies in the basal ganglia, brainstem, and cortex. Most cases are sporadic, though multiple genetic loci have been identified in familial cases. Mean age at onset is 60yrs. *Prevalence:* ↑ with age: 3.5% at 85–89yrs. *Clinical features:* The parkinsonian triad, plus non-motor symptoms such as: autonomic dysfunction (postural hypotension, constipation, urinary frequency/urgency, dribbling of saliva), sleep disturbance, and reduced sense of smell. Neuropsychiatric complications, such as depression, dementia, and psychosis, are common and debilitating. *Diagnosis:* Is clinical and based on the core features of bradykinesia with resting tremor and/or hypertonia; cerebellar disease and frontotemporal dementia should be excluded; a clinical response to dopaminergic therapy is supportive. ►Signs are invariably worse on one side—if symmetrical look for other causes. If an alternative cause is suspected then consider MRI to rule out structural pathology. Functional neuroimaging (DaTscan™, PET) is playing an emerging role. *Treatment:* Focuses on symptom control and does not slow disease progression (see BOX). Non-pharmacological options include deep brain stimulation (DBS, may help those who are partly dopamine-responsive) and surgical ablation of overactive basal ganglia circuits (eg subthalamic nuclei).

**Parkinson's plus syndromes** *Progressive supranuclear palsy:* (PSP, Steele-Richardson-Olszewski syndrome.) Early postural instability, vertical gaze palsy ± falls; rigidity of trunk >limbs; symmetrical onset; speech and swallowing problems; little tremor. *Multiple system atrophy:* (MSA; Shy-Drager.) Early autonomic features, eg impotence/incontinence, postural ↓BP; cerebellar + pyramidal signs; rigidity > tremor. *Cortico-basal degeneration:* (CBD.) Akinetic rigidity involving one limb; cortical sensory loss (eg astereognosis); apraxia (even autonomous interfering activity by affected limb—the 'alien limb' phenomenon). *Lewy body dementia:* See p486.

**Secondary causes** *Vascular parkinsonism:* (2.5–5% of parkinsonism, also called 'lower limb' parkinsonism). Eg diabetic/hypertensive patient with postural instability and falls (rather than tremor, bradykinesia, and festination). *Other secondary causes:* Drugs (neuroleptics, metoclopramide, prochlorperazine), toxins (manganese), Wilson's disease (p285), trauma (*dementia pugilistica*), encephalitis, neurosyphilis.

**Management** Requires input of a a multidisciplinary team (GP, neurologist, nurse specialist, social worker, carers, physio- and occupational therapist) to boost quality of life. Assess disability and cognition objectively and regularly, and monitor mood—depression is common. Involve palliative care services early on. Postural exercises and weight lifting may help. ►Don't forget the carers (p487): offer respite care.

### Pharmacological therapy in Parkinson's disease

A key decision is when to start supplementation of dopaminergic signalling with levodopa. Efficacy of this therapy reduces with time, requiring larger and more frequent dosing, with worsening SEs and response fluctuations (such as unpredictable freezing and pronounced end-of-dose reduced response: ~50% at 6yrs). Starting late may therefore be wise, eg when >70yrs or when PD seriously interferes with life: discuss pros and cons with the patient. ►Do not withdraw medication suddenly—risks acute akinesia and neuroleptic malignant syndrome. Be aware of situations where malabsorption could also have this effect (eg abdominal surgery, gastroenteritis).

*Levodopa:* Dopamine precursor, given combined with a dopa-decarboxylase inhibitor in co-beneldopa or co-careldopa. SEs: dyskinesia, painful dystonia. Non-motor SEs: psychosis; visual hallucinations, nausea and vomiting (give domperidone). Modified-release preparations should only be used in late disease.

*Dopamine agonists (DAs):* Ropinirole and pramipexole monotherapy can delay starting levodopa in early stages of PD, and allow lower doses of levodopa as PD progresses. Rotigotine transdermal patches are available as mono- or additive. SEs: drowsiness, nausea, hallucinations, compulsive behaviour (gambling, hypersexuality, p449). Ergot-derived DA-agonists (bromocriptine, pergolide, cabergoline) can cause fibrotic reactions, and are not favoured. Amantadine (weak DA) is used for drug-induced dyskinesias in late PD.

*Apomorphine:* Potent DA agonist used with continuous SC infusion to even out end-of-dose effects, or as a rescue-pen for sudden 'off' freezing. SE: injection-site ulcers.

*Anticholinergics:* (Eg benzhexol, orphenadrine.) Cause confusion in the elderly and have multiple SEs—limit to younger patients (but not 1st line).

*MAO-B inhibitors:* (Eg rasagiline, selegiline.) An alternative to dopamine agonists in early PD. SEs include postural hypotension and atrial fibrillation.

*COMT inhibitors:* (Eg entacapone, tolcapone.) May help motor complications in late disease. Lessen the 'off' time in those with end-of-dose wearing off. Tolcapone has better efficacy, but may cause severe hepatic complications and requires close monitoring of LFT.

# Multiple sclerosis (MS)

Inflammatory plaques of demyelination in the CNS *disseminated in space and time*; ie occuring at multiple sites, with ≥30d in between attacks. Demyelination heals poorly, eventually causing axonal loss; >80% of patients develop progressive disability. The exact cause of the disease remains unknown; it is most likely a combination of genetic and environmental factors. There is >30% concordance in identical twins, and unusual geographical distribution, with increasing incidence with latitude in some parts of the world (NB: adult migrants take their risk with them; children acquire the risk of where they settle)—leading to hypotheses of the roles of vitamin D and infection. *Mean age of onset* is 30yrs. ♀:♂ ≥3:1.

**Presentation** Usually monosymptomatic: ~20% present with unilateral optic neuritis (pain on eye movement and ↓rapid central vision). Corticospinal tract and bladder involvement are also common, and symptoms may worsen with heat (eg hot bath or exercise). Other symptoms/signs see table 10.8.

**Diagnosis** This is clinical, made by a consultant neurologist using established criteria (eg McDonald, see table 10.9) and after alternative diagnoses have been excluded. ►Early diagnosis and treatment reduce relapse rates and disability so refer to neurology as soon as MS is suspected.

**Tests** Depending on presenting symptoms, some patients may need extra supporting information to make a diagnosis (as per the McDonald criteria). ***MRI:*** Sensitive but not specific for plaque detection. It may also exclude other causes, eg cord compression. ***CSF:*** Oligoclonal bands of IgG on electrophoresis that are not present in serum suggest CNS inflammation. Delayed visual, auditory, and somatosensory *evoked potentials.*

**Progression** Most patients follow a relapsing-remitting course, with initial recovery in between relapses. With time, remission becomes incomplete, so disability accumulates (secondary progression). 10% of patients display steadily progressive disability in the absence of relapses (primary progressive MS), while a minority of patients experience no progressive disablement at all. *Poor prognostic signs:* Older ♂; motor signs at onset; many early relapses; many MRI lesions; axonal loss. *Pregnancy:* Does not alter the rate of progression: relapses may reduce during pregnancy and increase 3–6 months afterwards, but return to their previous rate thereafter.

**Management** As with all neurological conditions, requires the coordinated care of a multidisciplinary team and full involvement of the patient in all decisions.[8]

*Lifestyle advice:* Regular exercise, stopping smoking and avoiding stress may help.

*Disease-modifying drugs:* Dimethyl fumarate is an option for mild/moderate relapsing-remitting MS. The monoclonal antibodies alemtuzumab (acts against T cells) and natalizumab (acts against VLA-4 receptors that allow immune cells to cross the blood-brain barrier) are also approved for relapsing-remitting disease. Interferon beta and glatiramer are not recommended by NICE on the balance of clinical and cost-effectiveness. Azathioprine is not recommended due to its SE profile.

*Treating relapses:* Methylprednisolone, eg 0.5–1g/24h IV/PO for 3–5d shortens acute relapses; use sparingly (≲twice/yr; steroid SE, p377). It doesn't alter overall prognosis.

*Symptom control: Spasticity:* offer baclofen or gabapentin. Tizanidine or dantrolene are 2nd line; if these fail consider benzodiazepines. *Tremor:* botulinum toxin type A injections improve arm tremor and functioning. *Urgency/frequency:* if post-micturition residual urine >100mL, teach intermittent self-catheterization; if <100mL, try tolterodine. *Fatigue:* amantadine, CBT, and exercise may help.

Table 10.8 Clinical features of MS

| | | |
|---|---|---|
| *Sensory:* | • Dysaesthesia<br>• Pins and needles<br>• ↓Vibration sense<br>• Trigeminal neuralgia | *GI:* Swallowing disorders; constipation.<br>*Eye:* Diplopia; hemianopia; optic neuritis; visual phenomena (eg on exercise); bilateral internuclear ophthalmoplegia (p73); pupil defects. |
| *Motor:* | • Spastic weakness<br>• Myelitis | *Cerebellum:* Trunk and limb ataxia; intention tremor; scanning (ie monotonous) speech; falls. |
| *Sexual/GU:* | • Erectile dysfunction<br>• Anorgasmia; urine retention; incontinence | *Cognitive/visuospatial decline:* ►A *big* cause of unemployment, accidents, amnesia, ↓mood, ↓executive functioning. |

*NB* ↑T°, malaise, nausea, vomiting, positional vertigo, seizures, aphasia, meningism, bilateral optic neuritis, CSF leucocytosis and ↑CSF protein are rare in MS, and may suggest non-MS recurrent demyelinating disease, eg vasculitis or sarcoidosis.

## Diagnostic criteria for MS

Table 10.9 McDonald criteria for diagnosing MS (2010)

| Clinical presentation | Additional evidence needed for diagnosis |
|---|---|
| ≥2 attacks (relapses) with ≥2 objective clinical lesions | None |
| ≥2 attacks with 1 objective clinical lesion | • MRI: spatially disseminated lesions, or • +ve CSF *and* ≥2 MRI lesions, or • 2nd attack at a new site |
| 1 attack with ≥2 objective clinical lesions | Dissemination in time: • new lesion on repeat MRI after >3 months or • 2nd attack |
| 1 attack with 1 objective clinical lesion (monosymptomatic presentation) | Dissemination in space: • MRI *or* +ve CSF if ≥2 MRI lesions consistent with MS • *and* dissemination in time (by MRI *or* a 2nd clinical attack) |
| Insidious neurological progression suggestive of primary progressive MS | +ve CSF *and* dissemination in space on MRI/VEP *or* continued progression for ≥1yr |

►A careful history may reveal past episodes, eg brief unexplained visual loss, and detailed examination may show more than 1 lesion.

*Attacks* must last >1h, with >30d between attacks.

## Six MS eponyms

*Devic's syndrome:* (=Neuromyelitis optica, NMO.) MS variant with transverse myelitis, (loss of motor, sensory, autonomic, reflex, and sphincter function below the level of a lesion), optic atrophy, and anti-aquaporin 4 antibodies (p698).
*Lhermitte's sign:* Neck flexion causes 'electric shocks' in trunk/limbs. (Also +ve in cervical spondylosis, cord tumours and ↓$B_{12}$.)
*Uhthoff's phenomenon:* Worsening of symptoms with heat, eg in bath.
*Charles Bonnet syndrome:* (Rare.) ↓Acuity/temporary blindness ± complex visual hallucinations of faces, as well as animals, plants, and trees.
*Pulfrich effect:* Unequal eye latencies, causing disorientation in traffic as straight trajectories seem curved and distances are misjudged on looking sideways.
*Argyll Robertson pupil:* See p72.

## Space-occupying lesions (SOL)

**Signs** •↑*ICP:* (See p830.) Headache worse on waking, lying down, bending forward, or with coughing (p456); vomiting; papilloedema (only in 50% of tumours); ↓GCS.
• *Seizures:* Seen in ≤50%. Exclude SOL in all adult-onset seizures, especially if focal, or with a localizing aura or post-ictal weakness (Todd's palsy, p712).
• *Evolving focal neurology:* See BOX for localizing signs. ↑ICP causes *false localizing signs*: VIth nerve palsy is commonest (p70) due to its long intracranial course.
• *Subtle personality change:* Irritability, lack of application to tasks, lack of initiative, socially inappropriate behaviour.

**Causes** Tumour (primary or metastatic, later in topic), aneurysm, abscess (25% multiple); chronic subdural haematoma, granuloma (p197, eg tuberculoma), cyst (eg cysticercosis). *Tumours:* 30% are metastatic (eg breast, lung, melanoma). *Primaries:* astrocytoma, glioblastoma multiforme, oligodendroglioma, ependymoma. Also meningioma, primary CNS lymphoma (eg as non-infectious manifestation of HIV), and cerebellar haemangioblastoma.

**Differentials** Stroke, head injury, venous sinus thrombosis, vasculitis, MS, encephalitis, post-ictal, metabolic, or idiopathic intracranial hypertension.

**Tests** CT ± MRI (good for posterior fossa masses). Consider biopsy. Avoid LP before imaging (risks *coning*, ie cerebellar tonsils herniate through the foramen magnum).

**Tumour management** *Benign:* Remove if possible but some may be inaccessible. *Malignant:* Excision of gliomas is hard as resection margins are rarely clear, but surgery does give a tissue diagnosis, it debulks pre-radiotherapy, and makes a cavity for inserting carmustine wafers (delivers local chemotherapy). If a tumour is inaccessible but causing hydrocephalus, a ventriculo-peritoneal shunt can help. Chemo-radiotherapy is used post-op for gliomas or metastases, and as sole therapy if surgery is impossible. Oligodendroglioma with 1p/19q deletions is especially sensitive. In glioblastoma, temozolomide (alkylating agent) ↑survival. Seizure prophylaxis (eg phenytoin) is important, but often fails. Treat headache (eg codeine 60mg/4h PO). *Cerebral oedema:* Dexamethasone 4mg/8h PO; mannitol if ↑ICP acutely (p831). Plan meticulous palliative treatment (p534).

**Prognosis** Poor but improving (<50% survival at 5yrs) for CNS primaries; 40% 20yr survival for cerebellar haemangioblastoma; benign tumours are curable by excision.

**Third-ventricle colloid cysts** These congenital cysts declare themselves in adult life with amnesia, headache (often positional), obtundation (blunted consciousness), incontinence, dim vision, bilateral paraesthesiae, weak legs, and drop attacks. *℞:* Excision or ventriculo-peritoneal shunting.

## Idiopathic intracranial hypertension

Think of this in those presenting as if with a mass (headache, ↑ICP, and papilloedema)—*when none is found*. Most commonly seen in obese females in 3rd decade, who present with narrowed visual fields, blurred vision ± diplopia, VIth nerve palsy, and an enlarged blind spot, if papilloedema is present (it usually is). Consciousness and cognition are preserved.

**Associations** Endocrine abnormalities (Cushing's syndrome, hypoparathyroidism, ↑↓TSH), SLE, CKD, IDA, PRV, drugs (tetracycline, steroids, nitrofurantoin, and oral contraceptives).

**Management** Weight loss, acetazolamide or topiramate, loop diuretics, and prednisolone (start at ~40mg/24h PO; more SE than diuretics). Consider optic nerve sheath fenestration or lumbar-peritoneal shunt if drugs fail and visual loss worsens.

**Prognosis** Often self-limiting. Permanent significant visual loss in 10%.

## Localizing features

*Temporal lobe:* •Dysphasia (p86). •Contralateral homonymous hemianopia (or upper quadrantanopia if only Meyer's loop affected). •Amnesia. •Many odd or seemingly inexplicable phenomena, p491.

*Frontal lobe:* •Hemiparesis. •Personality change (indecent, indolent, indiscreet, facetious, tendency to pun). •Release phenomena such as the grasp reflex (fingers drawn across palm are grasped), significant only if unilateral. •Broca's dysphasia (p86), or more subtle difficulty with initiating and planning speech with intact repetition and no anomia—but loss of coherence. •Unilateral anosmia (loss of smell). •Perseveration (unable to switch from one line of thinking to another). •Executive dysfunction (unable to plan tasks). •↓Verbal fluency.

*Parietal lobe:* •Hemisensory loss. •↓2-point discrimination. •Astereognosis (unable to recognize an object by touch alone). •Sensory inattention. •Dysphasia (p86). •Gerstmann's syndrome (p700).

*Occipital lobe:* •Contralateral visual field defects. •Palinopsia (persisting images once the stimulus has left the field of view). •Polyopia (seeing multiple images).

*Cerebellum:* Remember **DANISH**: **d**ysdiadochokinesis (impaired *rapidly alternating* movements, p67) and **d**ysmetria (past-pointing); **a**taxia (limb/truncal—but if truncal ataxia is worse on eye closure, blame the dorsal columns); **n**ystagmus; **i**ntention tremor; **s**lurred speech (dysarthria); **h**ypotonia.

*Cerebellopontine angle:* (Eg acoustic neuroma/vestibular Schwannoma; p462.) Ipsilateral deafness, nystagmus, ↓corneal reflex, facial weakness (rare), ipsilateral cerebellar signs (above), papilloedema, VIth nerve palsy (p70).

*Midbrain:* (Eg pineal tumours or midbrain infarction.) Failure of up or down gaze; light-near dissociated pupil responses (p72), nystagmus on convergent gaze.

# Bell's palsy (idiopathic facial nerve palsy)



Affects 15–40/100 000/yr, ♂≈♀. Risk ↑ in pregnancy (×3) and in diabetes (×5).

**Clinical features** Abrupt onset (eg overnight or after a nap) with complete unilateral facial weakness at 24–72h; ipsilateral numbness or pain around the ear; ↓taste (ageusia); hypersensitivity to sounds (from stapedius palsy). On examination the patients will be unable to wrinkle their forehead, confirming LMN pathology (see p70), or whistle (tests buccinator). *Other symptoms of VIIth palsy (from any cause):* • Unilateral sagging of the mouth. • Drooling of saliva. • Food trapped between gum and cheek. • Speech difficulty. • Failure of eye closure may cause a watery or dry eye, ectropion (sagging and turning-out of the lower lid), injury from foreign bodies, or conjunctivitis.

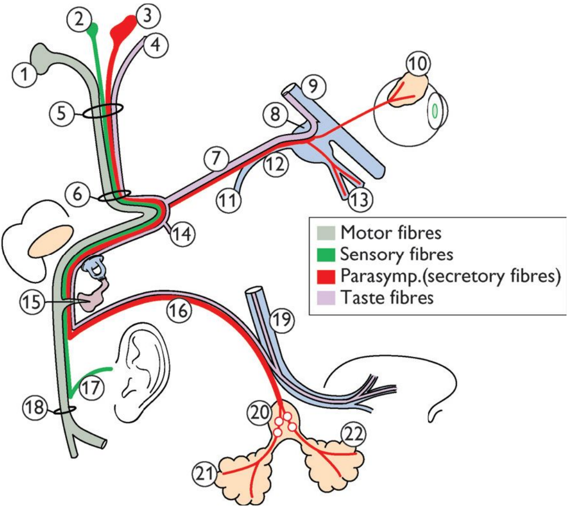
**Other causes of VIIth palsy** Account for ~30% of facial palsies. Think of these if: rashes, bilateral symptoms, UMN signs, other cranial nerve involvement, or limb weakness. *Infective:* Ramsay Hunt syndrome (BOX), Lyme disease, meningitis, TB, viruses (HIV, polio). *Brainstem lesions:* Stroke, tumour, MS. *Cerebellopontine angle tumours:* Acoustic neuroma, meningioma. *Systemic disease:* DM, sarcoidosis, Guillain-Barré. *Local disease:* Orofacial granulomatosis, parotid tumours, otitis media or cholesteatoma, skull base trauma.

►Lyme disease, Guillain-Barré, sarcoid, and trauma often cause bilateral weakness.

**Tests** Rule out the other causes: *Blood:* ESR; glucose; ↑*Borrelia* antibodies in Lyme disease, ↑VZV antibodies in Ramsay Hunt syndrome (BOX). *CT/MRI:* Space-occupying lesions; stroke; MS; *CSF:* (Rarely done) for infections.

**Prognosis** *Incomplete paralysis* without axonal degeneration usually recovers completely within a few weeks. Of those with *complete paralysis* ~80% make a full spontaneous recovery, but ~15% have axonal degeneration (~50% in pregnancy) in which case recovery is delayed, starting after ~3 months, and may be complicated by aberrant reconnections: *synkinesis,* eg eye blinking causes synchronous upturning of the mouth; misconnection of parasympathetic fibres (red in fig 10.26) can produce *crocodile tears* (gusto-lacrimal reflex) when eating stimulates unilateral lacrimation, not salivation.

**Management** *Drugs:* If given within 72h of onset, prednisolone (eg 60mg/d PO for 5d, tailing by 10mg/d) speeds recovery, with 95% making a full recovery. Antivirals (eg aciclovir) don't help; although some cases are thought to be associated with HSV-1, no one has shown actively replicating virus. There are little data to guide treatment if presenting after 72h of onset, but corticosteroids are widely used (though SE, p377). No advice on the use of steroids is universally agreed in pregnancy. *Protect the eye:* • Dark glasses and artificial tears (eg hypromellose) if evidence of drying. • Encourage regular eyelid closure by pulling down the lid by hand. • Use tape to close the eyes at night. *Surgery:* Consider if eye closure remains a long-term problem (lagophthalmos) or ectropion is severe.




**1** Facial nerve nucleus, deep in reticular formation of lower pons
**2** Spinal nucleus of V
**3** Superior salivary nuc.
**4** Solitary tract
**5** Porus acusticus internus
**6** Meatal foramen
**7** Large petrosal nerve
**8** Sphenopalatine ganglion
**9** Superior maxillary nerve
**10** Lacrimal gland
**11** Large deep petrosal nerve
**12** Vidian nerve
**13** Nose/palate gland nerves
**14** Small petrosal nerve at geniculate ganglion
**15** Stapedial nerve
**16** Chorda tympani
**17** Auricular branch
**18** Stylomastoid foramen
**19** Lingual nerve—and taste VII and general sensory from tongue ($V^3$)
**20** Submandibular ganglion (and gland, **21**)
**22** Sublingual gland

Fig 10.26 Facial nerve branches. The motor part moves the muscles of the face, scalp, and ears—also buccinator (puffs out the cheeks), platysma, stapedius, and the posterior belly of the digastric. It also contains the sympathetic motor fibres (vasodilator) of the submaxillary and sublingual glands (via the chorda tympani nerve). The sensory part contains the fibres of taste for the anterior ⅔ of the tongue and a few somatic sensory fibres from the middle ear region.

## Ramsay Hunt syndrome

Latent varicella zoster virus reactivating in the geniculate ganglion of the VIIth cranial nerve. *Symptoms:* Painful vesicular rash on the auditory canal ± on drum, pinna, tongue palate, or iris (→hyphaema, ie blood under the cornea) with ipsilateral facial palsy, loss of taste, vertigo, tinnitus, deafness, dry mouth and eyes. The rash may be subtle or even absent ('*herpes sine herpete*'=herpes without herpes). *Incidence:* ~5/100 000 (higher if >60yrs). *Diagnosis:* Clinical, as antiviral treatment is thought to be most effective within the 1st 72h, while the virus is replicating. *℞:* Antivirals (eg *aciclovir* 800mg PO 5×daily for 7d) + *prednisolone*, as for Bell's palsy. *Prognosis:* If treated within 72h, ~75% recover well; if not, ~⅓ make a good recovery, ⅓ a reasonable recovery, and ⅓ a poor recovery.

# Mononeuropathies

Lesions of individual peripheral or cranial nerves. Causes are usually local, such as trauma, or entrapment (eg tumour), except for carpal tunnel syndrome (see BOX 'Carpal tunnel syndrome').

**Median nerve C6-T1** The median nerve is the nerve of precision grip—muscles involved are easier to remember if you use your '**LOAF**' (two **l**umbricals, **o**pponens pollicis, **a**bductor pollicis brevis, and **f**lexor pollicis brevis). The clinical features depend on the location of the lesion: *At the wrist:* (Eg see BOX 'Carpal tunnel syndrome'.) Weakness of abductor pollicis brevis and sensory loss over the radial 3½ fingers and palm. *Anterior interosseous nerve lesions:* (Eg trauma.) Weakness of flexion of the distal phalanx of the thumb and index finger. *Proximal lesions:* (Eg compression at the elbow.) May show combined defects.

**Ulnar nerve C7-T1** Vulnerable to elbow trauma. *Signs:* Weakness/wasting of medial (ulnar side) wrist flexors, interossei (cannot cross the fingers in the good luck sign), and medial two lumbricals (claw hand, more marked in wrist lesions with digitorum profundus intact); hypothenar eminence wasting, weak 5th digit abduction, and 4th and 5th DIP joint flexion; sensory loss over medial 1½ fingers and ulnar side of the hand. *Treatment:* see BOX 'Managing ulnar mononeuropathies from entrapments'.

**Radial nerve C5-T1** This nerve opens the fist. It may be damaged by compression against the humerus. *Signs:* Test for wrist and finger drop with elbow flexed and arm pronated; sensory loss is variable—the dorsal aspect of the root of the thumb (the anatomical snuff box) is most reliably affected. Muscles involved: ('**BEAST**') **b**rachioradialis; **e**xtensors; **a**bductor pollicis longus; **s**upinator; **t**riceps.

**Brachial plexus** Pain/paraesthesiae and weakness in the affected arm in a variable distribution. *Causes:* Trauma, radiotherapy (eg for breast carcinoma), prolonged wearing of a heavy rucksack, cervical rib, thoracic outlet compression (also affects vasculature), or neuralgic amyotrophy (Parsonage–Turner syndrome: unilateral sudden, severe pain, followed over hours by profound weakness, resolving completely over days. May rarely involve the phrenic or lower cranial nerves.)

**Phrenic nerve C3-5** C3, 4, 5 keeps the diaphragm alive: lesions cause orthopnoea with a raised hemidiaphragm on CXR. *Causes:* Lung cancer, TB, paraneoplastic syndromes, myeloma, thymoma, cervical spondylosis/trauma, thoracic surgery, infections (HZV, HIV, Lyme disease), muscular dystrophy.

**Lateral cutaneous nerve of the thigh L2-L3** *Meralgia paraesthetica* is anterolateral burning thigh pain from entrapment under the inguinal ligament.

**Sciatic nerve L4-S3** Damaged by pelvic tumours or fractures to pelvis or femur. Lesions affect the hamstrings and all muscles below the knee (foot drop), with loss of sensation below the knee laterally.

**Common peroneal nerve L4-S1** Originates from sciatic nerve just above knee. Often damaged as it winds round the fibular head (trauma, sitting cross-legged). *Signs:* Foot drop, weak ankle dorsiflexion/eversion, sensory loss over dorsal foot.

**Tibial nerve L4-S3** Originates from sciatic nerve just above knee. Lesions lead to an inability to stand on tiptoe (plantarflexion), invert the foot, or flex the toes, with sensory loss over the sole.

**Mononeuritis multiplex** Describes the involvement of two or more peripheral nerves. Causes tend to be systemic: DM, connective tissue disorders (rheumatoid, SLE), vasculitis (granulomatosis with polyangiitis formerly Wegener's granulomatosis, PAN), and more rarely sarcoidosis, amyloid, leprosy. Electromyography (EMG) helps define the anatomic site of lesions.

### Carpal tunnel syndrome: the commonest mononeuropathy

The median nerve and nine tendons compete for space within the wrist. Compression is common, especially in women who have narrower wrists but similar-sized tendons to men.

*Clinical features:* Aching pain in the hand and arm (especially at night), and paraesthesiae in thumb, index, and middle fingers: relieved by dangling the hand over the edge of the bed and shaking it (remember 'wake and shake'). There may be sensory loss and weakness of abductor pollicis brevis ± wasting of the thenar eminence. Light touch, 2-point discrimination, and sweating may be impaired.

*Causes:* Anything causing swelling or compression of the tunnel: myxoedema; prolonged flexion (eg in a Colles' splint); acromegaly; myeloma; local tumours (lipomas, ganglia); rheumatoid arthritis; amyloidosis; pregnancy; sarcoidosis.

*Tests:* Neurophysiology helps by confirming the lesion's site and severity (and likelihood of improvement after surgery). Maximal wrist flexion for 1 min (*Phalen's test*) may elicit symptoms, and tapping over the nerve at the wrist can induce tingling (*Tinel's test*) but both are rather non-specific.

*Treatment:* Splinting, local steroid injection ± decompression surgery.

*NB:* There is also *tarsal* tunnel syndrome: unilateral burning sole pain following tibial nerve compression.

### Managing ulnar mononeuropathies from entrapments

*The ulnar nerve* asks for trouble in at least five places at the elbow, starting proximally at the arcade of Struthers (a musculofascial band ~8cm proximal to the medial epicondyle), and ending distally where it exits the flexor carpi ulnaris muscle in the forearm. Most often, compression occurs at the *epicondylar groove* or at the point where the nerve passes between the two heads of flexor carpi ulnaris (true *cubital tunnel syndrome*). Trauma can easily damage the nerve against its bony confines (the medial condyle of the humerus—the 'funny bone'). Normally, stretch and compression forces on the ulnar nerve at the elbow are moderated by its ability to glide in its groove. When normal excursion is restricted, irritation ensues. This may cause a vicious cycle of perineural scarring, consequent loss of excursion, and progressive symptoms—without antecedent trauma. Compressive ulnar neuropathies at the wrist (*Guyon's canal*—between the pisiform and hamate bones) are less common, but they can also result in disability.

*Treatment* centres on rest and avoiding pressure on the nerve, but if symptoms continue, night-time soft elbow splinting (to prevent flexion >60°) is warranted. A splint for the hand may help prevent permanent clawing of the fingers. For chronic neuropathy associated with weakness, or if splinting fails, a variety of *surgical procedures* have been tried. For moderately severe neuropathies, decompressions *in situ* may help, but often fail. Medial *epicondylectomies* are effective in ≤50% (but many will recur). Subcutaneous *nerve re-routing* (transposition) may be tried.

# Polyneuropathies

Motor and/or sensory disorder of multiple peripheral or cranial nerves: usually symmetrical, widespread, and often worse distally ('glove and stocking' distribution). They can be classified by: chronicity, function (*sensory, motor*, *autonomic*, *mixed*), or pathology (*demyelination*, *axonal degeneration*, or *both*). For example, Guillain-Barré syndrome (p702) is an acute, predominantly motor, demyelinating neuropathy, whereas chronic alcohol abuse leads to a chronic, initially sensory then mixed, axonal neuropathy.

**Diagnosis** The history is vital: be clear about the time course, the precise nature of the symptoms, and any preceding or associated events (eg D&V before Guillain-Barré syndrome; ↓weight in cancer; arthralgia from a connective tissue disease). Ask about travel, alcohol and drug use, sexual infections, and family history. If there is palpable nerve thickening think of leprosy or Charcot-Marie-Tooth. Examine other systems for clues to the cause, eg alcoholic liver disease.

**Tests** FBC, ESR, glucose, U&E, LFT, TSH, $B_{12}$, electrophoresis, ANA, ANCA, CXR, urinalysis, consider LP ± specific genetic tests for inherited neuropathies, lead level, antiganglioside antibodies. Nerve conduction studies distinguish demyelinating from axonal causes.

*Sensory neuropathy:* (Eg DM, CKD, leprosy.) Numbness; pins and needles, paraesthesiae; affects 'glove and stocking' distribution. Difficulty handling small objects such as buttons. Signs of trauma (eg finger burns) or joint deformation may indicate sensory loss. Diabetic and alcoholic neuropathies are typically painful.

*Motor neuropathy:* (Eg Guillain-Barré syndrome, lead poisoning, Charcot-Marie-Tooth syndrome.) Often progressive (may be rapid); weak or clumsy hands; difficulty in walking (falls, stumbling); difficulty in breathing (↓vital capacity). Signs: LMN lesion: wasting and weakness most marked in the distal muscles of hands and feet (foot or wrist drop). Reflexes are reduced or absent.

*Cranial nerves:* Swallowing/speaking difficulty; diplopia.

*Autonomic system:* See BOX.

**Management** *Treat the cause* (table 10.10). Involve physio and OT. Foot care and *shoe choice* are important in sensory neuropathies to minimize trauma. Splinting joints helps prevent contractures in prolonged paralysis. In Guillain-Barré and chronic inflammatory demyelinating polyradiculoneuropathy (CIDP: autoimmune demyelination of peripheral nerves), IV immunoglobulin helps. For vasculitic causes, steroids/immunosuppressants may help. Treat neuropathic pain with amitriptyline, duloxetine, gabapentin or pregabalin.

Table 10.10 Causes of polyneuropathy

| Metabolic | Vasculitides | Malignancy | Inflammatory |
|---|---|---|---|
| Diabetes mellitus<br>Renal failure<br>Hypothyroidism<br>Hypoglycaemia<br>Mitochondrial disorders | Polyarteritis nodosa<br>Rheumatoid arthritis<br>GPA | Paraneoplastic syndromes<br>Polycythaemia rubra vera | Guillain-Barré syndrome<br>Sarcoidosis; CIDP* |
| **Infections** | **Nutritional** | **Inherited syndromes** | **Drugs** |
| Leprosy<br>HIV<br>Syphilis<br>Lyme disease | ↓Vit $B_1$<br>↓Vit $B_{12}$/folate<br>↑Vit $B_6$<br>↓Vit E | Charcot-Marie-Tooth<br>Refsum's syndrome<br>Porphyria<br>Leucodystrophy | Vincristine<br>Cisplatin<br>Isoniazid<br>Nitrofurantoin<br>Phenytoin<br>Metronidazole |
| **Others** | | | |
| Paraproteinaemias, amyloidosis, lead, arsenic | | | |

*Chronic inflammatory demyelinating polyradiculoneuropathy (CIDP): autoimmune demyelination of peripheral nerves (distal onset of weakness/sensory loss in limbs + nerve enlargement + ↑CSF protein).

## Autonomic neuropathy

Sympathetic and parasympathetic neuropathies may be isolated or part of a generalized sensorimotor peripheral neuropathy.

**Causes** DM, amyloidosis, Guillain-Barré and Sjögren's syndromes, HIV, leprosy, SLE, toxic, genetic (eg porphyria), or paraneoplastic, eg paraneoplastic encephalomyeloneuropathies and Lambert-Eaton myasthenic syndrome (LEMS, p512).

**Signs** *Sympathetic:* Postural hypotension, ↓sweating, ejaculatory failure, Horner's syndrome (p702). *Parasympathetic:* Constipation, nocturnal diarrhoea, urine retention, erectile dysfunction, Holmes-Adie pupil (p72).

### Autonomic function tests

- *BP: Postural drop* of ≳20/10mmHg is abnormal.
- *ECG:* A variation of <10bpm with respiration is abnormal (check R-R interval).
- *Cystometry:* Bladder pressure studies.
- *Pupils:* Instil 0.1% adrenaline (dilates if post-ganglionic sympathetic denervation, not if normal); 2.5% cocaine (dilates if normal; not if sympathetic denervation); 2.5% methacholine (constricts if parasympathetic lesion)—rarely used.
- *Paraneoplastic antibodies:* Anti-HU, anti-YO, anti-RI, anti-amphiphysin, anti-CV2, anti-MA2. *Other Ab:* Antiganglionic acetylcholine receptor antibody presence shows that the cause may be autoimmune autonomic ganglionopathy.

**Primary autonomic failure** Occurs alone (autoimmune autonomic ganglionopathy), as part of multisystem atrophy (MSA, p494), or with Parkinson's disease, typically in a middle-aged/elderly man. Onset: insidious; symptoms as listed previously.

# Motor neuron disease (MND)

MND is a cluster of neurodegenerative diseases affecting 6/100 000 (♂:♀≈3:2), characterized by selective loss of neurons in motor cortex, cranial nerve nuclei, and anterior horn cells. Upper and lower motor neurons can be affected but there is *no* sensory loss or sphincter disturbance, thus distinguishing MND from MS and polyneuropathies. MND never affects eye movements, distinguishing it from myasthenia (p512). There are four clinical patterns:

1 *ALS/amyotrophic lateral sclerosis.* (Archetypal MND; up to 80%.) Loss of motor neurons in motor cortex *and* the anterior horn of the cord, so combined UMN + LMN signs (p446). Worse prognosis if: bulbar onset, ↑age; ↓FVC.

2 *Progressive bulbar palsy.* (10–20%.) Only affects cranial nerves IX–XII. See BOX 'Bulbar and corticobulbar ('pseudobulbar') palsy'.

3 *Progressive muscular atrophy.* (<10%.) Anterior horn cell lesion, so LMN signs only. Affects distal muscle groups before proximal. Better prognosis than ALS.

4 *Primary lateral sclerosis.* (Rare.) Loss of Betz cells in motor cortex: mainly UMN signs, marked spastic leg weakness and pseudobulbar palsy. No cognitive decline.

**Presentation** Think of MND in those >40yrs (median UK age at onset is 60) with stumbling spastic gait, foot-drop ± proximal myopathy, weak grip (door-handles don't turn) and shoulder abduction (hair-washing is hard), or aspiration pneumonia. Look for UMN signs: spasticity, brisk reflexes, ↑plantars; and LMN signs: wasting, fasciculation of tongue, abdomen, back, thigh. Is speech or swallowing affected (bulbar signs)? Fasciculation is not enough to diagnose an LMN lesion: look for weakness too. Frontotemporal dementia occurs in ~25% (see BOX 'Dignity and Dignitas').

**Diagnostic criteria** (See BOX 'Revised El Escorial diagnostic criteria for ALS'). There is no diagnostic test. Brain/cord MRI helps exclude structural causes, LP helps exclude inflammatory ones, and neurophysiology can detect subclinical denervation and help exclude mimicking motor neuropathies.[14]

**Prognosis** Poor, <3yrs post onset in half of patients.

**Management** Adopt a multidisciplinary approach: neurologist, palliative nurse, hospice, physio, OT, speech therapist, dietician, social services—all orchestrated by the GP. Riluzole, an inhibitor of glutamate release and NMDA receptor antagonist, is the only medication shown to improve survival. Mulitple other drugs that have shown promise in animal models have failed to prove benefit in clinical trials, including neurotrophic factors, anti-apoptotic agents, antioxidants, and immunomodulatory drugs. For supportive/symptomatic treatment: *Excess saliva:* Advise on positioning, oral care, and suctioning. Try an antimuscarinic (eg propantheline) or glycopyrronium bromide (can be given SC). Botulinum toxin A may help. *Dysphagia:* Blend food. Gastrostomy is an option—discuss early on. *Spasticity:* Exercise, orthotics. See MS for drugs (p496). *Communication difficulty:* Provide 'augmentative and alternative' communication equipment. *End-of-life care:* ►Involve palliative care team from diagnosis (p532). Consider opioids to relieve breathlessness and discuss non-invasive ventilation (see BOX 'Dignity and Dignitas').

14 If no UMN signs and distal arm muscles are affected in the distribution of individual nerves, suspect multifocal motor neuropathy with conduction block (diagnose on nerve conduction studies; ℞: IV Ig). Gynaecomastia, atrophic testes ± infertility suggests Kennedy syndrome (bulbospinal muscular atrophy).

## Bulbar and corticobulbar ('pseudobulbar') palsy

**Bulbar palsy** denotes diseases of the nuclei of cranial nerves IX–XII in the medulla. *Signs:* An *LMN lesion* of the tongue and muscles of talking and swallowing: flaccid, fasciculating tongue (like a sack of worms); jaw jerk is normal or absent, speech is quiet, hoarse, or nasal. *Causes:* MND, Guillain-Barré, polio, myasthenia gravis, syringobulbia (p516), brainstem tumours, central pontine myelinolysis (p672).

**Corticobulbar palsy** *UMN lesion* of muscles of swallowing and talking due to bilateral lesions above the mid-pons, eg corticobulbar tracts (MS, MND, stroke, central pontine myelinolysis). It is commoner than bulbar palsy. *Signs:* Slow tongue movements, with slow deliberate speech; ↑jaw jerk; ↑pharyngeal and palatal reflexes; pseudobulbar affect (PBA)—weeping unprovoked by sorrow or mood-incongruent giggling (emotional incontinence *without* mood change is also seen in MS, Wilson's, and Parkinson's disease, dementia, nitrous oxide use, and head injury). In some countries, dextromethorphan + quinidine is licensed for PBA.

## Revised El Escorial diagnostic criteria for ALS

**Definite** Lower + upper motor neuron signs in 3 regions.
**Probable** Lower + upper motor neuron signs in 2 regions.
**Probable with lab support** Lower + upper motor neuron signs in 1 region, or upper motor neuron signs in ≥1 region + EMG shows acute denervation in ≥2 limbs.
**Possible** Lower + upper motor neuron signs in 1 region.
**Suspected** Upper or lower motor neuron signs only—in 1 or more regions.

## Dignity and Dignitas

ALS is closely linked with frontotemporal dementia (FTD, p486) by increasing clinical, genetic, and molecular evidence (nucleotide repeat expansions in gene C9orf72 have been described in familial and sporadic ALS and in FTD). However, patients with MND often have no cognitive impairment in the early stages of the disease, and witness their inorexable physical decline with terrified awareness. For this reason it is imperative to plan for the future early: discuss their wishes for end-of-life care and in the eventuality of respiratory decline before it is too late for wishes to be communicated. These discussions are crucial but not binding: the patient can change their mind at any point, and also refuse any life-prolonging treatment, knowing the consequence is death. However, in most countries, including the UK, we cannot traverse that line that lies between management of a supported death following withdrawal of life-prolonging (or death-prolonging) interventions, and acting with the intention of causing death. Court battles ensue, with requests for assisted suicide that may frustrate and challenge ethicists, physicians, politicians, and the judiciary, but above all, patients and their families. Faced with such conflicting passions, perhaps our role is to be clear that we stand beside our patients, come what may.

# Cervical spondylosis

Degeneration of the cervical spine with age is inevitable, and has a wide clinical spectrum, ranging from asymptomatic to progressive spastic quadriparesis and sensory loss due to compression of the cord (myelopathy).

**Pathogenesis** Degeneration of the annulus fibrosus (the tough coating of the intervertebral discs), combined with osteophyte formation on the adjacent vertebra leads to narrowing of the spinal canal and intervertebral foramina (figs 10.27, 10.28). As the neck flexes and extends, the cord is dragged over these protruding bony spurs anteriorly and indented by a thickened ligamentum flavum posteriorly.

**Presenting complaint** Neck stiffness (but common in anyone >50yrs old), crepitus on moving neck, stabbing or dull arm pain (brachialgia), forearm/wrist pain.

**Signs** Limited, painful neck movement ± crepitus (examine gently). Neck flexion may produce tingling down the spine (Lhermitte's sign, p497). NB: this does not distinguish between cord or roots (or both) involvement.
*Root compression (radiculopathy):* Pain/'electrical' sensations in arms or fingers at the level of the compression (table 10.11), with numbness, dull reflexes, LMN weakness, and eventual wasting of muscles innervated by the affected root. NB: UMN signs below level of the affected root suggests cord compression.
►*Features of cord compression:* Progressive symptoms (eg ↑weak, clumsy hands; gait disturbance); UMN leg signs (spastic weakness, ↑plantars); LMN arm signs (wasting, hyporeflexia); incontinence, hesitancy, and urgency are late features.
**Which nerve root is affected?** See table 10.11.

**ΔΔ** MS; nerve root neurofibroma; subacute combined degeneration of the cord (↓$B_{12}$); compression by bone or cord tumours.

**Management** ►Urgent MRI and specialist referral guided by red flag symptoms (see p542). Bear in mind that although these are stressed in virtually every set of guidelines, no two lists are alike and review of evidence suggests that the accuracy of these features is low. ►Don't make referral decisions based upon the presence or absence of a single feature, but use these to inform your judgement. Otherwise: give analgesia (as per WHO ladder) and encourage gentle activity. Cervical collars may give respite during brief periods of increased pain, but restrict mobility, so may prolong symptoms: avoid where possible. If no improvement in 4–6 weeks then MRI and consider neurosurgical referral for: interlaminar cervical epidural injections, transforaminal injections or surgical decompression (via *anterior approach*, eg discectomy or *posterior approach*, eg laminectomy—fig 10.29, or laminoplasty—fig 10.30). There is no consensus or high-quality evidence to guide selection of approach or of patients, though interventions may be best reserved for those with progressive deterioration, myelopathy causing disabling neurologic deficits, or those at risk for deterioration (eg severe spinal cord compression on MRI).

Table 10.11 Clinical patterns of nerve root impingement

| Typical motor and sensory deficits from individual root involvement (C5-8) | | |
|---|---|---|
| **Nerve root** | **Motor and sensory deficit** | **Pain pattern** |
| **C5 (C4/C5 disc)** | Weak deltoid & supraspinatus; ↓supinator jerks; numb elbow. | Pain in neck/shoulder that radiates down front of arm to elbow. |
| **C6 (C5/C6 disc)** | Weak biceps & brachioradialis; ↓biceps jerks; numb thumb & index finger. | Pain in shoulder radiating down arm below elbow.* |
| **C7 (C6/C7 disc)** | Weak triceps & finger extension; ↓triceps jerks; numb middle finger. | Pain in upper arm and dorsal fore-arm. |
| **C8 (C7/T1 disc)** | Weak finger flexors & small muscles of the hand; numb 5th & ring finger. | Pain in upper arm and medial fore-arm. |
| ▶*Worrying symptoms:* Night pain, ↓weight, fever. | | |

*Passive head turning may exacerbate C6 radicular pain but not carpal tunnel syndrome (Spurling's manoeuvre).

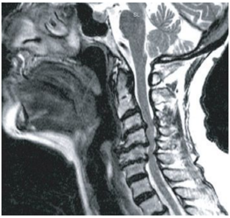


Fig 10.27 A T2-weighted MRI (∴ CSF looks bright). The cord is compressed between osteophytes anteriorly and the ligamentum flavum posteriorly.

©Prof P Scally.

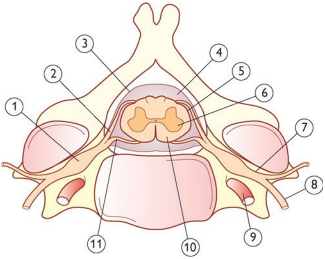



Fig 10.28 Cervical vertebra. **1** Dorsal root ganglion; **2** Dorsal root; **3** Dura mater; **4** Subarachnoid space; **5** Pia mater; **6** Grey matter; **7** Spinal nerve; **8** Ventral ramus; **9** Vertebral artery in the transverse foramen; **10** White matter; **11** Ventral spinal nerve.

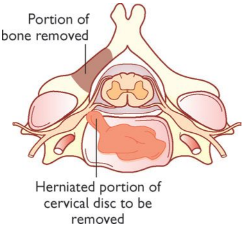



Fig 10.29 Laminectomy.

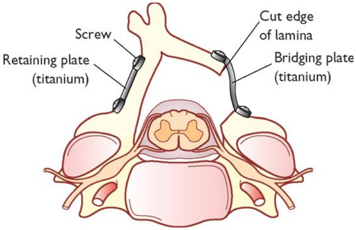



Fig 10.30 Laminoplasty (screws and plates).

# Myopathy

Primary disorder of muscle with gradual-onset symmetrical weakness; it may be confused clinically with neuropathy. *In favour of myopathy:* •Gradual onset of symmetric *proximal* weakness—difficulty combing hair and climbing stairs (NB: weakness is also *distal* in myotonic dystrophy). •Specific muscle groups affected (ie *selective* weakness on first presentation). •Preserved tendon reflexes. •No paraesthesiae or bladder problems. •No fasciculation (suggests anterior horn cell or root disease).

*Rapid onset* suggests a toxic, drug, or metabolic myopathy (or a neuropathy). *Excess fatigability* (↑weakness with exercise) suggests myasthenia (p512). Spontaneous *pain* at rest and local tenderness occurs in inflammatory myopathies. Pain on exercise suggests ischaemia or metabolic myopathy (eg McArdle's disease). *Oddly firm* muscles (due to infiltrations with fat or connective tissue) suggest pseudohypertrophic muscular dystrophies (eg Duchenne's).



**Tests** ESR, CK, AST, and LDH may be raised. Do EMG and tests relevant to systemic causes (eg TSH, p216). Muscle biopsy and genetic testing may help reach a diagnosis.

**Muscular dystrophies** A group of genetic diseases (see table 10.12) with progressive degeneration and weakness of specific muscle groups. The primary abnormality may be in the muscle membrane. There may be unusually firm muscles due to infiltration by fat or connective tissue, and marked variation in size of individual muscle fibres on histology. • *Duchenne's muscular dystrophy:* The commonest (3/1000 male live births). Presents at ~4yrs old with clumsy walking, then difficulty in standing, and respiratory failure. Pseudohypertrophy is seen, especially in the calves. Serum creatine kinase ↑ >40-fold. There is no specific treatment. Some survive beyond 20yrs. Home ventilation improves prognosis. Genetic counselling is vital. • *Becker's muscular dystrophy:* (~0.3/1000 ♂ births.) Presents similarly to Duchenne's but with milder symptoms, at a later age, and with a better prognosis. • *Facioscapulohumeral muscular dystrophy:* (FSHD, Landouzy–Dejerine.) Almost as common as Duchenne's. Onset is ~12–14yrs old, with inability to puff out the cheeks and difficulty raising the arms above the head. *Signs:* weakness of face ('ironed out' expression), shoulders, and upper arms (often asymmetric with deltoids spared), foot-drop, scapular winging (fig 10.31), scoliosis, anterior axillary folds, and horizontal clavicles. ≲20% need a wheelchair by 40yrs.

**Myotonic disorders** Cause tonic muscle spasm (myotonia), and demonstrate long chains of central nuclei within muscle fibres on histology. The commonest is *myotonic dystrophy* which is, in fact, clinically and genetically heterogeneous, with two major forms (DM1 and DM2—see table 10.12), both showing abnormal trinucleotide repeat expansions in regulatory (non-coding) genetic regions. DM1 is the commoner, more severe, and typically presents between 20–40 yrs old with *distal* weakness (hand/foot drop), weak sternomastoids, and myotonia. Facial weakness and muscle wasting give a long, haggard appearance. *Also:* cataracts, male frontal baldness, diabetes, testis/ovary atrophy, cardiomyopathy, and ↓cognition. Most DM1 patients die in late middle age of respiratory or cardiac complications. Mexiletine may help with disabling myotonia. Genetic counselling is important.

**Inflammatory myopathies** There may be spontaneous muscle pain at rest and local tenderness on palpation. *Inclusion body myositis* is the chief example if aged >50yrs. Weakness starts with quadriceps, finger flexors, or pharyngeal muscles. Ventral extremity muscle groups are more affected than dorsal or girdle groups. Response to therapy is poor and patients typically progress over a decade to require assistance with activites of daily living. Histology shows ringed vacuoles + intranuclear inclusions. *Polymyositis* and *dermatomyositis*, see p552.

**Metabolic myopathies** Eg *McArdle's disease* (glycogen storage disorder). Presents with muscle pain and weakness after exercise.

**Acquired myopathies of late onset** Often part of systemic disease—eg hyperthyroidism, malignancy, Cushing's, hypo- and hypercalcaemia.

**Drug causes** Alcohol; statins; steroids; chloroquine; zidovudine; vincristine; cocaine.

Fig 10.31 Winging of both scapulae in facioscapulohumeral muscular dystrophy, due to weakness of thoracoscapular muscles.

Reproduced from Donaghy, *Brain's Diseases of the Nervous System*, 12th edition, with permission from Oxford University Press.

Table 10.12 Genetics of some commoner congenital myopathies

| Condition | Inheritance | Chr | Gene | Pathogenesis |
|---|---|---|---|---|
| **Duchenne's muscular dystrophy** | X-linked recessive | X | Dystrophin (stabilizes muscle fibres) | Partial deletions or duplications in dystrophin render *non*-functional |
| **Becker's muscular dystrophy** | X-linked recessive | X | Dystrophin | Partial deletions or duplications in dystrophin render *hypo*-functional |
| **FSHD type 1** | Autosomal dominant | 4 | DUX4 (transcriptional activator) | Partial deletion of D4Z4 repeating unit releases normal repression of DUX4 expression |
| **FSHD type 2** | Autosomal dominant | 4 | DUX4 (transcriptional activator) | Hypomethylation of D4Z4 releases normal repression of DUX4 expression |
| **DM type 1** | Autosomal dominant | 19 | DMPK (serine-threonine kinase) | Expansion of short repetitive sequences of nucleotides; in both forms this expanded sequence is transcribed into RNA which then misfolds and sequesters other RNA binding proteins |
| **DM type 2** | Autosomal dominant | 3 | ZNF9 (transcriptional regulator) | |

# Myasthenia gravis (MG)

MG is an autoimmune disease mediated by antibodies to nicotinic acetylcholine receptors (AChR) on the post-synaptic side of the neuromuscular junction (fig 10.32). Both B and T cells are implicated.

**Presentation** Slowly increasing or relapsing muscular fatigue. Muscle groups affected, in order: extraocular; bulbar (swallowing, chewing); face; neck; limb girdle; trunk. *Signs:* Ptosis, diplopia, myasthenic snarl on smiling, 'peek sign' of orbicularis fatigability (eyelids begin to separate after manual opposition to sustained closure). On counting to 50, the voice fades (dysphonia is a rare presentation). Tendon reflexes are normal. *Symptoms exacerbated by:* Pregnancy, ↓$K^+$, infection, over-treatment, change of climate, emotion, exercise, gentamicin, opiates, tetracycline, quinine, β-blockers.

**Differentials** Polymyositis/other myopathies (p510); SLE; Takayasu's arteritis (fatigability of the extremities); botulism (see BOX).

**Associations** Include autoimmune disease (especially rheumatoid arthritis and SLE). If <50yrs, it is commoner in ♀ and associated with thymic hyperplasia; >50, it is commoner in men, and associated with thymic atrophy or thymic tumour.

**Tests** •*Antibodies:* ↑Anti-AChR antibodies in 90% (70% in MG variant confined to ocular muscles). If anti-AChR −ve look for MuSK antibodies (muscle-specific tyrosine kinase; especially in ♀). •*EMG:* Decremental muscle response to repetitive nerve stimulation ± ↑single-fibre jitter. •*Imaging:* CT to exclude thymoma (68% 5yr survival). •*Other:* Ptosis improves by >2mm after ice application to the eyelid for >2min—a neat, non-invasive test (but not diagnostic). The Tensilon® (edrophonium) test may not give clear answers and has dangers, so is rarely used.

**Treatment** •*Symptom control:* Anticholinesterase, eg pyridostigmine (60–120mg PO up to 6×daily; max 1.2g/d). Cholinergic SE: ↑salivation, lacrimation, sweats, vomiting, miosis. Other SE: diarrhoea, colic (controllable with propantheline 15mg/8h). • *Immunosuppression:* Treat relapses with prednisolone—start at 5mg on alternate days, ↑ by 5mg/wk up to 1mg/kg on each treatment day. ↓Dose on remission (may take months). Give osteoporosis prophylaxis. SE: weakness (hence low starting dose). Azathioprine, ciclosporin, and mycophenolate mofetil may also be used. •*Thymectomy:* Has beneficial effects, even in patients without a thymoma: consider especially in younger patients with onset <5yrs previously and poor response to medical therapy. A recent randomized controlled trial shows improved symptom scores sustained over 3yrs, with reduced need for immunosuppression. Surgery also prevents local invasion if thymoma is present.

**Myasthenic crisis** Life-threatening weakness of respiratory muscles during a relapse. ►Can be difficult to differentiate from cholinergic crisis (ie overtreatment—but this is rare, and usually only occurs in doses of pyridostigmine >960mg/d). Monitor forced vital capacity. *Ventilatory support* may be needed. Treat with plasmapheresis (removes AChR antibodies from the circulation) or IVIg and identify and treat the trigger for the relapse (eg infection, medications).

# Lambert-Eaton myasthenic syndrome (LEMS)

LEMS can be paraneoplastic (50% are associated with malignancies, in particular small-cell lung cancer) or autoimmune. Unlike MG, antibodies are to voltage-gated $Ca^{2+}$ channels on *pre*-synaptic membrane (see fig 10.33; anti-P/Q type VGCC antibodies are +ve in 85–95%).

**Clinical features** •Gait difficulty before eye signs. •Autonomic involvement (dry mouth, constipation, impotence). •Hyporeflexia and weakness, which improve after exercise. •Diplopia and respiratory muscle involvement are rare. •EMG shows similar changes to MG except amplitude increases greatly post-exercise.

**Treatment** Pyridostigmine, 3,4-diaminopyridine or IVIg (get specialist help). ►Do regular CXR/high-resolution CT as symptoms may precede the cancer by >4yrs.

## How synapses work—the neuromuscular junction

**1** Before transmission can occur, neurotransmitter must be packed into synaptic vesicles. At the neuromuscular junction (NMJ) this is acetylcholine (ACh). Each vesicle contains ~8000 ACh molecules.

**2** When an action potential arrives at the pre-synaptic terminal, depolarization opens voltage-gated $Ca^{2+}$ channels (VGCCs). In *Lambert-Eaton* syndrome, anti-P/Q type VGCC antibodies disrupt this stage of synaptic transmission.[15]

**3** Influx of $Ca^{2+}$ through the VGCCs triggers fusion of synaptic vesicles with the pre-synaptic membrane (a process that *botulinum toxin* interferes with), and neurotransmitter is released from the vesicles into the synaptic cleft.

**4** Transmitter molecules cross the synaptic cleft by diffusion and bind to receptors on the post-synaptic membrane, causing depolarization of the post-synaptic membrane (the end-plate potential). This change in the post-synaptic membrane triggers muscle contraction at the NMJ, or onward transmission of the action potential in neurons. In *myasthenia gravis*, antibodies block the post-synaptic ACh receptors, preventing the end-plate potential from becoming large enough to trigger muscle contraction—and muscle weakness ensues.

**5** Transmitter action is terminated by enzyme-induced degradation of transmitter (eg acetylcholinesterase), uptake into the pre-synaptic terminal or glial cells, or by diffusion away from synapse. Anticholinesterase treatments for myasthenia gravis, such as *pyridostigmine*, reduce the rate of degradation of ACh, increasing the chance that it will trigger an end-plate potential.

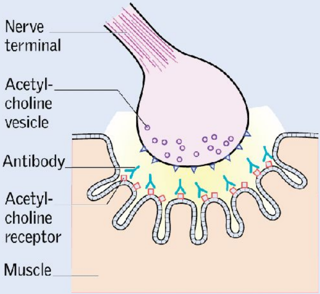



Fig 10.32 Myasthenia gravis features *post-synaptic* AChR antibodies. Tendon reflexes are normal because the synapses do not have time to become fatigued with such a brief muscle contraction. Ocular palsies are common (it's not exactly clear why).

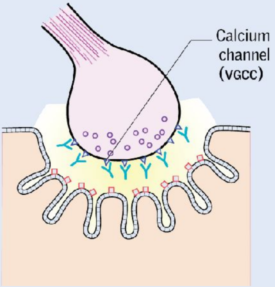



Fig 10.33 Lambert-Eaton syndrome features pre-synaptic $Ca^{2+}$-channel antibodies. Depressed tendon reflexes are common, because less transmitter is released, but reflexes may ↑ after maximum voluntary contraction due to a build of transmitter in the synaptic cleft (post-tetanic potentiation).

**15** Disruption of pre-synaptic transmission affects release of ACh in autonomic nervous system as well as at neuromuscular junction, explaining the prominence of dysautonomia in LEMS unlike in MG.

# Neurofibromatosis



## Type 1 neurofibromatosis (NF1, von Recklinghausen's disease)

Autosomal dominant inheritance (gene locus 17q11.2). Expression of NF1 is variable, even within a family. *Prevalence:* 1 in 2500, ♀:♂≈1:1; no racial predilection.

*Signs: Café-au-lait spots:* flat, coffee-coloured patches of skin seen in 1st year of life (clearest in UV light), increasing in size and number with age. Adults have ≥6, >15mm across. They do *not* predispose to skin cancer. *Freckling:* typically in skin-folds (axillae, groin, neck base, and submammary area), and usually present by age 10. *Dermal neurofibromas:* small, violaceous nodules, gelatinous in texture, which appear at puberty, and may become papillomatous. They are not painful but may itch. Numbers increase with age. *Nodular neurofibromas* arise from nerve trunks. Firm and clearly demarcated, they can give rise to paraesthesiae if pressed. *Lisch nodules* (fig 10.34) are tiny harmless regular brown/translucent mounds (hamartomas) on the iris (use a slit lamp) ≲2mm in diameter. They develop by 6yrs old in 90%. Also short stature and macrocephaly.

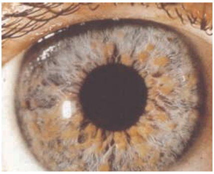


Fig 10.34 Multiple brown Lisch nodules on the iris. ©Jon Miles.

*Complications:* Occur in 30%. Mild learning disability is common. *Local effects of neurofibromas:* nerve root compression (weakness, pain, paraesthesiae); GI—bleeds, obstruction; bone—cystic lesions, scoliosis, pseudarthrosis. ↑BP from renal artery stenosis or phaeochromocytoma. Plexiform neurofibromas (large, subcutaneous swellings). *Malignancy* (5% patients with NF1): optic glioma, sarcomatous change in a neurofibroma. ↑Epilepsy risk (slight). *Rare association:* carcinoid syndrome (p271).

*Management:* Multidisciplinary team with geneticist, neurologist, surgeon, and physiotherapist, orchestrated by a GP. Yearly cutaneous survey and measurement of BP. Dermal neurofibromas are unsightly, and catch on clothing; if troublesome, excise, but removing all lesions is unrealistic. Genetic counselling is vital (*OHCS* p154).

## Type 2 neurofibromatosis (NF2)

Autosomal dominant inheritance, though 50% are *de novo*, with mosaicism in some (NF2 gene locus is 22q11). Rarer than NF1 with a prevalence of only 1 in 35 000.

*Signs: Café-au-lait* spots are fewer than in NF1. *Bilateral vestibular Schwannomas* (= acoustic neuromas; p462) are characteristic, becoming symptomatic by ~20yrs old when sensorineural hearing loss is the 1st sign. There may be tinnitus and vertigo. The rate of tumour growth is unpredictable and variable. The tumours are benign but cause problems by pressing on local structures and by ↑ICP. They may be absent in mosaic NF2. *Juvenile posterior subcapsular lenticular opacity* (a form of cataract) occurs before other manifestations and can be useful in screening those at risk.

*Complications:* Tender Schwannomas of cranial and peripheral nerves, and spinal nerve roots. Meningiomas (45% in NF2, often multiple). Glial tumours are less common. Consider NF2 in any young person presenting with one of these tumours in isolation.

*Management:* Hearing tests yearly from puberty in affected families, with MRI brain if abnormality is detected. A normal MRI in the late teens is helpful in assessing risk to any offspring. A clear scan at 30yrs (unless a family history of late onset) indicates that the gene has not been inherited. Treatment of vestibular Schwannomas is neurosurgical and complicated by hearing loss/deterioration and facial palsy. Mean survival from diagnosis is ~15yrs.

**Schwannomatosis** Multiple tender cutaneous Schwannomas without the bilateral vestibular Schwannomas that are characteristic of NF2. Indistinguishable from mosaic NF2, where vestibular Schwannomas are also absent, except by genetic analysis of tumour biopsies. There is typically a large tumour load, assessable only by whole-body MRI. Mutations in the tumour suppressor genes SMARCB1 and LZTR1 and spontaneous NF2 mutations have all been described. Life expectancy is normal.

## Diagnostic criteria for neurofibromatosis

*NF1 (von Recklinghausen's disease):*
Diagnosis is made if 2 of the following are found:

1. ≥6 *café-au-lait* macules >5mm (pre-pubertal) or >15mm (post-pubertal)
2. ≥2 neurofibromas of any type or 1 plexiform
3. Freckling in the axillary or inguinal regions
4. Optic glioma
5. ≥2 Lisch nodules
6. Distinctive osseous lesion typical of NF1, eg sphenoid dysplasia
7. First-degree relative with NF1 according to the above-listed criteria.

*Differential:* McCune-Albright syndrome (*OHCS* p650), multiple lentigines, urticaria pigmentosa.

*NF2:*
Diagnosis is made if either of the following are found:

1. Bilateral vestibular Schwannomas seen on MRI or CT
2. First-degree relative with NF2, and either:
   a) Unilateral vestibular Schwannoma; or
   b) One of the following:
   - Neurofibroma
   - Meningioma
   - Glioma
   - Schwannoma
   - Juvenile cataract (NF2 type).

*Differential:* NF1, Schwannomatosis.

*Causes of café-au-lait spots:* Normal (eg up to 5); NF1 (↑melanocyte density vs 'normal' *café-au-lait* spots); NF2; rare syndromes: Gaucher's; McCune-Albright; Russell-Silver; tuberous sclerosis; Wiskott-Aldrich.

# Syringomyelia



A syrinx is a tubular cavity in or close to the central canal of the cervical cord. *Mean age of onset:* 30yrs. *Incidence:* 8/100 000/yr. Symptoms may be static for years, but then worsen fast—eg on coughing or sneezing, as ↑pressure causes extension, eg into the brainstem (syringobulbia, see later in topic).

**Causes** *Typically,* blocked CSF circulation (without 4th ventricular communication), with ↓flow from basal posterior fossa to caudal space, eg Arnold–Chiari malformation (cerebellum herniates through foramen magnum); basal arachnoiditis (after infection, irradiation, subarachnoid haemorrhage); basilar invagination (in which the top of the odontoid process of C2 migrates upwards, causing foramen magnum stenosis ± medulla oblongata compression); masses (cysts, rheumatoid pannus, encephalocoele, tumours). *Less commonly*, a syrinx may develop after myelitis, cord trauma, or rupture of an AV malformation, or within spinal tumours (ependymoma or haemangioblastoma) due to fluid secreted from neoplastic cells or haemorrhage.

**Signs** *Dissociated sensory loss* (absent pain and T° sensation, with preserved light touch, vibration, and joint-position sense) due to pressure from the syrinx on the decussating anterolateral pathway (fig 10.35) in a root distribution reflecting the location of the syrinx (eg for typical cervical syrinx then sensory loss is over trunk and arms); *wasting/weakness* of hands ± *claw-hand* (then arms→shoulders→respiratory muscles). Anterior horn cells are also vulnerable. *Other signs:* Horner's syndrome (can be bilateral and therefore more difficult to spot); UMN leg signs; body asymmetry, limb hemihypertrophy, or unilateral odo- or chiromegaly (enlarged hand or foot), perhaps from release of trophic factors via anterior horn cells; Charcot's joints in the shoulder/wrist due to lost joint proprioception (see fig 5.11, p213).

**Syringobulbia** (Brainstem involvement.) Nystagmus, tongue atrophy, dysphagia, pharyngeal/palatal weakness, Vth nerve sensory loss.

**MRI imaging** How big is the syrinx? Any base-of-brain (Chiari) malformation?

**Surgery** Don't wait for gross deterioration to occur. Decompression at the foramen magnum may be tried in Chiari malformations to promote free flow of CSF, and so prevent syrinx dilatation. Surgery may reduce pain and progression.

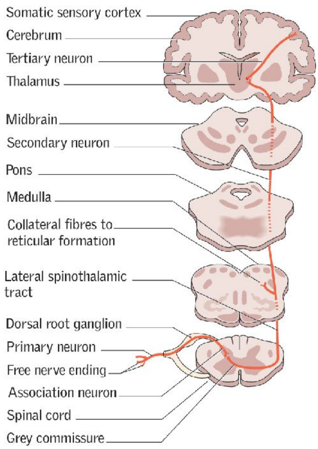



Fig 10.35 The anterolateral system.

## Retroviruses and neurology

**HIV and AIDS** (p398.) Can have multiple neurological manifestations: these conditions are part of the differential diagnosis of meningitis, intracranial mass lesions, dementia, encephalomyelitis, cord problems, and peripheral neuropathies.

*Acute infection:* May be associated with transient aseptic meningoencephalitis (typically self-limiting), myelopathy, and neuropathy.

*Opportunistic infections:* Arise during low CD4 counts, which allow unusual or atypical organisms to infect the nervous system: • *Toxoplasma gondii* (p400) is the main CNS pathogen in AIDS, causing cerebral abscesses which present with focal signs, eg seizures, hemiparesis. CT/MRI shows ring-shaped contrast-enhancing lesions. Treat with pyrimethamine (+folinic acid) + sulfadiazine or clindamycin for 6 months. Continue secondary prophylaxis until CD4 count >200. Pneumocystis prophylaxis also protects against toxoplasmosis. • *Cryptococcus neoformans* (fig 10.36) causes a chronic meningitis with fever and headache (neck stiffness may be absent). Cognition alters slowly, seizures and coma may follow. Treat with amphotericin followed by fluconazole. • *Cytomegalovirus (CMV)* can cause encephalopathy. • *Progressive multifocal leukoencephalopathy (PML)* is caused by the JC virus. There is progressive white matter inflammation. Mortality even with antiretroviral therapy is around 50% at 1yr. • Syphilis and TB may also cause meningitis.

*Tumours:* Affecting the CNS include primary cerebral lymphoma (associated with EBV) and B-cell lymphoma. CSF JC virus PCR is useful in distinguishing PML from lymphoma.

*Neuropathies:* Common in HIV, and may be a result of the disease itself or antiretroviral therapy. Up to 30% of patients have a peripheral neuropathy, which is painful and predominantly sensory. Other clinical pictures include polyradiculopathy, mononeuritis multiplex, and proximal myopathy.

*Chronic HIV-associated neurocognitive disorder (HAND):* While antiretroviral therapy (ART) has decreased the incidence of CNS complications in HIV/AIDS, people are living longer with the disease, and chronic complications such as HIV-associated dementia are increasing. This occurs in 7–15%, late in the disease, and usually when the CD4 count is <200. Progressive behavioural changes are seen along with subcortical features: memory loss, poor attention, and bradykinesia. Various encephalopathies may also contribute to this, eg PML.

**Human T-cell lymphotrophic virus (HTLV-1)** Is another retrovirus with neurological manifestations, though much more rarely than HIV (~0.5%). It causes: *Tropical spastic paraplegia,* a slowly progressing myelopathy, typically affecting the thoracic area. There may be paraesthesiae, sensory loss, and disorders of micturition. *Demyelinating polyneuropathy* and *ataxia* may also occur.

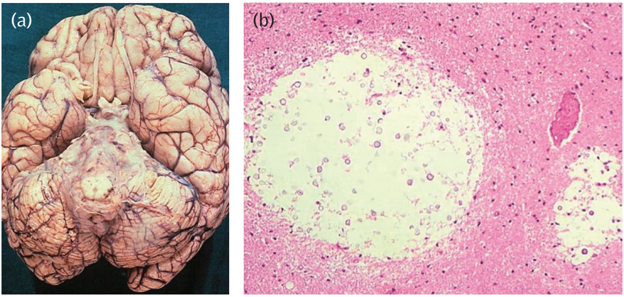



Fig 10.36 Cryptococcosis: (a) Chronic meningitis involving the basal leptomeningitis with multiple small intraparenchymal cysts seen in the cerebral cortex. (b) Under the microscope we see these cysts as dilatation of the perivascular space to form cavities filled with colonies of cryptococci, which appear as round basophilic structures.

Reproduced from Gray *et al.*, *Escourelle and Poirier's Manual of Basic Neuropathology*, 2013, by permission of Oxford University Press, USA.

# 11 Oncology and palliative care

**Contents**



**Fig 11.1** How not to break bad news. The third day of admission brings me some examples of doctor's communication skills being the worst I could possibly imagine under the most painful of circumstances…I'm laid in a hospital bed sobbing and scared, about at the most vulnerable a patient could be…a young gynaecology SHO I have never met enters my room…I can tell he has pulled the short straw… He nervously sits down next to me and out of the blue, after a cursory introduction tells me, 'Your MRI shows evidence of spread'. I am quite astounded at the lack of quality communication given the circumstances.

*The Other Side*, by Kate Granger MBE, FRCP, 1981–2016.

Kate Granger, a medical registrar then consultant geriatrician, was diagnosed with a desmoplastic small round cell tumour at the age of 29. This is a cancer that medical science has no answer to. But Kate had her own answer. She turned her terminal diagnosis on its head and began a dialogue on death and dying, offering her experience as an inspirational lesson in compassion and care. Before you are a patient, before you have cancer, but most importantly, before you are a doctor, you are simply a human being. And if your humanity is lost or forgotten, then you cannot care, even if medical science is able to provide an answer. hellomynameis.org.uk; #hellomynameis

Image and text reproduced courtesy of the family of Dr Kate Granger, MBE.

*We thank Professor Max Watson, our Specialist Reader, for his contribution to this chapter.*

## Looking after people with cancer

Cancer will affect 50% of people born after 1960 and >25% of all deaths in the UK are from cancer.[1] While many may not appreciate the poor prognosis attached to diagnoses such as liver failure or heart failure, 'cancer' has a widespread association with suffering and death. Yet 'cancer' is not a homogenous disease but a group of conditions with prognoses ranging from very good (98% 10yr survival for testicular cancer) to extremely poor (21% 1yr survival for pancreatic cancer).[1]

Communication[2] is the first step on a cancer pathway and underpins whatever that diagnosis may subsequently entail for the individual. A range of overwhelming feelings can surface upon receiving a cancer diagnosis: shock, numbness, denial, panic, anger, resignation ('I knew all along...'). Preconceptions, possibly derived vicariously from friends and family, may be deeply embedded leading to despair or inappropriate optimism. Without an understanding of your patient's starting point, you may fail to be effective in your guidance and support.

### Tips for the discussion of a cancer diagnosis

1. Set the environment up carefully. Choose a quiet place where you will not be disturbed. Make sure family or friends are present according to your patient's wishes. Anticipate likely questions and be sure of your facts.
2. Find out what the patient already knows and believes (often a great deal). 'What are you worried about today?'
3. Give some warning: 'There is some bad news for us to address'.
4. Ascertain how much the person wants to know. 'Are you someone who likes to know all the details about your condition?' Although information is a priority for the majority of cancer patients, this may change with the individual, and the course of the disease. 'Monitors' will seek information, 'blunters' will distract themselves.
5. Share information about diagnosis and treatments. Specifically list supporting people (oncology multidisciplinary team) and institutions (hospices). Break information down into manageable chunks and check understanding for each.
6. Invite questions patients may feel they cannot ask. 'Is there anything else you want me to explain?' Do not hesitate to go over the same ground repeatedly. Allow denial, don't force the pace, give time. Listen to any concerns raised, encourage the airing of feelings. Empathize.
7. Address prognosis. Be honest. Doctors are often too optimistic. Encourage an appropriate level of hope (see BOX 'Spiritual pain', p535), refer to an expert.
8. Make a plan. The desire to be involved in decisions about treatment is variable: your patient's locus of control can be internal (desire control of their own destiny) or external (passive acceptance). Decision-making can be immediate, deferred, panicked, or rationally deliberated. Time may be required to facilitate any style of decision-making: your plan may be simply to come back and talk again.
9. Summarize, and offer availability. Record details of your conversation including the language used.
10. Follow through. ►Leave your patient with the knowledge that you are with them, and that your unwritten contract will not be broken.

No rules guarantee success. Use whatever your patient gives you—closely observe both verbal and non-verbal cues. Getting to know your patient, seeking out the right expert for each stage of treatment, and making an agreed management plan, are all required.

For any situation which involves the communication of bad news, consider **SPIKES**:[3]

- **S**etting up the interview.
- Assess the patient's **P**erception of the situation.
- Obtain an **I**nvitation (asking the patient's permission to explain).
- Give **K**nowledge and information to the patient.
- Address the **E**motional response with **E**mpathy.
- **S**trategy and **S**ummary: aim for consensus with patient and family.

# The genetics of cancer



## How cancers develop

Human life requires cells which are capable of dividing millions of times. These cells need to be able to adapt and change so that different tissues and organs can be formed. They need to command their own blood supply. Without extensive mechanisms to control cell growth and prevent the replication of abnormal cells, these requirements for life become the basis for the development of a cancer. Failure of control mechanisms causes cancer.

Cancer is a genetic disease. Genetic changes occur in pathways associated with cell growth, cell differentiation, and cell death. Mutations can be inherited or acquired. Acquired or somatic errors occur due to age, exposure to carcinogens, and in unchecked rapid cell turnover. Mutations result in:

- 'gain of function' *oncogenes* that have pathological activity in the absence of a relevant signal. For example, ras is a protein involved in signal transduction. It is mutated in ~30% of human cancers. Oncogenes behave in a dominant manner: mutation to one allele results in unchecked activation.
- 'loss of function' *tumour suppressor genes* no longer act as inhibitors of pro-malignant processes. In most cases, mutations to both alleles must occur for a cancer phenotype. This can occur either as two separate somatic events, or in the case of predisposition genes, the first 'hit' is inherited and the second occurs somatically. Tumours therefore occur earlier and more frequently in familial cancers. p53 is a tumour suppressor gene mutated in ~50% of human cancers.

Most cancers arise from multiple mutations. This is perhaps best represented in the stepwise accumulation of mutations in colorectal cancer (fig 11.2). An understanding of the molecular biology of cancer facilitates drug development (fig 11.3).

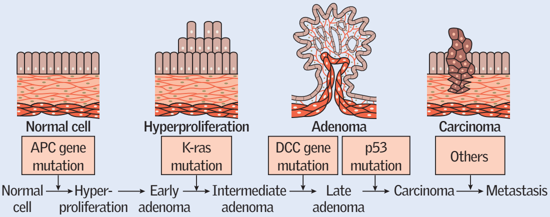



Fig 11.2 Cellular mutations and contributing genes in the development of colorectal cancer.

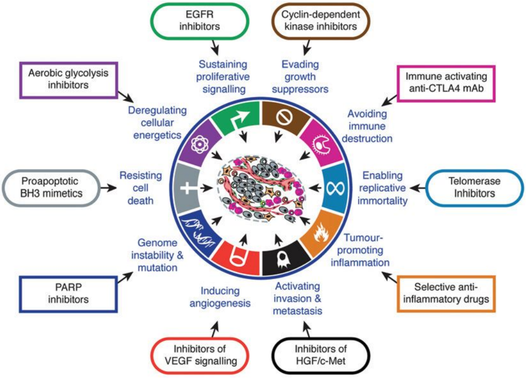



Fig 11.3 Therapeutic targeting in cancer.

## Hereditary cancer syndromes

A hereditary cancer[4] is suggested by:

- unusual early age or presentation (eg male breast cancer)
- multiple primary cancers or bilateral/multifocal cancers
- clustering of cancers in relatives
- cancers in multiple generations
- rare tumours (eg retinoblastoma) or histology (medullary thyroid cancer, p223)
- ethnicity (eg Ashkenazi heritage and breast cancer).

Genetic testing is appropriate if the sensitivity and specificity of the test are good enough, and if the result of the test will impact diagnosis and management (see p27).

## Breast/ovarian cancer

~5–10% of breast cancers are due to mutations in BRCA1 (17q) or BRCA2 (13q).[4,5] Both genes function as tumour suppressors although they are dominant: a cancer phenotype can be seen when one copy of the gene is normal. A BRCA1 mutation confers a 55–65% lifetime risk of breast cancer and a 39% risk of ovarian cancer. For BRCA2, the risk of breast cancer is 45% (6% in affected males), and 11% for ovarian cancer. Mutations are also linked to prostate, peritoneal, and pancreatic cancers. TP53 mutations (somatic >inherited) also confer a risk of breast cancer. Refer if:

- 1st-degree relative with: breast cancer <40yrs, male breast cancer, bilateral breast cancer <50yrs
- 1st- and 2nd-degree relative with breast cancer or ovarian cancer
- three 1st- or 2nd-degree relatives with breast cancer
- risk assessment calculation of >3% risk in 10yrs or lifetime risk ≥17%
- other: Ashkenazi ancestry, sarcoma <45yrs, multiple cancers at a young age.

Genetic counselling and testing for BRCA1, BRCA2, and TP53 mutations is offered if calculated risk of mutation is >10%. If known BRCA1/2 mutation, offer women annual MRI 30–49yrs and annual mammograpy 50–69yrs (MRI 20–69yrs if TP53 mutation). Prophylactic tamoxifen or raloxifene may be appropriate depending on tolerance and VTE/endometrial cancer risk. Surgical management (mastectomy/oophorectomy) should only be via a specialist MDT.

## Colorectal cancer

~25% have a family history. ~5% have identified mutations. Refer to specialist genetic service if: two 1st-degree relatives with colorectal cancer at average age <60yrs, or if criteria for an autosomal dominant colorectal cancer syndrome is met:

- *Lynch syndrome (hereditary non-polyposis colorectal cancer (HNPCC)):* 1–3% of colorectal cancer. Autosomal dominant due to mutations in mismatch repair genes. Lifetime risk of colorectal cancer up to ~80%. Increased risk of other 'Lynch cancers': endometrium, ovary, urinary tract, stomach, small bowel, hepatobiliary tract. Suspect if ≥3 affected relatives (one 1st-degree), from two successive generations, of whom one was affected <50yrs old. Colonoscopic surveillance (at least biennial) from 25–75yrs.
- *Familial adenomatous polyposis:* Due to mutations in the APC tumour suppressor gene (5q) (fig 11.2). <1% of colorectal cancer. Causes multiple colorectal adenomas (>100 in classical disease) which undergo malignant transformation. Gene penetrance approaches 100% by 50yrs. Surveillance sigmoidoscopy from 12yrs, with prophylactic surgery usually <25yrs guided by polyp number, size, and dysplasia.
- *Peutz–Jeghers syndrome:* 1 in 25 000–280 000. Hamartomatous polyps. 10–20% risk of colorectal cancer, ~50–60% risk of GI cancer, ~60% risk of breast cancer. Due to germline mutations in STK11, a tumour suppressor gene (19p14). Surveillance in all (see p708).

## Prostate cancer

5–10% (~50% disease <55yrs) estimated to be due to inherited factors. Genes include BRCA1, BRCA2, mismatch repair, and HOXB13 which interacts with androgen receptor. Age and race contribute. See p530 for screening.

**Other familial cancer syndromes** Von Hippel–Lindau (p320, p712), Carney complex (p223), MEN (p223), neurofibromatosis (p514).

## Cancer diagnosis



A variety of clinical signs and symptoms should alert you to the possible presence of malignancy. The following list is based on clinical features with a 3% positive predictive value for cancer.[6] It is by no means exhaustive and does not negate the value of clinical judgement. Urgent = within 2 weeks.

### Lung

- Admit if: symptomatic superior vena caval obstruction (p528), stridor.
- Urgent referral if: >40yrs with unexplained haemoptysis, CXR suggestive of cancer.
- Urgent CXR if >40yrs and:
  - persistent/recurrent chest infection
  - finger clubbing
  - supraclavicular/cervical lymphadenopathy
  - thrombocytosis
  - two of: cough, fatigue, SOB, chest pain, weight loss, ↓appetite, smoker, asbestos.

### Upper GI

- Urgent endoscopy if: dysphagia, or >55yrs with weight loss and upper abdominal pain/reflux/dyspepsia.
- Urgent referral if: >40yrs plus jaundice, or upper abdominal mass.
- Urgent CT of the pancreas if >60yrs plus weight loss plus any of: diarrhoea, back pain, abdominal pain, nausea, constipation, new-onset diabetes.
- Non-urgent endoscopy if:
  - >55yrs and one of: treatment-resistant dyspepsia, upper abdominal pain plus low Hb, ↑plts, or N&V plus upper GI symptoms/weight loss
  - haematemesis.

### Lower GI

PR examination and FBC in all.

- Urgent referral if: positive faecal occult blood, >40yrs with abdominal pain plus weight loss, >50yrs with unexplained rectal bleeding, >60yrs with iron-deficient anaemia or change in bowel habit.
- Consider urgent referral if: rectal/abdominal mass, anal ulceration, <50yrs with rectal bleeding plus lower GI symptoms or weight loss or iron-deficiency anaemia.
- Faecal occult blood testing if: >50yrs plus abdominal pain or weight loss, <60yrs with change in bowel habit or iron-deficiency anaemia, >60yrs and anaemia.

### Gynaecological

- Urgent referral if: ascites, pelvic mass (fibroid excluded), >55yrs with post-menopausal bleeding.

### Breast

- Urgent referral if: >30yrs with unexplained breast lump, >50yrs with symptoms or change to one nipple.
- Consider urgent referral if: skin changes, >30yrs with axillary lump.

### Urology

- Urgent referral if:
  - irregular prostate on PR, abnormal age-specific PSA (see p530)
  - >40yrs with unexplained visible haematuria, >60yrs with unexplained non-visible haematuria plus dysuria or ↑WCC
  - non-painful enlargement or change in shape/texture of testicle.

### Central nervous system

- Urgent MRI in progressive, sub-acute loss of central neurological function.

►Unexplained weight loss, ↓appetite, and DVT can be non-specific signs of cancer. Assess for any additional risk factors, symptoms, signs, and refer accordingly.

See also *haematology* (p352); *thyroid* (p600); *skin* (p596).

## Cancer and the multidisciplinary team

The care of all patients diagnosed with cancer is formally reviewed by a multidisciplinary team (MDT). The aim of the MDT is to coordinate high-quality diagnosis, treatment, and care. The MDT should make a recommendation on the best initial treatment for cancer. Note: an MDT can only 'recommend'; the decision must be made in consultation with the patient. The MDT is made up of healthcare professionals with expertise in treating and supporting patients with cancer. Members should include, but are not limited to:

- lead clinician and lead nurse specialist
- radiologists (see BOX 'Interventional oncology', p527)
- histopathologists
- expert surgeons, eg upper GI, colorectal, breast, plastics
- oncologists (medical and clinical)
- palliative care physicians
- nominated member to support ongoing clinical trials
- patient representative
- administrative support.

## Cancer staging

Staging systems are used to describe the extent of a cancer. This is vital to determine the most appropriate treatment, to assess prognosis, and to identify relevant clinical trials. A cancer is always referred to by the stage given at diagnosis. The TNM system is most widely used and is based on the extent of tumour (T), spread to lymph nodes (N), and the presence of metastases (M) (table 11.1).

Table 11.1 TNM cancer staging

| | | | |
|---|---|---|---|
| **Tx** | Primary tumour cannot be measured | **Nx** | Nodes cannot be assessed |
| **T0** | Primary tumour cannot be found | **N0** | No node involvement |
| **$T_{is}$** | Carcinoma *in situ* (abnormal cells present) | **N1–3** | Number/location of node metastases |
| **T1–4** | Size and/or extent of primary tumour (1=small tumour /minimal invasion; 4=large tumour/extensive invasion) | **M0** | No distant spread |
| | | **M1** | Distant metastases |

Other prefixes may also be used: c refers to clinical stage; p is the stage after pathological examination; y refers to stage after neoadjuvant therapy; r is used if a tumour is re-staged after a disease-free interval; a indicates stage at autopsy.

The TNM staging may be converted to an overall, less detailed classification of cancer stage: 0–IV. Stage 0 refers to carcinoma *in situ;* Stages I–III describe the size of cancer and/or nearby spread; Stage IV indicates metastatic disease.

Some cancers may have alternative staging systems such as Duke's classification for colorectal cancer (p616). See also lung cancer (p176); breast cancer (p602); oesophageal cancer (p618); bladder cancer (p647).

## Cancer imaging

Imaging is essential in oncology for diagnosis, prognosis, and to inform and guide treatment. As well as plain radiographs, ultrasound scans, CT, and MRI; there is a wealth of more specialist imaging including:

*PET-CT:* PET uses a non-specific radioactive tracer (FDG) which highlights areas of increased metabolism, cell proliferation, or hypoxia. It therefore accumulates in cancer cells >non-cancer cells. PET-CT is a powerful combination of anatomical (CT) and functional (PET) information allowing diagnosis, increased accuracy of staging, and assessment of treatment response.

*Monoclonal antibodies:* Radio-labelled tumour antibodies specific to the tumour under investigation, eg prostate specific membrane antigen, somatostatin (neuroendocrine tumours), oestrogen receptor (breast). They can offer better specificity than standard PET images. (For monoclonal antibodies in treatment see p524.)

*Bone scintigraphy (bone scan):* Detects abnormal metabolic activity in bones including bone metastases.

# Cancer treatment: medicine and surgery



## Chemotherapy

Chemotherapy[7] is the use of any chemical substance to treat disease. In modern-day use, the term refers primarily to the use of cytotoxic drugs in the treatment of cancer. The aim is to deliver enough cytotoxic drug to a cancer-cell target which is expressed differently compared to normal tissue. Cytotoxic drugs are given at intervals (cycles of treatment) to allow recovery of normal tissue. Chemotherapy is the only systemic treatment for cancer (surgery and radiotherapy are local treatments). This is important as most cancers are considered to be systemic either due to metastases, or the potential to metastasize in the future. ►*Chemotherapy should be prescribed and given only under expert guidance by people trained in its use.*

Includes:

- *Single-agent:* Rarely curative as genetically resistant cells are selected out.
- *Combination chemotherapy:* A combination of drugs with different mechanisms of action and different side-effect profiles reduces the likelihood of resistance and toxicity. The drugs used should have:
  - cytotoxic activity for that tumour, preferentially able to induce remission
  - different mechanisms of action, ideally additive or synergistic effects
  - non-overlapping toxicity to maximize benefit of full therapeutic doses
  - different mechanisms of resistance.
- *Adjuvant:* After other initial treatment to reduce the risk of relapse, eg following surgical removal of, eg breast, bowel cancer.
- *Neoadjuvant:* Used to shrink tumours prior to surgical or radiological treatment. May allow later treatment to be more conservative.
- *Palliative:* No curative aim, offers symptom relief, may prolong survival.

## Classes of cytotoxic drugs

- *Alkylating agents:* Anti-proliferative drugs that bind via alkyl groups to DNA leading to apoptotic cell death, eg cyclophosphamide, chlorambucil, busulfan.
- *Angiogenesis inhibitors:* Eg bevacizumab, aflibercept, sunitinib.
- *Antimetabolites:* Interfere with cell metabolism including DNA and protein synthesis, eg methotrexate, 5-fluorouracil.
- *Antioestrogens:* Aromatoase inhibitors (eg letrozole, anastrozole), oestrogen receptor antagonists (eg tamoxifen, raloxifene) used in breast cancer treatment.
- *Antitumour antibiotics:* Interrupt DNA function, eg dactinomycin, doxorubicin, mitomycin, bleomycin.
- *Monoclonal antibodies:* Antibodies to a specific tumour antigen can slow tumour growth by enhancing host immunity, or be conjugated with chemotherapy/radioactive isotopes to allow targeted treatment. Expect more of these in future.
- *Topoisomerase inhibitors:* Interrupt regulation of DNA winding, eg etoposide.
- *Vinca alkaloids and taxanes:* 'Spindle poisons' which target mechanisms of cell division, eg vincristine, vinblastine, docetaxel.

## Side-effects

Due to cytotoxic effects on non-cancer cells. Greatest effect seen on dividing cells, ie gut, hair, bone marrow, gametes (see BOX 'Fertility and cancer', p525).

- *Vomiting:* Prophylaxis given with most cytotoxic regimens (see p251).
- *Alopecia:* May profoundly impact quality of life. Consider 'cold-cap', wig services.
- *Neutropenia:* Most commonly seen 7–14d after chemotherapy. ►►Neutropenic sepsis is life-threatening and needs urgent assessment and empirical treatment (p352).

### Extravasation of chemotherapy

Extravasation[8] = inadvertent infiltration of a drug into subcutaneous/subdermal tissue. *Presentation:* Tingling, burning, pain, redness, swelling, no 'flashback'/resistance from cannula. *Management:* Stop and disconnect infusion. Aspirate any residual drug before cannula removed. Follow local policies (ask for the 'extravasation kit'). Follow any drug-specific recommendations. For DNA-binding drugs (anthracyclines, alkylating agents, antitumour antibiotics), use a dry cold compress to vasoconstrict and ↓ drug spread. For non-DNA-binding drugs (vinca alkaloids, taxanes, platin salts), use a dry warm compress to vasodilate and ↑ drug distribution.

## Surgery

- *Prevention:* Risk-reducing surgery, eg thyroidectomy in MEN (p223), colectomy in FAP (p521).
- *Screening:* Endoscopy, colposcopy.
- *Diagnosis and staging:* Fine needle aspiration, core needle biopsy, vacuum-assisted biopsy, excisional/incisional biopsy, sentinel lymph node biopsy, endoscopy, diagnostic/staging laparoscopy, laparascopic ultrasound.
- *Treatment:* Resection of solid tumour (may be combined with chemo/radiotherapy).
- *Reconstruction:* Eg following treatment for breast, head and neck cancers.
- *Palliation:* Bypass, stoma, stenting, pathological fractures.

## Clinical trials

- *Advantages:* Possibility of more effective treatment than currently available, close monitoring with direct access to a research team, reassurance from increased number of clinical encounters, gain from altruism.
- *Disadvantages:* Possibility of receiving therapy that is no better or worse than standard therapy, unknown toxicity from new agents, time-consuming, anxiety from increased number of clinical encounters.

You may look after patients who are participating in clinical trials. For many of these, you will not be familiar with the trial therapy or even know which therapy the patient is receiving: a new therapy, an old therapy, or placebo. ►Contact the research team to discuss any clinical concerns or change in treatment. Contact details should be recorded in the patient's notes. Look in the notes for, or ask the patient if they have a copy of the 'Participant Information Sheet' which is mandatory for all UK research studies.

Information on relevant trials for your patient is available:

- Cancer Research UK (www.cancerresearchuk.org/about-cancer/find-a-clinical-trial).
- UK Clinical Trials Gateway (www.ukctg.nihr.ac.uk).

## Fertility and cancer

Chemotherapy and radiotherapy may:

- damage spermatogonia causing impaired spermatogenesis or male sterility
- hasten oocyte depletion leading to premature ovarian failure.

If cancer treatment carries a risk of infertility, fertility preservation techniques[9] should be discussed prior to treatment being given.

- *Men:* Semen cryopreservation should be offered before treatment due to the risk of genetic damage in sperm after initiation of chemotherapy. Intracytoplasmic sperm injection means that even a small amount of banked sperm can be used successfully in the future.
- *Women:* Cryopreservation of:
  1. Embryos
  2. Oocytes: if ethical objections to embryo preservation or no partner
  3. Ovarian tissue: no ovarian stimulation required, experimental technique.

  Ovarian transposition (oophoropexy) may be possible prior to pelvic radiotherapy but protection is not guaranteed due to radiation scatter.

## Beau's lines

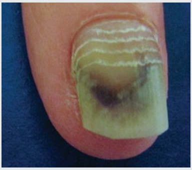


Fig 11.4 Beau's lines.

Beau's lines (fig 11.4) are horizontal depressions in the nail plate that run parallel to the moon-shaped portion of the nail bed. They result from a sudden interruption of nail keratin synthesis and may be due to local infection/trauma, systemic illness, or from medication (p76). Each line in this photo coincided with a round of chemotherapy for breast cancer.

# Cancer treatment: radiotherapy



Radiotherapy[10] is used in >50% of all cancer and forms part of treatment in 40% of those considered cured. It uses ionizing radiation to cause damage to DNA. This prevents cell division and leads to cell death. The aim of radiotherapy treatment is to inactivate cancer cells without causing a severe reaction in normal tissue.

**Radical treatment** Given with curative intent. Total dose ranges from 40–70 gray (Gy) in up to 40 fractions. Some regimens involve several smaller fractions a day with a gap of 6–8h. Combined chemoradiation is used in some sites, eg anus and oesophagus, to increase response rate.

**Palliative radiotherapy** Aims to relieve symptoms, may not impact on survival. Doses are smaller and given in fewer fractions to offer short-term tumour control with minimal side-effects. Palliation is used for brain metastases, spinal cord compression, visceral compression, and bleeding, eg haemoptysis, haematuria. Bone pain from metastases can be reduced or eliminated in 60% of cases.

## Early reactions

Occur ~2 weeks into treatment, peak ~2–4 weeks after treatment.

- *Tiredness:* ~80%. Improves ~4 weeks after treatment completed but chronic in ~30%. Advise patients to stay as active as possible.
- *Skin reactions:* Include erythema, dry desquamation, moist desquamation, and ulceration. Aqueous cream can be used on unbroken areas.
- *Mucositis:* All patients receiving head and neck treatment should have a dental check-up before therapy. Avoid smoking. Antiseptic mouthwashes may help. Aspirin gargle and other soluble analgesics can be tried. Treat oral thrush with fluconazole 50mg/24h PO, nystatin may exacerbate nausea.
- *Nausea and vomiting:* Occur when stomach, liver, or brain treated. Try metoclopramide 10mg/8h PO (dopamine antagonist), domperidone 10mg/8h PO (blocks the central chemoreceptor trigger zone), or ondansetron 4–8mg/8h PO/IV (serotonin $5HT_3$ antagonist) (see p251).
- *Diarrhoea:* Usually after abdominal or pelvic treatments. Maintain good hydration. Avoid high-fibre agents. Try loperamide 2mg PO after loose stools (max 16mg/24h).
- *Dysphagia:* Following thoracic treatments. Speech and language input, nutrition.
- *Cystitis:* After pelvic treatments. Drink plenty of fluid.

## Late reactions

Months-years after treatment.

- *CNS/PNS: Somnolence:* 4–6wks after brain radiotherapy. Consider ↑steroid dose. *Spinal cord myelopathy:* progressive weakness. MRI to exclude cord compression. *Brachial plexopathy:* numb, weak, or painful arm after axillary radiotherapy.
- *Lung: Pneumonitis* can occur 6–12wks after thoracic treatment causing dry cough ± dyspnoea. Bronchodilators and tapered steroids may help.
- *GI: Xerostomia* = reduced saliva. Dental care and nutrition important. Treat with water, saliva substitutes, salivary stimulants. *Benign strictures* of oesophagus or bowel. Treat with dilatation. Seek a specialist surgical opinion regarding *fistulae*. *Radiation proctitis* may be a problem after prostate irradiation.
- *GU: Urinary frequency:* small fibrosed bladder after pelvic treatment. *Vaginal stenosis*, *dyspareunia*, *erectile dysfunction* can occur after pelvic radiotherapy. ↓*fertility:* due to pelvic radiotherapy (see p525).
- *Endocrine: Panhypopituitarism* following radical treatment involving pituitary fossa. Check hormone profile in children: growth hormone replacement may be required. *Hypothyroidism* in ~50% after neck treatment: check TFTs annually.
- *Secondary cancers:* Risk (2–4 per 10 000 person-years) is usually insignificant compared to recurrence/death from primary lesion. More important for younger patients after curative treatment. Women <35yrs receiving radiotherapy for Hodgkin's lymphoma should be offered breast screening from 8yrs after treatment.

►↑ Cancer survival means ↑ numbers living with poor health or disability after treatment (~625 000 in UK). Remember the emotional and physical impact of cancer extends beyond the prescribed course of radiotherapy/chemotherapy.

### Methods of delivering radiotherapy

*Conventional external beam radiotherapy (EBRT):* Is the most common form of treatment. Delivers beams of ionizing radiation to the patient from an external linear accelerator.

*Stereotactic radiotherapy* is a highly accurate form of EBRT used to target small lesions with great precision—most frequently in treating intracranial tumours. It is often referred to by the manufacturer's name, eg *Gamma Knife®*, *Truebeam®*.

*Brachytherapy* involves a radiation source being placed within or close to a tumour, allowing a high local radiation dose. Implants may be placed within a cavity (eg uterus, post-surgical space) or within tissue (eg prostate, breast).

*Radioisotope therapy* uses tumour-seeking radionuclides to target specific tissues. For example, $^{131}$I (radioiodine) to ablate remaining thyroid tissue after thyroidectomy for thyroid cancer.

## Interventional oncology

Interventional oncology (IO) refers to interventional radiology procedures used in the treatment or palliation of patients with cancer. IO can be divided into disease-modifying and symptomatic procedures.

*Disease-modifying IO:*
Intended to modify cancer progression and/or to improve prognosis. Includes:
- Image-guided ablation, eg radiofrequency ablation, cryoablation, irreversible electroporation.
- Embolization, eg transarterial embolization, chemoembolization, selective internal radiation therapy.
- Image-guided brachytherapy.
- Isolated perfusion chemotherapy: uses occlusion techniques to protect normal tissue from high doses of chemotherapy.

*Symptomatic IO:*
Provides relief from cancer-related symptoms, but does not modify the underlying disease process. The techniques (table 11.2) can offer significantly improved quality of life, reduce admissions, and increase time spent outside of hospital.

Table 11.2 Interventional techniques available for cancer symptom control

| Clinical problem | Interventional treatment option |
|---|---|
| **Ascites** | Temporary/permanent image-guided ascitic drain |
| **Pleural effusion** | Temporary/permanent image-guided pleural drain |
| **Superior vena cava obstruction (p528)** | Superior vena cava stenting |
| **Oesophageal obstruction** | Oesophageal stenting |
| **Large bowel obstruction** | Colonic stenting |
| **Tumour-related haemorrhage** | Transarterial embolization |
| **Jaundice** | Biliary drainage and stenting |
| **Renal tract obstruction** | Nephrostomy, ureteric stenting |
| **Bone metastases** | Image-guided ablation |

►Talk to your interventional radiologist.

Emergencies[11,12] in oncology include:

### Neutropenic sepsis
Temperature >38°C and neutrophil count <$0.5\times10^9$/L. Suspect in all patients who are unwell and within 6wks of receiving chemotherapy. Localizing signs may be absent. Examine indwelling catheter sites. ►►Immediate treatment saves lives. Use local guidelines or treat empirically with piperacillin/tazobactam (see p352).

### Spinal cord compression
3–5% of cancer patients have spinal metastases. ~15% of those with advanced cancers develop metastatic spinal cord compression. Most commonly associated with lung, prostate, breast, myeloma, melanoma. ►►Urgent treatment is required to preserve neurological function and relieve pain.

*Causes:* Collapse or compression of a vertebral body due to metastases (common), direct extension of a tumour into vertebral column (rare).

*Signs and symptoms:* Back pain in ~95%. Ask about nocturnal pain and pain with straining. Worry if there is cervical/thoracic pain. Also limb weakness, difficulty walking, sensory loss, bowel/bladder dysfunction. Maintain a high index of suspicion.

*Management:* Admit for bed rest and arrange urgent (within 24h) MRI of the whole spine. Give dexamethasone 16mg/24h PO with prophylatic gastroprotection, eg PPI, and blood glucose monitoring. If reduced mobility consider thromboprophylaxis (compression stockings, LMWH). Refer urgently to clinical oncology/cancer MDT. Radiotherapy is the commonest treatment and should be given within 24 hours of MRI diagnosis. Decompressive surgery ± radiotherapy may be appropriate depending on prognosis. Patients with loss of motor function after >48h are unlikely to recover function. (See also p466.)

### Superior vena cava (SVC) syndrome
Reduced venous return from head, neck, and upper limbs. Due to extrinsic compression (most common), or venous thrombosis (consider if current or past central venous access). ►►SVC syndrome with airway compromise requires urgent treatment.

*Causes:* >90% of SVC syndrome results from malignancy. Most common cancers: lung (~75%), lymphoma, metastatic (eg breast), thymoma, germ cell.

*Signs and symptoms:* Diagnosis is made clinically. SOB, orthopnoea, stridor, plethora/cyanosis, oedema of face and arm, cough, headache, engorged neck veins (non-pulsatile ↑JVP), engorged chest wall veins. *Pemberton's test:*[1] elevation of the arms to the side of the head causes facial plethora/cyanosis.

*Management:* Prop up. Assess for hypoxia (pulse oximetry, blood gas) and give oxygen if needed. Dexamethasone 16mg/24h. CT is used to define the anatomy of the obstruction. Balloon venoplasty and SVC stenting provide the most rapid relief of symptoms (see BOX 'Interventional oncology', p527). Treat with radiotherapy or chemotherapy depending on the sensitivity of the underlying cancer.

### Malignancy-associated hypercalcaemia
Most common metabolic abnormality in cancer patients: ~10–20% of patients with cancer, ~40% of myeloma. It is a poor prognostic sign: 75% mortality within 3 months. Calcium is highly protein-bound and needs correcting to the serum albumin concentration. PTH levels should be suppressed (see pp676–7).

*Causes:* PTH-related protein produced by the tumour (see p529), local osteolysis, eg myeloma, tumour production of calcitriol.

*Signs and symptoms:* Weight loss, anorexia, nausea, polydipsia, polyuria, constipation, abdominal pain, dehydration, weakness, confusion, seizure, coma.

*Management:* Aggressive rehydration. Bisphosphonates (if eGFR ≥30), eg zoledronic acid IV, usually normalize calcium within 3 days and can be given as a repeated infusion. Calcitonin produces a more rapid (2h) but short-term effect and tolerance can develop. Long-term treatment is by control of the underlying malignancy.

1 Pemberton described this 'useful' sign of venous obstruction due to a goitre in 1946.

### Brain metastases

Affect up to ~40% of patients with cancer. Most commonly: lung, breast, colorectal, melanoma. Poor prognosis: median survival 1–2 months; better prognosis with single lesion, breast cancer (see also p830).

*Signs and symptoms:* Headache (~50%, often worse in the morning, when coughing or bending), focal neurological signs (~30%), ataxia (~21%), fits (~18%), nausea, vomiting, papilloedema.

*Management:* Urgent CT/MRI depending on underlying diagnosis, disease stage, and performance status. Dexamethasone 16mg/24h to reduce cerebral oedema. Stereotactic radiotherapy (see p527). Discuss with neurosurgery, especially if large lesion or associated hydrocephalus.

### Tumour lysis syndrome

Chemotherapy for rapidly proliferating tumours (leukaemia, lymphoma, myeloma) leads to cell death and ↑urate, ↑$K^+$, ↑phosphate, ↓calcium. Risk of arrhythmia and renal failure (see p314).

*Management:* Prevent with hydration and uricolytics, eg rasburicase, allopurinol.

### Paraneoplastic syndromes

Paraneoplastic syndromes[13] (table 11.3) consist of symptoms attributable to a malignancy mediated by hormones, cytokines, or the cross-reaction of tumour antibodies. They do not correlate with stage/prognosis and may pre-date other cancer symptoms.

Table 11.3 Examples of paraneoplastic syndromes

| Paraneoplastic syndrome | Comment | Malignancies | See |
|---|---|---|---|
| **Hypercalcaemia** | Parathyroid hormone-related protein secreted by tumour | Lung, oesophagus, skin, cervix, breast, kidney | p528 |
| **SIADH** | Excessive antidiuretic hormone (ADH) secretion causing ↓$Na^+$ | Lung, pancreas, lymphomas, prostate | p673 |
| **Cushing's syndrome** | Tumour secretes ACTH or CRF, causing adrenal to produce high levels of corticosteroid | Lung, pancreas, thymus, carcinoid | p224 |
| **Neuropathy** | Antibody-mediated neuronal degeneration: peripheral, autonomic, cerebellar | Lung, breast, myeloma, Hodgkin's, GI | p504 |
| **Lambert-Eaton myasthenic syndrome** | Antibody to voltage-gated ion channel on pre-synaptic membrane causes weakness (proximal leg most common) | Mostly lung. Also GI, breast, thymus | p512 |
| **Dermatomyositis & polymyositis** | Inflammation of the muscles +/- heliotrope rash | Lung, breast, ovary, GI | p552 |
| **Acanthosis nigricans** | Velvety, hyperpigmented skin (usually flexural) | GI | p562 |
| **Pemphigus** | Blisters to skin/mucous membranes | Lymphoma, thymus, Kaposi's sarcoma | |
| **Hypertrophic osteoarthropathy** | Periosteal bone formation, arthritis, and finger clubbing | Lung | |

### Trousseau's sign

Trousseau (fig 11.5) was probably the first to discover a paraneoplastic syndrome. He noticed that many patients with migratory thrombophlebitis ('Trousseau's sign') developed gastric cancer. Unfortunately, he developed migratory thrombophlebitis himself and correctly predicted his own death from GI malignancy.

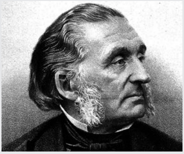


Fig 11.5 Armand Trousseau 1801–1867.
Wellcome Library, London. Armand Trousseau. Lithograph by JBA Lafosse, 1866, after P Petit.

# Tumour markers

Tumour markers[14] are specific molecules (usually glycoproteins) that may be found in higher concentrations in the serum, tissue, or urine in patients with certain cancers.

## Tumour markers in diagnosis

►*Tumour markers are insufficiently sensitive or specific to be diagnostic in isolation.*

- Many tumour markers are ↑ in several cancers and benign conditions (table 11.4).
- Measuring ≥1 tumour marker is unlikely to aid diagnosis unless suspecting a germ cell tumour.
- Do not make opportunistic requests for panels of tumour markers in patients with non-specific symptoms: they are not helpful and lead to potentially unnecessary investigation. This includes testing PSA in women and CA 125 in men.
- In carefully selected patients, in whom cancer is suspected, highly raised levels of a tumour marker may be helpful:
  - α-fetoprotein (αFP) and human chorionic gonadotrophin (hCG) in testicular/germ cell tumours.
  - CA 125 in combination with USS and menopausal status.
  - αFPin those at high risk of hepatocellular carcinoma.
  - PSA >100ng/mL usually indicates metastatic prostate cancer.

## Tumour markers in monitoring

The main value of tumour markers is in monitoring patients known to have cancer. This includes the course of the disease, the effectiveness of treatment, and the detection of cancer recurrence. The following markers may be useful:

- αFP and hCG in testicular/germ cell tumours.
- CEA in colorectal cancer.
- CA 125 in ovarian cancer.
- A cautious interpretation of PSA within the limits of its specificity and sensitivity.

## Screening for cancer

The UK has several well-established cancer screening programmes. Women are invited for mammography every 3yrs (50–70yrs) and offered cervical smear tests every 3–5yrs (25–64yrs). Men and women aged 60–74yrs are offered faecal occult blood testing every 2yrs.

Screening tests aim to pick out those who need further investigation to rule out or diagnose a cancer, in the hope that earlier diagnosis and treatment result in better outcomes. All screening tests come with risk: anxiety, harm/discomfort from the test, cost, false positives resulting in further invasive tests, false negatives conferring inappropriate reassurance when symptoms arise. When considering screening an asymptomatic population the potential risks and benefits need to be weighed carefully and the Wilson criteria (see p23) should be satisfied.

### Should PSA be used to screen for prostate cancer?

Most men with prostate cancer will have a high prostate-specific antigen (PSA). The higher the PSA, the more likely cancer is. However, PSA is non-specific and also raised in benign prostatic disease, BMI <25, recent ejaculation, recent rectal examination, prostatitis, and UTI. 76% of patients with a raised PSA do not have cancer. Following screening tests (see PROMIS study, 2017, for use of multi-parametric MRI), prostate biopsy is required for diagnosis. This has an inherent risk of complications including bleeding, infection, and urinary retention. ~1% will require hospital admission.

The risks of PSA testing and subsequent biopsy need to be counterbalanced by benefits from screening.[15] ~1 in 800 men avoid death from prostate cancer as a result of PSA screening. But screening picks up many cancers that will never become fatal. This 'overdiagnosis' is thought to occur in ~40% of positive screens with significant risks from treatment including urinary incontinence, erectile dysfunction, and IHD. This balance of risk versus benefit means that population screening for prostate cancer using PSA is not recommended. Despite this, any patient >50yrs (or >45yrs if high risk) can request PSA testing in primary care. ►Interpret any PSA result in conjunction with digital rectal examination and other risk factors.

**Table 11.4** Summary of tumour markers

| Tumour marker | Relevant cancer | Use | Other associated cancers | Associated benign conditions |
|---|---|---|---|---|
| Alpha-fetoprotein (αFP) | Germ cell/testicular<br>Hepatocellular | Diagnosis, monitoring treatment, detecting recurrence | Colorectal; gastric; hepatobiliary; lung | Cirrhosis; pregnancy; neural tube defects |
| Calcitonin | Medullary thyroid | Diagnosis, monitoring treatment, detecting recurrence | None known | C-cell hyperplasia |
| Cancer antigen (CA)125 | Ovarian | Monitoring ovarian cancer. Prognosis after chemotherapy | Breast; cervical; endometrial; hepatocellular; lung; non-Hodgkin's lymphoma; pancreatic; medullary thyroid carcinoma; peritoneal; uterine | Liver disease; cystic fibrosis; pancreatitis; urinary retention; diabetes; heart failure; pregnancy; SLE; sarcoid; RA; diverticulitis; IBS; endometriosis; fibroids |
| CA19–9 | Pancreatic | Monitoring pancreatic cancer | Colorectal; gastric; hepatocellular; oesophageal; ovarian | Acute cholangitis; cholestasis; pancreatitis; diabetes; IBS; jaundice |
| CA15–3 | Breast | Monitoring breast cancer | Hepatocellular; pancreatic | Cirrhosis; benign breast disease; in normal health |
| Carcinoembryonic antigen (CEA) | Colorectal | Monitoring adenocarcinomas | Breast; gastric; lung; mesothelioma; oesophageal; pancreatic | Smoking; chronic liver disease; chronic kidney disease; diverticulitis; jaundice |
| Human chorionic gonadotrophin (hCG) | Germ cell/testicular, gestational trophoblastic | Diagnosis, prognosis, monitoring of germ cell tumours | Lung | Pregnancy |
| Paraproteins | Myeloma | Diagnosis, monitoring treatment, detecting recurrence | None known | None known |
| Thyroglobulin | Thyroid (follicular/papillary) | Monitoring treatment, detecting recurrence | None known | None known |

Source data from 'Serum tumour markers: how to order and interpret them', Sturgeon C M, Lai L C, Duffy M J, 2012, BMJ Publishing Ltd.

# Palliative care: principles and pain

*You matter because you are you and you matter to the last moment of your life. We will do all we can to help you, not only to die peacefully, but to live until you die.*

Dame Cicely Saunders (1918–2005), founder of the modern hospice.

Palliative care is the active, holistic care of patients with advanced progressive illness. It combines management of pain and other symptoms, with the provision of psychological, social, and spiritual support.

►*Palliative care is not just for the end of life and it is not just for patients with cancer.*

Palliative care should run in parallel with other medical treatments. Good symptom control is important in any disease for improving quality of life and may even prolong survival.[16] Take time to find out exactly what is troubling your patient using a problem-based approach. Consider:

- physical
- psychological
- spiritual
- social.

Remember, each person comes with a set of emotions, preconceptions, and a family already attached. ►Most hospitals now have a dedicated palliative care team for help and advice (including out of hours). Use their expertise.

## Assessment of pain

Pain is one of the most feared sequelae of a terminal diagnosis and yet it is not inevitable. However, pain is a complex phenomenon. While the aim of management is for the patient to be pain free, this may not be achievable in all cases so do not promise this.

Do not assume a cause: detailed history and examination are needed to understand aetiology, which will guide subsequent treatment, eg pain from nerve infiltration or local pressure may respond better to agents other than opioids. History and examination are essential for all patients, including those at the end of life. Evaluate severity, nature, functional deficit, and psychological state as all of these contribute to the symptom burden.

## Management of pain

Aim to modify the underlying pathology where possible, eg radiotherapy, chemotherapy, surgery. Use analgesia to relieve background pain and provide additional PRN doses for 'breakthrough' pain. Effective analgesia is possible in the majority of patients by combining five principles:

1. *By the mouth*—give orally whenever possible.
2. *By the clock*—give at fixed intervals to offer continuous relief.
3. *By the ladder*—following the WHO stepwise approach (see fig 13.5, p575).
4. *For the individual*—there are no standard doses for opioids, needs vary.
5. *Attention to detail*—communicate, set times carefully, warn of side-effects.

### The WHO analgesic ladder

Increase and decrease the analgesia required according to the 'steps' on the ladder[17] (fig 13.5, p575):

1. Non-opioid, eg paracetamol.
2. Opioid for mild to moderate pain, eg codeine.
3. Opioid for moderate to severe pain, eg morphine, diamorphine, oxycodone.

- Persisting/increasing pain and side-effects inform the decision to step up and step down. Take one step at a time to achieve pain relief without toxicity (except in new, severe pain when step 2 may be omitted).
- Paracetamol (PO/PR/IV) at step 1 may have an opiate-sparing effect, and should be continued at steps 2 and 3. Stop step 2 opioids if moving to step 3.
- Use laxatives and anti-emetics with strong opioids.
- Adjuvants which can be added at all steps include: NSAIDs, amitriptyline, pregabalin, corticosteroids, nerve block, transcutaneous electrical nerve stimulation (TENS), radiotherapy.

## Opioids

The amount of opioid required to relieve pain varies and should be titrated on an individual basis. Oral morphine is 1st-line. If the oral route is unavailable, use morphine or diamorphine SC (see table 11.5, and tables 11.6, 11.7, p536). *Explanation and regular review are important.* Prescribe anti-emetics and laxatives for all patients.

*Start low, go slow:* For an opioid-naïve patient with moderate to severe pain, consider oral morphine 5mg every 4 hours plus 5mg PRN (maximum hourly). Consider a lower starting dose if elderly, ↓BMI, or renal impairment. If pain is not controlled, ↑ dose by 30–50% every 24h.

*Convert to modified release:* When pain is controlled, calculate the total daily dose *including* PRN and divide into two 12h doses of a modified-release preparation (eg MST Continus® 12h). Transdermal preparations are available: seek expert help for dose, check adhesion, and rotate site.

*Use a PRN dose for breakthrough pain:* 1/10th–1/6th of the total daily dose as an immediate-release preparation, eg Oramorph® or Sevredol®.

*Side-effects:* Drowsiness, nausea/vomiting (usually ↓ after 5 days), constipation, dry mouth. If difficulty tolerating morphine, or pain plus toxicity, consider an opioid switch (eg oxycodone) and ↓ dose by 25–30%.

*Toxicity:* Sedation, respiratory depression, visual hallucinations, myoclonic jerks, delirium. Be alert: recognizing toxicity early usually means naloxone is avoided. Monitor pulse oximetry, give oxygen if required. Consider ΔΔ: intracranial bleed, renal failure. ↓Opioids and sedating drugs. Consider hydration. Seek expert help if remains opioid-toxic or in pain. ►Naloxone is only indicated for life-threatening respiratory depression (see p842). In patients on regular opiates it can precipitate a pain crisis and potentially fatal acute withdrawal.[18]

*Renal failure:* Patients with renal impairment (eGFR <30) are at risk of toxicity due to accumulation of renally excreted opioids and metabolites. Monitor closely. Fentanyl, alfentanil, and buprenorphine have predominantly hepatic metabolism —seek expert advice.

*Concerns:* Patients may shrink from using opioids. Misconceptions are common: they are addictive, for the dying, if they use morphine now it will not work when they really need it. Respiratory depression is very rare when opioids are correctly titrated but opioids often get blamed when a patient deteriorates. *Reassure patients that opioids are effective and safe when used appropriately.*

*Morphine-resistant pain:* Seek expert help. Consider methadone, ketamine, and adjuvants such as NSAIDs, steroids, muscle relaxants, anxiolytics, nerve blocks. If neuropathic pain is suspected, try amitriptyline, pregabalin, or topical lidocaine. Consider the effect of psychological and spiritual well-being on pain (see p535).

*Rapid analgesia:* Most PRN medication takes time to have an effect. If this is a problem, seek expert help regarding rapid-release preparations (eg sublingual, intranasal, or buccal fentanyl). Try to pre-empt times of high pain (eg dressing changes) and give analgesia in advance.

Table 11.5 Opioid dose equivalents: conversions are not exact, potency can vary. If in doubt, use a dose below your estimate. Practice is variable: always defer to local guidelines first.

| | Relative potency | 4h dose (mg) | 24h dose (mg) |
|---|---|---|---|
| **Morphine PO** | 1 | 5 | 30 |
| **Morphine SC** | 2 | 2.5 | 15 |
| **Diamorphine SC** | 3 | 1.5–2 | 10 |
| **Oxycodone PO** | 2 | 2.5 | 15 |
| **Oxycodone SC** | 4 | 1.25 | 7.5 |
| **Alfentanil SC** | 30 | Too short-acting | 1 |
| **Codeine PO** | 0.1 | 60 (6h dose) | 240 |
| **Tramadol PO** | 0.1 | 100 (6h dose) | 400 |
| **Fentanyl patch** | 25mcg/h approximates to 60mg/24h oral morphine | | |

# Palliative care: other symptoms



Non-pain symptoms[19] include:

## Nausea and vomiting

*Causes:* Chemotherapy, constipation, hypercalcaemia, oral candidiasis, GI obstruction, drugs, severe pain, infection, renal failure.

*Management:* Treat reversible causes, eg laxatives for constipation, analgesia for pain, hypercalcaemia (see p528), fluconazole for oral candidiasis. Anti-emetic choice should be based on the likely mechanism of nausea. Consider the site of anti-emetic action, especially when using a combination of drugs. Oral absorption may be poor so consider alternative routes (SC/IV/PR). Options include the following:

- Cyclizine 50mg/8h: antihistamine, anticholinergic, central action so good for intracranial disorders.
- Metoclopramide 10–20mg/8h: blocks central chemoreceptor trigger zone, peripheral prokinetic effects so good in gastroparesis, monitor for extra-pyramidal side-effects.
- Domperidone 10–20mg/8h PO: peripheral antidopaminergic so no dystonic effects.
- Haloperidol 1.5mg PO initially 1–2 times daily: dopamine antagonist, effective in drug- or metabolically induced nausea, use lower doses IV/SC as twice as potent.
- Ondansetron 4–8mg/8h: serotonin antagonist, good for chemo/radiotherapy-related nausea, may cause constipation.
- Levomepromazine 6.25mg, initially 1–2 times daily: broad spectrum, but can sedate, may be very effective if fear/anxiety are contributing to symptoms.

Antisecretory drugs, such as hyoscine butylbromide or octreotide may be required for patients with vomiting and bowel obstruction: seek expert advice.

## Constipation

*Causes:* Very common side-effect of opioids. Better to prevent than treat so prescribe laxatives for all patients starting opioids. Also hypercalcaemia (see p528), dehydration, drugs, or intra-abdominal disease.

*Treatment:* Treat reversible causes. Good fluid intake. Ensure privacy and access to toilet. Medication options include the following:

- Stimulant (eg senna 2–4 tablets or bisacodyl 5–10mg) at night ± a softener (eg sodium docusate 100mg BD).
- Osmotic laxative (eg macrogol).
- Rectal treatments: bisacodyl/glycerol suppositories, phosphate enema.

## Breathlessness

*Causes:* Look for reversible causes including infection, effusion, anaemia, arrhythmia, thromboembolism. If stridor or signs of superior vena cava syndrome, treat urgently (see p528).

*Treatment:* Treat reversible causes as appropriate. Consider thoracocentesis ± pleurodesis for a pleural effusion. Recurrent pleural effusions may warrant a radiologically placed permanent drain (see p527). If the patient remains distressed, consider a trial of low-dose opioids. These reduce respiratory drive and the sensation of breathlessness. If opioid-naive, start with 2.5mg of an immediate-release morphine every 4h. If already taking an opioid, use the appropriate breakthrough dose (see p533). Benzodiazepines may help if associated anxiety, eg lorazepam 500mcg SL every 4–6h.

## Oral problems

*Causes:* Poor oral hygiene, radiation, drugs (anticholinergics, chemotherapy, diuretics), infection (candidiasis, herpes simplex).

*Treatment:* Oral candidiasis: topical miconazole, oral fluconazole 50mg OD but check for interactions (eg warfarin). Nystatin is often ineffective and may exacerbate nausea. Herpes simplex: oral gan/aciclovir. Good mouth care maintains comfort and the ability to communicate. Maintain fluid intake with frequent, small drinks. Simple measures are often ↑ effective: sugar-free chewing gum, normal saline mouthwashes, soft toothbrush. Products containing alcohol may sting. Salivary stimulants (rather than substitutes) can be helpful for dry mouth, eg pilocarpine eye drops 4%, 3 drops in the floor of mouth QDS. Severe mucositis may need admission and systemic opioids.

### Insomnia
*Causes:* Terminally ill patients may experience physical and emotional exhaustion. Often multifactorial. Poor sleep can increase symptom burden.
*Treatment:* Simple steps may make a big difference: appropriate room temperature, darkness, and quiet during the night (request a side room for in-patients). Give prescribed glucocorticoids in the morning. Avoid waking patients for late medications and routine observations. Discuss and address psychosocial issues. In some cases zopiclone or benzodiazepines may be used to help patients rest and re-establish normal sleep–wake cycles (may exacerbate delirium).

### Pruritus
*Causes:* Systemic disease (renal failure, hepatitis, polycythaemia), cancer-related (cholestasis, lymphoma, leukaemia, hepatoma, myeloma, paraneoplastic), primary-skin disease, drug reaction (opioids, SSRI, chemotherapy).
*Treatment:* Underlying causes where possible: cholestasis (biliary stenting, colestyramine, sertraline, rifampicin), opioid-induced (antihistamine, opioid switch), paraneoplastic (paroxetine). Topical emollients regularly and as a soap substitute. Avoid topical antihistamines due to risk of contact dermatitis.

### Venepuncture
Repeated venepuncture with the risk of painful extravasation and phlebitis may be avoided by use of a central catheter (eg Hickman® tunnelled line or PICC). Problems: infection, blockage (flush with 0.9% saline or dilute heparin every week), axillary thrombosis, and line slippage.

**Agitation** See 'Care in the last days of life', p536.

**Respiratory tract secretions** See 'Care in the last days of life', p536.

### Spiritual pain
*'The spiritual aspects of an illness concerns the human experiences of sickness (or 'dis-ease') and the search for meaning within it.'*

Peter W Speck

Spirituality is a means of experiencing life. It relates to the way in which people understand and live their lives. It is comprised of elements including meaning, purpose, and something greater than 'self'. It is distinct from faith, which is a religious experience, that may or may not be part of spirituality. Spiritual pain[20,21] or suffering is common when people are facing death. It can include feelings of hopelessness, guilt, isolation, meaninglessness, and confusion. Consider:
- the past: painful memories, guilt
- the present: isolation, anger
- the future: fear, hopelessness.

Reminiscence helps address the past, provides context, and offers recognition of the patient as an individual. Anger should be acknowledged. Fear of the imagined future may not change, but is potentially reduced through discussion. The nature of hope may need to be modified. If hope for a cure is inappropriate, it should not be the main or only hope. Realistic hopes include discharge from hospital, seeing family members happy, being remembered. Making a will, handing over responsibilities, and dealing with unfinished business facilitate control and may allow a sense of completion.

►Remember the whole person: history, coping mechanisms, state of well-being. Elements such as these will alter how disease affects the patient and how the patient responds to disease.

►Companionship is essential in spiritual support. At times a doctor needs to modify their role to simply accompany the dying patient. This is manageable within established professional boundaries and therapeutic. If you cannot do this, find someone who can: palliative care teams, Macmillan nurses, and chaplains (a listening ear for patients of all faiths and none) are all valuable resources.

►Spiritual pain is exacerbated by physical symptoms. These must be addressed if spiritual support is to be effective.

# Palliative care: care in the last days of life



Once it is recognized that a patient is entering the final days of their illness (see 'Diagnosing dying', p12), the focus of care should be the relief of distressing symptoms.[22]

►An individualized care plan should be made and discussed with your patient, their family, and relevant medical staff.

Continue to treat reversible problems as appropriate (eg urinary retention). Stop observations and blood tests (unless you are going to act on them). Rationalize medications but keep any that provide ongoing symptom benefit.

*Prescribe as required subcutaneous end of life drugs.* Prescribe PRN SC medications before they are needed, in anticipation of symptoms (see table 11.6).

*Start a syringe driver* when symptom control drugs are needed regularly (see table 11.7). Practice is variable, some drugs may be used outside of licensed indications. Always defer to local guidelines first. If pain relief is insufficient, review regular dose and recalculate the PRN requirement (1/10th–1/6th of 24h dose).

## Anticipatory end of life medication

Table 11.6 Typical anticipatory medications.

| Indication | Drug | Subcutaneous dose |
|---|---|---|
| Pain | Morphine | 2.5mg SC or 5mg PO (maximum every 1h)<br>If established on opioids use 1/10th–1/6th of daily dose (see p533 and table 11.5) |
| Agitation + N&V | Haloperidol | 2.5mg SC (maximum every 1h) |
| Agitation + anxiety | Midazolam | 2.5mg SC (maximum every 1h) |
| N&V | Levomepromazine | 6.25mg SC TDS |
| Troublesome respiratory secretions | Glycopyrronium | 200mcg SC every 4–8h |

## Syringe drivers

Syringe drivers (table 11.7) allow a continuous SC infusion of drugs, avoiding repeated cannulation and injection when the oral route is no longer feasible. Some medications should not be put in the same syringe—check interactions. Take into account regular doses when calculating requirements. If in doubt, seek expert help. Do not forget anticipatory prescribing in addition (table 11.6).

Table 11.7 Symptom control medication by SC infusion. Practice is variable, defer to local guidelines first.

| Indication | Drug | Subcutaneous dose |
|---|---|---|
| Pain | Morphine | If opioid naïve: 10–15mg/24h<br>If on opioids calculate daily opioid dose (consider reducing by 25–30%) then convert (table 11.5) to SC morphine over 24h |
| Anxiety, agitation, delirium | Midazolam | 5–20mg/24h |
| | Levomepromazine | 25–75mg/24h |
| | Haloperidol | 2–10mg/24h |
| N&V | Cyclizine | 150mg/24h |
| | Haloperidol | 1–3mg/24h |
| | Levomepromazine | 6.25–12.5mg/24h (sedation at higher doses) |
| Respiratory secretions | Hyoscine butylbromide | 60–120mg/24h. Also used for bowel colic |
| | Glycopyrronium | 600–1200mcg/24h |
| Seizures | Seizure prophylaxis: midazolam 20–30mg/24h (may sedate). Dexamethasone, midazolam and levetiracetam can be given by SC infusion. | |

*Manage agitation.* Look for reversible causes (pain, dehydration, urinary retention). Use an antipsychotic agent (eg haloperidol) to manage agitated delirium (see tables 11.6, 11.7). Try a benzodiazepine such as midazolam if there is a large element of anxiety. *Opioids should not be used to sedate a dying patient.* Seek early advice from palliative care if agitation is escalating or a significant problem.

*Manage excessive secretions.* Noise is generated by turbulent air flow and pooling of saliva in the hypopharynx. This may be more distressing for relatives and staff than the patient. There is little evidence that pharmacological agents are beneficial, though they are commonly used. Repositioning and intermittent suctioning may help. If you think the patient is distressed, consider a trial of an antisecretory drug (glycopyrronium or hyoscine butylbromide: see tables 11.6, 11.7).

*Hydration.* Many patients approaching the end of life are unable to eat/drink or have a very poor appetite for food/fluid. Helping to take sips and good mouth care may suffice. Fluid via non-oral routes (NG, SC, IV) is given for symptomatic benefit; the effect on survival is unknown. Any potential benefit must be weighed against the risk of symptomatic fluid overload. Discuss this with patients and families, explaining the pros and cons of hydration and the uncertainty of the effect of hydration on survival:[23] relatives may assume the patient is dying faster because of dehydration or that they will be suffering with thirst. Make decisions about giving fluids on a case-by-case basis. Review a patient's hydration status at least daily.

*Plan for death.* Ensure that a 'Do not attempt resuscitation' or 'Allow natural death', order has been made. Discuss this with the patient (where appropriate), their family, and/or others of importance to them. Document everything clearly. Do they want to die at home? This can usually be arranged at very short notice with help from district nursing teams and community palliative care. Discuss transfer to a hospice or nursing home if appropriate.

*Respond to changes in the clinical situation.* Patients who are thought to be at the end of their life occasionally improve. Be alert to signs of improvement, and be prepared to switch back to active treatment when appropriate.

*Communicate.*[23] *The importance of clear and regular written and verbal communication with dying patients and their families cannot be overemphasized.* Find out what is important to your patient. How much information do they know and want to know about their situation and prognosis? Be sensitive to social, religious, and cultural issues.

### On wanting to die

Physician-assisted suicide is the provision of drugs by a doctor for self-administration by a person to terminate their own life. This is distinct from euthanasia where a doctor administers the lethal drug. There have been repeated attempts to introduce physician-assisted dying (physician-assisted suicide only of the terminally ill) into UK law, but all have been rejected by Parliament.

Consider autonomy. Should competent patients have the right to determine their death, especially if their situation is unbearable, without prospect of improvement? It is a powerful argument. But bearable is subjective and prognosis is an inexact science. And beneficence is divided: is it merciful, or is it abandonment, to end suffering through death? And what of consent? Consent is key. Yet consent is complex. Could a legal process help? Not without a unique understanding of each patient and the means by which they experience life. A combination of law and medicine may offer false comfort without full accountability by either. Protection for the vulnerable is a valid concern for doctors and society.

Requests to hasten death are complex and include personal, psychological, spiritual, social, cultural, and demographic factors. ~10% of terminally ill patients will consider euthanasia or physician-assisted suicide. These wishes may or may not be fixed, with ~50% of patients changing their mind within 6 months. Discuss this. Ask your patient how they feel today and what they are afraid of feeling tomorrow. Listen. Answer questions. Offer palliative care. Palliative care is never futile. A wish to die is associated with a need for information, reassurance, and competence in symptom control. Provide these, or find someone who can.

# 12 Rheumatology

Contents




Fig 12.1 When William Pitt the Elder, British statesman, was struck by yet another attack of gout he was absent from Parliament in 1773 when its members were persuaded to levy a substantial tax on tea imports to the American colonies. The resulting Tea Act of 1773 was born. Colonists boarded ships of the East India Company in Boston Harbour and crates of tea were thrown overboard. In response, the British government sent troops to occupy Boston to control the colonists. The armed response to these occupying forces led to the American War of Independence. Thirteen colonies from the United Kingdom became independent. And so it is told that gout had a part to play in the beginning of the American Revolution!

## A rapidly advancing speciality

Rheumatology originates from the Greek word 'rheuma' meaning that which 'flows as a river or stream'. The British Society of Rheumatology defines rheumatology as a 'multidisciplinary branch of medicine that deals with the investigation, diagnosis and management of patients with arthritis and other musculoskeletal conditions...incorporating over 200 disorders affecting joints, bones, muscles and soft tissues, including inflammatory arthritis and other systemic autoimmune disorders, vasculitis, soft tissue conditions, spinal pain and metabolic bone disease'. Rheumatological diseases affect over 10 million UK adults and 12 000 children. Recent advances owe largely to new discoveries about the immunology of these disorders and the discovery of biologic DMARDs.


*We thank Professor Kevin Davies, our Specialist Reader, for his contribution to this chapter. We also thank Dr Susie Higgins for her contribution to this chapter.*

## The rheumatological history

In the assessment of an arthritic presentation, pay particular attention to the distribution of joint involvement (including spine) and the presence of symmetry. Also look for disruption of joint anatomy, limitation of movement (by pain or contracture), joint effusions and peri-articular involvement (see p540 for a fuller assessment). Ask about, and examine for, *extra-articular features:* skin and nail (see p76) involvement (include scalp, hairline, umbilicus, genitalia, and natal cleft—psoriasis can easily be missed); eye signs (see p560); lungs (eg fibrosis) (see p198); kidneys (see p314); heart; GI (eg mouth ulcers, diarrhoea); GU (eg urethritis, genital ulcers); and CNS.

### Three screening questions for musculoskeletal disease

1. Are you free of any pain or stiffness in your joints, muscles, or back?
2. Can you dress yourself without too much difficulty?
3. Can you manage walking up and down stairs?

If *yes* to all three, serious inflammatory muscle/joint disease is unlikely.

*Presenting symptoms:*
- Pattern of involved joints.
- Symmetry (or not).
- Morning stiffness >30min (eg RA).
- Pain, swelling, loss of function, erythema, warmth.

*Extra-articular features:*
- Rashes, photosensitivity (eg SLE).
- Raynaud's (SLE; systemic sclerosis; polymyositis and dermatomyositis).
- Dry eyes or mouth (Sjögren's).
- Red eyes, iritis (eg AS).
- Diarrhoea/urethritis (reactive arthritis).
- Nodules or nodes (eg RA; TB; gout).
- Mouth/genital ulcers (eg Behçet's, SLE).
- Weight loss (eg malignancy, any systemic inflammatory disease).

*Related diseases:*
- Crohn's/UC (in ankylosing spondylitis), preceding infections, psoriasis.

*Current and past drugs:*
- NSAIDs, DMARDs (p547).
- Biological agents (eg TNFα inhibitors).

*Family history:*
- Arthritis, psoriasis, autoimmune disease.

*Social history:*
- Age.
- Occupation.
- Sexual history.
- Ethnicity (eg SLE is commoner in African-Caribbeans and Asians).
- Ability to function (eg dressing, grooming, writing, walking).
- Domestic situation, social support, home adaptations.
- Smoking (may worsen RA).
- IBD.

## Arthritides

The pattern of joint involvement can provide clues to the underlying cause (table 12.1).

Table 12.1 Patterns of presentation of arthritis

| Monoarthritis | Oligoarthritis (≤5 joints) | Polyarthritis (>5 joints involved) | |
|---|---|---|---|
| Septic arthritis | Crystal arthritis | **Symmetrical** | **Asymmetrical** |
| Crystal arthritis (gout, CPPD) | Psoriatic arthritis | Rheumatoid arthritis | Reactive arthritis |
| Osteoarthritis | Reactive arthritis, eg *Yersinia, Salmonella, Campylobacter* | Osteoarthritis | Psoriatic arthritis |
| Trauma (haemarthrosis) | Ankylosing spondylitis | Viruses (eg hepatitis A, B, & C; mumps) | |
| | Osteoarthritis | Systemic conditions* (can be either) | |

*Connective tissue disease (eg SLE and relapsing polychondritis), sarcoidosis, malignancy (eg leukaemia), endocarditis, haemochromatosis, sickle-cell anaemia, familial Mediterranean fever, Behçet's.

▸▸Exclude septic arthritis in any acutely inflamed joint, as it can destroy a joint in under 24h (p544). Inflammation may be less overt if immunocompromised (eg from the many immunosuppressive drugs used in rheumatological conditions) or if there is underlying joint disease. Joint aspiration (p541) is the key investigation, and if you are unable to do it, find someone who can.

## Assessing the locomotor system

This aims to screen for rheumatological conditions primarily affecting mobility (as a consequence of underlying joint disease). It is based on the **GALS** locomotor screen (**G**ait, **A**rms, **L**egs, **S**pine).[1]

**Essence** 'Look, feel, and move' (active and passive). If a *joint looks normal* to you, *feels normal* to the patient, and has *full range of movement*, it usually is normal. Make sure the patient is comfortable, and obtain their consent before examination. The GALS screening examination should be done in light underwear.

*Spine: Observe from behind:* is muscle bulk normal (buttocks, shoulders)? Is the spine straight? Are paraspinal muscles symmetrical? Any swellings/deformities? *Observe from the side:* is cervical and lumbar lordosis normal? Any kyphosis? '*Touch your toes, please*': is lumbar spine flexion normal, eg Schober's test?[1] *Observe from in front:* '*Tilt your head*' (without moving the shoulders)—tests lateral neck flexion. Palpate for typical fibromyalgia tender points (see p558).

*Arms:* '*Try putting your hands behind your head*'—tests functional shoulder movement. '*Arms out straight*'—tests elbow extension and forearm supination/pronation. Examine the hands: any deformity (fig 12.2), wasting, or swellings? *Squeeze across 2nd–5th metacarpophalangeal joints.* Pain may denote joint or tendon synovitis. '*Put your index finger on your thumb*'—tests pincer grip. Assess dexterity, eg fastening a button or picking up a coin.

*Legs: Observe legs:* normal quadriceps bulk? Any swelling or deformity? *With patient lying supine:* any leg length discrepancy? *Internally/externally rotate each hip in flexion. Passively flex knee and hip to the full extent.* Is movement limited? Any crepitus? *Find any knee effusion* using the patella tap test. If there is fluid, consider aspirating and testing for crystals or infection. *With patient standing: observe feet*: any deformity? Are arches high or flat? Any callosities? These may indicate an abnormal gait of some chronicity. *Squeeze across metatarsophalangeal joints:* see as for arms. Also: although not in the GALS system, *palpate the heel and Achilles tendon* to identify plantar fasciitis and Achilles tendonitis often associated with seronegative rheumatological conditions. *Examine the patient's shoes* for signs of uneven wear.

*Gait: Observe walking:* is the gait smooth? Good arm swing? Stride length OK? Normal heel strike and toe off? Can they turn quickly?

**Range of joint movement** Is noted in degrees, with anatomical position being the neutral position—eg elbow flexion 0°–150° normally, but with fixed flexion and limited movement, range may be reduced to 30°–90°. A valgus deformity deviates laterally (away from the mid-line, fig 12.3); a varus deformity points towards the mid-line.

Fig 12.2 Swan-neck deformity.

Reproduced from Watts *et al.*, *Oxford Textbook of Rheumatology*, 2013, with permission from Oxford University Press.

Fig 12.3 Bilateral hallux valgus.

Reproduced from *British Medical Journal*, 'Hallux valgus', R Choa, R Sharp, K R Mahtani, 2010, with permission from BMJ Publishing Group Ltd.

1 Schober's test: make a mark on the lumbar spine at the level of the posterior iliac spine. Measure out a line from 5cm below to 10cm above the mark. Ask to bend forward as far as they can. If the line does not lengthen by at least 5cm in flexion, there is reduced lumbar flexion, eg in ankylosing spondylitis.

## Some important rheumatological investigations

*Joint aspiration:* The most important investigation in any monoarthritic presentation (table 12.2, see also *OHCS* p706). Send synovial fluid for urgent white cell count, Gram stain, polarized light microscopy (for crystals, p548), and culture. The risk of inducing septic arthritis, using sterile precautions, is <1:10 000.[2] Look for blood,[3] pus, and crystals (gout or CPPD crystal arthropathy; p548). ►Do not attempt joint aspiration through inflamed and potentially infected skin (eg through a psoriatic plaque or overlying cellulitis).

Table 12.2 Synovial fluid in health and disease

| | Appearance | Viscosity | WBC/mm$^3$ | Neutrophils |
|---|---|---|---|---|
| **Normal** | Clear, colourless | ↔ | ≤200 | None |
| **Osteoarthritis** | Clear, straw | ↑ | ≤1000 | ≤50% |
| **Haemorrhagic*** | Bloody, xanthochromic | Varies | ≤10 000 | ≤50% |
| **Acutely inflamed** | | | | |
| • *RA* | Turbid, yellow | ↓ | 1000–50 000 | Varies |
| • *Crystal* | Turbid, yellow | ↓ | 5000–50 000 | ~80% |
| **Septic** | Turbid, yellow | ↓ | 10 000–100 000 | >90% |

*Eg trauma, tumour, or haemophilia.

*Blood tests:* FBC, ESR, urate, U&E, CRP. Blood culture for septic arthritis. Consider rheumatoid factor, anti-CCP, ANA, other autoantibodies (p553), and HLA B27 (p551)—as guided by presentation. Consider causes of reactive arthritis (p551), eg viral serology, urine chlamydia PCR, hepatitis and HIV serology if risk factors are present.

*Radiology:* Look for erosions, calcification, widening or loss of joint space, changes in underlying bone of affected joints (eg periarticular osteopenia, sclerotic areas, osteophytes). Characteristic X-ray features for various arthritides are shown in figs 12.4–12.6. Irregularity of the sacroiliac joints is seen in spondyloarthritis. Ultrasound and MRI are more sensitive in identifying effusions, synovitis, enthesitis and infection than plain radiographs—discuss further investigations with a radiologist. Do a CXR for RA, vasculitis, TB, and sarcoid.

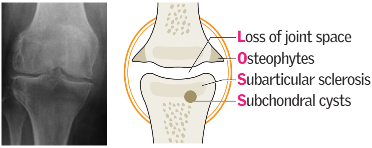



Fig 12.4 X-ray features of osteoarthritis.
Courtesy of Dr DC Howlett.

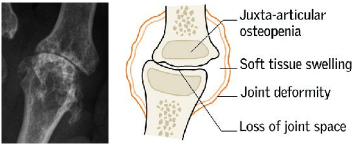



Fig 12.5 X-ray features of rheumatoid arthritis (MCPJ).
Courtesy of Dr DC Howlett.

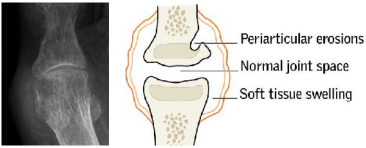



Fig 12.6 X-ray features of gout (1st MTPJ).
Courtesy of Dr DC Howlett.

# Back pain

Back pain is very common, and often self-limiting, but *be alert to sinister causes*, ie malignancy, infection, or inflammatory causes.

| Red flags for sinister causes of back pain | |
|---|---|
| ►Aged <20yrs or >55yrs old | ►Thoracic back pain |
| ►Acute onset in elderly people | ►Morning stiffness |
| ►Constant or progressive pain | ►Bilateral or alternating leg pain |
| ►Nocturnal pain | ►Neurological disturbance (incl. sciatica) |
| ►Worse pain on being supine | ►Sphincter disturbance |
| ►Fever, night sweats, weight loss | ►Current or recent infection |
| ►History of malignancy | ►Immunosuppression, eg steroids/HIV |
| ►Abdominal mass | ►Leg claudication or exercise-related leg weakness/numbness (spinal stenosis). |

**Examination**

1. With the patient standing, gauge the extent and smoothness of lumbar forward/lateral flexion and extension (see p540).
2. *Test for sacroiliitis:* palpate posteriorly down the length of the spine, including over spinous processes, paraspinal muscles, and the sacroiliac joints; examining for tenderness.
3. Neurological deficits (see BOX): test lower limb sensation, power, and deep tendon and plantar reflexes. Digital rectal examination for perianal tone and sensation.
4. Examine for nerve root pain (table 12.3): this is distributed in relevant dermatomes, and is worsened by coughing or bending forward. *Straight leg test* (L4, L5, S1): positive if raising the leg with the knee extended causes pain below the knee, which increases on foot dorsiflexion (Lasègue's sign). It suggests irritation to the sciatic nerve. The main cause is lumbar disc prolapse. Also *femoral stretch test* (L2–L4): pain in front of thigh on lifting the hip into extension with the patient lying face downwards and the knee flexed.
5. Signs of generalized disease—eg malignancy. Examine other systems (eg abdomen) as pain may be referred.

**Causes** Age determines the most likely causes:

15–30yrs: Prolapsed disc, trauma, fractures, ankylosing spondylitis (AS; p550), spondylolisthesis (a forward shift of one vertebra over another, which is congenital or due to trauma), pregnancy.

30–50yrs: Degenerative spinal disease, prolapsed disc, malignancy (primary or secondary from lung, breast, prostate, thyroid, or kidney ca).

>50yrs: Degenerative, osteoporotic vertebral collapse, Paget's (see p685), malignancy, myeloma (see p368), spinal stenosis.

*Rarer:* Cauda equina tumours, psoas abscess, spinal infection (eg discitis, usually staphylococcal but also *Proteus*, *E. coli*, *S. typhi*, and TB—there are often no systemic signs).

**Investigations** Arrange relevant tests if you suspect a specific cause, or if red flag symptoms: FBC, ESR, and CRP (myeloma, infection, tumour), U&E, ALP (Paget's), serum/urine electrophoresis (myeloma), PSA. *X-rays*—imaging may not always be necessary but can exclude bony abnormalities and fractures. Correlation between radiographic abnormalities and clinical features can be poor. ***MRI*** is the image of choice and can detect disc prolapse, cord compression (fig 12.7), cancer, infection, or inflammation (eg sacroiliitis).

**Management** ►►Urgent neurosurgical referral if any neurological deficit (see BOX). Keep the diagnosis under review. For non-specific back pain, focus on *education* and *self-management*. Advise patients to continue normal activites and be active. Regular paracetamol ± NSAIDs ± codeine. Consider low-dose amitriptyline/duloxetine if these fail (not SSRIs for pain). Offer *physiotherapy*, *acupuncture*, or an *exercise programme* if not improving.[4] Address *psychosocial issues*, which may predispose to developing chronic pain and disability (see p559). Referral to pain clinic or surgical options for patients with intractable symptoms.

## Neurosurgical emergencies

►*Acute cauda equina compression* Alternating or bilateral root pain in legs, saddle anaesthesia (perianal), loss of anal tone on PR, bladder ± bowel incontinence.

►*Acute cord compression* Bilateral pain, LMN signs (p446) at level of compression, UMN and sensory loss below, sphincter disturbance.

*Immediate urgent treatment* prevents irreversible loss, eg laminectomy for disc protrusions, radiotherapy for tumours, decompression for abscesses.

*Causes* (same for both): bony metastasis (look for missing pedicle on x-ray), large disc protrusion, myeloma, cord or paraspinal tumour, TB (p392), abscess.

**Table 12.3** Nerve root lesions

| Nerve root | Pain | Weakness | Reflex affected |
|---|---|---|---|
| L2 | Across upper thigh | Hip flexion and adduction | Nil |
| L3 | Across lower thigh | Hip adduction, knee extension | Knee jerk |
| L4 | Across knee to medial malleolus | Knee extension, foot inversion and dorsiflexion | Knee jerk |
| L5 | Lateral shin to dorsum of foot and great toe | Hip extension and abduction<br>Knee flexion<br>Foot and great toe dorsiflexion | Great toe jerk |
| S1 | Posterior calf to lateral foot and little toe | Knee flexion<br>Foot and toe plantar flexion<br>Foot eversion | Ankle jerk |

**Fig 12.7** Sagittal T2-weighted MRI of the lumbar spine showing a herniated L5–S1 disc.

Courtesy of Norwich Radiology Department.

# Osteoarthritis (OA)

Osteoarthritis is the most common joint condition worldwide, with a clinically significant impact on >10% of persons aged >60 years.[5] It is usually primary (generalized), but may be secondary to joint disease or other conditions (eg haemochromatosis, obesity, occupational).

**Signs and symptoms** *Localized disease* (often knee or hip): Pain and crepitus on movement, with background ache at rest. Worse with prolonged activity. Joints may 'gel' (brief stiffness after rest, usually 10–15 minutes or so). Joints may feel unstable, with a perceived lack of power due to pain. *Generalized disease:* 'Nodal OA' (typically DIP, PIP, CMC joints, and knees in post-menopausal females). There may be joint tenderness, derangement and bony swelling (Heberden's at DIP and Bouchard's at PIP), reduced range of movement and mild synovitis. Assess effect of symptoms on occupation, family duties, hobbies, and lifestyle expectations.

**Tests** Plain radiographs show: **L**oss of joint space, **O**steophytes, **S**ubarticular sclerosis and **S**ubchondral cysts (fig 12.4 p541). CRP may be slightly elevated.[6]

**Management** *Core treatments:* Exercise to improve local muscle strength and general aerobic fitness (irrespective of age, severity, or comorbidity). Weight loss if overweight.[7] *Analgesia:* Regular paracetamol ± topical NSAIDs. If ineffective use codeine or short-term oral NSAID (+PPI)—see BOX. Topical capsaicin (derived from chillies) may help. Intra-articular steroid injections temporarily relieve pain in severe symptoms. Intra-articular hyaluronic acid injections (viscosupplementation) are not NICE approved[8] Glucosamine and chondroitin products are not recommended, although patients may try them if they wish (can be bought over the counter). *Non-pharmacological:* Use a multidisciplinary approach, including physiotherapists and occupational therapists. Try heat or cold packs at the site of pain, walking aids, stretching/manipulation or TENS. *Surgery:* Joint replacement (hips, or knees) is the best way to deal with severe OA that has a substantial impact on quality of life.

# Septic arthritis

►►Consider septic arthritis in any acutely inflamed joint, as it can destroy a joint in under 24h and has a mortaility rate up to 11%. Inflammation may be less overt if immunocompromised (eg from medication) or if there is underlying joint disease. The knee is affected in >50% cases.

**Risk factors** Pre-existing joint disease (especially rheumatoid arthritis); diabetes mellitus, immunosuppression, chronic renal failure, recent joint surgery, prosthetic joints (where infection is particularly difficult to treat), IV drug abuse, age >80yrs.[9]

**Investigations** Urgent joint aspiration for synovial fluid microscopy and culture is the key investigation (p541), as plain radiographs and CRP may be normal. The main differential diagnoses are the crystal arthropathies (p548). Blood cultures are essential (prior to antibiotics).

*Ask yourself* 'How did the organism get there?' Is there immunosuppression, or another focus of infection, eg from indwelling IV lines, infected skin, or pneumonia (present in up to 50% of those with pneumococcal arthritis)?[10]

►►**Treatment** If in doubt start empirical IV antibiotics (after aspiration) until sensitivities are known. Common causative organisms are *Staph. aureus*, streptococci, *Neisseria gonococcus*, and Gram −ve bacilli. Follow local guidelines for antibiotic choice and contact microbiology for advice for all complex cases/immunosuppressed patients (eg HIV). Consider flucloxacillin 2g QDS IV (clindamycin if penicillin allergic); Vancomycin IV plus 2nd- or 3rd-generation cephalosporin, eg cefuroxime if MRSA risk; 2nd- or 3rd-generation cephalosporin if Gram −ve organisms suspected.[11] For suspected gonococcus or meningococcus, consider ceftriaxone. Antibiotics are required for a prolonged period, conventionally ~2 weeks IV, then if patient improving 2–4 weeks PO.[12] Consider orthopaedic review for arthrocentesis, washout, and debridement; ►►always urgently refer patients with prosthetic joint involvement.

## Prescribing NSAIDs: benefit vs risk profiling

Around 60% of patients will respond to any NSAID, but there is considerable variation in response and tolerance—if one isn't effective, try another. Mainly act as analgesics rather than modifying the disease process *per se*.

►►NSAIDs caused ~2000 UK deaths in 2011.[13] Individualized risk: benefit analysis for each patient (including indication, dose, proposed duration of use, and comorbidity) is crucial and needs careful and experienced thought. Follow local recommendations and national guidelines where available.

*NSAID side effects:* ►The main serious side effects are GI bleeding (and ulcers and perforation), cardiovascular events (MI and stroke), and renal injury. The risks are dose related, starting with the first dose, so always aim to use the lowest possible dose for the shortest period of time. Risks increase considerably with age, polypharmacy, history of peptic ulcers, and renal impairment.

*GI side effects:* NICE recommends co-prescription of PPI for any patient aged >45 years, and those with other risk factors for GI bleeding. Drug interactions can increase bleeding risks—avoid concomitant prescribing of anticoagulants, antiplatelet agents, SSRI, spironolactone, steroids, and bisphosphonates. Coxibs are slightly lower risk than non-selective NSAIDs.

*Cardiovascular side effects:* NSAIDs—all are associated with a small increased risk of MI and stroke (independent of cardiovascular risk factor or duration of use).[14] Risks are higher in those with concomitant IHD risk factors, eg diabetes and hypertension. Coxibs and diclofenac are higher risk, and are contraindicated if prior history of MI, PVD, stroke, or heart failure. Naproxen has the lowest cardiovascular risk. Low-dose celecoxib may be considered for patients on low-dose aspirin (if NSAID is required) as it does not interact with it.[15]

*Renal risks:* Higher for patients already on diuretics, ACE, or ARB. Risks are also increased in the elderly, those with hypertension and T2DM. Overall, naproxen (<1000mg/day) or ibuprofen (<1200mg/day) plus PPI may be the safest options.

*Alternatives to NSAIDs:* Paracetamol, topical NSAIDs, opioids. Strengthening exercises may be more beneficial than mild oral analgesics.

*Counselling patients:* Make sure patients understand about the drugs they are taking: *bleeding is more common in those who know less about their drugs.*[16]

- Only to take NSAIDs when they need them.
- Stop NSAIDs and seek urgent medical review if they develop abdominal pain or any symptoms of GI bleeding (eg report black stools ± faints immediately).
- Do not mix prescription NSAIDs with over-the-counter formulations: mixing NSAIDs can increase risks 20-fold.
- Smoking and alcohol increase risk profile of NSAIDs.

# Rheumatoid arthritis (RA)



RA is a chronic systemic inflammatory disease, characterized by a symmetrical, deforming, peripheral polyarthritis. It increases the risk of cardiovascular disease by 2–3 fold. **Epidemiology** Prevalence is ~1% (↑ in smokers). ♀:♂ >2:1. Peak onset: 5th–6th decade. HLA DR4/DR1 linked (associated with ↑severity).

**Presentation** *Typically:* Symmetrical swollen, painful, and stiff small joints of hands and feet, worse in the morning. This can fluctuate and larger joints may become involved. *Less common presentations:* •Sudden onset, widespread arthritis. •Recurring mono/polyarthritis of various joints (*palindromic RA*).[2] •Persistent monoarthritis (knee, shoulder, or hip). •Systemic illness with extra-articular symptoms, eg fatigue, fever, weight loss, pericarditis, and pleurisy, but initially few joint problems (commoner in ♂). •Polymyalgic onset—vague limb girdle aches. •Recurrent soft tissue problems (eg frozen shoulder, carpal tunnel syndrome, de Quervain's tenosynovitis).

**Signs** *Early:* (Inflammation, no joint damage.) Swollen MCP, PIP, wrist, or MTP joints (often symmetrical). Look for tenosynovitis or bursitis. *Later:* (Joint damage, deformity.) Ulnar deviation and subluxation of the wrist and fingers. Boutonnière and swan-neck deformities of fingers (fig 12.2 on p540) or Z-deformity of thumbs occur. Hand extensor tendons may rupture. Foot changes are similar. Larger joints can be involved. Atlanto-axial joint subluxation may threaten the spinal cord (rare).

**Extra-articular manifestations** Affect ~40% of RA patients. *Nodules:* Elbows, lungs, cardiac, CNS, lymphadenopathy, vasculitis. *Lungs:* Pleural disease, interstitial fibrosis, bronchiolitis obliterans, organizing pneumonia. *Cardiac:* IHD, pericarditis, pericardial effusion; carpal tunnel syndrome; peripheral neuropathy; splenomegaly (seen in 5%; only 1% have Felty's syndrome: RA + splenomegaly + neutropenia, see p698). *Eye:* Episcleritis, scleritis, scleromalacia, keratoconjunctivitis sicca (p560); osteoporosis; amyloidosis is rare (p370).[17,18]

**Investigations** Rheumatoid factor (RhF) is positive in ~70% (p553). High titres associated with severe disease, erosions, and extra-articular disease. Anticyclic citrullinated peptide antibodies (anti-CCP) are highly specific (~98%) for RA with a reasonable sensitivity (70–80%); they may also predict disease progression.[19] Anaemia of chronic disease, ↑platelets, ↑ESR, ↑CRP. *X-rays* show soft tissue swelling, juxta-articular osteopenia and ↓joint space. Later there may be bony erosions, subluxation, or complete carpal destruction (see fig 12.5 on p541). Ultrasound and MRI can identify synovitis more accurately, and have greater sensitivity in detecting bone erosions than conventional X-rays.[20]

**Diagnostic criteria** See table 12.4.

**Management** ►Refer early to a rheumatologist (before irreversible destruction).
- Disease activity is measured using the DAS28.[3] Treatment should be escalated until satisfactory control is achieved: 'treat to target'.
- Early use of DMARDs and biological agents improves long-term outcomes (see BOX 'Influencing biological events in RA').
- Steroids rapidly reduce symptoms and inflammation. Avoid starting unless appropriately experienced. Useful for acute exacerbations, eg IM depot *methylprednisolone* 80–120mg. Intra-articular steroids have a rapid but short-term effect (*OHCS* pp706–9). Oral steroids (eg *prednisolone* 7.5mg/d) may control difficult symptoms, but side effects preclude routine long-term use.
- NSAIDs (see p545) are good for symptom relief, but have no effect on disease progression. Paracetamol and weak opiates are rarely effective.
- Offer specialist physio- and occupational therapy, eg for aids and splints.
- Surgery may relieve pain, improve function, and prevent deformity.
- There is ↑risk of cardiovascular and cerebrovascular disease, as atherosclerosis is accelerated in RA.[21] Manage risk factors (p93). Smoking also ↑ symptoms of RA.

**2** In rheumatological palindromes, arthritis lasting hours or days runs to and fro, visiting and revisiting three or more sites, typically knees, wrists, and MCP joints. It may presage RA, SLE, Whipple's, or Behçet's disease. Remissions are (initially) complete, leaving no radiological mark.
**3** 28-joint Disease Activity Score—assesses tenderness and swelling at 28 joints (MCPs, PIPs, wrists, elbows, shoulders, knees), ESR/CRP, and patient's self-reported symptom severity.

Table 12.4 Criteria for diagnosing RA[22]

| | |
|---|---|
| | *When to suspect RA?* Those with ≥1 swollen joint and a suggestive clinical history, which is not better explained by another disease. Scores ≥6 are diagnostic. |
| **A** | *Joint involvement* (swelling or tenderness ± imaging evidence) |
| | 1 large joint =0 · 2–10 large joints =1 · 1–3 small* joints† =2 |
| | 4–10 small* joints† =3 · > 10 joints (at least 1 small joint) =5 |
| **B** | *Serology* (at least 1 test result needed) |
| | Negative RF *and* negative anti-CCP =0 · Low +ve RF *or* low +ve anti-CCP =2 |
| | High +ve RF *or* high +ve anti-CCP =3 |
| **C** | *Acute phase reactants* (at least 1 test result needed) |
| | Normal CRP and normal ESR =0 · Abnormal CRP or abnormal ESR =1 |
| **D** | *Duration of symptoms:* <6 weeks =0 ≥6 weeks =1 |

*= MCPJ, PIPJ, 2nd–5th MTPJ, wrists, and thumb IPJ; †= with or without involvement of large joints.

## Quality of life

Depression, disability, and pain are important quality of life predictors. Be mindful of the impact of disease on relationships, work, and hobbies and acknowledge and explore this with your patients. Patients may wish to investigate complementary therapies and may find benefit from support groups .

## Influencing biological events in RA

The chief biological event is inflammation. Over-produced cytokines and cellular processes erode cartilage and bone, and produce the systemic effects seen in RA.
**Disease-modifying antirheumatic drugs** (DMARDs) are 1st line and should ideally be started within 3 months of persistent symptoms. They can take 6–12 weeks for symptomatic benefit. Best results are often achieved with a combination of methotrexate, sulfasalazine, and hydroxychloroquine.[23] Leflunomide is another option.
►*Immunosuppression* is a potentially fatal SE of treatment (especially in combination with methotrexate) which can result in pancytopenia, ↑susceptibility to infection (including atypical organisms), and neutropenic sepsis (p352). Regular FBC, LFT monitoring.[24]
**Other SE** •Methotrexate—pneumonitis (pre treatment CXR), oral ulcers, hepatotoxicity, teratogenic. •Sulfasalazine—rash, ↓sperm count, oral ulcers, GI upset. •Leflunomide—teratogenicity (♂ and ♀), oral ulcers, ↑BP, hepatotoxicity. •Hydroxychloroquine—can cause retinopathy; pre treatment and annual eye screen required.
**Biological agents and NICE guidance** Initiated by specialists, for patients with active disease despite adequate trial of at least 2 DMARDs. Pre treatment screening for TB, hepatitis B/C, HIV essential.
1 *TNFα inhibitors:* Eg infliximab (p265), etanercept, adalimumab, are approved by NICE as 1st-line agents. Where methotrexate is contraindicated, can be used as monotherapy. Clinical response can be striking, with improved function and health outcomes, although response may be inadequate/unsustained.[25]
2 *B-cell depletion:* Eg rituximab, used in combination with methotrexate and approved by NICE for severe active RA where DMARDs and a TNFα blocker have failed.[26]
3 *IL-1 and IL-6 inhibition:* Eg tocilizumab (IL-6 receptor blocker), approved by NICE in combination with methotrexate where TNFα blocker has failed (or is contraindicated).[27] Monitor for hypercholesterolaemia.
4 *Inhibition of T-cell co-stimulation:* Eg abatacept—licensed for active RA where patients have not responded to DMARDs or TNFα blocker.[28]
**Side effects of biological agents** Serious infection, reactivation of TB (∴ screen and consider prophylaxis) and hepatitis B; worsening heart failure; hypersensitivity; injection-site reactions and blood disorders. ANA and reversible SLE-type illness may evolve. Data suggests there is no increased risk of solid organ tumours but skin cancers may be more common.[29] TNF inhibitors do not appear to be associated with a further increase in the already elevated lymphoma occurrence in RA.[30]

## Crystal arthropathies: gout

Gout[31] typically presents with an acute monoarthropathy with severe joint inflammation (fig 12.8). >50% occur at the metatarsophalangeal joint of the big toe (fig 12.12) (podagra). Other common joints are the ankle, foot, small joints of the hand, wrist, elbow, or knee. It can be polyarticular. It is caused by deposition of monosodium urate crystals in and near joints. Attacks may be precipitated by trauma, surgery, starvation, infection, or diuretics. It is associated with raised plasma urate. In the long term, urate deposits (= tophi, eg in pinna, tendons, joints; see fig 12.9) and renal disease (stones, interstitial nephritis) may occur. *Prevalence:* ~1%. ♂:♀≈4:1.

**Differential diagnoses** Exclude septic arthritis in any acute monoarthropathy (p544). Then consider reactive arthritis, haemarthrosis, CPPD (see following topic) and palindromic RA (p546).

**Risk factors** *Reduced urate excretion:* Elderly, men, post-menopausal females, impaired renal function, hypertension, metabolic syndrome, diuretics, antihypertensives, aspirin. *Excess urate production:* Dietary (alcohol, sweeteners, red meat, seafood), genetic disorders, myelo- and lymphoproliferative disorders, psoriasis, tumour-lysis syndrome, drugs (eg alcohol, warfarin, cytotoxics). *Associations:* Cardiovascular disease, hypertension, diabetes mellitus, and chronic renal failure (see p680).[31] Gout is an independent risk factor for mortality from cardiovascular and renal disease. Screen for and treat CKD, hypertension, dyslipidaemia, diabetes.

**Investigations** Polarized light microscopy of synovial fluid shows *negatively birefringent* urate crystals (fig 12.10). Serum urate (SUA) is usually raised but may be normal.[32] Radiographs show only soft-tissue swelling in the early stages. Later, well-defined 'punched out' erosions are seen in juxta-articular bone (see fig 12.6 on p541). There is no sclerotic reaction, and joint spaces are preserved until late.

**Treatment of acute gout** High-dose NSAID (see BOX, p545) or if CI use colchicine (500mcg BD) which is effective but slower to work (*BNF* states max 6mg per course although rheumatologists will often use more).[33] NB: in renal impairment, NSAIDs and colchicine are problematic. Steroids (oral, IM, or intra-articular) may also be used.[34] Rest and elevate joint. Ice packs and 'bed cages' can be effective.

**Prevention** Lose weight, avoid prolonged fasts, alcohol excess, purine-rich meats, and low-dose aspirin. *Prophylaxis:* Start if >1 attack in 12 months, tophi, or renal stones. The aim is to ↓ attacks and prevent damage caused by crystal deposition. Use *allopurinol* and titrate from 100mg/24h, increasing every 4 weeks until plasma urate <0.3mmol/L (max 300mg/8h). SE: rash, fever, ↓WCC. Allopurinol may trigger an attack so wait 3 weeks after an acute episode, and cover with regular NSAID (for up to 6 weeks) or colchicine (0.5mg/12h PO for up to 6 months). Avoid stopping allopurinol in acute attacks once established. *Febuxostat* (80mg/24h) is an alternative if allopurinol is CI or not tolerated. It ↓ uric acid by inhibiting xanthine oxidase (SE: ↑LFTs) and is more effective at reducing serum urate than allopurinol (number of acute attacks the same).[35] Uricosuric drugs ↑ urate excretion.

## Calcium pyrophosphate deposition (CPPD)

- **Acute CPPD crystal arthritis** Acute monoarthropathy usually of larger joints in elderly. Usually spontaneous but can be provoked by illness, surgery, or trauma.
- **Chronic CPPD** Inflammatory RA-like (symmetrical) polyarthritis and synovitis.
- **Osteoarthritis** with CPPD chronic polyarticular osteoarthritis with superimposed acute CPP attacks.

**Risk factors** Old age, hyperparathyroidism (see p222), haemochromatosis (see p288), hypophosphataemia (see p679). **Tests** Polarized light microscopy of synovial fluid shows weakly positively birefringent crystals (fig 12.11). It is associated with soft tissue calcium deposition on x-ray. **Management** Acute attacks: cool packs, rest, aspiration, and intra-articular steroids. NSAIDs (+PPI) ± colchicine 0.5–1.0mg/24h (used with caution) may prevent acute attacks. Methotrexate and hydroxychloroquine may be considered for chronic CPP inflammatory arthritis.[36]

Fig 12.8 Acute monoarthritis in gout.

Fig 12.9 Ulcerated tophi in gout.

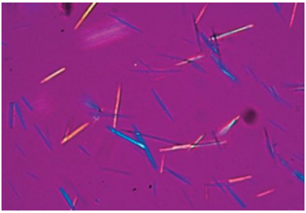


Fig 12.10 **N**eedle-shaped monosodium urate crystals found in gout, displaying **N**egative birefringence under polarized light.

Reproduced from Warrell *et al*., *Oxford Textbook of Medicine*, 2010, with permission from Oxford University Press.

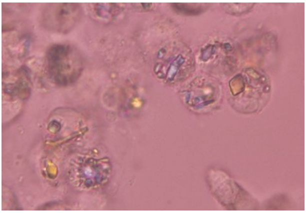


Fig 12.11 Rhomboid-shaped calcium pyrophosphate dihydrate crystals in **P**seudogout, showing **P**ositive birefringence in polarized light.

Image courtesy of Prof. Eliseo Pascual, Sección de Reumatología, Hospital General Universitario de Alicante.

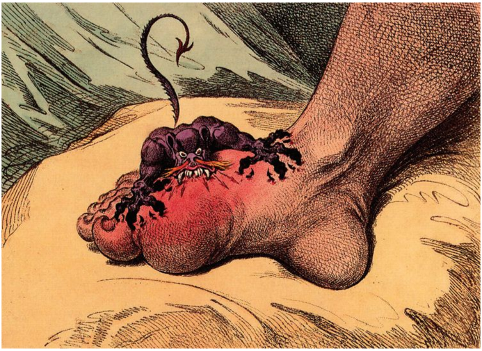


Fig 12.12 Don't underestimate the severity of pain caused by gout—as illustrated by satirical artist and gout sufferer James Gillray (1756–1815).

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# Spondyloarthritides

The spondyloarthropathies (SpA) are a group of related chronic inflammatory conditions. They tend to, although not always, affect the axial skeleton with shared clinical features:

1. Seronegativity (= rheumatoid factor −ve).
2. HLA B27 association—see BOX.
3. 'Axial arthritis': pathology in spine (spondylo-) and sacroiliac joints.
4. Asymmetrical large-joint oligoarthritis (ie <5 joints) or monoarthritis.
5. Enthesitis: inflammation of the site of insertion of tendon or ligament into bone, eg plantar fasciitis, Achilles tendonitis, costochondritis.
6. Dactylitis: inflammation of an entire digit ('sausage digit'), due to soft tissue oedema, and tenosynovial and joint inflammation.
7. Extra-articular manifestations: eg iritis (anterior uveitis), psoriaform rashes (psoriatic arthritis), oral ulcers, aortic valve incompetence, inflammatory bowel disease.

NB: Behçet's syndrome (p694) can also present with uveitis, skin lesions, and arthritis and is not always associated with gross oral or genital ulcerations.

1. **Ankylosing spondylitis (AS)** A chronic inflammatory disease of the spine and sacroiliac joints, of unknown aetiology (likely strong genetic/environmental interplay). *Prevalence:* 0.25–1%. *Men present earlier:* ♂:♀ ~6:1 at 16yrs old, and ~2:1 at 30yrs old. ~90% are HLA B27 +ve (see BOX). *Symptoms and signs:* The typical patient is a man <30yrs old with gradual onset of low back pain, worse during the night with spinal morning stiffness relieved by exercise. Pain radiates from sacroiliac joints to hips/buttocks, and usually improves towards the end of the day. There is progressive loss of spinal movement (all directions)—hence ↓thoracic expansion. See pp540–2 for tests of spine flexion and sacroiliitis. The disease course is variable; a few progress to kyphosis, neck hyperextension (question-mark posture; fig 12.13), and spino-cranial ankylosis. Other features include *enthesitis* (see BOX), especially Achilles tendonitis, plantar fasciitis, at the tibial and ischial tuberosities, and at the iliac crests. Anterior mechanical chest pain due to costochondritis and fatigue may feature. *Acute iritis* occurs in ~⅓ of patients and may lead to blindness if untreated (but may also have occurred many years before, so enquire directly). AS is also associated with osteoporosis (up to 60%), aortic valve incompetence (<3%), and pulmonary apical fibrosis. *Tests:* Diagnosis is clinical, supported by imaging.[4] MRI allows detection of active inflammation (bone marrow oedema) as well as destructive changes such as erosions, sclerosis, and ankylosis. X-rays can show SI joint space narrowing or widening, sclerosis, erosions, and ankylosis/fusion. Vertebral syndesmophytes are characteristic (often T11–L1 initially): bony proliferations due to enthesitis between ligaments and vertebrae. These fuse with the vertebral body above, causing ankylosis. In later stages, calcification of ligaments with ankylosis lead to a 'bamboo spine' appearance. *Also:* FBC (normocytic anaemia), ↑ESR, ↑CRP, HLA B27 +ve (+ve in 90–95% of cases but only 5% of patients HLA B27 +ve have AS). *Management: Exercise*, not rest, for backache, including intense exercise regimens to maintain posture and mobility—ideally with a specialist physiotherapist. NSAIDs usually relieve symptoms within 48h, and may slow radiographic progression.[37] *TNFα BLOCKERS* (eg etanercept, adalimumab) are indicated in severe active AS.[38] Local steroid injections provide temporary relief. Surgery includes hip replacement to improve pain and mobility if the hips are involved, and rarely spinal osteotomy. There is ↑risk of osteoporotic spinal fractures (consider bisphosphonates). *Prognosis:* There is not always a clear relationship between the activity of arthritis and severity of underlying inflammation (as for all the spondyloarthritides). Prognosis is worse if ESR >30; onset <16yrs; early hip involvement or poor response to NSAIDs.[39]
2. **Enteric arthropathy** *Associations:* Inflammatory bowel disease, GI bypass, coeliac and Whipple's disease (p716). Arthropathy often improves with the treatment of bowel symptoms (beware NSAIDs). Use DMARDs for resistant cases.

4 Sacroiliitis on imaging plus ≥1 SpA feature or HLA B27 positive plus ≥2 SpA features.

**3 Psoriatic arthritis** (*OHCS* p594.) Occurs in 10–40% with psoriasis and can present before skin changes. Patterns are: •symmetrical polyarthritis (like RA) •DIP joints •asymmetrical oligoarthritis •spinal (similar to AS) •psoriatic arthritis mutilans (rare, ~3%, severe deformity). *Radiology:* Erosive changes, with 'pencil-in-cup' deformity in severe cases. Associated with nail changes in 80%, synovitis (dactylitis), acneiform rashes and palmo-plantar pustulosis. *Management:* NSAIDs, sulfasalazine, methotrexate. Anti-TNF agents are also effective.

**4 Reactive arthritis** A condition in which arthritis and other clinical manifestations occur as an autoimmune response to infection elsewhere in the body—typically GI or GU, although the preceding infection may have resolved or be asymptomatic by the time the arthritis presents. *Other clinical features:* Iritis, keratoderma blenorrhagica (brown, raised plaques on soles and palms), circinate balanitis (painless penile ulceration secondary to *Chlamydia*), mouth ulcers, and enthesitis. Patients may present with a triad of urethritis, arthritis, and conjunctivitis (Reiter's syndrome). *Tests:* ↑ESR & ↑CRP. Culture stool if diarrhoea. Infectious serology. Sexual health review. X-ray may show enthesitis with periosteal reaction. *Management:* There is no specific cure. Splint affected joints acutely; treat with NSAIDs or local steroid injections. Consider sulfasalazine or methotrexate if symptoms >6 months. Treating the original infection may make little difference to the arthritis.

### HLA-B27 disease associations

The HLA system plays a key role in immunity and self-recognition. More than one hundred HLA B27 disease associations have been made[40], yet the actual role of HLA B27 in triggering an inflammatory response is not fully understood. ~5% of the UK population are HLA B27 positive—most do not have any disease. The chance of an HLA B27 positive person developing spondyloarthritis or eye disease is 1 in 4. Common associations include:

*Ankylosing spondylitis:* 85-95% of all those with AS are HLA B27 positive.
*Acute anterior uveitis:* 50–60% are HLA B27 positive.
*Reactive arthritis:* 60–85% are HLA B27 positive.
*Enteric arthropathy:* 50–60% are HLA B27 positive.
*Psoriatic arthritis:* 60–70% are HLA B27 positive.

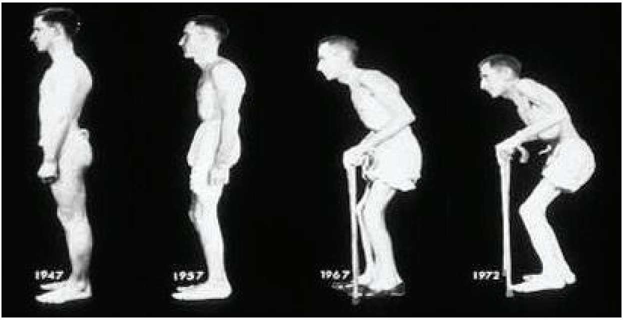


Fig 12.13 Progression of disease and effect on posture in severe ankylosing spondylitis.

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## Autoimmune connective tissue diseases

Included under this heading are SLE (p554), systemic sclerosis, Sjögren's syndrome (p710), idiopathic inflammatory myopathies (myositis—see following topic), mixed connective tissue disease, relapsing polychondritis, and undifferentiated connective tissue disease and overlap syndromes. They overlap with each other, affect many organ systems, and often require immunosuppressive therapies (p376). Consider as a differential in unwell patients with multi-organ involvement, especially if no infection.

**Systemic sclerosis** Features scleroderma (skin fibrosis), internal organ fibrosis, and microvascular abnormalities. Severe cases have a 40–50% mortality at 5 years. 90% are ANA positive and 30–40% have anticentromere antibodies (see BOX). Skin disease is limited or diffuse. *Limited* involves the face, hands, and feet (formally CREST syndrome). It is associated with anticentromere antibodies in 70–80%. Pulmonary hypertension is often present subclinically, and can become rapidly life-threatening, so should be looked for (℞: sildenafil, bosentan). *Diffuse* can involve the whole body. Antitopoisomerase-1 (SCL-70) antibodies in 40% and anti-RNA polymerase in 20%. Prognosis is often poor. Control BP meticulously. Perform annual echocardiogram and spirometry. Both limited and diffuse have the potential for organ fibrosis: lung, cardiac, GI, and renal (p314) but this occurs later in limited sub-set.

*Management:* Currently no cure. Immunosuppressive regimens, including IV cyclophosphamide, are used for organ involvement or progressive skin disease. Trials of antifibrotic tyrosine kinase inhibitors are ongoing.[41] Monitor BP and renal function. Regular ACE-i or A2RBs ↓ risk of renal crisis (p314). *Raynaud's phenomenon:* (see p708).

**Mixed connective tissue disease** Combines features of systemic sclerosis, SLE, and polymyositis and the presence of high titres of anti-U1-RNP antibodies.

**Relapsing polychondritis** Rare condition with recurrent episodes of cartilage inflammation and destruction. Affects pinna (floppy ears), nasal septum, larynx (stridor), tracheobronchial tree (infections), and joints. *Associations:* Aortic valve disease, polyarthritis, and vasculitis. 30% have underlying rheumatic or autoimmune disease. Diagnosis is clinical. *℞:* Steroids, DMARDs or CPAP/tracheostomy for airway involvement.

## Polymyositis and dermatomyositis

Rare conditions characterized by insidious onset of progressive symmetrical proximal muscle weakness and autoimmune-mediated *striated* muscle inflammation (myositis), associated with myalgia ± arthralgia. Muscle weakness may also cause dysphagia, dysphonia (ie poor phonation, *not* dysphasia), or respiratory weakness. The myositis (esp. in dermatomyositis) may be a paraneoplastic phenomenon, commonly from lung, pancreatic, ovarian, or bowel malignancy. Screen for cancers.

**Dermatomyositis** Myositis plus skin signs: •Macular rash (*shawl sign* is +ve if over back and shoulders). •Lilac-purple (*heliotrope*) rash on eyelids often with oedema (fig 12.26, p563). •Nailfold erythema (*dilated capillary loops*). •*Gottron's papules:* roughened red papules over the knuckles, also seen on elbows and knees (pathognomonic if ↑CK + muscle weakness). Malignancy in 30% cases.

**Extra-muscular signs** In both conditions include fever, arthralgia, Raynaud's, interstitial lung fibrosis and myocardial involvement (myocarditis, arrhythmias).

**Tests** Muscle enzymes (ALT, AST, LDH, CK, & aldolase) ↑ in plasma; EMG shows characteristic fibrillation potentials; muscle biopsy confirms diagnosis (and excludes mimicking conditions). MRI shows muscle oedema in acute myositis. *Autoantibody associations:* (see BOX) anti-Mi2, anti-Jo1—associated with acute onset and interstitial lung fibrosis that should be treated aggressively.

**Differential diagnoses** Carcinomatous myopathy, inclusion-body myositis, muscular dystrophy, PMR, endocrine/metabolic myopathy (eg steroids), rhabdomyolysis, infection (eg HIV), drugs (penicillamine, colchicine, statins, or chloroquine).

**Management** Start prednisolone. Immunosuppressives (p376) and cytotoxics are used early in resistant cases. Hydroxychloroquine/topical tacrolimus for skin disease.

## Plasma autoantibodies (Abs): disease associations

►Always interpret in the context of clinical findings: Different antibodies have different disease asssociations.

*Rheumatological:* Rheumatoid factor (RhF) positive in:

| | | | |
|---|---|---|---|
| Sjögren's syndrome | ≤100% | Mixed connective tissue disease | 50% |
| Felty's syndrome | ≤100% | SLE | ≤40% |
| RA | 70% | Systemic sclerosis | 30% |
| Infection (SBE/IE; hepatitis) | ≤50% | Normal | 2–10% |

*Anticyclic citrullinated peptide Ab (anti-CCP):*[5] rheumatoid arthritis (~96% specificity)

*Antinuclear antibody (ANA)* positive by immunofluorescence in:

| | | | |
|---|---|---|---|
| SLE | >95% | Systemic sclerosis | 96% |
| Autoimmune hepatitis | 75% | RA | 30% |
| Sjögren's syndrome | 68% | Normal | 0–2% |

ANA titres are expressed according to dilutions at which antibodies can be detected, ie 1:160 means antibodies can still be detected after the serum has been diluted 160 times. Titres of 1:40 or 1:80 may not be significant. The pattern of staining may indicate the disease (although these are not specific):

| | | | |
|---|---|---|---|
| • *Homogeneous* | SLE | • *Nucleolar* | Systemic sclerosis |
| • *Speckled* | Mixed CT disease | • *Centromere* | Limited systemic sclerosis |

*Anti-double-stranded DNA (dsDNA):* SLE (60% sensitivity, but highly specific).
*Antihistone Ab:* drug-induced SLE (~100%).
*Antiphospholipid Ab* (eg anti-cardiolipin Ab): antiphospholipid syndrome, SLE.
*Anticentromere Ab:* limited systemic sclerosis.
*Anti-extractable nuclear antigen (ENA) antibodies* (usually with +ve ANA):

| | |
|---|---|
| • Anti-Ro (SSA) | SLE, Sjögren's syndrome, systemic sclerosis. Associated with congenital heart block. |
| • Anti-La (SSB) | Sjögren's syndrome, SLE (15%). |
| • Anti-Sm | SLE (20–30%). |
| • Anti-RNP | SLE, mixed connective tissue disease. |
| • Anti Jo-1; Anti-Mi-2 | Polymyositis, dermatomyositis. |
| • Anti-Scl70 | Diffuse systemic sclerosis. |

*Gastrointestinal:* (For liver autoantibodies, see p284.)
*Antimitochondrial Ab (AMA):* primary biliary cholangitis (>95%), autoimmune hepatitis (30%), idiopathic cirrhosis (25–30%).
*Anti-smooth muscle Ab (SMA):* autoimmune hepatitis (70%), primary biliary cholangitis (50%), idiopathic cirrhosis (25–30%).
*Gastric parietal cell Ab:* pernicious anaemia (>90%), atrophic gastritis (40%), 'normal' (10%).
*Intrinsic factor Ab:* pernicious anaemia (50%).
*α-gliadin Ab, antitissue transglutaminase, anti-endomysial Ab:* coeliac disease.
*Endocrine:* *Thyroid peroxidase Ab:* Hashimoto's thyroiditis (~87%), Graves' (>50%). *Islet cell Ab (ICA), glutamic acid decarboxylase (GAD) Ab:* type 1 diabetes mellitus (75%).
*Renal:* *Glomerular basement membrane Ab (anti-GBM):* Goodpasture's disease (100%).
*Antineutrophil cytoplasmic Ab (ANCA):*

- *Cytoplasmic (cANCA)*, specific for *serine proteinase-3 (PR3 +ve)*. Granulomatosis with polyangiitis (Wegener's) (90%); also microscopic polyangiitis (30%), polyarteritis nodosa (11%).
- *Perinuclear (pANCA)*, specific for *myeloperoxidase (MPO +ve)*. Microscopic polyangiitis (45%), Churg–Strauss, pulmonary-renal vasculitides (Goodpasture's).

Unlike immune-complex vasculitis, in ANCA-associated vasculitis no complement consumption or immune complex deposition occurs (ie pauci-immune vasculitis).[42]
ANCA may also be +ve in UC/Crohn's, sclerosing cholangitis, autoimmune hepatitis, Felty's, RA, SLE, or drugs (eg antithyroid, allopurinol, ciprofloxacin).
*Neurological:* *Acetylcholine receptor Ab:* myasthenia gravis (90%)(see p512).
*Anti-voltage-gated $K^+$-channel Ab:* limbic encephalitis.
*Anti-voltage-gated $Ca^{2+}$-channel Ab:* Lambert-Eaton syndrome (see p512).
*Anti-aquaporin 4:* neuromyelitis optica (Devic's disease, p698).

5 Most centres now use anti-CCP antibodies for the initial workup of suspected RA.

# Systemic lupus erythematosus (SLE)



SLE is a multisystemic autoimmune disease. Autoantibodies are made against a variety of autoantigens (eg ANA) which form immune complexes . Inadequate clearance of immune complexes results in a host of immune responses which cause tissue inflammation and damage. Environmental triggers play a part (eg EBV p405).[43]

**Prevalence** ~0.2%. ♀:♂≈9:1, typically women of child-bearing age. Commoner in African-Caribbeans, Asians, and if HLA B8, DR2, or DR3 +ve. ~10% of patients have a 1st- or 2nd-degree relative with SLE.

**Clinical features** See BOX. Remitting and relapsing illness of variable presentation and course. Features often non-specific (malaise, fatigue, myalgia, and fever) or organ-specific and caused by active inflammation or damage. Other features include lymphadenopathy, weight loss, alopecia, nail-fold infarcts, non-infective endocarditis (Libman–Sacks syndrome), Raynaud's (30%; see p708), stroke, and retinal exudates.

**Immunology** >95% are ANA +ve. A high anti-double-stranded DNA (dsDNA) antibody titre is highly specific, but only +ve in ~60% of cases. ENA (p553) may be +ve in 20–30% (anti-Ro, anti-La, anti-Sm, anti-RNP); 40% are RhF +ve; antiphospholipid antibodies (anticardiolipin or lupus anticoagulant) may also be +ve. SLE may be associated with other autoimmune conditions: Sjögren's (15–20%), autoimmune thyroid disease (5–10%).

**Diagnosis** See BOX. **Monitoring activity** *Three best tests:* **1** Anti-dsDNA antibody titres. **2** Complement: ↓C3, ↓C4 (denotes consumption of complement, hence ↓C3 and ↓C4, and ↑C3d and ↑C4d, their degradation products). **3** ESR. *Also:* BP, urine for casts or protein (lupus nephritis, below), FBC, U&E, LFTs, CRP (usually normal) ►*think of SLE whenever someone has a multisystem disorder and ↑ESR but* CRP *normal.* If ↑CRP, think instead of infection, serositis, or arthritis. Skin or renal biopsies may be diagnostic.

**Drug-induced lupus** Causes (>80 drugs) include isoniazid, hydralazine (if >50mg/24h in slow acetylators), procainamide, quinidine, chlorpromazine, minocycline, phenytoin, anti-TNF agents. It is associated with antihistone antibodies in >95% of cases. Skin and lung signs prevail (renal and CNS are rarely affected). The disease remits if the drug is stopped. Sulfonamides or the oral contraceptive pill may worsen idiopathic SLE.

**Management** Refer: complex cases should involve specialist SLE/nephritis clinics.
- *General measures:* High-factor sunblock. Hydroxychloroquine, unless contraindicated, reduces disease activity and improves survival. Screen for co-morbidities and medication toxicity. For skin flares, first trial topical steroids.
- *Maintenance:* NSAIDs (unless renal disease) and hydroxychloroquine for joint and skin symptoms. Azathioprine, methotrexate, and mycophenolate as steroid-sparing agents. Belimumab (monoclonal antibody) used as an add-on therapy for autoantibody positive disease where disease activity is high.[44] (See BOX.)
- *Mild flares:* (No serious organ damage.) Hydroxychloroquine or low-dose steroids.
- *Moderate flares:* (Organ involvement.) May require DMARDs or mycophenolate.

►►*Severe flares: If life- or organ-threatening,* eg haemolytic anaemia, nephritis, severe pericarditis or CNS disease; urgent high-dose steroids, mycophenolate, rituximab, cyclophosphamide. MDT working vital for neuropsychiatric lupus (psychometric testing, lumbar puncture may be indicated).

*Lupus nephritis:* (p314.) May require more intensive immunosuppression with steroids and cyclophosphamide or mycophenolate. BP control vital (e.g. ACE-i). Renal replacement therapy (p306) may be needed if disease progresses; nephritis recurs in ~50% post-transplant, but is a rare cause of graft failure.[45]

*Prognosis:* ~80% survival at 15 years.[43] There is an increased long-term risk of CVD and osteoporosis.

**Antiphospholipid syndrome** Can be associated with SLE (20–30%). Often occurs as a primary disease. Antiphospholipid antibodies (anticardiolipin & lupus anticoagulant, anti-β 2 glycoprotein 1) cause **CLOTS**: **C**oagulation defect (arterial/venous), **L**ivedo reticularis (p557), **O**bstetric (recurrent miscarriage), **T**hrombocytopenia. Thrombotic tendency affects cerebral, renal, and other vessels. *Dx:* Persistent antiphosphlolipid antibodies with clinical features. *℞:* Anticoagulation; seek advice in pregnancy.[46]

## Systemic Lupus International Collaborating Clinics Classification

A favourite differential diagnosis, SLE mimics other illnesses, with wide variation in symptoms that may come and go unpredictably. Diagnose SLE[47] in an appropriate clinical setting if ≥4 criteria (at least 1 clinical and 1 laboratory) *or* biopsy-proven lupus nephritis with positive ANA or anti-DNA.

### Clinical criteria

1. *Acute cutaneous lupus:* Malar rash/butterfly. Fixed erythema, flat or raised, over the malar eminences, tending to spare the nasolabial folds (fig 12.14). Occurs in up to 50%. Bullous lupus, toxic epidermal necrolysis variant of SLE, maculopapular lupus rash, photosensitive lupus rash, or subacute cutaneous lupus (non-indurated psoriasiform and/or annular polycyclic lesions that resolve without scarring).
2. *Chronic cutaneous lupus:* Discoid rash, erythematous raised patches with adherent keratotic scales and follicular plugging ± atrophic scarring (fig 12.15). Think of it as a three-stage rash affecting ears, cheeks, scalp, forehead, and chest: erythema→pigmented hyperkeratotic oedematous papules→atrophic depressed lesions.
3. *Non scarring alopecia:* (In the absence of other causes.)
4. *Oral/nasal ulcers:* (In the absence of other causes.)
5. *Synovitis:* (Involving two or more joints *or* two or more tender joints with >30 minutes of morning stiffness.)
6. *Serositis:* a) Lung (pleurisy for >1 day, or pleural effusions, or pleural rub; b) pericardial pain for >1 day, or pericardial effusion, or pericardial rub, or pericarditis on ECG.
7. *Urinanalysis:* Presence of proteinuria (>0.5g/d ) *or* red cell casts.
8. *Neurological features:* Seizures; psychosis; mononeuritis multiplex; myelitis; peripheral or cranial neuropathy; cerebritis/acute confusional state in absence of other causes.
9. *Haemolytic anaemia.*
10. *Leucopenia:* (WCC <4.) At least once, or lymphopenia (lymphocytes <1) at least once.
11. *Thrombocytopenia:* (Platelets <100.) At least once.

### Laboratory criteria

1. +ve ANA (+ve in >95%).
2. Anti-dsDNA.
3. Anti-Smith antibodies present.
4. Antiphospholipid Abs present.
5. Low complement (C3, C4, or C50).
6. +ve Direct Coombs test.

Adapted from 'Derivation and validation of the Systemic Lupus International Collaborating Clinics classification criteria for systemic lupus erythematosus'. Petri M *et al.*, *Arthritis and Rheumatism*, vol 64, issue 8 (2012) 2677–2686.

Fig 12.14 Malar rash, with sparing of the nasolabial folds.

Courtesy of David F. Fiorentino, MD, PhD; by kind permission of *Skin & Aging*.

Fig 12.15 Discoid rash.

Courtesy of Amy McMichael, MD; by kind permission of *Skin & Aging*.

# Vasculitis



The vasculitides are inflammatory disorders of blood vessels; commonly classified using the modified Chapel Hill criteria.[48] They can affect any organ, and presentation depends on the organs involved. It may be a primary condition or secondary to other diseases, eg SLE, RA, hepatitis B & C, HIV. Categorized by size of blood vessels affected.

- **Large** Giant cell arteritis, Takayasu's arteritis (see p712).
- **Medium** Polyarteritis nodosa, Kawasaki disease (*OHCS* p646).
- **Small** • ANCA-associated: microscopic polyangiitis; granulomatosis with polyangiitis (Wegener's granulomatosis); and eosinophilic granulomatosis with polyangiitis (Churg Strauss syndrome; 40–60% are ANCA positive). • Immune complex vasculitis: Goodpasture's disease; cryoglobulinaemic vasculitis; IgA vasculitis (Henoch-Schonlein purpura).
- **Variable vessel vasculitis** Behçet's (p694) and Cogan's syndrome.

**Symptoms** Different vasculitides preferentially affect different organs, causing different patterns of symptoms (see BOX 'Features of vasculitis'), Often only overwhelming fatigue with ↑ESR/CRP. ▶▶*Consider vasculitis in any unidentified multisystem disorder.* If presentation does not fit clinically or serologically into a specific category consider malignancy-associated vasculitis. A severe vasculitis flare is a medical emergency. If suspected, seek urgent help, as organ damage may occur rapidly (eg critical renal failure <24h). **Tests** ↑ESR/CRP. ANCA may be +ve. ↑Creatinine if renal failure. Urine: proteinuria, haematuria, casts on microscopy. Angiography ± biopsy may be diagnostic. **Management** Large-vessel: steroids in most cases, may add steroid-sparing agents later. Medium/small: immunosuppression (steroids, ± another agent, eg cyclophosphamide if severe, or methotrexate/azathioprine depending on features).

**▶▶Giant cell arteritis (GCA) = temporal arteritis** Common in the elderly—consider Takayasu's if under 55yrs (p712). Associated with PMR in 50%. *Symptoms:* Headache, temporal artery and scalp tenderness (eg when combing hair), tongue/jaw claudication, amaurosis fugax, or sudden unliateral blindness. Extracranial symptoms: malaise, dyspnoea, weight loss, morning stiffness, and unequal or weak pulses.[49] The risk is irreversible bilateral visual loss, which can occur suddenly if not treated—ask an ophthalmologist. *Tests:* ESR & CRP are ↑↑, ↑platelets, ↑ALP, ↓Hb. Temporal artery biopsy within 14 days of starting steroids, or FDG-PET. Skip lesions occur, so don't be put off by a negative biopsy (up to 10%). *Management:* Start prednisolone 60mg/d PO *immediately* or IV methylprednisolone if evolving visual loss or history of amaurosis fugax.[50] *Prognosis:* Typically a 2-year course, then complete remission. Reduce prednisolone once symptoms have resolved and ↓ESR; ↑dose if symptoms recur. Main cause of death and morbidity in GCA is long-term steroid treatment so balance risks! Give PPI, bisphosphonate, calcium with colecalciferol, and consider aspirin.[6; 51]

**Polyarteritis nodosa (PAN)** PAN is a necrotizing vasculitis that causes aneurysms and thrombosis in medium-sized arteries, leading to infarction in affected organs with severe systemic symptoms. ♂:♀≈2:1. It may be associated with hepatitis B, and is rare in the UK. *Symptoms:* See BOX. Systemic features, skin (rash, 'punched out' ulcers, nodules), renal (main cause of death, renal artery narrowing, glomerular ischaemia, insufficiency, HTN), cardiac, GI, GU, neuro involvement. Usually spares lungs. Coronary aneurysms occur in Kawasaki disease (*OHCS* p646). *Tests:* Often ↑WCC, mild eosinophilia (in 30%), anaemia, ↑ESR, ↑CRP, ANCA −ve. Renal or mesenteric angiography (see fig 12.16), or renal biopsy can be diagnostic. *Treatment:* Control BP and refer. Steroids for mild cases and steroid-sparing agents if more severe. Hepatitis B should be treated (p278) after initial treatment with steroids.[52]

**Microscopic polyangiitis** A necrotizing vasculitis affecting small- and medium-sized vessels. *Symptoms:* Rapidly progressive glomerulonephritis usually features; pulmonary haemorrhage occurs in up to 30%; other features are rare. *Tests:* pANCA (MPO) +ve (p553). *Treatment:* Steroids plus eg methotrexate. For maintenance: methotrexate, rituximab, or azathioprine.

**Hypocomplementaemic urticarial vasculitis** A lupus-like illness with urticaria and antibodies to complement (C1q).

**6** Low-dose aspirin has been shown to decrease the rate of visual loss and cerebrovascular accidents in GCA but there are also conflicting reports regarding its efficacy at preventing ischaemic events in GCA.

## Features of vasculitis

The presentation of vasculitis will depend on the organs affected:

*Systemic:* Fever, malaise, weight loss, arthralgia, myalgia.
*Skin:* Purpura, ulcers, livedo reticularis (fig 12.17), nailbed infarcts, digital gangrene.
*Eyes:* Episcleritis, scleritis, visual loss.
*ENT:* Epistaxis, nasal crusting, stridor, deafness.
*Pulmonary:* Haemoptysis and dyspnoea (due to pulmonary haemorrhage).
*Cardiac:* Angina or MI (due to coronary arteritis), heart failure, and pericarditis.
*GI:* Pain or perforation (infarcted viscus), malabsorption (chronic ischaemia).
*Renal:* Hypertension, haematuria, proteinuria, casts, and renal failure (renal cortical infarcts; glomerulonephritis in ANCA +ve vasculitis).
*Neurological:* Stroke, fits, chorea, psychosis, confusion, impaired cognition, altered mood. Arteritis of the vasa nervorum (arterial supply to peripheral nerves) may cause mononeuritis multiplex or a sensorimotor polyneuropathy.[53]
*GU:* Orchitis—testicular pain or tenderness.

## Polymyalgia rheumatica (PMR)

PMR is not a true vasculitis and its pathogenesis is unknown. PMR and GCA share the same demographic characteristics and, although separate conditions, the two frequently occur together.

*Features:* Age >50yrs; subacute onset (<2 weeks) of bilateral aching, tenderness, and morning stiffness in shoulders, hips, and proximal limb muscles ± fatigue, fever, ↓weight, anorexia, and depression. There may be associated mild polyarthritis, tenosynovitis, and carpal tunnel syndrome (10%). Weakness is not a feature.

*Investigations:* ↑CRP, ESR typically >40 (but may be normal); AlkP is ↑ in 30%. Note creatinine kinase levels are normal (helping to distinguish from myositis/myopathies).

*Differential diagnoses:* Recent-onset RA, polymyositis, hypothyroidism, primary muscle disease, occult malignancy or infection, osteoarthritis (especially cervical spondylosis, shoulder OA), neck lesions, bilateral subacromial impingement (*OHCS* p666), spinal stenosis (*OHCS* p676).

*Management:* Prednisolone 15mg/d PO. Expect a dramatic response within 1 week and consider an alternative diagnosis if not. ↓ dose slowly, eg by 1mg/month (according to symptoms/ESR). Investigate apparent 'flares' during withdrawal—attributable to another condition? Most need steroids for ≥2yrs, so give bone protection. Addition of methotrexate may be considered, under specialist supervision, for patients at risk of relapse/prolonged therapy. NSAIDs are not effective. Inform patients to seek urgent review if symptoms of GCA develop.

Fig 12.16 Renal angiogram showing multiple aneurysms in PAN.

Courtesy of Dr William Herring.

Fig 12.17 Livedo reticularis: pink-blue mottling caused by capillary dilatation and stasis in skin venules. Causes: physiological, cold, or vasculitis.

# Fibromyalgia and chronic fatigue syndrome

Fibromyalgia and chronic fatigue syndrome are part of a diffuse group of overlapping syndromes, sharing similar demographic and clinical characteristics, in which chronic symptoms of fatigue and widespread pain feature prominently. Their existence as discrete entities is controversial, especially in the absence of clear pathology, and some find such dysfunctional diagnoses frustrating. However, a correct diagnosis enables the doctor to give appropriate counselling and advise appropriate therapies, and allows the patient to begin to accept and deal with their symptoms.



## Fibromyalgia

Fibromyalgia comprises up to 10% of new referrals to the rheumatology clinic.[54]

*Prevalence:* 0.5–4%. ♀:♂≈6:1 but varies depending on which diagnostic criteria are used .

*Risk factors:* BOX 'Risk factors'. Female sex, middle age, low household income, divorced, low educational status.

*Associations:* Other somatic syndromes such as chronic fatigue syndrome, irritable bowel syndrome (p266), and chronic headaches syndromes (see *OHCS* p502). Also found in ~25% of patients with RA, AS, and SLE.

*Features:* Diagnosis depends on pain that is *chronic* (>3 months) and widespread (involves left and right sides, above and below the waist, and the axial skeleton). Profound fatigue is almost universal with complaint of unrefreshing sleep and significant fatigue and pain with small increases in physical exertion. *Additional features:* morning stiffness (~80–90%), paraesthesiae (without underlying cause), headaches (migraine and tension), poor concentration, low mood, and sleep disturbance (~70%). Widespread and severe tender points.

*Investigations:* All normal. Diagnosis is clinical. Over-investigation can consolidate illness behaviour; but exclude other causes of pain and/or fatigue (eg RA, PMR see p557, vasculitis see p556, hypothyroidism see p220, myeloma see p368).

*Management:* Multidisciplinary with optimal results coming from full engagement of the patient who should be encouraged to remain as active as they feel able, and ideally to continue to participate in the workforce. New symptoms should be fully reviewed to exclude an alternative diagnosis. Patients should be advised that there is no one specific treatment that is guaranteed to work, but any of the following may help: graded exercise programmes, including both aerobic and strength-based training. Pacing of activity is vital to avoid over-exertion and consequent pain and fatigue. Long-term *graded exercise programmes* improve functional capacity.[55,56] Relaxation, rehabilitation and physiotherapy may also help. *Cognitive-behavioural therapy* (CBT) aims to help patients develop coping strategies and set achievable goals.[57]

*Pharmacotherapy:* Low-dose amitriptyline (eg 10–20mg at night) has been shown to help relieve pain and improve sleep. Pregabalin (150–300mg/12h PO) can be used if amitriptyline is ineffective. Duloxetine or an SSRI can be used for fibromyalgia with comorbid anxiety and depression. Steroids or NSAIDs are not recommended because there is no inflammation (if it does respond, reconsider your diagnosis!).

## Chronic fatigue syndrome (AKA myalgic encephalomyelitis)

Chronic fatigue syndrome[58] is defined as persistent disabling fatigue lasting >6 months, affecting mental and physical function, present >50% of the time, plus ≥4 of: myalgia (~80%), polyarthralgia, ↓memory, unrefreshing sleep, fatigue after exertion >24h, persistent sore throat, tender cervical/axillary lymph nodes. Management principles are similar to fibromyalgia and include graded exercise and CBT. No pharmacological agents have yet been proved effective for chronic fatigue syndrome (see also *OHCS* p502).

## Risk factors: yellow flags

Psychosocial risk factors for developing persisting chronic pain and long-term disability have been termed 'yellow flags':[59]

- ►Belief that pain and activity are harmful.
- ►Sickness behaviours such as extended rest.
- ►Social withdrawal.
- ►Emotional problems such as low mood, anxiety, or stress.
- ►Problems or dissatisfaction at work.
- ►Problems with claims for compensation or time off work.
- ►Overprotective family or lack of support.
- ►Inappropriate expectations of treatment, eg low active participation in treatment.

## An existential approach to difficult symptoms

The manner in which management is discussed is almost as important as the management itself, which should focus on education of the patient *and their family* and on developing coping strategies. Such a diagnosis may be a relief or a disappointment to the patient. *Explain* that fibromyalgia is a relapsing and remitting condition, with no easy cures, and that they will continue to have good and bad days. *Reassure* them that there is no serious underlying pathology, that their joints are not being damaged, and that no further tests are necessary, but be sympathetic to the fact that they may have been seeking a physical cause for their symptoms.

We all at some stage come across a patient with difficult symptoms and an exasperating lack of pathology to explain them. Investigations are all normal, and medications do not seem to work. It is tempting to dismiss such patients as malingerers, but often this conclusion comes from the clinician approaching the problem from the wrong angle. The patient has symptoms that are real and disabling to them, and that will not improve without help. Perhaps a more pragmatic approach is to take advice from the Danish philosopher Kierkegaard who wrote to a friend in 1835, '*What I really lack is to be clear in my mind what I am to do, not what I am to know ... The thing is to understand myself ... to find a truth which is true for me.*' Listen to the patient and accept their story. Then help them to focus on what they can *do* to improve their situation, and to move away from dwelling on finding a physical answer to their symptoms.

## Pathogenesis of fibromyalgia

The current hypothesis is that fibromyalgia is caused by aberrant peripheral and central pain processing. Two key features of the condition are *allodynia* (pain in response to a non-painful stimulus) and *hyperaesthesia* (exaggerated perception of pain in response to a mildly painful stimulus), examined for by palpation of tender points. Research is beginning to suggest that certain antidepressants can relieve pain and other symptoms, and especially those that have both serotonergic and noradrenergic activity (tricyclics and venlafaxine). Those acting on serotonergic receptors only are less effective. There is also some evidence to support the use of alternative therapies such as acupuncture and spa therapies, which have been postulated to act through similar spinal pain-modulatory pathways.[60] Thus far, trials have involved relatively small numbers of patients or short time periods, and lack the power to draw strong conclusions. However, it is interesting to note that the CSF of patients with fibromyalgia appears to have increased levels of substance P, while levels of noradrenaline and serotonin metabolites are decreased. All three are neurotransmitters involved in descending pain-modulatory pathways in the spinal cord.[61,62] Evidence from PET imaging suggests that patients with fibromyalgia may have an abnormal central dopamine response to pain.[63] The critical question is: is this cause or effect?

# Systemic conditions causing eye signs



*The eye is host to many diseases: the more you look, the more you'll see, and the more you'll enjoy, not least because the eye is as beautiful as its signs are legion.*

**Behçet's** (p694.) Systemic inflammatory disorder, HLA B27 association. Causes a uveitis amongst other systemic manifestations. Cause unknown.

**Granulomatous disorders** Syphilis, TB, sarcoidosis, leprosy, brucellosis, and toxoplasmosis may inflame either the front chamber (anterior uveitis/iritis) or back chamber (posterior uveitis/choroiditis). Refer to an ophthalmologist.

**Systemic inflammatory diseases** May manifest as *iritis* in ankylosing spondylitis and reactive arthritis; *uveitis* in Behçet's; *conjunctivitis* in reactive arthritis; *scleritis* or *episcleritis* in RA, vasculitis, and SLE. Scleritis in RA and granulomatosis with polyangiitis (Wegener's) may damage the eye. ►►Refer urgently if eye pain. GCA causes optic nerve ischaemia presenting as sudden blindness.

**Keratoconjunctivitis sicca** A reduction in tear formation, tested by the Schirmer filter paper test (<5mm in 5min). It causes a gritty feeling in the eyes, and a dry mouth (xerostomia from ↓saliva production). It is found on its own (Sjögren's syndrome), or with other diseases, eg SLE, RA, sarcoidosis. ℞: Artificial tears/saliva.

**Hypertensive retinopathy** ↑BP damages retinal vessels. Hardened arteries are shiny ('silver wiring'; fig 12.18) and 'nip' veins where they cross (AV nipping; fig 12.19). Narrowed arterioles may become blocked, causing localized retinal infarction, seen as cotton-wool spots. Leaks from these in severe hypertension manifest as hard exudates or macular oedema. Papilloedema (fig 12.20) or flame haemorrhages suggest accelerated hypertension (p138) requiring urgent treatment.

**Vascular occlusion** *Emboli* passing through the retinal vasculature may cause *retinal artery occlusion* (global or segmental retinal pallor) or *amaurosis fugax* (p476). Roth spots (small retinal infarcts occur in infective endocarditis. In dermatomyositis, there is orbital oedema with retinopathy showing cotton-wool spots (micro-infarcts). *Retinal vein occlusion* is caused by ↑BP, age, or hyperviscosity (p372). Suspect in any acute fall in acuity. If it is the central vein, the fundus is like a stormy sunset (those angry red clouds are haemorrhages). In branch vein occlusion, changes are confined to a wedge of retina. Get expert help.

**Haematological disorders** *Retinal haemorrhages* occur in leukaemia; comma-shaped *conjunctival haemorrhages* and retinal new vessel formation may occur in sickle-cell disease. *Optic atrophy* is seen in pernicious anaemia (and also MS).

**Metabolic disease** Diabetes mellitus: p210. Hyperthyroid exophthalmos: p219. Lens opacities are seen in hypoparathyroidism. Conjunctival and corneal calcification can occur in hypercalcaemia. In gout, conjunctival urate deposits may cause sore eyes.

**Systemic infections** Septicaemia may seed to the vitreous causing endophthalmitis. Syphilis can cause iritis (+ pigmented retinopathy if congenital). Systemic fungal infections may affect the eye, eg in the immunocompromised or in IV drug users, requiring intra-vitreal antibiotics. **AIDS and HIV** CMV retinitis (pizza-pie fundus—a mixture of cotton-wool spots, infiltrates, and haemorrhages, p438) may be asymptomatic but can cause sudden visual loss. If present, it implies AIDS (CD4 count <100 × $10^6$/L; p398). Cotton-wool spots on their own indicate HIV retinopathy and may occur in early disease. Kaposi's sarcoma may affect the lids (non-tender purple nodule) or conjunctiva (red fleshy mass).

Fig 12.18 Silver wiring.

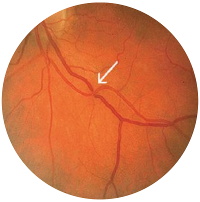
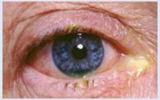



Fig 12.19 AV nipping.


Fig 12.20 Papilloedema.

Table 12.5 Differential diagnosis of a red eye

| | Conjunctiva | Iris | Pupil | Cornea | Anterior chamber | Intraocular pressure | Treatment | Appearance |
|---|---|---|---|---|---|---|---|---|
| Acute glaucoma | Both ciliary and conjunctival vessels injected. Entire eye is red. See *OHCS* p430. | Injected | Dilated, fixed, oval | Steamy, hazy | Very shallow | Very high | Refer. IV acetazolamide + pilocarpine drops (miotic); peripheral iridotomy. |  |
| Anterior uveitis (iritis) | Redness most marked around cornea, which doesn't blanch on pressure. Usually unilateral. *Causes:* AS, RA, Reiter's sarcoidosis, herpes simplex, herpes zoster, and Behçet's disease. NB: a similar scleral appearance but without papillary or anterior chamber involvement may be *scleritis* (eg RA, SLE, vasculitis). | Injected | Small, irregular due to adhesions between the anterior lens and the pupil margin | Normal | Turgid | Normal | Refer. Steroid eye drops (eg 0.5% prednisolone) + mydriatic (eg cyclopentolate 0.5%). |  |
| Conjunctivitis | Often bilateral. Conjunctival vessels injected, greatest toward fornices, but blanching on pressure. Mobile over sclera. Purulent discharge. | Normal | Normal | Normal | Normal | Normal | Most do not require treatment. Consider chloramphenicol ointment or drops. |  |
| Subconjunctival haemorrhage | Bright red sclera with white rim around limbus. *Causes:* ↑BP; leptospirosis; bleeding disorders; trauma; snake venom; haemorrhagic fevers. | Normal | Normal | Normal | Normal | Normal | Looks alarming but resolves spontaneously. Check BP if elderly; refer if traumatic; on warfarin? |  |

Images courtesy of Prof. Jonathan Trobe.

## Skin manifestations of systemic diseases



**Erythema nodosum** (fig 12.21) Painful, blue-red, raised lesions on shins (± thighs/arms). *Causes:* sarcoidosis, drugs (sulfonamides, contraceptive pill, dapsone), streptococcal infection. *Less common:* IBD, BCG vaccination, leptospirosis, *Mycobacterium* (TB, leprosy), *Yersinia*, or various viruses and fungi. Cause unknown in 30–50%.

**Erythema multiforme** (See *OHCS* p588.) (fig 12.22) 'Target' lesions: symmetrical ± central blister, on palms/soles, limbs, and elsewhere. *Stevens–Johnson syndrome* (p710): a rare, severe variant with fever and mucosal involvement (mouth, genital, and eye ulcers), associated with a hypersensitivity reaction to drugs (NSAIDs, sulfonamides, anticonvulsants, allopurinol), or infections (herpes, *Mycoplasma*, orf). Also seen in collagen disorders. 50% of cases are idiopathic. Get expert help in severe disease.

**Erythema migrans** (fig 12.23) Presents as a small papule at the site of a tick bite which develops into a spreading large erythematous ring, with central fading. It lasts from 48h to 3 months and there may be multiple lesions in disseminated disease. *Cause:* the rash is pathognomonic of Lyme disease and occurs in ~80% of cases (p422).

**Erythema marginatum** Pink coalescent rings on trunk which come and go. It is seen in rheumatic fever (or rarely other causes, eg drugs). See fig 3.42, p143.

**Pyoderma gangrenosum** (fig 12.24) Recurring nodulo-pustular ulcers, ~10cm wide, with tender red/blue overhanging necrotic edge, purulent surface, and healing with cribriform scars on leg, abdomen, or face. *Associations:* IBD, autoimmune hepatitis, granulomatosis with polyangiitis (Wegener's), myeloma, neoplasia. ♀>♂. *Treatment:* get help. Oral steroids ± ciclosporin should be 1st-line therapy.[64]

**Vitiligo** (fig 12.25) *Vitellus* is Latin for *spotted calf:* typically white patches ± hyperpigmented borders. Sunlight makes them itch. *Associations:* autoimmune disorders; premature ovarian failure. Treat by camouflage cosmetics and sunscreens (± steroid creams ± dermabrasion). UK Vitiligo Society: 0800 018 2631.

## Specific diseases and their skin manifestations

**Crohn's** Perianal/vulval/oral ulcers; erythema nodosum; pyoderma gangrenosum.

**Dermatomyositis** Gottron's papules (rough red papules on the knuckles/extensor surfaces); shawl sign; heliotrope rash on eyelids (fig 12.26). It may be associated with lung, bowel, ovarian, or pancreatic malignancy (p552).

**Diabetes mellitus** Ulcers, *necrobiosis lipoidica* (shiny yellowish area on shin ± telangiectasia; fig 12.27), *granuloma annulare* (*OHCS* p586), *acanthosis nigricans* (pigmented, rough thickening of axillary, neck, or groin skin with warty lesions; fig 12.28).

**Coeliac disease** ***Dermatitis herpetiformis:*** Itchy blisters, in groups on knees, elbows, and scalp. The itch (which can drive patients to suicide) responds to *dapsone* 25–200mg/24h PO within 48h—and this may be used as a diagnostic test. The maintenance dose may be as little as 50mg/wk. A gluten-free diet should be adhered to, but in 30% dapsone will need to be continued. SE (dose-related): haemolysis (CI: anaemia, G6PD-deficiency), hepatitis, agranulocytosis (monitor FBC and LFTs). There is an ↑ risk of small bowel lymphoma with coeliac disease *and* dermatitis herpetiformis—so surveillance is needed.

**Hyperthyroidism** ***Pretibial myxoedema:*** Red oedematous swellings above lateral malleoli, progressing to thickened oedema of legs and feet, *thyroid acropachy*—clubbing + subperiosteal new bone in phalanges. ***Other endocrinopathies:*** p203.

**Liver disease** Palmar erythema; spider naevi; gynaecomastia; decrease in pubic hair; jaundice; bruising; scratch marks.

**Malabsorption** Dry pigmented skin, easy bruising, hair loss, leuconychia.

**Neoplasia** ***Acanthosis nigricans:*** (See 'Diabetes mellitus' and fig 12.28.) Associated with gastric cancer. ***Dermatomyositis:*** see earlier in topic. ***Thrombophlebitis migrans:*** Successive crops of tender nodules affecting blood vessels throughout the body, associated with pancreatic cancer (especially body and tail). ***Acquired ichthyosis:*** Dry scaly skin associated with lymphoma. ***Skin metastases:*** Especially melanoma, and colonic, lung, breast, laryngeal/oral, or ovarian malignancy.

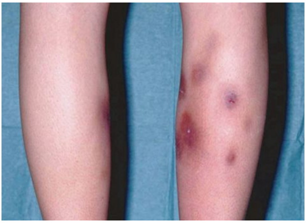


Fig 12.21 Erythema nodosum.

Fig 12.22 Erythema multiforme.

Fig 12.23 Erythema migrans.

Courtesy of CDC/James Gathany.

Fig 12.24 Pyoderma gangrenosum.

Reproduced from *BMJ*, 'Diagnosis and treatment of pyoderma gangrenosum', Brooklyn *et al.*, 333: 181-4, 2006; with permission from BMJ Publishing Group Ltd.

Fig 12.25 Vitiligo. Compare with fig 9.58, p441.

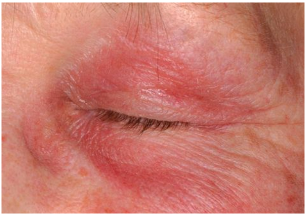


Fig 12.26 Heliotrope rash.

Courtesy of Nick Taylor, East Sussex Hospitals Trust.

Fig 12.27 Necrobiosis lipoidica.

Reproduced from Warrell *et al.*, *Oxford Textbook of Medicine*, 2010, with permission from Oxford University Press.

Fig 12.28 Acanthosis nigricans.

Reproduced from Lewis-Jones, *Paediatric Dermatology*, 2010, with permission from Oxford University Press.

# 13 Surgery



Contents



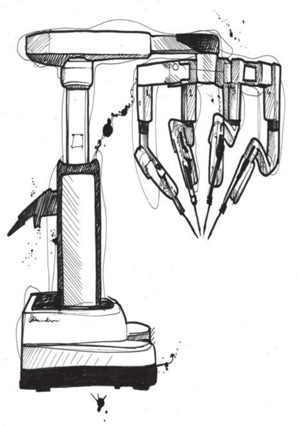


Fig 13.1 The Da Vinci robot, the first robotic surgery system to receive regulatory approval. With 4 arms, tiny, wristed tremor-free joints with multiple axes of rotation and high-resolution 3D imaging, the system offers the potential for significant advances in minimally invasive surgery, which are currently being realized across a range of fields. But where is the surgeon in this picture? At first glance, this technological *tour de force* may appear to supplant the skill of the human surgeon—yet the machine must still possess an operator who must train and achieve all of the skills necessary to perform this challenging surgery. The history of surgery is one of adaptation of surgical skills to new technologies, from anaesthesia and asepsis to organ transplantation and laparoscopy. How can training pathways adapt in turn to allow for acquisition of these new skills without unacceptable patient risk? And whatever the surgical approach, the old maxim remains the same: the art of surgery lies in selecting the right operation at the right time for the right patient.

*We thank Mr Antonio Foliaki, our Specialist Reader for this chapter, and Mr William Breakey, for their contribution to this chapter.*

## The language of surgery

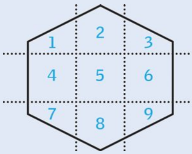



Fig 13.2 Abdominal areas.

**1** Right upper quadrant (RUQ) or hypochondrium.
**2** Epigastrium.
**3** Left upper quadrant (LUQ) or hypochondrium.
**4** Right flank (merges posteriorly with right loin, p57).
**5** Periumbilical or central area.
**6** Left flank (merges posteriorly with left loin, p57).
**7** Right iliac fossa (RIF).
**8** Suprapubic area.
**9** Left iliac fossa (LIF).

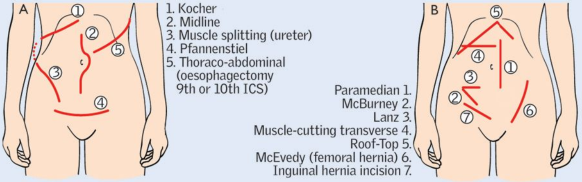



Fig 13.3 Incisions.

| | | | |
|---|---|---|---|
| **-ectomy** | Cutting something out. | | |
| **-gram** | A radiological image. | | |
| **-pexy** | Anchoring of a structure to keep it in position. | | |
| **-plasty** | Surgical refashioning in order to regain function/cosmesis. | | |
| **-scopy** | Procedure with instrumentation for looking into the body. | | |
| **-stomy** | An artificial union between a conduit and the outside or another conduit. | | |
| **-tomy** | Cutting something open to the outside world. | | |
| **-tripsy** | Fragmentation of an object. | | |
| angio- | Tube or vessel | lith- | Stone |
| appendic- | Appendix | mast- | Breast |
| chole- | Relating to gall/bile | meso- | Mesentery |
| colp- | Vagina | nephr- | Kidney |
| cyst- | Bladder | orchid- | Testicle |
| -doch- | Ducts | oophor- | Ovary |
| enter- | Small bowel | phren- | Diaphragm |
| eschar- | Dead tissue, eg from burn | pyloro- | Pyloric sphincter |
| gastr- | Stomach | pyel- | Renal pelvis |
| hepat- | Liver | proct- | Anal canal |
| hyster- | Uterus | salping- | Fallopian tube |
| lapar- | Abdomen | splen- | Spleen |

**abscess** A cavity containing pus. Remember: *if there is pus about, let it out*.

**colic** Intermittent pain from over-contraction/obstruction of a hollow viscus.

**cyst** Fluid-filled cavity lined by epi/endothelium.

**fistula** An abnormal connection between two epithelial surfaces. Fistulae often close spontaneously, but will not in the presence of malignant tissue, distal obstruction, foreign bodies, chronic inflammation, and the formation of a muco-cutaneous junction (eg stoma).

**hernia** The protrusion of a viscus/part of a viscus through a defect of the wall of its containing cavity into an abnormal position.

**ileus** Used in this book as a term for adynamic bowel.

**sinus** A blind-ending tract, typically lined by epithelial or granulation tissue, which opens to an epithelial surface.

**stent** An artificial tube placed in a biological tube to keep it open.

**stoma** (p582) An artificial union between conduits or a conduit and the outside.

**ulcer** (p660) Interruption in the continuity of an epi/endothelial surface.

**volvulus** (p611) Twisting of a structure around itself. Common GI sites include the sigmoid colon and caecum, and more rarely the stomach.

| | | | | | |
|---|---|---|---|---|---|
| epi- | Upon | pan- | Whole | peri- | Around |
| end- | Inside | para- | Alongside | sub- | Beneath |
| mega- | Enlarged | per- | Going through | trans- | Across |

# Pre-operative care

**Aims** To provide diagnostic and prognostic information. Ensures the patient understands the nature, aims, and expected outcome of surgery. To allay anxiety and pain:
- Ensure that the right patient gets the right surgery. Have the symptoms and signs changed? If so, inform the surgeon.
- Assess/balance risks of anaesthesia, and maximize fitness. Comorbidities? Drugs? Smoker? Optimizing oxygenation *before* major surgery improves outcome.
- Obtain informed consent (p568).
- Check proposed anaesthesia/analgesia with anaesthetist.

**Family history** May be relevant, eg in malignant hyperpyrexia (p572); dystrophia myotonica (p510); porphyria; cholinesterase problems; sickle-cell disease.

**Drugs** Any drug/plaster/antiseptic allergies? ►Inform the anaesthetist about *all* drugs even if 'over-the-counter'. ►Steroids: see p590; diabetes: see p588.
- *Antibiotics:* Tetracycline and neomycin may ↑ neuromuscular blockade.
- *Anticoagulants:* ►*Tell the surgeon*. Avoid epidural, spinal, and regional blocks. *Aspirin* should probably be continued unless there is a major risk of bleeding. Discuss stopping *clopidogrel* therapy with the cardiologists/neurologists. See p590.
- *Anticonvulsants:* Give as usual pre-op. Post-op, give drugs IV (or by NGT) until able to take orally. Valproate: give usual dose IV. Phenytoin: give IV slowly (<50mg/min, on cardiac monitor). IM phenytoin absorption is unreliable.
- *β-blockers:* Continue up to and including the day of surgery as this precludes a labile cardiovascular response.
- *Contraceptive pill:* See *BNF*. Stop 4wks before major/leg surgery; ensure alternative contraception is used. Restart 2wks after surgery, provided patient is mobile.
- *Digoxin:* Continue up to and including morning of surgery. Check for toxicity (ECG; plasma level); do plasma $K^+$ and $Ca^{2+}$ (suxamethonium can ↑$K^+$ and lead to ventricular arrhythmias in the fully digitalized).
- *Diuretics:* Beware hypokalaemia, dehydration. Do U&E (and bicarbonate).
- *Eye-drops:* β-blockers get systemically absorbed.
- *HRT:* As with contraceptive pill there may be an increased risk of DVT/PE.
- *Levodopa:* Possible arrhythmias when patient under GA.
- *Lithium:* Get expert help; may potentiate neuromuscular blockade and cause arrhythmias. See *OHCS* p349.
- *MAOIs:* Get expert help as interactions may cause hypotensive/hypertensive crises.
- *Thyroid medication:* see p600.
- *Tricyclics:* These enhance adrenaline (epinephrine) and arrhythmias.

**Preparation** ►Starve patient; NBM ≥2h pre-op for clear fluids and ≥6h for solids.[1]
- Is any bowel or skin preparation needed, or prophylactic antibiotics (p570)?
- Start DVT prophylaxis as indicated, eg graduated compression stockings (CI in peripheral arterial disease); LMWH (p350): eg moderate risk, 20mg ~2h pre-op then 20mg/24h; high risk (eg orthopaedic surgery), 40mg 12h pre-op then 40mg/24h; or heparin 5000U SC 2h pre-op, then every 8–12h SC for 7d or until ambulant.
- Ensure necessary premedications (p572), regular medications, analgesia, anti-emetics, antibiotics are all prescribed as appropriate. Confirm NBM.
- Book any pre-, intra-, or post-operative x-rays or frozen sections.
- Book post-operative physiotherapy.
- If needed, site IV cannula, catheterize (p762), and/or insert a Ryle's tube (p759).
- Meta-analyses have shown no benefit to patients from mechanical bowel cleansing before colonic surgery—this is no longer considered good practice. There is also no evidence for the use of enemas prior to rectal surgery.

## Pre-operative history, examination, and tests

It is the anaesthetist's responsibility to assess suitability for anaesthesia. The ward doctor assists with a good history and examination, should anticipate necessary tests, and can also reassure and inform the patient. The surgical team should consent the patient.

►The World Health Organization 'Surgical Safety Checklist' should be completed for every patient undergoing a surgical procedure, ensuring pre-operative preparation, intra-operative monitoring, and post-operative review.

**History** Assess past history of: MI[1], diabetes, asthma, hypertension, rheumatic fever, epilepsy, jaundice. Existing illnesses, drugs, and allergies? Be alert to chronic lung disease, ↑BP, arrhythmias, and murmurs. Assess any specific risks, eg is the patient pregnant? Is the neck/jaw immobile and teeth stable (intubation risk)? Has there been previous anaesthesia? Were there any complications (eg nausea, DVT)?

**Examination** Assess cardiorespiratory system, exercise tolerance. Is the neck unstable (eg arthritis complicating intubation)? ►Is DVT/PE prophylaxis needed (p578)? ►For 'unilateral' surgery, mark the correct arm/leg/kidney (surgeon).
**Tests** Be guided by the history and examination and local/NICE protocols.

- *U&E, FBC, and finger-prick blood glucose in most patients.* If Hb <100g/L tell anaesthetist. Investigate/treat as appropriate. U&E are particularly important if the patient is starved, diabetic, on diuretics, a burns patient, has hepatic or renal disease, has an ileus, or is parenterally fed.
- *Crossmatch:* Blood type is identified and units are allocated to the patient. *Group and save (G&S):* Blood type is identified and held, pending crossmatch (if required). Contact your lab to discuss requirements—this decreases wastage and allows increased efficiency of blood stocks.
- *Specific blood tests:* LFT in jaundice, malignancy, or alcohol abuse. Amylase in acute abdominal pain. Blood glucose if diabetic (p588). Drug levels as appropriate (eg digoxin, lithium). Clotting studies in liver or renal disease, DIC (p352), massive blood loss, or if on valproate, warfarin, or heparin. HIV, HBsAg in high-risk patients, after counselling. Sickle test in those from Africa, West Indies, or Mediterranean—and if origins are in malarial areas (including most of India). Thyroid function tests in those with thyroid disease.
- *CXR* if known cardiorespiratory disease, pathology, or symptoms or >65yrs old. Remember to check the film prior to surgery.
- *ECG* if >55yrs old or poor exercise tolerance, or history of myocardial ischaemia, hypertension, rheumatic fever, or other heart disease.
- *Echocardiogram* may be performed if there is a suspicion of poor LV function.
- *Pulmonary function tests* in known pulmonary disease/obesity.
- *Lateral cervical spine X-ray* (flexion and extension views) if history of rheumatoid arthritis/ankylosing spondylitis/Down's syndrome, to warn of difficult intubations.
- *MRSA screen:* Screen and decolonize nasal carriers according to local policy (eg nasal mupirocin ointment). Colonization is *not* a contraindication to surgery. Place patients last on the list to minimize transmission to others and cover with appropriate antibiotic prophylaxis, eg vancomycin or teicoplanin. Consider and document major blood-borne virus risk (HIV/HBV/HCV) according to local policies.

## American Society of Anesthesiologists (ASA) classification

| | |
|---|---|
| Class I | Normally healthy patient. |
| Class II | Mild systemic disease. |
| Class III | Severe systemic disease that limits activity but is not incapacitating. |
| Class IV | Incapacitating systemic disease which poses a constant threat to life. |
| Class V | Moribund: not expected to survive 24h even with operation. |

You will see a space for an ASA number on most anaesthetic charts. It is a health index at the time of surgery. The suffix E is used in emergencies, eg ASA 2E.

1 If within the last 6 months, the perioperative risk of re-infarction (up to 40%) makes most elective surgery too risky. ECHO, and stress testing (+ exercise ECG or MUGA scan, p741) should be done.

# Consent

*In which of the following situations would you seek 'informed written consent' from a patient?* **1** Feeling for a pulse. **2** Taking blood. **3** Inserting a central line. **4** Removing a section of small bowel during a laparotomy for division of adhesions. **5** Orchidectomy after a failed operation for testicular torsion.

English law states that *any* intervention or treatment needs consent—ie all of the above—yet, for different reasons. In fact, *written* consent itself is not required by law, but it does constitute 'good medical practice' in the best interests of the patient and practitioner. Sometimes actions and words can imply valid consent, eg by simply entering into conversation or holding out an arm. If the consequences are not clear and the patient has *capacity* to give consent, you should seek informed written consent as a record of your conversation.



## For consent to be valid

- It can be given any time before the intervention/treatment is initiated. Earlier is better as this will give the patient time to think about the risks, benefits, and alternatives—he or she may even bring forward questions on issues that you had not considered relevant. Think of consent as an ongoing process throughout the patient's time with you, not just the moment of signing the form.
- The proposed treatment or test must be clearly understood by the patient, taking into account the benefits, risks (including complication rates if known), additional procedures, alternative courses of action, and their consequences.
- It must be given *voluntarily*. This can be difficult to evaluate—eg when live organ donation is being considered—see BOX 'Special circumstances for consent' for other difficult situations.
- The doctor providing treatment or undertaking the test needs to ensure that the patient has given valid consent. The act of seeking consent is ultimately the responsibility of the doctor looking after the patient, though the task may be delegated to another health professional, as long as they are suitably trained and qualified.

**Capacity** relates to the ability to **1** understand, **2** retain, and **3** weigh up relevant information and **4** communicate the decision. Capacity is not a fixed state, but is specific to any given time and decision—a person may lack capacity to be involved in a particular complex decision but retain capacity to decide other aspects of their care, or recover capacity as they recover from acute illness. Therefore, do not label anyone as unable to make a decision unless you have taken all practicable steps to help them to do so without success—non-urgent decisions should always be deferred if there is a chance to recover capacity. When acting on behalf of a person who lacks capacity, do so in their best interests and involve family members or an appointed surrogate where clinical urgency allows.

## When taking consent

- Does the patient currently have capacity for the decision in question?
- Are you the right person to be obtaining consent?[2]
- Use words the patient understands and avoid jargon and abbreviations.
- Ensure that they believe your facts and can retain 'pros' and 'cons' long enough to inform their decision. Fact sheets/diagrams for individual operations help.
- Make sure their choice is free from pressure from others, and explain that they can withdraw consent at any time after the form is signed. Some patients may view the consent form as a contract from which they cannot *renege*.
- If the patient is illiterate, a witnessed mark does endorse valid consent. Similarly, if the patient is willing but physically unable to sign the consent form, then a witnessed entry into the medical notes stating so is valid.
- Remember to discuss further procedures that may become necessary during the proposed treatment. This avoids waking up to a nasty surprise (eg a missing testicle as in scenario 5 earlier in this topic).

## Special circumstances for consent

Consent is complex, but remember that it exists for the benefit of the patient *and* the doctor, giving you an opportunity to revisit expectations and involve the patient in their own care. There are some areas of treatment or investigation for which it may be advisable to seek specialist advice if it is not part of your regular practice[2]:

- Photography of a patient.
- Innovative or novel treatment.
- Living organ donation.
- Storage, use, or removal of human tissue (for any length of time).
- The storage, loss, or use of gametes.
- The use of patient records or tissue in research or teaching.
- In the presence of an advanced directive or living will, expressly refusing a particular treatment, investigation, or action.
- Consent if <16yrs (consent form 3 in NHS). In the UK, those >16yrs can give valid consent. Those <16yrs can give consent for a medical decision provided they understand what it involves—the concept of *Gillick* competence. It is still good practice to involve the parents in the decision, if the child is willing. If *<18yrs and refusing life-saving surgery*, talk to the parents and your senior; the law is unclear. You may need to contact the duty judge in the High Court.
- Consent in the incapacitated (NHS consent form 4). No one (parents, relatives, or even members of a healthcare team) is able to give consent on behalf of an adult in England, and the High Court may be required to give a ruling on the matters of lawfulness of a proposed procedure. Proceeding in a patient's best interest is decided by the clinician overseeing their care, with involvement of family members or a nominated advocate.

## The right to refuse treatment

*Theirs not to make reply,*
*Theirs not to reason why,*
*Theirs but to do and die.* Alfred, Lord Tennyson from *The Charge of the Light Brigade*, 1854.

The rights of a patient are something of an antithesis to this military macabre of Tennyson, and it is our responsibility to respect the legal and ethical rights of those we treat. We do this not only for the sake of the individual, but also for the sake of an enduring trust between the patient and doctor, remembering that it is the patient's right to refuse treatment (if a fully competent adult) even when this may result in the death of the patient, or even the death of an unborn child, whatever the stage of pregnancy. The only exceptions apply to treatment of mental health disorders (see eg in England and Wales the Mental Health Act 2007).

**2** If in any doubt, turn to: your senior/consultant; your employing organization; your legal defence organization; your national medical association; your local research ethics committee.

# Prophylactic antibiotics in surgery

Prophylactic antibiotics are given to counter the risk of wound infection (see table 13.1), which occurs in ~20% of elective GI surgery (up to 60% in emergency surgery). Antibiotics are also given if infection elsewhere, although unlikely, would have severe consequences (eg when prostheses are involved). A single dose given before surgery has been shown to be just as good as more prolonged regimens in biliary and colorectal surgery. Additional doses may be given if high-risk/prolonged procedures, or if major blood loss. ►Wound infections are not necessarily trivial since sepsis may lead to haemorrhage, wound dehiscence, and initiate a fatal chain of events, so take measures to minimize the risk of wound infection:

- Time administration correctly (eg IV prophylaxis should be given 30min prior to surgery to maximize skin concentration; metronidazole PR is given 2h before).
- Use antibiotics which will kill anaerobes and coliforms.
- Consider use of peri-operative supplemental oxygen. This is a practical method of reducing the incidence of surgical wound infections.
- Practise strictly sterile surgical technique. (Ask for a hand with scrubbing up if you are not sure—theatre staff will be more than pleased to help!)

**Antibiotic regimens** ►Adhere to local guidelines. BNF examples include:

- *Appendicectomy; colorectal resections and open biliary surgery:* A single dose of IV piperacillin/tazobactam 4.5g/8h *or* gentamicin 1.5mg/kg + metronidazole 500mg *or* co-amoxiclav 1.2g alone.
- *Oesophageal or gastric surgery:* 1 dose of IV gentamicin *or* piperacillin/tazobactam *or* co-amoxiclav (doses as for appendicectomy).
- *Vascular surgery:* 1 dose of IV piperacillin/tazobactam *or* flucloxacillin 1–2g + gentamicin. Add metronidazole if risk of anaerobes (eg amputations, gangrene, or diabetes).
- *MRSA:* For high-risk patients add teicoplanin or vancomycin to the above-listed regimens.

Table 13.1 Classification of surgical procedures and wound infection risk

| Category | Description | Infection risk |
|---|---|---|
| Clean | Incising uninfected skin without opening a viscus | <2% |
| Clean-contaminated | Intra-operative breach of a viscus (but not colon) | 8–10% |
| Contaminated | Breach of a viscus + spillage or opening of colon | 12–20% |
| Dirty | The site is already contaminated with pus or faeces, or from exogenous contagion, eg trauma | 25% |

Data from *MRCS Core Modules: Essential Revision Notes*, S. Andrews, Pastest.

# Sutures

Sutures are central to the art of surgery. No single suture fits the bill for every occasion, and so suture selection (including size) depends on the job in hand (see tables 13.2, 13.3). In their broadest sense they are absorbable or non-absorbable, synthetic or natural, and their structure may be divided into monofilament, twisted, or braided. Monofilament sutures are quite slippery but minimize infection and produce less reaction. Braided sutures have plaited strands and provide secure knots, but they may allow infection to occur in surrounding tissue between their strands. Twisted sutures have 2 twisted strands and similar qualities to braided sutures. Sizes are denoted according to a scale running down from #5 (heavy braided suture). Most common modern sutures are smaller than size #0, hence rising numbers with a -0 suffix correspond to progressively finer grades of suture up to 11-0 (fine ophthalmic monofilaments). 3-0 or 4-0 are the best sizes for skin closure. Timing of suture removal depends on site and the general health of the patient. Face and neck sutures may be removed after 5d (earlier in children), scalp and back of neck after 5d, abdominal incisions and proximal limbs (including clips) after ~10d, distal extremities after 14d. In patients with poor wound healing, eg steroids, malignancy, infection, cachexia (p35), the elderly, or smokers, sutures may need ~14d.

## Some commonly encountered suture materials

### Absorbable

Table 13.2 Absorbable suture materials

| Name | Material | Construction | Use |
|---|---|---|---|
| Monocryl® | Poliglecaprone | Monofilament | Subcuticular skin closure |
| PDS® | Polydioxanone | Monofilament | Closing abdominal wall |
| Vicryl® | Polyglactin | Braided multifilament | Tying pedicles; bowel anastomosis; subcutaneous closure |
| Dexon® | Polyglycolic acid | Braided multifilament | Very similar to Vicryl® |

### Non-absorbable

Table 13.3 Non-absorbable suture materials

| Name | Material | Construction | Use |
|---|---|---|---|
| Ethilon® | Polyamide | Monofilament | Closing skin wounds |
| Prolene® | Polypropylene | Monofilament | Arterial anastomosis |
| Mersilk®N | Silk | Braided multifilament | Securing drains |
| Metal | Eg steel | Clips or monofilament | Skin wound/sternotomy closure |

N = natural; other natural materials (eg cotton and catgut) are rarely used these days.

## Surgical drains in the post-operative period

The decision when to insert and remove drains may seem to be one of the great surgical enigmas—but there are basically three types to get a grip of:

1 To drain the area of surgery and are often put under suction or −ve pressure (Redivac® uses a 'high vacuum'). These are removed when they stop draining. They protect against collection, haematoma, and seroma formation (in breast surgery this can cause overlying skin necrosis).
2 To protect sites where leakage may occur in post-operative period, eg bowel anastomoses. These form a tract and are removed after about 1wk.
3 To collect red blood cells from the site of the operation, which can then be autotransfused within 6h, protecting from the hazards of allotransfusion—it is used commonly in orthopaedics. (eg Bellovac®).

'Shortening a drain' means withdrawing it (eg by 2cm/d) to allow the tract to seal, bit by bit. ►Check the surgeon's wishes before altering a drain. ►If a drain 'falls out' on the ward, avoid re-siting it because it is now covered in skin flora. Put a collecting bag over the wound and contact surgical registrar.

# Anaesthesia

Before anaesthesia, explain to the patient what will happen and where they will wake up, otherwise the recovery room or ITU will be frightening. Explain that they may feel ill on waking. The premedication aims to allay anxiety and to make the anaesthesia itself easier to conduct (see BOX). Typical regimens might include:

- *Anxiolytics:* Benzodiazepines, eg lorazepam 2mg PO; temazepam 10–20mg PO. In children, use oral premeds as first choice, eg midazolam 0.5mg/kg (tastes bitter so often put in paracetamol suspension). Give oral premedication 2h before surgery (1h if IM route used).
- *Analgesics:* See p574. Pre-emptive analgesia is not often used and effects are hard to determine. The aim is to dampen pain signals before they arrive. In children or anxious adults, local anaesthetic cream may be used on a few sites before inserting an IVI (►the anaesthetist may prefer to site the cannula themselves!).
- *Anti-emetics:* Post-operative nausea and vomiting is experienced by ~25% of all patients. $5HT_3$ antagonists (eg ondansetron 4mg IV/IM) are the most effective agents; others, eg metoclopramide 10mg/8h IV/IM/PO, are also used—see p251.
- *Antacids:* Ranitidine 50mg IV or omeprazole 40mg PO/IV if aspiration risk.
- *Antisialogues:* Glycopyrronium (200–400mcg in adults, 4–8mcg/kg in children; given IV/IM 30–60min before induction) is sometimes used to decrease secretions that may cause respiratory obstruction in smaller airways.
- *Antibiotics:* See p570.

### Side-effects of anaesthetic agents

- *Hyoscine, atropine:* Anticholinergic, ∴ tachycardia, urinary retention, glaucoma, sedation (especially in the elderly).
- *Opioids:* Respiratory depression, ↓cough reflex, nausea and vomiting, constipation.
- *Thiopental:* (Induction agent.) Laryngospasm.
- *Propofol:* (Induction agent.) Respiratory/cardiac depression, pain on injection.
- *Volatile agents, eg isoflurane:* Nausea and vomiting, cardiac depression, respiratory depression, vasodilation, hepatotoxicity (see *BNF*).

### The complications of anaesthesia are due to loss of:

- *Pain sensation:* Urinary retention, pressure necrosis, local nerve injuries (eg radial nerve palsy from arm hanging over the table edge).
- *Consciousness:* Cannot communicate 'wrong leg/kidney'. NB: in some patients (eg 0.15%) *retained* consciousness is the problem. Awareness under GA sounds like a contradiction of terms, but remember that anaesthesia is a process rather than an event. Such awareness can lead to ill-defined, delayed neuroses and post-traumatic stress disorder (*OHCS* p353).
- *Muscle power:* Corneal abrasion (∴ tape the eyes closed), no respiration, no cough (leads to pneumonia and atelectasis—partial lung collapse causing shunting ± impaired gas exchange: it starts minutes after induction, and may be related to the use of 100% $O_2$, supine position, surgery, age, and to loss of power).

**Local/regional anaesthesia** If unfit/unwilling to undergo general anaesthesia, local nerve blocks (eg brachial plexus) or spinal blocks (contraindication: anticoagulation, local infection) using long-acting local anaesthetics such as bupivacaine may be indicated. See table 13.4 for doses and toxicity effects.

**Drugs complicating anaesthesia** ►Inform anaesthetist. See p566 for lists of specific drugs, and actions to take.

**Malignant hyperpyrexia** This is a rare complication, precipitated by any volatile agent, eg halothane, or suxamethonium. It exhibits autosomal dominant inheritance. There is a rapid rise in temperature (>1°C every 30min); masseter spasm may be an early sign. Complications include hypoxaemia, hypercarbia, hyperkalaemia, metabolic acidosis, and arrhythmias. ►Get expert help immediately. Prompt treatment with dantrolene[3] (skeletal muscle relaxant), active cooling and ITU care can reduce mortality significantly.

**3** Give 1mg/kg every 5min IV—up to 10mg/kg in total (*OHCS* p628).

## Principles and practical conduct of anaesthesia

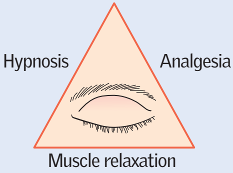



Fig 13.4 Principles of anaesthesia.

The conduct of anaesthesia (fig 13.4) typically involves:

- *Induction:* Either intravenous (eg propofol 1.5–2.5mg/kg IV at a rate of 20–40mg every 10s; thiopental is an alternative) or, if airway obstruction or difficult IV access, gaseous (eg sevoflurane or nitrous oxide, mixed in $O_2$).
- *Airway control:* Either using a facemask, an oropharyngeal (Guedel) airway or by intubation. The latter usually requires muscle relaxation with a depolarizing/non-depolarizing neuromuscular blocker (*OHCS* p622).
- *Maintenance of anaesthesia:* Either a volatile agent added to $N_2O/O_2$ mixture, or high-dose opiates with mechanical ventilation, or IV infusion anaesthesia (eg propofol 4–12mg/kg/h IVI).
- *Recovery:* Change inspired gases to 100% oxygen only, then discontinue any anaesthetic infusions and reverse muscle paralysis. Extubate once spontaneously breathing, place patient in recovery position, and give oxygen by facemask.
- For further details, see the *Anaesthesia* chapter in *OHCS* (p612).

## ⚠ Local anaesthetic toxicity and maximum doses

Anaesthetists are masters of the drug dose by weight! It is important to remember the maximum doses for local anaesthetics, not least because we use them so frequently, but because the effects of overdose can be lethal.

Handy to remember (though it can be worked out with a pen, paper, and SI units) is that a 1% concentration is equivalent to 10mg/mL. Local anaesthetics are also basic, and so do not work well in acidic environments, eg abscesses.

Table 13.4 Example of maximum doses for local anaesthetic

| % concⁿ | Lidocaine concⁿ (mg/mL) | Approx. allowable volume (mL/kg) | Approx. allowable volume for 70kg adult (mL) |
|---|---|---|---|
| 0.25% | 2.5 | 1.12 | ≤80 |
| 0.5% | 5 | 0.56 | ≤40 |
| 1% | 10 | 0.28 | ≤20 |
| 2% | 20 | 0.14 | ≤10 |

Local anaesthetic toxicity starts with peri-oral tingling and paraesthesiae, progressing to drowsiness, seizures, coma, and cardiorespiratory arrest. If suspected (the patient feels 'funny' and develops early signs) then stop administration immediately and commence ABC resuscitation as required. ►►Treatment is with lipid emulsion. Find out where this is stored in your hospital.

# The control of pain

Humans are the most exquisite devices ever made for experiencing pain: the richer our inner lives, the greater the varieties of pain there are for us to feel, and the more resources we have for dealing with pain. If we can connect with patients' inner lives we may make a real difference. *Never forget how painful pain* is, nor how fear magnifies pain. Try not to let these sensations, so often interposed between your patient and recovery, be invisible to you as he or she bravely puts up with them.

**Approach to management** (fig 13.5 and p532.) Review and chart each pain carefully and individually.

- Identify and treat the underlying pathology wherever possible.
- Give *regular* doses rather than on an 'as-required' basis.
- Choose the best route: PO, PR, IM, epidural, SC, inhalation, or IV.
- Explanation and reassurance contribute greatly to analgesia.
- Allow the patient to be in charge. This promotes wellbeing, and does not lead to overuse. Patient-controlled continuous IV morphine delivery systems are useful.
- Liaise with the Acute Pain Service, if possible.



**Non-narcotic (simple) analgesia** Paracetamol 0.5–1.0g/4h PO (up to 4g daily; 15mg/kg/4h IV over 15min in children <50kg; up to 60mg/kg/d). Caution in liver impairment. NSAIDs, eg ibuprofen 400mg/8h PO (see *BNFc* for dosing in children) are good for musculoskeletal pain and renal or biliary colic. CI: peptic ulcer, clotting disorders, anticoagulants. Cautions: asthma, renal or hepatic impairment, heart failure, IHD, pregnancy, and the elderly. Aspirin is contraindicated in children due to the risk of Reye's syndrome (*OHCS* p652).

**Opioid drugs for severe pain** Morphine (eg 10–15mg/2–4h IV/IM) or diamorphine (5–10mg/2–4h PO, SC, or slow IV, but you may need much more) are best. NB: these are 'controlled' drugs. For palliative care, see p532. Alternative delivery routes include transdermal (once baseline requirements are established) or sublingual. *Side-effects of opioids:* These include nausea (so give with an anti-emetic, p251), respiratory depression, constipation, cough suppression, urinary retention, ↓BP, and sedation (do not use in hepatic failure or head injury). Dependency is rarely a problem. *Naloxone* (eg 100–200mcg IV, followed by 100mcg increments, eg every 2min until responsive) may be needed to reverse the effects of excess opioids (p842).

**Epidural analgesia** Opioids and anaesthetics are given into the epidural space by infusion or as boluses. Ask the advice of the Pain Service. SEs are thought to be less, as the drug is more localized: watch for respiratory depression and local anaesthetic-induced autonomic blockade (↓BP).

**Adjuvant treatments** Eg radiotherapy for bone cancer pain; anticonvulsants, antidepressants, gabapentin or steroids for neuropathic pain, antispasmodics, eg hyoscine butylbromide[4] (Buscopan® 10–20mg/8h PO/IM/IV) for intestinal or renal tract colic. If brief pain relief is needed (eg for changing dressings or exploring wounds), try inhaled nitrous oxide (with 50% $O_2$—as Entonox®) with an 'on-demand' valve. Transcutaneous electrical nerve stimulation (TENS), local heat, local or regional anaesthesia, and neurosurgical procedures (eg excision of neuroma) may be tried but can prove disappointing. Treat conditions that exacerbate pain (eg constipation, depression, anxiety).

4 Not to be confused with *hyoscine hydrobromide*; used for drying secretions and in motion sickness.

WHO's Pain relief ladder

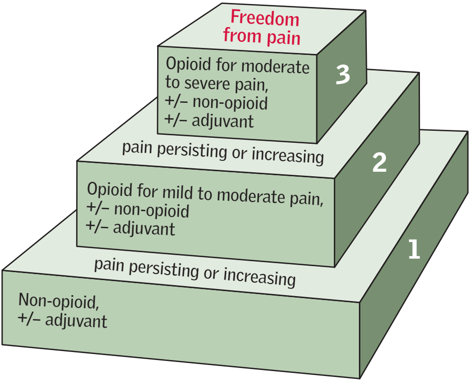



**Fig 13.5** World Health Organization pain ladder.

# General post-operative complications

**Pyrexia** Mild pyrexia in the 1st 48h is often from atelectasis (needs prompt physio, not antibiotics), tissue damage/necrosis, or even from blood transfusions, but still have a low threshold for infection screen. Consider evidence for peritonism, chest, urinary, wound, or cannula site infections, as well as possible endocarditis, meningism, or DVT (also causes ↑°T). Send blood for FBC, U&E, CRP, and cultures (±LFT). Dipstick the urine. Consider MSU, CXR, and abdominal ultrasound/CT depending on clinical findings.

**Confusion** may manifest as agitation, disorientation, and attempts to leave hospital, especially at night. Gently reassure the patient in well-lit surroundings. See p484 for a full work-up. Common causes are:

- hypoxia (pneumonia, atelectasis, LVF, PE)
- drugs (opiates, sedatives, and many others)
- urinary retention
- MI or stroke
- infection (see earlier)
- alcohol withdrawal (p280)
- liver/renal failure.

Occasionally, sedation is necessary to examine the patient; consider lorazepam 1mg PO/IM (antidote: flumazenil) or haloperidol 0.5–2mg IM. Reassure relatives that post-op confusion is common (seen in up to 40%) and reversible.

**Dyspnoea or hypoxia** Any previous lung disease? Sit patient up and give $O_2$, monitor peripheral $O_2$ sats by pulse oximetry (p162). Examine for evidence of: •pneumonia, pulmonary collapse, or aspiration •LVF (MI; fluid overload) •pulmonary embolism (p190) •pneumothorax (p190; due to CVP line, intercostal block, or mechanical ventilation). *Tests:* FBC; ABG; CXR; ECG. Manage according to findings.

**↓BP** If severe, tilt bed head-down and give $O_2$. Check pulse and BP yourself; compare it with pre-op values. Post-op ↓BP is often from hypovolaemia resulting from inadequate fluid input, so check fluid chart and replace losses. Monitor urine output (may need catheterization). A CVP line can help monitor fluid resuscitation (normal is 0–5cm$H_2O$ relative to sternal angle). Hypovolaemia may also be caused by haemorrhage so review wounds, drains, and abdomen. If unstable, return to theatre for haemostasis. Beware cardiogenic and neurogenic causes and look for evidence of MI or PE. Consider sepsis and anaphylaxis. *Management:* p790.

**↑BP** may be from pain, urinary retention, idiopathic hypertension (eg missed medication), or inotropic drugs. Oral cardiac medications (including antihypertensives) should be continued throughout the perioperative period even if NBM. Treat the cause, consider increasing the regular medication, or if not absorbing orally try 50mg labetalol IV over 1min (see p140).

**↓Urine output (oliguria)** Aim for output of >30mL/h in adults (or >0.5mL/kg/h). *Anuria* may reflect a blocked or malsited catheter (see p763) rather than AKI. Flush or replace catheter. *Oliguria* is usually due to too little replacement of lost fluid. Treat by increasing fluid input. ►Acute kidney injury may follow shock, drugs, transfusion, pancreatitis, or trauma (see p300 for classification and management of AKI).

- Review fluid chart and examine for signs of volume depletion.
- Urinary retention is also common, so examine for a palpable bladder.
- Establish normovolaemia (a CVP line may help); you may need 1L/h IVI for 2–3h. A 'fluid challenge' of 250–500mL over 30min may also help.
- Catheterize bladder (for accurate monitoring)—see p762; check U&E.
- If intrinsic renal failure is suspected, stop nephrotoxic drugs (eg NSAIDs, ACE-i) and refer to a nephrologist early.

**Nausea/vomiting** Any mechanical obstruction, ileus, or emetic drugs (opiates, digoxin, anaesthetics)? Consider AXR, NGT, and an anti-emetic (►*not* metoclopramide because of its prokinetic property). See p251 for choice of anti-emetics.

**↓$Na^+$** Pre-op level? Excess IV fluids may exacerbate the situation. Correct slowly (p672). SIADH (p673) can be precipitated by pain, nausea, opioids, and chest infection.

## Post-operative bleeding

*Primary haemorrhage:* Continuous bleeding, starting during surgery. Replace blood loss. If severe, return to theatre for adequate haemostasis. Treat shock vigorously (p790–804).

*Reactive haemorrhage:* Haemostasis appears secure until BP rises and bleeding starts. Replace blood and re-explore wound.

*Secondary haemorrhage* (caused by infection) occurs 1–2wks post-op.

## Talking about post-op complications...

When asked to give your thoughts on the complications of an operation—maybe with an examiner or a patient—a good starting point is to divide them up accordingly (and for each of the following stratify as immediate, early, and late):

- *From the anaesthetic:* (p572.) Eg respiratory depression from induction agents.
- *From surgery in general:* (p576.) Eg wound infection, haemorrhage, neurovascular damage, DVT/PE.
- *From the specific procedure:* Eg saphenous nerve damage in stripping of the long varicose vein.

Tailor the discussion towards the individual who, eg if an arteriopath, may have a significant risk of cardiac ischaemia during hypotensive episodes while under the anaesthetic. For some other post-op complications, see:

• Pain (p574) • DVT (p578) • Pulmonary embolus (p190; massive, p818) • Wound dehiscence (p580) • Complications in post-gastric surgery (p622) • Other complications of specific operations (p580).

# Deep vein thrombosis (DVT)

DVTs occur in 25–50% of surgical patients, and many non-surgical patients. All hospital inpatients should be assessed for DVT/PE risk and offered prophylaxis if appropriate. 65% of below-knee DVTs are asymptomatic; these rarely embolize to the lung.

**Risk factors** ↑Age, pregnancy, synthetic oestrogen, trauma, surgery (especially pelvic/orthopaedic), past DVT, cancer, obesity, immobility, thrombophilia (p374).

**Signs** •Calf warmth/tenderness/swelling/erythema. •Mild fever. •Pitting oedema.

**ΔΔ** Cellulitis; ruptured Baker's cyst. Both may coexist with a DVT.

**Tests** ►Calculate Wells score (see table 13.5) before ordering *D-dimer*. D-dimer is sensitive but not specific for DVT (also ↑ in infection, pregnancy, malignancy, and post-op).

**Wells score:** *≤1 point = DVT unlikely:* Perform D-dimer. If negative, DVT excluded. If positive, proceed to USS (if USS negative, DVT excluded; if positive, treat as DVT).

*≥2 points = DVT likely:* Do D-dimer and USS. If both negative, DVT excluded. If USS positive, treat as DVT. If D-dimer positive and USS negative, repeat USS in 1 week.

Do *thrombophilia tests* (p374) *before* commencing anticoagulant therapy if there are no predisposing factors, in recurrent DVT, or if DVT in unusual site. Look for *underlying malignancy:* Urine dip; FBC, LFT, $Ca^{2+}$; CXR ± CT abdomen/pelvis (and mammography in ♀) if >40yrs.

**Prevention** •Stop the oral contraceptive pill 4wks pre-op. •Mobilize early. •LMWH eg enoxaparin 20mg/24h SC, ↑ to 40mg for high-risk patients (p375) (caution if eGFR less than 30mL/min/1.73m²). •Graduated compression stockings ('thromboembolic deterrent (TED) stockings'; CI: ischaemia) and intermittent pneumatic compression devices ↓ risk of DVT by ~70% in surgical patients. •Fondaparinux (a factor Xa inhibitor) ↓ risk of DVT over LMWH in eg major orthopaedic surgery without ↑ risk of bleeding.

**Treatment** LMWH (eg enoxaparin 1.5mg/kg/24h SC) or fondaparinux. LMWH is superior to unfractionated heparin (used in renal failure or if ↑risk of bleeding; dose guided by APTT, p350). Cancer patients should receive LMWH for 6 months (then review). In others, start warfarin simultaneously with LMWH (warfarin is prothrombotic for the first 48h). Stop heparin when INR is 2–3; treat for 3 months in most (longer in some cases—see p351). *Direct oral anticoagulants* (DOACs p190), eg dabigatan, apixaban, rivoraxaban, are newer alternatives licensed for the treatment of DVT with benefits relating to simpler dosing and monitoring and ↓ bleeding risk. *Inferior vena caval filters* may be used in active bleeding, or when anticoagulants fail, to minimize risk of PE. *Post-phlebitic change* (pain, swelling, and skin changes) can be seen in 10–30%—graduated compression stockings may help.

Surgery

### Pretest clinical probability scoring for DVT: the Wells score

In patients with symptoms in both legs, the more symptomatic leg is used.

Table 13.5 Wells score

| Clinical features | Score |
|---|---|
| Active cancer (treatment within last 6 months or palliative) | 1 point |
| Paralysis, paresis, or recent plaster immobilization of leg | 1 point |
| Recently bedridden for >3d or majory surgery in last 12wks | 1 point |
| Local tenderness along distribution of deep venous system | 1 point |
| Entire leg swollen | 1 point |
| Calf swelling >3cm compared with asymptomatic leg (measured 10cm below tibial tuberosity) | 1 point |
| Pitting oedema (greater in the symptomatic leg) | 1 point |
| Collateral superficial veins (non-varicose) | 1 point |
| Previously documented DVT | 1 point |
| Alternative diagnosis at least as likely as DVT | −2 points |

## Swollen legs

**Bilateral oedema** implies systemic disease with ↑venous pressure (eg right heart failure) or ↓intravascular oncotic pressure (any cause of ↓albumin, so test the urine for protein). It is *dependent* (distributed by gravity), which is why legs are affected early, but severe oedema extends above the legs. In the bed-bound, fluid moves to the new dependent area, causing a sacral pad. The exception is the local increase in venous pressure occurring in IVC obstruction: the swelling neither extends above the legs nor redistributes. *Causes:* •Right heart failure (p134). •↓Albumin (p686, eg renal or liver failure). •Venous insufficiency: acute, eg prolonged sitting, or chronic, with haemosiderin-pigmented, itchy, eczematous skin ± ulcers. •Vasodilators, eg *nifedipine, amlodipine*. •Pelvic mass (p57, p604). •Pregnancy—if ↑BP + proteinuria, diagnose pre-eclampsia (*OHCS* p48): find an obstetrician urgently. In all the above, both legs need not be affected to the same extent.

**Unilateral oedema** Pain ± redness implies DVT or inflammation, eg cellulitis or insect bites (any blisters?). Bone or muscle may be to blame, eg tumours; necrotizing fasciitis (p660); trauma (check for sensation, pulses, and severe pain esp. on passive movement: ►a *compartment syndrome* with ischaemic necrosis needs prompt fasciotomy). Impaired mobility suggests trauma, arthritis, or a Baker's cyst (p694). *Non-pitting oedema* is oedema you cannot indent: see p35.

**Nine questions to ask**

| | | |
|---|---|---|
| **1** Is it *both* legs? | **2** Is she pregnant? | **3** Is she mobile? |
| **4** Any trauma? | **5** Any pitting (p35)? | **6** Past diseases/on drugs? |
| **7** Any pain? | **8** Any skin changes? | **9** Any oedema elsewhere? |

**Tests** ►Look for proteinuria (+hypoalbuminaemia ≈nephrotic syndrome). CCF?

**Treatment of leg oedema** Treat the cause. Diuretics for all is not an answer. Elevating legs for dependent oedema (ankles higher than hips—do not just use footstools); raise the foot of the bed. Graduated support stockings may help (CI: ischaemia).

### Travel and DVT

Long-distance travel appears to be a risk factor for the development of venous thromboembolism (VTE). Data suggests this is not confined to air travel, increases with the duration of travel, and results in clinical thrombosis more often in travellers with pre-existing risk factors. Dehydration, immobilization, decreased oxygen tension, and prolonged sitting have all been suggested as contributory factors.
The risk of developing a DVT from a long-distance flight has been estimated at 1 in 10 000 to 1 in 40 000 for the general population.

- The incidence of DVT in *high-risk* groups has been shown to be 4–6% for flights >10h. Travellers with ≥1 risk factor should consider compression stockings. For high-risk individuals consider a single dose of prophylactic LMWH for flights >6h.
- There is ↑risk of PE associated with long-distance air travel.
- Compression stockings may ↓ risk of DVT.
- There is no evidence to support the use of prophylactic aspirin.
- Risk reduction measures: leg exercises, increased water intake, and refraining from alcohol or caffeine during the journey.

## Specific post-operative complications



**Laparotomy** Wound may break down from a few days to a few weeks post-op (incidence ≈3.5%). Particular risk in the elderly, malnourished (eg cancer, IBD), if infection, uraemia, or haematoma is present, or in repeat laparotomies. Warning sign is a pink serous discharge. Always assume the defect involves the whole of the wound. Wound dehiscence may lead to a 'burst abdomen' with evisceration of bowel (mortality 15–30%). If you are on the ward when this happens, call your senior, put the viscera back into the abdomen, place a sterile dressing over the wound, and give IV antibiotics (eg piperacillin/tazobactam; see local guidelines). Allay anxiety, give parenteral pain control, set up an IVI, and return patient to theatre. *Incisional hernia* is a common late problem (20%), repairable by mesh insertion (if necessary).

**Biliary surgery** *Early:* Iatrogenic bile duct injury, cholangitis, bile leakage, bleeding into the biliary tree (haemobilia—may lead to melaena or haematemesis); pancreatitis. Retained stones may be removed by ERCP (p742); if this is not available a 'T-tube' left in the bile duct at the time of closure allows free drainage to the exterior; unrelieved distal obstruction of the bile duct may result in fistula formation and chronic leakage of bile. If jaundiced, maintain a good urine output, monitor coagulation, and consider antibiotics. *Late:* Bile duct stricture; post-cholecystectomy syndrome (symptoms arising from alterations in bile flow due to loss of the reservoir function of the gallbladder).

**Thyroid surgery** *Early:* Recurrent (± superior) laryngeal nerve palsy (→hoarseness) can occur permanently in 0.5% and transiently in 1.5%—warn the patient that *their voice will be different* for a few days post-op because of intubation and local oedema (NB: pre-operative fibreoptic laryngoscopy should be performed to exclude pre-existing vocal cord dysfunction); thyroid storm (symptoms of severe hyperthyroidism—see p834); tracheal obstruction due to haematoma in the wound: ►►relieve by immediate removal of stitches or clips using the cutter/remover that should remain at the beside; may require urgent surgery; hypoparathyroidism (p222); ►check plasma $Ca^{2+}$ daily; transient drops in serum concentration are common, permanent in 2.5%. *Late:* Hypothyroidism; recurrent hyperthyroidism.

**Mastectomy** Arm lymphoedema in up to 20% of those undergoing axillary node sampling or dissection. The risk of lymphoedema increases with the level of axillary dissection: risk is lower with level 1 dissection (remains inferior to *pec. minor*) compared to level 3 dissection (goes superior to *pec. minor*, rarely done); skin necrosis.

**Arterial surgery** Bleeding; thrombosis; embolism; graft infection; MI; AV fistula formation. *Complications of aortic surgery:* Gut ischaemia; renal failure; respiratory distress; trauma to ureters or anterior spinal artery (leading to paraplegia); ischaemic events from distal emboli from dislodged thrombus; aorto-enteric fistula.

**Colonic surgery** *Early:* Sepsis; ileus; fistulae; anastomotic leak (11% for radical rectal surgery); haemorrhage; trauma to ureters or spleen. *Late:* Adhesional obstruction (BOX).

**Small bowel surgery** Short gut syndrome (best defined *functionally*, as malabsorption due to insufficient residual small bowel; adults with ≤150cm at risk). Diarrhoea and malabsorption (particularly of fats) lead to a number of metabolic abnormalities including deficiency in vitamins A, D, E, K, and $B_{12}$, hyperoxaluria (causing renal stones), and bile salt depletion (causing gallstones). The management of short bowel syndrome is complex, aiming to correct metabolic abnormalities, optimize residual bowel function, and support nutrition (using parenteral route if necessary).

**Tracheostomy** Mediastinitis; surgical emphysema. Later: stenosis.

**Splenectomy** (p373.) Acute gastric dilatation (a serious consequence of not using a NGT, or to check that the one in place is working); thrombocytosis; sepsis. ►Lifetime sepsis risk is partly preventable with pre-op vaccines—ie *Haemophilus* type B, meningococcal, and pneumococcal (p407 & p167) and prophylactic penicillin.

**Genitourinary surgery** Septicaemia (from instrumentation in the presence of infected urine)—consider a stat dose of gentamicin; urinoma—rupture of a ureter or renal pelvis leading to a mass of extravasated urine.

**Gastrectomy** See p622. **Prostatectomy** p642. **Haemorrhoidectomy** p632.

### Adhesions—legacy of the laparotomy, bane of the surgeon

When re-operating on the abdomen, the struggle against adhesions tests the farthest and darkest boundaries of patience of the abdominal surgeon and the assistant. The skill and persistence required to gently and atraumatically tease apart these fibrous bands that restrict access and vision makes any progression, no matter how slight, cause for subdued celebration. Perseverance is the name of this game.

Surgical division of adhesions is known as ***adhesiolysis***. Any surgical procedure that breaches the abdominal or pelvic cavities can predispose to the formation of adhesions, which are found in up to 90% of those with previous abdominal surgery; this is why we do not rush to operate on small bowel obstruction: the operation predisposes to yet more adhesions. Handling of the serosal surface of the bowel causes inflammation, which over weeks to years can lead to the formation of fibrous bands that tether the bowel to itself or adjacent structures—though adhesions can also form secondary to infection, radiation injury, and inflammatory processes such as Crohn's disease. Their main sequelae are intestinal obstruction (the cause in ~60% of cases—see p610) and chronic abdominal or pelvic pain. Studies have shown that adhesiolysis may help relieve chronic pain, though for a small proportion of patients the pain never improves or even worsens after directed intervention.

As far as prevention is concerned, the best approach is to avoid operating; laparoscopy compared with laparotomy reduces the rate of local adhesions. Insertion of synthetic films (eg hyaluronic acid/carboxymethyl membrane) to prevent adhesions to the anterior abdominal wall reduces incidence, extent, and severity of adhesions, but not incidence of obstruction or operative re-intervention.

## Stoma care

A stoma (Greek=*mouth*) is an artificial union between a conduit and the outside world—eg a colostomy, in which faeces are made to pass through an opening in the abdominal wall when a loop of colon is brought out onto the skin. NB: a stoma can also be made between two internal conduits (eg a choledochojejunostomy).

**Colostomies** (Usually left illiac fossa and flush with the skin—fig 13.8.) May be temporary or permanent. Are they suitable for a laparoscopic operation?

- *Loop colostomy:* A loop of colon is exteriorized and partially divided, forming two stomas that are joined together (the proximal end passes stool, the distal end passes mucus, see fig 13.6). A rod under the loop prevents retraction and may be removed after 7d. A loop colostomy is often temporary and performed to protect a distal anastomosis, eg after anterior resection.
- *End colostomy:* The bowel is divided and the proximal end brought out as a stoma; the distal end may be: 1 *resected*, eg abdominoperineal (AP) resection (inspect the perineum for absent anus when examining a stoma) 2 *closed* and left in the abdomen (Hartmann's procedure) 3 *exteriorized*, forming a 'mucous fistula'.
- *Paul-Mikulicz colostomy:* A double-barrelled colostomy in which the colon is divided completely (eg to excise a section of bowel). Each end is exteriorized as two separate stomas.

*Output:* Colostomies ideally pass 1–2 formed motions/day into an adherent plastic pouch. Some may be managed with irrigation, thus avoiding a pouch.
*Incidence:* 21000 stomas/yr in UK (>50% are permanent). Most manage their stomas well. The cost for appliances is ~£1500/yr. If there is a skin reaction to the adhesive or pouch, a change of device may be all that is needed. Contact the stoma nurse.

**Ileostomies** (Usually right illiac fossa.) Protrude from the skin and emit frequent fluid motions which contain active enzymes (so the skin needs protecting—see fig 13.7). Loop ileostomies can be formed to defunction the colon as a temporary measure eg during control of difficult perianal Crohn's disease. End ileostomy follows total or subtotal colectomy, eg for UC; subsequent formation of *ileal pouch-anal anastomosis* (pouch of ileum is joined to the upper anal canal) can allow for stoma reversal.

**Alternative (non-stoma forming) surgery** *Low/ultralow anterior resection:* All or part of the rectum is excised and the proximal colon anastomosed to the top of the anal canal (the lower the level of anastomosis, the higher the risk of complication).
*Transanal endoscopic microsurgery:* Allows excision of small tumours within the rectum with preservation of sphincter function.

**Urostomies** are fashioned after total cystectomy, bringing urine from the ureters to the abdominal wall via an *ileal conduit* that is usually incontinent. Formation of a catheterizable valvular mechanism may retain continence. Advances in urological surgery have seen an increase in continence-saving procedures such as orthotopic neobladder reconstruction, with good long-term continence rates.

**When choosing a stoma site, avoid:**

- Bony prominences (eg anterior superior iliac spine, costal margins).
- The umbilicus.
- Old wounds/scars—there may be adhesions beneath.
- Skin folds and creases.
- The waistline.
- The site should be assessed pre-operatively by the stoma nurse, with the patient both lying and standing.

## Complications of stomas

►Liaise early with the stoma nurse, starting pre-operatively.

*Early:*

- Haemorrhage at stoma site.
- Stoma ischaemia—colour progresses from dusky grey to black.
- High output (can lead to ↓$K^+$)—consider loperamide ± codeine to thicken output.
- Obstruction secondary to adhesions (see p581).
- Stoma retraction.

*Delayed:*

- Obstruction (failure at operation to close lateral space around stoma).
- Dermatitis around stoma site (worse with ileostomy).
- Stoma prolapse.
- Stomal intussusception.
- Stenosis.
- Parastomal hernia (risk increases with time). NB: prophylactic mesh insertion at the time of stoma formation reduces this risk.
- Fistulae.
- Psychological problems.

## Psychological aspects of stoma care

The physical and psychological aspects of stoma care must not be undervalued. Be alert to any vicious cycle in which a skin reaction leads to leakage and precipitates a fear of going out, or a fear of eating. This in turn may lead to poor nutrition and further skin reactions, resulting in further leakage and depression. These cycles can be circumvented by the *stoma nurse*, who is *the* expert in fitting secure, odourless devices and providing patients with a wealth of physical and psychological support, both pre and post operative (explaining what is going to happen, what the stoma will be like, and troubleshooting post-op problems). ►*Early referral prevents problems*. Without input from the stoma nurse, a patient may reject their colostomy, never attend to it, and develop deep-seated psychological and psychiatric problems.

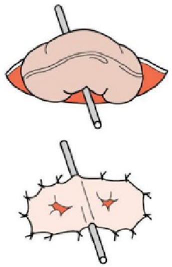


Fig 13.6 A loop colostomy with double-barrelled stoma and supporting ostomy rod.

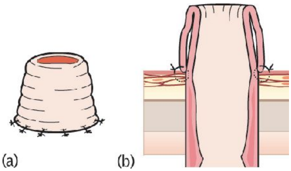



Fig 13.7 An ileostomy sits proud, has prominent mucosal folds, and is often right-sided.

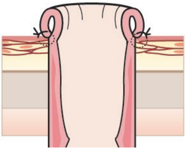


Fig 13.8 A colostomy sits flush with the skin and is typically sited in the left iliac fossa.

# Nutritional support in hospital

►Over 25% of hospital inpatients may be malnourished. Hospitals can become so focused on curing disease that they ignore the foundations of good health—malnourished patients recover more slowly and experience more complications. See table 13.6.



### Why are so many hospital patients malnourished?

1. Increased nutritional requirements (eg sepsis, burns, surgery).
2. Increased nutritional losses (eg malabsorption, output from stoma).
3. Decreased intake (eg dysphagia, nausea, sedation, coma).
4. Effect of treatment (eg nausea, diarrhoea).
5. Enforced starvation (eg prolonged periods nil by mouth).
6. Missing meals (eg due to investigations—minimize meal time disruption).
7. Difficulty with feeding (eg lost dentures; no one available to assist).
8. Unappetizing food.

**Identifying at-risk patients** Assess nutrition state (using eg Malnutrition Universal Screening Tool[3]) and weight on admission; reassess weekly thereafter. Involve dieticians early in those at risk.

- *History:* Recent ↓weight (>20%, accounting for fluid balance); recent reduced intake; diet change (eg recent change in consistency of food); nausea, vomiting, pain, diarrhoea which might have led to reduced intake.
- *Examination:* State of hydration (p666): dehydration can go hand-in-hand with malnutrition, and overhydration can mask malnutrition. Evidence of malnutrition: skin hanging off muscles (eg over biceps); no fat between fold of skin; hair rough and wiry; pressure sores; sores at corner of mouth. Calculate body mass index (p244); BMI <18.5kg/m$^2$ suggests malnourishment. Anthropomorphic indices, eg mid-arm circumference, skin fold measures, and grip strength are also used.
- *Investigations:* Generally unhelpful. Low albumin suggestive, but is affected by many things other than nutrition. ↑Albumin can be helpful in monitoring recovery.

**Enteral nutrition** (Ie nutrition given into gastrointestinal tract.) If at all possible, give nutrition by mouth. An all-fluid diet can meet requirements (but get advice from dietician). If danger of choking or aspiration (eg after stroke), consider semi-solid diet. Early post-op enteral nutrition has been shown to benefit patients (eg after GI surgery) and may reduce complications. *Tube feeding:* Liquid nutrition via a tube: Nasogastric typically placed without guidance (p759); nasojejunal tubes require endoscopic placement (used if gastric outlet obstruction, delayed gastric emptying, post-gastrectomy, or pancreatitis). Alternatively, gastric or jejunal tubes may be inserted radiologically or surgically (ie gastrostomy/jejunostomy). Use for nutritionally complete, commercially prepared feeds. ►Close liaison with a dietician is essential. *Polymeric* feeds consist of undigested proteins, starches, and long-chain fatty acids (eg Nutrison standard®, Osmolite®). Normally contain ~1kCal/mL and 4–6g protein per 100mL. Typical requirements met with 2L/24h. *Elemental* feeds consist of individual amino acids, oligo- and monosaccharides needing minimal digestion. Feed is typically initiated at a slow, continuous rate (nausea and vomiting less problematic) but patients may build up to shorter, bolus feeds, freeing them from pumps between.

### Guidelines for success

- Use fine-bore (9Fr) nasogastric feeding tube when possible.
- Check position of nasogastric tube (pH testing) before starting feeding (p759); the positioning of a nasojejunal tube can be checked on abdominal X-ray.
- Build up feeds gradually to avoid diarrhoea and distension.
- Weigh at least weekly.
- Check blood glucose and plasma electrolytes (monitor for refeeding syndrome if previously malnourished—p587).
- Treat underlying conditions vigorously, eg sepsis may impede +ve nitrogen balance.

### Nil by mouth (NBM) before theatre

If in doubt about what is acceptable oral intake prior to induction for general anaesthesia (eg for GI surgery), it is best to liaise with the anaesthetist concerned. However, guidelines have been published by many colleges and societies to outline what is safe in the perioperative period:

- For *adult elective surgery* in healthy patients without GI comorbidity:
    - Water or clear fluids (eg black tea/coffee) are allowed up to 2h beforehand.
    - All other intake up to 6h beforehand.
- In *emergency surgery*, ≥6h NBM prior to theatre is best—but poor scheduling of an emergency list is not an excuse for starving patients for days.

**Table 13.6** Daily energy and nutritional requirements

| Substance | Requirement (/kg/d) | Notes |
|---|---|---|
| Energy | 20–40kCal | Normal adult requirements will be 2000–2500kCal/d; even catabolic patients rarely require >2500kCal/d. |
| | 84–168kJ | Multiply kCal by a factor of 4.2. |
| Nitrogen | 0.2–0.4g | 6.25g of enteral protein gives 1g of nitrogen. Considering nitrogen balance is important because although catabolism is inevitable, replenishment is vital. |
| Protein | 0.5g | Contains 5kCal/g. |
| Fat | 3g | Contains 10kCal/g. |
| Carbohydrate | 2g | Contains 4kCal/g. |
| Water | 25–30mL | +500mL/d for each °C of pyrexia. |
| Na/K/Cl | 1.0mmol each | Electrolytes need to be considered, even if not on IVI. |

# Parenteral (intravenous) nutrition

Do not undertake parenteral feeding lightly: it has risks. Specialist advice is vital. It should only be considered if the patient is likely to become malnourished without it—this normally means that the gastrointestinal tract is not functioning (eg bowel obstruction), and is unlikely to function for at least 7d. Parenteral feeding may supplement other forms of nutrition (eg in short bowel syndrome or active Crohn's disease, when nutrition cannot be sufficiently absorbed in the gut) or it can be used alone (total parenteral nutrition—TPN). ►Even if there is GI disease, studies show that enteral nutrition is safer, cheaper, and at least as efficacious as parenteral nutrition in the perioperative period.[5]

**Administration** Nutrition must be given via a dedicated central venous line (or peripherally inserted central catheter—PICC line) or via a dedicated lumen of a multi-lumen catheter (see figs 13.9 and 13.10).

**Requirements** There are many different regimens for parenteral feeding. Most provide 2000kCal and 10–14g nitrogen in 2–3L; this usually meets a patient's daily requirements (see table 13.6, p585). ~50% of calories are provided by fat and ~50% by carbohydrate. Regimens comprise vitamins, minerals, trace elements, and electrolytes; these will normally be included by the pharmacist.



### Complications

- *Sepsis:* (eg *Staphylococcus epidermidis* and *Staphylococcus aureus; Candida; Pseudomonas;* infective endocarditis.) Look for spiking pyrexia and examine wound at tube insertion point. Stop PN, take line *and* peripheral cultures and give antibiotics via the line. If central venous line-related sepsis is suspected, the safest course of action is always to remove the line. Do not attempt to salvage a line when *Staph. aureus* or *Candida* infection has been identified.
- *Thrombosis:* Central vein thrombosis may occur, resulting in pulmonary embolus or superior vena caval obstruction (p528).
- *Metabolic imbalance:* Electrolyte abnormalities—see BOX 'Refeeding syndrome'; deranged plasma glucose; hyperlipidaemia; deficiency syndromes (table 6.9, p268); acid-base disturbance (eg hypercapnia from excessive $CO_2$ production).
- *Mechanical:* Pneumothorax; embolism of IV line tip.

### Guidelines for success

►Liaise closely with line insertion team, nutrition team, and pharmacist.
- Meticulous sterility. Do not use central venous lines for uses other than nutrition. Remove the line if you suspect infection. Culture its tip.
- Review fluid balance at least twice daily, and requirements for energy and electrolytes daily.
- Check weight, fluid balance, and urine glucose daily during establishment of parenteral nutrition. Check plasma glucose, creatinine and electrolytes (including calcium and phosphate), and FBC daily until stable. Check LFT and lipid clearance three times a week until stable. Check zinc and magnesium weekly.
- Do not rush. Achieve the maintenance regimen in small steps.
- Treat underlying conditions vigorously—eg sepsis may impede +ve nitrogen balance.

5 Enteral feeding promotes integrity of the gut mucosal barrier, thus preventing bacterial and endotoxin translocation across the gut wall, which can lead to multiple organ dysfunction and perpetuation of a systemic inflammatory response—even when the gut is not the primary source of pathology.

## Refeeding syndrome

►This is a life-threatening metabolic complication of refeeding via any route after a prolonged period of starvation. At-risk patients include those initiating artificial feeding (enteral or parenteral) after prolonged starvation, or with malignancy, anorexia nervosa, or alcoholism. As the body turns to fat and protein metabolism in the starved state, there is a drop in the level of circulating insulin (because of the paucity of dietary carbohydrates). The catabolic state also depletes intracellular stores of phosphate, although serum levels may remain normal (0.85–1.45mmol/L). When refeeding begins, the level of insulin rises in response to the carbohydrate load, and one of the consequences is to increase cellular uptake of phosphate.

A hypophosphataemic state (<0.50mmol/L) normally develops within 4d and is mostly responsible for the features of '*refeeding syndrome*', which include: rhabdomyolysis; red and white cell dysfunction; respiratory insufficiency; arrhythmias; cardiogenic shock; seizures; sudden death.

**Prevention** Give high-dose Pabrinex® during re-feeding window. Identify at-risk patients, assess and monitor closely during refeeding (glucose, lipids, sodium, potassium, phosphate, calcium, magnesium, and zinc). Close involvement of a nutritionist is required.

**Treatment** is of the complicating features and includes parenteral phosphate administration (eg 18mmol/d) in addition to oral supplementation.

## The venous system at the thoracic outlet

When trying to judge the position of a central venous line tip on CXR (see fig 13.10) it helps to know the anatomical landmarks of the venous system (fig 13.9). The subclavian veins join the internal jugular veins behind the sternoclavicular joints to form the brachiocephalic veins. These come together behind the right 1st sternocostal joint to form the superior vena cava (SVC), which runs from this point to the right 3rd sternocostal joint. The right atrium starts here.

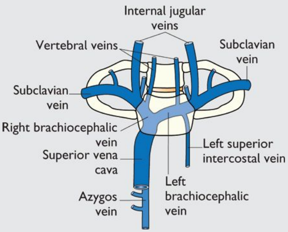



Fig 13.9 Neck veins.

Fig 13.10 Right arm PICC (peripherally inserted central catheter) still with a wire in the lumen. This is a radiograph at the time of insertion to determine if placement is correct. The tip lies in the SVC—ie good positioning for TPN or long-term antibiotic therapy. The tip of a Hickman line, for cytotoxic administration, is better in the right atrium, to avoid possible irritation of the SVC and consequent thrombosis or stenosis. NB: this image was acquired in the angiography room, where radio-opaque material appears black (it is easier to see contrast media against a white background). A similar effect may be achieved by digitally inverting a standard X-ray.

Image courtesy of Prof. Peter Scally.

# Diabetic patients undergoing surgery

Over 10% of surgical patients will have diabetes. This group face a greater risk of post-operative infection and cardiac complications. Tight glycaemic control is therefore essential and improves outcome. Aim to achieve an HbA1c of <69mmol/mol prior to elective surgery. Patients are often well informed about their diabetes—involve them fully in managing their diabetic care. Check your hospital's policy for managing diabetic patients who will be NBM before surgery. A general guide follows.

## Insulin-treated diabetes mellitus

- Try to place the patient first on the list in order to minimize the fasting period.
- Give all usual insulin the night before surgery.
- Long-acting (basal) insulin is usually continued at normal times (eg glargine; detemir), even when patients are on a variable rate intravenous insulin infusion (VRIII)—previously known as a 'sliding scale' (see BOX).
- If on AM list, ensure no subcutaneous rapid-acting (bolus) or mixed insulin is given on the morning of surgery. If PM list, give the normal morning bolus insulin, or half the mixed insulin dose.
- If eating and drinking post-operatively, resume the usual insulin with evening meal. If AM list (or early PM) and eating a late lunch, give half the morning insulin dose with this meal. If not eating until evening, a VRIII may be needed if the capillary glucose readings are high.
- Omit all rapid-acting and mixed insulin while the patient is on a VRIII.
- It not eating or drinking post-op, start a VRIII 2 hrs prior to surgery. Aim for serum glucose levels of 6–10mmol/L and check finger-prick glucose every 2 hrs. When ready to eat, give normal dose of rapid acting or mixed insulin with the 1st meal and stop the VRIII 30–60min later.
- IV fluid is required while the patient is on a VRIII: see BOX.
- A glucose–potassium–insulin (GKI) infusion can be used as an alternative to a VRIII, although it is no longer used as standard in the UK.

## Tablet-treated diabetes mellitus

- If diabetes is poorly controlled (eg fasting glucose >10mmol/L), treat as for patients on insulin (see earlier in topic).
- Give usual medications the night before surgery, except long-acting sulfonylureas (eg glibenclamide) which can cause prolonged hypoglycaemia when fasting and may need to be substituted 2–3 days pre-operatively. Discuss with the diabetes team.
- If eating and drinking post-operatively: on AM list, omit morning medication and take any missed drugs with lunch, after surgery. If PM list, take normal medications with breakfast, omit midday doses, and take any missed drugs with a late lunch. The dose of these may need reducing, depending on dietary intake.
- If not eating or drinking post-op, start a VRIII 2 hours prior to surgery. Once eating and drinking, oral hypoglycaemics can be restarted.
- Some patients may need a phase of subcutaneous insulin following major surgery—refer to the diabetes team if serum glucose levels are persistently raised.
- *Metformin and iodine IV contrast:* Metformin can be continued after IV contrast in patients with normal serum creatinine and/or eGFR >60mL/min. To minimize the risk of nephrotoxicity, if serum creatinine is raised or eGFR <60mL/min, stop metformin for 48h after contrast and check renal function has returned to baseline before restarting.

**Diet-controlled diabetes** There are usually no problems; patients should be treated as if not diabetic (and do not need to be first on the list). Check capillary blood glucose peri-operatively. Avoid 5% glucose IVI as this increases blood glucose levels.

**Peri-operative morbidity and mortality** Diabetes mellitus is classed as an intermediate risk factor for increased peri-operative cardiovascular risk by the American Heart Association, so screen for the presence of asymptomatic cardiac and renal disease (p567) and be aware of possible 'silent' myocardial ischaemia. Long-term post-op survival has been found to be poorer for patients with diabetes; however, peri-operative cardiovascular morbidity and mortality were only increased in the presence of congestive heart failure and haemodialysis—ie *not* diabetes alone.

## How to write up a variable rate intravenous insulin infusion (VRIII)

*Variable rate intravenous insulin infusion (VRIII)* is more accurate a term than the previously used 'sliding scale'. Prescribe 50 units of short-acting insulin in 50mL of 0.9% saline to infuse at the rate shown in table 13.7 (according to blood glucose levels). NB: this is a guide only—infusions may vary between institutions and no one infusion rate is suitable for all patients.

Table 13.7 Guide to VRIII according to blood glucose levels

| Capillary blood glucose (mmol/L) | IV soluble insulin (rate of infusion) |
|---|---|
| <4.0 | 0.5 units/h (0.0 if long-acting insulin continued) |
| 4.1–7.0 | 1 unit/h |
| 7.1–9.0 | 2 units/h |
| 9.1–11.0 | 3 units/h |
| 11.1–14.0 | 4 units/h |
| 14.1–17.0 | 5 units/h |
| 17.1–20 | 6 units/h |
| >20 | 6 units/h; request urgent diabetic review |

Fluids should be prescribed to run with the VRIII (through the same cannula via a non-return valve). Ideally use 0.45% sodium chloride with 5% glucose *and* either 0.15% potassium chloride (KCl) (=20mmol/L) or 0.3% KCl (=40mmol/L). This provides a constant supply of substrate, but it is not widely available.

Alternatively, use 10% glucose + KCl. This has a lower risk of hypoglycaemia and hyponatraemia than 5% glucose. If capillary glucose >15mmol/L when starting the VRIII use 0.9% saline until <15mmol/L, then use 10% glucose.

Fluid should infuse at 83–125mL/h (ie 1L over 8–12 hours). Omit potassium if there is renal impairment or hyperkalaemia and slow the rate of infusion in heart failure.

## Jaundiced patients undergoing surgery

Avoid operating in patients with obstructive jaundice—consider prior ERCP to relieve. There is ↑risk of bleeding, peri-operative infection, and renal failure.

**Coagulopathy** Vitamin K↓ in obstruction (requires bile in order to be absorbed. If no history of chronic liver disease, give parenteral vitamin K (consider even if clotting is normal). FFP may be required in liver disease or active bleeding.

**Sepsis** ↑Risk due to •↑bacterial translocation •bacterial colonization of biliary tree •↓neutrophil function. If cholangitis present, give antibiotics. Antibiotic prophylaxis for ERCP not recommended unless biliary decompression fails, or there is a history of biliary disorders; liver transplantation; presence of a pancreatic pseudocyst; or neutropenia, in which case give oral ciprofloxacin or IV gentamicin (check local policy).

**Renal failure** ↑Risk post-op due to ↑intestinal absorption of endotoxin (normally limited by the detergent effect of bile). This causes ↑renal vasoconstriction and acute tubular necrosis (see p298). The use of lactulose or bile salts pre-op may help. Ensure adequate IV fluids pre- and post-operatively to maintain good urine output. Monitor urine output every 2h. Consider central line, inotropes, and furosemide if output poor despite adequate hydration. Measure and correct U&E daily.

## Surgery in those on anticoagulants

►Inform the surgeon and anaesthetist. Risks and benefits must be individualized.
**Warfarin** *Minor surgery* can be undertaken without stopping (if INR <3.5 it may be safe to proceed). In *major surgery*, stop for 3–5d pre-op. Vitamin K ± FFP or Beriplex® may be needed for emergency reversal of INR. One elective option is conversion to heparin (stop 6h prior to surgery, and monitor APTT perioperatively). When re-warfarinizing, give LMWH until INR is therapeutic, as warfarin is initially prothrombotic.
**DOACs** Decision to stop will be based upon the patient's risk of having a thromboembolic event and bleeding risk associated with the procedure.[4] Where procedure has *no clinically important bleeding risk* it can be performed just before the next DOAC dose/18–24h after last dose and dosing restarted 6h post-op. *Low bleeding risk procedure*, omit DOAC 24h pre-op. *High bleeding risk procedure*,omit DOAC 48h pre-op. ►If renal function impaired, may require longer periods of omission pre-op.
**Antiplatelets** Decision to stop is complex and best discussed with the treating team (eg cardiologist or neurologist). Premature discontinuation of clopidogrel in patients with drug-eluting stents can lead to stent thrombosis. The bleeding effects of aspirin are reversed 5d after stopping—check local policy to see if cessation required.

## Surgery in those on steroids

Patients on steroids may not be able to mount an appropriate adrenal response to meet the stress of surgery due to suppression of the hypothalamic–pituitary–adrenal (HPA) axis. Extra corticosteroid cover may be required, depending on the type of surgery. Consider cover for any patient taking >5mg/d of prednisolone (or equivalent) for more than 2 weeks or any patient who has had their long-term steroid reduced in the last 2–4 weeks. There is also potential for HPA suppression in patients taking long-term, high-dose inhaled or topical corticosteroids. A guide to supplementation follows. Patients should take their normal morning steroid dose.

- *Minor procedures under local anaesthetic:* No supplementation required.
- *Moderate procedures:* (Eg joint replacement.) Give 50mg hydrocortisone before induction and 25mg every 8h for 24h. Resume normal dose thereafter.
- *Major surgery:* Give 100mg hydrocortisone before induction and 50mg every 8h for 24h. After 24h, halve this dose each day until the level of maintenance.

Patients with primary adrenal insufficiency will need extra cover—discuss with an endocrinologist. The major risk with adrenal insufficiency is hypotension, so if this is encountered without an obvious cause, consider a stat dose of hydrocortisone. ►See p836 for treatment of Addisonian crisis and *BNF* section 6.3 for steroid dose equivalents.

# Minimally invasive and day case surgery

Laparoscopy was developed within gynaecology and is now in widespread use for diagnostic purposes and surgical procedures such as appendicectomy, fundoplication, splenectomy, adrenalectomy, hernia repair, colectomy, prostatectomy, and nephrectomy. Minimally invasive surgery is also used for thyroidectomy and parathyroidectomy.

As a rule of thumb, whatever can be done by laparotomy can also be done with the laparoscope. This does not mean that it *should* be done, but if the surgeon is adequately trained, and the patient feels better sooner, has less post-operative pain, can return to work earlier, and has fewer complications, then there are obvious advantages. Pre-procedure counselling should always discuss the complications of laparoscopic surgery (eg accidental damage to other intra-abdominal organs) as well as the risk of conversion to an open procedure.



**Challenges of minimal access surgery** The 2-dimensional visual representation and different surgical approach alters the normal appearance of familiar anatomy. Palpation is impossible and it may be harder to locate lesions prior to resection. As a result, pre-operative imaging may need to be more extensive. A new skill has to be learned and taught.

**Day-case surgery** Advances in surgical techniques as well as perioperative care mean better results for the patient (shorter waiting lists, fewer infections, fewer days off work, and ↑patient satisfaction). Many operations can now be performed as day cases. Theoretically any procedure is suitable, provided the time under general anaesthetic does not exceed ~1h. The use of regional anaesthesia helps to avoid the SE of nausea and disorientation that may accompany a general anaesthetic, thus facilitating discharge.

Bear in mind that the following groups of patients may not be suitable for day-case surgery: •Severe dementia. •Severe learning difficulties. •Living alone (and no helpers). •Children if supervision difficult—changes in expectation, delays, and pain relief can be problematic. •BMI >32 (p244). •ASA category ≥III (p567) thus potentially unstable comorbidities—discuss with the anaesthetist. •Infection at the site of the operation.

### Exposing patients to our learning curves? The jury is still out...

All surgeons get better over time (for a while), as they perform new techniques with increasing ease and confidence. When Wertheim did his first hysterectomies, his first dozen patients died—but then one survived. He assumed it was a good operation, and pressed ahead. He was a brave man, and thousands of women owe their lives to him. But had he tried to do this today, he would have been stopped. The UK's General Medical Council (GMC) and other august bodies tell us that we must protect the public by reporting doctors whose patients have low survival rates. The reason for this is partly ethical, and partly to preserve self-regulation.

We have the toughest codes of practice and disciplinary procedures of any group of workers. It is assumed that doctors are loyal to each other out of self-interest, and that this loyalty is bad. This has never been tested formally, and is not evidence-based. We can imagine two clinical worlds: one of constant 'reportings' and recriminatory audits, and another of trust and team-work. Both are imperfect, but we should not assume that the first world would be better for our patients.

When patients are sick with fear, they do not, perhaps, want to know everything. We may tell to protect ourselves. We may *not* tell to protect ourselves. Perhaps what we should do is, in our hearts, appeal to those 12 dead women-of-Wertheim—a jury as infallible as sacrificial—and try to hear their reply. And to those who complain that in doing so we are playing God, it is possible to reply with some humility that, whatever it is, it does not seem like play.

'*It is amazing what little harm doctors do when one considers all the opportunities they have.*' M. Twain.

# Lumps



►Examine the regional lymph nodes as well as the lump. If the lump is a node, examine its area of drainage. Always examine the circulation and nerve supply distal to any lump.

**History** How long has it been there? Does it hurt? Any other symptoms, eg itch? Any other lumps? Is it getting bigger? Ever been abroad? Otherwise well?

**Physical exam** Remember the 6 **S**'s: **s**ite, **s**ize, **s**hape, **s**moothness (consistency), **s**urface (contour/edge/colour), and **s**urroundings. *Other questions:* Does it transilluminate (see next paragraph)? Is it fixed/tethered to skin or underlying structures (see BOX)? Is it fluctuant/compressible? Temperature? Tender? Pulsatile (US duplex may help)?

**Transilluminable lumps** After eliminating as much external light as possible, place a bright, thin 'pen' torch on the lump, from behind (or at least to the side), so the light is shining through the lump towards your eye. If the lump glows red it is said to transilluminate—a fluid-filled lump such as a hydrocele is a good example.

**Lipomas** These benign fatty lumps, occurring wherever fat can expand (ie *not* scalp or palms), have smooth, imprecise margins, a hint of fluctuance, and are not fixed to skin or deeper structures. Symptoms are only caused via pressure. Malignant change very rare (suspect if rapid growth/hardening/vascularization). Multiple scattered lipomas, which may be painful, occur in Dercum's disease, typically in post-menopausal women.

**Sebaceous cysts** Refer to either *epidermal* (fig 13.11) or *pilar cysts* (they are not of sebaceous origin and contain keratin, not sebum). They appear as firm, round, mobile subcutaneous nodules of varying size. Look for the characteristic central punctum. Infection is quite common, and foul pus exits through the punctum. They are common on the scalp, face, neck, and trunk. *Treatment:* Excision of cyst and contents.

**Lymph nodes** Causes of enlargement: *Infection:* Glandular fever; brucellosis; TB; HIV; toxoplasmosis; actinomycosis; syphilis. *Infiltration:* Malignancy (carcinoma, lymphoma); sarcoidosis.

**Cutaneous abscesses** Staphylococci are the most common organisms. Haemolytic streptococci only common in hand infections. *Proteus* is a common cause of non-staphylococcal axillary abscesses. Below the waist, faecal organisms are common (aerobes and anaerobes). *Treatment:* Incise and drain. *Boils (furuncles)* are abscesses involving a hair follicle and associated glands. A *carbuncle* is an area of subcutaneous necrosis which discharges itself on to the surface through multiple sinuses. Think of *hidradenitis suppurativa* if recurrent inguinal or axillary abscesses.

**Rheumatoid nodules** (fig 13.12) Collagenous granulomas which appear in established rheumatoid arthritis on the extensor aspects of joints—especially the elbows (fig 13.12).

**Ganglia** Degenerative cysts from an adjacent joint or synovial sheath commonly seen on the dorsum of the wrist or hand and dorsum of the foot. May transilluminate. 50% disappear spontaneously. Aspiration may be effective, especially when combined with instillation of steroid and hyaluronidase. For the rest, treatment of choice is excision rather than the traditional blow from your bible (the *Oxford Textbook of Surgery*)! See fig 13.13.

**Fibromas** These may occur anywhere in the body, but most commonly under the skin. These whitish, benign tumours contain collagen, fibroblasts, and fibrocytes.

**Dermoid cysts** Contain dermal structures and are found at the junction of embryonic cutaneous boundaries, eg in the midline or lateral to the eye. See fig 13.14.

**Malignant tumours of connective tissue** Fibrosarcomas, liposarcomas, leiomyosarcomas (smooth muscle), and rhabdomyosarcomas (striated muscle). These are staged using modified TNM system including tumour grade. Needle-core (Trucut®) biopsies of large tumours precede excision. Any lesion suspected of being a sarcoma should not be simply enucleated. ►Refer to a specialist.

**Neurofibromas** See p514.

**Keloids** Caused by irregular hypertrophy of vascularized collagen forming raised edges at sites of previous scars that extend outside the scar (fig 13.15). Common in dark skin. Treatment can be difficult. Intralesional steroid injections are a mainstay.

## In or under the skin?

**Intradermal**
- Sebaceous cyst
- Abscess
- Dermoid cyst
- Granuloma.

**Subcutaneous**
- Lipoma
- Ganglion
- Neuroma
- Lymph node.

If a lump is intradermal, you cannot draw the skin over it, while if the lump is subcutaneous, you should be able to manipulate it independently from the skin.

Fig 13.11 Epidermal cyst.
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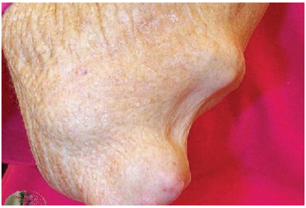

Fig 13.12 Rheumatoid nodule.
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Fig 13.13 Ganglion.
Courtesy of John M Erikson, MD, Raleigh Hand Centre.

Fig 13.14 Dermoid cyst.
Reproduced from Lewis-Jones, *Paediatric Dermatology*, 2010, with permission from Oxford University Press.

Fig 13.15 Keloid scar.
Courtesy of East Sussex Hospitals Trust.

## Malignant tumours

**1** *Malignant melanoma:* (See fig 13.16.) ♀:♂ ≈ 1.3:1. UK incidence: ≥10:100 000/yr (up ≥200% in last 20yrs). Commonly affects younger patients ∴ early diagnosis is vital. Short periods of intense UV exposure is a major cause, particularly in the early years. May occur in pre-existing moles. If smooth, well-demarcated, and regular, it is unlikely to be a melanoma but diagnosis can be tricky. Most melanomas have features described by Glasgow 7-point checklist (table 13.8) and ABCDE critera (BOX), but not all. ►If in doubt, refer.

Table 13.8 Glasgow 7-point checklist (refer if ≥3 points, or with 1 point if suspicious)

| Major (2 pts each) | Minor (1 pt each) | |
|---|---|---|
| • Change in size | • Inflammation | • Crusting or bleeding |
| • Change in shape | • Sensory change | |
| • Change in colour | • Diameter >7mm (►unless growth is in the vertical plane) | |

*Superficial spreading melanomas* (70%) grow slowly, metastasize later, and have better prognosis than *nodular melanomas* (10–15%) which invade deeply and metastasize early. Nodular lesions may be amelanotic in ~5%. *Others: acral melanomas* occur on palms, soles, and subungual areas; *lentigo maligna melanoma* evolves from pre-exisiting lentigo maligna. Breslow thickness (depth in mm), tumour stage, and presence of ulceration are important prognostic factors. ℞: urgent excision can be curative. Chemotherapy gives a response in 10–30% with metastatic disease (*OHCS* p592). Ipilimumab, a human monoclonal antibody that blocks CTLA-4, an inhibitory T-cell receptor, has been shown to improve survival in patients with metastatic melanoma.[5,6]

**2** *Squamous cell cancer:* Usually presents as an ulcerated lesion, with hard, raised edges, in sun-exposed sites. May begin in solar keratoses (see later in topic), or be found on the lips of smokers or in long-standing ulcers (=Marjolin's ulcer). Metastasis to lymph nodes is rare, local destruction may be extensive. ℞: excision + radiotherapy to treat recurrence/affected nodes. See fig 13.17. NB: the condition may be confused with a keratoacanthoma—a fast-growing, benign, self-limiting papule plugged with keratin.

**3** *Basal cell carcinoma:* (AKA *rodent ulcer*) *Nodular:* typically a pearly nodule with rolled telangiectatic edge, on the face or a sun-exposed site. May have a central ulcer. See fig 13.18. Metastases are very rare. It slowly causes local destruction if left untreated. *Superficial:* lesions appear as red scaly plaques with a raised smooth edge, often on the trunk or shoulders. *Cause:* (most frequently) UV exposure. ℞: excision; cryotherapy; for superficial BCCs topical flurouracil or imiquimod (see as for 'Solar keratoses').

## Pre-malignant tumours

**1** *Solar (actinic) keratoses* appear on sun-exposed skin as crumbly, yellow-white crusts. Malignant change to squamous cell carcinoma may occur after several years. *Treatment:* cryotherapy; 5% fluorouracil cream or 5% imiquimod—work by causing: erythema → vesiculation → erosion → ulceration → necrosis → healing epithelialization, leaving healthy skin unharmed. Warn patients of expected inflammatory reaction. See *BNF* for dosing. Alternatively: diclofenac gel (3%, use thinly twice-daily for ≤90d).

**2** *Bowen's disease:* Slow-growing red/brown scaly plaque, eg on lower legs. *Histology:* full-thickness dysplasia (carcinoma *in situ*). It infrequently progresses to squamous cell cancer. Penile Bowen's disease is called Queyrat's erythroplasia. *Treatment:* cryotherapy, topical fluorouracil (see as for 'Solar keratoses') or photodynamic therapy.

**3** *See also* Kaposi's sarcoma (p702); Paget's disease of the breast (p708).

**Others** •*Secondary carcinoma:* Most common metastases to skin are from breast, kidney, or lung. Usually a firm nodule, most often on the scalp. See also acanthosis nigricans (p562). •*Mycosis fungoides:* Cutaneous T-cell lymphoma usually confined to skin. Causes itchy, red plaques (Sézary syndrome-variant also associated with erythroderma). •*Leucoplakia:* This appears as white patches (which may fissure) on oral or genital mucosa (where it may itch). Frank carcinomatous change may occur. •*Leprosy:* Suspect in any anaesthetic hypopigmented lesion (p441). •*Syphilis:* Any genital ulcer is syphilis until proved otherwise. Secondary syphilis: papular rash—including, unusually, on the palms (p412).

**ABCDE criteria for diagnosis of melanoma**

**A**symmetry
**B**order—irregular
**C**olour—non-uniform
**D**iameter >7mm
**E**levation

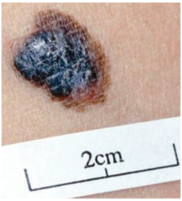


Fig 13.16 Melanoma.

Fig 13.17 Squamous cell cancer.

Fig 13.18 Basal cell carcinoma (BCC).

# Lumps in the neck

▶Don't biopsy lumps until tumours within the head and neck have been excluded by an ENT surgeon. Culture all biopsied lymph nodes for TB.

**Diagnosis** (See fig 13.19.) First, ask how long the lump has been present. If <3wks, self-limiting infection is the likely cause and extensive investigation is unwise. Next ask yourself where the lump is. Is it intradermal—eg sebaceous cyst with a central punctum (p594)? Is it a lipoma (p594)? If the lump is not intradermal, and is not of recent onset, you are about to start a diagnostic hunt over complicated terrain. 85% of neck swellings are *lymph nodes* (examine areas which they serve). Consider TB, viruses such as HIV or EBV (infectious mononucleosis), any chronic infection, or, if >20yrs, consider lymphoma (hepatosplenomegaly?) or metastases (eg from GI or bronchial or head and neck neoplasia), 8% are goitres (p600), and other diagnoses account for 7%.

**Tests** Do virology and TB tests (p394). US shows lump consistency: cystic, solid, complex, vascular. CT defines masses in relation to their anatomical neighbours. CXR may show malignancy or, in sarcoid, reveal bilateral hilar lymphadenopathy. Consider fine-needle aspiration (FNA).

**Midline lumps** •If patient is <20yrs old, likely diagnosis is *dermoid cyst* (p594). •If it moves *up* on tongue protrusion and is below the hyoid, likely to be a *thyroglossal cyst*, a fluid-filled sac resulting from incomplete closure of the thyroid's migration path. ℞: Surgery; they are the commonest congenital cervical cystic lump. •If >20yrs old, it is probably a *thyroid isthmus* mass. •If it is bony hard, the diagnosis may be a *chondroma* (benign cartilaginous tumour).

**Submandibular triangle** (Bordered by the mental process, mandible, and the line between the two angles of the mandible.) •If <20yrs, self-limiting lymphadenopathy is likely. If >20yrs, exclude *malignant lymphadenopathy* (eg firm and non-tender). ▶Is TB likely? •If it is not a node, think of submandibular *salivary stone*, *sialadenitis*, or *tumour* (see BOX for *Salivary gland pathology*).

**Anterior triangle** (Between midline, anterior border of sternocleidomastoid, and the line between the two angles of the mandible.) •*Branchial cysts* emerge under the anterior border of sternocleidomastoid where the upper third meets the middle third (age <20yrs). Due to non-disappearance of the cervical sinus (where 2nd branchial arch grows down over 3rd and 4th). Lined by squamous epithelium, their fluid contains cholesterol crystals. Treat by excision. There may be communication with the pharynx in the form of a fistula. •If lump in the supero-posterior area of the anterior triangle, is it a *parotid tumour* (more likely if >40yrs)? •*Laryngoceles* are an uncommon cause of anterior triangle lumps. They are painless and may be made worse by blowing. These cysts are classified as external, internal, or mixed, and may be associated with laryngeal cancer. If pulsatile may be: •*Carotid artery aneurysm*, •*Tortuous carotid artery*, or •*Carotid body tumours* (chemodectoma). These are very rare, move from side to side but not up and down, and splay out the carotid bifurcation. They are usually firm and occasionally soft and pulsatile. They do not usually cause bruits. They may be bilateral, familial, and malignant (5%). Suspect in any mass just anterior to the upper third of sternomastoid. Diagnose by duplex USS (splaying at the carotid bifurcation) or digital computer angiography. ℞: Extirpation by vascular surgeon.

**Posterior triangle** (Behind sternocleidomastoid, in front of trapezius, above clavicle.) •*Cervical ribs* may intrude into this area. These are enlarged costal elements from C7 vertebra. The majority are asymptomatic but can cause Raynaud's syndrome by compressing subclavian artery and neurological symptoms (eg wasting of 1st dorsal interosseous) from pressure on lower trunk of the brachial plexus. •*Pharyngeal pouches* can protrude into the posterior triangle on swallowing (usually left-sided). •*Cystic hygromas* (usually infants) arise from jugular lymph sac. These macrocystic lymphatic malformations transilluminate brightly. Treat by surgery or hypertonic saline sclerosant injection. Recurrence can be troublesome. •*Pancoast's tumour* (see p708). •*Subclavian artery aneurysm* will be pulsatile.

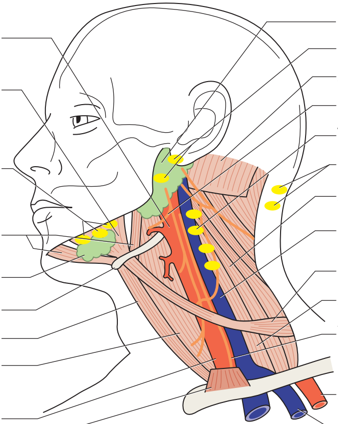


Fig 13.19 Important structures in the head and neck.

## Salivary gland pathology

There are three pairs of major salivary glands: parotid, submandibular, and sublingual (there are many minor glands). *History:* Lumps; swelling related to food; pain. *Examination:* Note external swelling; look for secretions; bimanual palpation for stones. Examine VIIth nerve and regional lymph nodes. *Cytology:* Do FNA.

*Acute swelling* Think of mumps and HIV. *Recurrent unilateral pain and swelling* is likely to be from a stone. 80% are submandibular. The classical story is of pain and swelling on eating—with a red, tender, swollen, but uninfected gland. The stone may be seen on plain x-ray or by sialography (fig 13.20). Distal stones are removed via the mouth but deeper stones may require excision of the gland. *Chronic bilateral symptoms* may coexist with dry eyes and mouth and autoimmune disease, eg hypothyroidism, Mikulicz's or Sjögren's syndrome (p706 & p710)—also bulimia or alcohol excess. *Fixed swelling* may be from a tumour/ALL (fig 8.49, p355), sarcoid, amyloid, granulomatosis with polyangiitis , or be idiopathic.

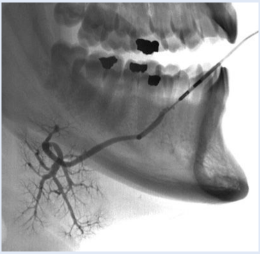

Fig 13.20 Normal sialogram of the submandibular gland. Wharton's (submandibular) duct opens into the mouth near the frenulum of the tongue.

**Salivary gland tumours** (table 13.9) '80% are in the parotid, 80% of these are pleomorphic adenomas, 80% of these are in the superficial lobe.' Deflection of the ear outwards is a classic sign. ►Remove any salivary gland swelling for assessment if present for >1 month. VIIth nerve palsy means malignancy.

Table 13.9 Types of salivary gland tumours

| Benign or malignant | Malignant | Malignant |
|---|---|---|
| Cystadenolymphoma | Mucoepidermoid | Squamous or adeno ca |
| Pleomorphic adenoma | Acinic cell | Adenoid cystic ca |

Pleomorphic adenomas often present in middle age and grow slowly. Remove by superficial parotidectomy. Adenolymphomas (Warthin's tumour): usually older men; soft; treat by enucleation. Carcinomas: rapid growth; hard fixed mass; pain; facial palsy. Treatment: surgery + radiotherapy.

# Lumps in the thyroid

If the thyroid (fig 13.21) is enlarged (=goitre), ask yourself: **1** Is the thyroid diffusely enlarged or nodular? **2** Is the patient euthyroid, thyrotoxic (p218), or hypothyroid (p220)?

**Diffuse goitre:** *Causes:* Endemic (iodine deficiency); congenital; secondary to goitrogens (substances that ↓ iodine uptake); acute thyroiditis (de Quervain's); physiological (pregnancy/puberty); autoimmune (Graves' disease; Hashimoto's thyroiditis).

**Nodular goitre:** •*Multinodular goitre (MNG):* The most common goitre in the UK. 50% who present with a single nodule actually have MNG. Patients are usually euthyroid, but may become hyperthyroid ('toxic'). MNG may be retro- or substernal. Hypothyroidism and malignancy within MNG are rare. Plummer's disease is hyperthyroidism with a single toxic nodule (uncommon). •*Fibrotic goitre:* Eg Reidel's thyroiditis. •*Solitary thyroid nodule:* typically cyst, adenoma, discrete nodule in MNG or malignant (~10%).

**Investigations** Check TSH and USS (solid, cystic, complex or part of a group of lumps). If abnormal consider: •$T_4$, autoantibodies (p216, eg if Hashimoto's/Graves', suspected). •CXR with thoracic inlet view (tracheal goitres and metastases?). •Radionuclide scans (fig 13.22) may show malignant lesions as hypofunctioning or 'cold', whereas a hyperfunctioning 'hot' lesion suggests adenoma. •FNA (fine-needle aspiration) and cytology—will characterize lesion. ►A FNA finding of a follicular neoplasm can be challenging (15-30% malignant)—discuss with cytopathologist and perform molecular diagnostics where available; if any doubt, refer for surgery.

**What should you do if high-resolution ultrasound shows impalpable nodules?**
Such thyroid nodules can usually be observed provided they are:
- <1cm across (which accounts for most; ultrasound can detect lumps <2mm; such 'incidentalomas' occur in 46% of routine autopsies) and are asymptomatic.
- There is no past history of thyroid cancer or neck irradiation.
- No family history of medullary cancer (if present, do USS-guided FNA).

**Thyroid cancer**
1. *Papillary:* (60%.) Often in younger patients. Spread: lymph nodes and lung (jugulodigastric node metastasis is the so-called lateral aberrant thyroid). ℞: total thyroidectomy to remove non-obvious tumour ± node excision ± radioiodine ($^{131}$I) to ablate residual cells. Give levothyroxine to suppress TSH. Prognosis: better if young and ♀.
2. *Follicular:* (≤25%.) Occurs in middle-age and spreads early via blood (bone, lungs). Well-differentiated. ℞: total thyroidectomy + $T_4$ suppression + radioiodine ablation.
3. *Medullary:* (5%.) Sporadic (80%) or part of MEN syndrome (p223). May produce calcitonin which can be used as a tumour marker. They do not concentrate iodine. ►Perform a phaeochromocytoma screen pre-op. ℞: thyroidectomy + node clearance. External beam radiotherapy may prevent regional recurrence.
4. *Lymphoma:* (5%.) ♀:♂≈3:1. May present with stridor or dysphagia. Do full staging pre-treatment (chemoradiotherapy). Assess histology for mucosa-associated lymphoid tissue (MALT) origin (associated with a good prognosis).
5. *Anaplastic:* Rare. ♀:♂≈3:1. Elderly, poor response to any treatment. In the absence of unresectable disease, excision + radiotherapy may be tried.

**Thyroid surgery** Plays a significant role in the management of thyroid disease. Operations include partial lobectomy or lobectomy (for isolated nodules); and thyroidectomy (for cancers, MNG, or Graves'). *Indications:* Pressure symptoms, relapse hyperthyroidism after >1 failed course of drug treatment, carcinoma, cosmetic reasons, symptomatic patients planning pregnancy. *Pre-operative management:* Render euthyroid pre-op with antithyroid drugs (eg carbimazole up to 20mg/12h PO or propylthiouracil 200mg/12h PO but stop 10d prior to surgery as these increase vascularity). Propranolol up to 80mg/8h PO can be used to control tachycardia or tremor associated with hyperthyroidism (continue for 5d post-op). Check vocal cords by indirect laryngoscopy pre- and post-op (risk of recurrent laryngeal nerve injury). Check serum $Ca^{2+}$ (and PTH if abnormal). *Complications:* see p580.

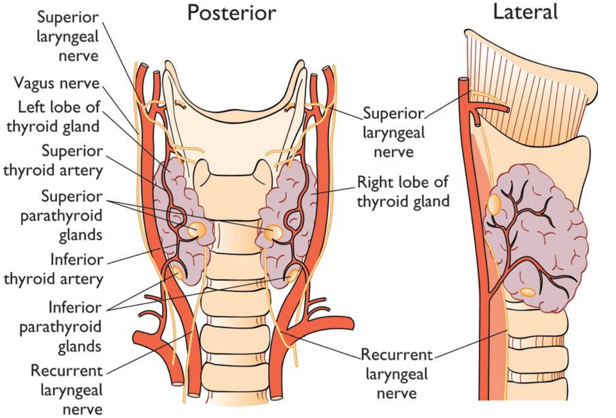


**Fig 13.21** The anatomy of the region of the thyroid gland. The important structures that must be considered when operating on the thyroid gland include:

- Recurrent laryngeal nerve
- Superior laryngeal nerve
- Parathyroid glands
- Trachea
- Common carotid artery
- Internal jugular vein (not depicted—see fig 13.23).

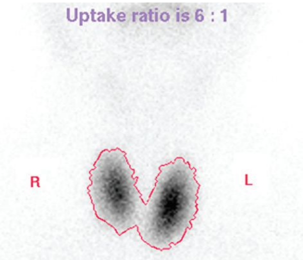


**Fig 13.22** Radionuclide study of the thyroid showing changes consistent with Graves' disease (see also *hot and cold nodules* (p216) and *nuclear medicine*, p738). There is increased uptake of the radionuclide trace diffusely throughout both lobes of the gland.

Image courtesy of Norwich Radiology Department.

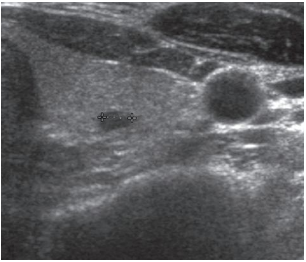

**Fig 13.23** Transverse ultrasound of the left lobe of the thyroid showing a small low-reflectivity cyst within higher-reflectivity thyroid tissue. Note the proximity to the gland of the common carotid artery and internal jugular vein (the latter compressed slightly by pressure from the probe), both seen beneath the body of sternocleidomastoid muscle.

Image courtesy of Norwich Radiology Department.

# Breast carcinoma



**Epidemiology** Affects 1 in 8 ♀; nearly 60 000 new cases per year in UK (incidence increasing). Rare in men (~1% of all breast cancers).

**Risk factors** Risk is related to family history, age, and uninterrupted oestrogen exposure, hence: nulliparity; 1st pregnancy >30yrs old, early menarche; late menopause; HRT; obesity; BRCA genes (p521); not breastfeeding; past breast cancer (metachronous rate ≈2%, synchronous rate ≈1%).

**Pathology** Non-invasive ductal carcinoma *in situ* (DCIS) is premalignant and seen as microcalcification on mammography (unifocal or widespread). Non-invasive lobular CIS is rarer and tends to be multifocal. Invasive ductal carcinoma is most common (~70%) whereas invasive lobular carcinoma accounts for 10–15% of breast cancers. Medullary cancers (~5%) tend to affect younger patients while colloid/mucoid (~2%) tend to affect the elderly. Others: papillary, tubular, adenoid-cystic and Paget's (p708). 60–70% of breast cancers are oestrogen receptor +ve, conveying better prognosis. ~30% over-express HER2 (growth factor receptor gene) associated with aggressive disease and poorer prognosis.

**Investigations** (See p82 for history and examination.) ►All lumps should undergo *'triple' assessment:* Clinical examination + histology/cytology + mammography/ultrasound; see fig 13.24.

**Staging**: *Stage 1:* Confined to breast, mobile. *Stage 2:* Growth confined to breast, mobile, lymph nodes in ipsilateral axilla. *Stage 3:* Tumour fixed to muscle (but not chest wall), ipsilateral lymph nodes matted and may be fixed, skin involvement larger than tumour. *Stage 4:* Complete fixation of tumour to chest wall, distant metastases. Also *TNM staging:* (p523) T1<2cm, T2, 2–5cm, T3 >5cm, T4, fixity to chest wall or *peau d'orange*; N1, mobile ipsilateral nodes; N2, fixed nodes; M1, distant metastases.

**Treating local disease** (Stage 1–2.)[7] •*Surgery:* Removal of tumour by wide local excision (WLE) or mastectomy ± breast reconstruction + axillary node sampling/surgical clearance or sentinel node biopsy (BOX 'Sentinel node biopsy'). •*Radiotherapy:* Recommended for all patients with invasive cancer after WLE. Risk of recurrence decreases from 30% to <10% at 10yrs and increases overall survival. Axillary radiotherapy used if lymph node +ve on sampling and surgical clearance not performed (↑risk of lymphoedema and brachial plexopathy). SE: pneumonitis, pericarditis, and rib fractures. •*Chemotherapy:* Adjuvant chemotherapy improves survival and reduces recurrence in most groups of women (consider in all except excellent prognosis patients), eg epirubicin + 'CMF' (cyclophosphamide + methotrexate + 5-FU). Neoadjuvant chemotherapy has shown no difference in survival but may facilitate breast-conserving surgery. •*Endocrine agents:* Aim to ↓oestrogen activity and are used in oestrogen receptor (ER) or progesterone receptor (PR) +ve disease. The ER blocker tamoxifen is widely used, eg 20mg/d PO for 5yrs post-op (may rarely cause uterine cancer so warn to report vaginal bleeding). Aromatase inhibitors (eg anastrozole) targeting peripheral oestrogen synthesis are also used (may be better tolerated). They are only used if post-menopausal. If pre-menopausal and an ER+ve tumour, ovarian ablation (via surgery or radiotherapy) or GnRH analogues (eg goserelin) ↓ recurrence and ↑ survival. •*Support:* Breastcare nurses •*Reconstruction options:* Eg tissue expanders/implants/nipple tattoos, latissimus dorsi flap, TRAM (transverse rectus abdominis myocutaneous) flap.

**Treating distant disease** (Stage 3–4.)[8] Long-term survival is possible and median survival is >2yrs. Staging investigations should include CXR, bone scan, liver USS, CT/MRI or PET-CT (p739), + LFTs and $Ca^{2+}$. Radiotherapy (p526) to painful bony lesions (*bisphosphonates*, p677, may ↓pain and fracture risk). Tamoxifen is often used in ER+ve; if relapse after initial success, consider chemotherapy. Trastuzumab should be given for HER2 +ve tumours, in combination with chemotherapy. CNS surgery for solitary (or easily accessible) metastases may be possible; if not—radiotherapy. Get specialist help for arm lymphoedema (try decongestive methods first).

**Preventing deaths** •Promote awareness. •*Screening:* 2-view mammography every 3yrs for women aged 47–73 in UK has ↓breast cancer deaths by 30% in women >50yrs.

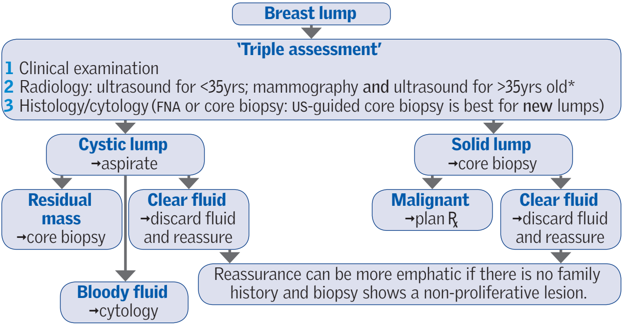


**Fig 13.24** Triple assessment and investigation of a breast lump.
* US is more accurate at detecting invasive breast cancer, though mammography remains most accurate at detecting ductal carcinoma *in situ* (DCIS). MRI is used in the assessment of multifocal/bilateral disease and patients with cosmetic implants who are identified as high risk.

## Sentinel node biopsy

Decreases needless axillary clearances in lymph node −ve patients.
- Patent blue dye and/or radiocolloid injected into periareolar area or tumour.
- A gamma probe/visual inspection is used to identify the sentinel node.
- The sentinel node is biopsied and sent for histology ± immunohistochemistry; further clearance only if sentinel node +ve.

Sentinel node identified in 90%. False −ve rates <5% for experienced surgeons.

## Prognostic factors in breast cancer

Tumour size, grade, lymph node status, ER/PR status, presence of vascular invasion all help assess prognosis. Nottingham Prognostic Index (NPI) is widely used to predict survival and risk of relapse, and to help select appropriate adjuvant systemic therapy. NPI = *0.2 × tumour size (cm) + histological grade + nodal status*[6]

If treated with surgery alone, 10yr survival rates are: NPI <2.4: 95%; NPI 2.4–3.4: 85%; NPI 3.4–4.4: 70%; NPI 4.4–5.4: 50%; NPI >5.4: 20%.

## Benign breast disease

*Fibroadenoma:* Usually presents <30yrs but can occur up to menopause. Benign overgrowth of collagenous mesenchyme of one breast lobule. Firm, smooth, mobile lump, the 'breast mouse'. Painless. May be multiple. ⅓ regress, ⅓ stay the same, ⅓ get bigger. ℞: observation and reassurance, but if in doubt refer for USS (usually conclusive) ± FNA. Surgical excision if large.

*Breast cysts:* Common >35yrs, esp. perimenopausal. Benign, fluid-filled rounded lump. Not fixed to surrounding tissue. Occasionally painful. ℞: diagnosis confirmed on aspiration (perform only if trained).

*Infective mastitis/breast abscesses:* Infection of mammary duct often associated with lactation (usually *Staph. aureus*). Abscess presents as painful, hot swelling of breast segment. ℞: antibiotics. Open incision or percutaneous drainage if abscess.

*Duct ectasia:* Typically around menopause. Ducts become blocked and secretions stagnate. Present with nipple discharge (green/brown/bloody) ± nipple retraction ± lump. Refer for confirmation of diagnosis. Usually no ℞ needed. Advise to stop smoking.

*Fat necrosis:* Fibrosis and calcification after injury to breast tissue. Scarring results in a firm lump. Refer for triple assessment. No ℞ once diagnosis confirmed.

6 Nodal status is scored 1–3: 1 = node −ve; 2 = 1–3 nodes +ve; 3 = >3 nodes +ve for breast cancer. Histological grade is also scored 1–3.

# Abdominal masses

As with any mass (see p594), determine size, site, shape, and surface. Find out if it is pulsatile and if it is mobile. Examine supraclavicular and inguinal nodes. Is the lump ballotable (like bobbing an apple up and down in water)?

| Right iliac fossa masses | | |
|---|---|---|
| • Appendix mass/abscess | • Intussusception | • Transplanted kidney |
| • Caecal carcinoma | • TB mass | • Kidney malformation |
| • Crohn's disease | • Amoebic abscess | • Tumour in an undescended testis. |
| • Pelvic mass (see later in topic) | • Actinomycosis (p389) | |

**Abdominal distension** Flatus, fat, fluid, faeces, or fetus (p57)? Fluid may be outside the gut (ascites) or sequestered in bowel (obstruction; ileus). To demonstrate ascites elicit signs of a fluid thrill and/or shifting dullness (p61).

| Causes of ascites | | Ascites with portal hypertension | |
|---|---|---|---|
| • Malignancy | • CCF; pericarditis | • Cirrhosis | • Portal nodes |
| • Infections—esp. TB | • Pancreatitis | • Budd-Chiari syndrome (p696) | |
| • ↓Albumin (eg nephrosis) | • Myxoedema. | • IVC or portal vein thrombosis. | |

*Tests:* Aspirate ascitic fluid (p764) for cytology, culture and albumin;[7] US.

**Left upper quadrant mass** Is it spleen, stomach, kidney, colon, pancreas, or a rare cause (eg neurofibroma)? Pancreatic cysts may be true (congenital; cystadenomas; retention cysts of chronic pancreatitis; cystic fibrosis) or pseudocysts (fluid in lesser sac from acute pancreatitis).

**Splenomegaly** Causes are often said to be infective, haematological, neoplastic, etc, but grouping by associated feature is more useful clinically:

| Splenomegaly with fever | With lymphadenopathy | With purpura |
|---|---|---|
| • Infection[HS] (malaria, SBE/IE, hepatitis,[HS] EBV,[HS] TB, CMV, HIV)<br>• Sarcoid; malignancy.[HS] | • Glandular fever[HS]<br>• Leukaemias; lymphoma<br>• Sjögren's syndrome. | • Septicaemia; typhus<br>• DIC; amyloid[HS]<br>• Meningococcaemia. |
| **With arthritis** | **With ascites** | **With a murmur** |
| • Sjögren's syndrome<br>• Rheumatoid arthritis; SLE<br>• Infection, eg Lyme (p422)<br>• Vasculitis/Behçet's (p556). | • Carcinoma<br>• Portal hypertension.[HS] | • SBE/IE<br>• Rheumatic fever<br>• Hypereosinophilia<br>• Amyloid[HS] (p370). |
| **With anaemia** | **With ↓weight + *CNS* signs** | **Massive splenomegaly** |
| • Sickle-cell;[HS] thalassaemia[HS]<br>• Leishmaniasis;[HS] leukaemia[HS]<br>• Pernicious anaemia (p334)<br>• POEMS syn. (p220). | • Cancer; lymphoma<br>• TB; arsenic poisoning<br>• Paraproteinaemia.[HS] | • Malaria (hyper-reactivity after chronic exposure)<br>• Myelofibrosis; CML[HS]<br>• Gaucher's syndrome[HS]<br>• Leishmaniasis. |

[HS]=causes of hepatosplenomegaly.

**Smooth hepatomegaly** Hepatitis, CCF, sarcoidosis, early alcoholic cirrhosis (a small liver is typical later); tricuspid incompetence (→ pulsatile liver).

**Craggy hepatomegaly** Secondaries or 1° hepatoma. (Nodular cirrhosis typically causes a small, shrunken liver, not an enlarged craggy one.)

**Pelvic masses** Fibroids, fetus, bladder, ovarian cysts or malignancies. *Is it truly pelvic?*—Yes, if by palpation you cannot get 'below it'.

**Investigating lumps** Check FBC (with film); CRP; U&E; LFT; Ca$^{2+}$; tumour markers only as appropriate. Imaging by CT or US (transvaginal approach may be useful); MRI also has a role, eg in assessment of liver masses (p286). *Others:* TB tests (p394). Biopsy to give a tissue diagnosis may be obtained using a fine needle guided by CT, US, or endoscopy.

7 Subtract fluid albumin from serum albumin to obtain serum-ascites albumin gradient (SAAG). Gradient <11g/L suggests malignancy, infections, or pancreatitis.

**The first successful laparotomy...**

In 1809, an American surgeon by the name of Ephraim McDowell performed an astonishing operation: the first successful elective laparotomy for an abdominal tumour. It was an ovariotomy for an ovarian mass in a 44-year-old who, prior to physical examination by McDowell, was believed to be gravid. Not only was this feat performed in the age before anaesthesia and antisepsis, but it was also performed on a table in the front room of McDowell's Kentucky home, at that time on the frontier of the West in the United States. His account of the operation makes fascinating reading. While the strength of his diagnostic convictions combined with his speed and skill at operating is to be admired (the operation took 25 minutes), there is an even more laudable part played in this story. The patient, Mrs Jane Todd-Crawford, was fully willing to be involved with what can only be described as experimental surgery in the face of uncertainty. She defied pain simply by reciting psalms and hymns, and was back at home within 4 weeks with no complications, ultimately living another 33 years. We would be well served in remembering the exceptional commitment of Mrs Todd-Crawford. In the rush and hurry of our daily tasks perhaps it is all to easy to forget that the undertaking of surgery today may be no less fear-provoking for patients than it was 200 years ago.

# The acute abdomen

Someone who becomes acutely ill and in whom symptoms and signs are chiefly related to the abdomen has an acute abdomen. Prompt laparotomy is sometimes essential: *repeated examination is the key to making the decision*.

### Clinical syndromes that usually require laparotomy

1 *Rupture of an organ* (Spleen, aorta, ectopic pregnancy.) Shock is a leading sign—see table 13.10 for assessment of blood loss. Abdominal swelling may be seen. Any history of trauma: blunt trauma → spleen; penetrating trauma → liver? *Delayed* rupture of the spleen may occur weeks after trauma. Peritonism may be mild.

2 *Peritonitis* (Perforation of peptic ulcer/duodenal ulcer, diverticulum, appendix, bowel, or gallbladder.) Signs: prostration, shock, lying still, +ve cough test (p62), tenderness (± rebound/percussion pain, p62), board-like abdominal rigidity, guarding, and no bowel sounds. Erect CXR may show gas under the diaphragm (fig 13.26). NB: acute pancreatitis (p636) causes these signs, but does *not* require a laparotomy so don't be caught out and ►*always check serum amylase*.

### Syndromes that may not require a laparotomy

*Local peritonitis:* Eg diverticulitis, cholecystitis, salpingitis, and appendicitis (the latter *will* need surgery). If abscess formation is suspected (swelling, swinging fever, and ↑WCC) do US or CT. Drainage can be percutaneous (US or CT-guided), or by laparotomy. Peritoneal inflammation can cause localized ileus with a 'sentinel loop' of intraluminal gas visible on plain AXR (p729).

*Colic* is a regularly waxing and waning pain, caused by muscular spasm in a hollow viscus, eg gut, ureter, salpinx, uterus, bile duct, or gallbladder (in the latter, pain is often dull and constant). Colic, unlike peritonitis, causes restlessness and the patient may well be pacing around when you go to review!

**Obstruction of the bowel** See p610.

**Tests** U&E; FBC; amylase; LFT; CRP; lactate (is there mesenteric ischaemia?); urinalysis. ►►Urine and serum hCG is vital to exclude ectopic pregnancy. Erect CXR (fig 13.26), AXR may show Rigler's sign (p728). Laparoscopy may avert open surgery. CT can be helpful provided it is readily available and causes no delay (pp732–3); US may identify perforation or free fluid (appropriate performer training is important).

**Pre-op** ►Don't rush to theatre. *Anaesthesia compounds shock*, so resuscitate properly first (p790) unless blood being lost faster than can be replaced, eg ruptured ectopic pregnancy, (*OHCS* p262), aneurysm leak (p654), trauma.

**Plan** Bed rest, keep NBM; assess volume status (BOX) and treat shock (p790); cross-match/group and save; analgesia (p574); arrange imaging; consider need for IVI, blood cultures, and antibiotics (eg piperacillin/tazobactam 4.5g/8h IV); ECG.

**The medical acute abdomen** Irritable bowel syndrome (p266) is the chief cause, so always ask about episodes of pain associated with loose stools, relieved by defecation, bloating, and urgency (but *not* blood—this may be UC). Other causes:

| | | |
|---|---|---|
| ►►Myocardial infarction | Pneumonia (p166) | Sickle-cell crisis (p341) |
| Gastroenteritis or UTI | Thyroid storm (p834) | Phaeochromocytoma (p837) |
| Diabetes mellitus/DKA (p206) | Zoster (p404) | Malaria (p416) |
| Bornholm disease | Tuberculosis (p393) | Typhoid fever (p415) |
| Pneumococcal peritonitis | Porphyria (p692) | Cholera (p430) |
| Henoch–Schönlein (p702) | Narcotic addiction | *Yersinia enterocolitica* (p431) |
| Tabes dorsalis (p412) | PAN (p556) | Lead colic |

**Hidden diagnoses** ►►Mesenteric ischaemia (p620), ►►acute pancreatitis (p636), and ►►leaking AAA (p654) are the *Unterseeboote* of the acute abdomen—unsuspected, undetectable unless carefully looked for, and underestimatedly deadly. They may have non-specific symptoms and signs that are surprisingly mild, so always think of them when assessing the acute abdomen and hopefully you will 'spot' them! ►Finally: *always exclude pregnancy (± ectopic?) in females.*

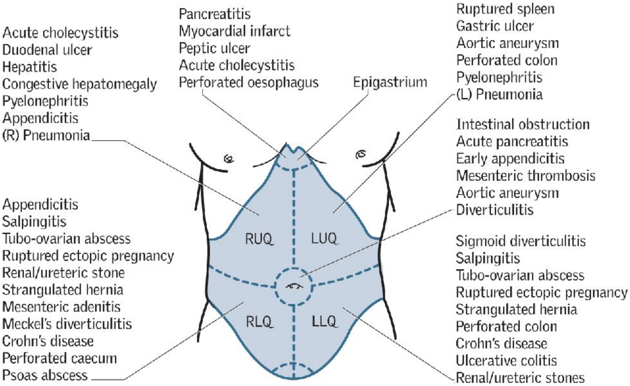


**Fig 13.25** Causes of abdominal pain.

## Assessing hypovolaemia from blood loss

►►Treat suspected shock rather than wait for BP to fall. The most likely cause of shock in a surgical patient is hypovolaemia. Check urine output, GCS, and capillary refill (CR) as measures of renal, brain, and skin perfusion.

When there is any blood loss, assess the status of the following:

**Table 13.10** Estimating blood loss based on patient's initial presentation

| Parameter | Class I | Class II | Class III | Class IV |
|---|---|---|---|---|
| Blood loss | <750mL | 750–1500mL | 1500–2000mL | >2000mL |
| | <15% | 15–30% | 30–40% | >40% |
| Pulse | <100bpm | >100bpm | >120bpm | >140bpm |
| BP | ↔ | ↔ | ↓ | ↓ |
| Pulse pressure | ↔ or ↑ | ↓ | ↓ | ↓ |
| Respirations | 14–20/min | 20–30/min | 30–40/min | >35/min |
| Urine output | >30mL/h | 20–30mL/h | 5–15mL/h | Negligible |
| Mental state | Slightly anxious | Anxious | Confused → | →Lethargic |
| Fluid to give | Crystalloid | Crystalloid | Crystalloid + blood | |

Assumes a body mass of 70kg.

An adaptation of 'Estimated blood loss based on initial presentation' table from the 9th edition of the *Advanced Trauma Life Support Manual*. Adapted with permission from the American College of Surgeons.

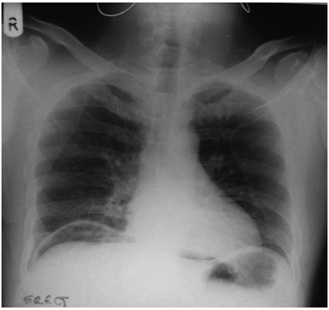


**Fig 13.26** Erect CXR showing air beneath the right hemidiaphragm, indicating presence of a pneumoperitoneum. *Causes:*

- Bowel perforation (visible only in 75%) (fig 13.25).
- Gas-forming infection, eg *C. perfringens*.
- Iatrogenic, eg laparoscopic surgery (detectable on CXR up to 10d post-op).
- *Per vaginam* (eg sexual activity).
- Interposition of bowel between liver and diaphragm (Chilaiditi sign—not true free air).

Image courtesy of Mr P. Paraskeva.

# Acute appendicitis

**Incidence** Most common surgical emergency (lifetime incidence = 6%). Can occur at any age, though highest incidence is between 10–20yrs.[8] It is rare before age 2 because the appendix is cone shaped with a larger lumen.

**Pathogenesis** Gut organisms invade the appendix wall after lumen obstruction by lymphoid hyperplasia, faecolith, or filarial worms. This leads to oedema, ischaemic necrosis, and perforation.

**Presentation** Classically periumbilical pain that moves to the RIF. Associated signs may include tachycardia, fever, peritonism with guarding and rebound or percussion tenderness in RIF. Pain on right during PR examination suggests an inflammed, low-lying pelvic appendix. Anorexia is an important feature; vomiting is rarely prominent—*pain normally precedes vomiting* in the surgical abdomen. Constipation is usual, though diarrhoea may occur. *Additional signs: Rovsing's sign* (pain > in RIF than LIF when the LIF is pressed). *Psoas sign* (pain on extending hip if retrocaecal appendix). *Cope sign* (pain on flexion and internal rotation of right hip if appendix in close relation to obturator internus).



**Investigations** Blood tests may reveal neutrophil leucocytosis and elevated CRP. US may help, but the appendix is not always visualized. CT has high diagnostic accuracy and is useful if diagnosis is unclear: it reduces −ve appendicectomy rate.

### Variations in the clinical picture

- Inflammation in a retrocecal/retroperitoneal appendix (2.5%) may cause flank or RUQ pain; its only sign may be ↑tenderness on the right on PR.
- The child with vague abdominal pain who will not eat their favourite food.
- The shocked, confused octogenarian who is not in pain.
- Appendicitis occurs in ~1/1000 pregnancies. Mortality is higher, especially from 20wks' gestation. Perforation is more common, and increases fetal mortality. Pain is often less well localized (may be RUQ) and signs of peritonism less obvious.

### Hints

- If a child is anxious, use their hand to press their tummy.
- Check for recent viral illnesses and lymphadenopathy—mesenteric adenitis?
- Don't *start* palpating in the RIF (makes it difficult to elicit pain elsewhere).
- Expect diagnosis to be wrong half the time. If diagnosis is uncertain, re-examine often. A normal appendix is removed in up to 20% of patients.

**Treatment** Prompt *appendicectomy* (fig 13.27). *Antibiotics:* piperacillin/tazobactam 4.5g/8h, 1 to 3 doses IV starting 1h pre-op, reduces wound infections. Give a longer course if perforated. *Laparoscopy:* Has diagnostic and therapeutic advantages (if surgeon experienced), especially in women and the obese. It is not recommended in cases of suspected gangrenous perforation as the rate of abscess formation may be higher.

### Complications

- *Perforation* is commoner if a faecolith is present and in young children, as the diagnosis is more often delayed.
- *Appendix mass* may result when an inflamed appendix becomes covered with omentum. US/CT may help with diagnosis. Some advocate early surgery. Alternatively, initial conservative management—NBM and antibiotics. If the mass resolves, some perform an interval (ie delayed) appendicectomy. Exclude a colonic tumour (laparotomy or colonoscopy), which can present as early as the 4th decade.
- *Appendix abscess* May result if an appendix mass fails to resolve but enlarges and the patient gets more unwell. Treatment usually involves drainage (surgical or percutaneous under US/CT-guidance). Antibiotics alone may bring resolution.

8 There is a second peak between 60–70yrs; older adults may present later with atypical symptoms.

## Explaining the patterns of abdominal pain

Internal organs and the visceral peritoneum have no somatic innervation, so the brain attributes the visceral (splanchnic) signals to a physical location whose dermatome corresponds to the same entry level in the spinal cord. Importantly, there is no laterality to the visceral unmyelinated C-fibre pain signals, which enter the cord bilaterally and at multiple levels. Division of the gut according to embryological origin is the important determinant here: see table 13.11.

Table 13.11 Somatic referral of abdominal pain

| Gut | Division points | Somatic referral | Arterial supply |
|---|---|---|---|
| Fore | Proximal to 2nd part of duodenum | Epigastrium | Coeliac axis |
| Mid | Above to ⅔ along transverse colon | Periumbilical | Superior mesenteric |
| Hind | Distal to above | Suprapubic | Inferior mesenteric |

Early inflammation irritates the structure and walls of the appendix, so a colicky pain is referred to the mid-abdomen—classically periumbilical. As the inflammation progresses and irritates the parietal peritoneum (especially on examination), the somatic, lateralized pain settles at McBurney's point, ⅔ of the way along from the umbilicus to the right anterior superior iliac spine.

These principles also help us understand patterns of *referred pain*. In pneumonia, the T9 dermatome is shared by the lung and the abdomen. Also, irritation of the underside of the diaphragm (sensory innervation is from above through the phrenic nerve, C3–5) by an inflamed gallbladder or a subphrenic abscess refers pain to the right shoulder: dermatomes C3–5.

## ΔΔ

- Ectopic (►►do a pregnancy test!)
- UTI (test urine!)
- Mesenteric adenitis
- Cystitis.
- Cholecystitis
- Diverticulitis
- Salpingitis/PID
- Dysmenorrhoea
- Crohn's disease
- Perforated ulcer
- Food poisoning
- Meckel's diverticulum

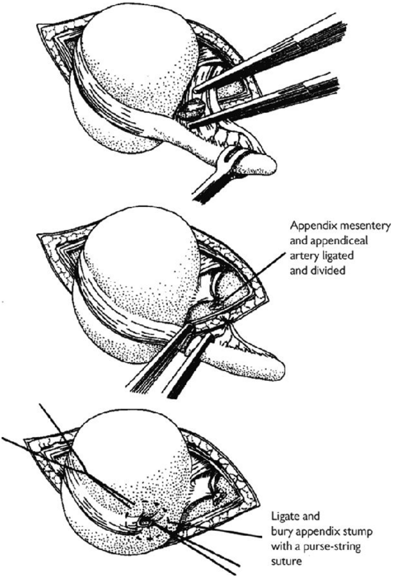


Fig 13.27 Appendicectomy.

Reproduced from McLatchie *et al.*, *Operative Surgery*, 2006, with permission from Oxford University Press.

# Obstruction of the bowel

**Cardinal features of intestinal obstruction** •*Vomiting*,[9] nausea and anorexia. •*Colic* occurs early (↓ in long-standing obstruction). •*Constipation* may be absolute (ie no faeces or flatus passed) in distal obstruction; less pronounced if obstruction is high. •*Abdominal distension* ↑ as the obstruction progresses with active, 'tinkling' bowel sounds.

## The key decisions

1. *Is it obstruction of the small or large bowel?* In small bowel obstruction, vomiting occurs early, distension is less, and pain is higher in the abdomen; in large bowel obstruction, pain is more constant. The AXR plays a key role (fig 13.28 & p728).
2. *Is there an ileus or mechanical obstruction?* Ileus is functional obstruction from ↓bowel motility (see BOX 'Paralytic ileus or pseudo-obstruction?' & p728). Bowel sounds are absent; pain tends to be less.
3. *Is the obstructed bowel simple/closed loop/strangulated? Simple:* one obstructing point and no vascular compromise. *Closed loop:* obstruction at two points (eg sigmoid volvulus) forming a loop of grossly distended bowel at risk of perforation. *Strangulated:* blood supply is compromised and the patient is iller than you would expect. There is sharper, more constant, and *localized* pain. Peritonism is the cardinal sign. There may be fever + ↑WCC with other signs of mesenteric ischaemia (p620).

**Causes** See table 13.12.

Table 13.12 Causes of bowel obstruction

| Causes: small bowel | Causes: large bowel | Rarer causes |
|---|---|---|
| • Adhesions (p581) | • Colon ca (p616) | • Crohn's stricture |
| • Hernias (p612) | • Constipation (p260) | • Gallstone ileus (p634) |
| | • Diverticular stricture | • Intussusception |
| | • Volvulus | • TB (developing world) |
| | • Sigmoid (see BOX 'Sigmoid volvulus') | • Foreign body |
| | • Caecal | |

## Management

- *General principles:* Cause, site, speed of onset, and completeness of obstruction determine definitive therapy: strangulation and large bowel obstruction require surgery; ileus and incomplete small bowel obstruction can be managed conservatively, at least initially.
- *Immediate action:* ►'Drip and suck'—NGT and IV fluids to rehydrate and correct electrolyte imbalance (p668). Being NBM does not give adequate rest for the bowel because it can produce up to 9L of fluid/d. Also: analgesia, blood tests (inc. amylase, FBC, U&E), AXR, erect CXR, catheterize to monitor fluid status.
- *Further imaging:* CT to establish the cause of obstruction (may show dilated, fluid-filled bowel and a transition zone at the site of obstruction—figs 13.29, 13.30). Oral Gastrografin® prior to CT can help identify level of obstruction and may have mild therapeutic action against mechanical obstruction. Consider investigating the cause of large bowel obstruction by colonoscopy but beware risk of perforation.
- *Surgery:* ►Strangulation needs emergency surgery. Closed loop obstruction may be managed with surgery or endoscopic decompression attempted. Endoscopic stenting may be used for obstructing large bowel malignancies either in palliation or as a bridge to surgery in acute obstruction (p616). Small bowel obstruction secondary to adhesions should rarely lead to surgery—see BOX, p581.

9 Fermentation of the intestinal contents in established obstruction causes 'faeculent' vomiting. True 'faecal' vomiting is found when there is a colonic fistula with the proximal gut.

## Paralytic ileus or pseudo-obstruction?

**Paralytic ileus** is adynamic bowel due to the absence of normal peristaltic contractions. Contributing factors include abdominal surgery, pancreatitis (or any localized peritonitis), spinal injury, hypokalaemia, hyponatraemia, uraemia, peritoneal sepsis and drugs (eg tricyclic antidepressants).

**Pseudo-obstruction** resembles mechanical GI obstruction but with no obstructing lesion. *Acute* colonic pseudo-obstruction is called Ogilvie's syndrome (p706), and clinical features are similar to that of mechanical obstruction. Predisposing factors: puerperium; pelvic surgery; trauma; cardiorespiratory and neurological disorders. *Treatment:* Neostigmine or colonoscopic decompression are sometimes useful. In chronic pseudo-obstruction weight loss from malabsorption is a problem.

## Sigmoid volvulus

Sigmoid volvulus occurs when the bowel twists on its mesentery, which can produce severe, rapid, strangulated obstruction (fig 13.28c). It tends to occur in the elderly, constipated, and comorbid patient, and is managed by insertion of a flatus tube or sigmoidoscopy. Sigmoid colectomy is sometimes required. ►If not treated successfully, it can progress to perforation and fatal peritonitis.

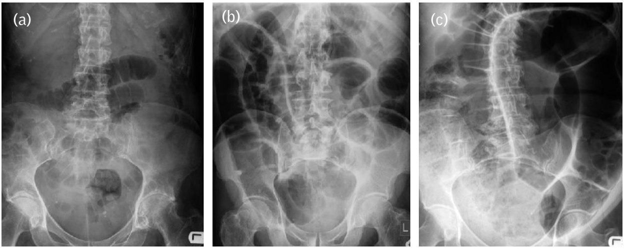


Fig 13.28 (a) Small bowel obstruction: AXR shows central gas shadows with *valvulae conniventes* that completely cross the lumen and no gas in the large bowel. (b) Large bowel obstruction: AXR shows peripheral gas shadows proximal to the blockage (eg in caecum) but not in the rectum. (c) Sigmoid volvulus: there is a characteristic AXR with an 'inverted U' loop of bowel that looks a bit like a coffee bean.

Images (a), (b), and (c) reproduced from Darby *et al.*, *Oxford Handbook of Medical Imaging*, 2011, with permission from Oxford University Press.

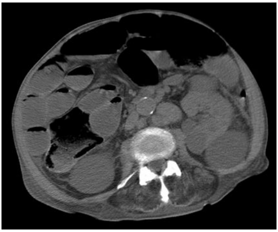

Fig 13.29 Unenhanced axial CT of the abdomen showing multiple loops of dilated, fluid-filled small bowel in a patient with small bowel obstruction.

Image courtesy of Norwich Radiology Dept.

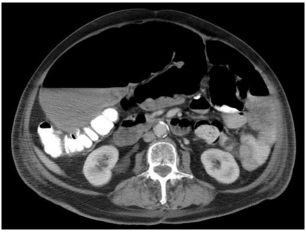

Fig 13.30 Axial CT of the abdomen post-oral contrast showing dilated loops of fluid and air-filled large bowel (contrast medium is in the small bowel).

Image courtesy of Norwich Radiology Dept.

# Abdominal hernias

**Definition** The protrusion of a viscus or part of a viscus through a defect of the walls of its containing cavity into an abnormal position. See fig 13.31. *Terminology:*
- *Irreducible:* contents cannot be pushed back into place (see p614 for technique).
- *Obstructed:* bowel contents cannot pass—features of intestinal obstruction (p610).
- *Strangulated:* ischaemia occurs—the patient requires urgent surgery.
- *Incarceration:* contents of the hernial sac are stuck inside by adhesions.

Care must be taken with reduction as it is possible to push an incarcerated hernia back into the abdominal cavity, giving the initial appearance of successful reduction.

**Inguinal hernia** The commonest type in both ♂ & ♀ (but ♂>>♀), p614.

**Femoral hernia** Bowel enters the femoral canal, presenting as a mass in the upper medial thigh or above the inguinal ligament where it points down the leg, unlike an inguinal hernia which points to the groin. They occur more often in ♀ especially in middle age and the elderly. They are likely to be irreducible and to strangulate due to the rigidity of the canal's borders. *Anatomy:*See fig 13.32 *Differential diagnosis:* (See p651.) **1** Inguinal hernia. **2** Saphena varix. **3** An enlarged Cloquet's node (p615). **4** Lipoma. **5** Femoral aneurysm. **6** Psoas abscess. *Treatment:* Surgical repair is recommended. *Herniotomy* is ligation and excision of the sac, *herniorrhaphy* is repair of the hernial defect.

**Paraumbilical hernias** occur just above or below the umbilicus. Risk factors are obesity and ascites. Omentum or bowel herniates through the defect. Surgery involves repair of the rectus sheath (Mayo repair).

**Epigastric hernias** pass through linea alba above the umbilicus.

**Incisional hernias** follow breakdown of muscle closure after surgery (11–20%). If obese, repair is not easy. Mesh repair has ↓recurrence but ↑infection over sutures.

**Spigelian hernias** occur through the linea semilunaris at the lateral edge of the rectus sheath, below and lateral to the umbilicus.

**Lumbar hernias** occur through the inferior or superior lumbar triangles in the posterior abdominal wall.

**Richter's hernias** involve bowel wall only—not the whole lumen.

**Maydl's hernias** involve a herniating 'double loop' of bowel. The strangulated portion may reside as a single loop *inside* the abdominal cavity.

**Littré's hernias** are hernial sacs containing strangulated Meckel's diverticulum.

**Obturator hernias** occur through the obturator canal. Typically there is pain along the medial side of the thigh in a thin woman.

**Sciatic hernias** pass through the lesser sciatic foramen (a way through various pelvic ligaments). GI obstruction + a gluteal mass suggests this rare possibility.

**Sliding hernias** contain a partially extraperitoneal structure (eg caecum on the right, sigmoid colon on the left). The sac does not completely surround the contents.

**Paediatric hernias** include *Umbilical hernias:* (3% of live births). Are a result of a persistent defect in the transversalis fascia. Surgical repair rarely needed as most resolve by the age of 3. *Indirect inguinal hernias* (~4% of all ♂ infants due to patent *processus vaginalis*—prematurity is a risk factor; uncommon in ♀ infants—consider testicular feminization.) Surgical repair is required. *Gastroschisis:* Protrusion of the abdominal contents through a defect in the anterior abdominal wall to the right of the umbilicus. Prompt surgical repair required. *Exomphalos:* Abdominal contents are found outside the abdomen, covered in a three-layer membrane consisting of peritoneum, Wharton's jelly, and amnion. Surgical repair less urgent because the bowel is protected by these membranes.

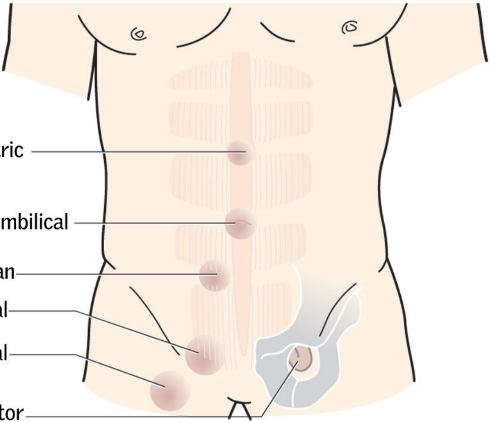


Fig 13.31 Some examples of hernias.

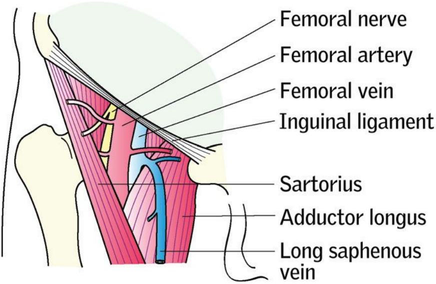


Fig 13.32 The boundaries of the femoral canal are anteriorly the inguinal ligament; medially the lacunar ligament (and pubic bone); laterally the femoral vein (and iliopsoas); and posteriorly the pectineal ligament and pectineus. The canal contains fat and Cloquet's node. The neck of the hernia is felt inferior and lateral to the pubic tubercle (inguinal hernias are superior and medial to this point).

# Inguinal hernias



*Indirect* hernias pass through the internal inguinal ring and, if large, out through the external ring (fig. 13.33). *Direct* hernias push their way *directly* forward through the posterior wall of the inguinal canal, into a defect in the abdominal wall (Hesselbach's triangle; medial to the inferior epigastric vessels and lateral to the rectus abdominus). *Predisposing conditions:* males (♂:♀≈8:1), chronic cough, constipation, urinary obstruction, heavy lifting, ascites, past abdominal surgery (eg damage to the iliohypogastric nerve during appendicectomy). There are two landmarks to identify: *the deep (internal) ring* may be defined as being the mid-point of the inguinal ligament, ~1½cm above the femoral pulse (which crosses the mid-inguinal point); *the superficial (external) ring* is a split in the external oblique aponeurosis just superior and medial to the pubic tubercle (the bony prominence forming the medial attachment of the inguinal ligament).

**Examination** Look for previous scars; feel the other side (more common on the right); examine the external genitalia. Then ask: •Is the lump visible? If so, ask the patient to reduce it—if he cannot, make sure that it is not a scrotal lump. Ask him to cough. Appears *above and medial to the pubic tubercle*. •If no lump is visible, feel for a cough impulse. •Repeat the examination with the patient standing. *Distinguishing direct from indirect hernias:* This is loved by examiners but is of little clinical use—not least because repair is the same for both (see 'Repairs' later in topic). The best way is to reduce the hernia and occlude the deep (internal) ring with two fingers. Ask the patient to cough or stand—if the hernia is restrained, it is indirect; if not, it is direct. The 'gold standard' for determining the type of inguinal hernia is at surgery: direct hernias arise medial to the inferior epigastric vessels; indirect hernias are lateral.

| Indirect hernias: | Direct hernias: | Femoral hernias: |
|---|---|---|
| • Common (80%)<br>• Can strangulate. | • Less common (20%)<br>• Reduce easily<br>• Rarely strangulate. | • More frequent in females<br>• Frequently irreducible<br>• Frequently strangulate. |

**Irreducible hernias** You may be called because a long-standing hernia is now irreducible and painful. It is always worth trying to reduce these yourself to prevent strangulation and necrosis (demanding prompt laparotomy). Learn how to do this from an expert, ie one of your patients who has been reducing his hernia for years. Then you will know how to act correctly when the emergency presents. Notice that such patients use the flat of the hand, directing the hernia from below, up towards the contralateral shoulder. Sometimes, as the hernia obstructs, reduction requires perseverance, which may be rewarded by a gurgle from the retreating bowel and a kiss from the attending spouse who had thought that surgery was inevitable.

**Repairs** Weight loss (if over-weight) and stop smoking pre-op. Warn that hernias may recur and patients should be counselled about possibility of chronic pain post-operatively. Mesh techniques (eg Lichtenstein repair) have replaced older methods. In mesh repairs, a polypropylene mesh reinforces the posterior wall. Recurrence rate is less than with other methods (eg <2% vs 10%). (CI: strangulated hernias, contamination with pus/bowel contents.) Local anaesthetic techniques and day-case 'ambulatory' surgery may halve the price of surgery. This is important because this is one of the most common operations (>100 000 per year in the UK). *Laparoscopic repair* gives similar recurrence rates. Methods include *transabdominal pre-peritoneal* (TAPP) in which the peritoneum is entered and the hernia repaired, and *totally extraperitoneal* (TEP), which decreases the risk of visceral injury. For benefits of laparoscopic surgery see p592.

*Return to work:* Will depend upon surgical approach and patient—discuss this pre-operatively. Rest for 4wks and convalescence over 8wks with open approaches, but laparoscopic repairs may allow return to manual work (and driving) after ≤2wks if all is well.

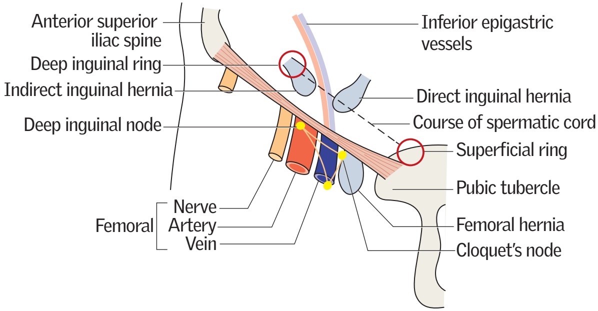


Fig 13.33 Anatomy of the inguinal canal. *Floor:* Inguinal ligament and lacunar ligament medially; *Roof:* Fibres of transversalis, internal oblique; *Anterior:* External oblique aponeurosis + internal oblique for the lateral ⅓; *Posterior:* Laterally, transversalis fascia; medially, conjoint tendon.

## The contents of the inguinal canal in the male

- The external spermatic fascia (from external oblique), cremasteric fascia (from internal oblique and transverses abdominus), and internal spermatic fascia (from transversalis fascia) covering the cord.
- The spermatic cord:
  - Vas deferens, obliterated processus vaginalis, and lymphatics.
  - Arteries to the vas, cremaster, and testis.
  - The pampiniform plexus and the venous equivalent of the above.
  - The genital branch of the genitofemoral nerve and sympathetic nerves.
- The ilioinguinal nerve, which enters the inguinal canal via the anterior wall and runs anteriorly to the cord.

NB: in the female the round ligament of the uterus is in place of the male structures. A hydrocele of the canal of Nuck is the female equivalent of a hydrocele of the cord.

# Colorectal carcinoma



This is the 3rd most common cancer and 2nd most common cause of UK cancer deaths (16 000 deaths/yr). Usually adenocarcinoma. 86% of presentations are in those >60yrs old. Lifetime UK incidence: ♂ = 1:15; ♀ = 1:19.

**Predisposing factors** Neoplastic polyps (see BOX & p520); IBD (UC and Crohn's); genetic predisposition (<8%), eg FAP and HNPCC (see p521); diet (low-fibre; ↑red and processed meat); ↑alcohol; smoking; previous cancer. *Prevention:* While routine chemoprevention is not currently recommended due to gastrointestinal SEs, aspirin ≥75mg/d reduces incidence and mortality.

**Presentation** depends on site: *Left-sided:* Bleeding/mucus PR; altered bowel habit or obstruction (25%); tenesmus; mass PR (60%). *Right:* ↓Weight; ↓Hb; abdominal pain; obstruction less likely. *Either:* Abdominal mass; perforation; haemorrhage; fistula. See p522 for a guide to urgent referral criteria. See fig 13.34 for distribution.

**Tests** FBC (microcytic anaemia); faecal occult blood (FOB, see BOX); sigmoidoscopy or colonoscopy (figs 6.7 & 6.8, p249), which can be done 'virtually' by CT (fig 16.31, p743); LFT; liver MRI/US. CEA (p531) may be used to monitor disease and effectiveness of treatment. If family history of FAP, refer for DNA test once >15yrs old.

**Spread** Local, lymphatic, by blood (liver, lung, bone) or transcoelomic. The TNM system (Tumour, Node, Metastases see table 13.13 and p523) is used to stage disease and is preferred to the older Dukes' classification (Dukes A: limited to muscularis mucosae; Dukes B: extension through muscularis mucosae; Dukes C: involvement of regional lymph nodes).

**Surgery** aims to cure and may ↑ survival times by up to 50%. In elective surgery, anastomosis is typically achieved at the 1st operation. *Laparoscopic surgery* has revolutionized surgery for colon cancer. It is as safe as open surgery and there is no difference in overall survival or disease recurrence. •Right hemicolectomy for caecal, ascending, or proximal transverse colon tumours. •Left hemicolectomy for tumours in distal transverse or descending colon. •Sigmoid colectomy for sigmoid tumours. •Anterior resection for low sigmoid or high rectal tumours. •Abdomino-perineal (AP) resection for tumours low in the rectum (≲8cm from anus): permanent colostomy and removal of rectum and anus. •Hartmann's procedure in emergency bowel obstruction, perforation, or palliation (p582). •Transanal endoscopic microsurgery allows local excision through a wide proctoscope for localized rectal disease. *Endoscopic stenting* should be considered for palliation in malignant obstruction and as a bridge to surgery in acute obstruction. Stenting ↓ need for colostomy, has less complications than emergency surgery, shortens intensive care and total hospital stays, and prevents unnecessary operations. Surgery with liver resection may be curative if single-lobe hepatic metastases and no extrahepatic spread.

**Radiotherapy** is mostly used in palliation for colonic cancer. It is occasionally used pre-op in rectal cancer to allow resection. Post-op radiotherapy is only used in patients with rectal tumours at high risk of local recurrence.

**Chemotherapy** Adjuvant chemotherapy for stage 3 disease has been shown to reduce disease recurrence by 30% and mortality by 25%. Benefits for stage 2 disease are more marginal and warrant an individualized approach. The FOLFOX regimen has become standard (fluorouracil, folinic acid and oxaliplatin). Chemotherapy is also used in palliation of metastatic disease. *Biological therapies:* Bevacizumab (anti-VEGF antibody) improves survival when added to combination therapy in advanced disease. Cetuximab and panitumumab (anti-EGFR agents) improve response rate and survival in KRAS wild-type metastatic colorectal cancer.

**Prognosis** Survival is dependent on age and stage; for stage 1 disease, 5yr survival is ~75% but this drops to just 5% with diagnosis at stage 4, hence the imperative for effective screening (BOX).

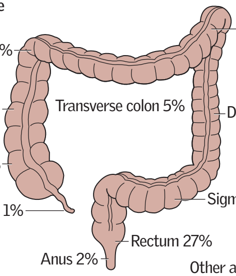


Fig 13.34 Distribution of colorectal carcinomas. These are averages: black females tend to have more proximal neoplasms. White men tend to have more distal neoplasms.

## TNM staging in colorectal cancer

Table 13.13 Colorectal cancer: TNM staging

| | | | |
|---|---|---|---|
| Tx | Primary tumour cannot be assessed | Nx | Nodes cannot be assessed |
| $T_{is}$ | Carcinoma *in situ* | N0 | No node spread |
| T1 | Invading submucosa | N1 | Metastases in 1–3 regional nodes |
| T2 | Invading muscularis propria | N2 | Metastases in >3 regional nodes |
| T3 | Invading subserosa and beyond (not other organs) | M0 | No distant spread |
| T4 | Invasion of adjacent structures | M1 | Distant metastasis |

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TNM status used to define overall stage. This is complex with several important subtypes, but in essence, stage 1 disease is T1 or T2/N0/M0; stage 2 is T3 or T4/N0/M0; stage 3 is characterized by N1 or N2 but still M0; stage 4 is M1.

## Polyps, the challenges of screening, and the NHS

Polyps are growths that appear above the mucosa and can be inflammatory, hamartomatous, or neoplastic. Left *in situ*, polyps carry a risk of malignant transformation that will relate to size and histology (tubular or villous adenomas, esp. if >2cm). Patients with polyps may have no symptoms and thus a colonoscopy is required to detect and remove. Colonscopy allows the opportunity to detect colorectal cancer at an earlier stage when treatment may be more effective.

However, population-based colonoscopic screening is costly and some studies have suggested that the test does not impact on deaths from right-sided cancers which are rarer and harder to detect (fig 13.34). Therefore, the NHS has introduced a one-off screening flexible sigmoidoscopy offered to all people in their 55th year. Trial results have shown the incidence of colorectal cancer in the intervention (screening) group is reduced by 33% and mortality from colorectal cancer is reduced by 43%. Number needed to screen to prevent one diagnosis (=191); or death (=489).

In parallel, the NHS Bowel Cancer Screening Programme (introduced in 2006) offers colonoscopy to all men and women aged 60–75 who test positive for faecal occult blood (FOB) using a home testing kit performed every 2 years. This FOB-stratification targets screening to those in the highest risk groups, permitting detection of more advanced adenomas and early stage cancers. The relative risk of death from colorectal cancer in patients undergoing screening is reduced by 16%. A 11% increase in incidence rates since 2006 for people aged 60–69 is almost certainly due to earlier detection through the screening programme.

# Carcinoma of the oesophagus

**Incidence** Australia <5/100 000/yr; UK <9; Iran >100. *Risk factors:* Diet, alcohol excess, smoking, achalasia, reflux oesophagitis ± Barrett's oesophagus (p695); obesity, hot drinks, nitrosamine exposure, Plummer–Vinson syndrome (p250). ♂:♀≈5:1.

**Site** 20% occur in the upper part, 50% in the middle, and 30% in the lower part. They may be squamous cell (proximal) or adenocarcinomas (distal; incidence rising).

**Presentation** Dysphagia; ↓weight; retrosternal chest pain. *Signs from the upper third of the oesophagus:* Hoarseness; cough (may be paroxysmal if aspiration pneumonia). *ΔΔ:* See 'Dysphagia', p250.

**Tests** Oesophagoscopy with biopsy is the investigation of choice ± EUS, CT/MRI for staging (fig 13.35), or laparoscopy if significant infra-diaphragmatic component. *Staging:* See table 13.14.



**Treatment** Survival rates are poor with or without treatment. If localized T1/T2 disease, radical curative oesophagectomy may be tried. Pre-op chemotherapy (cisplatin + fluorouracil) for localized disease may improve survival, but causes some morbidity. If surgery is *not* indicated, then chemoradiotherapy may be better than radiotherapy alone. Palliation in advanced disease aims to restore swallowing with chemo/radiotherapy, stenting, and laser use.

## TNM staging in oesophageal cancer

Spread of oesophageal cancer is direct, by submucosal infiltration and local spread—or to nodes, or, later, via the blood.

Table 13.14 Oesophageal cancer: TNM staging

| | | | |
|---|---|---|---|
| **$T_{is}$** | Carcinoma *in situ* | **Nx** | Nodes cannot be assessed |
| **T1** | Invading lamina propria/submucosa | **N0** | No node spread |
| **T2** | Invading muscularis propria | **N1–N3** | Regional node metastases |
| **T3** | Invading adventitia | **M0** | No distant spread |
| **T4** | Invasion of adjacent structures | **M1** | Distant metastasis |

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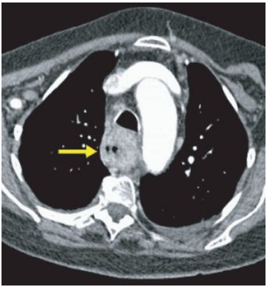

Fig 13.35 Axial CT of the chest after IV contrast medium showing concentric thickening of the oesophagus (arrow); the diagnosis here is oesophageal carcinoma. Loss of the fatty plane around the oesophagus suggests local invasion. Anterior to the oesophagus is the trachea and next to it is the arch of the aorta.

Image courtesy of Dr Stephen Golding.

## Carcinoma of the stomach

Incidence of adenocarcinoma at the gastro-oesophageal junction is increasing in the West, though incidence of distal and gastric body carcinoma has fallen sharply. It remains a tumour notable for its gloomy prognosis and non-specific presentation.

**Incidence** 23/100 000/yr in the UK, but there are unexplained wide geographical variations; it is especially common in Japan, as well as Eastern Europe, China, and South America. ♂:♀≈2:1. *Risk factors:* Pernicious anaemia, blood group A, *H. pylori* (p252), atrophic gastritis, adenomatous polyps, lower social class, smoking, diet (high nitrate, high salt, pickling, low vitamin C), nitrosamine exposure.

**Pathology** A range of clinical and histological classifications are in use. Of note, 'early' gastric carcinoma (confined to mucosa and submucosa) carries a better prognosis with endoscopic resection often possible.

**Presentation** *Symptoms:* Often non-specific. Dyspepsia (p59; age ≥55yrs with treatment-refractory symptoms demands investigation), ↓weight, vomiting, dysphagia, anaemia. *Signs* suggesting incurable disease: epigastric mass, hepatomegaly; jaundice, ascites (p604); large left supraclavicular (Virchow's) node (=Troisier's sign); acanthosis nigricans (p562). Most patients in the West present with locally advanced (inoperable) or metastatic disease. *Spread* is local, lymphatic, blood-borne, and transcoelomic, eg to ovaries (Krukenberg tumour).

**Tests** Gastroscopy + multiple ulcer edge biopsies. ►*Aim to biopsy all gastric ulcers as even malignant ulcers may appear to heal on drug treatment*. Endoscopic ultrasound (EUS) can evaluate depth of invasion; CT/MRI helps staging. Staging laparoscopy is recommended for locally advanced tumours. Cytology of peritoneal washings can help identify peritoneal metastases.

**Treatment** See p622 for a description of surgical resections. Early gastric cancers may be resectable endoscopically (endoscopic mucosal resection). Partial gastrectomy may suffice for more advanced distal tumours. If proximal, total gastrectomy may be needed. Combination chemotherapy (eg epirubicin, cisplatin and fluorouracil) appears to increase survival in advanced disease. If given perioperatively in operable disease it improves survival compared to surgery alone. Surgical palliation is often needed for obstruction, pain, or haemorrhage. In locally advanced and metastatic disease, chemotherapy increases quality of life and survival. Targeted therapies are likely to have an increasing role, eg trastuzumab for HER-2-positive tumours.

**5yr survival** <10% overall, but nearly 20% for patients undergoing radical surgery. The prognosis is much better for 'early' gastric carcinoma.

### Bile duct and gallbladder cancers

All are rare, have an overall poor prognosis, and are difficult to diagnose. They account for ~3% of all GI cancers worldwide, but there is geographical variation (↑ in north-east Thailand, Japan, Korea, and Eastern Europe). Most are adenocarcinomas. Primary sclerosing cholangitis (p282) is the commonest predisposing factor in the West. *Presentation:* Varies according to location and may include obstructive jaundice, pruritus, abdominal pain, weight loss, and anorexia. *Investigations:* US, CT, and ERCP. MRI has a role for determining extent of invasion in bile duct cancers.

*Treatment:*
- *Bile duct cancer:* surgical resection is the only potentially curative treatment yet ~80% present with inoperable disease. Palliation includes biliary stenting and chemotherapy.
- *Gallbladder cancer:* again, radical surgery is the only chance of cure. Patients with a calcified ('porcelain') gallbladder have an increased risk of cancer—prophylactic surgery should be considered. Palliative treatment of inoperable disease includes biliary stenting and chemotherapy.

# Bowel ischaemia



There are three main types of bowel ischaemia: ►AF with abdominal pain should always prompt thoughts of mesenteric ischaemia.

**1 Acute mesenteric ischaemia** almost always involves the small bowel and may follow superior mesenteric artery (SMA; fig 13.36) thrombosis (~35%) or embolism (~35%), mesenteric vein thrombosis (~5%; younger patients with hypercoagulable states—tends to affect smaller lengths of bowel), or non-occlusive disease (~20%; occurs in low-flow states and usually reflects poor cardiac output, though there may be other factors such as recent cardiac surgery or renal failure). Other causes include trauma, vasculitis (p556), radiotherapy, or strangulation (volvulus or hernia, p612).

*Presentation* is a classical clinical triad: ►acute severe abdominal pain; no/minimal abdominal signs; rapid hypovolaemia→shock. Pain tends to be constant, central, or around the RIF. The degree of illness is often far out of proportion with clinical signs.

*Tests:* There may be ↑Hb (due to plasma loss), ↑WCC, modestly raised plasma amylase, and a persistent metabolic acidosis (high lactate). Early on, the abdominal x-ray shows a 'gasless' abdomen. CT/MR may show evidence of ischaemia with CT/MR angiography or formal arteriography if doubt remains. Often the diganosis is made on finding a nasty, necrotic bowel at laparotomy.

*Treatment:* The main life-threatening complications secondary to acute mesenteric ischaemia are **1** septic peritonitis and **2** progression of a systemic inflammatory response syndrome (SIRS) to multi-organ failure, mediated by bacterial translocation across the dying gut wall. Resuscitation with fluid, antibiotics (eg piperacillin/tazobactam, see table 9.3, p386), and, usually, LMWH/heparin are required. If arteriography is done, thrombolytics may be infused locally via the catheter. At surgery, dead bowel must be removed. Revascularization may be attempted on potentially viable bowel but it is a difficult process and often needs a 2nd laparotomy.

*Prognosis:* Poor for arterial thrombosis and non-occlusive disease (<40% survive), though not so bad for venous and embolic ischaemia.

**2 Chronic mesenteric ischaemia** (AKA intestinal angina.) The triad of severe, colicky post-prandial abdominal pain ('gut claudication'), ↓weight (eating hurts), and an upper abdominal bruit may be present ± PR bleeding, malabsoprtion, and N&V. Typically brought about through a combination of a low-flow state with atheroma (95% due to diffuse atherosclerotic disease in all three mesenteric arteries). It is rare and difficult to diagnose. *Tests:* CT angiography and contrast-enhanced MR angiography are replacing traditional angiography. *Treatment:* Once diagnosis is confirmed, surgery should be considered due to the ongoing risk of acute infarction. Percutaneous transluminal angioplasty and stent insertion has replaced open revascularization. It is associated with less post-operative morbidity and mortality, but has higher restenosis rates.

**3 Chronic colonic ischaemia** (AKA ischaemic colitis) usually follows low flow in the inferior mesenteric artery (IMA) territory and ranges from mild ischaemia to gangrenous colitis. *Presentation:* Lower left-sided abdominal pain ± bloody diarrhoea. *Tests:* CT may be helpful but lower GI endoscopy is 'gold-standard'. *Treatment:* Usually conservative with fluid replacement and antibiotics. Most recover but subsequent development of ischaemic strictures is common. Gangrenous ischaemic colitis (presenting with peritonitis and hypovolaemic shock) requires prompt resuscitation followed by resection of the affected bowel and stoma formation. Mortality is high.

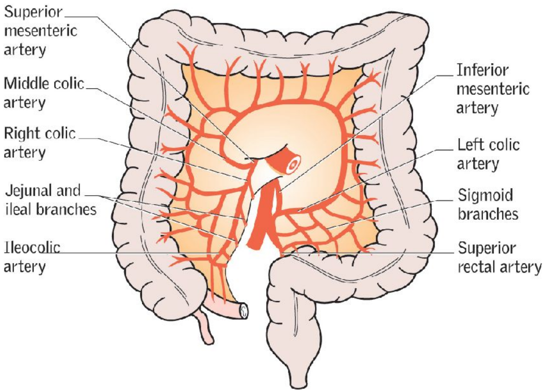


**Fig 13.36** The arterial supply to the colon.

# Gastric surgery and its aftermath



►Indications for gastric surgery include gastric cancer (p619) and perforated/haemorrhaging peptic ulcers. Medical therapy (p252) for peptic ulcers has made elective surgery exceedingly rare/redundant. ►►*Emergency surgery* may be needed for haemorrhage or perforation. Haemorrhage is usually treated by under-running the bleeding ulcer base or excision of the ulcer. If the former is done, then a biopsy should be taken to exclude malignancy. Perforation is usually managed by excision of the hole for histology, then closure.

**Gastric carcinoma** Localized disease may be treated by curative gastrectomy. Lesions in the proximal third or extensive infiltrative disease require total gastrectomy, while lesions in the distal two-thirds can be treated with a partial gastrectomy. Laparoscopic surgery may be as effective and safe as open surgery in specialist centres.

**Surgery:** *Billroth I:* Partial gastrectomy with simple gastroduodenal re-anastomosis. *Billroth II (AKA Polya) gastrectomy:* (fig 13.37) Partial gastrectomy with gastrojejunal anastomosis. The duodenal stump is oversewn (leaving a blind afferent loop), and anastomosis is achieved by a longitudinal incision into the proximal jejunum. *Roux-en-Y:* (fig 13.38) Following total or subtotal gastrectomy, the proximal duodenal stump is oversewn, the proximal jejunum is divided from the distal duodenum and connects with the oesophagus (or proximal stomach after subtotal gastrectomy), while the distal duodenum is connected to the distal jejunum.

*Lymph node clearance* is a controversial area. RCTs and meta-analyses suggest there may be limited benefit and increased morbidity associated with extended lymph node resections ($D_2$ or $D_3$) over resection limited to the perigastric nodes ($D_1$).

### Physical complications of gastrectomy

- *Abdominal fullness:* Feeling of early satiety (± discomfort and distension) improving with time. Advise to take small, frequent meals.
- *Afferent loop syndrome:* Post-gastrectomy (eg Billroth II), the afferent loop may fill with bile after a meal, causing upper abdominal pain and bilious vomiting. This is difficult to treat—but often improves with time.
- *Diarrhoea:* May be disabling after vagotomy. Codeine phosphate may help.
- *Gastric tumour:* A rare complication of any surgery which ↓acid production.
- *↑Amylase:* If with abdominal pain, this may indicate afferent loop obstruction after Billroth II surgery and requires emergency surgery.

### Metabolic complications

- *Dumping syndrome:* Fainting and sweating after eating due to food of high osmotic potential being dumped in the jejunum, causing oligaemia from rapid fluid shifts. 'Late dumping' is due to rebound hypoglycaemia and occurs 1–3h after meals. Both tend to improve with time but may be helped by eating less sugar, and more guar gum and pectin (slows glucose absorption). Acarbose may also help to reduce the early hyperglycaemic stimulus to insulin secretion.
- *Weight loss:* Often due to poor calorie intake.
- *Bacterial overgrowth ± malabsorption* (blind loop syndrome) may occur.
- *Anaemia:* Usually from lack of iron, hypochlorhydria, and stomach resection. $B_{12}$ levels are frequently low but megaloblastic anaemia is rare.
- *Osteomalacia:* There may be pseudofractures which look like metastases.

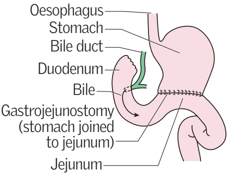


Fig 13.37 Billroth II.

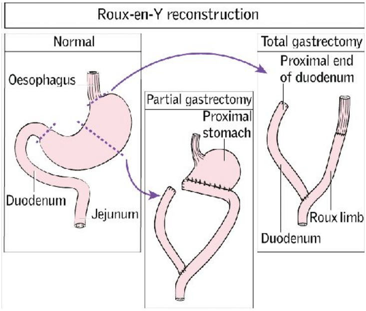


Fig 13.38 The Roux-en-Y reconstruction.

**Theodor Billroth**

Theodor Billroth was a surgeon of German-Austrian origin, whose name lives on as a set of operations on the stomach. He was a pioneer of abdominal surgery and the use of aseptic techniques, performing the first Billroth I procedure in 1881 for the resection of a pyloric gastric carcinoma. Among the many of his remarkable achievements is included the first laryngectomy. He was also a talented musician (a close friend of Brahms) and a dedicated educator with something of a realist's view of the world:

'*The pleasure of a physician is little, the gratitude of patients is rare, and even rarer is material reward, but these things will never deter the student who feels the call within him.*' Theodor Billroth (1829–94).

# Fundoplication for gastro-oesophageal reflux

Laparoscopic fundoplication is the surgical procedure of choice when symptoms of GORD are refractory to medical therapy *and* there is severe reflux (confirmed by pH-monitoring)—see p254. Symptoms may be complicated by a hiatus hernia, which is repaired during the procedure.

**Surgery** The defect in the diaphragm is repaired by tightening the crura. Reflux is prevented by wrapping the gastric fundus around the lower oesophageal sphincter—see fig 13.39. There are various types of procedure, eg Nissen (360° wrap), Toupet (270° posterior wrap), Watson (anterior hemifundoplication). Laparoscopic surgery is at least as effective at controlling reflux as open surgery but with a lower mortality and morbidity. Wound infections and respiratory complications are also more common in open surgery, though the incidence of dysphagia is similar for the two procedures—but see p592.

**Complications** Dysphagia (if the wrap is too tight), 'gas-bloat syndrome' (inability to belch/vomit), and new-onset diarrhoea.

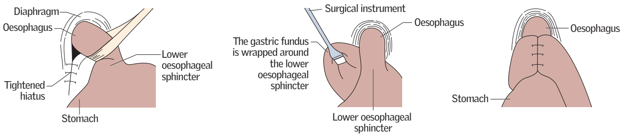


Fig 13.39 Laparoscopic Nissen fundoplication.

# Oesophageal rupture

**Causes** •*Iatrogenic*, eg endoscopy/biopsy/dilatation (accounts for 85–90% of perforations). •*Trauma*, eg penetrating injury/ingestion of foreign body. •*Carcinoma* •*Boerhaave syndrome*—rupture due to violent vomiting. •*Corrosive ingestion*.

**Clinical features** Odynophagia, tachypnoea, dyspnoea, fever, shock, surgical emphysema (a crackling sensation felt on palpating the skin over the chest or neck caused by air tracking from the lungs. ΔΔ: Pneumothorax).

**℞** Iatrogenic perforations are less prone to mediastinitis and sepsis and may be managed conservatively with NG tube, PPI, and antibiotics. Others require resuscitation, PPI, antibiotics, antifungals, and surgery (debridement of mediastinum and placement of T-tube for drainage and formation of a controlled oesophago-cutaneous fistula).

# Surgical management of obesity

Severe obesity is increasing in prevalence worldwide and is associated with type 2 diabetes mellitus (T2DM); hypertension; ischaemic heart disease; sleep apnoea; osteoarthritis; and depression. Bariatric surgery has become very successful at weight reduction, symptom improvement,and improving quality of life. Surgery increases life expectancy by around 3 years (but may not prolong survival in high-risk men).

*Indications:* According to NICE guidelines,[9] weight-loss surgery in adults should be considered if *all* the following criteria are met:

1. BMI ≥40 (or ≥35 with significant comorbidities that could improve with ↓weight).
2. Failure of non-surgical management to achieve and maintain clinically beneficial weight loss for 6 months.
3. Fitness for surgery and anaesthesia.
4. Intensive management in tier 3 service (provides guidance on diet, physical activity, and psychosocial concerns, as well as lifelong medical monitoring).
5. The patient must be well informed and motivated.

If BMI ≥50, or in newly diagnosed T2DM with BMI ≥30, surgery is recommended as first-line treatment.

**Comparison with medical therapy** Surgery is more effective in achieving weight loss than non-surgical management and weight loss is more likely to be maintained in the longer term. Adverse events are more common following surgery, and vary from one procedure to another.

**Procedures** There are two main mechanisms causing weight loss: **1** Restriction of calorie intake by reducing stomach capacity. **2** Malabsorption of nutrients by reducing the length of functional small bowel. NB: This also affects the levels of circulating gut peptides (eg PYY and GLP-1), which are thought to play a role in the mechanism of satiety and weight loss. Choose surgical intervention jointly with patient:

- *Laparoscopic adjustable gastric banding (LAGB):* This restrictive technique creates a pre-stomach pouch by placing a silicone band around the top of the stomach, which serves as a new smaller stomach. The band can be adjusted by addition or removal of saline through a subcutaneous port (see fig 13.40). LAGB is associated with improvements in comorbidities and quality of life. Weight loss is slower and less than with gastric bypass but there is lower mortality and fewer complications. Relatively non-invasive and band removal possible. *Complications:* pouch enlargement, band slip, band erosion, and port infection/breakage.
- *Sleeve gastrectomy:* (fig 13.41) Involves division of the stomach vertically, reducing it in size by about 75%. The pyloric valve at the bottom of the stomach is left intact so function and digestion are unaltered. The procedure is not reversible and may be a first stage for progression to Roux-en-Y gastric bypass or duodenal switch in very obese patients where a single-stage procedure would be technically difficult or unsafe.
- *Roux-en-Y gastric bypass:* (fig 13.42) Laparoscopic or open. A portion of the jejunum is attached to a small stomach pouch to allow food to bypass the distal stomach, duodenum, and proximal jejunum. It can be performed laparoscopically and works by both restriction and malabsorption. Mean excess weight loss at 5 years is 62.8%.Current evidence demonstrates greater weight loss, greater resolution of comorbidities, and lower reoperation rates compared to LAGB. *Complications:* micronutrient deficiency (requires vitamin supplementation and lifelong follow-up/blood tests), dumping syndrome, wound infection, hernias, malabsorption, diarrhoea, and a mortality of <0.5% (at experienced centres).

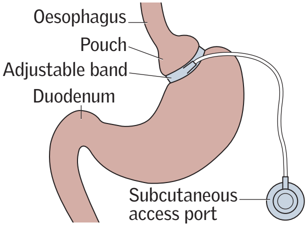


Fig 13.40 Adjustable gastric band.

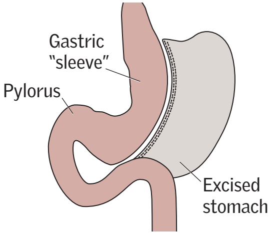


Fig 13.41 Vertical sleeve gastrectomy.

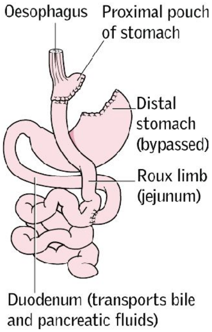


Fig 13.42 Gastric bypass.

# Diverticular disease

A GI *diverticulum* is an outpouching of the gut wall, usually at sites of entry of perforating arteries. *Diverticulosis* means that diverticula are present, and *diverticular disease* implies they are symptomatic. *Diverticulitis* refers to inflammation of a diverticulum. Diverticula can be aquired or congenital and may occur elsewhere, but the most important are acquired colonic diverticula, to which this page refers.

**Pathology** Most within sigmoid colon with 95% of complications at this site, but right-sided and massive single diverticula can occur. High intraluminal pressures (due, perhaps, to lack of dietary fibre) force the mucosa to herniate through the muscle layers of the gut at weak points adjacent to penetrating vessels. 30% of Westerners have diverticulosis by age 60. The majority are asymptomatic.

**Diagnosis** Diverticula are a common incidental finding at colonoscopy (fig 6.11, p249). CT abdomen is best to confirm acute diverticulitis and can identify extent of disease and any complications (eg colovesical fistulae). Colonoscopy risk perforation in acute setting. AXR may identify obstruction or free air (perforation).

**Diverticular disease** Altered bowel habit ± left-sided colic relieved by defecation; nausea and flatulence. High-fibre diets do not help symptoms; try antispasmodics, eg mebeverine 135mg/8h PO. Surgical resection occasionally resorted to.

**Diverticulitis** features above + pyrexia, ↑WCC, ↑CRP/ESR, a tender colon ± localized or generalized peritonism. Mild attacks can be treated at home with bowel rest (fluids only) ± antibiotics. If fluids and pain not tolerated, admit for analgesia, NBM, IV fluids and IV antibiotics. Most attacks settle but complications include abscess formation (necessitating percutaneous CT-guided drainage), or perforation. ►Beware diverticulitis in immunocompromised patients (eg on steroids) who often have few symptoms and may present late.

**Surgery** The need for surgery is reflected by the degree of infective complications:

| | | |
|---|---|---|
| Stage 1 | Pericolic or mesenteric abscess | Surgery rarely needed |
| Stage 2 | Walled off or pelvic abscess | May resolve without surgery |
| Stage 3 | Generalized purulent peritonitis | Surgery required |
| Stage 4 | Generalized faecal peritonitis | Surgery required |

Indications for elective surgery include stenosis, fistulae, or recurrent bleeding.

**Complications** ►►*Perforation:* There is ileus, peritonitis ± shock. Mortality: 40%. Manage as for an acute abdomen. At laparotomy a Hartmann's procedure may be performed (p582). Primary anastomosis is possible in selected patients. Emergency laparoscopic management is an emerging alternative.

- *Haemorrhage* is usually sudden and painless. It is a common cause of big rectal bleeds (p629). Embolization (at angiography) or colonic resection only necessary if ongoing massive bleeding and colonoscopic haemostasis has been unsuccessful.
- *Fistulae:* Enterocolic, colovaginal, or colovesical (pneumaturia ± intractable UTIs). Treatment is surgical, eg colonic resection.
- *Abscesses*, eg with swinging fever, leucocytosis, and localizing signs, eg boggy rectal mass (pelvic abscess—drain rectally). If no localizing signs, remember the aphorism: *pus somewhere, pus nowhere* = *pus under the diaphragm*. A subphrenic abscess is a horrible way to die, so do an urgent ultrasound. Antibiotics ± ultrasound/CT-guided drainage may be needed.
- *Post-infective strictures* may form in the sigmoid colon.

## Rectal bleeding—an acute management plan

**Causes** Diverticulitis, colorectal cancer, haemorrhoids, IBD, perianal disease, angiodyplasia (submucosal arteriovenous malformations, typically elderly). Rarities: trauma, ischaemia colitis, radiation proctitis, aorto-enteric fistula.

**An acute management plan** for this common surgical event:

- ▶▶ *ABC* resuscitation, if necessary.
- ▶▶ *History and examination.*
- ▶▶ *Blood tests:* FBC, U&E, LFT, clotting, amylase (always thinking of pancreatitis), CRP, group and save—await Hb result before crossmatching unless unstable and bleeding.
- ▶▶ *Imaging:* May only need plain AXR, but if there are signs of perforation (eg sepsis, peritonism) or if there is cardiorespiratory comorbidity, then request an erect CXR.
- ▶▶ *Fluid management:* Insert 2 cannulae (≥18G) into the antecubital fossae. Insert a urinary catheter if there is a suspicion of haemodynamic compromise—there is no absolute indication, but remember that you are weighing up the risks and benefits. Give crystalloid as replacement and maintenance IVI. Blood transfusion only if significant blood loss (table 13.10, p607).
- ▶▶ *Clotting:* Withold ± reverse anticoagulation and antiplatelet agents (p351).
- ▶▶ *Antibiotics* may occasionally be required if there is evidence of sepsis or perforation, eg piperacillin/tazobactam 4.5g/8h IV.
- ▶▶ *Keep bedbound:* The patient may feel the need to get out of bed to pass stool, but this could be another large bleed, resulting in collapse if they try to walk. ▶Don't allow them to mobilize and inform the nursing staff of this.
- ▶▶ *Start a stool chart* to monitor volume and frequency of motions. Send a sample for MC&S (x3 if known to have compromising comorbidity such as IBD).
- ▶▶ *Diet:* Keep on clear fluids so that they can have something, yet the colon will be as clear as possible if colonoscopy required.
- ▶▶ *Interventions* if bleeding not settling with conservative management: *Angiography* may allow localization of bleeding (eg sigmoid diverticulum or right sided angiodysplasia) as well as therapeutic embolization; *CT angiography* is a non-invasive alternative (without interventional options); *colonoscopy* may permit endoscopic haemostasis.
- ▶▶ *Surgery:* The main indication for this is unremitting, massive bleeding that is not controlled by other means.

# Perianal problems



**Pruritus ani** Itch occurs if the anus is moist/soiled; fissures, incontinence, poor hygiene, tight pants, threadworm, fistula, dermatoses, lichen sclerosis, anxiety, contact dermatitis (perfumed goods). Cause is often unknown. *℞:* Avoid scratching, perianal hygiene, avoid foods which loosen stools. Soothing ointment, mild topical corticosteroid if perinanal inflammation (max 2wks), oral antihistamine for noctural itch.

**Fissure-in-ano** Painful tear in the squamous lining of the lower anal canal—often, if chronic, with a 'sentinel pile' or mucosal tag at the external aspect. 90% are posterior (anterior ones follow parturition). ♂>♀. *Causes:* Most are due to hard faeces. Spasm may constrict the inferior rectal artery, causing ischaemia, making healing difficult and perpetuating the problem. Rare causes (multiple ± lateral): syphilis; herpes; trauma; Crohn's; anal cancer; psoriasis. Groin nodes suggest a complicating factor (eg immunosuppression/HIV). *℞:* 5% lidocaine ointment + GTN ointment (0.2–0.4%) *or* topical diltiazem (2%); dietary fibre, fluids, stool softener, and hygiene advice. Botulinum toxin injection (2nd line) and topical diltiazem (2%) are at least as effective as GTN with fewer side-effects. If conservative measures fail, surgical options include *lateral partial internal sphincterotomy*.

**Fistula-in-ano** A track communicates between the skin and anal canal/rectum. Blockage of deep intramuscular gland ducts is thought to predispose to the formation of abscesses, which discharge to form the fistula. *Goodsall's rule* determines the path of the fistula track: if anterior, the track is in a straight line (radial); if posterior, the internal opening is *always* at the 6 o'clock position, taking a tortuous course. *Causes:* perianal sepsis, abscesses (see later in topic), Crohn's disease, TB, diverticular disease, rectal carcinoma, immunocompromise. *Tests:* MRI; endoanal US scan. *℞:* Fistulotomy + excision. High fistulae (involving continence muscles of anus) require 'seton suture' tightened over time to maintain continence; low fistulae are 'laid open' to heal by secondary intention—division of sphincters poses no risk to continence.

**Anorectal abscesses** Usually caused by gut organisms (rarely staphs or TB). ♂:♀≈1:8. Perianal (~45%), ischiorectal (≤30%), intersphincteric (>20%), supralevator (~5%) (fig 13.43). *℞:* Incise & drain under GA. *Associations:* DM, Crohn's, malignancy, fistulae.

**Perianal haematoma** (AKA thrombosed external pile—see p633). Strictly, it is actually a clotted venous saccule. It appears as a 2–4mm 'dark blueberry' under the skin at the anal margin. It may be evacuated under LA or left to resolve spontaneously.

**Pilonidal sinus** Obstruction of natal cleft hair follicles ~6cm above the anus. Ingrowing of hair excites a foreign body reaction and may cause secondary tracks to open laterally ± abscesses, with foul-smelling discharge. (Barbers get these between fingers.) ♂:♀≈10:1. Obese Caucasians and those from Asia, the Middle East, and Mediterranean at ↑risk. *℞:* Excision of the sinus tract ± primary closure. Consider pre-op antibiotics. Complex tracks can be laid open and packed individually, or skin flaps can be used to cover the defect. Offer hygiene and hair removal advice.

**Rectal prolapse** The mucosa (partial/type 1), or all layers (complete/type 2—more common), may protrude through the anus. Incontinence in 75%. It is due to a lax sphincter, prolonged straining, and related to chronic neurological and psychological disorders. *℞: Abdominal approach*: fix rectum to sacrum (rectopexy) ± mesh insertion ± rectosigmoidectomy. Laparoscopic rectoplexy is as effective as open repair. *Perineal approach*: Delorme's procedure (resect close to dentate line and suture mucosal boundaries), anal encirclement with a Thiersch wire.

**Perianal warts** Condylomata acuminata (viral warts) are treated with podophyllotoxin or imiquimod or cryotherapy/surgical excision. Giant condylomata acuminata of Buschke & Loewenstein may evolve into verrucous cancers (low-grade, non-metastasizing). Condylomata lata secondary to syphilis is treated with penicillin.

**Proctalgia fugax** Idiopathic, intense, brief, stabbing/crampy rectal pain, often worse at night. The mainstay of treatment is reassurance. Inhaled salbutamol or topical GTN (0.2–0.4%) or topical diltiazem (2%) may help.

**Anal ulcers** Consider Crohn's, anal cancer, lymphogranuloma venerum, TB, syphilis.

**Skin tags** Seldom cause trouble but are easily excised.

## Anal cancer

*Incidence:* 1233 new cases of anal cancer in the UK (2013). *Risk factors:* Anoreceptive intercourse; HPV (HPV 16 associated with worse prognosis); HIV. *Histology:* Squamous cell (85%); rarely basaloid, melanoma, or adenocarcinoma. Anal margin tumours are usually well-differentiated, keratinizing lesions with a good prognosis. Anal canal tumours arise above dentate line, are poorly differentiated and non-keratinizing with a poorer prognosis. *Spread:* Tumours above the dentate line spread to pelvic lymph nodes; those below spread to the inguinal nodes. *Presentation:* Bleeding, pain, bowel habit change, pruritus ani, masses, stricture. *ΔΔ:* Perianal warts; leucoplakia; lichen sclerosis; Bowen's disease; Crohn's disease. *Treatment:* Chemo-irradiation (radiotherapy + fluorouracil + mitomycin/cisplatin) is usually preferred to anorectal excision & colostomy; 75% retain normal anal function.

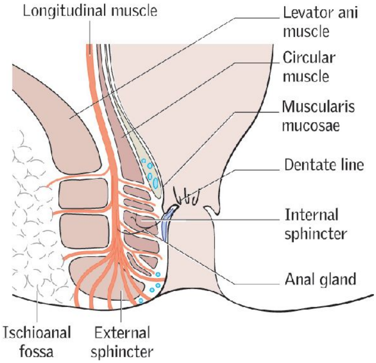


**Fig 13.43** Anatomy of the anal canal. Perianal abscesses present as tender, inflamed, localized swellings at the anal verge. Ischiorectal abscesses are also tender but cause a diffuse, indurated swelling in the ischioanal fossa area. You will find your patient waiting anxiously for you, pacing about, or on the edge of their chair: avoiding all pressure is imperative. NB: above the dentate line = visceral nerve innervation (hence no pain sensation); below = somatic innervation (very sensitive to pain).

# Haemorrhoids (piles)



**Definition** Haemorrhoids (≈*running blood* in Greek) are disrupted and dilated anal cushions. The anus is lined mainly by discontinuous masses of spongy vascular tissue—the anal cushions, which contribute to anal closure. Viewed from the lithotomy position, the three anal cushions are at 3, 7, and 11 o'clock (where the three major arteries that feed the vascular plexuses enter the anal canal). They are attached by smooth muscle and elastic tissue, but are prone to displacement and disruption, either singly or together. The effects of gravity (standing), increased anal tone (?stress), and the effects of straining at stool may make them become both bulky and loose, and so to protrude to form piles (Latin *pila*, meaning a ball). They are vulnerable to trauma (eg from hard stools) and bleed readily from the capillaries (bright red blood) of the underlying lamina propria. NB: piles are *not* varicose veins.

As there are no sensory fibres above the dentate line (squamomucosal junction), piles are not painful unless they thrombose when they protrude and are gripped by the anal sphincter, blocking venous return. See fig 13.44.

**Differential diagnosis** Perianal haematoma; anal fissure; abscess; tumour; proctalgia fugax. ►Never ascribe rectal bleeding to piles without examination or investigation.

**Causes** Constipation with prolonged straining is a key factor. In many the bowel habit may be normal. Congestion from a pelvic tumour, pregnancy, CCF, or portal hypertension are important in only a minority of cases. ►Elicit red flags in history.

**Pathogenesis** There is a vicious circle: vascular cushions protrude through a tight anus, become more congested, and hypertrophy to protrude again more readily. These protrusions may then strangulate. See table 13.15 for classification.

**Symptoms** Bright red rectal bleeding, often coating stools, on the tissue, or dripping into the pan after defecation. There may be mucous discharge and pruritus ani. Severe anaemia may occur. Symptoms such as weight loss, tenesmus, and change in bowel habit should prompt thoughts of other pathology. ►In all rectal bleeding do:
- An abdominal examination to rule out other diseases.
- PR exam: prolapsing piles are obvious. Internal haemorrhoids are not palpable.
- Colonoscopy/flexible sigmoidoscopy to exclude proximal pathology if ≳50 years old.

**Treatment** 1 *Medical:* (1st-degree.) ↑Fluid and fibre is key ± topical analgesics & stool softener (bulk forming). Topical steroids for short periods only.
2 *Non-operative:* (2nd & 3rd degree, or 1st degree if medical therapy failed.) •*Rubber band ligation.* Cheap, but needs skill. Banding produces an ulcer to anchor the mucosa (SE: bleeding, infection; pain). It has the lowest recurrence rate. •*Sclerosants.* (1st- or 2nd-degree.) 2mL of 5% phenol in oil is injected into the pile above the dentate line, inducing fibrotic reaction. Recurrence higher (SE: impotence; prostatitis). •*Infra-red coagulation.* Applied to localized areas of piles, it works by coagulating vessels and tethering mucosa to subcutaneous tissue. It is as successful as banding and may be less painful. •*Bipolar diathermy and direct current electrotherapy.* Causes coagulation and fibrosis after local application of heat. Success rates are similar to those of infrared coagulation, and complication rates are low.
3 *Surgery:* •*Excisional haemorrhoidectomy* is the most effective treatment (excision of piles ± ligation of vascular pedicles, as day-case surgery, needing ~2wks off work). Scalpel, electrocautery, or laser may be used. •*Stapled haemorrhoidopexy* (procedure for prolapsing haemorrhoids) may result in less pain, a shorter hospital stay, and quicker return to normal activity than conventional surgery. It is used when there is a large internal component, but has a higher recurrence and prolapse rate than excisional. *Surgical complications* include constipation; infection; stricture; bleeding.

**Prolapsed, thrombosed piles** Analgesia, ice packs, and stool softeners. Pain usually resolves in 2-3wks. Some advocate early surgery.

**Table 13.15** Classification of haemorrhoids

| | |
|---|---|
| **1st degree** | Remain in the rectum |
| **2nd degree** | Prolapse through the anus on defecation but spontaneously reduce |
| **3rd degree** | As for 2nd-degree but require digital reduction |
| **4th degree** | Remain persistently prolapsed |

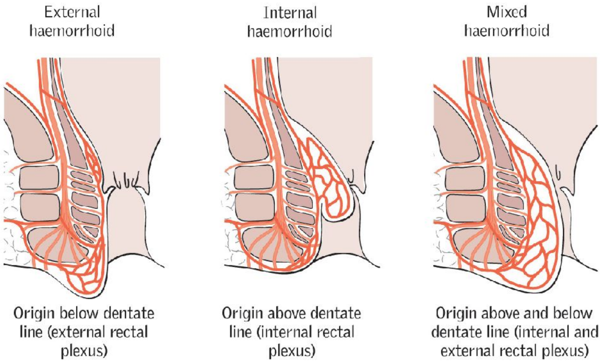


**Fig 13.44** Internal and external haemorrhoids.

# Gallstones

Bile contains cholesterol, bile pigments (from broken down Hb), and phospholipids. If the concentrations vary, different stones may form. *Pigment stones:* Small, friable, and irregular. *Causes:* haemolysis. *Cholesterol stones:* Large, often solitary. *Causes:* ♀, age, obesity (Admirand's triangle: ↑risk of stone if ↓lecithin, ↓bile salts, ↑cholesterol). *Mixed stones:* Faceted (calcium salts, pigment, and cholesterol). *Gallstone prevalence:* 8% of those over 40yrs. 90% remain asymptomatic. Risk factors for stones becoming symptomatic: smoking; parity.

**Biliary colic** Gallstones are symptomatic with cystic duct obstruction or if passed into the common bile duct (CBD[10]). RUQ pain (radiates → back) ± jaundice. *℞:* Analgesia (see p636), rehydrate, NBM. Elective laparoscopic cholecystectomy (see BOX 'Early or delayed cholecystectomy?'). Do urinalysis, CXR, and ECG.

**Acute cholecystitis** follows stone or sludge impaction in the neck of the gallbladder (GB[10]), which may cause continuous epigastric or RUQ pain (referred to the right shoulder—see p609), vomiting, fever, local peritonism, or a GB mass. The main difference from *biliary colic* is the inflammatory component (local peritonism, fever, WCC↑; see table 13.16). If the stone moves to the CBD, obstructive jaundice and cholangitis may occur—see BOX 'Complications of gallstones'. *Murphy's sign:* lay 2 fingers over the RUQ; ask patient to breathe in. This causes pain & arrest of inspiration as an inflamed GB impinges on your fingers. It is only +ve if the same test in the LUQ does not cause pain. A *phlegmon* (RUQ mass of inflamed adherent omentum and bowel) may be palpable. *Tests:* ↑WCC, US—a thick-walled, shrunken GB (also seen in chronic disease), pericholecystic fluid, stones, CBD (dilated if >6mm). Plain AXR only shows ~10% of gallstones; it may identify a 'porcelain' GB (associated risk of cancer). *Treatment:* NBM, pain relief, IVI, and antibiotics, eg *co-amoxiclav* 625mg/8h IV. Laparoscopic cholecystectomy is the treatment of choice for all patients fit for GA. Open surgery is required if there is GB perforation. If elderly or high risk/unsuitable for surgery, consider percutaneous cholecystostomy; cholecystectomy can still be done later. Cholecystostomy is also the preferred treatment for acalculous cholecystitis.

**Chronic cholecystitis** Chronic inflammation ± colic. 'Flatulent dyspepsia': vague abdominal discomfort, distension, nausea, flatulence, and fat intolerance (fat stimulates cholecystokinin release and GB contraction). US to image stones and assess CBD diameter. MRCP (p742) is used to find CBD stones. *℞:* Cholecystectomy. If US shows a dilated CBD with stones, ERCP (p742) + sphincterotomy before surgery. If symptoms persist post-surgery consider hiatus hernia/IBS/peptic ulcer/chronic pancreatitis/tumour.

## Other presentations

- *Obstructive jaundice with CBD stones:* (See p272.) If LFT worsening, ERCP with sphincterotomy ± biliary trawl, then cholecystectomy may be needed, or open surgery with CBD exploration. If CBD stones are suspected pre-operatively, they should be identified by MRCP (p742).
- *Cholangitis:* (Bile duct infection.) Causing RUQ pain, jaundice, and rigors (Charcot's triad, BOX 'Complications of gallstones'). Treat with, eg piperacillin/tazobactam 4.5g/8h IV.
- *Gallstone ileus:* A stone erodes through the GB into the duodenum; it may then obstruct the terminal ileum. AXR shows: air in CBD (= pneumobilia), small bowel fluid levels, and a stone. Duodenal obstruction is rarer (Bouveret's syndrome).
- *Pancreatitis:* See p636.
- *Mucocoele/empyema:* Obstructed GB fills with mucus (secreted by GB wall)/pus.
- *Silent stones:* Do elective surgery on those with sickle cell, immunosuppression, (debatably diabetes) as well as all calcified/porcelain GBs.
- *Mirizzi's syndrome:* A stone in the GB presses on the bile duct causing jaundice.
- *Gallbladder necrosis:* Rare because of dual blood supply (hepatic artery via cystic artery, and from small branches of the hepatic artery in the GB fossa).
- *Other:* Causes of cholecystitis and biliary symptoms other than gallstones are rare. Consider infection (typhoid, cryptosporidiosis, and brucellosis); cholecystokinin release; parenteral nutrition; anatomical abnormality; polyarteritis nodosa (p556).

**10** Common abbreviations used in this section: CBD, common bile duct; GB, gallbladder.

## Complications of gallstones

*In the gallbladder & cystic duct:*
- Biliary colic
- Acute and chronic cholecystitis
- Mucocoele
- Empyema
- Carcinoma
- Mirizzi's syndrome.

*In the bile ducts:*
- Obstructive jaundice
- Cholangitis
- Pancreatitis.

*In the gut:*
- Gallstone ileus.

Table 13.16 Biliary colic, cholecystitis, or cholangitis?

| | RUQ pain | Fever / ↑WCC | Jaundice |
|---|---|---|---|
| Biliary colic | ✓ | ✗ | ✗ |
| Acute cholecystitis | ✓ | ✓ | ✗ |
| Cholangitis | ✓ | ✓ | ✓ |

## Early or delayed cholecystectomy?

**For acute cholecystitis** Laparoscopic cholecystectomy for acute cholecystitis has traditionally been performed 6–12wks after the acute episode due to anticipated increased mortality and conversion to open procedure. Early laparoscopic cholecystectomy, within 7d of symptom onset, is now the treatment of choice. Early surgery reduces the duration of hospital admission compared with delayed surgery, but does not reduce mortality or complications. Up to one-quarter of people scheduled for delayed surgery may require urgent operations because of recurrent or worsening symptoms.[10]

**For biliary colic** Patients with biliary colic due to gallstones waiting for an elective laparoscopic cholecystectomy may develop significant complications, such as acute pancreatitis (p636) during the waiting period. One high-bias trial found early laparoscopic cholecystectomy (within 24h of an acute episode) decreased potential complications that may develop during the wait for elective surgery.

## Acute pancreatitis

This unpredictable disease (mortality ~12%) is characterized by self-perpetuating pancreatic enzyme-mediated autodigestion; oedema and fluid shifts cause hypovolaemia, as extracellular fluid is trapped in the gut, peritoneum, and retroperitoneum (worsened by vomiting). Although pancreatitis is mild in 80% of cases; 20% develop severe complicated and life-threatening disease: progression may be rapid from mild oedema to necrotizing pancreatitis. ~50% of cases that advance to necrosis are further complicated by infection.

**Causes** The one mnemonic we can all agree on: '**GET SMASHED**'. **G**allstones (~35%), **E**thanol (~35%), **T**rauma (~1.5%), **S**teroids, **M**umps, **A**utoimmune (PAN), **S**corpion venom, **H**yperlipidaemia, hypothermia, hypercalcaemia, **E**RCP (~5%) and emboli, **D**rugs. Also pregnancy and neoplasia or no cause found (~10–30%).

**Symptoms** Gradual or sudden severe epigastric or central abdominal pain (radiates to back, sitting forward may relieve); vomiting prominent.

**Signs** ►May be subtle in serious disease. ↑HR, fever, jaundice, shock, ileus, rigid abdomen ± local/general tenderness, periumbilical bruising (Cullen's sign) or flanks (Grey Turner's sign) from blood vessel autodigestion and retroperitoneal haemorrhage.

**Tests** Raised serum *amylase* (>1000U/mL or around 3-fold upper limit of normal). The degree of elevation is not related to severity of disease. ►Amylase may be normal even in severe pancreatitis (levels starts to fall within 24–48h). It is excreted renally so renal failure will ↑ levels. Cholecystitis, mesenteric infarction, and GI perforation can cause lesser rises. Serum *lipase* is more sensitive and specific for pancreatitis (especially when related to alcohol), and rises earlier and falls later. *ABG* to monitor oxygenation and acid-base status. *AXR:* No psoas shadow (↑retroperitoneal fluid), 'sentinel loop' of proximal jejunum from ileus (solitary air-filled dilatation). *Erect CXR* helps exclude other causes (eg perforation). *CT* is the standard choice of imaging to assess severity and for complications. *US* (if gallstones + ↑AST). *ERCP* if LFTs worsen. *CRP* >150mg/L at 36h after admission is a predictor of severe pancreatitis.

**Management** Severity assessment is essential (see BOX and table 13.17).
- Nil by mouth, consider NJ feeding (decrease pancreatic stimulation). Set up IVI and give lots of crystalloid, to counter third-space sequestration, until vital signs are satisfactory and urine flow stays at >30mL/h. Insert a urinary catheter and consider CVP monitoring.
- Analgesia: pethidine 75–100mg/4h IM, or morphine (may cause Oddi's sphincter to contract more, but it is a better analgesic and not contraindicated).
- Hourly pulse, BP, and urine output; daily FBC, U&E, $Ca^{2+}$, glucose, amylase, ABG.
- If worsening: ITU, $O_2$ if ↓$P_aO_2$. In suspected abscess formation or pancreatic necrosis (on CT), consider parenteral nutrition ± laparotomy & debridement ('necrosectomy'). Antibiotics may help in severe disease.
- ERCP + gallstone removal may be needed if there is progressive jaundice.
- Repeat imaging (usually CT) is performed in order to monitor progress.

**ΔΔ** Any acute abdomen (p606), myocardial infarct.

**Early complications** Shock, ARDS (p186), renal failure (►give lots of fluid!), DIC, sepsis, ↓$Ca^{2+}$, ↑glucose (transient; 5% need insulin).

**Late complications** (>1wk.) *Pancreatic necrosis* and *pseudocyst* (fluid in lesser sac, fig 13.45), with fever, a mass ± persistent ↑amylase/LFT; may resolve or need drainage. *Abscesses* need draining. *Bleeding* from elastase eroding a major vessel (eg splenic artery); embolization may be life-saving. *Thrombosis* may occur in the splenic/gastroduodenal arteries, or colic branches of the SMA, causing bowel necrosis. *Fistulae* normally close spontaneously. If purely pancreatic they do not irritate the skin. Some patients suffer *recurrent oedematous pancreatitis* so often that near-total pancreatectomy is contemplated. ►It can all be a miserable course.

## Modified Glasgow criteria for predicting severity of pancreatitis

►Three or more positive factors detected within 48h of onset suggest severe pancreatitis, and should prompt transfer to ITU/HDU. Mnemonic: **PANCREAS**.

Table 13.17

| | |
|---|---|
| **P**$_aO_2$ | <8kPa |
| **A**ge | >55yrs |
| **N**eutrophilia | WBC >15 x $10^9$/L |
| **C**alcium | <2mmol/L |
| **R**enal function | Urea >16mmol/L |
| **E**nzymes | LDH >600iu/L; AST >200iu/L |
| **A**lbumin | <32g/L (serum) |
| **S**ugar | Blood glucose >10mmol/L |



These criteria have been validated for pancreatitis caused by gallstones and alcohol; Ranson's criteria are valid for alcohol-induced pancreatitis, and can only be fully applied after 48h, which does have its disadvantages. Other criteria for assessing severity include the Acute Physiology and Chronic Health Examination (APACHE)-II, and the Bedside Index for Severity in Acute Pancreatitis (BISAP).

Fig 13.45 Axial CT of the abdomen (with IV and PO contrast media) showing a pancreatic pseudocyst occupying the lesser sac of the abdomen posterior to the stomach. It is called a 'pseudocyst' because it is not a true cyst, rather a collection of fluid in the lesser sac (ie not lined by epi/endothelium). It develops at ≥6wks. The cyst fluid is of low attenuation compared with the stomach contents because it has not been enhanced by the contrast media.

Image courtesy of Dr Stephen Golding.

# Urinary tract calculi (nephrolithiasis)



Renal stones (calculi) consist of crystal aggregates. Stones form in collecting ducts and may be deposited anywhere from the renal pelvis to the urethra, though classically at: **1** Pelviureteric junction **2** Pelvic brim **3** Vesicoureteric junction.

**Prevalence** Common: lifetime incidence up to 15%. *Peak age:* 20–40yr. ♂:♀≈3:1.

**Types** •Calcium oxalate (75%). •Magnesium ammonium phosphate (struvite/triple phosphate; 15%). •Also: urate (5%), hydroxyapatite (5%), brushite, cystine (1%), mixed.

**Presentation** Asymptomatic or: **1** *Pain:* Excruciating spasms of renal colic 'loin to groin' (or genitals/inner thigh), with nausea/vomiting. Often cannot lie still (differentiates from peritonitis). *Obstruction of kidney:* felt in the loin, between rib 12 and lateral edge of lumbar muscles (like intercostal nerve irritation pain; the latter is not colicky, and is worsened by specific movements/pressure on a trigger spot). *Obstruction of mid-ureter:* may mimic appendicitis/diverticulitis. *Obstruction of lower ureter:* may lead to symptoms of bladder irritability and pain in scrotum, penile tip, or labia majora. *Obstruction in bladder or urethra:* causes pelvic pain, dysuria, strangury (desire but inability to void) ± interrupted flow. **2** *Infection:* Can coexist (↑risk if voiding impaired), eg UTI; pyelonephritis (fever, rigors, loin pain, nausea, vomiting); pyonephrosis (infected hydronephrosis) **3** *Haematuria.* **4** *Proteinuria.* **5** *Sterile pyuria.* **6** *Anuria.*

**Examination** Usually no tenderness on palpation. May be renal angle tenderness especially to percussion if there is retroperitoneal inflammation.

**Tests** FBC, U&E, $Ca^{2+}$, $PO_4^{3-}$, glucose, bicarbonate, urate. *Urine dipstick:* Usually +ve for blood (90%). *MSU:* MC&S. *Further tests for cause:* Urine pH; 24h urine for: calcium, oxalate, urate, citrate, sodium, creatinine; stone biochemistry (sieve urine & send stone).

*Imaging:* Non-contrast CT is investigation of choice for imaging stones (99% visible) & helps exclude differential causes of an acute abdomen. ►A ruptured abdominal aortic aneurysm may present similarly. 80% of stones are visible on KUB XR (kidneys + ureters + bladder). Look along ureters for calcification over the transverse processes of the vertebral bodies. US an alternative for hydronephrosis or hydroureter.

℞ *Initially:* Analgesia, eg diclofenac 75mg IV/IM, or 100mg PR. (If CI: *opioids*) + IV fluids if unable to tolerate PO; antibiotics (eg piperacillin/tazobactam 4.5g/8h IV, or gentamicin) if infection. *Stones <5mm in lower ureter:* ~90–95% pass spontaneously. ↑Fluid intake. *Stones >5mm/pain not resolving: Medical expulsive therapy:* ►start at presentation; nifedipine 10mg/8h PO or α-blockers (tamsulosin 0.4mg/d) promote expulsion and reduce analgesia requirements. Most pass within 48h (>80% after ~30d). If not, try extracorporeal shockwave lithotripsy (ESWL) (if <1cm), or ureteroscopy using a basket. ESWL: US waves shatter stone. SE: renal injury, may also cause ↑BP and DM. *Percutaneous nephrolithotomy (PCNL):* keyhole surgery to remove stones, when large, multiple, or complex. Open surgery is rare.

►*Indications for urgent intervention* (*delay kills glomeruli*): Presence of infection *and* obstruction—a percutaneous nephrostomy or ureteric stent may be needed to relieve obstruction (p640); urosepsis; intractable pain or vomiting; impending AKI; obstruction in a solitary kidney; bilateral obstructing stones.

**Prevention** *General:* Drink plenty. *Normal* dietary $Ca^{2+}$ intake (low $Ca^{2+}$ diets increase oxalate excretion). *Specifically:* •*Calcium stones:* in hypercalciuria, a thiazide diuretic is used to ↓$Ca^{2+}$ excretion. •*Oxalate:* ↓oxalate intake; pyridoxine may be used (p295). •*Struvite (phosphate mineral):* treat infection promptly. •*Urate:* allopurinol (100–300mg/24h PO). Urine alkalinization may also help, as urate is more soluble at pH>6 (eg with potassium citrate or sodium bicarbonate). •*Cystine:* vigorous hydration to keep urine output >3L/d and urinary alkalinization (as above-mentioned). Penicillamine is used to chelate cystine, given with pyridoxine to prevent vitamin $B_6$ deficiency.

## Questions to address when confronted by a stone

*What is its composition?* (See table 13.18.)

**Table 13.18** Types, causes, and X-ray appearance of renal stones

| Type | Causative factors | Appearance on X-ray |
|---|---|---|
| Calcium oxalate (fig 13.46) | Metabolic or idiopathic | Spiky, radio-opaque |
| Calcium phosphate | Metabolic or idiopathic | Smooth, may be large, radio-opaque |
| Magnesium ammonium phosphate (fig 13.47) | UTI (proteus causes alkaline urine and calcium precipitation and ammonium salt formation) | Large, horny, 'staghorn', radio-opaque |
| Urate (p680) | Hyperuricaemia | Smooth, brown, radiolucent |
| Cystine (fig 13.48) | Renal tubular defect | Yellow, crystalline, semi-opaque |

*Why has he or she got this stone now?*

- *Diet:* chocolate, tea, rhubarb, strawberries, nuts, and spinach all ↑oxalate levels.
- *Season:* variations in calcium and oxalate levels are thought to be mediated by vitamin D synthesis via sunlight on skin.
- *Work:* can he/she drink freely at work? Is there dehydration?
- *Medications:* precipitating drugs include: diuretics, antacids, acetazolamide, corticosteroids, theophylline, aspirin, allopurinol, vitamin C and D, indinavir.

*Are there any predisposing factors?* For example:

- *Recurrent UTIs* (in magnesium ammonium phosphate calculi).
- *Metabolic abnormalities:*
    - Hypercalciuria/hypercalcaemia (p676): hyperparathyroidism, neoplasia, sarcoidosis, hyperthyroidism, Addison's, Cushing's, lithium, vitamin D excess.
    - Hyperuricosuria/↑plasma urate: on its own, or with gout.
    - Hyperoxaluria.
    - Cystinuria (p321).
    - Renal tubular acidosis (pp316–7).
- *Urinary tract abnormalities:* eg pelviureteric junction obstruction, hydronephrosis (renal pelvis or calyces), calyceal diverticulum, horseshoe kidney, ureterocele, vesicoureteric reflux, ureteral stricture, medullary sponge kidney.[11]
- *Foreign bodies:* eg stents, catheters.

*Is there a family history?* ↑Risk of stones 3-fold. Specific diseases include X-linked nephrolithiasis and Dent's disease (proteinuria, hypercalciuria, and nephrocalcinosis).

►*Is there infection above the stone?* Eg fever, loin tender, pyuria? This needs urgent intervention.

**Fig 13.46** Calcium oxalate monohydrate.
Image courtesy of Dr Glen Austin.

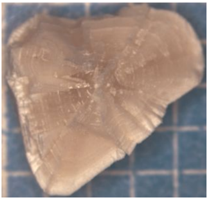

**Fig 13.47** Struvite stone.
Image courtesy of Dr Glen Austin.

**Fig 13.48** Cystine stone.
Image courtesy of Dr Glen Austin.

**11** Medullary sponge kidney is a typically asymptomatic developmental anomaly of the kidney mostly seen in adult females, where there is dilatation of the collecting ducts, which if severe leads to a sponge-like appearance of the renal medulla. *Complications/associations:* UTIs, nephrolithiasis, haematuria and hypercalciuria, hyperparathyroidism (if present, look for genetic markers of MEN type 2A, see p223).

# Urinary tract obstruction



►Urinary tract obstruction is common and should be considered in any patient with impaired renal function. ►►Damage can be permanent if the obstruction is not treated promptly. Obstruction may occur anywhere from the renal calyces to the urethral meatus, and may be *partial* or *complete*, *unilateral* or *bilateral*. Obstructing lesions are *luminal* (stones, blood clot, sloughed papilla, tumour: renal, ureteric, or bladder), *mural* (eg congenital or acquired stricture, neuromuscular dysfunction, schistosomiasis), or *extra-mural* (abdominal or pelvic mass/tumour, retroperitoneal fibrosis, or iatrogenic—eg post surgery). Unilateral obstruction may be clinically silent (normal urine output and U&E) if the other kidney is functioning. ►Bilateral obstruction or obstruction with infection requires urgent treatment. See p641.

## Clinical features

- *Acute upper tract obstruction:* Loin pain radiating to the groin. There may be superimposed infection ± loin tenderness, or an enlarged kidney.
- *Chronic upper tract obstruction:* Flank pain, renal failure, superimposed infection. Polyuria may occur due to impaired urinary concentration.
- *Acute lower tract obstruction:* Acute urinary retention typically presents with severe suprapubic pain ± acute confusion (elderly); often acute on chronic (hence preceded by chronic symptoms , see next bullet point). Clinically: distended, palpable bladder containing ~600mL, dull to percussion. Causes include prostatic obstruction (usual cause in older ♂), urethral strictures, anticholinergics, blood clots eg from bladder lesion ('clot retention'), alcohol, constipation, post-op (pain/inflammation/anaesthetics), infection (p296), neurological (cauda equina syndrome, see p466).
- *Chronic lower tract obstruction: Symptoms:* urinary frequency, hesitancy, poor stream, terminal dribbling, overflow incontinence. *Signs:* distended, palpable bladder (capacity may be >1.5L) ± large prostate on PR. Complications: UTI, urinary retention, renal failure (eg bilateral obstructive uropathy—see BOX 'Obstructive uropathy'). Causes include prostatic enlargement (common); pelvic malignancy; rectal surgery; DM; CNS disease, eg transverse myelitis/MS; zoster (S2–S4).

**Tests** *Blood:* U&E, creatinine, FBC, and prostate-specific antigen (PSA, p530).[12] *Urine:* Dipstick and MC&S. *Ultrasound* (p744) is the imaging modality of choice for investigating upper tract obstruction: If there is hydronephrosis or hydroureter (distension of the renal pelvis and calyces or ureter), arrange a *CT scan.* This will determine the level of obstuction. NB: in ~5% of cases of obstruction, no distension is seen on US. *Radionuclide imaging* enables functional assessment of the kidneys.

**Treatment** *Upper tract obstruction:* Nephrostomy or ureteric stent. NB: stents may cause significant discomfort and patients should be warned of this and other risks (see BOX 'Problems of ureteric stenting'). α-blockers help reduce stent-related pain (↓ureteric spasm). Pyeloplasty, to widen the PUJ, may be performed for idiopathic PUJ obstruction.

*Lower tract obstruction:* Insert a urethral or suprapubic catheter (p762) to relieve acute retention. In chronic obstruction only catheterize patient if there is pain, urinary infection, or renal impairment; intermittent self-catheterization is sometimes required (p763). If in clot retention the patient will require a 3-way catheter and bladder washout. If >1L residual check U&E and monitor for post-obstructive diuresis (see BOX 'Obstructive uropathy'). Monitor weight, fluid balance, and U&E closely. Treat the underlying cause if possible, eg if prostatic obstruction, start an α-blocker (see p642). After 2–3 days, trial without catheter (TWOC, p763) may work (especially if <75yrs old and <1L drained or retention was triggered by a passing event, eg GA).

12 Do venepuncture for PSA **before** PR, as PR can ↑ total PSA by ~1ng/mL (free PSA ↑ by 10%). It's difficult to know if acute retention raises PSA, but relieving obstruction does cause it to drop.

## Problems of ureteric stenting (depend on site)

*Common:*
- Stent-related pain
- Trigonal irritation
- Haematuria
- Fever
- Infection
- Tissue inflammation
- Encrustation
- Biofilm formation.

*Rare:*
- Obstruction
- Kinking
- Ureteric rupture
- Stent misplacement
- Stent migration (especially if made of silicone)
- Tissue hyperplasia
- Forgotton stents.

## Obstructive uropathy

In chronic urinary retention, an episode of *acute* retention may go unnoticed for days and, because of their background symptoms, may only present when overflow incontinence becomes a nuisance—pain is not necessarily a feature. After diagnosing acute on chronic retention and placing a catheter, the bladder residual can be as much as 1.5L of urine. Don't be surprised to be called by the biochemistry lab to be told that the serum creatinine is 1000μmol/L! The good news is that renal function usually returns to baseline after a few days (there may be mild background impairment). Ask for an urgent renal US (fig 13.49) and consider the following in the acute plan to ensure a safe course:

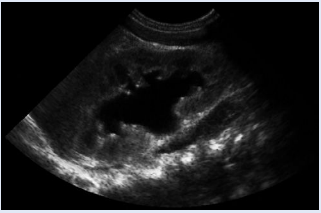

**Fig 13.49** Ultrasound of an obstructed kidney showing hydronephrosis. Note dilatation of renal pelvis and ureter, and clubbed calyces.

Image courtesy of Norwich Radiology Department.

- **Hyperkalaemia** See p301.
- **Metabolic acidosis** On ABG there is likely to be a respiratory compensated metabolic acidosis. Concerns should prompt discussion with a renal specialist (a good idea anyway), in case haemodialysis is required (p306).
- **Post-obstructive diuresis** In the acute phase after relief of the obstruction, the kidneys produce *a lot* of urine—as much as a litre in the first hour. It is vital to provide resuscitation fluids and then match input with output. ►Fluid depletion rather than overload is the danger here.
- **Sodium- and bicarbonate-losing nephropathy** As the kidney undergoes diuresis, $Na^+$ and bicarbonate are lost in the urine in large quantities. Replace 'in for out' (as mentioned above) with isotonic 1.26% sodium bicarbonate solution—this should be available from ITU. Some advocate using 0.9% saline, though the chloride load may exacerbate acidosis. Withhold any nephrotoxic drugs.
- **Infection** Treat infection, bearing in mind that the ↑WCC and ↑CRP may be part of the stress response. Send a sample of urine for MC&S.

# Benign prostatic hyperplasia

**Benign prostatic hyperplasia (BPH)** is common (24% if aged 40–64; 40% if older). *Pathology:* Benign nodular or diffuse proliferation of musculofibrous and glandular layers of the prostate. Inner (transitional) zone enlarges in contrast to peripheral layer expansion seen in prostate carcinoma. *Features: Lower urinary tract symptoms* (LUTS) = nocturia, frequency, urgency, post-micturition dribbling, poor stream/flow, hesitancy, overflow incontinence, haematuria, bladder stones, UTI. *Management:* Assess severity of symptoms and impact on life. PR exam. *Tests:* MSU; U&E; ultrasound (large residual volume, hydronephrosis—fig 13.49), PSA (prior to PR exam; see also BOX 'Advice to asymptomatic men', p645), transrectal US ± biopsy. Then consider:

- *Lifestyle:* Avoid caffeine, alcohol (to ↓urgency/nocturia). Relax when voiding. Void twice in a row to aid emptying. Control urgency by practising distraction methods (eg breathing exercises). Train the bladder by 'holding on' to ↑time between voiding.
- *Drugs* are useful in mild disease, and while awaiting surgery. • *α-blockers* are 1st line (eg tamsulosin 400mcg/d PO; also alfuzosin, doxazosin, terazosin). ↓Smooth muscle tone (prostate and bladder). SE: drowsiness; depression; dizziness; ↓BP; dry mouth; ejaculatory failure; extra-pyramidal signs; nasal congestion; ↑weight. • *5α-reductase inhibitors:* can be added, or used alone, eg finasteride 5mg/d PO (↓conversion of testosterone to the more potent androgen dihydrotestosterone). Excreted in semen, so use condoms; females should avoid handling. SE: impotence; ↓libido. ↓prostate size over 3–6mths and ↓ long-term retention risk.
- *Surgery:*
  - *Transurethral resection of prostate* (TURP) ≤14% become impotent (see BOX). Crossmatch 2U. Beware bleeding, clot retention, and post TURP syndrome: absorption of washout causing CNS & CVS disturbance. ~12% need redoing within 8yrs.
  - *Transurethral incision of the prostate* (TUIP) involves less destruction than TURP, and less risk to sexual function, gives similar benefit. Relieves pressure on the urethra. Maybe best surgical option for those with small glands <30g.
  - *Retropubic prostatectomy* is an open operation (if prostate very large).
  - *Transurethral laser-induced prostatectomy* (TULIP) may be as good as TURP.
  - *Robotic prostatectomy* is gaining popularity as a less traumatic and minimally invasive treatment option.

## Advice for patients concerning transurethral prostatectomy (TURP)

Pre-op consent issues may centre on risks of the procedure, eg:

- Haematuria/haemorrhage
- Haematospermia
- Hypothermia
- Urethral trauma/stricture
- Post TURP syndrome (↓T°; ↓$Na^+$)
- Infection; prostatitis
- Erectile dysfunction ~10%
- Incontinence ≤10%
- Clot retention near strictures
- Retrograde ejaculation (common).

**Post-operative advice**

- Avoid driving for 2wks after the operation.
- Avoid sex for 2wks after surgery. Then get back to normal. The amount ejaculated may be reduced (as it flows backwards into the bladder—harmless, but may cloud the urine). It means you may be infertile. Erections may be a problem after TURP, but do not expect this: in some men, erections improve. Rarely, orgasmic sensations are reduced.
- Expect to pass blood in the urine for the first 2wks. A small amount of blood colours the urine bright red. Do not be alarmed.
- At first you may need to urinate *more* frequently than before. Do not be despondent. In 6wks things should be much better—but the operation cannot be guaranteed to work (8% fail, and lasting incontinence is a problem in 6%; 12% may need repeat TURPs within 8yrs, compared with 1.8% of men undergoing open prostatectomy).
- If feverish, or if urination hurts, take a sample of urine to your doctor.

## Retroperitoneal fibrosis

**Causes** Idiopathic retroperitoneal fibrosis (RPF), inflammatory aneurysms of the abdominal aorta, and perianeurysmal RPF. With idiopathic RPF there is an associated inflammatory response resulting in fibrinoid necrosis of the vasa vasorum, affecting the aorta and small and medium retroperitoneal vessels. The ureters get embedded in dense, fibrous tissue resulting in progressive bilateral ureteric obstruction. Secondary causes of RPF include malignancy, typically lymphoma.

**Associations** Drugs (eg β-blockers, bromocriptine, methysergide, methyldopa), autoimmune disease (eg thyroiditis, SLE, ANCA+ve vasculitis), smoking, asbestos.

**Typical patient** Middle-aged ♂ with vague loin, back, or abdominal pain, ↑BP.

**Tests** *Blood:* ↑Urea and creatinine; ↑ESR; ↑CRP; anaemia. *Ultrasound:* Dilated ureters (hydronephrosis). *CT/MRI:* Periaortic mass (fig 13.50). Biopsy under imaging guidance is used to rule out malignancy.

**Treatment** Retrograde stent placement to relieve obstruction (removed after 12 months) ± ureterolysis (dissection of the ureters from the retroperitoneal tissue). Immunosuppression (in idiopathic RPF) with low-dose steroids has good long-term results.

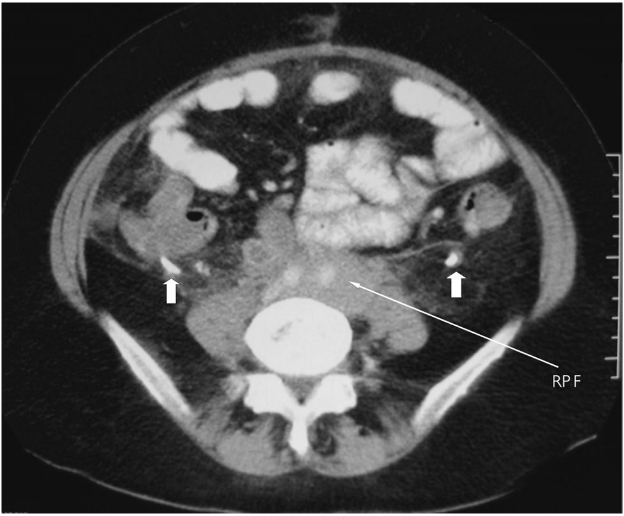


Fig 13.50 CT scan of retroperitoneal fibrosis (RPF), with subsequent obstruction and dilatation of the ureters (thick arrows).

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# Urinary tract malignancies

**Renal cell carcinoma (RCC)** arises from proximal renal tubular epithelium. *Epidemiology:* Accounts for 90% of renal cancers; mean age 55yrs. ♂:♀≈2:1. 15% of haemodialysis patients develop RCC. *Features:* 50% found incidentally. Haematuria, loin pain, abdominal mass, anorexia, malaise, weight loss, PUO—often in isolation. Rarely, invasion of left renal vein compresses left testicular vein causing a varicocele. Spread may be direct (renal vein), via lymph, or haematogenous (bone, liver, lung). 25% have metastases at presentation. *Tests:* BP: ↑from renin secretion. *Blood:* FBC (polycythaemia from erythropoietin secretion); ESR; U&E, ALP (bony mets?). *Urine:* RBCs; cytology. *Imaging:* US (p744); CT/MRI; CXR ('cannon ball' metastases). *℞:* Radical nephrectomy (nephron-sparing surgery is as good for T1 tumours + preserves renal function). Cryotherapy and radiofrequency ablation is an option for patients unfit or unwilling to undergo surgery. RCC is generally radio- & chemoresistant. In those with unresectable or metastatic disease, options include: high-dose IL-2 and other T-cell activation therapies; anti-angiogenesis agents (eg pazopanib, sunitinib, axitinib, or bevacizumab); mTOR inhibitors, eg temsirolimus. The *Mayo prognostic risk score (SSIGN)* was developed to predict survival and uses information on tumour stage, size, grade, and necrosis. *Prognosis:* 10yr survival ranges from 96.5% (scores 0–1) to 19.2% (scores ≥10).

**Transitional cell carcinoma (TCC)** may arise in the bladder (50%), ureter, or renal pelvis. *Epidemiology:* Age >40yrs; ♂:♀≈4:1. *Risk factors:* p646. *Presentation:* Painless haematuria; frequency; urgency; dysuria; urinary tract obstruction. *Diagnosis:* Urine cytology; IVU; cystoscopy + biopsy; CT/MRI. *℞:* See 'Bladder tumours', p646. *Prognosis:* Varies with clinical stage/histological grade: 10–80% 5yr survival.

**Wilms' tumour** (nephroblastoma) is a childhood tumour of primitive renal tubules and mesenchymal cells. *Prevalence:* 1:100 000—the chief abdominal malignancy in children. It presents with an abdominal mass and haematuria. *℞:* *OHCS* p133.

**Prostate cancer** The commonest male malignancy. *Incidence:* ↑with age: 80% in men >80yrs (autopsy studies). *Associations:* +ve family history (x2–3 ↑risk, p521), ↑testosterone. Most are adenocarcinomas arising in peripheral prostate. Spread may be local (seminal vesicles, bladder, rectum) via lymph, or haematogenously (sclerotic bony lesions). *Symptoms:* Asymptomatic or nocturia, hesitancy, poor stream, terminal dribbling, or obstruction. ↓Weight ± bone pain suggests mets. *DRE exam of prostate:* may show hard, irregular prostate. *Diagnosis:* ↑PSA (normal in 30% of small cancers); transrectal US & biopsy; bone scan; CT/MRI. *Staging:* MRI. *Treatment: Disease confined to prostate:* options depend on prognosis (see BOX 'Prognostic factors'), patient preference, and comorbidities. •*Radical prostatectomy* if <70yrs gives excellent disease-free survival (laparoscopic surgery is as good). The role of adjuvant hormonal therapy is being explored. •*Radical radiotherapy* (± neoadjuvant & adjuvant hormonal therapy) is an alternative curative option that compares favourably with surgery (no RCTs). It may be delivered as external beam or brachytherapy. •*Hormone therapy alone* temporarily delays tumour progression but refractory disease eventually develops. Consider in elderly, unfit patients with high-risk disease. •*Active surveillance*—particularly if >70yrs and low-risk. *Metastatic disease:* •*Hormonal drugs* may give benefit for 1–2yrs. LHRH agonists, eg 12-weekly goserelin (10.8mg SC) first stimulate, then inhibit pituitary gonadotrophin. NB: risks tumour 'flare' when first used—start anti-androgen, eg cyproterone acetate, in susceptible patients. The LHRH antagonist degarelix is also used in advanced disease. *Symptomatic ℞:* Analgesia; treat hypercalcaemia; radiotherapy for bone mets/spinal cord compression. *Prognosis:* 10% die in 6 months, 10% live >10yrs. *Screening:* DRE of prostate; transrectal US; PSA (see BOX 'Advice to asymptomatic men').

**Penile cancer** *Epidemiology:* Rare in UK, more common in Far East and Africa, very rare in circumcised. Related to chronic irritation, viruses, smegma. *Presentation:* Chronic fungating ulcer, bloody/purulent discharge, 50% spread to lymph at presentation *℞:* Radiotherapy & irridium wires if early; amputation & lymph node dissection if late.

## Advice to asymptomatic men asking for a PSA blood test

- Many men over 50 consider a PSA test to detect prostatic cancer. *Is this wise?*
- The test is not very accurate, and we cannot say that those having the test will live longer—even if they turn out to have prostate cancer. Most men with prostate cancer die from an unrelated cause.
- If the test is falsely positive, you may needlessly have more tests, eg prostate sampling via the back passage (causes bleeding and infection in 1–5% of men).
- Only one in three of those with a high PSA level will have cancer.
- You may be worried needlessly if later tests put you in the clear.
- If a cancer is found, there's no way to tell *for sure* if it will impinge on health. You might end up having a bad effect from treatment that wasn't needed.
- There is much uncertainty on treating those who *do* turn out to have prostate cancer: options are radical surgery to remove the prostate (risks erectile dysfunction and incontinence), radiotherapy, or hormones.
- Screening via PSA has shown conflicting results. Some RCTs have shown no difference in the rate of death from prostate cancer, others have found reduced mortality, eg 1 death prevented per 1055 men invited for screening (if 37 cancers detected).

►Ultimately, you must decide for yourself what you want.

## Prognostic factors in prostate cancer

A number of prognostic factors help determine if 'watchful waiting' or aggressive therapy should be advised: •Pre-treatment PSA level. •Tumour stage (as measured by the TNM system; p523). •Tumour grade—Gleason score. Gleason grading is from 1 to 5, with 5 being the highest grade, and carrying the poorest prognosis. Gleason grades are decided by analysing histology from two separate areas of tumour specimen, and adding them to get the total Gleason score for the tumour, from 2 to 10. Scores 8–10 suggest an aggressive tumour; 5–7: intermediate; 2–4: indolent.

## Benign diseases of the penis

**Balanitis** Acute inflammation of the foreskin and glans. Associated with strep and staph infections. More common in diabetics. Often seen in young children with tight foreskins *℞:* Antibiotics, circumcision, hygiene advice.

**Phimosis** The foreskin occludes the meatus. In young boys this causes recurrent balanitis and ballooning, but time (+ trials of gentle retraction) may obviate the need for circumcision. In adulthood presents with painful intercourse, infection, ulceration, and is associated with balanitis xerotica obliterans.

**Paraphimosis** Occurs when a tight foreskin is retracted and becomes irreplaceable, preventing venous return leading to oedema and even ischaemia of the glans. Can occur if the foreskin is not replaced after catheterization. ►►*℞:* Ask patient to squeeze glans. Try applying a 50% glucose-soaked swab (oedema may follow osmotic gradient). Ice packs and lidocaine gel may also help. May require aspiration/dorsal slit/circumcision.

## Prostatitis

May be acute or chronic. Usually those >35yrs. *Acute prostatitis* is caused mostly by *S. faecalis* and *E. coli*, also *Chlamydia* (and previously TB). *Features:* UTIs, retention, pain, haematospermia, swollen/boggy prostate on DRE. *℞:* Analgesia; levofloxacin 500mg/24h PO for 28d. *Chronic prostatitis* may be bacterial or non-bacterial. Symptoms as for acute prostatitis, but present for >3 months. Non-bacterial chronic prostatitis does not respond to antibiotics. Anti-inflammatory drugs, α-blockers, and prostatic massage all have a place.

# Bladder tumours

>90% are transitional cell carcinomas (TCCs) in the UK. Adenocarcinomas and squamous cell carcinomas are rare in the West (the latter may follow schistosomiasis). UK incidence ≈ 1:6000/yr. ♂:♀≈5:2. Histology is important for prognosis: *Grade 1*—differentiated; *Grade 2*—intermediate; *Grade 3*—poorly differentiated. 80% are confined to bladder mucosa, and only ~20% penetrate muscle (increasing mortality to 50% at 5yrs).

**Presentation** Painless haematuria; recurrent UTIs; voiding irritability.

**Associations** Smoking; aromatic amines (rubber industry); chronic cystitis; schistosomiasis (↑risk of squamous cell carcinoma); pelvic irradiation.

**Tests**
- Cystoscopy with biopsy is diagnostic.
- Urine: microscopy/cytology (cancers may cause sterile pyuria).
- CT urogram is both diagnostic and provides staging.
- Bimanual EUA helps assess spread.
- MRI or lymphangiography may show involved pelvic nodes.

**Staging** See table 13.19.

**Treating TCC of the bladder**
- *Tis/Ta/T1:* (80% of all patients) Diathermy via transurethral cystoscopy/transurethral resection of bladder tumour (TURBT). Consider a regimen of intravesical *BCG* (which stimulates a non-specific immune response) for multiple small tumours or high-grade tumours. Alternative chemotherapeutic agents include mitomycin, epirubicin and gemcitabine. 5yr survival ≈ 95%.
- *T2–3:* Radical cystectomy is the 'gold standard'. Radiotherapy gives worse 5yr survival rates than surgery, but preserves the bladder. 'Salvage' cystectomy can be performed if radiotherapy fails, but yields worse results than primary surgery. Post-op chemotherapy (eg M-VAC: methotrexate, vinblastine, doxorubicin, and cisplatin) is toxic but effective. Neoadjuvant chemotherapy with M-VAC or GC (gemcitabine and cisplatin) has improved survival compared to cystectomy or radiotherapy alone. Methods to preserve the bladder with transurethral resection or partial cystectomy + systemic chemotherapy have been tried, but long-term results are disappointing. If the bladder neck is not involved, orthotopic reconstruction rather than forming a urostoma is an option (both using ~40cm of the patient's ileum), but adequate tumour clearance must not be compromised. ►The patient should have all these options explained by a urologist and an oncologist.
- *T4:* Usually palliative chemo/radiotherapy. Chronic catheterization and urinary diversions may help to relieve pain.

**Follow-up** History, examination, and regular cystoscopy: •*High-risk tumours:* Every 3 months for 2yrs, then every 6 months. •*Low-risk tumours:* First follow-up cystoscopy after 9 months, then yearly.

**Tumour spread** Local → to pelvic structures; lymphatic → to iliac and para-aortic nodes; haematogenous → to liver and lungs.

**Survival** This depends on age at surgery. For example, the 3yr survival after cystectomy for T2 and T3 tumours is 60% if 65–75yrs old, falling to 40% if 75–82yrs old (operative mortality is 4%). With unilateral pelvic node involvement, only 6% of patients survive 5yrs. The 3yr survival with bilateral or para-aortic node involvement is nil.

**Complications** Cystectomy can result in sexual and urinary malfunction. Massive bladder haemorrhage may complicate treatment or be a feature of disease treated palliatively. Determining the cause of bleeding is key. Consider alum solution bladder irrigation (if no renal failure) as 1st-line treatment for intractable haematuria in advanced malignancy: it is an inpatient procedure.

Table 13.19 TNM staging of bladder cancer (See also p523)

| | |
|---|---|
| **Tis** | Carcinoma *in situ* |
| **Ta** | Tumour confined to epithelium |
| **T1** | Tumour in submucosa or lamina propria |
| **T2** | Invades muscle |
| **T3** | Extends into perivesical fat |
| **T4** | Invades adjacent organs |
| **N0** | No LN involved |
| **N1-N3** | Progressive LN involvement |
| **M0** | No metastases |
| **M1** | Distant metastasis |

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## Is asymptomatic non-visible haematuria significant?

Dipstick tests are often done routinely for patients on admission. If non-visible (previously microscopic) haematuria is found, but the patient has no related symptoms, what does this mean? Before rushing into a barrage of investigations, consider:

- One study found that incidence of urogenital disease (eg bladder cancer) was no higher in those with asymptomatic microhaematuria than in those without.
- Asymptomatic non-visible haematuria is the sole presenting feature in only 4% of bladder cancers, and there is no evidence that these are less advanced than malignancies presenting with macroscopic haematuria.
- When monitoring those with treated bladder cancer for recurrence, non-visible-haematuria tests have a sensitivity of only 31% in those with superficial bladder malignancy, in whom detection would be most useful.
- Although 80% of those with flank pain due to a renal stone have microscopic haematuria, so do 50% of those with flank pain but no stone.

The conclusion is not that urine dipstick testing is useless, but that results should not be interpreted in isolation. ►Unexplained non-visible haematuria in those >50yrs should be referred under the 2-week rule. Smokers and those with +ve family history for urothelial cancer may also be investigated differently from those with no risk factors. It is worth considering, that in a young, fit athlete, the diagnosis is more likely to be exercise-induced haematuria. Wise doctors work collaboratively with their patients. 'Shall we let sleeping dogs lie?' is a reasonable question for *some* patients.

# Urinary incontinence



►Think twice before inserting a urinary catheter.
►Carry out rectal examination to exclude faecal impaction.
►Is the bladder palpable after voiding (retention with overflow)?
►Is there neurological comorbidity: eg MS; Parkinson's disease; stroke; spinal trauma?

**Incontinence in men** Enlargement of the prostate is the major cause of incontinence: urge incontinence (see later in topic) or dribbling may result from partial retention of urine. TURP (p642) & other pelvic surgery may weaken the bladder sphincter and cause incontinence. Troublesome incontinence needs specialist assessment.

**Incontinence in women** Often under-reported with delays before seeking help.

- *Functional incontinence:* Ie when physiological factors are relatively unimportant. The patient is 'caught short' and too slow in finding the toilet because of (for example) immobility, or unfamiliar surroundings.
- *Stress incontinence:* Leakage from an incompetent sphincter, eg when intra-abdominal pressure rises (eg coughing, laughing). Increasing age and obesity are risk factors. The key to diagnosis is the loss of small (but often frequent) amounts of urine when coughing etc. Examine for pelvic floor weakness/prolapse/pelvic masses. Look for cough leak on standing and with full bladder. Stress incontinence is common in pregnancy and following birth. It occurs to some degree in ~50% of post-menopausal women. In elderly women, pelvic floor weakness, eg with uterine prolapse or urethrocele (*OHCS* p290), is a very common association.
- *Urge incontinence/overactive bladder syndrome:* The urge to urinate is quickly followed by uncontrollable and sometimes complete emptying of the bladder as the detrusor muscle contracts. Urgency/leaking is precipitated by: arriving home (latchkey incontinence, a conditioned reflex); cold; the sound of running water; caffeine; and obesity. *Δ:* urodynamic studies. *Cause:* detrusor overactivity (see table 13.20), eg from central inhibitory pathway malfunction or sensitization of peripheral afferent terminals in the bladder; or a bladder muscle problem. Check for organic brain damage (eg stroke; Parkinson's; dementia). *Other causes:* urinary infection; diabetes; diuretics; atrophic vaginitis; urethritis.

In both sexes incontinence may result from confusion or sedation. Occasionally it may be purposeful (eg preventing admission to an old people's home) or due to anger.

**Management** ►Effective treatment can have a huge impact on quality of life.

*Check for:* UTI; DM; diuretic use; faecal impaction; palpable bladder; GFR.

- *Stress incontinence:* Pelvic floor exercises are 1st line (8 contractions ×3/d for 3 months). Intravaginal electrical stimulation may also be effective, but is not acceptable to many women. A ring pessary may help uterine prolapse, eg while awaiting surgical repair. *Surgical options* (eg tension-free vaginal tape) aim to stabilize the mid-urethra. Urethral bulking also available. Medical options: duloxetine 40mg/12h PO (50% have ≥50% ↓ in incontinence episodes). SE = nausea.
- *Urge incontinence:* The patient (or carer) should complete an 'incontinence' chart for 3d to define the pattern of incontinence. Examine for spinal cord and CNS signs (including cognitive test, p64); and for vaginitis (if postmenopausal). Vaginitis can be treated with topical oestrogen therapy for a limited period. Bladder training (may include pelvic floor exercises) and weight loss are important. Drugs may help reduce night-time incontinence (see BOX) but can be disappointing. Consider aids, eg absorbent pad. If ♂ consider a condom catheter.

►Do urodynamic assessment (cystometry & urine flow rate measurement) before any surgical intervention to exclude detrusor overactivity or sphincter dyssynergia.

Table 13.20 Managing detrusor overactivity in urge incontinence

| Agents for detrusor overactivity | Notes |
|---|---|
| Antimuscarinics: eg tolterodine *SR* 4mg/24h; SE: dry mouth, eyes/skin, drowsiness, constipation, tachycardia, abdominal pain, urinary retention, sinusitis, oedema, ↑weight, glaucoma precipitation. Up to 4mg/12h may be needed (unlicensed). | Improves frequency & urgency. Alternatives: solifenacin 5mg/24h (max 10mg); oxybutynin, but more SE unless transdermal route or modified release used; trospium or fesoterodine (prefers M3 receptors). Avoid in myasthenia, and if glaucoma or UC are uncontrolled. |
| Topical oestrogens | Post-menopausal urgency, frequency + nocturia may occasionally be improved by raising the bladder's sensory threshold. Systemic therapy worsens incontinence. |
| β3 adrenergic agonist: mirabegron 50mg/24h; SE tachycardia; CI: severe HTN; Caution if renal/hepatic impairment. | Consider if antimuscarinics are contraindicated or clinically ineffective, or if SE unacceptable. |
| Intravesical botulinum toxin (Botox®) | Consider if above medications ineffective. |
| Percutaneous posterior tibial nerve stimulation (PTNS). (A typical treatment consists of ×12 weekly 30 min sessions.) | Consider if drug treatment ineffective and Botox® not wanted. PTNS delivers neuromodulation to the S2–S4 junction of the sacral nerve plexus. |
| Neuromodulation via transcutaneous electrical stimulation | Sacral nerve stimulation inhibits the reflex behaviour of involuntary detrusor contractions. |
| Modulation of afferent input from bladder | Gabapentin (unlicensed). |
| Hypnosis, psychotherapy, bladder training* | (These all require good motivation.) |
| Surgery (eg clam ileocystoplasty) | Reserved for troublesome or intractable symptoms. The bladder is bisected, opened like a clam, and 25cm of ileum is sewn in. |

NB: desmopressin nasal spray 20mcg nocte reduces urine production and ∴ nocturia in overactive bladder. Unsuitable if elderly (SE: fluid retention, heart failure, ↓$Na^+$).
*Mind over bladder: •Void when you DON'T have urge; DON'T go to the bathroom when you do have urge. •Gradually extend the time between voiding. •Schedule your trips to toilet. •Stretch your bladder to normal capacity. • When urge comes, calm down and make it go using mind over bladder tricks.

## Not all male urinary symptoms are prostate-related!

**Detrusor overactivity** Men get this as well as women. Pressure-flow studies help diagnose this (as does detrusor thickness ≥2.9mm on US).
**Primary bladder neck obstruction** A condition in which the bladder neck does not open properly during voiding. Studies in men and women with voiding dysfunction show that it is common. The cause may be muscular or neurological dysfunction or fibrosis. *Diagnosis:* Video-urodynamics, with simultaneous pressure-flow measurement, and visualization of the bladder neck during voiding. *Treatment:* Watchful waiting; α-blockers (p642); surgery.

**Urethral stricture** This may follow trauma or infection (eg gonorrhoea)—and frequently leads to voiding symptoms, UTI, or retention. Malignancy is a rare cause. *Imaging:* Retrograde urethrogram or antegrade cystourethrogram if the patient has an existing suprapubic catheter. *Internal urethrotomy* involves incising the stricture transurethrally using endoscopic equipment—to release scar tissue. *Stents* incorporate themselves into the wall of the urethra and keep the lumen open. They work best for short strictures in the bulbar urethra (anterior urethral anatomy, from proximal to distal: prostatic urethra→posterior or membranous urethra→bulbar urethra→penile or pendulous urethra→fossa navicularis→meatus).

# Lumps in the groin and scrotum

►Testicular lump = cancer until proved otherwise.
►Acute, tender enlargement of testis = torsion (p652) until proved otherwise.

**Diagnosing scrotal masses** (fig 13.51)
**1** *Can you get above it?* **2** *Is it separate from the testis?* **3** *Cystic or solid?*
- Cannot get above: inguinoscrotal hernia (p614) or hydrocele extending proximally.
- Separate and cystic: epididymal cyst.
- Separate and solid: epididymitis/varicocele.
- Testicular and cystic: hydrocele.

►Testicular and solid—*tumour*, haematocele, granuloma (p196), orchitis, gumma (p412). US may help.

**Epididymal cysts** Usually develop in adulthood and contain clear or milky (spermatocele) fluid. They lie above and behind the testis. Remove if symptomatic.

**Hydroceles** (Fluid within the tunica vaginalis.) *Primary* (associated with a patent processus vaginalis, which typically resolves during the 1st year of life) or *secondary* to testis tumour/trauma/infection. Primary hydroceles are more common, larger, and usually in younger men. Can resolve spontaneously. *℞:* Aspiration (may need repeating) or surgery: plicating the tunica vaginalis (Lord's repair)/inverting the sac (Jaboulay's repair). ►Is the testis normal after aspiration? If any doubt, do US.

**Epididymo-orchitis** *Causes*: *Chlamydia* (eg if <35yrs); *E. coli*; mumps; *N. gonorrhoeae*; TB. *Features:* Sudden-onset tender swelling, dysuria, sweats/fever. Take '1st catch' urine sample; look for urethral discharge. Consider STI screen. Warn of possible infertility and symptoms worsening before improving. *℞:* If <35yrs; doxycycline 100mg/12h (covers chlamydia; treat sexual partners). If gonorrhoea suspected add ceftriaxone 500mg IM stat. If >35yrs (mostly non-STI), associated UTI is common so try ciprofloxacin 500mg/12h or ofloxacin 200mg/12h. Antibiotics should be used for 2–4wks. Also: analgesia, scrotal support, drainage of any abscess.

**Varicocele** Dilated veins of pampiniform plexus. Left side more commonly affected. Often visible as distended scrotal blood vessels that feel like 'a bag of worms'. Patient may complain of dull ache. Associated with subfertility, but repair (via surgery or embolization) seems to have little effect on subsequent pregnancy rates.

**Haematocele** Blood in tunica vaginalis, follows trauma, may need drainage/excision.

**Testicular tumours** The commonest malignancy in ♂ aged 15–44; 10% occur in undescended testes, even after orchidopexy. A contralateral tumour is found in 5%. *Types:* Seminoma, 55% (30–65yrs); non-seminomatous germ cell tumour, 33% (NSGCT; previously teratoma; 20–30yrs); mixed germ cell tumour, 12%; lymphoma.

*Signs:* Typically painless testis lump, found after trauma/infection ± haemospermia, secondary hydrocele, pain, dyspnoea (lung mets), abdominal mass (enlarged nodes), or effects of secreted hormones. 25% of seminomas & 50% of NSGCTs present with metastases. *Risk factors:* Undescended testis; infant hernia; infertility.

*Staging:* **1** No evidence of metastasis. **2** Infradiaphragmatic node involvement (spread via the para-aortic nodes *not* inguinal nodes). **3** Supradiaphragmatic node involvement. **4** Lung involvement (haematogenous).

*Tests:* (Allow staging.) CXR, CT, excision biopsy. α-FP (eg >3IU/mL)[13] and β-hCG are useful tumour markers and help monitor treatment (p531); check *before & during* ℞.

*℞:* Radical orchidectomy (inguinal incision; occlude the spermatic cord before mobilization to ↓risk of intra-operative spread). Options are constantly updated (surgery, radiotherapy, chemotherapy). Seminomas are exquisitely radiosensitive. Stage 1 seminomas: orchidectomy + radiotherapy cures ~95%. Do close follow-up to detect relapse. Cure of NSGCT, even if metastases are present, is achieved by 3 cycles of bleomycin + etoposide + cisplatin. 5yr survival >90% in all groups.
►Encourage regular self-examination (prevents late presentation).

13 αFP is *not* ↑ in pure seminoma; may also be ↑ in: hepatitis, cirrhosis, liver cancer, open neural tube defect.

## Diagnosing groin lumps: lateral to medial thinking

- Psoas abscess—may present with back pain, limp, and swinging pyrexia.
- Neuroma of the femoral nerve.
- Femoral artery aneurysm.
- Saphena varix—like a hernia, it has a cough impulse.
- Lymph node.
- Femoral hernia.
- Inguinal hernia.
- Hydrocele or varicocele.
- Also consider an undescended testis (cryptorchidism).

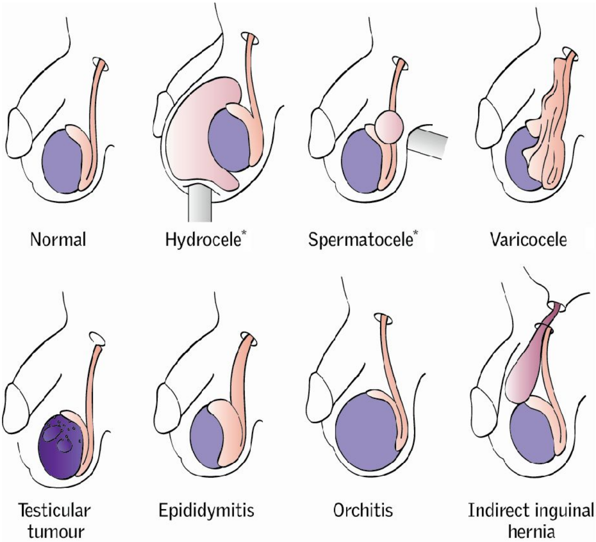


Fig 13.51 Diagnosis of scrotal masses. (*=transilluminates: position of pen torch shown in image).

## ►►Torsion of the testis

The aim is to recognize this condition before the cardinal signs and symptoms are fully manifest, as prompt surgery saves testes. If surgery is performed in <6h the salvage rate is 90–100%; if >24h it is 0–10%.

►*If in any doubt, surgery is required. If suspected refer immediately to urology.*

*Symptoms:* Sudden onset of pain in one testis, which makes walking uncomfortable. Pain in the abdomen, nausea, and vomiting are common.

*Signs:* Inflammation of one testis—it is very tender, hot, and swollen. The testis may lie high and transversely. Torsion may occur at any age but is most common at 11–30yrs. With *intermittent torsion* the pain may have passed on presentation, but if it was severe, and the lie is horizontal, prophylactic fixing may be wise.

*ΔΔ:* The main one is epididymo-orchitis (p650) but with this the patient tends to be older, there may be symptoms of urinary infection, and more gradual onset of pain. Also consider tumour, trauma, and an acute hydrocele. NB: *torsion of testicular or epididymal appendage* (the hydatid of Morgagni—a remnant of the Müllerian duct)—usually occurs between 7–12yrs, and causes less pain. Its tiny blue nodule may be discernible under the scrotum. It is thought to be due to the surge in gonadotrophins which signal the onset of puberty. *Idiopathic scrotal oedema* is a benign condition usually between ages 2 and 10yrs, and is differentiated from torsion by the absence of pain and tenderness.

*Tests:* Doppler US may demonstrate lack of blood flow to testis. Only perform if diagnosis equivocal—do not delay surgical exploration.

*Treatment:* ►Ask consent for possible orchidectomy + *bilateral* fixation (orchidopexy)—see p568. At surgery expose and untwist the testis. If its colour looks good, return it to the scrotum and fix *both* testes to the scrotum.

## Undescended testes

**Incidence** About 3% of boys are born with at least one undescended testis (30% of premature boys) but this drops to 1% after the first year of life. Unilateral is four times more common than bilateral. (If bilateral then should have genetic testing.)

- *Cryptorchidism:* Complete absence of the testis from the scrotum (anorchism is absence of both testes).
- *Retractile testis:* The genitalia are normally developed but there is an excessive cremasteric reflex. The testis is often found at the external inguinal ring. ℞: reassurance (examining while in a warm bath, for example, may help to distinguish from maldescended/ectopic testes).
- *Maldescended testis:* May be found anywhere along the normal path of descent from abdomen to groin.
- *Ectopic testis:* Most commonly found in the superior inguinal pouch (anterior to the external oblique aponeurosis) but may also be abdominal, perineal, penile, and in the femoral triangle.

**Complications of maldescended and ectopic testis** Infertility; ×40 increased risk of testicular cancer (risk remains after surgery but in cryptorchidism may be ↓ if orchidopexy performed before aged 10), increased risk of testicular trauma, increased risk of testicular torsion. Also associated with hernias (due to patent processus vaginalis in >90%, p613) and other urinary tract anomalies.

**Treatment of maldescended and ectopic testis** restores (potential for) spermatogenesis; the increased risk of malignancy remains but becomes easier to diagnose.

*Surgery:* Orchidopexy, usually dartos pouch procedure, is performed in infancy: testis and cord are mobilized following a groin incision, any processus vaginalis or hernial sac is removed and the testis is brought through a hole made in the dartos muscle into the resultant subcutaneous pouch where the muscle prevents retraction.

*Hormonal:* Hormonal therapy, most commonly human chorionic gonadotrophin (hCG), is sometimes attempted if an undescended testis is in the inguinal canal.

# ►►Aneurysms of arteries

An artery with a dilatation >50% of its original diameter has an aneurysm; remember this is an ongoing process. True aneurysms are abnormal dilatations that involve all layers of the arterial wall. False aneurysms (pseudoaneurysms) involve a collection of blood in the outer layer only (adventitia) which communicates with the lumen (eg after trauma). Aneurysms may be fusiform (eg most AAAs) or sac-like (eg Berry aneurysms; fig 10.17 p479).

**Causes** Atheroma, trauma, infection (eg mycotic aneurysm in endocarditis; tertiary syphilis—especially thoracic aneurysms), connective tissue disorders (eg Marfan's, Ehlers-Danlos), inflammatory (eg Takayasu's aortitis, p712).

**Common sites** Aorta (infrarenal most common), iliac, femoral, and popliteal arteries.

**Complications** Rupture; thrombosis; embolism; fistulae; pressure on other structures.

**Screening** All ♂ at age 65yr are invited for screening in UK, decreases mortality from ruptured AAA.

**Ruptured abdominal aortic aneurysm (AAA)** Death rates/year from ruptured AAAs rise with age: 125 per million in those aged 55–59; 2728 per million if over 85yrs. *Symptoms & signs:* Intermittent or continuous abdominal pain (radiates to back, iliac fossae, or groins; ►don't dismiss this as renal colic), collapse, an *expansile* abdominal mass (it expands and contracts, unlike swellings that are purely pulsatile, eg nodes overlying arteries), and shock. If in doubt, assume a ruptured aneurysm.

**Unruptured AAA** *Definition:* >3cm across. *Prevalence:* 3% of those >50yrs. ♂:♀ >3:1. Less common in diabetics. *Cause:* Degeneration of elastic lamellae and smooth muscle loss. There is a genetic component. *Symptoms:* Often none, they *may* cause abdominal/back pain, often discovered incidentally on abdominal examination (see BOX). *Monitoring:* RCTs have failed to demonstrate benefit from early endovascular repair (EVAR, see later in paragraph) of aneurysms <5.5cm (where rupture rates are low). Risk of rupture below this size is <1%/yr, compared to ~25%/yr for aneurysms >6cm across. ~75% of aneurysms under monitoring will eventually need repair. Rupture is more likely if: •↑BP •Smoker •♀ •Positive family history. Modify risk factors if possible at diagnosis. *Elective surgery:* Reserve for aneurysms ≥5.5cm or expanding at >1cm/yr, or symptomatic aneurysms. Operative mortality: ~5%; complications include spinal or visceral ischaemia and distal emboli from dislodged thrombus debris. Studies show that age >80yrs should not, in itself, preclude surgery. *Stenting (EVAR):* Major surgery can be avoided by inserting an endovascular stent via the femoral artery. EVAR has less early mortality but higher graft complications, eg failure of stent-graft to totally exclude blood flow to the aneurysm—'endoleak'. See fig 13.52.

## Emergency management of a ruptured abdominal aneurysm

Mortality—treated: 41% and improving; untreated: ~100%.

- ►►Summon a vascular surgeon and an experienced anaesthetist; warn theatre.
- ►►Do an ECG, and take blood for amylase, Hb, crossmatch (10–40U may eventually be needed). Catheterize the bladder.
- ►►Gain IV access with 2 large-bore cannulae. Treat shock with O Rh−ve blood (if not cross matched), but keep systolic BP ≤100mmHg to avoid rupturing a contained leak (NB: *raised* BP is common early on).
- ►►Take the patient straight to theatre. Don't waste time on X-rays: fatal delay may result, though CT can help in a stable patient with an uncertain diagnosis.
- ►►Give prophylactic antibiotics, eg co-amoxiclav 625mg IV.
- ►►Surgery involves clamping the aorta above the leak, and inserting a Dacron® graft (eg 'tube graft' or, if significant iliac aneurysm also, a 'trouser graft' with each 'leg' attached to an iliac artery).

## ►►Thoracic aortic dissection

Blood splits the aortic media with sudden tearing chest pain (± radiation to back). As the dissection extends, branches of the aorta occlude sequentially leading to hemiplegia (carotid artery), unequal arm pulses and BP, or acute limb ischaemia, paraplegia (anterior spinal artery), and anuria (renal arteries). Aortic valve incompetence, inferior MI, and cardiac arrest may develop if dissection moves proximally. *Type A* (70%) dissections involve the ascending aorta, irrespective of site of the tear, while if the ascending aorta is not involved it is called *type B* (30%). ►►All patients with type A thoracic dissection should be considered for surgery: *get urgent cardiothoracic advice*. Definitive treatment for type B is less clear and may be managed medically, with surgery reserved for distal dissections that are leaking, ruptured, or compromising vital organs. *Management:* •Crossmatch 10U blood. •ECG & CXR (expanded mediastinum is rare). •CT or transoesophageal echocardiography (TOE). Take to ITU; hypotensives: keep systolic at ~100–110mmHg: labetalol (p140) or esmolol ($t½$ is ultra-short) by IVI is helpful here (calcium-channel blockers may be used if β-blockers contraindicated). Acute operative mortality: <25%.

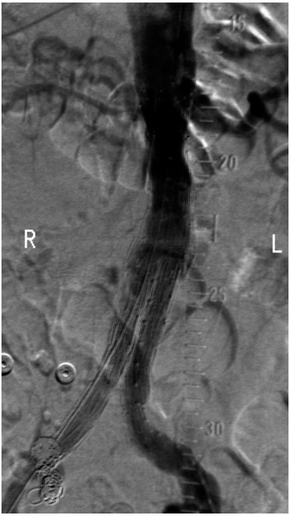


Fig 13.52 Stenting: not an open or closed case … this is a digital subtraction angiogram showing correct positioning of an endovascular stent at the end of the procedure. Although less invasive than open repair, some are unsuited to this method, owing to the anatomy of their aneurysm. Lifelong monitoring is needed: stents may leak and the aneurysm progress (the risk can be reduced by coiling the internal iliac arteries, as shown).

Image courtesy of Norwich Radiology Dept.

## Peripheral arterial disease (PAD)

Occurs due to atherosclerosis causing stenosis of arteries (fig 13.53) via a multi-factorial process involving modifiable and non-modifiable risk factors. 65% have coexisting clinically relevant cerebral or coronary artery disease. ►Cardiovascular risk factors should be identified and treated aggressively. The chief feature of PAD is intermittent claudication (= *to limp*). Prevalence = 10%.

**Symptoms** Cramping pain in the calf, thigh, or buttock after walking for a given distance (the claudication distance) and relieved by rest (calf claudication suggests femoral disease while buttock claudication suggests iliac disease). Ulceration, gangrene (p660), and foot pain at rest—eg burning pain at night relieved by hanging legs over side of bed—are the cardinal features of *critical ischaemia*. Buttock claudication ± impotence imply Leriche's syndrome (p704). Young, heavy smokers are at risk from Buerger's disease (thromboangiitis obliterans, p696).

*Fontaine classification for PAD:* **1** Asymptomatic. **2** Intermittent claudication. **3** Ischaemic rest pain. **4** Ulceration/gangrene (critical ischaemia).

**Signs** Absent femoral, popliteal, or foot pulses; cold, white leg(s); atrophic skin; punched out ulcers (often painful); postural/dependent colour change; Buerger's angle (angle that leg goes pale when raised off the couch) of <20° and capillary filling time >15s are found in severe ischaemia.

**Tests** Exclude DM, arteritis (ESR/CRP). FBC (anaemia, polycythaemia); U&E (renal disease); lipids (dyslipidaemia); ECG (cardiac ischaemia). Do thrombophilia screen and serum homocysteine if <50 years. *Ankle-brachial pressure index (ABPI):* Normal = 1–1.2; PAD = 0.5–0.9; critical limb ischaemia <0.5 or ankle systolic pressure <50mmHg. Beware falsely high results from incompressible calcified vessels in severe atherosclerosis, eg DM.

**Imaging** *Colour duplex US* 1st line. If considering intervention *MR/CT angiography* (fig 13.54) for extent and location of stenoses and quality of distal vessels ('run-off').

**℞** **1** *Risk factor modification:* Quit smoking (vital). Treat hypertension and high cholesterol. Prescribe an antiplatelet agent (unless contraindicated), to prevent progression and to reduce cardiovascular risk. *Clopidogrel* is recommended as 1st-line.

**2** *Management of claudication:* •*Supervised exercise programmes* reduce symptoms by improving collateral blood flow (2h per wk for 3 months). Encourage patients to excercise to the point of maximal pain. •*Vasoactive drugs*, eg naftidrofuryl oxalate, offer modest benefit and are recommended only in those who do not wish to undergo revascularization and if exercise fails to improve symptoms.

If conservative measures have failed and PAD is severely affecting a patient's lifestyle or becoming limb-threatening, intervention is required.

- *Percutaneous transluminal angioplasty (PTA)* is used for disease limited to a single arterial segment (a balloon is inflated in the narrowed segment). 5-year patency is 79% (iliac) and 55% (femoral). Stents can be used to maintain artery patency.
- *Surgical reconstruction:* If atheromatous disease is extensive but distal run-off is good (ie distal arteries filled by collateral vessels), consider arterial reconstruction with a bypass graft (fig 13.54). Procedures include femoral-popliteal bypass, femoral-femoral crossover, and aorto-bifemoral bypass grafts. Autologous vein grafts are superior to prosthetic grafts (eg Dacron® or PTFE) when the knee joint is crossed.
- *Amputation* <3% of patients with intermittent claudication require major amputation within 5 years (↑ in diabetes, p212). Amputation may relieve intractable pain and death from sepsis and gangrene. A decision to amputate must be made by the patient, usually against a background of failed alternative strategies. The knee should be preserved where possible as it improves mobility and rehabilitation potential (this must be balanced with the need to ensure wound healing). Rehabilitation should be started early with a view to limb fitting. Gabapentin (regimen on p504) can be used to treat the gruelling complication of phantom limb pain.
- *Future therapies:* Early-phase clinical trials have demonstrated the safety and benefit of gene therapy (eg hepatocyte growth factor) in critical limb ischaemia.

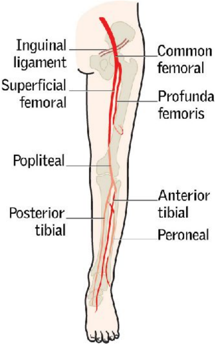


**Fig 13.53** Leg arteries.


**Fig 13.54** CT angiogram showing (a) normal lower limb vasculature and (b) heavily diseased arteries with previous left femoral anterior tibial bypass.

Reproduced from 'Diagnosis and management of peripheral arterial disease', *BMJ*, Peach *et al*., 345:e5208, 2012, with permission from BMJ Publishing Group Ltd.

## Acute limb ischaemia

▶▶Surgical emergency requiring revascularization within 4–6h to save the limb. May be due to thrombosis *in situ* (~40%), emboli (38%), graft/angioplasty occlusion (15%), or trauma. Thrombosis more likely in known 'vasculopaths'; emboli are sudden, eg in those without previous vessel disease; they can affect multiple sites, and there may be a bruit. Mortality: 22%. Amputation rate: 16%.

- *Symptoms and signs:* The 6 'P's of acute ischaemia: **p**ale, **p**ulseless, **p**ainful, **p**aralysed, **p**araesthetic, and '**p**erishingly cold'. Onset of fixed mottling implies irreversibility. Emboli commonly arise from the heart (AF; mural thrombus) or aneurysms. ▶In patients with *known* PAD, sudden deterioration of symptoms with deep duskiness of the limb may indicate acute arterial occlusion. This appearance is due to extensive pre-existing collaterals and must not be misdiagnosed as gout/cellulitis.
- *Management:* ▶This is an emergency and may require urgent open surgery or angioplasty. If diagnosis is in doubt, do urgent arteriography. If the occlusion is embolic, the options are surgical embolectomy (Fogarty catheter) or local thrombolysis, eg tissue plasminogen activator (t-PA, p345), balancing the risks of surgery with the haemorrhagic complications of thrombolysis.
- Anticoagulate with heparin after either procedure and look for the source of emboli. ▶Be aware of possible post-op reperfusion injury and subsequent compartment syndrome.

# Varicose veins (VVs)

Long, tortuous, & dilated veins of the superficial venous system (see fig 13.55).

**Pathology** Blood from superficial veins of the leg passes into deep veins via perforator veins (perforate deep fascia) and at the saphenofemoral and saphenopopliteal junctions. Valves prevent blood from passing from deep to superficial veins. If they become incompetent there is venous hypertension and dilatation of the superficial veins occurs. *Risk factors:* Prolonged standing, obesity, pregnancy, family history, and contraceptive pill. *Causes:* Primary mechanical factors (in ~95%); secondary to obstruction (eg DVT, fetus, pelvic tumour), arteriovenous malformations, overactive muscle pumps (eg cyclists); rarely congenital valve absence.

**Symptoms** 'My legs are ugly.' Pain, cramps, tingling, heaviness, and restless legs. But studies show these symptoms are only slightly commoner in those with VVs.

**Signs** Oedema; eczema; ulcers; haemosiderin; haemorrhage; phlebitis; *atrophie blanche* (white scarring at the site of a previous, healed ulcer); lipodermatosclerosis (skin hardness from subcutaneous fibrosis caused by chronic inflammation and fat necrosis). On their own VVs don't cause DVTs (except possibly *proximally spreading thrombophlebitis* of the long saphenous vein).

**Examination** See p78.

**Treatment** ►NICE guidelines[11] suggest that the criteria for specialist referral of patients with VVs should be: bleeding, pain, ulceration, superficial thrombophlebitis, or 'a severe impact on quality of life' (ie not for cosmetic reasons alone).

- *Treat any underlying cause.*
- *Education*—avoid prolonged standing and elevate leg(s) whenever possible; support stockings (compliance is a problem); lose weight; regular walks (calf muscle action aids venous return).
- *Endovascular treatment* (less pain and earlier return to activity than surgery.)
    - *Radiofrequency ablation (VNUS Closure®)*—a catheter is inserted into the vein and heated to 120°C destroying the endothelium and 'closing' the vein. Results are as good as conventional surgery at 3 months.
    - *Endovenous laser ablation (EVLA)* is similar but uses a laser. Outcomes are similar to surgical repair after 2yrs (in terms of quality of life and recurrence).
    - *Injection sclerotherapy*—either liquid or foam can be used. Liquid sclerosant is indicated for varicosities below the knee if there is no gross saphenofemoral incompetence. It is injected at multiple sites and the vein compressed for a few weeks to avoid thrombosis (intravascular granulation tissue obliterates the lumen). Alternatively foam sclerosant is injected under ultrasound guidance at a single site and spreads rapidly throughout the veins, damaging the endothelium. Ultrasound monitoring prevents inadvertent spread of foam into the femoral vein. It achieves ~80% complete occlusion but is not more effective than liquid sclerotherapy or surgery.
    - *Surgery*—there are several choices, depending on vein anatomy and surgical preference, eg saphenofemoral ligation (Trendelenburg procedure); multiple avulsions; stripping from groin to upper calf (stripping to the ankle is not needed, and may damage the saphenous nerve). *Post-op:* bandage legs tightly and elevate for 24h. Surgery is more effective than sclerotherapy in the long term. ►Before surgery and after venous mapping, ensure that all varicosities are indelibly marked *to either side* (to avoid tattooing if the incision is made through inked skin).

**Saphena varix** Dilatation in the saphenous vein at its confluence with the femoral vein (the SFJ). It transmits a cough impulse and may be mistaken for an inguinal or femoral hernia, but on closer inspection it may have a bluish tinge.

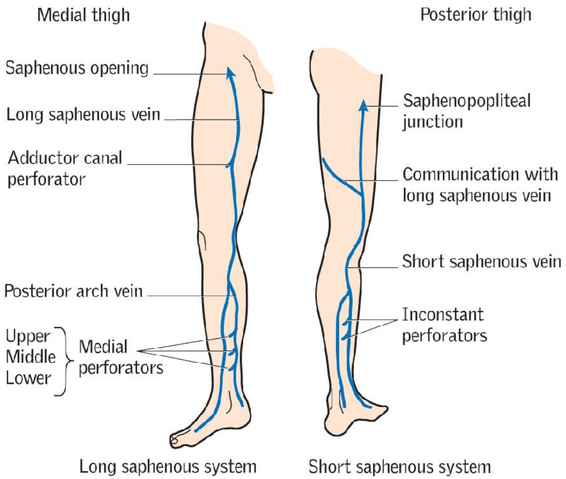


**Fig 13.55** The superficial veins of the leg.

## When do varicose veins become an illness?

Perhaps when they hurt? Or is this too simple? 'Certain illnesses are desirable: they provide a compensation for a functional disorder...' (Albert Camus); *this is known to be common with VVs*. Perhaps many opt for surgery as a displacement activity to confronting deeper problems. We adopt the sickness role when we want sympathy. Somatization is hard to manage; here is one approach to consider:

- Give time; don't dismiss these patients as 'just the "worried well"'.
- Explore factors perpetuating illness behaviour (misinformation, social stressors).
- Agree a plan that makes sense to the patient's holistic view of themself.
- Treat any underlying depression (drugs and cognitive therapy, *OHCS* p344).

## Gangrene and necrotizing fasciitis

**Definitions** Gangrene is death of tissue from poor vascular supply and is a sign of critical ischaemia (see p656). Tissues are black and may slough. *Dry gangrene* is necrosis in the absence of infection. Note a line of demarcation between living and dead tissue.[14] ℞: restoration of blood supply ± amputation. *Wet gangrene* is tissue death and infection (associated with discharge) occurring together (p213, fig 5.10). ℞: analgesia; broad-spectrum IV antibiotics; surgical debridement ± amputation. *Gas gangrene* is a subset of necrotizing myositis caused by spore-forming clostridial species. There is rapid onset of myonecrosis, muscle swelling, gas production, sepsis, and severe pain. Risk factors include diabetes, trauma, and malignancy. ℞: remove all dead tissue (eg amputation). Give benzylpenicillin ± clindamycin. *Hyperbaric $O_2$* can improve survival and ↓ the number of debridements.

►►**Necrotizing fasciitis** is a rapidly progressive infection of the deep fascia causing necrosis of subcutaneous tissue. Prompt recognition (difficult in the early stages) and aggressive treatment is required. ►*In any atypical cellulitis, get early surgical help.* There is intense pain over affected skin and underlying muscle. Group A β-haemolytic streptococci is a major cause, although infection is often polymicrobial. Fournier's gangrene is necrotizing fasciitis localized to the scrotum and perineum. ℞: Radical debridement ± amputation; IV antibiotics, eg benzylpenicillin and clindamycin.

## Skin ulcers

Ulcers are abnormal breaks in an epithelial surface. Leg ulcers affect ~2% in developed countries.

**Causes** May be multiple. For leg ulcers, venous disease accounts for 70%, mixed arterial and venous disease for 15%, and arterial disease alone for 2%. Other contributory factors include neuropathy (eg in DM), lymphoedema, vasculitis, malignancy (p596), infection (eg TB, syphilis), trauma (eg pressure sores: see fig 10.16, p473), pyoderma gangrenosum, drugs (eg nicorandil, hydroxyurea).

**History** Ask about number, pain, trauma. Explore comorbidities—eg VVs, peripheral arterial disease, diabetes, vasculitis. Length of history? Is the patient taking steroids? Are self-induced ulcers, *dermatitis artefacta,* a possibility? Has a biopsy been taken?

**Examination** Note features such as site, number, surface area, depth, edge, base, discharge, lymphadenopathy, sensation, and healing (see BOX). If in the legs, note features of venous insufficiency or arterial disease and, if possible, apply a BP cuff to perform ankle-brachial pressure index (ABPI).

**Tests** Skin and ulcer biopsy may be necessary—eg to assess for vasculitis (will need immunohistopathology) or malignant change in an established ulcer (Marjolin's ulcer = SCC presenting in chronic wound). If ulceration is the first sign of a suspected systemic disorder then further screening tests will be required.

**Management** Managing ulcers is often difficult and expensive. Treat the cause(s) and focus on prevention. Optimize nutrition. Are there adverse risk factors (drug addiction, or risk factors for arteriopathy, eg smoking)? Get expert nursing care. Consider referral to specialist community nurse or leg ulcer/tissue viability clinic:

- 'Charing-Cross' 4-layer compression bandaging is better than standard bandages (use only if arterial pulses OK: ABPI (p656) should be >0.8). Honey dressings can improve healing in mild-moderate burns, but as an adjuvant to compression bandaging for leg ulcers they do not significantly improve healing rate.Negative pressure wound therapy (eg VAC®) helps heal diabetic ulcers.
- Surgery, larval therapy, and hydrogels are used to debride sloughy necrotic tissue (avoid hydrogels in diabetic ulcers due to ↑risk of wet gangrene).

►Routine use of antibiotics does not improve healing. Only use if there is infection (not colonization).

14 'The first sign of his approaching end was when one of my old aunts, while undressing him, removed a toe with one of his old socks.' Graham Greene, *A Sort of Life*, 1971, Simon & Schuster.

## Features of skin ulceration to note on examination

**Site** Above the medial malleolus ('gaiter' area) is the favourite place for *venous ulcers* (fig 13.56; mostly related to superficial venous disease, but may reflect venous hypertension via damage to the valves of the deep venous system, eg 2° to DVT). Venous hypertension leads to the development of superficial varicosities and skin changes (*lipodermatosclerosis* = induration, pigmentation, and inflammation of the skin). Minimal trauma to the leg leads to ulceration which often takes many months to heal. Ulcers on the sacrum, greater trochanter, or heel suggest *pressure sores* (*OHCS* p604), particularly if the patient is bed-bound with suboptimal nutrition.

**Temperature** The ulcer and surrounding tissues are cold in an ischaemic ulcer. If the skin is warm and well perfused then local factors are more likely.

**Surface area** Draw a map of the area to quantify and time any healing (a wound >4wks old is a chronic ulcer as distinguished from an acute wound).

**Shape** Oval, circular (cigarette burns), serpiginous (*Klebsiella granulomatis*, p412); unusual morphology can be secondary to mycobacterial infection, eg cutaneous tuberculosis or scrofuloderma (*tuberculosis colliquativa cutis*, where an infected lymph node ulcerates through to the skin).

**Edge** Shelved/sloping ≈ healing; punched-out ≈ ischaemic or syphilis; rolled/everted ≈ malignant; undermined ≈ TB.

**Base** Any muscle, bone, or tendon destruction (malignancy; pressure sores; ischaemia)? There may be a grey-yellow slough, beneath which is a pale pink base. *Slough* is a mixture of fibrin, cell breakdown products, serous exudate, leucocytes, and bacteria—it need not imply infection, and can be part of the normal wound healing process. *Granulation tissue* is a deep pink gel-like matrix contained within a fibrous collagen network and is evidence of a healing wound.

**Depth** If not uncomfortable for the patient (eg in neuropathic ulceration), a probe can be used to gauge how deep the ulceration extends.

**Discharge** Culture before starting any antibiotics (which usually don't work). A watery discharge is said to favour TB; ►bleeding can ≈ malignancy.

**Associated lymphadenopathy** Suggests infection or malignancy.

**Sensation** Decreased sensation around the ulcer implies neuropathy.

**Position in phases of extension/healing** Healing is heralded by granulation, scar formation, and epithelialization. Inflamed margins ≈ extension.

**Fig 13.56** Venous ulcer in the gaiter area in an obese woman.

Reproduced from Burge *et al*., *Oxford Handbook of Medical Dermatology*, 2016, with permission from Oxford University Press.

# 14 Clinical chemistry

Contents



Fig 14.1 Rosalyn Yalow (1921–2011). Trained in nuclear physics, and made good use of this when she developed the radioimmunoassay technique to allow trace amounts of peptides to be measured in serum using antibodies. This revolutionized laboratory medicine, making reliable assays for hormones widely available, and underpinned the development of endocrinology. It was also employed to screen blood against a range of pathogens, and the technique is still commonly used today. She was awarded the Nobel prize for her work in 1977, and commented to a group of children that: 'Initially, new ideas are rejected...Later they become dogma, if you're right. And if you're really lucky you can publish your rejections as part of your Nobel presentation.'

## On being normal in the society of numbers

Laboratory medicine reduces our patients to a few easy-to-handle numbers: this is the discipline's great attraction—and its greatest danger. The normal range (reference interval) is usually that which includes 95% of a given population (given a normal distribution, see p750). If variation is randomly distributed, 2.5% of our results will be 'too high', and 2.5% 'too low' on an average day, when dealing with apparently normal people. This statistical definition of normality is the simplest. Other definitions may be *normative*—ie stating what an upper or lower limit *should* be. The upper end of the reference interval for plasma cholesterol may be given as 6mmol/L because this is what biochemists state to be the *desired* maximum. 40% of people in some populations will have a plasma cholesterol greater than 6mmol/L and thus may be at increased risk. The WHO definition of anaemia in pregnancy is an Hb of <110g/L, which makes 20% of mothers anaemic. This 'lax' criterion has the presumed benefit of triggering actions that result in fewer deaths from haemorrhage. So do not just ask 'What is the normal range?'—also enquire about who set the range, for what population, and for what reason.

*We thank Dr Petra Sulentic, our Specialist Reader, for her contribution to this chapter.*

# The essence of laboratory medicine

## General principles

- Laboratory testing may contribute to four aspects of medicine:
  - diagnosis (eg TSH in hypothyrodism)
  - prognosis (eg clotting in liver failure)
  - monitoring disease activity or progression (eg creatinine in chronic kidney disease)
  - screening (eg phenylketonuria in newborn babies).
- Only do a test if the result will influence management. Make sure you look at the result.
- ►Always interpret laboratory results in the context of the patient's clinical picture.
- If a result does not fit with the clinical picture, trust clinical judgement and repeat the test. Could it be an artefact? The 'normal' range for a test (reference interval) is usually defined as the interval, symmetrical about the mean, containing 95% of results in a given population (p751). The more tests you run, the greater the probability of an 'abnormal' result of no significance (see p751).
- Laboratory staff like to have contact with you. They are an excellent source of help and information for both requests and results.
- ►Involve the patient. Don't forget to explain to them where the test fits into their overall management plan.

## Getting the best out of the lab—a laboratory decalogue

1. Interest someone from the laboratory in your patient's problem.
2. Fill in the request form fully.
3. Give clinical details, not your preferred diagnosis.
4. Ensure that the lab knows who to contact.
5. Label specimens as well as the request form.
6. Follow the hospital labelling routine for crossmatching.
7. Find out when analysers run, especially batched assays.
8. Talk with the lab before requesting an unusual test.
9. Be thoughtful: at 16:30h the routine results are being sorted.
10. Plot results graphically: abnormalities show sooner.

## Artefacts and pitfalls in laboratory tests

- Do not take blood samples from an arm that has IV fluid running into it.
- Repeat any unexpected and inconsistent result before acting on it.
- For clotting time do not sample from a heparinized IV catheter.
- Serum $K^+$ is overestimated if the sample is old or haemolysed (this occurs if venepuncture is difficult).
- If using Vacutainers, fill *plain* tubes first—otherwise, anticoagulant contamination from previous tubes can cause errors.
- Total calcium results are affected by albumin concentration (p676).
- INR may be overestimated if citrate bottles are underfilled.
- Drugs may cause *analytic* errors (eg prednisolone cross-reacts with cortisol). Be suspicious if results are unexpected.
- Food may affect result, eg bananas raise urinary HIAA (p271).

### Ford Madox Ford 1915 '*The Good Soldier*'

Normal values can have hidden historical, social, and political desiderata—just like the normal values novelists ascribe to their characters: '*Conventions and traditions I suppose work blindly but surely for the preservation of the normal type; for the extinction of proud, resolute and unusual individuals... Society must go on, I suppose, and society can only exist if the normal, if the virtuous, and the slightly-deceitful flourish, and if the passionate, the headstrong, and the too-truthful are condemned to suicide and to madness. Yes, society must go on; it must breed, like rabbits. That is what we are here for ... But, at any rate, there is always Leonora to cheer you up; I don't want to sadden you. Her husband is quite an economical person of so normal a figure that he can get quite a large proportion of his clothes ready-made. That is the great desideratum of life.*'

Oxford World's Classics, pp181–92

# Biochemistry—some major disease patterns



**Dehydration** ↑Urea (disproportionate relative to smaller ↑ in creatinine),[1] ↑albumin (also useful to plot change in a patient's condition), ↑haematocrit (PCV); also ↓urine volume and ↓skin turgor.

**Abnormal kidney function** There are two major biochemical pictures (table 14.1): *Low GFR:* Usually oliguric. *Causes:* early acute oliguric renal failure (p298), chronic kidney disease (p302). In chronic kidney disease also ↓Hb, ↑PTH, and renal bone disease. *Tubular dysfunction:* Results from damage to tubules. Diagnosis is made by testing renal concentrating ability (p241). May be polyuric with ↑urinary glucose, amino acids, proteins (lysozyme, $\beta_2$-microglobulin), or phosphate. *Causes:* recovery from acute kidney injury, hypercalcaemia, hyperuricaemia, myeloma, pyelonephritis, hypokalaemia, Wilson's disease, galactosaemia, and heavy metal poisoning.

Table 14.1 Biochemical profile in abnormal kidney function

| | Low GFR | Tubular dysfunction |
|---|---|---|
| Urea & creatinine | ↑ | Normal |
| $K^+$ & urate | ↑ | ↓ |
| $H^+$ | ↑ | ↑ |
| $HCO_3^-$ | ↓ | ↓ |
| $PO_4^{3-}$ | ↑ | ↓ |
| $Ca^{2+}$ | ↓ | Normal |

**Thiazide and loop diuretics** ↓$Na^+$, ↓$K^+$, ↑$HCO_3^-$, ↑urea.

**Bone disease** See table 14.2 for typical biochemical patterns.

Table 14.2 Serum biomarkers in common diseases of bone

| | $Ca^{2+}$ | $PO_4^{3-}$ | ALP |
|---|---|---|---|
| Osteoporosis (p682) | Normal | Normal | Normal |
| Osteomalacia (p684) | ↓ | ↓ | ↑ |
| Paget's | Normal | Normal | ↑↑ |
| Myeloma | ↑ | ↑, normal | Normal |
| Bone metastases | ↑ | ↑, normal | ↑ |
| 1° Hyperparathyroidism | ↑ | ↓, normal | Normal, ↑ |
| Hypoparathyroidism | ↓ | ↑ | Normal |
| Renal failure (low GFR) | ↓ | ↑ | Normal, ↑ |

**Hepatocellular disease** ↑Bilirubin, ↑↑AST, ALP ↑ slightly, ↓albumin. Also ↑clotting times. For details of the differences between AST and ALT, see p291.

**Cholestasis** ↑Bilirubin, ↑↑γGT, ↑↑ALP, ↑AST.

**Excess alcohol intake** Evidence of hepatocellular disease. Early evidence if ↑γGT, ↑MCV, and ethanol in blood before lunch.

**Myocardial infarction** ↑Troponin, ↑CK, ↑AST, ↑LDH (p118).

**Addison's disease** ↑$K^+$, ↓$Na^+$, ↑urea.

**Cushing's syndrome** May show ↓$K^+$, ↑$HCO_3^-$, ↑$Na^+$.

**Conn's syndrome** May show ↓$K^+$, ↑$HCO_3^-$. $Na^+$ normal or ↑. (Also hypertension.)

**Diabetes mellitus** ↑Glucose, (↓$HCO_3^-$ if acidotic).

**Diabetes insipidus** ↑$Na^+$, ↑plasma osmolality, ↓urine osmolality. (Both hypercalcaemia and hypokalaemia may cause nephrogenic diabetes insipidus.)

**Inappropriate ADH secretion (SIADH)** (See p673.) ↓$Na^+$ with ↓↔ urea and creatinine, ↓plasma osmolality. ↑Urine osmolality (>plasma osmolality), ↑urine $Na^+$ (>20mmol/L).

**Some immunodeficiency states** Normal serum albumin but *low* total protein (because immunoglobulins are missing. Also makes crossmatching difficult because expected haemagglutinins are absent; *OHCS* p198).

1 Dehydration affects urea more than creatinine because in dehydration a greater proportion of filtered urea is reabsorbed by the kidney. Creatinine is hardly reabsorbed at all.

## Laboratory results: ▸▸when to take action NOW

- On receiving a dangerous result, first check the name and date.
- Go to the bedside. If the patient is conscious, turn off any IVI (until fluid is checked: a mistake may have been made) and ask the patient how he or she is. *Any fits, faints, collapses, or unexpected symptoms?*
- Be sceptical of an unexpectedly, wildly abnormal result with a well patient. Compare with previous values. Could the specimens have got muddled up; whose is it? Is there an artefact? Was the sample taken from the 'drip' arm? Is a low calcium due to a low albumin (p676)? Perhaps the lab is using a new analyser with a faulty wash cycle? ▸*When in doubt, seek help and repeat the test.*

The following values are somewhat arbitrary and must be taken as a guide only. Many results less extreme than those listed will be just as dangerous if the patient is old, immunosuppressed, or has some other pathology such as pneumonia.

*Plasma biochemistry:*

▸▸The main risks when plasma electrolytes are dangerously abnormal (table 14.3) are of cardiac arrhythmias and CNS events such as seizures.

Table 14.3 Dangerous levels of the common serum elecrolytes

| Electrolyte | Lower limit | Upper limit | Relevant pages |
|---|---|---|---|
| $Na^+$ | <120mmol/L | >155mmol/L | p672 |
| $K^+$ | <2.5mmol/L | >6.5mmol/L | p674, p301 |
| Corrected $Ca^{2+}$ | <2.0mmol/L | >3.5mmol/L | p676, p678 |
| Glucose | <2.0mmol/L | >20mmol/L | p832, p834 |

*Blood gases:*

- $P_aO_2$ <8.0kPa = *Severe hypoxia. Give $O_2$.* See p188.
- pH <7.1 = *Dangerous acidosis. See* p670 *to determine the cause.*

*Haematology results:*

- Hb <70g/L with low mean cell volume (<75fL) or history of bleeding. This patient may need urgent transfusion (no spare capacity)—ask about symptoms, co-morbidities, and baseline Hb. Check haematinics before transfusion. See p324.
- Platelets <40×$10^9$/L. May need a platelet transfusion; call a haematologist.
- *Plasmodium falciparum* seen on blood film. *Start antimalarials now*. See p418.
- ESR >30mm/h + headache. *Could there be giant cell arteritis?* See p556.

*CSF results:*

▸▸Never delay treatment when bacterial meningitis is suspected.

- >1 neutrophil/mm³. *Is there meningitis: usually >1000 neutrophils?* See p822.
- Positive Gram stain. *Talk to a microbiologist; urgent blind therapy*. See p822.

*Conflicting, equivocal, or inexplicable results:* ▸Get prompt help.

**Fluid requirement** Roughly 2–2.5L in a normal person (~70kg) over 24h. Normal daily losses are through urine (1500mL), stool (200mL), and insensible losses (800mL). This requirement is normally met through food (1000mL) and drink (1500mL).

**Intravenous fluids** Given if sufficient fluids cannot be given orally. About 2–2.5L of fluid containing roughly 70mmol $Na^+$ and 70mmol $K^+$ per 24h are required.[1] Thus, a good regimen is 2–2.5L of 0.18% glucose with sodium chloride with 20–30mmol of $K^+$ per litre of fluid. Alternative routes are via a central venous line or subcutaneously. However, remember that all cannulae carry a risk of MRSA infection: femoral > jugular > subclavian > peripheral, so always resume oral fluid intake as soon as possible.

►In a sick patient, don't forget to include additional sources of fluid loss when calculating daily fluid requirements, such as drains, fevers, or diarrhoea (see BOX 'Special cases'). Daily weighing helps to monitor overall fluid balance, as will fluid balance charts.

►Examine patients regularly to assess fluid balance (see BOX 'Assessing fluid balance').

### Fluid compartments and types of IV fluid

For a 70kg man, *total bodily fluid* is ~42L (60% body weight). Of this, ⅔ is intracellular (28L) and ⅓ is extracellular (14L). Of the extracellular compartment, ⅓ is intravascular, ie blood (5L). Different types of IV fluid will equilibrate with the different fluid compartments depending on the osmotic content of the given fluid.

*5% glucose:* (=Dextrose.) Isotonic, but contains only a small amount of glucose (50g/L) and so provides little energy (~10% daily energy per litre). The liver rapidly metabolizes all the glucose leaving only water, which rapidly equilibrates throughout all fluid compartments. It is, therefore, useless for fluid resuscitation (only 1/9 will remain in the intravascular space), but suitable for maintaining hydration. Excess 5% glucose IV may lead to water overload and hyponatraemia (p672).

*0.9% saline:* ('Normal saline.') Has about the same $Na^+$ content as plasma (150mmol/L) and is isotonic with plasma. 0.9% saline will equilibrate rapidly throughout the extracellular compartment only, and takes longer to reach the intracellular compartment than 5% glucose. It is, therefore, appropriate for fluid resuscitation, as it will remain predominantly in the extracellular space (and thus ⅓ of the given volume in the intravascular space), as well as for maintaining hydration. Hypertonic and hypotonic saline solutions are also available, but are for specialist use only.

*Colloids:* (eg Gelofusine®.) Have a high osmotic content similar to that of plasma and therefore remain in the intravascular space for longer than other fluids, making them appropriate for fluid resuscitation, but not for general hydration. Colloids are expensive, and may cause anaphylactic reactions. In reality, effective fluid resuscitation will use a combination of colloid and 0.9% saline.

*Hypertonic glucose (10% or 50%):* May be used in the treatment of hypoglycaemia. It is irritant to veins, so care in its use is needed. Infusion sites should be inspected regularly, and flushed with 0.9% saline after use.



*Glucose with sodium chloride:* (One-fifth 'normal saline.') Isotonic, containing 0.18% saline (30mmol/L of $Na^+$) and 4% glucose (222mmol/L). It has roughly the quantity of $Na^+$ required for normal fluid maintenance, when given 10-hourly in adults, but is now most commonly used in a paediatric setting.

*Hartmann's solution:* Contains $Na^+$ 131mmol, Cl− 111mmol, lactate 29mmol, $K^+$ 5mmol, $HCO_3^-$ 29mmol, and $Ca^{2+}$ 2mmol per litre of fluid. It is an alternative to 0.9% saline, and some consider it more physiological.

## Assessing fluid balance

*Underfilled:*
- Tachycardia
- Postural drop in BP (low BP is a late sign of hypovolaemia)
- ↓capillary refill time
- ↓urine output
- Cool peripheries
- Dry mucous membranes
- ↓skin turgor
- Sunken eyes.

*Overfilled:*
- ↑JVP (p43)
- Pitting oedema of the sacrum, ankles, or even legs and abdomen
- Tachypnoea
- Bibasal crepitations
- Pulmonary oedema on CXR (fig 16.3, p723).

See also p134 for signs of heart failure.

The JVP is a substitute marker of central venous pressure, and when assessing fluid balance is difficult, a CVP line may help to guide fluid management.

## Special cases

*Acute blood loss:* Resuscitate with colloid or 0.9% saline via large-bore cannulae until blood is available.
*Children:* Use glucose with sodium chloride for fluid maintenance: 100mL/kg for the first 10kg, 50mL/kg for the next 10kg, and 20mL/kg thereafter—all per 24h.
*Elderly:* May be more prone to fluid overload, so use IV fluids with care (smaller fluid bolus).
*GI losses:* (Diarrhoea, vomiting, NG tubes, etc.) Replace lost $K^+$ as well as lost fluid volume.
*Heart failure:* Use IV fluids with care to avoid fluid overload (p134).
*Liver failure:* Patients often have a raised total body sodium, so use salt-poor albumin or blood for resuscitation, and avoid 0.9% saline for maintenance.
*Acute pancreatitis:* Aggressive fluid resuscitation is required due to large amounts of sequestered 'third-space' fluid (p636).
*Poor urine output:* Aim for >1 mL/kg/h; the minimum is >0.5mL/kg/h. Give a fluid challenge, eg 500mL 0.9% saline over 1h (or half this volume in heart failure or the elderly), and recheck the urine output. If not catheterized, exclude retention; if catheterized, ensure the catheter is not blocked!
*Post-operative:* Check the operation notes for intraoperative losses, and ensure you chart and replace added losses from drains, etc.
*Shock:* Resuscitate with colloid or 0.9% saline via large-bore cannulae. Identify the type of shock (p790).
*Transpiration losses:* (Fever, burns.) Beware the large amounts of fluid that can be lost unseen through transpiration. Severe burns in particular may require aggressive fluid resuscitation (p846).

## Potassium in IV fluids

- Potassium ions can be given with 5% glucose or 0.9% saline, usually 20mmol/L or 40mmol/L.
- $K^+$ may be retained in renal failure, so beware giving too much IV.
- Gastrointestinal fluids are rich in $K^+$, so increased fluid loss from the gut (eg diarrhoea, vomiting, high-output stoma, intestinal fistula) will need increased $K^+$ replacement.

►The maximum concentration of $K^+$ that is safe to infuse via a peripheral line is 40mmol/L, at a maximum rate of 20mmol/h in a cardiac monitored patient. Fluid-restricted patients may require higher concentrations or rates in life-threatening hypokalaemia. Faster rates risk cardiac dysrhythmias and asystole, and higher concentrations thrombophlebitis, depending on the size of the vein, so give concentrated solutions >40mmol/L via a central venous catheter, and use ECG monitoring for rates >10mmol/h. For symptoms and signs of hyper- and hypokalaemia see p674.

# Electrolyte physiology and the kidney



**The kidney** Controls the homeostasis of a number of serum electrolytes (including $Na^+$, $K^+$, $Ca^{2+}$, and $PO_4^{3-}$), helps to maintain acid–base balance, and is responsible for the excretion of many substances. It also makes erythropoietin and renin, and hydroxylates 25-hydroxyvitamin D to 1,25-dihydroxyvitamin D (see p676 for $Ca^{2+}$ and $PO_4^{3-}$ physiology). All of these functions can be affected in chronic kidney disease (p302), but it is the biochemical effects of kidney failure that are used to monitor disease progression.

**The renin-angiotensin-aldosterone system** Plasma is filtered by the glomeruli, and $Na^+$, $K^+$, $H^+$, and water are reabsorbed from this filtrate under the control of the renin–angiotensin–aldosterone system. *Renin* is released from the juxtaglomerular apparatus (fig 7.13, p316) in response to low renal flow and raised sympathetic tone, and catalyses the conversion of *angiotensinogen* (a peptide made by the liver) to *angiotensin I*. This is then converted by angiotensin-converting enzyme (ACE), which is located throughout the vascular tree, to *angiotensin II*. The latter has several important actions including efferent renal arteriolar constriction (thus ↑perfusion pressure), peripheral vasoconstriction, and stimulation of the adrenal cortex to produce *aldosterone*, which activates the $Na^+/K^+$ pump in the distal renal tubule leading to reabsorption of $Na^+$ and water from the urine, in exchange for $K^+$ and $H^+$. Glucose spills over into the urine when the plasma concentration > renal threshold for reabsorption (≈10mmol/L, but this varies between people, and is ↓ in pregnancy).

**Control of sodium** Control is through the action of aldosterone on the distal convoluted tubule (DCT) and collecting duct to increase $Na^+$ reabsorption from the urine. The natriuretic peptides ANP, BNP, and CNP (p137) contribute to $Na^+$ homeostasis by reducing $Na^+$ reabsorption from the DCT and inhibiting renin. A *high GFR* (see later in this topic) results in increased $Na^+$ loss, and *high renal tubular flow* and haemodilution decrease $Na^+$ reabsorption in the proximal tubule.

**Control of potassium** Most $K^+$ is intracellular, and thus serum $K^+$ levels are a poor reflection of total body potassium. The concentrations of $K^+$ and $H^+$ in extracellular fluid tend to vary together. This is because these ions compete with each other in the exchange with $Na^+$ that occurs across most cell membranes and in the distal convoluted tubule of the kidney, where $Na^+$ is reabsorbed from the urine. Thus, if the $H^+$ concentration is high, less $K^+$ will be excreted into the urine. Similarly, $K^+$ will compete with $H^+$ for exchange across cell membranes and extracellular $K^+$ will accumulate. Insulin and catecholamines both stimulate $K^+$ uptake into cells by stimulating the $Na^+/K^+$ pump.

**Serum osmolality** A laboratory measurement of the number of osmoles per *kilogram* of solvent. It is approximated by *serum osmolarity* (the number of osmoles per *litre* of solution) using the equation $2(Na^+ + K^+)$ + Urea + Glucose, since these are the predominant serum electrolytes. Normal serum osmolarity is 280–300mmol/L, which will always be a little less than the laboratory-measured osmolality—the *osmolar gap*. However, if the osmolar gap is greater than 10mmol/L, this indicates the presence of additional solutes: consider diabetes mellitus or high blood ethanol, methanol, mannitol, or ethylene glycol.



**Control of water** Control is mainly via serum $Na^+$ concentration, since water intake and loss are regulated to hold the extracellular concentration of $Na^+$ constant. Raised plasma osmolality (eg dehydration or ↑glucose in diabetes mellitus) causes thirst through the hypothalamic thirst centre and the release of antidiuretic hormone (ADH) from the posterior pituitary. ADH increases the passive water reabsorption from the renal collecting duct by opening water channels to allow water to flow from the hypotonic luminal fluid into the hypertonic renal interstitium. Low plasma osmolality inhibits ADH secretion, thus reducing renal water reabsorption.

**Glomerular filtration rate (GFR)** Defined as the volume of fluid filtered by the glomeruli per minute (units mL/min), and is one of the primary measures of disease progression in chronic kidney disease. It can be estimated in a number of different ways (see BOX).

## Estimating GFR

Calculating GFR is useful because it is a more sensitive indication of the degree of renal impairment than serum creatinine. Subjects with low muscle mass (eg the elderly, women) can have a 'normal' serum creatinine, despite a significant reduction in GFR. This can be important when prescribing nephrotoxic drugs, or drugs that are renally excreted, which may therefore accumulate to toxic levels in the serum.

A number of methods for estimating GFR exist, all relying on a calculation of the clearance of a substance that is renally filtered and then not reabsorbed in the renal tubule. For example, the rate of clearance of creatinine can be used as a marker for the rate of filtration of fluid and solutes in the glomerulus because it is only slightly reabsorbed from the renal tubule. The more of the filtered substance that is reabsorbed, however, the less accurate the estimate of GFR.

*MDRD (Modification of Diet in Renal Disease Study Group):* This provides an estimate of GFR from four simple parameters: *serum creatinine*, *age*, *gender*, and *race (black/non-black)*. It is one of the best validated for monitoring patients with established moderately severe renal impairment,[2] and most labs now routinely report estimated GFR (eGFR) using the MDRD equation on all U&E reports:

$$\text{eGFR} = 32\,788 \times \text{serum creatinine}^{-1.154} \times \text{age}^{-0.203} \times [1.212\ \textit{if black}] \times [0.742\ \textit{if female}]$$

However, a number of caveats exist, so that it is best used in monitoring declining renal function rather than labelling elderly patients with mild renal impairment:

- It is not validated for mild renal impairment, and therefore its use for screening *general* populations is questionable.
- Inter-individual variations (and thus confidence intervals) are wide, although for each individual variations are small so that a decline in eGFR over a number of serum samples is always significant.
- Single results may be affected by variations in serum creatinine, such as after a protein-rich meal.

*Cockcroft-Gault equation:* This provides an estimate of creatinine clearance. It is an improvement on the MDRD equation because it also takes into account the patient's weight. However, 10% of creatinine is actively excreted in the tubules, and therefore creatinine clearance overestimates true GFR and underestimates renal impairment. Moreover, the equation assumes ideal body weight and is thus unreliable in the obese or oedematous. Also unreliable in unstable renal function.

$$\text{Creatinine clearance} = \frac{(140 - \text{age}) \times \text{weight (kg)} \times [0.85\ \textit{if female}] \times [1.212\ \textit{if black}]}{0.813 \times \text{serum creatinine } (\mu\text{mol/L})}$$

Creatinine clearance can also be calculated by measuring the excreted creatinine in a *24h urine collection* and comparing it with the serum creatinine concentration. However, the accuracy of collection is vital but often poor, making this an unreliable and inconvenient method.

GFR can also be measured by injection of a radioisotope followed by sequential blood sampling ($^{51}$*Cr-EDTA*) or by an isotope scan (eg DTPA $^{99}$Tc, p190). These methods allow a more accurate estimate of GFR than creatinine clearance, since smaller proportions of these substances are reabsorbed in the tubules. They also have the advantage of being able to provide split renal function.

*Inulin clearance:* The gold standard for calculating GFR, because 100% of filtered inulin (not insulin) is retained in the luminal fluid and therefore reflects exactly the rate of filtration of water and solutes in the glomerulus. However, measuring inulin clearance again requires urine collection over several hours, and also a constant IV infusion of inulin, and is therefore inconvenient to perform.

# Acid-base balance



Arterial blood pH is closely regulated in health to 7.40 ± 0.05 by various mechanisms including bicarbonate, other plasma buffers such as deoxygenated haemoglobin, and the kidney. Acid-base disorders needlessly confuse many people, but if a few simple rules are applied, then interpretation and diagnosis are easy. The key principle is that primary changes in $HCO_3^-$ are *metabolic* and in $CO_2$ *respiratory*. See fig 14.2.

## A simple method

1 *Look at the pH*, is there an acidosis or alkalosis?
- pH <7.35 is an acidosis; pH >7.45 is an alkalosis.

2 *Is the $CO_2$ abnormal?* (Normal range 4.7–6.0kPa.)
If so, is the change in keeping with the pH?
- $CO_2$ is an acidic gas—is $CO_2$ raised with an acidosis, lowered with an alkalosis? If so, it is in keeping with the pH and thus caused by a *respiratory* problem. If there is no change, or an opposite one, then the change is compensatory.

3 *Is the $HCO_3^-$ abnormal?* (Normal concentration 22–28mmol/L.)
If so, is the change in keeping with the pH?
- $HCO_3^-$ is alkaline—is $HCO_3^-$ raised with an alkalosis, lowered with an acidosis? If so, the problem is a *metabolic* one.

4 *Is the $P_aO_2$ abnormal?* Interpret in the context of the $FiO_2$.

**An example** Your patient's blood gas shows: pH 7.05, $CO_2$ 2.0kPa, $HCO_3^-$ 8.0mmol/L. There is an *acidosis*. The $CO_2$ is low, and thus it is a compensatory change. The $HCO_3^-$ is low and is thus the primary change, ie a *metabolic* acidosis.

**The anion gap** Estimates unmeasured plasma anions ('fixed' or organic acids such as phosphate, ketones, and lactate—hard to measure directly). It is calculated as the difference between plasma cations ($Na^+$ and $K^+$) and anions ($Cl^-$ and $HCO_3^-$). Normal range: 10–18mmol/L. It is helpful in determining the cause of a metabolic acidosis.

**Metabolic acidosis** ↓pH, ↓$HCO_3^-$

*Causes of metabolic acidosis and an increased anion gap:*
Due to increased production, or reduced excretion, of fixed/organic acids. $HCO_3^-$ falls and unmeasured anions associated with the acids accumulate.
- Lactic acid (shock, infection, tissue ischaemia).
- Urate (renal failure).
- Ketones (diabetes mellitus, alcohol).
- Drugs/toxins (salicylates, biguanides, ethylene glycol, methanol).

*Causes of metabolic acidosis and a normal anion gap:*
Due to loss of bicarbonate or ingestion of $H^+$ ions ($Cl^-$ is retained).
- Renal tubular acidosis.
- Diarrhoea.
- Drugs (acetazolamide).
- Addison's disease.
- Pancreatic fistula.
- Ammonium chloride ingestion.



**Metabolic alkalosis** ↑pH, ↑$HCO_3^-$
- Vomiting.
- $K^+$ depletion (diuretics).
- Burns.
- Ingestion of base.

**Respiratory acidosis** ↓pH, ↑$CO_2$
- Type 2 respiratory failure due to any lung, neuromuscular, or physical cause (p188).
- Most commonly chronic obstructive pulmonary disease (COPD). Look at the $P_aO_2$. It will probably be low. Is oxygen therapy required? Use controlled $O_2$ (Venturi connector) if COPD is the underlying cause, as too much oxygen may make matters worse (p189).

►► Beware exhaustion in asthma, pneumonia, and pulmonary oedema, which can present with this picture when close to respiratory arrest. A normal or high $P_aCO_2$ is worrying. These patients require urgent ITU review for ventilatory support.

**Respiratory alkalosis** ↑pH, ↓$CO_2$
A result of hyperventilation of any cause. *CNS causes:* Stroke; subarachnoid bleed; meningitis. *Others:* Mild/moderate asthma; anxiety; altitude; ↑T°; pregnancy; pulmonary emboli (reflex hyperventilation); drugs, eg salicylates.

**Terminology** To aid understanding, we have used the terms acidosis and alkalosis, where a purist would sometimes have used acidaemia and alkalaemia. Technically acidaemia is the state of having a low blood pH, whereas acidosis refers to the processes which generate $H^+$, leading to the acidaemia.

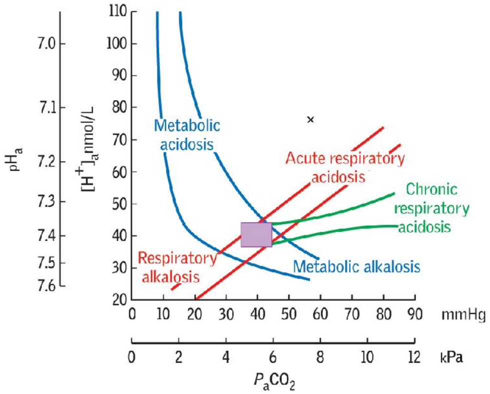


Fig 14.2 The shaded area represents normality. This method is very powerful. The result represented by point ×, for example, indicates that the acidosis is in part respiratory and in part metabolic. Seek a cause for each.

# Hypernatraemia

**Signs and symptoms** Lethargy, thirst, weakness, irritability, confusion, coma, and fits, along with signs of dehydration (p666). *Laboratory features:* ↑$Na^+$, ↑PCV, ↑alb, ↑urea.

## Causes

Usually due to water loss in excess of $Na^+$ loss:
- Fluid loss without water replacement (eg diarrhoea, vomit, burns).
- Diabetes insipidus (p240). Suspect if large urine volume. This may follow head injury, or CNS surgery, especially pituitary.
- Osmotic diuresis (for diabetic coma, see p832).
- Primary aldosteronism: rarely severe, suspect if ↑BP, ↓$K^+$, alkalosis (↑$HCO_3^-$).
- Iatrogenic: incorrect IV fluid replacement (excessive saline).

**Management** Give water orally if possible. If not, give glucose 5% IV slowly (1L/6h) guided by urine output and plasma $Na^+$. Use 0.9% saline IV if hypovolaemic, since this causes less marked fluid shifts and is hypotonic in a hypertonic patient. Avoid hypertonic solutions.



# Hyponatraemia

Plasma $Na^+$ concentration depends on the amount of both $Na^+$ and water in the plasma. Hyponatraemia therefore does not necessarily imply $Na^+$ depletion. Assessing fluid status is the key to diagnosis (see fig 14.3).

**Signs and symptoms** Look for anorexia, nausea, and malaise initially, followed by headache, irritability, confusion, weakness, ↓GCS, and seizures, depending on the severity and rate of change in serum $Na^+$. Cardiac failure or oedema may help to indicate the cause. Hyponatraemia also increases the risk of falls in the elderly.[3]

**Causes** See fig 14.3. Artefactual causes include: •blood sample was from a drip arm •high serum lipid/protein content causing ↑serum volume, with ↓$Na^+$ concentration but normal plasma osmolality •if hyperglycaemic (≳20mmol/L) add ~4.3mmol/L to plasma $Na^+$ for every 10mmol/L rise in glucose above normal.

**Iatrogenic hyponatraemia** If 5% glucose is infused continuously without adding 0.9% saline, the glucose is quickly used, rendering the fluid hypotonic and causing hyponatraemia, esp. in those on thiazide diuretics, women (esp. pre-menopausal), and those undergoing physiological stress (eg post-operative, septic). In some patients, only marginally low plasma $Na^+$ levels cause serious effects (eg ~128mmol/L)—don't attribute odd CNS signs to non-existent strokes/TIAs if ↓$Na^+$.

## Management

- Correct the underlying cause; never base treatment on $Na^+$ concentration alone. The presence of symptoms, the chronicity of the hyponatraemia, and state of hydration are all important. Replace $Na^+$ and water at the same rate they were lost.
- *Asymptomatic chronic hyponatraemia*, fluid restriction is often sufficient if asymptomatic, although demeclocycline (ADH antagonist) may be required. If hypervolaemic (cirrhosis, CCF), treat the underlying disorder first.
- *Acute or symptomatic hyponatraemia*, or if *dehydrated*, cautious rehydration with 0.9% saline may be given, but do not correct changes rapidly as *central pontine myelinolysis*[2] may result. Maximum rise in serum $Na^+$ 15mmol/L per day if chronic, or 1mmol/L per hour if acute. Consider using furosemide when not hypovolaemic to avoid fluid overload.
- *Vasopressor receptor antagonists* ('vaptans', eg tolvaptan) promote water excretion without loss of electrolytes, and appear to be effective in treating hypervolaemic and euvolaemic hyponatraemia but are expensive.[4]

▸▸ *In emergency:* (Seizures, coma) seek expert help. Consider hypertonic saline (eg 1.8% saline) at 70mmol $Na^+$/h ± furosemide. Aim for a gradual increase in plasma $Na^+$ to ≈125mmol/L. Beware heart failure and central pontine myelinolysis.[2]



**2** Central pontine myelinolysis: irreversible and often fatal pontine demyelination seen in malnourished alcoholics or rapid correction of ↓$Na^+$. There is subacute onset of lethargy, confusion, pseudobulbar palsy, para- or quadriparesis, 'locked-in' syndrome, or coma.

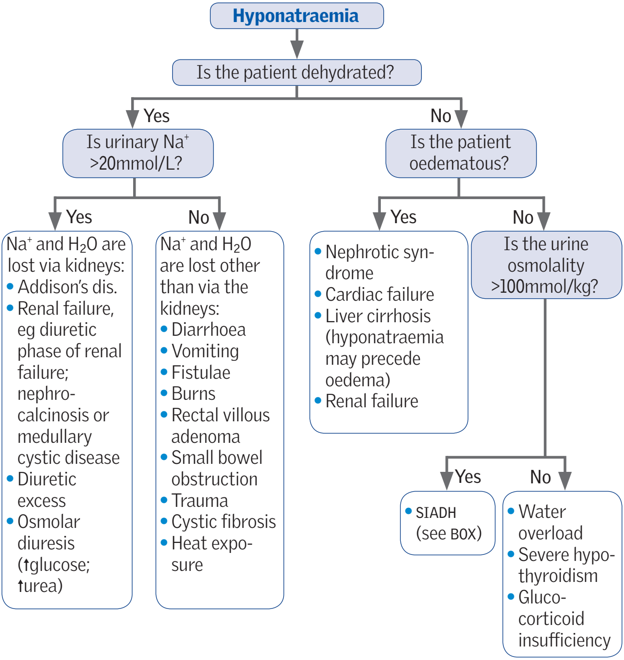


Fig 14.3 Hyponatraemia.

## Syndrome of inappropriate ADH secretion (SIADH)

An important, but over-diagnosed, cause of hyponatraemia. The diagnosis requires concentrated urine ($Na^+$ > 20mmol/L and osmolality > 100mosmol/kg) in the presence of hyponatraemia (plasma $Na^+$ < 125mmol/L) and low plasma osmolality (< 260mosmol/kg), in the absence of hypovolaemia, oedema, or diuretics.

*Causes:*

- *Malignancy:* lung small-cell, pancreas, prostate, thymus, or lymphoma.
- *CNS disorders:* meningoencephalitis, abscess, stroke, subarachnoid or subdural haemorrhage, head injury, neurosurgery, Guillain–Barré, vasculitis, or SLE.
- *Chest disease:* TB, pneumonia, abscess, aspergillosis, small-cell lung cancer.
- *Endocrine disease:* hypothyroidism (not true SIADH, but perhaps due to excess ADH release from carotid sinus baroreceptors triggered by ↓ cardiac output).
- *Drugs:* opiates, psychotropics, SSRIs, cytotoxics.
- *Other:* acute intermittent porphyria, trauma, major abdominal or thoracic surgery, symptomatic HIV.

*Treatment:* Treat the cause and restrict fluid. Consider salt ± loop diuretic if severe. Demeclocycline is used rarely. Vasopressin receptor antagonists ('vaptans', p672) are an emerging class of drug used in SIADH and other types of hyponatraemia.

# Hyperkalaemia

▶▶A plasma potassium >6.5mmol/L is a potential emergency and needs urgent assessment (see p301). The worry is of myocardial hyperexcitability leading to ventricular fibrillation and cardiac arrest. First assess the patient—do they look unwell, is there an obvious cause? If not, could it be an artefactual result?

**Concerning signs and symptoms** Include a fast irregular pulse, chest pain, weakness, palpitations, and light-headedness. ECG: (see fig 14.4) tall tented T waves, small P waves, a wide QRS complex (eventually becoming sinusoidal), and ventricular fibrillation.

*Artefactual results:* If the patient is well, and has none of the above-mentioned findings, repeat the test urgently as it may be artefactual, caused by: •haemolysis (difficult venepuncture; patient clenched fist) •contamination with potassium EDTA anticoagulant in FBC bottles (do FBCs *after* U&Es) •thrombocythaemia ($K^+$ leaks out of platelets during clotting) •delayed analysis ($K^+$ leaks out of RBCs; a particular problem in a primary care setting due to long transit times to the lab).[5]

## Causes

- Oliguric renal failure.
- $K^+$-sparing diuretics.
- Rhabdomyolysis (p319).
- Metabolic acidosis (DM).
- Excess $K^+$ therapy.
- Addison's disease (see p226).
- Massive blood transfusion.
- Burns.
- Drugs, eg ACE-i, suxamethonium.
- Artefactual result (see earlier 'Artefactual results').

## Treatment in non-urgent cases

Treat the underlying cause; review medications.

- Polystyrene sulfonate resin (eg Calcium Resonium® 15g/8h PO) binds $K^+$ in the gut, preventing absorption and bringing $K^+$ levels down over a few days. If vomiting prevents PO administration, give a 30g enema, followed at 9h by colonic irrigation.

## Emergency treatment

▶▶If there is evidence of myocardial hyperexcitability, or $K^+$ is >6.5mmol/L, get senior assistance, and treat as an emergency (see p301).

# Hypokalaemia

If $K^+$ <2.5mmol/L, urgent treatment is required. Note that hypokalaemia exacerbates digoxin toxicity.

**Signs and symptoms** Muscle weakness, hypotonia, hyporeflexia, cramps, tetany, palpitations, light-headedness (arrhythmias), constipation.

**ECG** Small or inverted T waves, prominent U waves (after T wave), a long PR interval, and depressed ST segments.

## Causes

- Diuretics.
- Vomiting and diarrhoea.
- Pyloric stenosis.
- Rectal villous adenoma.
- Intestinal fistula.
- Cushing's syndrome/steroids/ACTH.
- Conn's syndrome.
- Alkalosis.
- Purgative and liquorice abuse.
- Renal tubular failure (p316 & p664).

If on diuretics, ↑$HCO_3^-$ is the best indication that hypokalaemia is likely to have been long-standing. $Mg^{2+}$ may be low, and hypokalaemia is often difficult to correct until $Mg^{2+}$ levels are normalized. Suspect Conn's syndrome if hypertensive, hypokalaemic alkalosis in someone not taking diuretics (p228).

In *hypokalaemic periodic paralysis*, intermittent weakness lasting up to 72h appears to be caused by $K^+$ shifting from extra- to intracellular fluid. See *OHCS* p652.

**Treatment** *If mild:* (>2.5mmol/L, no symptoms.) Give oral $K^+$ supplement (≥80mmol/24h, eg Sando-K® 2 tabs/8h). Review $K^+$ after 3 days. If taking a thiazide diuretic, and $K^+$ >3.0 consider repeating and/or $K^+$-sparing diuretic. *If severe:* (<2.5mmol/L, and/or dangerous symptoms.) Give IV potassium cautiously, not more than 20mmol/h, and not more concentrated than 40mmol/L. Do not give $K^+$ if oliguric. ▶▶*Never* give $K^+$ as a fast stat bolus dose.

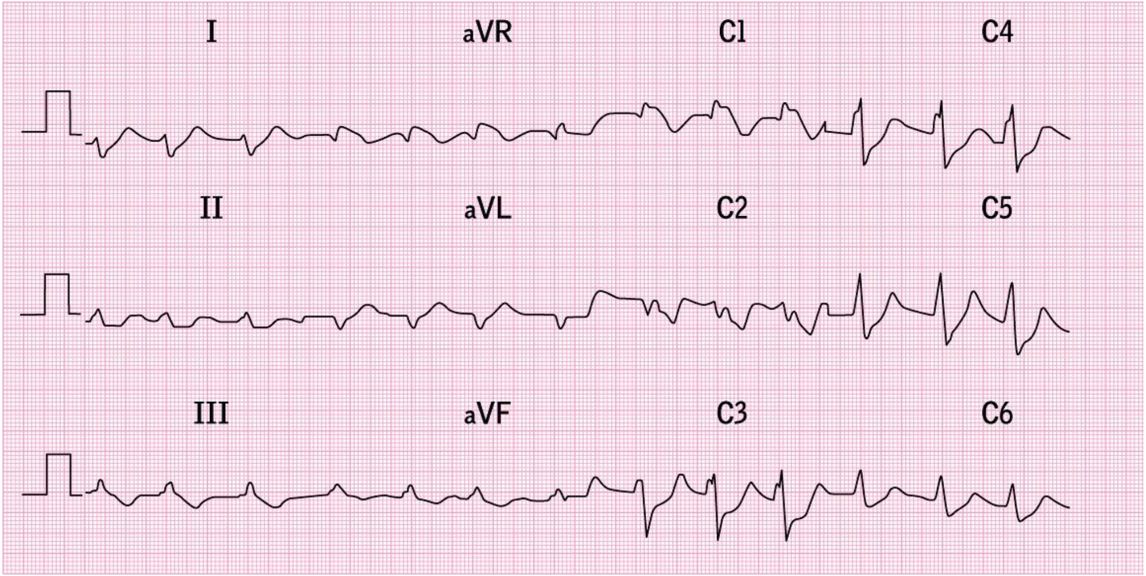


Fig 14.4 Hyperkalaemia—note the flattening of the P waves, prominent T waves, and widening of the QRS complex.

## Calcium and phosphate physiology



**Calcium and phosphate homeostasis is maintained through:**

*Parathyroid hormone (PTH):* Overall effect is ↑$Ca^{2+}$ & ↓$PO_4^{3-}$. Secretion by four parathyroid glands is triggered by ↓serum ionized $Ca^{2+}$; controlled by −ve feedback loop. Actions are: •↑osteoclast activity releasing $Ca^{2+}$ and $PO_4^{3-}$ from bones •↑$Ca^{2+}$ & ↓$PO_4^{3-}$ reabsorption in the kidney •↑renal production of 1,25-dihydroxy-vitamin $D_3$.

*Vitamin D and calcitriol:* Vit D is hydroxylated first in the liver to 25-hydroxy-vit D, and again in the kidney to 1,25-dihydroxy-vit D (calcitriol), the biologically active form, and 24,25-hydroxy-vit D (inactive). Calcitriol production is stimulated by ↓$Ca^{2+}$, ↓$PO_4^{3-}$, and ↑PTH. *Actions are:* •↑$Ca^{2+}$ and ↑$PO_4^{3-}$ absorption from the gut •inhibition of PTH release •enhanced bone turnover •↑$Ca^{2+}$ and ↑$PO_4^{3-}$ reabsorption in the kidney. Cholecalciferol (vit $D_3$—from animal sources) and ergocalciferol (vit $D_2$—from vegetables) are biologically identical in their activity. Disordered regulation of calcitriol underlies familial normocalcaemic hypercalciuria, which is a major cause of calcium oxalate renal stone formation (p638).

*Calcitonin:* Made in C-cells of the thyroid, this causes ↓$Ca^{2+}$ and ↓$PO_4^{3-}$, but its physiological role is unclear. It can be used as a marker of recurrence or metastasis in medullary carcinoma of the thyroid.

*Magnesium:* ↓$Mg^{2+}$ prevents PTH release, and may cause hypocalcaemia.

*Plasma binding:* Labs usually measure total plasma $Ca^{2+}$. ~40% is bound to albumin, and the rest is free ionized $Ca^{2+}$ which is the physiologically important amount (often available on blood gas analyser). Therefore, *correct total $Ca^{2+}$ for albumin* as follows: add 0.1mmol/L to $Ca^{2+}$ level for every 4g/L that albumin is below 40g/L, and a similar subtraction for raised albumin. However, many other factors affect binding (eg other proteins in myeloma, cirrhosis, individual variation) so be cautious in your interpretation. If in doubt over a high $Ca^{2+}$, take blood specimens uncuffed (remove tourniquet after needle in vein, but before taking blood sample), and with the patient fasted.

## Hypercalcaemia

**Signs and symptoms** 'Bones, stones, groans, and psychic moans.' Abdominal pain; vomiting; constipation; polyuria; polydipsia; depression; anorexia; weight loss; tiredness; weakness; hypertension, confusion; pyrexia; renal stones; renal failure; ectopic calcification (eg cornea—see BOX); cardiac arrest. ***ECG:*** ↓QT interval.

**Causes** (See fig 14.5.) Most commonly malignancy (eg from bone metastases, myeloma, PTHrP) or primary hyperparathyroidism. Others include sarcoidosis, vit D intoxication, thyrotoxicosis, lithium, tertiary hyperparathyroidism, milk-alkali syndrome, and familial benign hypocalciuric hypercalcaemia (rare; defect in calcium-sensing receptor). HIV can cause both ↑ & ↓$Ca^{2+}$ (perhaps from PTH-related bone remodelling).[6]

**Investigations** The main distinction is malignancy vs 1° hyperparathyroidism. Pointers to malignancy are ↓albumin, ↓$Cl^-$, alkalosis, ↓$K^+$, ↑$PO_4^{3-}$, ↑ALP. ↑PTH indicates hyperparathyroidism. Also FBC, protein electrophoresis, CXR, isotope bone scan, 24h urinary $Ca^{2+}$ excretion (for familial hypocalciuric hypercalcaemia).



### Causes of metastatic (ectopic) calcification 'PARATTHORMONE'

**P**arathormone (PTH)↑ (p222) and other causes of ↑$Ca^{2+}$, eg sarcoidosis; **A**myloidosis; **R**enal failure (relates to ↑$PO_4^{3-}$); **A**ddison's disease (adrenal calcification); **T**B nodes; **T**oxoplasmosis (CNS); **H**istoplasmosis (eg in lung); **O**verdose of vitamin D; **R**aynaud's-associated diseases (eg SLE; systemic sclerosis p552; dermatomyositis); **M**uscle primaries/leiomyosarcomas); **O**ssifying metastases (**o**steosarcoma) or **o**varian mets (to peritoneum); **N**ephrocalcinosis; **E**ndocrine tumours (eg gastrinoma).

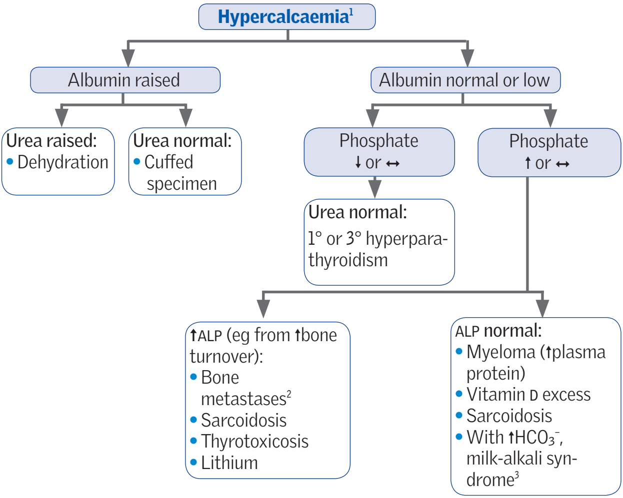


**Fig 14.5** Hypercalcaemia.

**1** This diagram is only a guide: use in conjunction with the clinical picture.
**2** Most common primary: breast, kidney, lung, thyroid, prostate, ovary, colon.
**3** Ingesting too much calcium and alkali (eg in milk) can cause hypercalcaemia with metastatic calcification and renal failure. Thyrotoxicosis causes alkalaemia because of hyperventilation.[7]

## Treating acute hypercalcaemia

Diagnose and treat the underlying cause. If $Ca^{2+}$ >3.5mmol/L and symptomatic:

1 *Correct dehydration:* If dehydrated give IV 0.9% saline.

2 *Bisphosphonates:* These prevent bone resorption by inhibiting osteoclast activity. A single dose of pamidronate lowers $Ca^{2+}$ over 2–3d; maximum effect is at 1wk. *Infuse slowly*, eg 30mg in 300mL 0.9% saline over 3h via a largish vein. Max dose 90mg (see table 14.4). Zoledronic acid is significantly more effective in reducing serum $Ca^{2+}$ than previously used bisphosphonates.[8] Usually, a single dose of 4mg IV (diluted to 100mL, over 15min) will normalize plasma $Ca^{2+}$ within a week. *SE*, flu symptoms, ↓$PO_4^{3-}$, bone pain, myalgia, nausea, vomiting, headache, lymphocytopenia, ↓$Mg^{2+}$, ↓$Ca^{2+}$, seizures.

3 *Further management:* Chemotherapy may help in malignancy. Steroids are used in sarcoidosis, eg prednisolone 40–60mg/d. Salmon calcitonin acts similarly to bisphosphonates, and has a quicker onset of action, but is now rarely used. NB: the use of furosemide is contentious, as supporting RCT evidence is scant.[9,10] It helps to promote renal excretion of $Ca^{2+}$, but can exacerbate hypercalcaemia by worsening dehydration. Thus it should only be used once fully rehydrated, and with concomitant IV fluids (eg 0.9% saline 1L/4–6h). Avoid thiazides.

**Table 14.4** Disodium pamidronate doses

| Calcium (mmol/L; corrected) | Single-dose pamidronate (mg) |
|---|---|
| <3 | 15–30 |
| 3–3.5 | 30–60 |
| 3.5–4 | 60–90 |
| >4 | 90 |

## Hypocalcaemia

►Apparent hypocalcaemia may be an artefact of hypoalbuminaemia (p676).

**Signs and symptoms** See BOX.[11] *Mild:* cramps, perioral numbness/paraesthesiae. *Severe:* carpopedal spasm (especially if brachial artery compressed, *Trousseau's sign*; see fig 14.6), laryngospasm, seizures. Neuromuscular excitability may also be demonstrated by tapping over parotid (facial nerve) causing facial muscles to twitch (*Chvostek's sign*; see fig 14.7). Cataract if chronic hypocalcaemia. *ECG:* Long QT interval.

**Causes** *With ↑$PO_4^{3-}$*
- Chronic kidney disease (p302).
- Hypoparathyroidism (incl thyroid or parathyroid surgery, p222).
- Pseudohypoparathyroidism (p222).
- Acute rhabdomyolysis.
- Hypomagnesaemia.

*With ↔ or ↓$PO_4^{3-}$*
- Vitamin D deficiency.
- Osteomalacia (↑ALP).
- Acute pancreatitis.
- Over-hydration.
- Respiratory alkalosis (total $Ca^{2+}$ is normal, but ↓ionized $Ca^{2+}$ due to ↑pH ∴ symptomatic).

**Treatment**
- *Mild symptoms:* Give calcium 5mmol/6h PO, with daily plasma $Ca^{2+}$ levels.
- *In chronic kidney disease:* See p302. May require **alfacalcidol**, eg 0.5–1mcg/24h PO.
- *Severe symptoms:* Give 10mL of **10% calcium gluconate** (2.25mmol) IV over 30min, and repeat as necessary. If due to respiratory alkalosis, correct the alkalosis.

**Features of hypocalcaemia 'SPASMODIC'**

**S**pasms (carpopedal spasms = Trousseau's sign)
**P**erioral paraesthesiae
**A**nxious, irritable, irrational
**S**eizures
**M**uscle tone ↑ in smooth muscle—hence colic, wheeze, and dysphagia
**O**rientation impaired (time, place, and person) and confusion
**D**ermatitis (eg atopic/exfoliative)
**I**mpetigo herpetiformis (↓$Ca^{2+}$ and pustules in pregnancy—rare and serious)
**C**hvostek's sign; **c**horeoathetosis; **c**ataract; **c**ardiomyopathy (long QT interval on ECG).

Fig 14.6 Trousseau's sign: on inflating the cuff, the wrist and fingers flex and draw together (carpopedal spasm).

Fig 14.7 Chvostek's sign: the corner of the mouth twitches when the facial nerve is tapped over the parotid.

## Phosphate





**Hypophosphataemia** Common and of little significance unless severe (<0.4mmol/L). *Causes:* Vitamin D deficiency, alcohol withdrawal, refeeding syndrome (p587), inadequate oral intake, severe diabetic ketoacidosis, renal tubular dysfunction and 1° hyperparathyroidism. *Signs and symptoms:* Muscle weakness or rhabdomyolysis, red cell, white cell and platelet dysfunction, and cardiac arrest or arrhythmias. *Treatment:* Oral or parenteral phosphate supplementation, eg Phosphate Polyfusor® IVI (100mmol $PO_4^{3-}$ in 500mL). Never give IV phosphate to a patient who is hypercalcaemic or oliguric.

**Hyperphosphataemia** Most commonly due to chronic kidney disease, when it is treated with phosphate binders, eg sevelamer 800mg/8h PO during meals. Also catabolic states such as tumour lysis syndrome (p529).

## Magnesium

Magnesium is distributed 65% in bone and 35% in cells; plasma concentration tends to follow that of $Ca^{2+}$ and $K^+$.

**Hypomagnesaemia** Causes paraesthesiae, ataxia, seizures, tetany, arrhythmias. Digitalis toxicity may be exacerbated. *Causes:* Diuretics, severe diarrhoea, ketoacidosis, alcohol abuse, total parenteral nutrition (monitor weekly), ↓$Ca^{2+}$, ↓$K^+$, and ↓$PO_4^{3-}$. *Treatment:* If needed, give magnesium salts, PO or IV (eg 8mmol $MgSO_4$ IV over 3min to 2h, depending on severity, with frequent $Mg^{2+}$ levels).

**Hypermagnesaemia** Rarely requires treatment unless severe (>7.5mmol/L). *Causes:* Renal failure or iatrogenic (eg excessive antacids). *Signs:* If severe: neuromuscular depression, ↓BP, ↓pulse, hyporeflexia, CNS & respiratory depression, coma.

## Zinc

**Zinc deficiency** This may occur in parenteral nutrition or, rarely, from a poor diet (too few cereals and dairy products; anorexia nervosa; alcoholism). Rarely it is due to a genetic defect. *Symptoms:* Alopecia, dermatitis (look for red, crusted skin lesions especially around nostrils and corners of mouth), night blindness, diarrhoea. *Diagnosis:* Therapeutic trial of zinc (plasma levels are unreliable as they may be low, eg in infection or trauma, without deficiency).

## Selenium

An essential element present in cereals, nuts, and meat. Low soil levels in some parts of Europe and China cause deficiency states. Required for the antioxidant glutathione peroxidase, which ↓ harmful free radicals. Selenium is also antithrombogenic, and is required for sperm motility proteins. Deficiency may increase risk of neoplasia and atheroma, and may lead to a cardiomyopathy or arthritis. Serum levels are a poor guide. Toxic symptoms may also be found with over-energetic replacement.

# Urate and the kidney



**Causes of hyperuricaemia** High levels of urate in the blood (hyperuricaemia) may result from increased turnover (15%) or reduced excretion of urate (85%). Either may be drug induced.

- *Drugs:* Cytotoxics, thiazides, loop diuretics, pyrazinamide.
- *Increased cell turnover:* Lymphoma, leukaemia, psoriasis, haemolysis, muscle death (rhabdomyolysis, p319; tumour lysis syndrome, p529).
- *Reduced excretion:* Primary gout (p548), chronic kidney disease, lead nephropathy, hyperparathyroidism, pre-eclampsia (*OHCS* p48).
- *Other:* Hyperuricaemia may be associated with hypertension and hyperlipidaemia. Urate may be raised in disorders of purine synthesis such as the *Lesch–Nyhan syndrome* (*OHCS* p648).

**Hyperuricaemia and renal failure** Severe renal failure from any cause may be associated with hyperuricaemia, and rarely this may give rise to gout. Sometimes the relationship of cause and effect is reversed so that it is the hyperuricaemia that causes the renal failure. This can occur following cytotoxic treatment (tumour lysis syndrome, p529), and in muscle necrosis.

*How urate causes renal failure:* Urate is poorly soluble in water, so over-excretion can lead to crystal precipitation. Renal failure occurs most commonly because urate precipitates in the renal tubules. This may occur at plasma levels ≥1.19mmol/L. In some instances, ureteric obstruction from urate crystals may occur. This responds to retrograde ureteric catheterization and lavage.

*Prevention of renal failure:* Before starting chemotherapy, ensure good hydration and initiate allopurinol (xanthine oxidase inhibitor) or rasburicase (recombinant urate oxidase), which prevent a sharp rise in urate following chemotherapy (see p528). There is a remote risk of inducing xanthine nephropathy.

*Treatment of hyperuricaemic acute kidney injury:* Exclude bilateral ureteric obstruction, then give prompt rehydration ± loop diuretic to wash uric acid crystals out of the renal tubules, and correct electrolyte abnormalities. Once oliguria is established, haemodialysis is required (in preference to peritoneal dialysis). There is no evidence for either preventing (see previous paragraph) or treating hyperuricaemic renal failure.

**Gout** See p548.

**Urate renal stones** Urate stones (fig 14.8) comprise 5–10% of all renal stones and are radiolucent.

**Fig 14.8** Urate stone.

©Dr G. Austin.

*Incidence:* ~5–10% in temperate climates (double if confirmed gout),[12] but up to 40% in hot, arid climates. ♂:♀≈4:1. But most urate stone formers have no detectable abnormalities in urate metabolism.

*Risk factors:* Acidic or strongly concentrated urine; ↑urinary excretion of urate; chronic diarrhoea; distal small bowel disease or resection (regional enteritis); ileostomy; obesity; diabetes mellitus; chemotherapy for myeloproliferative disorders; inadequate caloric or fluid intake.



*Treatment:* Hydration to increase urine volume (aim >2L/d). Unlike most other renal calculi, existing uric acid stones can often be dissolved with either systemic or topical alkalinizing agents. Potassium citrate or potassium bicarbonate at a dose titrated to alkalinize the urine to a pH of 6–7 dissolves some urate stones. If hyperuricosuria, consider dietary management ± allopurinol (xanthine oxidase inhibitor).

# Metabolic bone diseases: osteoporosis



Osteoporosis implies reduced bone mass. It may be 1° (age-related) or 2° to another condition or drugs. If trabecular bone is affected, crush fractures of vertebrae are common (hence the 'littleness' of little old ladies and their dowager's hump); if cortical bone is affected, long bone fractures are more likely, eg femoral neck: the big cause of death and orthopaedic expense (80% hip fractures in the UK occur in women >50yrs).

**Prevalence** (In those >50yrs): ♂ 6%, ♀ 18%. Women lose trabeculae with age, but in men, although there is reduced bone formation, numbers of trabeculae are stable and their lifetime risk of fracture is less.

**Risk factors** Age-independent risk factors for 1° osteoporosis: parental history, alcohol >4 units daily, rheumatoid arthritis, BMI <19, prolonged immobility, and untreated menopause. See BOX 'Osteoporosis risk factors' for other risk factors, including for 2° osteoporosis.

**Investigations** *X-ray* (low sensitivity/specificity, often with hindsight after a fracture). Bone densitometry (DEXA—see BOX 'DEXA bone densitometry'; table 14.5). *Bloods:* $Ca^{2+}$, $PO_4^{3-}$, and ALP normal. Consider specific investigations for 2° causes if suggestive history. *Biopsy* is unreliable and unnecessary with non-invasive techniques available.

**Management** Loss of bone mineral density may not be entirely irreversible. Age, number of risk factors, and bone mineral density (DEXA scan; see BOX 'DEXA bone densitometry') guide the pharmacological approach (eg FRAX, which is a WHO risk assessment tool for estimating 10-yr risk of osteoporotic fracture in untreated patients; see www.shef.ac.uk/frax),[13,14] although DEXA is not necessary if age >75yrs. Lifestyle measures should apply to all (including those at risk but not yet osteoporotic).

*Lifestyle measures:*
- Quit smoking and reduce alcohol consumption.
- Weight-bearing exercise may increase bone mineral density.[15]
- Balance exercises such as tai chi reduce risk of falls.
- Calcium and vitamin D-rich diet (use supplements if diet is insufficient—see 'Pharmacological measures' later in this topic).
- Home-based fall-prevention programme, with visual assessment and a home visit. NB: hip-protectors are unreliable for preventing fractures.[16]

*Pharmacological measures:*
- *Bisphosphonates:* alendronic acid is 1st line (10mg/d or 70mg/wk; not if eGFR <35). Use also for prevention in long-term steroid use. If intolerant, try etidronate or risedronate. Tell patient to swallow pills with *plenty* of water while remaining upright for >30min and wait 30min before eating or other drugs. (SE: photosensitivity; GI upset; oesophageal ulcers—stop if dysphagia or abdo pain; rarely, jaw osteonecrosis).
- *Calcium and vitamin D:* rarely used alone for prophylaxis, as questionable efficacy and some evidence of a small ↑CV risk. Offer if evidence of deficiency, eg calcium 1g/d + vit D 800U/d. Target serum 25-hydroxy-vitamin D level ≥75nmol/L.
- *Strontium ranelate:* due to an increased risk of cardiac problems it should only be used in those with severe intolerance of other agents and without cardiovascular disease.
- *Hormone replacement therapy (HRT)* can prevent (not treat) osteoporosis in postmenopausal women. Relative risk of breast cancer is 1.4 if used >10yrs; ↑CV risk.
- *Raloxifene* is a selective oestrogen receptor modulator (SERM) that acts similarly to HRT, but with ↓ breast cancer risk.
- *Teriparatide* (recombinant PTH) is useful in those who suffer further fractures despite treatment with other agents. There is a potential ↑ risk of renal malignancy.
- *Calcitonin* may reduce pain after a vertebral fracture.
- *Testosterone* may help in hypogonadal men by promoting trabecular connectivity.
- *Denosumab*, a monoclonal Ab to RANK ligand, given SC twice yearly ↓ reabsorption.

## DEXA bone densitometry: WHO osteoporosis criteria

It is better to scan the hip than the lumbar spine. Bone mineral density (g/cm$^2$) is compared with that of a young healthy adult. The 'T-score' is the number of standard deviations (SD, p751) the bone mineral density (BMD) is from the youthful average. Each decrease of 1 SD in BMD ≈ 2.6-fold ↑ in risk of hip fracture.

Table 14.5 Interpreting DEXA bone scan results

| | |
|---|---|
| T-score >0 | BMD is better than the reference. |
| 0 to −1 | BMD is in the top 84%: no evidence of osteoporosis. |
| −1 to −2.5 | Osteopenia. Risk of later osteoporotic fracture. Offer lifestyle advice. |
| −2.5 or worse | Osteoporosis. Offer lifestyle advice and treatment (p682). Repeat DEXA in 2yrs. |

*Some indications for DEXA:*

- NICE suggests DEXA if previous low-trauma fracture, or for women ≥ 65yrs with one or more risk factors for osteoporosis, or younger if two or more. The benefits of universal screening for osteoporosis remain unproven, but some authorities recommend this for men and women over 70—and earlier if risk factors are present.[17]
- DEXA is not needed pre-treatment for women over 75yrs if previous low-trauma fracture, or ≥ 2 present of rheumatoid arthritis, alcohol excess, or positive family history.
- Prior to giving long-term prednisolone (eg ≳3 months at >5mg/d). Steroids cause osteoporosis by promoting osteoclast bone resorption, ↓muscle mass, and ↓$Ca^{2+}$ absorption from the gut.
- Men or women with osteopenia if low-trauma, non-vertebral fracture.
- Bone and bone-remodelling disorders (eg parathyroid disorders, myeloma, HIV, esp. if on protease inhibitors).

## Osteoporosis risk factors: 'SHATTERED'

**S**teroid use of >5mg/d of prednisolone.
**H**yperthyroidism, hyperparathyroidism, hypercalciuria.
**A**lcohol and tobacco use ↑.
**T**hin (BMI <18.5).
**T**estosterone ↓ (eg antiandrogen ca prostate ℞).
**E**arly menopause.
**R**enal or liver failure.
**E**rosive/inflammatory bone disease (eg myeloma or rheumatoid arthritis).
**D**ietary ↓$Ca^{2+}$/malabsorption; diabetes mellitus type 1.

$Ca^{2+}$ $Na^+$ $K^+$

# Metabolic bone diseases: osteomalacia



In osteomalacia, there is a normal amount of bone but its mineral content is low (there is excess uncalcified osteoid and cartilage). This is the reverse of osteoporosis in which mineralization is unchanged, but there is overall bone loss. Rickets is the result if this process occurs during the period of bone growth; osteomalacia is the result if it occurs after fusion of the epiphyses.

## Signs and symptoms

*Rickets:* Growth retardation, hypotonia, apathy in infants. Once walking: knock-kneed, bow-legged, and deformities of the metaphyseal–epiphyseal junction (eg the rachitic rosary). Features of ↓$Ca^{2+}$—often mild (p678). Children with rickets are ill.

*Osteomalacia:* Bone pain and tenderness; fractures (esp. femoral neck); proximal myopathy (waddling gait), due to ↓$PO_4^{3-}$ and vitamin D deficiency per se.

## Causes

*Vitamin D deficiency:* Due to malabsorption (p266), poor diet, or lack of sunlight.

*Renal osteodystrophy:* Renal failure leads to 1,25-dihydroxy-cholecalciferol deficiency [1,25(OH)$_2$-vitamin D deficiency]. See also *renal bone disease* (p312).

*Drug-induced:* Anticonvulsants may induce liver enzymes, leading to an increased breakdown of 25-hydroxy-vitamin D.

*Vitamin D resistance:* A number of mainly inherited conditions in which the osteomalacia responds to high doses of vitamin D (see 'Treatment' later in this topic).

*Liver disease:* Due to reduced hydroxylation of vitamin D to 25-hydroxy-cholecalciferol and malabsorption of vitamin D, eg in cirrhosis (p276).

*Tumour-induced osteomalacia:* (Oncogenic hypophosphataemia.) Mediated by raised tumour production of phosphatonin fibroblast growth factor 23 (FGF-23) which causes hyperphosphaturia. ↓serum $PO_4^{3-}$ often causes myalgia and weakness.[18]

## Investigations

*Plasma:* Mildly ↓$Ca^{2+}$ (but may be severe); ↓$PO_4^{3-}$; ↑ALP; PTH high; ↓25(OH)-vitamin D, except in vitamin D resistance. In renal failure, ↓1,25(OH)$_2$-vitamin D (p312).

*Biopsy:* Bone biopsy shows incomplete mineralization. Muscle biopsy (if proximal myopathy) is normal.

*X-ray:* In osteomalacia, there is a loss of cortical bone; also, apparent partial fractures without displacement may be seen especially on the lateral border of the scapula, inferior femoral neck, and medial femoral shaft (Looser's zones; see fig 14.9). Cupped, ragged metaphyseal surfaces are seen in rickets (fig 14.10).

## Treatment

- In dietary insufficiency, give vitamin D, eg as one calcium $D_3$ forte tablet/12h PO.
- In malabsorption or hepatic disease, give vitamin $D_2$ (ergocalciferol), up to 40 000U (=1mg) daily, or parenteral calcitriol, eg 7.5mg monthly.
- If due to renal disease or vitamin D resistance, give alfacalcidol (1α-hydroxy-vitamin $D_3$) 250ng–1mcg daily, or calcitriol (1,25-dihydroxy-vitamin $D_3$) 250ng–1mcg daily, and adjust dose according to plasma $Ca^{2+}$. ►►Alfacalcidol and calcitriol can cause dangerous hypercalcaemia.
- Monitor plasma $Ca^{2+}$, initially weekly, and if nausea/vomiting.

**Vitamin D-resistant rickets** Exists in two forms. Type I has low renal 1α-hydroxylase activity, and type II has end-organ resistance to 1,25-dihydroxy-vitamin $D_3$, due to a point mutation in the receptor. Both are treated with large doses of calcitriol.

**X-linked hypophosphataemic rickets** Dominantly inherited—due to a defect in renal phosphate handling (due to mutations in the PEX or PHEX genes which encode an endopeptidase). Rickets develops in early childhood and is associated with poor growth. Plasma $PO_4^{3-}$ is low, ALP is high, and there is phosphaturia. Treatment is with high doses of oral phosphate, and calcitriol.

# Metabolic bone diseases: Paget's disease of bone

Also called *osteitis deformans*, there is increased bone turnover associated with increased numbers of osteoblasts and osteoclasts with resultant remodelling, bone enlargement, deformity, and weakness. Rare in the under-40s. Incidence rises with age (3% over 55yrs old). Commoner in temperate climates, and in Anglo-Saxons.

**Clinical features** Asymptomatic in ~70%. Deep, boring pain, and bony deformity and enlargement—typically of the pelvis, lumbar spine, skull, femur, and tibia (classically a bowed sabre tibia; fig 14.11). *Complications* include pathological fractures, osteoarthritis, ↑$Ca^{2+}$, nerve compression due to bone overgrowth (eg deafness, root compression), high-output CCF (if >40% of skeleton involved), and osteosarcoma (<1% of those affected for >10yrs—suspect if sudden onset or worsening of bone pain).[19]

**Radiology x-ray** Localized enlargement of bone. Patchy cortical thickening with sclerosis, osteolysis, and deformity (eg *osteoporosis circumscripta* of the skull). Affinity for axial skeleton, long bones, and skull. Bone scan may reveal 'hot spots'.

**Blood chemistry** $Ca^{2+}$ and $PO_4^{3-}$ normal; ALP markedly raised.

**Treatment** If analgesia fails, alendronic acid may be tried to reduce pain and/or deformity. It is more effective than etidronate or calcitonin, and as effective as IV pamidronate. Follow expert advice.


Fig 14.9 Osteomalacia. Cortical bone lucency and Looser's zones are seen in both forearms of a patient with osteomalacia.

Image courtesy of Dr Ian Maddison.

Fig 14.10 Rickets. Typical ragged metaphyseal surfaces are seen in the knee and ankle joints of a child with rickets, with bowing of the long bones.

Image courtesy of Dr Ian Maddison.



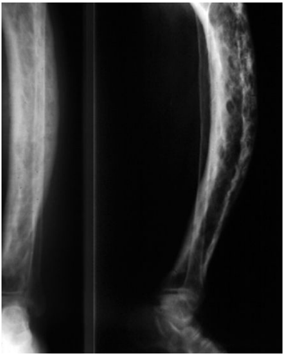

Fig 14.11 Paget's disease. The 'sabre tibia' seen in Paget's disease, with multiple sclerotic lesions.

Image courtesy of Dr Ian Maddison.

## Plasma proteins

The plasma contains a number of proteins including albumin, immunoglobulins, $\alpha_1$-antitrypsin, $\alpha_2$-macroglobulin, caeruloplasmin, transferrin, low-density lipoprotein (LDL), fibrinogen, complement, and factor VIII. The most abundant is albumin (see fig 14.12).

**Albumin** Synthesized in the liver; $t\frac{1}{2}\approx$20d. It binds bilirubin, free fatty acids, $Ca^{2+}$, and some drugs. *Low albumin:* Results in oedema, and is caused by: •↓*synthesis:* liver disease, acute phase response (due to ↑vascular permeability—eg sepsis, trauma, surgery), malabsorption, malnutrition, malignancy •↑*loss:* nephrotic syndrome, protein-losing enteropathy, burns •*haemodilution:* late pregnancy, artefact (eg from 'drip' arm). Also posture (↑5g/L if upright) and genetic variations. *High albumin:* Causes are dehydration; artefact (eg stasis).

**Immunoglobulins** (Antibodies) are synthesized by B cells. Five isoforms Ig A,D,E,G,M exist in humans, and IgG is the most abundant circulating form. *Specific monoclonal band* in paraproteinaemia (see p370). *Diffusely raised* in chronic infections, TB, bronchiectasis, liver cirrhosis, sarcoidosis, SLE, RA, Crohn's disease, 1° biliary cirrhosis, hepatitis, and parasitaemia. *Low* in nephrotic syndrome, malabsorption, malnutrition, and immune deficiency states (eg severe illness, renal failure, diabetes mellitus, malignancy, or congenital).

**Acute phase response** The body responds to a variety of insults with, among other things, the synthesis, by the liver, of a number of proteins (normally present in serum in small quantities)—eg $\alpha_1$-antitrypsin, fibrinogen, complement, haptoglobin, and CRP. A concomitant reduction in albumin level, is characteristic of conditions such as infection, malignancy (especially $\alpha_2$-fraction), trauma, surgery, and inflammatory disease.

**CRP** So called because it binds to a polysaccharide (fraction C) in the cell wall of pneumococci. Levels help monitor inflammation/infection (normal <8mg/L). Like the ESR, it is raised in many inflammatory conditions, but changes more rapidly. It increases in hours and begins to fall within 2–3d of recovery; thus it can be used to follow disease activity (eg Crohn's disease) or the response to therapy (eg antibiotics). CRP values in mild inflammation 10–50mg/L; active bacterial infection 50–200mg/L; severe infection or trauma >200mg/L; see table 14.6.

## Urinary proteins

Urinary protein loss >150mg/d is pathological (p294).

**Albuminuria** Usually caused by renal disease (p294). *Microalbuminuria:* Urinary protein loss between 30 and 300mg/d (so not visible on normal dipstick) and may be seen with diabetes mellitus, ↑BP, SLE, and glomerulonephritis (see p314 for role in DM). Can also be quantified by measuring the urinary *albumin:creatinine ratio* (A:CR), usually a first-in-the-morning spot urine sample. A level >30mg/mmol indicates albuminuria, and microalbuminuria is defined as >2.5mg/mmol in men and >3.5 in women. This is a useful screening test in diabetics, and subjects with reduced eGFR. Note some labs measure total urinary protein not albumin—a P:CR of 50, is equivalent to an A:CR of 30.[20]

**Bence Jones protein** Consists of light chains excreted in excess by some patients with myeloma (p368). They are not detected by dipsticks and may occur with normal serum electrophoresis.

**Haemoglobinuria** Caused by intravascular haemolysis (p336).

**Myoglobinuria** Caused by rhabdomyolysis (p319).

Table 14.6 C-reactive protein (CRP)

| Marked elevation | Normal-to-slight elevation |
|---|---|
| Bacterial infection | Viral infection |
| Abscess | Steroids/oestrogens |
| Crohn's disease | Ulcerative colitis |
| Connective tissue diseases (except SLE) | SLE |
| Neoplasia | Morbid obesity |
| Trauma | Atherosclerosis |
| Necrosis (eg MI) | |

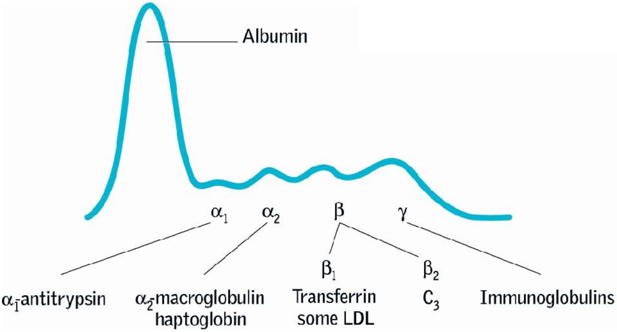


Fig 14.12 A normal electrophoretic scan.

# Plasma enzymes

▶Reference intervals vary between laboratories. See p752 for a guide to normal values.

Raised levels of specific enzymes can be a useful indicator of a disease. However, remember that most can be raised for other reasons too. Levels may be raised due to cellular damage, ↑cell turnover, cellular proliferation (malignancy), enzyme induction, and ↓clearance. The major causes of *raised enzymes:*

**Alkaline phosphatase** (Several distinguishable isoforms exist, eg liver and bone.)
- Liver disease (suggests cholestasis; also cirrhosis, abscess, hepatitis, or malignancy).
- Bone disease (isoenzyme distinguishable, reflects osteoblast activity) especially Paget's, growing children, healing fractures, bone metastases, osteomalacia, osteomyelitis, chronic kidney disease, and hyperparathyroidism.
- Congestive cardiac failure (moderately raised).
- Pregnancy (placenta makes its own isoenzyme).

**Alanine and aspartate aminotransferase (ALT and AST)**
- Liver disease (suggests hepatocyte damage).
- AST also ↑ in MI, skeletal muscle damage (especially crush injuries), and haemolysis.

**α-Amylase**
- Acute pancreatitis (smaller rise in chronic pancreatitis as less tissue remaining).
- *Also:* severe uraemia, diabetic ketoacidosis, severe gastroenteritis, and peptic ulcer.

**Creatine kinase (CK)** ▶A raised CK does not necessarily mean an MI.
- Myocardial infarction (p118; isoenzyme 'CK-MB'. Diagnostic if CK-MB >6% of total CK, or CK-MB mass >99 percentile of normal). CK returns to baseline within 48h (unlike troponin, which remains raised for ~10 days), ∴ useful for detecting re-infarction.
- Muscle damage (rhabdomyolysis, p319; prolonged running; haematoma; seizures; IM injection; defibrillation; bowel ischaemia; myxoedema; dermatomyositis, p552)—and *drugs* (eg statins).

**Gamma-glutamyl transferase (GGT, γGT)**
- Liver disease (particularly alcohol-induced damage, cholestasis, drugs).

**Lactate dehydrogenase (LDH)**
- Myocardial infarction (p118).
- Liver disease (suggests hepatocyte damage).
- Haemolysis (esp. sickle cell crisis), pulmonary embolism, and tumour necrosis.

**Troponin**
- Subtypes troponin T and troponin I are used clinically.
- Cardiac damage or strain (MI—p118, pericarditis, myocarditis, PE, sepsis, CPR).
- Chronic kidney disease (troponin T only; elevation less marked; aetiology unknown).

## Enzyme inducers and inhibitors

Hepatic drug metabolism is mainly by conjugation or oxidation. The oxidative pathways are catalysed by the family of cytochrome P450 isoenzymes, the most important of which is the CYP 3A4 isoenzyme. The cytochrome P450 pathway may be either induced or inhibited by a range of commonly used drugs and foods (table 14.7).

This can lead to important interactions or side-effects. For example, phenytoin reduces the effectiveness of the contraceptive pill due to more rapid oestrogen metabolism, and ciprofloxacin retards the metabolism of methylxanthines (aminophylline) which leads to higher plasma levels and potentially more side-effects. The *BNF* contains a list of the major interactions between drugs.

**Table 14.7** Common inhibitors and inducers of cytochrome P450 isoenzymes

| Enzyme inducers | Enzyme inhibitors | |
|---|---|---|
| Phenytoin | SSRIs | Amiodarone |
| Rifampicin | Ciprofloxacin | Diltiazem |
| Carbamazepine | Isoniazid | Verapamil |
| Alcohol | Macrolides | Omeprazole |
| St John's wort | HIV protease inhibitors | Grapefruit juice |
| Barbiturates | Imidazole and triazole antifungal agents | |

# Hyperlipidaemia

Lipids travel in blood packaged with proteins as lipoproteins. There are four classes: chylomicrons and VLDL (mainly triglyceride), LDL (mainly cholesterol), and HDL (mainly phospholipid) (for abbreviations see footnote[3]). The evidence that cholesterol is a major risk factor for cardiovascular disease (CVD) is undisputed ('4S' STUDY,[21] WOSCOPS,[22] CARE STUDY,[23] HEART PROTECTION STUDY[24]) and indeed it may even be the 'green light' that allows other risk factors to act.[25] Half the UK population have a serum cholesterol putting them at significant risk of CVD. HDL appears to correlate inversely with CVD.

## Who to screen for hyperlipidaemia

►NB: full screening requires a fasting lipid profile.

*Those at risk of hyperlipidaemia:* •Family history of hyperlipidaemia. •Corneal arcus <50yrs old. •Xanthomata or xanthelasmata (fig 14.13).

*Those at risk of CVD:* •Known CVD. •Family history of CVD <60yrs old. •DM or impaired glucose tolerance. •Hypertension. •Smoker. •↑BMI. •Low socioeconomic or Indian Asian background.

## Types of hyperlipidaemia

*Common primary hyperlipidaemia:* Accounts for 70% of hyperlipidaemia. ↑LDL only.

*Familial primary hyperlipidaemias:* Multiple phenotypes exist (see table 14.8). *Risk of ↑↑CVD*, although evidence suggests protection from CVD is achieved with lower doses of statin than for common primary hyperlipidaemia.[26] Refer to specialist.

*Secondary hyperlipidaemia:* Causes include: Cushing's syndrome, hypothyroidism, nephrotic syndrome, or cholestasis. ↑LDL. Treat the cause first.

*Mixed hyperlipidaemia:* Results in ↑ in both LDL and triglycerides. Caused by type 2 diabetes mellitus, metabolic syndrome, alcohol abuse, and chronic renal failure.

## Management

Identify familial or 2° hyperlipidaemias, as Rx may differ. Give lifestyle advice; aim for BMI of 20–25; encourage a Mediterranean-style diet—↑fruit, vegetables, fish, unsaturated fats; and ↓red meat; ↑exercise. Top Rx priority are those with known CVD (there is no need to calculate their risk: *ipso facto* they already have high risk). Second Rx priority is primary prevention in patients with chronic kidney disease or type-1 diabetes, and those with a 10-yr risk of CVD >10%, *irrespective of baseline lipid levels*.

- *1st-line therapy:* Atorvastatin 20mg PO at night, for primary prevention, and 80mg for secondary prevention and primary prevention in those with kidney disease.[27] Simvastatin 40mg, is an alternative. ↓cholesterol synthesis in the liver by inhibiting HMGCOA reductase. CI: porphyria, cholestasis, pregnancy. SE: myalgia ± myositis (stop if ↑CK ≥10-fold; if any myalgia, check CK; risk is 1 per 100 000 treatment-years),[28] abdominal pain, and ↑LFTs (stop if AST ≥100U/L). Cytochrome P450 inhibitors (p689) ↑serum concentrations (200mL of grapefruit juice ↑simvastatin concentration by 300%, and atorvastatin ↑80%, but pravastatin is almost unchanged).Current guidelines suggest a target plasma cholesterol reduction of ≥40 % in those with CVD.
- *2nd-line therapy:* Ezetimibe—a cholesterol absorption inhibitor, may be used in statin intolerance or combination with statins to achive target reduction.
- *3rd-line therapy:* Alirocumab—a monoclonal antibody against PCSK9 (acts to reduce hepatocyte LDL receptor expression). Very effective in reducing LDL,[29] but expensive and needs to be given by injection every 2 weeks. Others: fibrates, eg bezafibrate (useful in mixed hyperlipidaemias); anion exchange resins, eg colestyramine; nicotinic acid (↑HDL; ↓LDL; SE: severe flushes; aspirin 300mg ½h pre-dose helps this).
- *Hypertriglyceridaemia:* Responds best to fibrates, nicotinic acid, or fish oil.

**Xanthomata** These yellow lipid deposits may be: *eruptive* (itchy nodules in crops in hypertriglyceridaemia); *tuberous* (plaques on elbows and knees); or *planar*—also called palmar (orange streaks in palmar creases), 'diagnostic' of remnant hyperlipidaemia; or in tendons (p38), eyelids (*xanthelasma*, see fig 14.13), or cornea (*arcus*, p39).

**3** Abbreviations: (V)LDL = (very) low-density lipoprotein; IDL = intermediate-density lipoprotein; HDL = high-density lipoprotein; chol = cholesterol; trig = trigylcerides.

## Primary hyperlipidaemias

**Table 14.8** Classification of primary hyperlipidaemias

| | | | |
|---|---|---|---|
| Familial hyperchylomicronaemia (lipoprotein lipase deficiency or apoCII deficiency)[I] | Chol <6.5<br>Trig 10–30<br>Chylomicrons | ↑ | Eruptive xanthomata; lipaemia retinalis; hepatosplenomegaly |
| Familial hypercholesterolaemia[II] (LDL receptor defects) | Chol 7.5–16<br>Trig <2.3 | ↑LDL | Tendon xanthoma; corneal arcus; xanthelasma |
| Familial defective apolipoprotein B-100[IIa] | Chol 7.5–16<br>Trig <2.3 | ↑LDL | Tendon xanthoma; arcus; xanthelasma |
| Common hypercholesterolaemia[IIa] | Chol 6.5–9<br>Trig <2.3 | ↑LDL | *The commonest 1° lipidaemia*; may have xanthelasma or arcus |
| Familial combined hyperlipidaemia[IIb, IV, OR V] | Chol 6.5–10<br>Trig 2.3–12 | ↑LDL ↑VLDL<br>↓HDL | *Next commonest 1° lipidaemia*; xanthelasma; arcus |
| Dysbetalipoproteinaemia (remnant particle disease)[III] | Chol 9–14<br>Trig 9–14 | ↑IDL<br>↓HDL ↓LDL | Palmar striae; tubero-eruptive xanthoma |
| Familial hypertriglyceridaemia[IV] | Chol 6.5–12<br>Trig 3.0–6.0 | ↑VLDL | |
| Type V hyperlipoproteinaemia | Trig 10–30; chylomicrons found | | Eruptive xanthomata; lipaemia retinalis; hepatosplenomegaly |

Blue superscript numbers = WHO phenotype; chol/trig levels given in mmol/L.

### *Primary HDL abnormalities:*
- Hyperalphalipoproteinaemia: ↑HDL, chol >2.
- Hypoalphalipoproteinaemia (Tangier disease): ↓HDL, chol <0.92.

### *Primary LDL abnormalities:*
- Abetalipoproteinaemia (ABL): trig <0.3, chol <1.3, missing LDL, VLDL, and chylomicrons. Autosomal recessive disorder of fat malabsorption causing vitamin A & E deficiency, with retinitis pigmentosa, sensory neuropathy, ataxia, pes cavus, and acanthocytosis.
- Hypobetalipoproteinaemia: chol <1.5, ↓LDL, ↓HDL. Autosomal codominant disorder of apolipoprotein B metabolism. ↑longevity in heterozygotes. Homozygotes present with a similar clinical picture to ABL.

Fig 14.13 Xanthelasma. *Xanthos* is Greek for yellow, and *elasma* means plate. Xanthelasmata are lipid-laden yellow plaques, typically a few millimetres wide. They congregate around the lids, or just below the eyes, and signify hyperlipidaemia.

# The porphyrias



The porphyrias are a heterogenous group of rare diseases caused by various errors of haem biosynthesis (produced when iron is chelated into protoporphyrin IXα), which may be genetic or acquired. Depending on the stage in haem biosynthesis that is faulty, there is accumulation of either porphyrinogens, which are unstable and oxidize to porphyrins, or their precursors, porphobilinogen and δ-aminolaevulinic acid. Porphyrin precursors are neurotoxic, while porphyrins themselves induce photosensitivity and the formation of toxic free radicals.

- Alcohol, lead, and iron deficiency cause abnormal porphyrin metabolism.
- Genetic counselling (*OHCS* p154) should be offered to all patients and their families.

**Acute porphyrias** Occur when the accumulation of porphyrinogen precursors predominates, and are characterized by acute neurovisceral crises, though some forms have additional photosensitive cutaneous manifestations.

*Acute intermittent porphyria:* ('*The Madness of King George*.') A low-penetrant autosomal dominant condition (porphobilinogen deaminase gene); 28% have no family history (*de novo* mutations). ~10% of those with the defective gene have neurovisceral symptoms. Attacks are intermittent, more common in women and those aged 18–40, and may be precipitated by drugs. Urine porphobilinogens are raised during attacks (the urine may go deep red on standing) and also, in ~50%, between attacks. Faecal porphyrin levels are normal. There is never cutaneous photosensitivity. It is the commonest form of porphyria—prevalence in UK: 1–2/100 000.

*Variegate porphyria and hereditary coproporphyria:* Autosomal dominant, characterized by photosensitive blistering skin lesions and/or acute attacks. The former is prevalent in Afrikaners in South Africa. Porphobilinogen is high only during an attack, and other metabolites may be detected in faeces.

*Triggers of an acute attack:* Include infection, starvation (including pre-operative 'nil-by-mouth'), reproductive hormones (pregnancy, premenstrual), smoking, anaesthesia, and cytochrome P450 enzyme inducers (alcohol, and other drugs—see BOX).

*Features of an acute attack:*

- *Gastrointestinal:* abdominal pain, vomiting, constipation.
- *Neuropsychiatric:* peripheral neuropathy (weakness, hypotonia, pain, numbness), seizures (often associated with severe ↓$Na^+$), psychosis (or other odd behaviour).[4]
- *Cardiovascular:* hypertension, tachycardia, shock (due to sympathetic overactivity).
- *Other:* fever, ↓$Na^+$, ↓$K^+$, proteinuria, urinary porphobilinogens, discoloured urine. Rare but serious complications include bulbar and respiratory paralysis.

▸▸Beware the 'acute abdomen' in acute intermittent porphyria: colic, vomiting, fever, and ↑WCC—so mimicking an acute surgical abdomen. Anaesthesia could be disastrous.

*Treatment of an acute attack:*

- Remove precipitants (review medications; treat intercurrent illness/infection).
- IV fluids to correct electrolyte imbalance.
- High carbohydrate intake (eg Hycal®) by NG tube, or IV if necessary.
- IV haematin is 1st-line (inhibits production of porphyrinogen precursors).
- Nausea controlled with prochlorperazine 12.5mg IM.
- Sedate if necessary with chlorpromazine 50–100mg PO/IM.
- Pain control with opiate or opioid analgesia (avoid oxycodone).
- Seizures can be controlled with diazepam (although this will prolong the attack).
- Treat tachycardia and hypertension with a β-blocker.

**Non-acute porphyrias**

*Porphyria cutanea tarda (PCT)*, *erythropoietic protoporphyria*, and *congenital erythropoietic porphyria* are characterized by cutaneous photosensitivity alone, as there is no overproduction of porphyrinogen precursors, only porphyrins. PCT presents in adults with blistering skin lesions ± facial hypertrichosis and hyperpigmentation. Total plasma porphyrins and LFTs are ↑. Screen for associated disorders: hep C, HIV, iron overload, hepatocellular ca. ℞: phlebotomy, iron chelators, chloroquine, sunscreens.

## Drugs to avoid in acute intermittent porphyria

There are many, many drugs that may precipitate an acute attack ± quadriplegia, and this is by no means an exhaustive list (see BNF/*Oxford Textbook of Medicine*).

►►For an up-to-date list of drugs considered safe in acute porphyria see www.wmic.wales.nhs.uk/porphyria-safe-list-may-2016/

- Diclofenac
- Alcohol
- Oral contraceptive pill & HRT
- Tricyclic antidepressants
- Benzodiazepines
- Anaesthetic agents (barbiturates, halothane)
- Antibiotics (cephalosporins, sulfonamides, macrolides, tetracyclines, rifampicin, trimethoprim, chloramphenicol, metronidazole)
- Metoclopramide
- ACE-inhibitors
- $Ca^{2+}$-channel blockers
- Statins
- Anticonvulsants
- Furosemide
- Sulfonylureas
- Lidocaine
- Gold salts
- Antihistamines
- Amphetamines.



4 Be sure I looked at her eyes
   Happy and proud; at last I knew
Porphyria worshipped me; surprise
   Made my heart swell, and still it grew
   While I debated what to do.
That moment she was mine, mine, fair,
   Perfectly pure and good: I found
A thing to do, and all her hair
   In one long yellow string I wound
   Three times her little throat around,
And strangled her ...

From *Porphyria's Lover* by Robert Browning.


**Fig 15.1** 'No scientific discovery is named after its original discoverer', asserts Professor Stephen Stigler in '*Stigler's Law of Eponymy*', and in doing so, names the sociologist RK Merton as its discoverer—deliberately making Stigler's law exemplify itself. Is the same true in medicine? At least six others described Alzheimer's disease before Alois Alzheimer in 1906, and Tetralogy of Fallot (named after Étienne-Louis Fallot in 1888) was first described in 1672 by Niels Stenson. Not all medical eponyms obey Stigler's law. Forty years after it was first described, the French neurologist Jean-Marie Charcot (himself associated with at least 15 medical eponyms) attributed the name 'Parkinson's Disease' to the illness outlined in James Parkinson's 1817 essay '*The Shaking Palsy*'. Monochromatic doctors may try to abolish eponyms by regimenting them to histologically driven disease titles. But classifications vary as facts emerge, and as a result the renaming of non-eponyms becomes essential. Eponyms, however, carry on forever, because they imply nothing about causes.

**Alice in Wonderland syndrome** Altered perception in size and shape of body parts or objects ± an impaired sense of passing time—as experienced by *Alice* in Lewis Carroll's novel. Seen in epilepsy, migraine, and cerebral lesions.[1,2] *Alice Pleasance Liddell, 1865–1934*

**Arnold-Chiari malformation** Malformed cerebellar tonsils and medulla herniate through the foramen magnum. This may cause infantile hydrocephalus with mental retardation, optic atrophy, ocular palsies, and spastic paresis of the limbs. Spina bifida, syringomyelia (p516), or focal cerebellar and brainstem signs may occur (p499). There may be bony abnormalities of the base of the skull. Often presents in early adulthood. MRI aids diagnosis. *Julius Arnold, 1835–1915 (German pathologist); Hans Chiari, 1851–1916 (Austrian pathologist)*

**Baker's cyst** Fluid from a knee effusion escapes to form a popliteal cyst (often swollen and painful) in a sub-gastrocnemius bursa.[3] Usually secondary to degeneration. *ΔΔ:* DVT (exclude if calf swelling); sarcoma. *Imaging:* USS; MRI. *℞:* None if asymptomatic. NSAIDs/ice if painful. Spontaneous resolution may take 10–20 months. Arthroscopy + cystectomy may be needed. *William M Baker, 1838–1896 (British surgeon)*

**Bazin's disease** (*Erythema induratum.*) Localized areas of fat necrosis that produce painful, firm nodules ± ulceration and an indurated rash, characteristically on adolescent girls' calves. It is associated with TB. *Nodular vasculitis* is a variant unrelated to TB.[4] *Pierre-Antoine-Ernest Bazin, 1807–1878 (French dermatologist)*

**Behçet's disease** A systemic inflammatory disorder of unknown cause, associated with HLA-B5. It is most common along the old Silk Road, from the Mediterranean to China. *Features:* Recurrent oral and genital ulceration, uveitis, skin lesions (eg erythema nodosum, papulopustular lesions); arthritis (non-erosive large joint oligoarthropathy); thrombophlebitis; vasculitis; myo/pericarditis; CNS involvement (pyramidal signs); and colitis. *Diagnosis:* Mainly clinical. *Pathergy test:* needle prick leads to papule formation within 48hrs. *℞:* Colchicine for orogenital ulceration; steroids, azathioprine/cyclophosphamide for systemic disease. Infliximab has a role in ocular disease unresponsive to topical steroids.[5] *Hulusi Behçet, 1889–1948 (Turkish dermatologist)*



**Berger's disease** (*IgA nephropathy*, p311.) Ranges from invisible haematuria to rapidly progressive glomerulonephritis. Biopsy shows mesangial IgA deposition. Usually indolent disease, but progression to end-stage renal failure occurs. *℞:* ACE-i/ARB if ↑BP or proteinuria. Immunosuppression considered for progressive disease.[6] *Jean Berger, 1930–2011 (French nephrologist)*

**Bickerstaff's brainstem encephalitis** Ophthalmoplegia, ataxia, areflexia, and extensor plantars ± tetraplegia ± coma, and a *reversible* brain death picture (but there is *no* structural damage). *MRI:* hyperintense brainstem signals. GQ1b antibodies +ve.[7] Plasmapheresis may help. *Edwin R Bickerstaff, 1920–2008 (British physician)*

*We thank Dr Simon Eyre, our Specialist Reader, for his contribution to this chapter.*

## Barrett's oesophagus

Barrett's oesophagus is metaplasia of the normal stratified squamous epithelium of the distal oesophagus to a columnar epithelium, as a result of chronic GORD (p254). Estimates of prevalence vary widely, but in patients with a history of symptomatic GORD, rates of ≈8% have been reported. Importantly, screening studies in *asymptomatic* individuals have reported rates of ≈6%. General population-based screening is not recommended, although screening endoscopy may be considered in individuals with chronic GORD symptoms and *multiple* risk factors (>50 years old, obesity, ♂, white race, family history of Barrett's or oesophageal adenocarcinoma). *Diagnosis:* Biopsy of endoscopically visible columnarization allows histological corroboration. The length should be recorded (using the Prague classification). *Management:* ►Focus on detecting and preventing the most significant associated morbidity: oesophageal adenocarcinoma. The risk of progression is low (0.1–0.4% per patient per year, much lower than previously suggested). Risk factors for malignant transformation include ↑age, ♂, long segment of oesophagus involved, and evidence of dysplasia. Endoscopic surveillance for dysplasia is controversial and the evidence base is lacking. Current guidelines suggest that patients without dysplasia and in whom the length of involved oesophagus is <3cm should be considered for discharge from surveillance programmes, depending on the precise histology.[8] For those with more extensive disease, endoscopic assessment every 2–3 years is appropriate. If high-grade dysplasia or intramural carcinoma is detected, ***endoscopic resection*** or mucosal radiofrequency ablation (RFA) is recommended. If low-grade dysplasia is detected, it should be confirmed by repeat examination after 6 months and by an independent pathologist, prior to RFA.[9]

*Norman Rupert Barrett, 1903–1979 (British surgeon)*

## Brugada syndrome

Note right bundle branch block and the unusual morphology of the raised ST segments in $V_1$–$V_3$ (fig 15.2; there are three ECG variants of this pattern). This predominently autosomal dominant condition causing faulty sodium channels predisposes to fatal arrhythmias (eg ventricular fibrillation), typically in young males (eg triggered by a fever).[10] It is preventable by implanting a defibrillator. ►*Consider primary electrical cardiac disease in all with unexplained syncope*. Programmed electrical stimulation may be needed. Relatives of those with sudden unexplained death may undergo unmasking of arrhythmias by IV ajmaline tests—but some results are false +ve. Use judgement in subjecting those with ST abnormalities but no symptoms to electrophysiological tests, right ventricular myocardial biopsy, and MRI. Mutations in the SCN5A gene (encodes the cardiac voltage-gated $Na_v1.5$ channel) are found in 15–20%. Other mutations have also been described.[11]

*Pedro & Josep Brugada, described 1992 (Spanish cardiologists).*

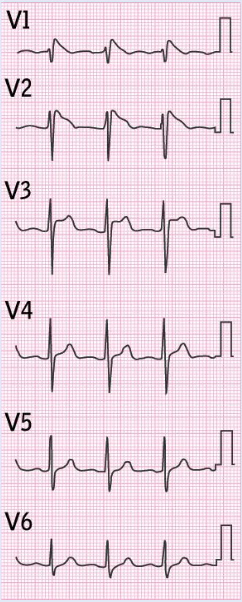


**Fig 15.2** Note right bundle branch block and ST morphology in leads $V_{1-3}$.

Courtesy of Dr Shayashi.

ABC

**Brown-Séquard syndrome** A lesion in one half of the spinal cord (due to hemisection or unilateral cord lesion) causes: •Ipsilateral UMN weakness below the lesion (severed corticospinal tract, causing spastic paraparesis, brisk reflexes, extensor plantars). •Ipsilateral loss of proprioception and vibration (dorsal column severed). •Contralateral loss of pain and temperature sensation (severed spinothalamic tract which has crossed over; fig 10.35 p516). *Causes:* Bullet, stab, tumour, disc hernia, myelitis,[12] septic emboli. *Imaging:* MRI. *Charles-Édouard Brown-Séquard, 1817–1894 (Mauritian neurologist)*

**Budd-Chiari syndrome** Hepatic vein obstruction by thrombosis or tumour causes congestive ischaemia and hepatocyte damage. Abdominal pain, hepatomegaly, ascites, and ↑ALT occur. Portal hypertension occurs in chronic forms. *Causes:* Include hypercoagulable states (combined OCP, pregnancy, malignancy, paroxysmal nocturnal haemoglobinuria, polycythaemia, thrombophilia), TB, liver, renal, or adrenal tumour. *Tests:* USS + Dopplers, CT, or MRI. Angioplasty or a transjugular intrahepatic portosystemic shunt (TIPSS) may be needed. Anticoagulate (lifelong) unless there are varices. Consider liver transplant in fulminant hepatic necrosis or cirrhosis.[13]

*George Budd, 1808–1882 (British physician); Hans Chiari, 1851–1916 (Austrian pathologist)*

**Buerger's disease** (*Thromboangiitis obliterans.*) Non-atherosclerotic smoking-related inflammation and thrombosis of veins and middle-sized arteries causing thrombophlebitis and ischaemia (→ulcers, gangrene). *Cause:* Unknown. Stopping smoking is vital. Most patients are men aged 20–45yrs (see BOX 'Poisoning your boss'). *Leo Buerger, 1879–1943 (US physician)*

**Caplan's syndrome** Multiple lung nodules in coal workers with RA, caused by an inflammatory reaction to anthracite (also associated with silica or asbestos exposure). CXR: bilateral peripheral nodules (0.5-5cm). *ΔΔ:* TB. *Anthony Caplan, 1907–1976 (British physician)*

**Charcot-Marie-Tooth syndrome** (*Peroneal muscular atrophy.*) This inherited neuropathy starts in puberty with weak legs and foot drop + variable loss of sensation and reflexes. The peroneal muscles atrophy, leading to an inverted champagne bottle appearance. Atrophy of hand and arm muscles also occurs. The most common form, CMT1A (PMP22 myelin gene mutation on chr. 17), has AD inheritance. Quality of life is good; *total* incapacity rare. Hand pain/paraesthesiae may respond to nerve release.

*Jean-Marie Charcot, 1825–1893; Pierre Marie, 1853–1940 (French neurologists); Howard H Tooth, 1856–1926 (British physician)*

**Churg-Strauss syndrome** *(Eosinophilic granulomatosis with polyangiitis.)* A triad of adult-onset asthma, eosinophilia, and vasculitis (± vasospasm ± MI ± DVT), affecting lungs, nerves, heart, and skin. A septic-shock picture/systemic inflammatory response syndrome may occur (with glomerulonephritis/renal failure, esp. if ANCA +ve). *℞:* Steroids; biological agents if refractory disease, eg rituximab.[14]

*Jacob Churg, 1910–2005; Lotte Strauss, 1913–1985 (US pathologists)*

**Creutzfeldt-Jakob disease (CJD)** The cause is a prion (PrPSc), a misfolded form of a normal protein (PrPc), that can transform other proteins into prion proteins (hence its infectivity). ↑PrPSc leads to spongiform changes (tiny cavities ± tubulovesicular structures) in the brain.[15] Most cases are *sporadic* (incidence: 1–3/million/yr). *Variant* CJD (vCJD; ≈225 cases worldwide)[16] is transmitted via contaminated CNS tissue affected by bovine spongiform encephalopathy (BSE) (see BOX 'Signs that may distinguish variant CJD'). *Inherited forms:* (eg Gerstmann–Sträussler–Scheinker syndrome, P102L mutation in PRNP gene with ataxia ± self-mutilation), the 'normal' protein is too unstable, readily transforming to PrPSc. *Iatrogenic causes:* Contaminated surgical instruments, corneal transplants, growth hormone from human pituitaries, and blood (vCJD only).[17] Prion protein resists sterilization. *Signs:* Progressive dementia, focal CNS signs, myoclonus (present in 95%),[18] depression, eye signs (diplopia, supranuclear palsies, complex visual disturbances, homonymous field defects, hallucinations, cortical blindness).[19] *Tests:* Tonsil/olfactory mucosa biopsy;[20] CSF gel electrophoresis; MRI. *Treatment:* None proven. Death occurs in ~6 months in sporadic CJD (a little slower in variant CJD). *Prevention:* Regulations to ↓spread of BSE and transmission to humans + ↓iatrogenic transmission.

*Hans G Creutzfeldt, 1885–1964 (German pathologist); Alfons M Jakob 1884–1931 (German neurologist)*

**Crigler-Najjar syndrome** Two rare syndromes of inherited unconjugated hyperbilirubinaemia presenting in the 1st days of life with jaundice ± CNS signs. *Cause:* Mutation in UGT enzyme activity causing absent (type 1) or impaired (type 2; mild) ability to excrete bilirubin. *℞:* T1: phototherapy and plasmapheresis to control jaundice; liver transplant before irreversible kernicterus (*OHCS* p115) develops.[21] T2: usually no ℞ needed. *John F Crigler 1919–2002; Victor A Najjar b1914 (US paediatricians)*

ABC

## Signs that may distinguish variant CJD from sporadic CJD (sCJD)

- An earlier age at presentation (median 29yrs vs 60yrs in sporadic CJD).
- Longer survival and later dementia (median 14 months vs 4 for sporadic CJD).
- Psychiatric features are an early sign (anxiety, withdrawal, apathy, agitation, a permanent look of fear in the eyes, depression, personality change, insomnia). Hallucinations and delusions may occur—before akinetic mutism.
- Painful sensory symptoms are commoner (eg foot pain hyperaesthesia).
- More normal EEG (sporadic CJD has a characteristic spike and wave pattern).
- Mean CSF tau-pT181/tau protein ratio is 10-fold higher in vCJD than in sCJD.[22]
- Homozygosity for methionine at codon 129 of the PRP gene is typical.

## Fame and infamy in the search for lost youth

After his neurological experiments, Brown-Séquard, the most visionary of all neuroanatomists and the grandfather of HRT, proclaimed he had found the secret of perpetual youth after injecting himself with a concoction of testicular blood, semen, and testicular extracts from dogs and guinea pigs. In the 1880s, over 12 000 doctors were queuing up to use his special extracts on patients, which he gave away free, provided results were reported back to him. 314 out of 405 cases of spinal syphilis improved, and his own urinary flow rate rose by 25%. Endocrinologists never forgave him for bringing their science into disrepute. To this day, no one really knows if his (literally) seminal work has given us anything of any practical value. But he might be pleased to know that testosterone is now known to have the urodynamic benefits he anticipated, at least in men with hypogonadism.[23]

Like many brilliant men, he had a cruel streak, backing clitoridectomy for preventing blindness and other imaginary complications of 'masturbatory melancholia'. Had he not been blinded by 19th-century ideas about female sexuality, could he have found a marvellous use for his concoctions, for 21st-century 'hypoactive sexual desire disorder'? Possibly, but only if he relied on placebo responses.[24,25]

## Poisoning your boss

In 1931, Buerger's disease caused gangrene in the toes of Harvey Cushing (p224)—the most cantankerous (and greatest) neurosurgeon ever. He had to be wheeled to the operating theatre to carry on his brilliant art (and to continue terrifying his assistants).[26] He had to retire partially, whereupon his colleagues presented him with a magnificent silver cigarette box, containing 2000 cigarettes (to which he was addicted)—one for each brain tumour he had removed during his long career, so verifying the truth that although we owe everything to our teachers, we must eventually kill them to move out from under their shadow.[1]

## Why bother studying rare diseases? The Liberski imperative...

For centuries, kuru was no bigger than a man's hand; a cloud barely visible on our horizon; a rare disease in cannibals beyond the Pacific. But meticulous work on kuru led to knowledge of prion diseases *before* the 1990s epidemic of vCJD. If in the 1950s, Gajdusek and Zigas had not been intrigued as to why kuru affected women and children more than men (their strange neural diet was the culprit), the discovery of vCJD would have been delayed, as no surveillance would have been in place. Neural tissue might still be in our food chain, with dreadful consequences. But further than this, the notion of 'protein-misfolding diseases'[2] would have been delayed by decades. So this is the lesson: ►*let curiosity flourish*. This is Liberski's imperative.[27] So now let's scan our horizon for other intriguing clouds.



**1** *Der Vogel kämpft sich aus dem Ei. Das Ei ist die Welt. Wer geboren werden will, muss eine Welt zerstören*. The bird struggles out of the egg. The egg is the world. Whoever will be born, must first destroy a world. (Hermann Hesse. *Demian*; 1917.)

**2** Cystic fibrosis (misfolded CFTR protein), Marfan's (misfolded fibrillin),[28] Fabry (misfolded α-galactosidase), Gaucher's (misfolded β-glucocerebrosidase), retinitis pigmentosa 3 (misfolded rhodopsin); some cancers may be caused by misfolding of tumour suppressor proteins (von Hippel–Lindau protein).

**Devic's syndrome** (*Neuromyelitis optica*; NMO.) Inflammatory demyelination causes attacks of optic neuritis ± myelitis.[29] Abnormal CSF (may mimic bacterial meningitis) and serum anti-AQP4 antibody (in 65%) help distinguish it from MS[30] (table 15.1). *℞:* IV steroids; plasma exchange. Azathioprine and rituximab[31] help prevent relapses. *Prognosis:* Variable; complete remission may occur. *Eugène Devic, 1858–1930 (French neurologist)*

**Dressler's syndrome** This develops 2–10wks after an MI, heart surgery (or even pacemaker insertion). It is thought that myocardial injury stimulates formation of autoantibodies against heart muscle. *Symptoms:* Recurrent fever and chest pain ± pleural or pericardial rub (from serositis). Cardiac tamponade may occur, so avoid anticoagulants. *℞:* Aspirin, NSAIDs, or steroids. *William Dressler, 1890–1969 (US cardiologist)*

**Dubin-Johnson syndrome** There is defective hepatocyte excretion of conjugated bilirubin. Typically presents in late teens with intermittent jaundice ± hepatosplenomegaly (autosomal recessive). *Tests:* ↑Bilirubin; ALT and AST are normal; bilirubinuria on dipstick; ↑ratio of urinary coproporphyrin I to III. Liver biopsy: diagnostic pigment granules.[32] *℞:* Usually none needed. *Isadore N Dubin, 1913–1981; Frank B Johnson, b1919 (US pathologists)*

**Dupuytren's contracture** (fig 15.3) Progressive shortening and thickening of the palmar fascia causing finger contracture and loss of extension (often 5th finger). *Prevalence:* ~10% of ♂ >65yrs (↑ if +ve family history). *Associations:* Smoking, alcohol use, heavy manual labour, trauma, DM, phenytoin, HIV. Peyronie's may coexist (p708). It is thought to be caused by local hypoxia. *℞:* Collagenase injections.[33] Surgery may be needed. *Baron Guillaume Dupuytren, 1777–1835 (French surgeon, famed also for treating Napoleon's haemorrhoids)*

**Ekbom's syndrome** (*Restless legs.*) Criteria: **1** Compelling desire to move legs. **2** Worse at night. **3** Relieved by movement. **4** Unpleasant leg sensations (eg shootings or tinglings) worse at rest. *Mechanism:* Endogenous opioid system fault causes altered central processing of pain. *Prevalence:* 1–3%. ♀:♂≈2:1. *Associations:* Iron deficiency, uraemia, pregnancy, DM, polyneuropathy, RA, COPD. *Exclude:* Cramps, positional discomfort, and local leg pathology. *℞:* Dopamine agonists are commonly used; also, anticonvulsants, opioids, and benzodiazepines.[34]

*Karl Axel Ekbom, 1907–1977 (Swedish neurologist)*

**Fabry disease** X-linked lysosomal storage disorder caused by abnormalities in the GLA gene, leading to a deficiency in α-galactosidase A. There is accumulation of glycosphingolipids in skin (angiokeratoma classically in a 'swimming trunk' distribution), eyes (corneal verticillata), heart (hypertrophy, mitral valve prolapse, dilated aortic root, arrhythmias, angina), kidneys (renal failure, p320), CNS (stroke) and nerves (neuropathy/acroparaesthesia). Prior to enzyme replacement, premature death in the 6th decade was due to CV and renal disease. *℞:* Enzyme replacement therapy with α or β human agalsidase.[35] *Johannes Fabry, 1860–1930 (German dermatologist)*

**Fanconi anaemia** Autosomal recessive, defective stem cell repair & chromosomal fragility leads to aplastic anaemia, ↑risk of AML and breast ca (BRCA2), skin pigmentation, absent radii, short stature, microcephaly, syndactyly, deafness, ↓IQ, hypopituitarism, and cryptorchidism. *℞:* Stem-cell transplant. *Guido Fanconi, 1892–1979 (Swiss paediatrician)*

**Felty's syndrome** A triad of rheumatoid arthritis + ↓WCC + splenomegaly (±hypersplenism, causing anaemia and ↓platelets), recurrent infections, skin ulcers, and lymphadenopathy. 95% are Rh factor +ve. Splenectomy may raise the WCC. *℞:* DMARDs (p547) ± *rituximab* if refractory.[36] *Augustus Roi Felty, 1895–1964 (US physician)*



**Fitz-Hugh-Curtis syndrome** Liver capsule inflammation causing RUQ pain due to transabdominal spread of chlamydial or gonococcal infection, often with PID ± 'violin-string' adhesions. *℞:* Antibiotics for PID (+ treat sexual partners) ± laparoscopic division of adhesions. *Thomas Fitz-Hugh, 1894–1963 (US physician); Arthur H Curtis, 1881–1955 (US gynaecologist)*

**Foster Kennedy syndrome** Optic atrophy of one eye due to optic nerve compression (most commonly from an olfactory groove meningioma), with papilloedema of the other eye secondary to ↑ICP. There is also central scotoma and anosmia.

*Robert Foster Kennedy, 1884–1952 (British neurologist)*

**Friedreich's ataxia** Expansions of the trinucleotide repeat GAA in the frataxin gene (recessive) causes degeneration of many nerve tracts: spinocerebellar tracts degenerate causing cerebellar ataxia, dysarthria, nystagmus, and dysdiadochokinesis. Loss of corticospinal tracts occurs (weakness and extensor plantar response) with peripheral

## Devic's syndrome and multiple sclerosis

Table 15.1 Distinguishing Devic's syndrome from multiple sclerosis

| | Devic's syndrome | Multiple sclerosis |
|---|---|---|
| **Course** | Monophasic or relapsing | Relapsing usually; see p496 |
| **Attack severity** | Usually severe | Often mild |
| **Respiratory failure** | ~30%, from cervical myelitis | Rare |
| **MRI head** | Usually normal | Many periventricular white-matter lesions |
| **MRI cord lesions** | Longitudinal, central | Multiple, small, peripheral |
| **CSF oligoclonal bands** | Absent | Present |
| **Permanent disability** | Unusual, and attack-related | In late progressive disease |
| **Other autoimmunities** | In ≤50% (eg Sjögren's) | Uncommon |

**Diagnostic criteria for Devic's** Optic neuritis, myelitis, and ≥2 out of 3 of: •MRI evidence of a continuous cord lesion for ≥3 segments. •Brain MRI at onset non-diagnostic for MS. •NMO-IgG (anti-AQP4) serum or CSF positivity (poorer prognosis).

NB: CNS involvement beyond the optic nerves and cord is compatible with NMO.

Fig 15.3 Dupuytren's contracture of the 5th finger. Note scar from previous surgery to the index finger.

nerve damage, so tendon reflexes are paradoxically depressed (differential diagnosis p446). There is also dorsal column degeneration, with loss of positional and vibration sense. Pes cavus and scoliosis occur. Cardiomyopathy may cause CCF. Typical age at death: ~50yrs. *℞:* There is no cure. Treat CCF, arrhythmias, and DM.

*Nikolaus Friedreich, 1825–1882 (German neurologist)*

**Froin's syndrome** ↑ CSF protein + xanthochromia with normal cell count—a sign of blockage in spinal CSF flow (eg from a spinal tumour). *Georges Froin, 1874–1932 (French physician)*

**Gardner's syndrome** A dominant variant of familial adenomatous polyposis, caused by mutations in the APC gene (5q21). There are multiple colon polyps (which inevitably become malignant; p520),[37] benign bone osteomas, epidermal cysts, dermoid tumours, fibromas, and neurofibromas. *Fundoscopy* reveals black spots (congenital hypertrophy of retinal pigment epithelium); this helps pre-symptomatic detection. *Presentation:* Can present from 2–70yrs with colonic (eg bloody diarrhoea) or extracolonic symptoms. Prophylactic surgery (eg proctocolectomy) is the only curative treatment. Endoscopic polypectomy with long-term celecoxib therapy has been used to postpone prophylactic colectomy.[38] *Eldon J Gardner, 1909–1989 (US physician)*

**Gélineau's syndrome** (*Narcolepsy*.) The patient, usually a young man, succumbs to irresistible attacks of inappropriate sleep ± vivid hypnogogic hallucinations, cataplexy (sudden hypotonia), and sleep paralysis (paralysis of speech and movement, while fully alert, at sleep onset or on waking). *Hypothesis:* Mutations lead to loss of hypothalamic hypocretin-containing neurons, via autoimmune destruction.[39] 95% are +ve for HLA DR2. *℞:* Stimulants (eg methylphenidate) may cause dependence ± psychosis. Modafinil may be better. SE: anxiety, aggression, dry mouth, euphoria, insomnia, ↑BP, dyskinesia, ↑ALP. *Jean-Baptiste-Édouard Gélineau, 1828–1906 (French physician)*

**Gerstmann's syndrome** A constellation of symptoms suggesting a dominant parietal lesion: finger agnosia (inability to identify fingers), agraphia (inability to write), acalculia (inability to calculate), and left-right disorientation.

*Josef Gerstmann, 1887–1969 (Austrian neurologist)*

**Gilbert's syndrome** A common cause of *unconjugated* hyperbilirubinaemia due to ↓ UGT-1 activity (the enzyme that conjugates bilirubin with glucuronic acid). *Prevalence:* 1–2%; 5–15% have a family history of jaundice. It may go unnoticed for many years and usually presents in adolescence with intermittent jaundice occuring during illness, exercise or fasting. *Diagnosis:* Mild ↑bilirubin; normal FBC and reticulocytes (ie no haemolysis). It is a benign condition. *Nicolas Augustin Gilbert, 1858–1927 (French physician)*

**Gilles de la Tourette syndrome** Tonic, clonic, dystonic, or phonic tics: jerks, blinks, sniffs, nods, spitting, stuttering, irrepressible explosive obscene verbal ejaculations (coprolalia, in 20%) or gestures (coprophilia, 6%),[40] grunts, squeaks, burps, twirlings, and nipping others ± tantrums. There may be a witty, innovatory, phantasmagoric picture, with mimicry (echopraxia), antics, impishness, extravagance, audacity, dramatizations, surreal associations, uninhibited affect, speed, 'go', vivid imagery and memory, and hunger for stimuli. *The tic paradox:* Tics are *voluntary*, but often *unwanted*: the desire to tic stems from the relief of the odd sensation that builds up prior to the tic and is relieved by it, 'like scratching a mosquito bite, tics lead to more tics'.[41] *Mean age of onset:* 6yrs. ♂:♀≈4:1. *Pathogenesis:* Unknown; multiple genetic loci implicated and neuroanatomical abnormalities reported on MRI. *Associations:* Obsessive-compulsive disorder; attention deficit hyperactivity disorder. *℞:* (None may be wanted.) Risperidone, haloperidol, or pimozide. Habit-reversal training.[42] Deep brain stimulation is rarely indicated, but may help.

ABC

*Marquis Georges Albert Édouard Brutus Gilles de la Tourette, 1857–1904 (French neurologist)*

**Goodpasture's disease** *(A pulmonary-renal syndrome.)* Acute glomerulonephritis + lung symptoms (haemoptysis/diffuse pulmonary haemorrhage) caused by antiglomerular basement membrane antibodies (binding kidney's basement membrane and alveolar membrane). *Tests:* CXR: infiltrates due to pulmonary haemorrhage, often in lower zones. Kidney biopsy: crescentic glomerulonephritis. *℞:* ►►Treat shock. Vigorous immunosuppressive treatment and plasmapheresis.

*Ernest William Goodpasture, 1886–1960 (US pathologist)*

### Cataplexy is highly specific for narcolepsy/Gélineau's syndrome

Daytime sleepiness has many causes, but if it occurs with cataplexy the diagnosis 'must' be narcolepsy. Cataplexy is bilateral loss of tone in antigravity muscles provoked by emotions such as laughter, startle, excitement, or anger. Associated phenomena include: falls, mouth opening, dysarthria, mutism, and phasic muscle jerking around the mouth. Most attacks are brief, but injury can occur (eg if several attacks per day). It is comparable to the atonia of rapid eye movement sleep *but without loss of awareness*. ΔΔ: bradycardia, migraine, atonic/akinetic epilepsy, delayed sleep phase syndrome, conversion disorder, malingering, and psychosis.

Don't confuse cata*plexy* with cata*lepsy*—a waxy flexibility where involuntary statue-like postures are effortlessly maintained (frozen) despite looking most uncomfortable.

**Guillain-Barré syndrome** (*Acute inflammatory demyelinating polyneuropathy.*)[43,44] (table 15.2) *Incidence:* 1–2/100 000/yr. *Signs:* A few weeks after an infection a symmetrical ascending muscle weakness starts. *Triggers: Campylobacter jejuni*, CMV, mycoplasma, zoster, HIV, EBV, vaccinations. The trigger causes antibodies which attack nerves. In 40%, no cause is found. It may advance quickly, affecting all limbs at once, and can lead to paralysis. There is a progressive phase of up to 4 weeks, followed by recovery. Unlike other neuropathies, *proximal* muscles are more affected, eg trunk, respiratory, and cranial nerves (esp. VII). Pain is common (eg back, limb) but sensory signs may be absent. *Autonomic dysfunction:* Sweating, ↑pulse, BP changes, arrhythmias. *Nerve conduction studies:* Slow conduction. *CSF:* ↑Protein (eg >5.5g/L), normal CSF white cell count. Respiratory involvement (the big danger) requires transfer to ITU. Do forced vital capacity (FVC) 4-hourly. ►*Ventilate sooner rather than later*, eg if FVC <1.5L, $P_aO_2$ <10kPa, $P_aCO_2$ >6kPa. *℞:* IV immunoglobulin 0.4g/kg/24h for 5d. Plasma exchange is good too (?more SE).[45] Steroids have no role. *Prognosis:* Good; ~85% make a complete or near-complete recovery. 10% are unable to walk alone at 1yr. *Complete paralysis is compatible with complete recovery. Mortality:* 10%.

*George C Guillain, 1876–1961; Jean-Alexandre Barré, 1880–1967 (French neurologists)*

**Henoch-Schönlein purpura (HSP)** (fig 15.5) A small vessel vasculitis, presenting with purpura (non-blanching purple papules due to intradermal bleeding), often over buttocks and extensor surfaces, typically affecting young ♂. There may be glomerulonephritis (p310), arthritis, and abdominal pain (± intussusception), which may mimic an 'acute abdomen'. *℞:* Mostly supportive.

*Eduard H Henoch, 1820–1910 (German paediatrician); Johann L Schönlein, 1793–1864 (German physician)*

**Horner's syndrome** A triad of **1** *miosis* (pupil constriction, fig 15.4) **2** partial *ptosis* (drooping upper eyelid) + *apparent enophthalmos* (sunken eye) **3** *anhidrosis* (ipsilateral loss of sweating). Due to interruption of the face's sympathetic supply, eg at the brainstem (demyelination, vascular disease), cord (syringomyelia), thoracic outlet (Pancoast's tumour, p708), or on the sympathetic's trip on the internal carotid artery into the skull (fig 15.6), and orbit. *Johann Friedrich Horner, 1831–1886 (Swiss ophthalmologist)*

**Huntington's disease** Incurable, progressive, autosomal dominant, neurodegenerative disorder presenting in middle age, often with prodromal phase of mild symptoms (irritability, depression, incoordination). Progresses to chorea, dementia ± fits & death (within ~15yrs of diagnosis). *Pathology:* Atrophy and neuronal loss of striatum and cortex. *Genetic basis:* Expansion of CAG repeat on Chr. 4. *℞:* (p87.) No treatment prevents progression. Counselling for patient and family.[46]

*George Huntington, 1850–1916 (US physician)*

**Jervell and Lange-Nielsen syndrome** Congenital, bilateral, autosomal recessive, sensorineural deafness, and long QT interval (p96, hence syncope, VT, *torsades*, ± sudden death—50% by age 15 if untreated). KCNQ1 or KCNE1 gene mutation causes $K^+$ channelopathy. *℞:* β-blocker, pacemaker, ICD, cochlear implants.[47]

*Anton Jervell, 1901–1987; Fred Lange-Nielsen, 1919–1989 (Norwegian physicians)*

ABC

**Kaposi's sarcoma (KS)** A spindle-cell tumour derived from capillary endothelial cells, caused by human herpes virus 8 (=Kaposi's sarcoma-associated herpes virus KSHV). It presents as purple papules (½–1cm) or plaques on skin (fig 15.7) and mucosa (look in mouth, but any organ). It metastasizes to nodes. There are four types: **1** Classic, a rare disease of the elderly. **2** Endemic, a disease of children documented prior to HIV. **3** Iatrogenic KS due to immunosuppression, eg organ transplant recipients. **4** AIDS-associated KS. Usually presents with low CD4 count and can indicate failure of HAART (p402). However ⅓ presents in HIV with near normal CD4 counts and an undetectable viral load. Initiation of HAART with rapid immune system constitution can precipitate KS. Lung KS may present in HIV +ve men and women as dyspnoea and haemoptysis. Bowel KS may cause nausea, abdominal pain. *Rare sites:* CNS, larynx, eye, glands, heart, breast, wounds, or biopsy sites. *Δ:* Biopsy. *℞:* Optimize HAART, local radiotherapy, surgical excision, intralesional therapy (vincristine, bleomycin), topical retinoids, interferon alfa, interleukin-12. Current research includes thalidomide, VEGF monoclonal antibodies, and sirolimus. *Moricz Kaposi, 1837–1902 (Hungarian dermatologist)*

## Guillain-Barré polyneuritis

**Table 15.2** Diagnostic criteria

| **Features required for** | **Features supporting diagnosis** |
| --- | --- |
| Progressive weakness of >1 limb<br>Areflexia<br>**Features making diagnosis doubtful**<br>• Sensory level<br>• Marked, persistent asymmetry of weakness<br>• Severe bowel and bladder dysfunction<br>• CSF WCC >50 | • Progression over days, up to 4wks<br>• Near symmetry of sytoms<br>• Sensory symptoms/signs only mild<br>• CN involvement (eg bilateral facial weakness)<br>• Recovery starts ~2wks after the period of progression has finished<br>• Autonomic dysfunction<br>• Absence of fever at onset<br>• CSF protein ↑ with CSF WCC <10×10⁶/L<br>• Typical electrophysiological tests |

Variants of Guillain-Barré syndrome include:

*Chronic inflammatory demyelinating polyradiculopathy (CIDP):* Characterized by a slower onset and recovery.

*Miller Fisher syndrome:* Comprises of ophthalmoplegia, ataxia, and areflexia. Associated with anti-GQ1b antibodies in the serum.

**Fig 15.4** Right Horner's: everything reduces: pupil, eye, sweating, etc.

**Fig 15.5** Henoch-Schönlein vasculitis.

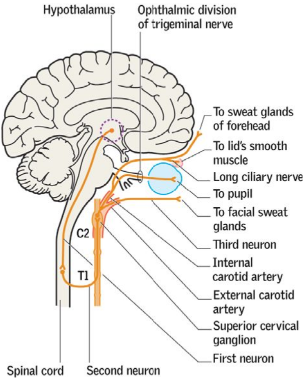


**Fig 15.6** Pathways in Horner's syndrome.

**Fig 15.7** Kaposi's sarcoma.

**Klippel-Trénaunay syndrome** A triad of port wine stain, varicose veins, and limb hypertrophy, due to vascular malformation. Usually sporadic (although AD inheritance has been reported).[48] *Maurice Klippel, 1858–1942; Paul Trénaunay, 1875–1938 (French physicians)*

**Korsakoff's syndrome** Hypothalamic damage & cerebral atrophy due to thiamine (vitamin $B_1$) deficiency (eg in alcoholics). May accompany Wernicke's encephalopathy. There is ↓ability to acquire new memories, confabulation (invented memory, owing to retrograde amnesia), lack of insight & apathy. *℞:* See *Wernicke's*, p714; patients rarely recover. *Sergei Sergeievich Korsakoff, 1853–1900 (Russian neuropsychiatrist)*

**Langerhans cell histiocytosis** (*Histiocytosis x.*) A group of single- (73%, eg bone) or multisystem (27%) disorders, with infiltrating granulomas containing dendritic (Langerhans) cells. ♂:♀ ≈1.5:1; at-risk organs are liver, lung, spleen, marrow. Pulmonary disease presents with pneumothorax or pulmonary hypertension. CXR/CT: nodules and cysts + honeycombing in upper and middle zones. *Δ:* Biopsy (skin, lung). *℞:* Local excision, steroids, vinblastine ± etoposide if severe.[49] *OHCS* p644. *Paul Langerhans, 1847–1888 (German pathologist)*

**Leriche's syndrome** Absent femoral pulse, claudication/wasting of the buttock, a pale cold leg, and erectile dysfunction from aorto-iliac occlusive disease, eg a saddle embolus at the aortic bifurcation. Surgery may help. *René Leriche, 1879–1955 (French surgeon)*

**Löffler's eosinophilic endocarditis** Restrictive cardiomyopathy + eosinophilia (eg $120 \times 10^9$/L). It may be an early stage of tropical endomyocardial fibrosis (and overlaps with hypereosinophilic syndrome, p330) but is distinct from eosinophilic leukaemia. *Signs:* Heart failure (75%) ± mitral regurgitation (49%) ± heart block. *℞:* Suppress the eosinophilia (prednisolone ± hydroxycarbamide), and then treat with anti-heart failure medication. *Wilhelm Löffler, 1887–1972 (Swiss physician)*

**Löffler's syndrome** (*Pulmonary eosinophilia.*) An allergic infiltration of the lungs by eosinophils. Allergens include: *Ascaris lumbricoides*, *Trichinella spiralis*, *Fasciola hepatica*, *Strongyloides*, *Ankylostoma*, *Toxocara*, *Clonorchis sinensis*, sulfonamides, hydralazine, and nitrofurantoin. Often symptomless with incidental CXR (diffuse fan-shaped shadows), or cough, fever, eosinophilia (in ~20%) & larval migrans (p433). *℞:* Eradicate cause. Steroids (if idiopathic). *Wilhelm Löffler, 1887–1972 (Swiss physician)*

**Lown-Ganong-Levine syndrome** A pre-excitation syndrome, similar to Wolf-Parkinson-White (WPW, p133), characterized by a short PR interval (<0.12sec), a normal QRS complex (as opposed to the δ-waves of WPW), and risk of supraventricular tachycardia (but not AF/flutter). The cause is not completely understood, but may be due to paranodal fibres that bypass all or part of the atrioventricular node. The patient may complain of intermittent palpitations.[50]

*Bernard Lown, b1921 (US cardiologist); William F Ganong, 1924–2007 (US physiologist); Samuel A Levine, 1891–1966 (US cardiologist)*

**McArdle's glycogen storage disease (type V)** Absence of muscle phosphorylase enzyme with resulting inability to convert glycogen into glucose (eg R50X mutation of PYGM gene; autosomal recessive). Fatigue & crises of cramps ± hyperthermia. Rhabdomyolysis/myoglobinuria follow exercise. *Tests:* ↑↑CK. Muscle biopsy is diagnostic (necrosis and atrophy). *℞:* Moderate aerobic exercise helps (by utilizing alternative fuel substrates).[51] Avoid heavy exertion and statins. Sucrose pre-exercise improves performance, as does a carbohydrate-rich diet. Low-dose creatine and ramipril (only if D/D ACE phenotype) may be of minimal benefit.[52] *Brian McArdle, 1911–2002 (British paediatrician)*

ABC

**Mallory-Weiss tear** Persistent vomiting/retching *causes* haematemesis via an oesophageal mucosal tear. *George K Mallory, 1900–1986 (US pathologist); Soma Weiss, 1898–1942 (US physician)*

**Marchiafava-Bignami syndrome** Corpus callosum demyelination and necrosis, most often secondary to chronic alcoholism. Type A is characterized by coma, stupor, and pyramidal tract features involving the entire corpus callosum. In type B, symptoms are mild and the corpus callosum is partially affected.[53] *Δ:* MRI. *℞:* As for Wernicke's, p714 (see BOX 'Adverse effects of alcohol on the CNS').

*Ettore Marchiafava, 1847–1935; Amico Bignami, 1862–1929 (Italian pathologists)*

**Marchiafava-Micheli syndrome** (*Paroxysmal nocturnal haemoglobinuria, PNH.*) An acquired clonal expansion of a multipotent stem cell manifesting with haemolytic anaemia (from complement-mediated intravascular haemolysis), large vessel thromboses and deficient haematopoiesis (ranging from mild to pancytopenia). See BOX, 'Paroxysmal nocturnal haemoglobulinuria' fig 15.8, and p338.

*Ettore Marchiafava, 1847–1935 (Italian pathologist); Ferdinando Micheli, 1872–1936 (Italian physician)*

## Paroxysmal nocturnal haemoglobinuria: the darkest hour

In paroxysmal nocturnal haemoglobinuria (PNH), surface proteins are missing in all blood cells due to a somatic mutation in the X-linked PIG-A gene. Cells lack the glycosyl-phosphatidylinositol (GPI) anchor that binds the surface proteins to cell membranes. This causes uncontrolled amplification of the complement system and leads to destruction of the RBC membrane and release of haemoglobin into the circulation (fig 15.8). NB: the phenomenom of haemoglobinuria[3] is not all that reliable. A much better test even than a marrow biopsy (right-hand panel, showing a clone of PNH cells) is flow cytometric analysis of GPI-anchored proteins on peripheral blood cells. This can determine the size of the PNH clone and type of GPI deficiency (complete or partial). ℞: Most benefit from supportive measures—but allogeneic stem cell transplantation is the only cure. Eculizumab is a monoclonal antibody that targets the C5 protein of the complement system. Blockade prevents activation of the complement distal pathway, reducing haemolysis, stabilizing haemoglobin, and reducing transfusion requirements.[54]

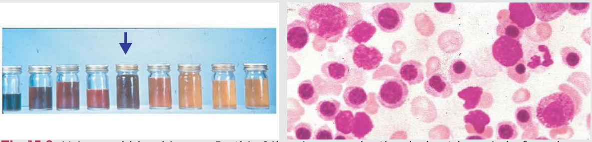

Fig 15.8 Urine and blood in PNH. In this 24h urine sample, the darkest hour is before dawn. Haemolysis occurs throughout the day and night, but the urine concentrated overnight produces the dramatic change in colour.

Courtesy of the Crookston Collection.

## Adverse effects of alcohol on the CNS

- ↓Inhibitions (risk taking; ↑unsafe sex)
- Wernicke's encephalopathy
- Korsakoff's syndrome
- Hepatic encephalopathy
- Cerebral atrophy (dementia)
- Central pontine myelinolysis
- Cerebellar atrophy (eg falls)
- Stroke (ischaemic and haemorrhagic)
- Seizures
- Marchiafava-Bignami syndrome.

ABC

3 In haemoglobinuria, urine dipstick will be positive for blood but microscopy of urine does not show RBCs (thus differentiating it from haematuria, but not myoglobinuria—where CK ± AST will be high).

**Marfan's syndrome** (table 15.3) Autosomal dominant disorder (fibrillin-1) with ↓extracellular microfibril formation; but ~25% have no family history. *Major criteria:* (Diagnostic if >2): lens dislocation (*ectopia lentis*; fig 15.9); aortic dissection/dilatation; dural ectasia; *skeletal features:* arachnodactyly (long spidery fingers), armspan > height, pectus deformity, scoliosis, pes planus. *Minor signs:* Mitral valve prolapse, high-arched palate, joint hypermobility. Diagnosis is clinical; MRI for dural ectasia. *℞:* The danger is aortic dissection: β-blockers slow dilatation of the aortic root. Annual echos, surgical repair when aortic diameter is >5cm. In pregnancy ↑risk of dissection. Homocystinuria has similar skeletal deformities.

*Antoine Bernard-Jean Marfan, 1858–1942 (French paediatrician)*

Table 15.3 Comparing Marfan's and homocystinuria

| Marfan's vs | Homocystinuria |
|---|---|
| Upwards lens dislocation | Downwards lens dislocation |
| Aortic valve incompetence | Heart rarely affected |
| Normal intelligence | Mental retardation |
| Scoliosis, flat feet, herniae | Recurrent thromboses, osteoporosis |
| Life expectancy is lower due to cardiovascular risks | Positive urine cyanide-nitroprusside test |
| | Response to treatment with pyridoxine |

**Meckel's diverticulum** The distal ileum contains embryonic remnants of gastric and pancreatic tissue. There may be gastric acid secretion, causing GI pain & occult bleeding. Δ: Radionucleotide scan; laparotomy. *Johann Friedrich Meckel, 1781–1833 (German anatomist)*

**Meigs' syndrome** A triad of **1** benign ovarian tumour (fibroma) **2** pleural effusion (R>L) & **3** ascites. It resolves on tumour resection. *Joe Vincent Meigs, 1892–1963 (US gynaecologist)*

**Ménétrier's disease** Giant gastric mucosal folds up to 4cm high, in the fundus, with atrophy of the glands + ↑mucosal thickness + hypochlorhydria + protein-losing gastropathy (hence hypoalbuminaemia ± oedema). *Causes:* CMV, *strep*, *H. pylori*. There may be epigastric pain, vomiting, ± ↓weight. It is pre-malignant. *℞:* Treat *H. pylori* or CMV if present; give high-dose PPI; if this fails, consider epidermal growth factor blockade with cetuximab or gastrectomy (eg if intractable symptoms or malignant change). Epidermal growth factor receptor blockade with cetuximab is 1st-line treatment.[55] Surgery if intractable symptoms or malignant change. *Pierre Eugène Ménétrier, 1859–1935 (French pathologist)*

**Meyer-Betz syndrome** (*Paroxysmal myoglobinuria.*) Rare idiopathic condition causing necrosis of exercising muscles. There is muscle pain, weakness, and discoloured urine: pink→brown (as ↑myoglobin is excreted). Acute kidney injury can result from myoglobinuria (p319). DIC is associated. *Tests:* ↑WCC, ↑LFT, ↑LDH, ↑CPK, ↑urine myoglobin. *Diagnosis:* Muscle biopsy, ↑CPK, and ↑serum myoglobin. Exertion should be avoided. *Friedrich Meyer-Betz, described 1910 (German physician)*

**Mikulicz's syndrome** Benign persistent swelling of lacrimal and parotid (or submandibular) glands due to lymphocytic infiltration. Exclude other causes (sarcoidosis, TB, viral infection, lymphoproliferative disorders). It is thought to be an IgG4-related plasmacytic systemic disease.[56] *Johann Freiherr von Mikulicz-Radecki, 1850–1905 (Polish-Austrian surgeon)*

ABC

**Milroy disease** 1° congenital lymphoedema. Mutations in the VEGFR3 gene (dominant) cause lymphatic malfunction with lower leg swelling from birth (fig 15.10). Δ: Lymphoscintigraphy; genetic testing. *℞:* •Compression hosiery/bandages. •Encourage exercise. •Good skin hygiene. •Treat cellulitis actively. *William Forsyth Milroy, 1855–1942 (US physician)*

**Münchausen's syndrome** Vivid liars, who are addicted to institutions, flit from hospital to hospital, feigning illness, eg hoping for a laparotomy or mastectomy, or they complain of awful bleeding, odd eye movements, curious fits, sexual assaults, throat closings, false asthma, or heart attacks. Münchausen-by-proxy entails injury to a dependent person by a carer (eg mother) to gain medical attention.

*Karl Friedrich Hieronymus, Freiherr von Münchausen, 1720–1797 (German aristocrat). Described by RAJ Asher in 1951*[57]

**Ogilvie's syndrome** *(Acute colonic pseudo-obstruction.)* Colonic obstruction in the absence of a mechanical cause, associated with recent severe illness or surgery. *℞:* Correct U&E. Colonoscopy allows decompression, and excludes mechanical causes. Neostigmine is also effective, suggesting parasympathetic suppression is to blame.[58] Surgery is rarely needed (eg if perforation). *William Heneage Ogilvie, 1887–1971 (British surgeon)*

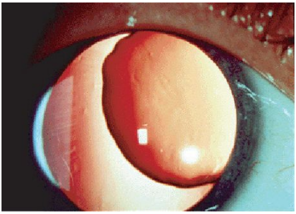

Fig 15.9 Lens dislocation in Marfan's syndrome: here the lens is dislocated superiorly and medially.

Courtesy of Prof Jonathan Trobe.

Fig 15.10 Milroy disease. Lymphoedema may be primary, as in Milroy or Meige disease, and is a feature in both Turner and Noonan syndromes. More commonly it is secondary to other conditions, eg cancer (after surgery, lymph node dissection, radiotherapy, or from direct tumour effect), cellulitis, varicose veins, or immobility/dependency. Filariasis (p421) is a common cause in tropical regions.

## Who was Baron Münchausen?

Baron Karl Münchausen was an 18th-century German aristocrat and fabulist, whose tall tales became first a popular book, then a byword for circular logic, and finally a medical syndrome of self-delusion. He is famous for riding cannonballs, travelling to the moon, and pulling himself out of a swamp by his own hair. ►►In emergencies (we've all had that sinking feeling…), this method may save your life, for example in your final exams (fig 15.11):

*Examiner:* 'What is ITP?'

*You:* 'ITP is idiopathic thrombocytopenic purpura.' (you have scored 50% already).

*Examiner:* 'And what is idiopathic thrombocytopenic purpura?'

*You:* 'It's when a cryptogenic cause of a low platelet count leads to purpura.'

You have deployed your skills with logical brilliance, without adding a single insight. For this Münchausen circularity you may be awarded 100%—unless your examiner is a philosopher, when the right answer would be 'What is ITP? I don't know—and nor do you'—but don't try this too often. You see, you must never forget that medicine is marvellously scientific, and no one is popular who dares cast doubt on this article of faith.

Fig 15.11 Münchausen during his finals.

**Ortner's cardiovocal syndrome** Recurrent laryngeal nerve palsy from a large left atrium (eg from mitral stenosis) or aortic dissection. *Norbert Ortner, 1865–1935 (Austrian physician)*

**Osler-Weber-Rendu syndrome** (*Hereditary telangiectasia.*) Autosomal dominant telangiectasia of skin & mucous membranes (causing epistaxis and GI bleeds), see fig 15.12. Associated with pulmonary, hepatic, and cerebral arteriovenous malformations.
*William Osler, 1849–1919 (Canadian); Frederick Weber, 1863–1962 (British); Henri Rendu, 1844–1902 (French)—physicians*

**Paget's disease of the breast (PDB)** Intra-epidermal spread of an intraduct cancer, which can look just like eczema. ►Any red, scaly lesion at the nipple (see fig 15.13) *must* suggest PDB: do a biopsy. *℞:* Breast-conserving surgery + radiotherapy. Sentinel node biopsy should be performed. *Sir James Paget, 1814–1899 (British surgeon)*

**Pancoast's syndrome** Apical lung ca invades the sympathetic plexus in the neck (→ipsilateral Horner's, p702) ± brachial plexus (→arm pain ± weakness) ± recurrent laryngeal nerve (→hoarse voice/bovine cough). *Henry Pancoast, 1875–1939 (US radiologist)*

**Parinaud's syndrome** (*Dorsal midbrain syndrome.*) Upward gaze palsy + pseudo-Argyll Robertson pupils (p72) ± bilateral papilloedema. *Causes:* Pineal or midbrain tumours; upper brainstem stroke; MS. *Henry Parinaud, 1844–1905 (French neuro-ophthalmologist)*

**Peutz-Jeghers' syndrome** Dominant germline mutations of tumour suppressor gene STK11 (in 66–94%) cause mucocutaneous dark freckles on lips (fig 15.14), oral mucosa, palms and soles, + multiple GI polyps (hamartomas), causing obstruction, intussusception, or bleeds. There is a 15-fold ↑risk of developing GI cancer∴ perform colonoscopy (from age 18yrs) and OGD (from age 25yrs) every 3yrs. NB: hamartomas are excessive focal overgrowths of normal cells in an organ composed of the same cell type. *Johannes LA Peutz, 1886–1957 (Dutch physician); Harold J Jeghers, 1904–1990 (US physician)*

**Peyronie's disease** (*Penile angulation.*) *Pathogenesis:* A poorly understood connective tissue disorder most commonly attributed to repetitive microvascular trauma during sexual intercourse, resulting in penile curvature and painful erectile dysfunction (in 50%; p230). *Prevalence:* 3–9%. *Typical age:* >40yrs. *Associations:* Dupuytren's (p698); atheroma; radical prostatectomy. *ΔΔ:* Haemangioma. *Tests:* Ultrasound/MRI. *℞:* Oral potassium para-aminobenzoate (Potaba®), intralesional verapamil, clostridial collagenase, or interferon α*2B*; topical verapamil 15% gel; iontophoresis with verapamil and dexamethasone. All have various success.[59] *Surgery:* (If disease stable for >3 months) tunica plication ± penile prostheses. Manage associated depression (seen in 48%). Penile rehabilitation can help (p230).[60] *Francois Gigot de la Peyronie, 1678–1747 (French surgeon)*

**Pott's syndrome** (*Spinal TB.*) Rare in the West, this is usually from an extra-spinal source, eg lungs. *Features:* Backache, and stiffness of *all* back movements. Fever, night sweats, and weight loss occur. Progressive bone destruction leads to vertebral collapse and gibbus (sharply angled spinal curvature). Abscess formation may lead to cord compression, causing paraplegia, and bowel/bladder dysfunction (p466). *X-rays:* (fig 15.15) Narrow disc spaces and vertebral osteoporosis, leading to destruction with wedging of vertebrae. Lesions in the thoracic spine often lead to kyphosis. Abscess formation in the lumbar spine may track down to the psoas muscle, and erode through the skin. *℞:* Anti-TB drugs (p394). *Sir Percival Pott, 1714–1788 (British surgeon)*

ABC

**Prinzmetal (variant) angina** Angina from coronary artery spasm, which may lead to MI, ventricular arrhythmias or sudden death. Severe chest pain occurs *without* physical exertion. Triggers include hyperventilation, cocaine and tobacco use. *ECG:* ST elevation. *℞:* Establish the diagnosis. GTN treats angina. Use $Ca^{2+}$-channel blockers (p114) and long-acting nitrates as prophylaxis. *Myron Prinzmetal, 1908–1987 (US cardiologist)*

**Raynaud's syndrome** This is peripheral digital ischaemia due to paroxysmal vasospasm, precipitated by cold or emotion. Fingers or toes ache and change colour: pale (ischaemia) →blue (deoxygenation) →red (reactive hyperaemia). It may be idiopathic (Raynaud's *disease*—prevalence: 3–20%; ♀:♂ >1:1) or have an underlying cause (Raynaud's *phenomenon*; fig 15.16). *Tests:* Exclude an underlying cause (see BOX 'Conditions in which Raynaud's phenomenon may be exhibited'). *℞:* Keep warm (eg hand warmers); stop smoking.[4] Nifedipine 5–20mg/8h PO helps, as may evening primrose oil, sildenafil, and epoprostenol (for severe attacks/digital gangrene). Relapse is common. Chemical or surgical (lumbar or digital) sympathectomy may help in those with severe disease. *AG Maurice Raynaud, 1834–1881 (French physician)*

## Prinzmetal angina and vascular hyperreactivity

Coronary spasm causes Prinzmetal angina and also contributes to coronary heart disease in general, eg acute coronary syndrome (esp. in Japan). Coronary spasm can be induced by ergonovine, acetylcholine, and methacholine (the former is used diagnostically).[5] These cause vasodilation by endothelium-derived nitric oxide when vascular endothelium is functioning normally, whereas they cause vasoconstriction if the endothelium is damaged. In the light of these facts, patients with coronary spasm are thought to have a disturbance in endothelial function as well as local hyperreactivity of the coronary arteries.

If full anti-anginal therapy does not reduce symptoms, stenting or intracoronary radiation (20Gy brachytherapy) to vasospastic segments may be tried. Prognosis is good (especially if non-smoker, no past MI, and no diabetes; progress to infarction is quite rare);[61] β-blockers and large doses of aspirin are contraindicated.

Prinzmetal angina is associated with vascular hyperreactivity/vasospastic disorders such as Raynaud's phenomenon and migraine. It is also associated with circle of Willis occlusion from intimal thickening (moyamoya disease).

Fig 15.12 Telangiectasia in Osler-Weber-Rendu syndrome.

Reproduced from Cox and Roper, *Clinical Skills*, 2005, with permission from Oxford University Press.

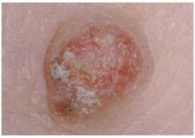

Fig 15.13 Paget's disease of the breast.

Fig 15.14 Perioral pigmentation, seen in Peutz–Jeghers' syndrome.

Reproduced from Cox and Roper, *Clinical Skills*, 2005, with permission from Oxford University Press.

Fig 15.15 TB of axis: soft tissue swelling displaces the retropharyngeal air–tissue boundary forwards. There is an anterior defect in the vertebra, below the axis peg.

Courtesy of Dr Ian Maddison, myweb.lsbu.ac.uk.

## Conditions in which Raynaud's phenomenon may be exhibited

*Connective tissue disorders:* Systemic sclerosis, SLE, rheumatoid arthritis, dermatomyositis/polymyositis.

*Occupational:* Using vibrating tools.

*Obstructive:* Thoracic outlet obstruction, Buerger's disease, atheroma.

*Blood:* Thrombocytosis, cold agglutinin disease, polycythaemia rubra vera (p366), monoclonal gammopathies.

*Drugs:* β-blockers.

*Others:* Hypothyroidism.

Fig 15.16 Raynaud's phenomenon in SLE.

Courtesy of the Crookston Collection.

4 Patient information on Raynaud's is available from www.raynauds.org.uk.

5 Since Prinzmetal angina is not a 'demand-induced' symptom, but a supply (vasospastic) abnormality, exercise tolerance tests don't help. The most sensitive and specific test is IV ergonovine; 50mcg at 5min intervals in a specialist lab until a +ve result or 400mcg is given. When positive, the symptoms and ↑ST should be present. Nitroglycerin rapidly reverses the effects of ergonovine if refractory spasm occurs.[62]

**Refsum disease** Phytanic acid accumulates in tissues and serum, due to PHYH or PEX7 gene mutation (recessive). This leads to anosmia (a universal finding) and early-onset retinitis pigmentosa, with variable combinations of neuropathy, deafness, ataxia, ichthyosis, and cardiomyopathy. *Tests:* ↑Plasma phytanic acid. *℞:* Restrict foods containing phytanic acid (animal fats, dairy products, green leafy vegetables); plasmapheresis is used for severe symptoms.[63] *Sigvald Bernhard Refsum, 1907–1991 (Norwegian physician)*

**Romano-Ward syndrome** A dominant mutation in a $K^+$ channel subunit causes long QT syndrome ± episodic VT, VF, *torsades,* ± sudden death. (Jervell and Lange-Nielsen syn is similar, p702.) *Cesarino Romano, 1924–2008 (Italian paediatrician); Owen C Ward, b1923 (Irish paediatrician)*

**Rotor syndrome** A rare, benign, autosomal recessive disorder. Primary non-haemolytic conjugated hyperbilirubinaemia, with almost normal hepatic histology (no pigmentation, in contrast to DJS, p698). Typically presents in childhood with mild jaundice. Cholescintigraphy reveals an 'absent' liver. *Arturo Belleza Rotor, 1907–1988 (Filipino physician)*

**Sister Mary Joseph nodule** An umbilical metastatic nodule from an intra-abdominal malignancy (fig 15.17). *Sister Mary Joseph Dempsey, 1856–1939 (US catholic nun & Dr William Mayo's surgical assistant)*

**Sjögren's syndrome** A chronic inflammatory autoimmune disorder, which may be primary (♀:♂≈9:1, onset 4th–5th decade) or secondary, associated with connective tissue disease (eg RA, SLE, systemic sclerosis). There is lymphocytic infiltration and fibrosis of exocrine glands, especially lacrimal and salivary glands. *Features:* ↓Tear production (dry eyes, keratoconjunctivitis sicca), ↓salivation (xerostomia—dry mouth, caries), parotid swelling. Other glands are affected causing vaginal dryness, dyspareunia, dry cough, and dysphagia. Systemic signs include polyarthritis/arthralgia, Raynaud's, lymphadenopathy, vasculitis, lung, liver, and kidney involvement, peripheral neuropathy, myositis, and fatigue. It is associated with other autoimmune diseases (eg thyroid disease, autoimmune hepatitis, PBC) and an ↑risk of non-Hodgkin's B-cell lymphoma. *Tests:* Schirmer's test measures conjunctival dryness (<5mm in 5min is +ve). Rose Bengal staining may show keratitis (use a slit-lamp). Anti-Ro (SSA; in 40%) & anti-La (SSB; in 26%) antibodies may be present (in pregnancy, these cross the placenta and cause fetal congenital heart block in 5%). ANA is usually +ve (74%); rheumatoid factor is +ve in 38%. There may be hypergammaglobulinaemia. Biopsy shows focal lymphocytic aggregation. *℞:* Treat sicca symptoms: eg hypromellose (artificial tears), frequent drinks, sugar-free pastilles/gum. NSAIDs and hydroxychloroquine are used for arthralgia. Immunosuppressants may be indicated in severe systemic disease.[64] *Henrik Conrad Samuel Sjögren, 1899–1986 (Swedish ophthalmologist)*

**Stevens-Johnson syndrome** A severe form of erythema multiforme (p562), and a variant of toxic epidermal necrolysis. It is caused by a hypersensitivity reaction, usually to drugs (eg salicylates, sulfonamides, penicillin, barbiturates, carbamazepine, phenytoin), but is also seen with infections or cancer. There is ulceration of the skin and mucosal surfaces (see fig 15.18). Typical target lesions develop, often on the palms or soles with blistering in the centre. There may be a prodromal phase with fever, malaise, arthralgia, myalgia ± vomiting and diarrhoea. *℞:* Mild disease is usually self-limiting—remove any precipitant and give supportive care (eg calamine lotion for the skin). Steroid use is controversial—trials have been variable, so ask a dermatologist and ophthalmologist. IV immunoglobulin has shown benefit. Plasmapheresis and immunosuppressive agents may have a role.[65] *Prognosis:* Mortality ~5%. May be severe for the first 10d before resolving over 30d. Damage to the eyes may persist and blindness can result. *Albert M Stevens 1884–1945; Frank C Johnson, 1894–1934 (US paediatricians)*

ABC

**Sturge-Weber syndrome (SWS)** Essential features: **1** Facial cutaneous capillary malformation (port wine stain; PWS) in the ophthalmic dermatome (V1 ± V2/V3). **2** Clinical signs or radiologic evidence of a leptomeningeal vascular malformation. 75% of patients with unilateral involvement develop seizures by age 1yr (95% if bilateral)—due (in part) to the increased metabolic demand of a developing brain in the setting of vascular compromise. Early management of seizures is critical to minimize brain injury. Some patients have severe cognitive and neurologic deficits beyond simple seizure activity. Screen early for glaucoma (50%). EEG and MRI help establish early diagnosis and treatment in patients at risk for SWS. Treat the PWS early with pulsed dye laser. *William A Sturge, 1850–1919; Frederick P Weber, 1863–1962 (British physicians)*

### Causes of a long QT interval

Many conditions and drugs (check *BNF*) cause a long QT interval. Brugada syndrome (p695) is similar, predisposing to sudden cardiac death.

*Congenital:* Romano-Ward syndrome (autosomal dominant). Jervell and Lange-Nielsen syndrome (autosomal recessive) with associated deafness (p702).

*Cardiac:* Myocardial infarction or ischaemia; mitral valve prolapse.

*HIV:* May be a direct effect of the virus or from protease inhibitors.

*Metabolic:* ↓$K^+$; ↓$Mg^{2+}$; ↓$Ca^{2+}$; starvation; hypothyroidism; hypothermia.

*Toxic:* Organophosphates.

*Anti-arrhythmic drugs:* Quinidine; amiodarone; procainamide; sotalol.

*Antimicrobials:* Erythromycin; levofloxacin; pentamidine; halofantrine.

*Antihistamines:* Terfenadine; astemizole.

*Motility drugs:* Domperidone.

*Psychoactive drugs:* Haloperidol; risperidone; tricyclics; SSRIs.

*Connective tissue diseases:* Anti-Ro/SSA antibodies (p552).

*Herbalism:* Ask about Chinese folk remedies (may contain unknown amounts of arsenic). Cocaine, quinine, and artemisinins (and other antimalarials) are examples of herbalism-derived products that can prolong the QT interval.

**Fig 15.17** Sister Mary Joseph nodule.

Reproduced from *Postgraduate Medical Journal,* 'Sister Joseph nodule', J E Clague, 78(917), 174, 2002 with permission from BMJ Publishing Group Ltd.

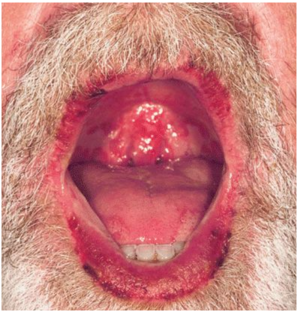

**Fig 15.18** Stevens–Johnson syndrome.

Reproduced from Emberger *et al*, Stevens–Johnson syndrome associated with anti-malarial prophylaxis. *Clinical Infectious Diseases* (2003) 37:1, with permission from Oxford University Press.

**Takayasu's arteritis** (*Aortic arch syndrome; pulseless disease.*) Rare outside of Japan, this systemic vasculitis affects the aorta and its major branches. Granulomatous inflammation causes stenosis, thrombosis, and aneurysms. It often affects women aged 20–40yrs. Symptoms depend on the arteries involved. The aortic arch is often affected, with cerebral, ophthalmological, and upper limb symptoms, eg dizziness, visual changes, weak arm pulses. Systemic features are common—eg fever, weight loss, and malaise. ↑BP is often a feature, due to renal artery stenosis. Complications include aortic valve regurgitation, aortic aneurysm and dissection; ischaemic stroke (↑BP and thrombus); and ischaemic heart disease. *Diagnosis:* ↑ESR and CRP; MRI/PET allows earlier diagnosis than standard angiography. *℞:* Prednisolone (1mg/kg/d PO). Methotrexate or cyclophosphamide have been used in resistant cases. BP control is essential to ↓risk of stroke. Angioplasty ± stenting, or bypass surgery is performed for critical stenosis. *Prognosis:* ~95% survival at 15 years.

*Mikito Takayasu, 1860–1938 (Japanese ophthalmologist)*

**Tietze's syndrome** (*Idiopathic costochondritis.*) Localized pain/tenderness at the costosternal junction, enhanced by motion, coughing, or sneezing. The 2nd rib is most often affected. The diagnostic key is *localized* tenderness which is marked (flinches on prodding). *Treatment:* Simple analgesia, eg NSAIDs. Its importance is that it is a benign cause of what at first seems to be alarming, eg cardiac pain. In lengthy illness, local steroid injections may be used. *Alexander Tietze, 1864–1927 (German surgeon)*

**Todd's palsy** Transient neurological deficit (paresis) after a seizure. There may be face, arm, or leg weakness, aphasia, or gaze palsy, lasting from ~30min–36h. The aetiology is unclear. *Robert Bentley Todd, 1809–1860 (Irish-born physician)*

**Vincent's angina** (*Necrotizing ulcerative gingivitis.*) Mouth infection with ulcerative gingivitis from *Borrelia vincentii* (a spirochaete) + fusiform bacilli, often affecting young ♂ smokers with poor oral hygiene. Try amoxicillin 500mg/8h and metronidazole 400mg/8h PO, + chlorhexidine mouthwash. *Jean Hyacinthe Vincent, 1862–1950 (French physician)*

**Von Hippel-Lindau syndrome** A dominant germline mutation of a tumour suppressor gene. It predisposes to bilateral renal cysts and clear cell renal carcinoma (p320), retinal & cerebellar haemangioblastoma, and phaeochromocytoma. See figs 15.19, 15.20. It may present with visual impairment or cerebellar signs (eg unilateral ataxia). *Eugen von Hippel, 1867–1939 (German ophthalmologist); Arvid Lindau, 1892–1958 (Swedish pathologist)*

**Von Willebrand's disease (VWD)** Von Willebrand's factor (VWF) has three roles in clotting: **1** To bring platelets into contact with exposed subendothelium. **2** To make platelets bind to each other. **3** To bind to factor VIII, protecting it from destruction in the circulation. There are >22 types of VWD; the commonest are:
*Type I:* (60–80%) ↓Levels of of VWF. Symptoms are mild. Autosomal dominant.
*Type II:* (20–30%) Abnormal VWF, with lack of high-molecular-weight multimers. Usually autosomal dominant inheritance. Bleeding tendency varies. There are 4 subtypes.
*Type III:* (1–5%) Undetectable VWF levels (autosomal recessive with gene deletions). VWF antigen is lacking and there is ↓factor VIII. Symptoms can be severe.
*Signs* are of a platelet-type disorder (p344): bruising, epistaxis, menorrhagia, ↑bleeding post-tooth extraction. *Tests:* ↑APTT, ↑bleeding time, ↓factor VIIIC (clotting activity), VWF ↓Ag; ↔INR and platelets. *℞:* Get expert help. Desmopressin is used in mild bleeding, VWF-containing factor VIII concentrate for surgery or major bleeds. Avoid NSAIDs. *Erik Adolf von Willebrand, 1870–1949 (Finnish physician)*

ABC

**Wallenberg's lateral medullary syndrome** This relatively common syndrome comprises lesions to multiple CNS nuclei, caused by posterior or inferior cerebellar artery occlusion leading to brainstem infarction (fig 15.21). *Features:* • Dysphagia, dysarthria (IX and X nuclei). • Vertigo, nausea, vomiting, nystagmus (vestibular nucleus). • Ipsilateral ataxia (inferior cerebellar peduncle). • Ipsilateral Horner's syndrome (descending sympathetic fibres). • Loss of pain and temperature sensation on the ipsilateral face (V nucleus) and contralateral limbs (spinothalamic tract). There is no limb weakness as the pyramidal tracts are unaffected.

In the rarer *medial medullary syndrome*, vertebral or anterior spinal artery occlusion causes ipsilateral tongue paralysis (XII nucleus) with contralateral limb weakness (pyramidal tract, sparing the face) and loss of position sense.

*Adolf Wallenberg, 1862–1949 (German neurologist)*

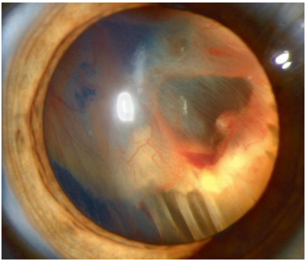

**Fig 15.19** Von Hippel–Lindau syndrome showing retinal detachment.



**Fig 15.20** Von Hippel–Lindau syndrome showing a retinal tumour.



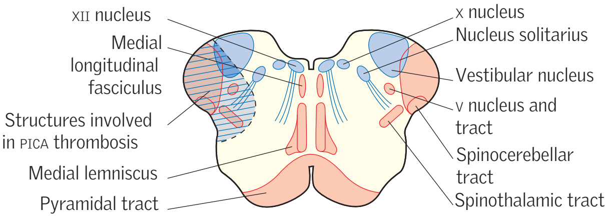


**Fig 15.21** Cross section of the medulla showing structures involved in Wallenberg's lateral medullary syndrome (posterior inferior cerebellar artery thrombosis).

**Waterhouse-Friderichsen's (WhF) syndrome** Bilateral adrenal cortex haemorrhage, often occurring in rapidly deteriorating meningococcal sepsis, alongside widespread purpura, meningitis, coma, and DIC (fig 15.22). The meningococcal endotoxin acts as a potent initiator of inflammatory and coagulation cascades. Other causes include *H. influenzae*, pneumococcal, streptococcal, and staphylococcal sepsis. Adrenal failure causes shock, as normal vascular tone requires cortisol to set activity of α- and β-adrenergic receptors, and aldosterone is needed to maintain extracellular fluid volume. *Treatment:* ►►Antibiotics, eg ceftriaxone (p822) and hydrocortisone 200mg/4h IV for adrenal support. ICU admission.

*Rupert Waterhouse, 1873–1958 (British physician); Carl Friderichsen, 1886–1979 (Danish paediatrician)*

**Weber's syndrome** (*Superior alternating hemiplegia*.) Ipsilateral oculomotor nerve palsy with contralateral hemiplegia, due to infarction of one-half of the midbrain, after occlusion of the paramedian branches of the basilar or posterior cerebral arteries. *Herman David Weber, 1823–1918 (German-born physician whose son described Sturge-Weber syndrome)*

**Wegener's granulomatosis** This has been renamed ***granulomatosis with polyangiitis (GPA)***, in part because of concerns over the suitability of Friedrich Wegener, a member of the Nazi party during WWII, to be the source of an eponym. GPA is a multisystem disorder of unknown cause characterized by necrotizing granulomatous inflammation and vasculitis of small and medium vessels. It has a predilection for the upper respiratory tract, lungs, and kidneys. *Features:* Upper airways disease is common, with nasal obstruction, ulcers, epistaxis, or destruction of the nasal septum causing a characteristic 'saddle-nose' deformity.[6] Sinusitis is often a feature. Renal disease causes rapidly progressive glomerulonephritis with crescent formation, proteinuria, or haematuria. Pulmonary involvement may cause cough, haemoptysis (severe if pulmonary haemorrhage), or pleuritis. There may also be skin purpura or nodules, peripheral neuropathy, mononeuritis multiplex, arthritis/arthralgia, or ocular involvement, eg keratitis, conjunctivitis, scleritis, episcleritis, uveitis. *Tests:* cANCA directed against PR3 is most specific and raised in the majority of patients (p553). Some patients express pANCA specific for MPO. ↑ESR/CRP. Urinalysis should be performed to look for proteinuria or haematuria. If these are present, consider a renal biopsy. CXR may show nodules ± fluffy infiltrates of pulmonary haemorrhage. CT may reveal diffuse alveolar haemorrhage. Atypical cells from cytology of sputum/BAL can be confused with bronchial carcinoma.[66] *Treatment:* Depends on the extent of disease. Severe disease (eg biopsy-proven renal disease) should be treated with corticosteroids and cyclophosphamide (or rituximab) to induce remission. Azathioprine and methotrexate are usually used as maintenance. Indications for plasma exchange include patients presenting with severe renal disease (eg creatinine >500µmol/L) and those with pulmonary haemorrhage. Co-trimoxazole should be given as prophylaxis against *Pneumocystis jirovecii* and staphylococcal colonization.

*Friedrich Wegener, 1907–1990 (German pathologist)*

**Wernicke's encephalopathy** Thiamine (vitamin $B_1$) deficiency with a classical triad of **1** confusion **2** ataxia and **3** ophthalmoplegia (nystagmus, lateral rectus, or conjugate gaze palsies). There is inadequate dietary intake, ↓GI absorption, and impaired utilization of thiamine resulting in focal areas of brain damage, including periaqueductal punctate haemorrhages (mechanism unclear). Always consider this diagnosis in alcoholics: it may also present with memory disturbance, hypotension, hypothermia, or reduced consciousness.[67] *Recognized causes:* Chronic alcoholism, eating disorders, malnutrition, prolonged vomiting, eg with chemotherapy, GI malignancy, or hyperemesis gravidarum. *Diagnosis:* Primarily clinical. Red cell transketolase activity is decreased (rarely done). *Treatment: Urgent* replacement to prevent irreversible Korsakoff's syndrome (p704). Give thiamine (Pabrinex®), 2 pairs of high-potency ampoules IV/IM/8h over 30min for 2d, then 1 pair OD for a further 5d. Oral supplementation (100mg OD) should continue until no longer 'at risk' (+ give other B vitamins). Anaphylaxis is rare. If there is coexisting hypoglycaemia (often the case in this group of patients), make sure thiamine is given before glucose, as Wernicke's can be precipitated by glucose administration to a thiamine-deficient patient. *Prognosis:* Untreated, death occurs in 20%, and Korsakoff's psychosis occurs in 85%—a quarter of whom will require long-term institutional care. *Karl Wernicke, 1848–1905 (German neurologist)*

ABC

Fig 15.22 Meningococcal sepsis with purpura.

ABC

6 Common causes of a 'saddle-nose' deformity are trauma and iatrogenic (eg post-rhinoplasty). Rarer causes (popular with some finals examiners): GPA, relapsing polychondritis, syphilis, and leprosy.

**Whipple's disease** A rare disease[68] featuring GI malabsorption which usually occurs in middle-aged white males, most commonly in Europe. It is fatal if untreated and is caused by *Tropheryma whippelii*, which, combined with defective cell-mediated immunity, produces a systemic disease. *Features:* Often starts insidiously with arthralgia (chronic, migratory, seronegative arthropathy affecting mainly peripheral joints). GI symptoms commonly include colicky abdominal pain, weight loss, steatorrhea/diarrhoea, which leads to malabsorption (p266). Systemic symptoms such as chronic cough, fever, sweats, lymphadenopathy, and skin hyperpigmentation also occur. Cardiac involvement may lead to endocarditis, which is typically blood culture negative. CNS features include a reversible dementia, ophthalmoplegia, and facial myoclonus (if all together, they are highly suggestive)—also hypothalamic syndrome (hyperphagia, polydipsia, insomnia). NB: CNS involvement may occur without GI involvement. *Tests:* Diagnosis requires a high level of clinical suspicion. Jejunal biopsy shows stunted villi. There is deposition of macrophages in the lamina propria-containing granules which stain positive for periodic acid-Schiff (PAS). Similar cells may be found in affected samples, eg CSF, cardiac valve tissue, lymph nodes, synovial fluid. The bacteria may be seen within macrophages on electron microscopy. PCR of bacterial RNA can be performed on serum or tissue. MRI may demonstrate CNS involvement. *Ɽ:* Should include antibiotics which cross the blood-brain barrier. Current recommendations: IV ceftriaxone (or penicillin+streptomycin) for 2wks then oral co-trimoxazole for 1 year. Shorter courses risk relapse. A rapid improvement in symptoms usually occurs.

*George Hoyt Whipple, 1878–1976 (US pathologist)*

**Zellweger syndrome** (*Cerebrohepatorenal syndrome*.) A rare recessive disorder characterized by absent peroxisomes (intracellular organelles required for many cellular activities including lipid metabolism). The syndrome has a similar molecular basis to infantile Refsum's syndrome, and although more severe, exhibits comparable biochemical abnormalities (p710). Clinical features include craniofacial abnormalities, severe hypotonia and mental retardation, glaucoma, cataracts, hepatomegaly, and renal cysts. A number of causative PEX gene mutations have been identified. Life expectancy is usually a few months only.

*Hans Zellweger, 1909–1990 (US paediatrician)*

**Zollinger-Ellison syndrome** This is the association of peptic ulcers with a gastrin-secreting adenoma (gastrinoma). Gastrin excites excessive gastric acid production, which may produce multiple ulcers in the duodenum and stomach. The adenoma is usually found in the pancreas, although it may arise in the stomach or duodenum. Most cases are sporadic; 20% are associated with multiple endocrine neoplasia, type 1 (MEN1, p223). 60% are malignant; metastases are found in local lymph nodes and the liver. *Symptoms:* Include abdominal pain and dyspepsia, from the ulcer(s), and chronic diarrhoea due to inactivation of pancreatic enzymes (also causes steatorrhoea) and damage to intestinal mucosa. *Incidence:* ~0.1% of patients with peptic ulcer disease. Suspect in those with multiple peptic ulcers, ulcers distal to the duodenum, or a family history of peptic ulcers (or of islet cell, pituitary, or parathyroid adenomas). *Tests:* (fig 15.23) ↑Fasting serum gastrin level (>1000pg/mL). Measure three fasting levels on different days. Hypochlorhydria (reduced acid production, eg in chronic atrophic gastritis) should be excluded as this also causes a raised gastrin level: gastric pH should be <2. The secretin stimulation test is useful in suspected cases with only mildly raised gastrin levels (100–1000pg/mL). The adenoma is often small and difficult to image; a combination of somatostatin receptor scintigraphy, endoscopic ultrasound, and CT is used to localize and stage the adenoma. OGD evaluates gastric/duodenal ulceration. *Ɽ:* High-dose proton pump inhibitors, eg omeprazole: start with 60mg/d and adjust according to response. Measuring intragastric pH helps determine the best dose (aim to keep pH at 2–7). All gastrinomas have malignant potential—and surgery is better sooner than later (with lymph node clearance generally recommended if >2cm in size). Surgery may be avoided in MEN1, as adenomas are often multiple, and metastatic disease is rare. If well-differentiated (G1 and G2) somatostatin analogues may be 1st-line and chemotherapy with streptozotocin (if available) + doxorubicin/*5-FU* is 2nd-line. In G3, etoposide + cisplatin is possible.[69] Selective embolization may be done for hepatic metastases. *Prognosis:* 5yr survival: 80% if single resectable lesion, ~20% with hepatic metastases. Screen all patients for MEN1.

*Robert M Zollinger, 1903–1992; Edwin H Ellison, 1918–1970 (US surgeons)*

ABC

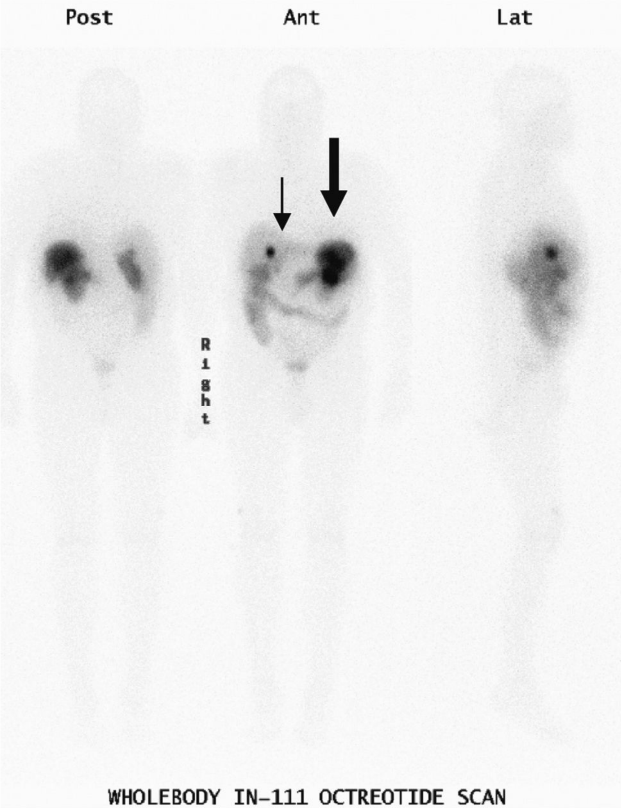


Fig 15.23 OctreoScan in patient with metastatic MEN 1 gastrinoma. Solitary hepatic metastatic deposit (thin arrow), gastric neuroendocrine tumour (thick arrow).

Reproduced from Wass *et al.*, *Oxford Textbook of Endocrinology and Diabetes*, 2011, with permission from Oxford University Press.

ABC

### Epilogue

25% of patients with rare diseases have to wait from 5–30 years for a diagnosis. 40% are misdiagnosed resulting in inappropriate drugs or psychological treatments—eg 20% of people with Ehlers-Danlos syndrome (p149) had to consult over 20 doctors before the diagnosis was made,[70] causing understandable loss of confidence in our profession. Lack of appropriate referral and rejection because of disease complexity are common problems. Let us cultivate our networks with each other and approach 'unexplained symptoms' with an open mind.

# 16 Radiology

**Contents**

*Principles in radiology:*




**Fig 16.1** The first X-ray, taken by Wilhelm Röntgen of his wife's hand (for which he won the first Nobel Prize in Physics in 1901). Upon seeing the ghostly image she cried 'I have seen my death!' Could this exclamation have prophesized the radiation-induced malignancies that have plagued the recipients (and administrators) of X-rays since their conception (see BOX)? Despite the obvious commercial potential, Röntgen declined to take out any patents on his new technology, preferring to see it developed for the benefit of humanity. Fate did not repay this generosity: his personal wealth depreciated away in the German hyperinflation of the 1920s, and he spent his later years in bankruptcy.




*We thank Professor Peter Scally, Dr Dean McCoombe, and Dr Paul Thomas, our Specialist Readers for this chapter.*

## Typical effective doses

The effective dose of an examination is calculated as the weighted sum of the doses to different body tissues. The weighting factor for each tissue depends on its sensitivity. The effective dose thus provides a single dose estimate related to the total radiation risk, no matter how the radiation dose is distributed around the body. This table is certainly not to be learnt; rather it serves as a reminder of the relative exposures to radiation that we prescribe in practice. ►Remember that US and MRI involve no radiation, would they provide the answer?

Table 16.1 Radiation doses in common radiological investigations

| Procedure | Typical effective dose (mSv) | CXR equivalents | Approx. equivalent period of background radiation |
|---|---|---|---|
| X-ray examinations | | | |
| Limbs and joints | <0.01 | <1 | <2 days |
| Chest (PA) | 0.015 | 1 | 2.5 days |
| Abdomen | 0.4 | 30 | 2 months |
| Lumbar spine | 0.6 | 40 | 3 months |
| CT head | 1.4 | 90 | 7.5 months |
| CT chest | 6.6 | 440 | 3 years |
| CT abdo/pelvis | 6.7 | 450 | 3 years |
| Radionuclide studies | | | |
| Lung ventilation | 0.4 | 30 | 9 weeks |
| Lung perfusion | 1 | 70 | 6 months |
| Bone | 3 | 200 | 1.4 years |
| PET head | 7 | 460 | 3.2 years |
| PET-CT | 18 | 1200 | 8.1 years |

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## Justifying exposure to ionizing radiation

The very nature of ionizing radiation that gives us vision into the human body also gives it lethal properties. In considering the decision to expose patients to radiation, the clinical benefits should outweigh the risks of genetic mutation and cancer induction. These risks can be hard to quantify (and estimates vary wildly—extrapolation of effects from doses associated with nuclear explosions are likely unreliable) but even with strict guidelines we still have a tendency to over-exposure in medical practice. Perhaps the best advice is to be certain of the importance of every dose of radiation that you sanction, and mindful of the comparative doses involved (see table 16.1).

The responsibility lies with us not to rely too heavily on radiology. ►Don't request examinations to comfort patients (or appease their consultants), to replace images already acquired elsewhere (or lost) simply to avoid medico-legal issues, or when the result will not affect management. To give an idea of relative doses, a CT of the abdomen and pelvis gives a typical effective dose of 500 times as much radiation as a CXR. This important factor also tells us about the preference of ultrasound over CT when investigating abdominal and pelvic complaints such as acute appendicitis, especially given the youthful demographics of this diagnosis.

►Unwitting exposure of the unborn fetus to radiation is inexcusable at any stage of gestation—unless the mother's life is in immediate danger—and it is the responsibility of the referring clinician, as well as the radiographer and the radiologist, to ensure that this is avoided.

# The art of the request

One of the most nerve-wracking moments that you can encounter as a recently qualified doctor is having to request an investigation from a seasoned consultant radiologist. What information do you need to give? How much? Who do you ask? Put yourself on the other side: what does the radiologist need to know to decide who needs what imaging and when? Keep the following in mind when requesting (never ordering!) an investigation:

**Patient details** Get the patient's name right! Include hospital number and date of birth on all requests.

**Clinical details** Think of your clinical question, what answer are you hoping radiology will provide? It can:

- *Confirm* a suspected diagnosis.
- *'Exclude'* something important (though remember that exclusion is never 100%).
- *Define* the extent of a disease.
- *Monitor* the progress of a disease.

► Don't forget to mention pertinent facts that may change the way the investigation is carried out: an agitated or confused patient may need sedation prior to an MRI of their head. A CT scan on a patient with an acutely raised creatinine may need to be done without contrast medium. Insertion of a drain on a patient with deranged clotting may need to wait while this is corrected. Include recent creatinine, Hb, and clotting on the form if appropriate. Don't forget to mention anticoagulants, eg warfarin, LMWH, and aspirin for intervention requests.

**Investigation details** What scan do you think is required, and how soon do you need it? Different clinical questions require different procedures. If you think the patient has a collection, would you like them to drain it? Always state whether intervention is required (eg US abdo ± drain insertion). Remember that the radiologist ultimately decides what imaging or procedure they undertake based on the information you have provided.

**Tips**

- Know your patient well, but keep your request brief and accurate.
- Know the clinical question and how the answer will change your management.
- Look up previous imaging before you go; asking for a CT on a patient who had one yesterday makes you look foolish and will not go down well.
- If in doubt, or if the investigation is very urgent, go down to the department in person. Regard this as an important opportunity: involving a radiologist will result in the best selection of imaging technique for your clinical question, will help expedite urgent requests, and should be of educational value for you.

**If your request is turned down** Don't be afraid to (politely) ask why. If you or your team still feel it is warranted, look back at the request; did you miss a relevant piece of information that would change the mind of the radiologist? If you still draw a blank, try speaking to a radiologist who specializes in that particular technique. Many teams have clinical radiology meetings; think about approaching someone who appreciates why you are asking that particular question. Alternatively, go back to your team; speak to your senior, who may have a better understanding of why the investigation is needed and be able to convey this to the radiologist.

►Remember that there is a patient at the heart of this, and you are their advocate. If the results of an investigation will change their management then explain this to the radiologist. Moreover, don't forget to explain it to your patient. Being whisked off to the department for investigation and intervention can be particularly terrifying if you aren't expecting it.

## Interpreting an image

You won't always be able to get an immediate radiologist's interpretation so it is important to know how to review an image. ►First make sure the image you are looking at is of your patient. Check its date. And remember:

- *Practice makes perfect*—always look at the image before checking the report, learning how to distinguish normal from abnormal.
- *Understand how the scan is done*—this makes interpretation easier and helps you appreciate which scan will give the answer you need. It also gives practical clues to the result—eg a routine CXR is performed in the postero-anterior (PA) direction (the source posterior to the patient to minimize the cardiac shadow).
- *Use a systematic approach*—so that you don't miss subtleties.
- *Understand your anatomy*—virtually all investigations yield a 2D image from a 3D structure. An understanding of anatomical relationships of the area in question will help reconstruct the images in your mind.
- *Orientation*—for axial cross-sectional imaging this is as if you are looking up at the supine patient ►*from the feet*. For images with non-conventional orientations (eg MRCP) look on the image for clue markings, or rely on your knowledge of anatomy—it can be tricky to visualize oblique sections!
- *Remember the patient*—an investigation is only one part of the clinical work-up, don't rely solely on the investigation result for your management decisions. Go back to see the patient after looking at the investigation and reading the radiologist's report: you might notice something that you didn't before.

## Presenting an image

Everyone has their own method for presenting, and the right way is *your own way*. As long as you cover everything systematically—because we all get 'hot-seat amnesia' at some point—the particulars will take care of themselves. Continue to polish your own method and remember a few extra tips for when an image is presented expectantly by your consultant/examiner and the floor is yours. A brief silence with a thoughtful expression as you analyse the image is fine, then:

- State the written details: name, date of birth, where and how the imaging was taken. Look for clues: weighting of an MRI, a '+ C' indicating that contrast medium has been used, the phase of the investigation (arterial/venous/portal), or even the name of the organ printed on an ultrasound.
- State the type, mode, and technical quality of investigation—not always easy!

*Going through this list also gives you a bit of thinking time.* Then:

- Start with life-threatening or very obvious abnormalities. *Then be systematic:*
- Is the patient's position adequate? Any lines, leads, or tubes? Note their position.
- Just like the bedside clues in a physical examination, there are clues in radiology examinations. Note oxygen masks, ECG leads, venous access, infusion apparatus, and invasive devices. Identifying what they are also helps you to look through what may otherwise appear to be a cluttered mess.
- Note any abnormalities and try to contextualize these with whatever you already know of the patient. The abnormality may be hiding in plain sight, but if struggling, step back and note any asymmetry or areas that just 'look different'.
- With cross-sectional imaging, scan through adjacent sections noting the anatomy of one organ system or structure at a time (this may mean going up and down through a CT abdomen multiple times).
- Giving a differential diagnosis is good practice, as not all findings are diagnostic.
- If there is additional clinical information that would help you to make a diagnosis, don't be afraid to ask. After all, we treat patients and not images!

*Remember:* •X-ray=*radiodensity* (lucency/opacity). •CT=*attenuation*. •US=*echogenicity*. •MRI=*signal intensity*.

## Chest x-ray—part 1



Images are usually taken on inspiration with the x-ray source behind the patient (postero-anterior, PA). Mobile images may be antero-posterior (AP, fig 16.6), magnifying heart size. If supine, distribution of air and fluid in lungs and pleural cavities is altered and the diaphragm is elevated.

Acclimatize yourself to the four cardinal elements of the chest radiograph, memorably (albeit slightly inaccurately) termed bone, air, fat, and 'water'/soft tissue. Each has its own radiographic density. ►*A border is only seen at an interface of two densities,* eg heart (soft tissue) and lung (air); this 'silhouette' is lost if air in the lung is replaced by consolidation ('water'). The silhouette sign localizes pathology (eg middle lobe pneumonia or collapse causing loss of clarity of the right heart border, fig 16.2). When interpreting a CXR use a systematic approach that works for you, eg:

**Technical quality**

- *Rotation:* The sternal ends of the clavicles should symmetrically overlie the transverse processes of the 4th or 5th thoracic vertebrae. A rotated image can alter the position of structures, eg rotation to the right projects the aortic arch vessels over the right upper zone, appearing as though there is a mass.
- *Inspiration:* There should be 5 to 7 ribs visible anteriorly (or 10 posteriorly). Hyperinflation can be abnormal, eg COPD. Poor inspiration can mimic cardiomegaly, as the heart is usually pulled down (hence elongated) with inspiration, and crowding of vessels at the lung bases can mimic consolidation or collapse. This is common in patients who are acutely unwell, particularly those in pain or unconscious. Take care in interpreting these images.
- *Exposure:* An under-exposed image will be too white and an over-exposed image will be too black. Both cause a loss of definition and quality although some compensation can be made with standard viewing software.
- *Position:* The entire lung margin should be visible.

**Trachea** Normally central or just to the right. Deviated by collapse (towards the lesion), expansion (away from the lesion), or patient rotation.

**Mediastinum** May be: *Widened* by mediastinal fat; retrosternal thyroid; aortic aneurysm/unfolding; lymph node enlargement (sarcoidosis, lymphoma, metastases, TB); tumour (thymoma, teratoma); cysts (bronchogenic, pericardial); paravertebral mass (TB). *Shifted* towards a collapsed lung or away from processes that add volume (eg a large mass or a tension pneumothorax).

There are three bulges normally visible on the left border of the mediastinum that help identify pathology if abnormal. From superior to inferior they are: **1** Aortic knuckle. **2** Pulmonary outflow tract. **3** Left ventricle.

**Hila** The left hilum is higher than the right or at the same level (not lower); they should be the same size and density. The hila may be: *Pulled up or down* by fibrosis or collapse. *Enlarged* by: pulmonary arterial hypertension; bronchogenic ca; lymph nodes. ►Sarcoidosis, TB, and lymphoma can give *bilateral* hilar lymphadenopathy. *Calcified* due to: sarcoid, past TB; silicosis; histoplasmosis (p408).

**Heart** Normally less than half of the width of the thorax (cardiothoracic ratio <0.5). ⅓ should lie to the right of the vertebral column, ⅔ to the left. It may appear elongated if the chest is hyperinflated (COPD); or enlarged if the image is AP or if there is LV failure (fig 16.3), or a pericardial effusion. Are there calcified valves?

**Diaphragm** The right side is often slightly higher (due to the liver). *Causes of raised hemidiaphragm:* Trouble above the diaphragm—lung volume loss or inflammation. Trouble with the diaphragm—stroke; phrenic nerve palsy (causes, p504; any mediastinal mass?). Trouble below the diaphragm—hepatomegaly; subphrenic abscess. NB: subpulmonic effusion (effusions having a similar contour to the diaphragm without a characteristic meniscus) and diaphragm rupture give apparent elevation. NB: bilateral palsies (polio, muscular dystrophy) cause hypoxia.

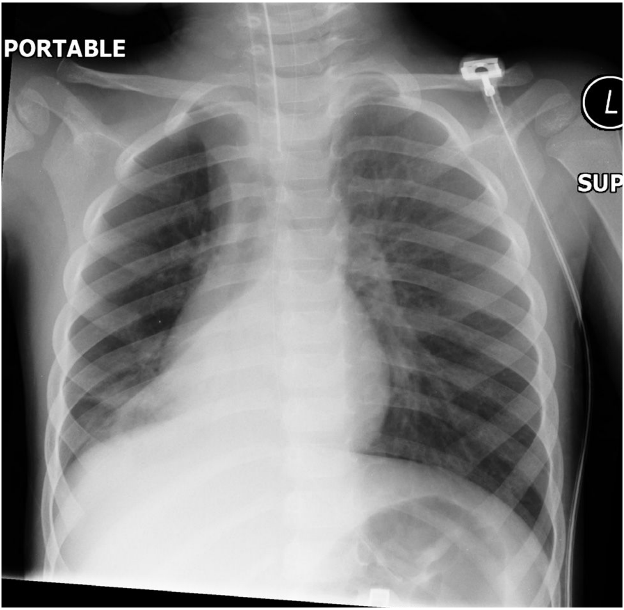



Fig 16.2 Lower lobe collapse (right lung). The right heart border is obscured. Volume loss in the right lower zone results in a hyper-expanded right upper lobe that is more radiolucent than the left upper lobe.

Courtesy of Dr Edmund Godfrey.

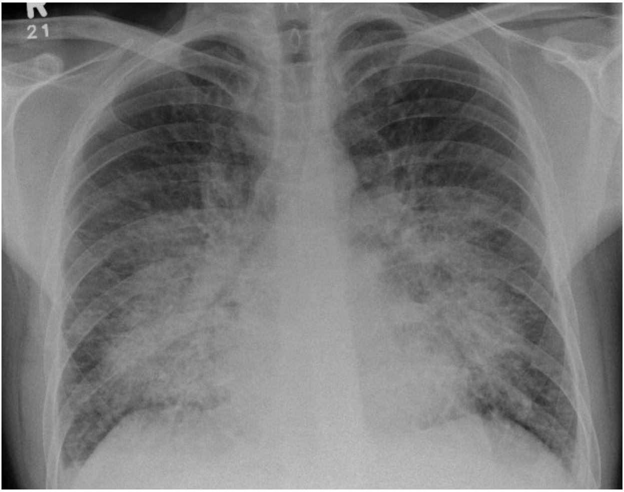



Fig 16.3 'Bat's wing', peri-hilar pulmonary oedema indicating heart failure and fluid overload.

Courtesy of Dr Edmund Godfrey.

## Chest x-ray—part 2: the lungs

The apex of the lower lobe rises up to the 4th rib posteriorly, so it is difficult to ascribe the true location of a lobe on a PA image without additional information from a lateral view. It may therefore be better to use the term 'zone' rather than lobe when localizing a lesion.

**Opacification** Lung opacities are described as nodular, reticular (network of fine lines, interstitial), or alveolar (fluffy). A single nodule may be called a space-occupying lesion (SOL).

*Nodules:* (If >3cm across, the term pulmonary mass is used instead.)
- Neoplasia: metastases (often missed if small), lung cancer, hamartoma, adenoma.
- Infections: varicella pneumonia, septic emboli, abscess (eg as an SOL), hydatid.
- Granulomas: miliary TB, sarcoidosis (see GPA, p714), histoplasmosis.
- Pneumoconioses (except asbestosis), Caplan's syndrome (p696).

*Reticular opacification:* =Lung parenchymal changes.
- Acute interstitial oedema.
- Infection: acute (viral, bacterial), chronic (TB, histoplasmosis).
- Fibrosis: usual interstitial pneumonia (UIP), non-specific interstitial pneumonia (NSIP), drugs (eg methotrexate, bleomycin, crack cocaine), connective tissue disorders (rheumatoid arthritis—p546, GPA—p714, SLE, PAN, systemic sclerosis—p552, sarcoidosis), industrial lung diseases (silicosis, asbestosis).
- Malignancy (lymphangitis carcinomatosa).

*Alveolar opacification:* = Airspace opacification, can be due to any material filling the alveoli:
- Pus—pneumonia.
- Blood—haemorrhage, DIC (p352).
- Water—heart, renal, or liver failure (p302, p274), ARDS (p186), smoke inhalation (p847), drugs (heroin), $O_2$ toxicity, near drowning (*OHCS* p768).
- Cells—lymphoma, adenocarcinoma.
- Protein—alveolar proteinosis, ARDS, fat emboli (~7d post fracture).

*'Ring' opacities:* Either airways seen end-on (bronchitis; bronchiectasis) or cavitating lesions, eg abscess (bacterial, fungal, amoebic), tumour, or pulmonary infarct (wedge-shaped with a pleural base).

*Linear opacities:* Septal lines (Kerley B lines, ie interlobular lymphatics seen with fluid, tumour, or dusts); atelectasis; pleural plaques (asbestos exposure).

*White-out of whole hemithorax:* (fig 16.4) Pneumonia, large pleural effusion, ARDS, post-pneumonectomy.

**Gas outside the lungs** Check for a pneumothorax (hard to spot if apical or in a supine image, can you see vascular markings right out to the periphery?), surgical emphysema (trauma, iatrogenic), and gas under the diaphragm (surgery, perforated viscus, trauma). *Pneumomediastinum:* Air tracks along mediastinum, into the neck. Due to rupture of alveolar wall (eg asthma or pulmonary barotrauma) or bronchial or oesophageal trauma (can be iatrogenic, eg from endoscope). *Pneumopericardium:* Rare (usually iatrogenic).

**Bones** Check the *clavicles* for fracture, *ribs* for fractures and lesions (eg metastases), *vertebral column* for degenerative disease, collapse, or destruction, and *shoulders* for dislocation, fracture, and arthritis.

**An apparently normal CXR?** Check for tracheal compression, absent breast shadow (mastectomy), double left heart border (left lower lobe collapse, fig 16.5), fluid level behind the heart (hiatus hernia, achalasia), and paravertebral abscess (TB).

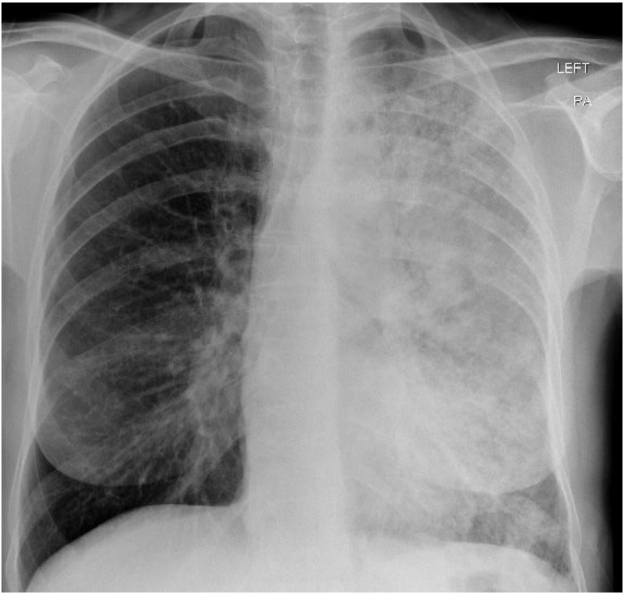



Fig 16.4 Opacification of the left hemithorax from consolidation.

Courtesy of Dr Edmund Godfrey.

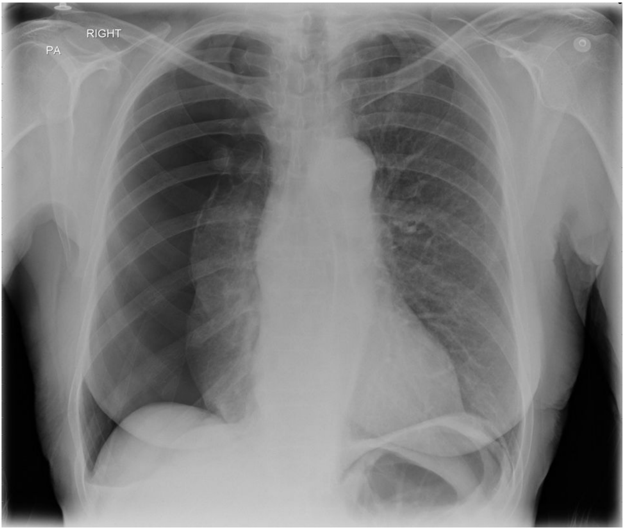



Fig 16.5 Large right-sided pneumothorax; note the trachea remains central, suggesting this is a simple pneumothorax, not a tension pneumothorax.

Courtesy of Dr Edmund Godfrey.

# Chest X-ray—part 3

Confirming the position of various tubes, lines, and leads on a CXR can be a daunting task, as incorrect positioning can have deadly consequences: an NG tube which is misplaced can cause aspiration pneumonia, or a poorly positioned CVC can lead to fatal arrhythmias. However, this can be a straightforward task if you recall some basic anatomy (figs 16.6, 16.7; table 16.2).

►If you are unsure, always ask a senior.

Table 16.2 Radiological confirmation of device placement

| Line/tube/lead | Correct position for tip(s) |
|---|---|
| CVC (p774) | In the SVC or brachiocephalic vein |
| PICC | In the SVC or brachiocephalic vein |
| Tunnelled line, eg Hickman | At the junction of the SVC and right atrium |
| Endotracheal | 3–7 cm above the carina (in adult) |
| Nasogastric (p759) | 10cm beyond the gastro-oesophageal junction |
| Chest drain (p766) | In the pleural space tracking either up (for pneumothorax) or down (for effusion) |
| Cardiac pacemaker/temporary pacing wire (p776) | Atrial lead—in the right appendage<br>Ventricular lead—in the apex of the right ventricle |

## Normal anatomy

- The SVC begins at the right 1st anterior intercostal space.
- The right atrium lies at the level of the 3rd intercostal space.
- The carina should be visible at the level of T5–T7 thoracic vertebrae.
- The right atrial appendage sits at the level of the 3rd intercostal space.

## Common bleeps from nursing staff

**1** *Central line not aspirating:*
- Is the tip in the right place (see earlier in topic) or has it gone up into the internal jugular, too far in (sitting against the tricuspid valve, does the patient have an arrhythmia?) or not far enough in (sitting against a venous valve)?
- Is the tip kinked, suggesting it may be in a side vessel or against the vessel wall?
- If the line looks appropriately positioned, consider flushing gently, could the line be blocked?

**2** *Patient not ventilating well:*
- Is the ET tube down the right main bronchus (causes left lung collapse, or rarely right pneumothorax)? Retract tube to correct position (see earlier in topic).
- Is the ET tube blocked? Most have a secondary port allowing ventilation even if the main hole is blocked, get anaesthetic assistance!

**3** *Chest drain not bubbling/swinging:*
- Is it correctly positioned (see earlier in topic)—if not in the pleural space, it cannot drain the air/fluid. Common problems include sitting in the soft tissue of the chest wall, or sitting above the effusion, below the pneumothorax, or in the oblique fissure.
- Is it blocked? If draining an effusion and correctly positioned, consider gently flushing with 10mL of sterile saline, then aspirating. If not successful, obtain senior advice.
- Has the effusion/pneumothorax resolved? Pneumothoraces can rapidly resolve with a correctly positioned drain.

**4** *Unable to aspirate from NG tube:*
- Is NG tube not far enough in/coiled in oesophagus? Tip is radio-opaque and should be visible below the diaphragm, if it is coiled it may lie in the pharynx or anywhere along the mediastinum.
- Is NG tube passing down the trachea and into the bronchus? The oesophagus is (generally speaking) a straight vertical line, if the tube veers off to left or right before it goes below the diaphragm, assume it is in the bronchus and replace it.

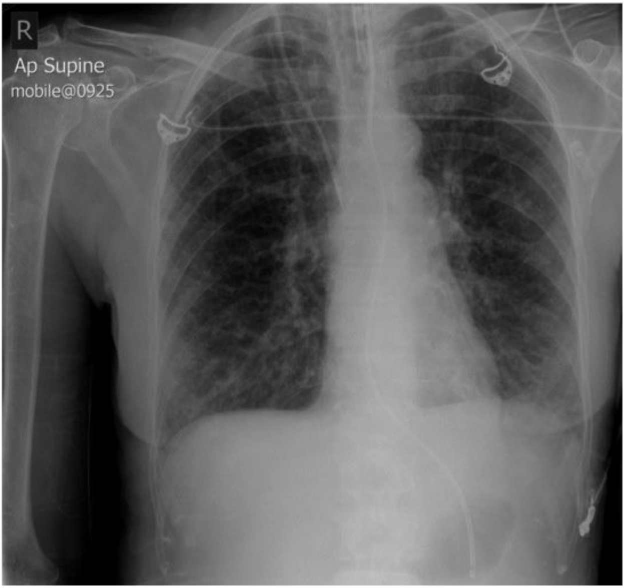



Fig 16.6 Image from ICU showing ET tube, CVC, and NG tube *in situ* with ECG leads placed across the chest. Image courtesy of Dr Elen Thomson, Leeds Teaching Hospitals.

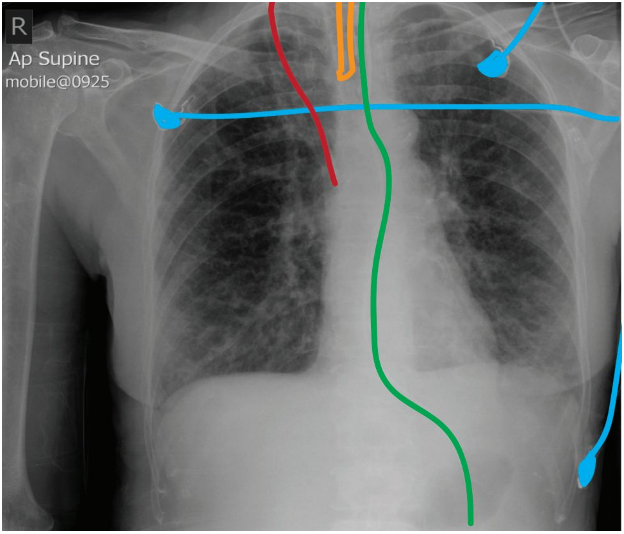



Fig 16.7 Knowing where lines and tubes should be placed is an essential skill. An ET tube (orange) should sit 3–7cm above the carina; this one is slightly high. The tip of the CVC (red, here a right internal jugular line) should lie in the SVC, as seen here, or just in the right atrium. The tip of the NG tube (green) must be seen below the diaphragm to ensure it is placed in the oesophagus, not the trachea. Do not confuse external leads (blue) with internal lines.

# Plain abdominal x-ray

These are rarely diagnostic and involve a radiation dose equivalent to 50 CXRs. Indications for AXR with acute abdominal symptoms:

- Suspicion of obstruction (or intussusception, eg in paediatrics).
- Acute flare of inflammatory bowel disease (eg to confirm/exclude megacolon).
- Renal colic with known renal stones (if first presentation, CT KUB is better).
- Ingestion of a sharp or poisonous foreign body (eg lithium battery).

Bowel gas pattern is best assessed on supine images and free intraperitoneal gas (signifying perforation) is best seen on an erect CXR (fig 13.26, p607).

**Gas patterns** Look for: an abnormal quantity of gas in the stomach, small intestine, or colon. Decide whether you are looking at small or large bowel (fig 16.8; table 16.3).

Small bowel diameter is normally ~2.5cm, the colon ~5cm, the caecum up to 10cm. Dilated small bowel is seen in obstruction and paralytic ileus. Dilated large bowel (≳6cm) is seen in both these, and also in 'toxic dilatation', and, in the elderly, in benign hypotonicity. Grossly dilated segments of bowel (coffee bean sign) are seen in sigmoid and caecal volvulae—fig 13.28c, p611. Loss of normal mucosal folds and bowel wall thickening are seen in inflammatory colitis (eg IBD)—fig 16.9. 'Thumb-printing' is protrusion of thickened mural folds into the lumen, seen in large bowel ischaemia and colitis.

Table 16.3 Radiological gas patterns in the bowel

| Small bowel | Large bowel | Ileus |
|---|---|---|
| • Smaller calibre | • Larger calibre | • Both small and large bowel visible |
| • Central; multiple loops | • Peripheral | • There is no clear transition point that corresponds to an obstructing lesion |
| • *Valvuli conniventes:* folds that go from wall to wall, all the way across the lumen; more regular and finer than haustra | • *Semi-lunar folds:** don't go all the way across the lumen, but may appear to do so if viewed from an angle | |
| • Grey (contains air and fluid) | • Blacker (contains gas)† | |

*Semi-lunar folds (plicae semilunares) lie in between adjacent haustra.
†The ascending colon contains liquid faeces, but the descending colon contains faecal pellets (*scybala*).

**Gas outside the lumen** You must explain any gas outside the lumen of the gut. It could be: **1** Pneumoperitoneum; signs on the supine AXR include: gas on both sides of the bowel wall (Rigler's sign), a triangle of gas in the RUQ trapped beneath the falciform ligament, and a circle of gas beneath the anterior abdominal wall. Seen with bowel perforation but also after laparoscopic surgery. **2** Gas in the urinary tract—eg in the bladder from a fistula. **3** Gas in the biliary tree (see next paragraph), or rarely **4** Intramural gas, found in bowel necrosis.

**Biliary tree** *Any stones:* ~10% visible on plain AXR. *Any gas:* (Pneumobilia.) Caused by: •post-ERCP/sphincterotomy •post-surgery (eg Whipple's) •recent stone passage •anaerobic cholangitis (rare) •gallbladder–bowel fistula: gallstone migrates directly into the bowel (Rigler's triad is seen in 25%: pneumobilia, small bowel obstruction, an ectopic gallstone). *Calcification:* ('Porcelain gallbladder'.) Chronic inflammation from gallstones (associated with gallbladder cancer).

**Urinary tract** Check for calculi (visible in 90% of cases)[1] and normal anatomy: *Kidneys:* Length equivalent to 2½–3½ vertebral bodies, slope inferolaterally. Right is lower than the left ('pushed down' by the liver). Their outline can usually be seen due to surrounding layer of perinephric fat. *Ureters:* Pass near the tips of the lumbar transverse processes, cross the sacroiliac joints, down to the ischial spines, and turn medially to join the bladder.

**Other soft tissues** Look for size/position of: liver, spleen, and bladder. A big liver will push bowel to the left side of the abdomen. An enlarged spleen displaces bowel and stomach bubble to the right. A big bladder elevates these.

**Medical devices** Double-J and biliary stents, nephrostomy and gastrostomy tubes, intrauterine devices, laparoscopic clips, and peritoneal dialysis catheters can be seen.

**Bones and joints** Plain AXR is not ideal, but there may be important abnormalities. In the lumbar spine, look for scoliosis and degeneration (osteophytes, joint space narrowing) as well as bone metastases or sacroiliitis.

1 Don't get confused by other calcifications—eg phleboliths: harmless calcifications found in the perivesical veins (rounded with a radiolucent centre).

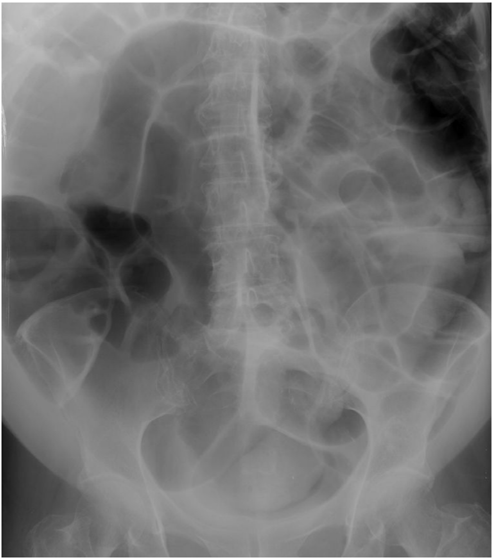


Fig 16.8 Multiple dilated air-filled loops of large and small bowel. This pattern is seen in ileus. Courtesy of Norwich Radiology Department.

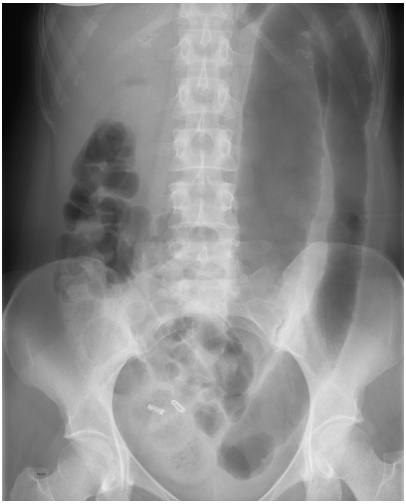


Fig 16.9 Abdominal image showing toxic megacolon associated with ulcerative colitis, note colon wall thickening and loss of mucosal folds.
Courtesy of Dr Edmund Godfrey.

# Computed tomography (CT)

Can give whole-body images in under one breath (thanks to continuous, helical data acquisition). Within a single slice (eg 0.5 or 5mm thick), CT records the attenuation (=loss of energy from, eg absorption or reflection) of different tissues to ionizing radiation and calculates a mean value for a given volume of tissue (a 'voxel'). This value is represented in greyscale as a single point, called a pixel, in the final 2D image (or 3D 'reconstruction'—fig 16.13). The greyscale of the pixel is measured on the Hounsfield scale (see fig 16.10) relative to the attenuation of water, 0 Hounsfield units (HU), and air, −1000HU. The human eye and display systems have a limited greyscale range, so different settings (levels and 'windows') are used to focus on differences in attenuation in ranges typical for tissues of different density, eg bone or lung (fig 16.11).

▶CTs are responsible for up to 40% of iatrogenic radiation in high-use settings, which could account for ~1% of all cancers: always balance benefits of CT vs other modalities with less or no radiation (ultrasound; MRI), particularly in the young or in those with chronic disease likely to undergo multiple imaging investigations. Discuss with a radiologist—there are several technical aspects of imaging that can limit radiation dose whilst still providing clinically useful information.

## Imaging of choice for

- Staging and monitoring most malignant disease.
- Intracranial pathology, eg stroke, trauma, ↑ICP, and space-occupying lesions.
- Trauma.
- Pre-operative assessment of complex masses.
- Assessment of acute abdomen (figs 16.15, 16.16). NB ultrasound increasingly used (p736).
- Following abdominal surgery.

**Contrast medium** (p748) Enhance anatomical detail by use of a high- or low (water)-attenuating medium to fill the lumen of a structure. *Give IV* to image vascular anatomy (fig 16.12) and vascular structures (including highly perfused tumours). Images acquired at different times ('phases') after injection will show the agent in arterial or venous structures or during 'washout' (=clearing). *Perfusion CT* maps cerebral blood flow by acquiring serial images after contrast administration then combines these into a colour-coded image of perfusion times (fig 16.14). Ensure IV cannulae secure and sufficient gauge to allow for rapid injection of agent as bolus—extravasation of contrast can cause significant tissue damage. *Give PO* eg 1–12h before imaging bowel. *Give PR* for examining distal colonic lumen.

Contrast-enhanced CTs may include a pre-contrast series. Unenhanced imaging alone reduces radiation exposure and may be adequate for images of the brain, spine, lung, and musculoskeletal system or necessary in those with renal failure (contrast is nephrotoxic).

**Streak artefact** Remember that the CT slice image is a matrix representation of the attenuation produced by rotating around the patient. High-attenuation items such as metal fillings, clips, and prostheses (and even bone) can cause interference.

**CT combined with PET** (See p739.) Combines the anatomical detail of CT with the metabolic information of PET, to aid assessment of, eg neoplastic lesions. Radiation doses are much higher than CT alone.

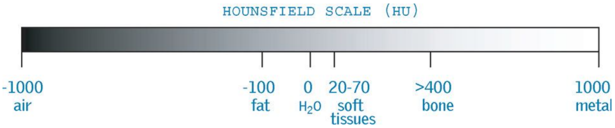



Fig 16.10 The Hounsfield scale.

Courtesy of Dr T Turmezei.

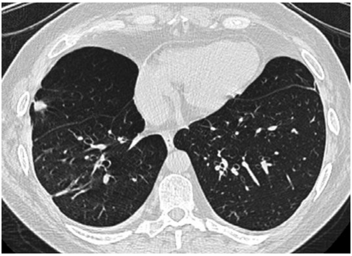


Fig 16.11 Axial high-resolution CT chest on a lung window algorithm; note solitary lesion in the right lung (in this case, from GPA).

Courtesy of Norwich Radiology Dept.

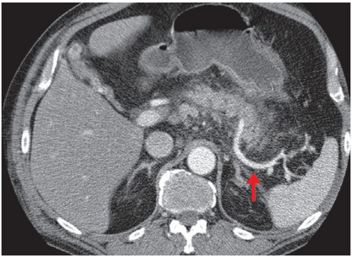


Fig 16.12 Axial CT of the abdomen after IV contrast (arterial phase). The tortuous splenic artery is enhanced (arrow)—so is the aorta, but not the inferior vena cava (compare to water in the stomach).

Courtesy of Norwich Radiology Dept.

Fig 16.13 Surface rendered 3D CT reconstruction of the pelvis. The posterior aspect of the right acetabulum is fractured. The right femur has been digitally removed for better viewing.

Courtesy of Norwich Radiology Dept.

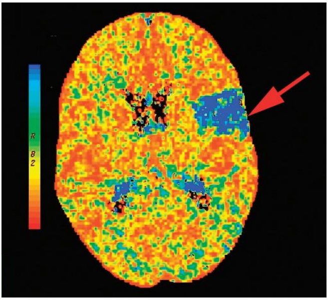


Fig 16.14 Cerebral perfusion CT showing ischaemia around the Sylvian fissure (arrow).

Courtesy of Dr C Cousens.

# CT images of two patients with acute abdominal pain

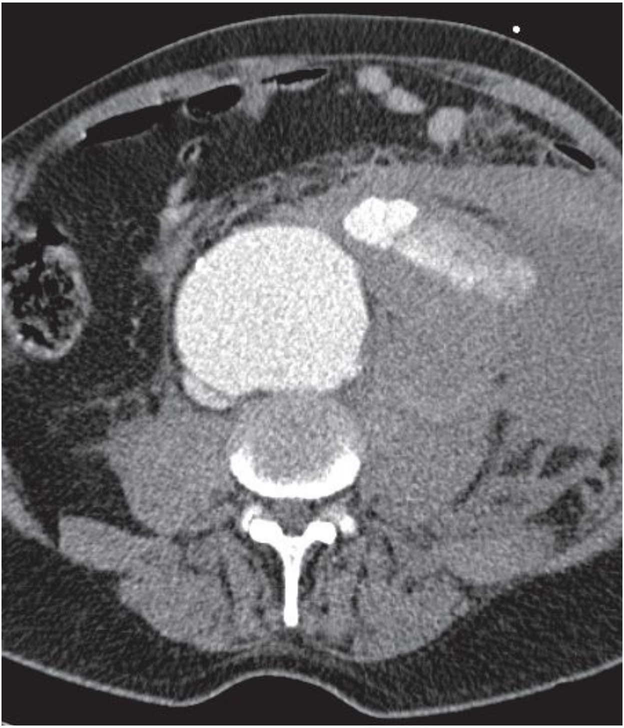


Fig 16.15 The history was of central abdominal pain with a non-peritonitic abdomen. The CT shows a leaking AAA. Under fluoroscopic screening, this can be repaired using stents, inserted via femoral arterial puncture and deployed in the aneurysm (p654). This kind of endovascular aneurysm repair (EVAR) is commonly used in the treatment of leaking AAA as well as elective repair of intact but enlarging aneurysms.



## The changing roles of surgeons and CT in the acutely unwell

In the days when general surgeons did their rounds towards the end of an on-call day, there would be wards of patients with undiagnosed abdominal pain having 'drip-and-suck' regimens (IVI and NGT) while awaiting improvement or a change in their clinical condition that revealed the need for surgery. On opening up, the surgeon would try to deal with whatever pathology was found. With increased subspecialization and accurate emergency imaging (CT and US), patients are now matched to a team best equipped to deal with their condition. In this context, drip-and-suck is on the ebb, giving way to imaging, early intervention, rapid discharge, or onward referral. With increasing pressures to safeguard surgical beds for elective cases and on junior surgeons to polish their surgical logbooks in decreased training hours, can come attempts to deflect away from surgical teams the care of patients in whom imaging or clinical circumstances suggest no current requirement for an operation. But is this always appropriate? Do we expect on-call surgeons to be practitioners of medicine, assessing and managing patients with surgical pathology, even if a trip to the operating theatre is not currently called for, or simply technicians restricted to cutting?

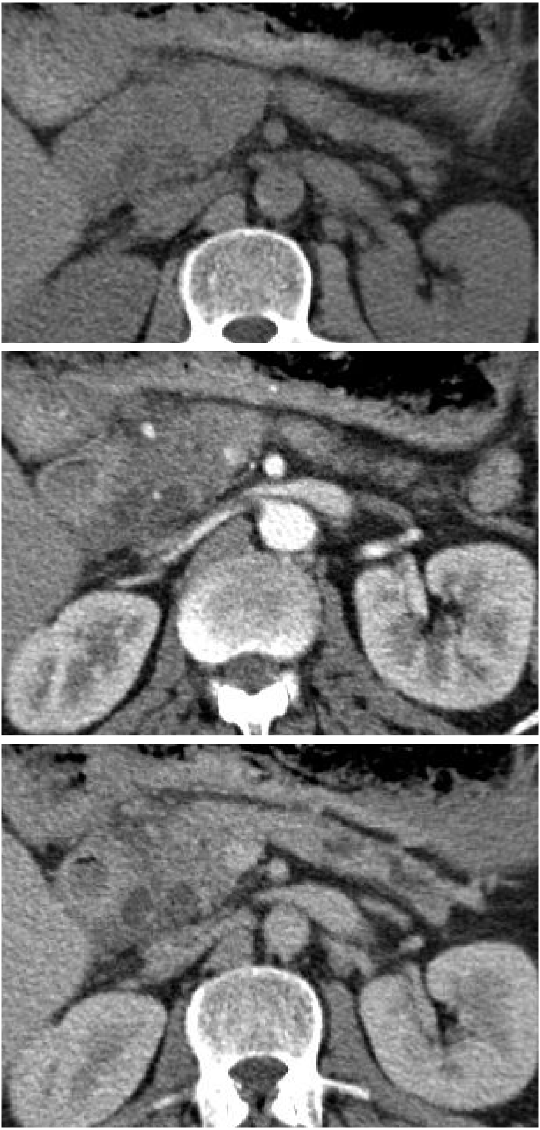


**Fig 16.16** Triple-phase CT abdomen, cropped to show the pancreas. Top panel—unenhanced image, middle panel—arterial phase of contrast medium to look for pseudocysts and parenchymal enhancement, bottom panel—portal venous phase to look at veins. The history here was also central abdominal pain with a non-peritonitic abdomen. The CT shows an enlarged pancreatic head with fat stranding around the duodenopancreatic groove ('groove pancreatitis'), and two small areas of fluid attenuation posteriorly, likely to be pseudocysts.

Top panel courtesy of Dr Edmund Godfrey.

# Magnetic resonance imaging (MRI)

1 A large proportion of the human body is fat or water (~80%).
2 Fat and water contain a large number of hydrogen nuclei (unpaired protons).
3 The spin of a positively charged hydrogen nucleus gives it magnetic polarity.

**Thus...**

- Placing the human body in a magnetic field aligns its hydrogen nuclei either with (parallel) or against (anti-parallel) the field.
- A radiofrequency (RF) pulse at the resonant frequency flips a few nuclei away from their original alignment by an angle depending on the amount of energy they absorb.
- When the RF pulse stops, the nuclei flip back (or *relax*) into their original alignment, emitting the energy (called an *echo*) that was absorbed from the RF pulse.
- Measuring and plotting the energy of the returning signal according to location (provided the nuclei haven't moved) gives a picture of fat, tissue, and water as distributed throughout the body.
- The hydrogen nuclei in flowing blood move after receiving the RF pulses. The echo is not detected, and so the vessel lumen appears black (flow void).

Rather than radiodensity or attenuation, the correct descriptive terminology for the greyscale seen in MRI is signal intensity: high signal appears white and low signal black (see table 16.4). Weighting is a quality of MRI that is dependent on the time between the RF pulses (repetition time, TR) and the time between an RF pulse and the echo (echo time, TE). MR images are most commonly T1-weighted (good for visualizing anatomy) or T2-weighted (good for visualizing disease) but can also be a mixture of both, called proton density (PD) weighting. FLAIR sequences produce heavily T2-weighted images. A good way to determine the weighting of an MR image is to look for water—eg in the aqueous humour of the eye, CSF, or synovial fluid (see table 16.4; fig 16.17).

Table 16.4 MRI sequence characteristics

| | T1-weighted | T2-weighted |
|---|---|---|
| **TR**<br>**TE** | Short (<1000ms)<br>Short (<30ms) | Long (>2000ms)<br>Long (>80ms) |
| **Low signal** ■ | Water<br>Flowing Hb<br>Fresh Hb<br>Haemosiderin | Bone<br>Flowing Hb<br>DeoxyHb<br>Haemosiderin<br>Melanin |
| **High signal** □ | Bone marrow<br>Fat<br>Cholesterol<br>Gadolinium (p762)<br>MetHb | Water<br>Cholesterol<br>Fresh Hb<br>MetHb |

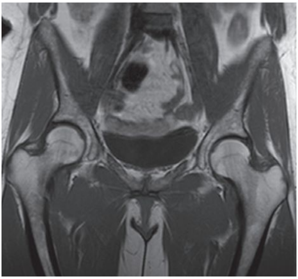


Fig 16.17 T1-weighted MRI of the hips. Normal adult bone marrow is high signal due to fat; note also low signal from urine in the bladder.

Courtesy of Norwich Radiology Department

**Advantages** MRI's great bonus is that it does not involve ionizing radiation. It has no known long-term adverse effects. It is *excellent for imaging soft tissues* (water- and hence proton-dense) and is preferred over CT for musculoskeletal disorders and for many intracranial, head, and neck pathologies (figs 16.18–16.20). Multiplanar acquisition of images can provide multiple views and 3D reconstruction from one scan. MR angiography is also excellent for reconstructing vascular anatomy. This avoids the need for invasive angiography with femoral puncture or CT contrast in patients with renal impairment.

**Disadvantages** Long acquisition times. Poor imaging of lung parenchyma. Claustrophobic. Incompatible with some metal implants. High cost and specialized interpretation (=limited availability).

**Contraindications** *Absolute:* •Pacemakers; other implanted electrical devices. •Metallic foreign bodies, eg intra-ocular (consider orbital X-ray to exclude), shrapnel. •Non-compatible surgical clips/coils/heart valves. *Relative:* •If unable to complete the pre-scan questionnaire. •Cochlear implants. NB: orthopaedic prostheses and extracranial metallic clips are generally safe. ►If uncertain, ask a radiologist. *Contrast:* •Renal impairment (gadolinium can cause systemic fibrosis). •Allergy. •Pregnancy.

Radiology

Fig 16.18 T2-weighted sagittal MRI of the cervical spine. There is impingement of the spinal cord at the C4/5 and C5/6 levels caused by degenerative disease. C2 (axis) is identifiable from the odontoid peg, which is embryologically derived from the body of C1 (atlas).

Courtesy of Norwich Radiology Department.

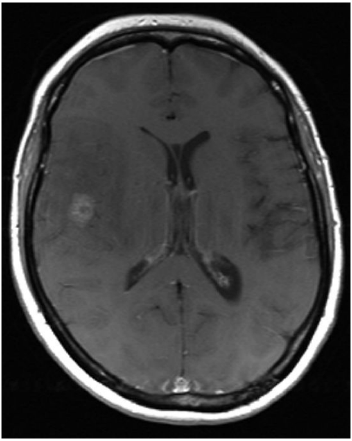


Fig 16.19 Axial T1-weighted MRI of the brain post-IV gadolinium. In the right temporo-parietal region there is a small area of high signal enhancement with a more central area of low signal, surrounded by a region of low signal (presumably vasogenic cerebral oedema) in comparison to the normal brain tissue. This is all causing mass effect with effacement of the sulci and adjacent right frontal horn of the lateral ventricle. There is very subtle midline shift.

Courtesy of Norwich Radiology Department.

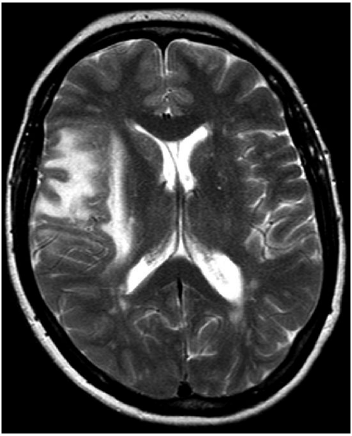


Fig 16.20 Axial T2-weighted MRI of the same patient at the same level as fig 16.19. The high signal in the temporo-parietal region is the oedema causing mass effect. The diagnosis was of a solitary metastasis. In this T2-weighted image the oedema and the cerebrospinal fluid are of high signal due to their water content.

Courtesy of Norwich Radiology Department.

# Ultrasound (US)



Unlike the other methods of imaging, US doesn't use electromagnetic radiation. Instead, it relies on properties of longitudinal sound waves. This has made it a popular and safe form of imaging with increasingly widespread applications. High-frequency sound waves (3–15MHz) are generated in the transducer (transmitter and receiver) by the vibrations of a piezo-electric quartz crystal as a voltage is applied. Passage of sound waves through tissue is affected by *attenuation* and *reflection*. Attenuation disperses waves out of the receiver's range, but it is the waves reflected back to the transducer that determine the image. Its quality depends on the difference in *acoustic impedance* between adjacent soft tissues.

**Processing** With the help of software a real-time 2D image is made. During processing, an average attenuation value is assumed throughout the tissue examined, so if a higher-than-average attenuation structure is in the superficial tissues (eg fibrous tissue, calcification, or gas), then everything deep to it will be in a low intensity (black) acoustic shadow. If a lower-than-average attenuation object (eg fluid-filled/cystic structure) is in the superficial tissues then everything deep to it will be high intensity (white) or enhanced. If a tissue interface is strongly disparate, eg gas in the intestine, then all the waves are reflected back, making it impossible to image beyond it. See also figs 16.21–16.24.

**Modes** *B:* (Brightness) is the most common, giving 2D slices that map the different magnitudes of echo in greyscale. *M:* (Movement) traces the movement of structures within the line of the sound beam. It is used in imaging, eg heart valves (p110).

**Duplex ultrasonography (flow and morphology)** By combining Doppler effects (shifts in wavelength caused by movement of a source or reflecting surface) with B-mode ultrasound technology, flow characteristics of blood can be inferred (fig 16.21). This is extremely useful in arterial and venous studies, and echocardiography.

**Advantages** Portable; fast; non-ionizing; cheap; real-time; can be used with intervention; can enter organs, eg rectum, vagina, bowel. *Endoscopic US* can be used to stage and biopsy lung and GI tract cancers, eg stomach, pancreas, and also image the heart = transoesophageal echocardiogram or TOE, p110.

**Disadvantages** Operator dependent—interoperator variability high; poor quality if patient is obese; interference from bone, bowel gas, calculi, or superimposed organs can limit depth and quality of imaging.

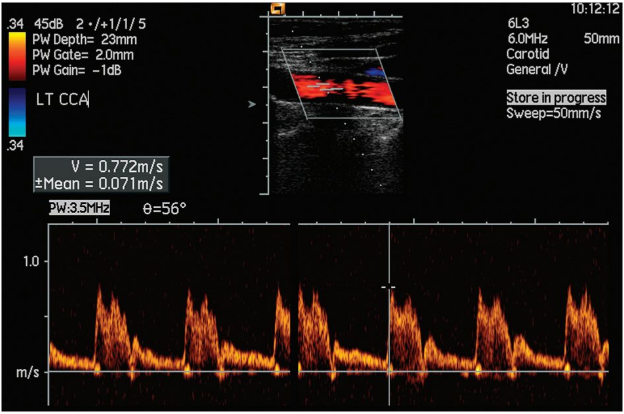



Fig 16.21 A normal Duplex US of the right common carotid artery with a flow rate=77cm/s. The Doppler trace (orange) is displayed below the main image.

Courtesy of Norwich Radiology Department.

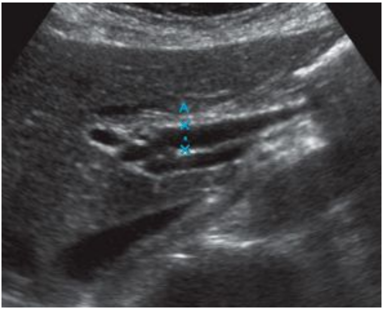


Fig 16.22 Ultrasound of the liver shows the common bile duct (CBD) to be dilated. Distal obstruction of the CBD causes proximal dilatation of the duct. It is important to correlate the width of the CBD with the ALP, as the normal diameter varies with age and previous interventions. Also check that the distal CBD tapers as it enters the duodenum. NB: the portal vein lies posterior to the duct (along with the hepatic artery) in the free edge of the lesser omentum. Next, ask 'What is causing the obstruction?' and 'Where can I get that information?'

Courtesy of Norwich Radiology Department.

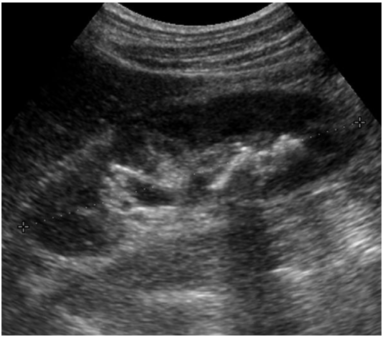


Fig 16.23 Ultrasound of the kidney. At first the image may seem normal but there is a wedge of posterior acoustic shadow cast by the object which is causing increased echogenicity in the lower pole calyces. Acoustic shadows in the kidney suggest stones—as here—or nephrocalcinosis.

Courtesy of Norwich Radiology Department.

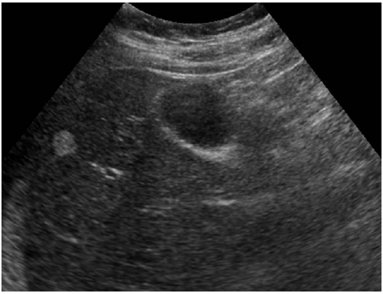


Fig 16.24 Longitudinal ultrasound of the right lobe of the liver showing a well-defined small area of echogenicity. This is the typical appearance of a liver haemangioma, a common benign liver lesion.

Courtesy of Norwich Radiology Department.

# Nuclear medicine



The majority of medical imaging is concerned with passing external waves (eg radiation) through the patient to a detector, and measuring scatter, slowing, or other alterations by various tissues. Nuclear medicine is the opposite; it measures emitted radiation from an internal source, introduced into the patient via injection, inhalation, or ingestion. It can be diagnostic (eg PET scanning) or therapeutic (eg radioiodine ($I^{131}$) ablation in thyrotoxicosis).

Because molecules labelled with radioisotopes are introduced into the patient, there is exposure to ionizing radiation, though doses are usually less than those from CT (see table 16.1, p719). The selection of molecule for labelling depends on the tissue of interest, as it should be something that will be readily taken up by that tissue, eg bisphosphonates for bone, glucose for fast-turnover tissue. Examples include:

**Ventilation/perfusion (VQ) scan** Uses inhaled technetium (Tc) or, less commonly, xenon-133 ($^{133}$Xe) plus injected $^{99m}$Tc macro-aggregates, which lodge in lung capillaries. Normal perfusion excludes PE but ventilation component requires a normal CXR for comparison (figs 16.25, 16.26). Due to the large number of 'indeterminate' scans VQ is considered inferior to CTPA for the investigation of PE except in pregnancy (see p192).

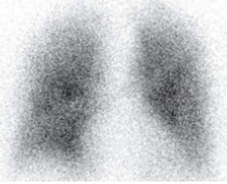


Fig 16.25 Ventilation scintigram. Courtesy of Norwich Radiology Department.

Fig 16.26 Perfusion scintigram showing mismatches with fig 16.25. Courtesy of Norwich Radiology Department.

Fig 16.27 Bone scintigram showing metastases. Courtesy of Norwich Radiology Department.

**Bone scintigraphy** $^{99m}$Tc-labelled bisphosphonates are readily taken up by bone, and concentrate in areas of pathology, eg tumours, fractures. It is much more sensitive in identifying metastases than X-ray, where lesions may not appear until >50% of bone matrix has been destroyed (fig 16.27).

**Thyroid disease** $TcO_4$ is used for differentiating Graves', toxic multinodular goitre, and subacute thyroiditis (fig 13.22, p601) as well as identifying ectopic tissue, functioning nodules, and residual/recurrent thyroid tissue after surgery. ~15% of cold (non-functioning) nodules are malignant. Hot nodules are often toxic adenomas.

**Phaeochromocytoma** Iodine-123 ($^{123}$I) meta-iodobenzylguanidine (MIBG) is taken up by sympathetic tissues, and indicates functioning, ectopic, and metastatic adrenal medullary (+other neural crest) tumours. $^{131}$I-MIBG is also used for treatment.

**Hyperparathyroidism** $^{99m}$Tc-methoxyisobutyl isonitrile (MIBI) scans can detect parathyroid adenomas.

**Haemorrhage** Red cells are removed from the patient and labelled with $^{99m}$Tc, then re-injected to allow localization of a bleeding point. Used in both acute (after endoscopy and CT) and chronic GI bleeding, red cell scans are more sensitive than CT angiography and useful in intermittent bleeding, although localization can be challenging.

**Renal function** Chromium-51 ($^{51}Cr$) EDTA or DTPA ($^{99m}Tc$, p669) is used to assess GFR. $^{99m}Tc$-mercapto-acetyltriglycine (MAG3) technique assesses relative (left-right) renal function and renal transit time (eg in renovascular disease). $^{99m}Tc$-dimercaptosuccinic acid (DMSA) scanning (fig 16.28) is the gold standard for evaluation of renal scarring that occurs, eg in reflux nephropathy.

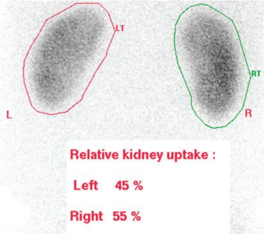



Fig 16.28 DMS showing relative renal function of each kidney.

Courtesy of Norwich Radiology Department.

**Positron emission tomography (PET)** One of the key investigations in malignancy, but also has a wide range of other uses. If the tracer chosen is $^{18}F$-fluorodeoxyglucose (FDG), a short half-life glucose analogue, it becomes concentrated in metabolically active tissues. FDG decays rapidly to produce a positron that, after travelling a few millimetres through tissue, annihilates with an electron to produce a pair of high-energy photons (gamma rays), which PET detects. Normal high uptake of FDG occurs in brain, liver, kidney, bladder, larynx, and lymphoid tissue of pharynx and must be considered when assessing images. *Neoplasms* have high uptake of FDG with hotspots suggesting primary disease or metastases. Since inflammatory lesions will also show high uptake, there is a risk of false-positive results (eg sarcoid, TB); diagnosis must be confirmed with histology of suspicious lesions. PET allows *staging* of many solid organ malignancies (lung, melanoma, oesophageal) as well as lymphomas, and is particularly useful for *planning* of radiotherapy and surgery for both primary disease and metastases. PET can also be used to image occult sources of infection. PET can be combined with CT or MRI to provide high-quality images combining anatomy with physiology. A range of alternative tracers are now entering clinical use with radiotracers conjugated to other tissue-specific substrates (eg $^{11}C$-labelled metomidate to detect tumors of adrenocortical origin, somatostatin tracers in neuroendocrine tumours, and amyloid tracers in Alzheimer's disease).

**Single photon emission computed tomography (SPECT)** Similar to PET but rather than using positron emission, it uses a radioisotope-labelled molecule as per conventional nuclear imaging, but with two gamma cameras for detection. The images produced are of lower resolution than PET but the isotopes used are longer lived and more easily available. Examples include myocardial perfusion scanning (p741).

**CT** *Cardiac CT:* Modern CT scanners can acquire images with sufficient speed and resolution to image coronary arteries and exclude significant disease with a negative predictive value of 97–99%. It can also visualize CABG patency, provide coronary artery $Ca^{2+}$ scoring (a risk factor for coronary artery disease, p117), demonstrate cardiac anatomy including congenital anomalies, and estimate ventricular function. *Vascular CT:* Has become routine in emergency assessment of suspected dissections, ruptured aneurysms, and arterial and venous thromboses (fig 16.29). CT angiography has overtaken invasive angiography in the assessment of many conditions such as stable angina and renal artery stenosis.

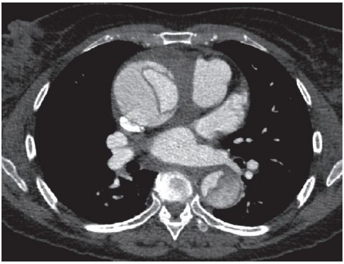


Fig 16.29 CT angiogram showing type A (ascending) aortic dissection with haemopericardium.

Courtesy of Dr C Cousins.

**Catheter angiography** Wherever intervention may be required, contrast studies such as angiography provide both image clarity and the possibility of proceeding to intervention, eg angioplasty or stenting of vessels, endovascular repair of aneurysms, clipping/coiling of aneurysms (see p746). Remember that these have a high burden of both radiation and contrast medium, so check renal function before requesting. *Complications* include those of arterial puncture (bleeding, infection, thrombosis, dissection, pseudoaneurysm formation) plus cholesterol emboli, thromboemboli, and vasospasm.

**MRI** *Cardiac MRI* using ECG-gating to acquire the imaging data and relate it to the position in the cardiac cycle (best when the patient is in sinus rhythm) can reduce movement artefact and lead to excellent resolution images for functional assessment. This, coupled with a lack of radiation, makes it ideal for the assessment of a wide range of structural and functional heart diseases. Flow velocities can be measured and, because the flow is proportional to the pressure differences, degrees of stenosis and regurgitation across heart valves can be calculated. Myocardial infarction, perfusion, and viability can also be imaged with the use of IV gadolinium contrast (p748). *Vascular MRI* is used to limit radiation exposure where multiple investigations may be required over a long time period, eg follow-up of intracranial aneurysm coiling, aortic root size in a young patient with Marfan's syndrome (p706), or Takayasu's arteritis (p712).

**Ultrasound** Non-invasive, relatively low cost, and with no radiation, US is excellent for assessing the heart and vasculature particularly in acute settings where the test can be performed at the bedside. *Cardiac US* (=echocardiography) evaluates myocardial and valvular anatomy and function (p110). The use of exercise or pharmacological agents for 'stress echocardiography' can permit more detailed functional assessment. *Vascular US* Doppler ultrasonography is widely used for detection of thrombotic disease (eg DVT p578, portal vein thrombosis p276) and carotid atherosclerosis (p472).

**Multiple gated acquisition** (MUGA) scanning is a non-invasive way to measure left ventricle ejection fraction. After injection of $^{99m}Tc$-labelled RBCs, a dynamic image of the left ventricle is obtained for a few hundred heartbeats by gamma camera. Since estimates of LVEF show less inter-operator variation than with echocardiograpy, uses include the detailed serial assessment of LVEF in patients undergoing cardiotoxic chemotherapy (eg anthracyclines, trastuzumab).

**Myocardial perfusion imaging** A non-invasive method of assessing regional myocardial blood flow and the cellular integrity of myocytes. The technique uses radionuclide tracers which cross the myocyte membrane and are trapped intracellularly. Thallium-201 ($^{201}$Tl), a $K^+$ analogue, is distributed via regional myocardial blood flow and requires cellular integrity for uptake. Newer technetium-99 ($^{99}$Tc)-based agents are similar to $^{201}$Tl but have improved imaging characteristics, and can be used to assess myocardial perfusion and LV performance in the same study (fig 16.30). Myocardial territories supplied by unobstructed coronary vessels have normal perfusion whereas regions supplied by stenosed coronary vessels have poorer relative perfusion, a difference that is accentuated by exercise. For this reason, exercise tests are used in conjunction with radionuclide imaging to identify areas at risk of ischaemia/infarction. Exercise scans are compared with resting views: *reversible* (ischaemia) or *fixed defects* (infarct) can be seen and the coronary artery involved reliably predicted. Drugs (eg adenosine, dobutamine, and dipyridamole) can also be used to induce perfusion differences between normal and underperfused tissues.

Myocardial perfusion imaging adds information in patients presenting with acute MI (to determine the amount of myocardium salvaged by thrombolysis) and in diagnosing acute chest pain in those without classical ECG changes (to define the presence of significant perfusion defects).

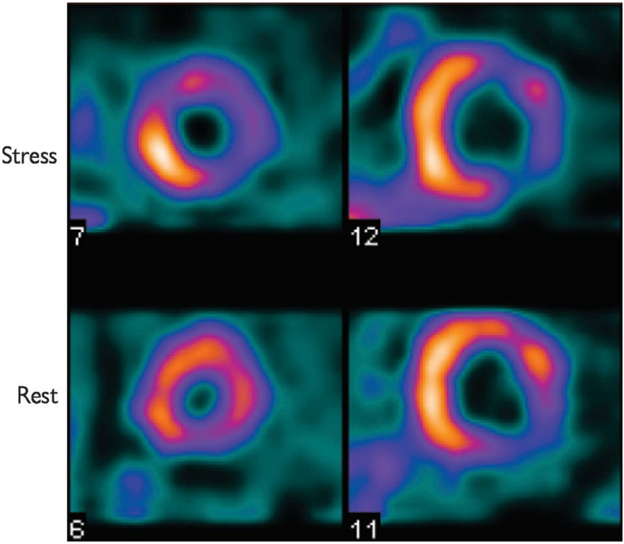



Fig 16.30 $^{99}$Tc perfusion study showing perfusion defect in the left ventricle anterior and lateral walls at stress which is partially reversible (difference between stress and rest images). This study is good for small vessel disease such as in diabetes; CT and coronary angiography do not show small vessel disease well.

Courtesy of Dr C Cousins.

**Ultrasound** Widely used for imaging all intra-abdominal organs, including an emerging role in small bowel imaging (though overlying bowel gas can cast acoustic shadows). US is the 1st-line imaging choice for abnormal LFTs, jaundice, hepatomegaly, renal dysfunction, and abdominal masses. Ensure the patient is 'nil by mouth' for 4 hours beforehand (aids gallbladder filling). Pelvic US needs a full bladder (consider clamping the catheter if appropriate). US may also guide diagnostic biopsy and therapeutic aspiration of cysts or collections.

**CT** plays an important role in the investigation of acute abdominal pain (see pp730–3). It is unparalleled in the detection of free gas and intra-abdominal collections, and allows good visualization of the colon and retroperitoneal areas. Oral or IV contrast medium enhances definition (p730). The big disadvantage is the radiation dose. *CT colonography* (*CTC*; fig 16.31) uses rectal air or $CO_2$ insufflation, usually coupled with an oral 'stool tagging' agent to visualize the colonic mucosa in those unfit for endoscopic evaluation or in whom endoscopic evaluation has failed (eg in a stenosing tumour, where it can be used to assess the proximal colon and allow assessment of liver and nodal metastases at the same time). A negative test can be regarded as definitive but if polyps or masses are seen then patients will usually require a colonoscopy.

*Wireless capsule endoscopy:* See p248.

**Magnetic resonance imaging (MRI)** This gives excellent soft tissue imaging, giving it an important role in imaging the liver, biliary system, pancreas, and pancreatic duct (MRCP—magnetic resonance cholangiopancreatography; fig 16.32). As well as assessing potential malignant disease, MRCP is the imaging modality of choice for detection of common bile duct stones that can be missed on US. MRI performed after fluid loading of the small bowel (fluid delivered orally=MRI enterography; fluid delivered via nasoduodenal tube=MRI enteroclysis) permits assessment of small bowel inflammation (eg Crohn's) and lesions that can be challenging to reach with conventional endoscopy.

**Endoscopic retrograde cholangiopancreatography** (**ERCP**; fig 16.33) *Indications:* No longer routinely used for diagnosis, it still has a significant therapeutic role: sphincterotomy for common bile duct stones; stenting of benign or malignant strictures and obtaining brushings to diagnose the nature of a stricture. *Method:* A catheter is advanced from a side-viewing duodenoscope via the ampulla into the common bile duct. Contrast medium is injected and images taken to show lesions in the biliary tree and pancreatic ducts. *Complications:* Pancreatitis; bleeding; cholangitis; perforation. Mortality <0.2% overall; 0.4% if performing stone removal.

**Endoscopic ultrasound** (**EUS**; see p736.) Commonly used in diagnosis of upper GI abnormalities, and is excellent for diagnosis of oesophageal, gastric, and pancreatic cancers. It allows staging by assessing depth of invasion, as well as histological diagnosis by biopsy of lesions.

**Contrast studies** (fig 16.34) These can help in dysphagia (p250) and assessing integrity of anastomoses post-op. Real-time fluoroscopic imaging studies assess swallowing function. Barium gives better contrast but iodine-based water-soluble contrast medium is used if there is a concern of perforation. *Contrast enemas* are increasingly obsolete and now used to exclude a leak following a low anterior resection, for proctograms, and not much else.

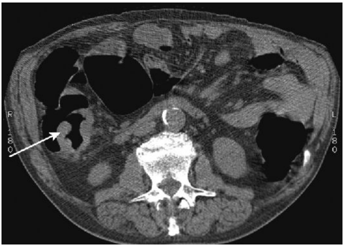


Fig 16.31 Axial CT colonogram: mural thickening (?ascending colon tumour).

Courtesy of Norwich Radiology Department.

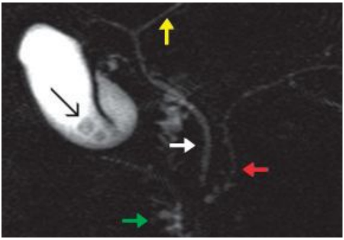


Fig 16.32 MRCP of the biliary system showing: left hepatic duct (yellow arrow); multiple gallstones in the gallbladder (black arrow); common bile duct (white arrow); pancreatic duct (red arrow); duodenum (green arrow).

Courtesy of Norwich Radiology Department.

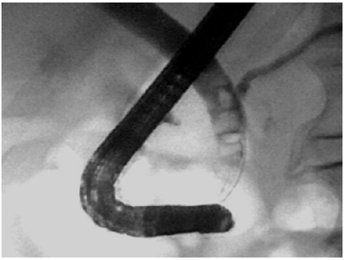


Fig 16.33 The ERCP shows a dilated common bile duct. The multiple filling defects are calculi within and obstructing the duct.

Courtesy of Norwich Radiology Department.

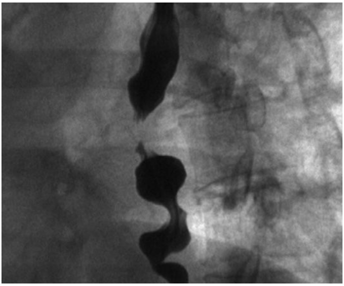


Fig 16.34 Barium swallow: note 'corkscrew' appearance of the oesophagus found in some motility disorders.

Courtesy of Norwich Radiology Department.

## Ultrasound

Imaging modality of choice for genitourinary problems. Can be used to assess:

*Kidneys:*

- Renal size—small in chronic kidney disease, large in renal masses, cysts, hypertrophy if other kidney missing, polycystic kidney disease (fig 16.35), and rarities (eg amyloidosis, p370).
- Hydronephrosis, which may indicate ureteric obstruction or reflux (fig 13.49, p641).
- Perinephric collections (trauma, post-biopsy).
- Renal perfusion (assessment of renovascular disease: Doppler US of renal arteries).
- Transplanted kidneys (collections, obstruction, perfusion).

*Lower urinary tract:*

- Bladder volume: useful in assessment of the need to catheterize (see p640) or for assessment of adequacy of bladder emptying (post-micturition residual volume).
- Prostate: transrectal ultrasound enables US-guided biopsy of focal lesions. NB: prostate size does not correlate with symptoms.

*Other:*

- Ovarian cysts, size, infections (pyosalpinx), uterine fibroids and other masses.
- Testicular masses, hydrocele, varicocele.

*Advantages:* Fast; cheap; independent of renal function; no IV contrast or radiation risk. *Disadvantages:* Intraluminal masses (transitional cell ca) in the upper tracts may not be seen; not a functional study; only suggests obstruction if there is dilatation of the collecting system (95% of obstructed kidneys) and so can miss obstruction from, eg retroperitoneal fibrosis.

**CT** (fig 16.36) First choice in renal colic. Performed without intravenous contrast so safe in renal impairment; such unenhanced images miss <2% of stones, but can show other pathologies. With IV contrast, CT can delineate masses (cystic or solid, contrast enhancement, calcification, local/distant extension, renal vein involvement); assess renal trauma (presence of two kidneys; haemorrhage; devascularization; laceration; urine leak); and show retroperitoneal lesions. CT has all but replaced intravenous urography and the radiation dose is similar.

**Plain abdominal x-ray** Can be used to look at the kidneys, the paths of the ureters, and bladder. However, in practice it is only useful for monitoring known renal calculi.

**Contrast studies** *Retrograde pyelography/ureterograms* are good at showing pelvi-calyceal, ureteric anatomy, and transitional cell carcinomas (TCCs). Contrast medium is injected via a ureteric catheter. With the advent of cystoscopy, allowing immediate intervention, these are rarely done in isolation. However, contrast medium is routinely used in cystoscopic placement of retrograde stents for obstruction.

*Percutaneous nephrostomy.* Used in obstruction to decompress the renal pelvis, which is punctured under local anaesthetic with imaging guidance. Images are obtained following contrast injection (antegrade pyelogram). A nephrostomy tube is then placed to allow decompression, sometimes followed by an antegrade stent if there is no easily treatable cause of obstruction.

**Renal arteriography** (fig 16.37) Therapeutic indications: angioplasty; stenting; embolization (bleeding tumour, trauma, AV malformation).

**Magnetic resonance imaging (MRI)** Soft tissue resolution can help clarify equivocal CT findings. Magnetic resonance angiography (MRA) helps image renal artery anatomy/stenosis (fig 16.38) and is also used in the assessment of potential live donors for kidney transplant, as well as to monitor patients following embolization of tumours, arteriovenous malformations, and aneurysms.

**Radionuclide imaging** See p738.

Fig 16.35 Ultrasound of the kidney showing multiple simple cysts.

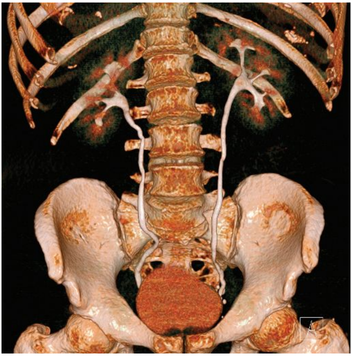


Fig 16.36 3D reconstruction of CT urogram showing normal appearances of both kidneys, ureters, and bladder.

Courtesy of Dr Edmund Godfrey.

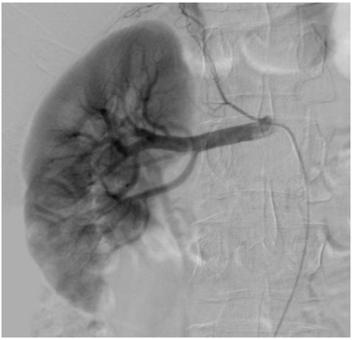


Fig 16.37 Renal artery digital subtraction angiogram (DSA; DSA is the final arbiter of renal artery stenosis). It is possible to tell that this is a DSA as no other structure has any definition or contrast in the image. There is, however, some interference from overlying bowel gas, which is not an uncommon problem. GI tract peristalsis can be diminished during the examination by using IV buscopan.


Fig 16.38 Coronal 3D MRA of the kidneys showing two renal arteries supplying the left kidney. This is important information pre-transplant. Anomalous renal arteries are common and, like the normal renal arteries, are end arteries, hence the consequence of infarction if tied at surgery.

**CT** (fig 16.39) Imaging modality of choice for patients presenting with acute neurological symptoms suggestive of a stroke. It is better than MRI at showing *acute haemorrhage* and *fractures*, and is much easier to do in ill or anaesthetized patients, and so is good in emergencies. The attenuation of biological soft tissues is in a narrow range from about +80 for blood and muscle, to 0 for CSF, and down to −100 for fat (Hounsfield units, p730). IV contrast medium initially gives an angiographic effect, whitening the vessels. Later, if there is a defect in the blood-brain barrier (eg tumours or infection), contrast medium will opacify a lesion's margins, giving enhancing white areas.

- Some CNS areas, eg pituitary gland, choroid plexus, have no blood-brain barrier and enhance normally.
- Fresh blood is of higher attenuation (ie whiter) than brain tissue.
- In old haematomas, Hb breaks down and loses attenuation, so a subacute subdural haematoma at 2wks may be of the same attenuation as adjacent brain.
- A chronic subdural haematoma will be of relatively low attenuation.

CT is often used in *acute stroke* to exclude haemorrhage (eg pre-antiplatelets) and with perfusion scanning (fig 16.14) to aid management decisions regarding thrombolysis. The actual area of infarction/ischaemia may not show up for a day or so, and will be low-attenuation cytotoxic oedema (affecting both white and grey matter—look for loss of grey matter definition).

*Tumours and abscesses* appear similar, eg a ring-enhancing mass, surrounding vasogenic oedema, and mass effect. Vasogenic oedema (from leaky capillaries) is extracellular and spreads through the white matter (grey matter spared). Mass effect causes compression of the sulci and ipsilateral ventricles, and may also cause herniation (subfalcine, transtentorial, or tonsillar). ►See p483 (and also fig 16.40).

Another indication for CT is acute, *severe headache*, eg suggestive of subarachnoid haemorrhage (p478). An unenhanced CT may show fresh blood, hydrocephalus or ↑ICP, any of which could make LP unsafe.

CT angiography gives excellent mapping of the cerebral circulation (fig 16.41), and can be done directly after unenhanced CT, looking for an aneurysm if the unenhanced CT shows subarachnoid haemorrhage.

**MRI** (MRI in stroke: fig 10.19, p481) The chief image sequences are:

- *T1-weighted images:* Give good anatomical detail to which the T2 image can be compared. Fat is brightest (↑signal intensity); other tissues are darker to varying degrees. Flowing blood is low signal. Gadolinium contrast (p748) usually results in an increase in signal intensity. See table 16.4.
- *T2-weighted images:* These provide the best detection of most lesions as they usually contain some oedema or fluid and therefore appear white (eg fig 16.20, p735). Fat and fluid appear brightest. Flowing blood is again low signal.

*Magnetic resonance angiography* maps carotid, vertebrobasilar, and cerebral arterial circulations (and sinuses, veins). Functional MRI can image local blood flow.

**Catheter angiography** (fig 16.42) Less commonly used since the advent of MRA and CT angiography and perfusion techniques, though it has the advantage of allowing immediate therapy—eg coil embolization of saccular aneurysms.

**Radionuclide imaging** (p738) PET is mostly used as a research tool in dementia, but perfusion scintigraphy scan be used in the assessment of Alzheimer's disease, other dementias, and localizing epileptogenic foci. SPECT to visualize uptake of $^{123}$I-FP-CIT (DaTSCAN™) can be used to assess reduced striatal dopaminergic transport in Parkinson's disease.

Fig 16.39 Unenhanced axial CT head: note the old infarct in the left middle cerebral artery territory.

Courtesy of Norwich Radiology Department.

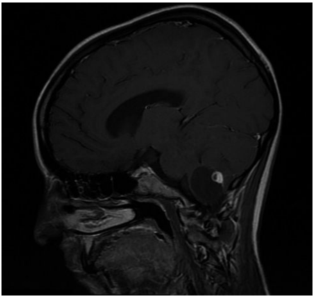


Fig 16.40 T1-weighted MRI of the brain showing a haemangioblastoma in a patient with Von Hippel-Lindau syndrome (p712). Note enhancement with contrast medium.

Courtesy of Dr Edmund Godfrey.

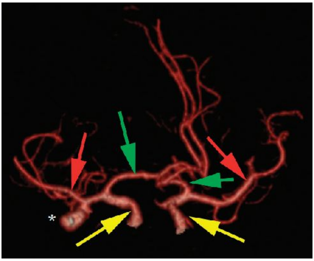




Fig 16.41 A 3D reconstruction of a CT angiogram of the paired internal carotid arteries (yellow arrows) and their branches (anterior cerebral arteries—green arrows, middle cerebral arteries—red arrows), seen from the front and slightly to the right. There is an aneurysm of the right middle cerebral artery (*).

Courtesy of Norwich Radiology Department.

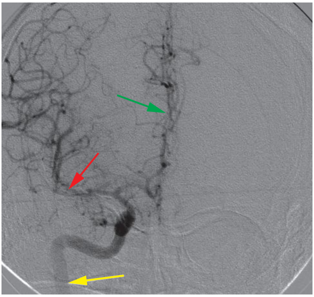


Fig 16.42 Digital subtraction angiogram (DSA). The right internal carotid artery (yellow arrow), anterior cerebral artery (green arrow), and middle cerebral artery (red arrow) are shown.

Courtesy of Norwich Radiology Department.

# Contrast media in imaging



The use of a contrast medium can alter the electron density of two previously similar tissues, thus allowing them to be distinguished. Contrast medium is usually administered by the following routes:

- *PO:* Barium- or iodine-based agents for, eg swallow or enhancing visualization of bowel lumen on CT.
- *Inhaled:* Technetium or xenon used in ventilation scintigraphy.
- *IV:* (Most widespread clinical application.) Iodine or gadolinium.
- *PR:* Air or $CO_2$ can be introduced to the colon for CT colonography, iodinated contrast medium is used for water-soluble enemas.

**Iodine-based contrast agents** Iodine is used because of its relatively high electron density and good physiological tolerance. When used with CT, the examination is said to be contrast enhanced—look for '+ c' amongst the scan details. *Exercise caution in:* renal or cardiac impairment; myeloma; diabetes; sickle cell disease; elderly; infants; the acutely unwell. ►Avoid iodine-based agents in active hyperthyroidism.

►Have renal function to hand in these patients (see p315). Minor reactions include nausea, vomiting, and a sensation of warmth. More severe reactions include urticaria, bronchospasm, angioedema, and low BP (1:250); theoretical risk of death for 1:150 000.

►Metformin should be withheld for 48h after IV contrast administration because of the risk of lactic acidosis.

**Barium sulfate** Used in examination of the GI tract. Water-insoluble particles of 0.6–1.4μm diameter are mixed with large organic molecules such as pectin and gum to promote good flow, mucosal adherence, and high density in thin layers. *Complications:* Chemical pneumonitis or peritonitis. Never administer if you suspect perforated viscus.

**Water-soluble, non-ionic, iodine-based contrast agents** Used instead of barium where there is a risk of peritoneal contamination (eg fistula, megacolon, ulceration, diverticulitis, bowel anastomosis, acute intestinal haemorrhage). Gastrograffin should not be used.

►Contains iodine so establish allergy history and thyroid status.

**Air** In CT colonography, air (or $CO_2$) is insufflated as a negative contrast medium after barium administration to enhance mucosal definition. Water can also be used PO and PR to outline the lumen of the gut.

**Gadolinium** A lanthanide series element with paramagnetic qualities that is administered intravenously (as gadolinium-DTPA) to enhance the contrast of certain structures in MRI. It works by reducing the time to relaxation (TR) of hydrogen nuclei in its proximity and appears as high signal on T1-weighted scans. It does not cross the blood-brain barrier so is useful in enhancing isointense extra-axial tumours such as meningiomas. It can also highlight areas where the blood-brain barrier has broken down secondary to inflammatory or neoplastic processes. It is renally excreted: ►check eGFR: if significantly reduced, gadolinium is contraindicated, as up to 30% may develop progressive nephrogenic systemic fibrosis/nephrogenic fibrosing dermopathy which causes generalized fibrosis which impairs movement and breathing—and which may be fatal. Aberrations in calcium-phosphate metabolism and erythropoietin treatment seem to increase risk. Other adverse reactions include headache, nausea, and local irritation at the site of injection, with idiosyncratic reaction reported in less than 1%.

## Imaging the acutely unwell patient

Asking yourself '*Does this investigation need to be done right now?*' will often yield the answer '*No!*', yet there are a few occasions when early imaging can provide vital diagnostic information and influence the prognosis for a patient:

- Acute cauda equina syndrome (p466): ►►MRI lumbar spine.
- Suspected thoracic aorta dissection (p655): ►►CT thorax + IV contrast, MRI or transoesophageal echo (TOE). The mediastinum is rarely widened on CXR.
- Suspected leaking abdominal aortic aneurysm (p654): ►►CT aorta.
- Acute kidney injury (p298): ►►US of renal tract to exclude obstruction.
- Acute pulmonary oedema: ►►portable CXR: don't delay to get an ideal film.
- Acute abdomen with signs of peritonism: ►►erect CXR to find intraperitoneal free gas (fig 13.26, p607; ≈GI perforation). Remember: post-op there will be detectable gas (air/$CO_2$) in the abdomen for ~10 days. ►►CT if suspicion of intra-abdominal source for sepsis or pathology requiring prompt surgery (eg appendicitis). US for ectopic pregnancy.
- Any patient with post-traumatic midline cervical spine tenderness—not just for the emergency department! ►►Collar and backboard immobilization followed by a CT. All the vertebrae down to the top of T1 must be visualized and cleared before it is safe to take the collar off.
- Sudden-onset focal neurology, worst-ever headache, deteriorating GCS: ►►CT head, then LP if no evidence of ↑ICP.

Remember that imaging—or re-imaging for a poor quality film—should never delay the definitive treatment of an emergency condition, eg:

- Tension pneumothorax (p814 and fig 16.43): ►►decompression *not* CXR.
- Intra-abdominal haemorrhage or viscus rupture (p606): ►►laparotomy.
- High clinical suspicion of torsion of testis (p652): ►►surgery *not* Doppler US.

Prior to the advent of interventional radiology, a collapsed, shocked patient with an acute abdomen would have skipped CT and gone straight for a laparotomy. However, you should bear in mind that ruptured aneurysms are increasingly being managed by endovascular repair under fluoroscopic guidance (p654) so this is one area where rapid imaging may be preferable to immediate intervention.

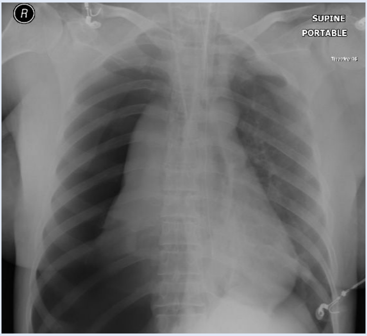



Fig 16.43 This is a great educational image from ICU. The inexperienced doctor could be distracted by the poor quality image, missing the lung bases: technicians do their best under difficult conditions. To ask for a new CXR here would be a mistake ►►note the large right-sided tension pneumothorax needing immediate decompression! *Lungs:* The right lung field is too black compared to the left, the right hemidiaphragm is depressed, and the right lung is seen collapsed against the mediastinum. *Mediastinum:* Left-shifted, obstructing venous return—so ↓cardiac output, and a threat to life. Is it being pushed or pulled? Check hila, bones, and soft tissues. Since the right lung is collapsed and mediastinum shifted to left this suggests ►►right tension pneumothorax. Needle thoracocentesis decompression and a chest drain are needed now.

Courtesy of Dr Edmund Godfrey.

# 17 Reference intervals, etc.

**Contents**



Fig 17.1 Carl Friedrich Gauss (1777–1855) was a German astronomer and mathematician who made many contributions to science, not least of which was establishing the normal (Gaussian) distribution (see p751). After his death, Gauss's brain was examined by anatomist Rudolph Wagner, to test the popular theory that intellectual ability correlated with physical properties of the brain such as weight and surface markings. While his brain was noted to display an unusually intricate pattern of sulci, lack of similarly patterned sulci among the brains of other intellectuals cast doubts on the theory. Further lack of evidence to support Wagner's hypothesis came when Gauss' brain was weighed: it would perhaps have amused Gauss to learn that the weight of his brain, whilst slightly above average, lay very much within the central region of a Gaussian distribution. After some rather sloppy housekeeping, Gauss' brain was stored in a mislabelled pot at the University in Göttingen, accidentally switched with that of his contemporary, the physician Conrad Fuchs. Over 150 years later, some careful searching of the archives and a MRI scanner revealed the switch.

# Statistics in medicine

- *The normal (Gaussian) distribution curve.* This bell-shaped graph (fig 17.2) is the theoretical basis of reference intervals, and explains 'lab error'—why some tests repeated at close intervals may reveal slightly different values. Hb for example, has a lab error of ~5g/L. This emphasizes the importance of the *clinical picture* in decision-making, rather than treating the numbers alone: don't subject anaemic patients to blood transfusions unless they have a clinical need. See p324.
- *Range* is the lowest and highest value of all observations in the set being studied.
- *Arithmetic mean* is the sum of all observations ÷ by the number of observations.
- *Median* is the middle value (eg 9 data points are higher and 9 are lower). If their distribution is Normal, then the median coincides with the mean.
- *Standard deviation (SD)* is the square root of the variance (the average of the square of the distance of each data point from the mean). When the distribution of the observations is Normal, 95% of observations are located in the interval 'mean ± 1.96 SD'. This is the basis of the reference interval.
- *Standard error of the mean* gives an estimate of the reliability of the mean of a sample representing the mean of the population from which the sample was taken, and is the SD of the sample ÷ by the square root of the number of observations in the sample. Thus the larger the sample size, the smaller the standard error of the mean—the basis of ensuring that clinical trial evidence is based upon enough observations to be confident that differences seen between groups do not occur by chance alone.

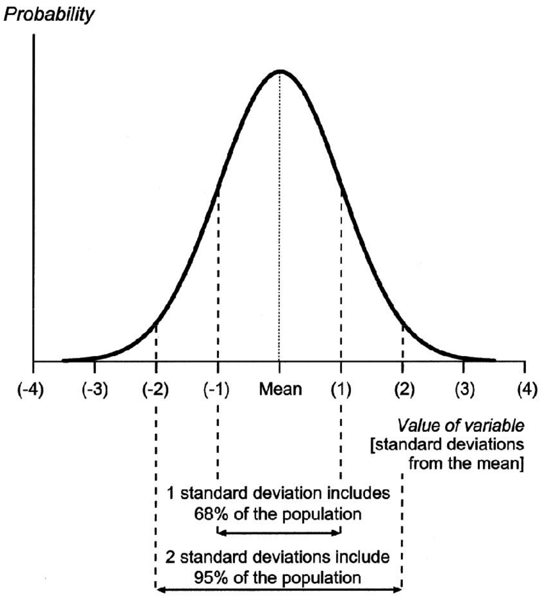



Fig 17.2 Normal (Gaussian) distribution curve.

Reproduced from Bhopal, *Concepts of Epidemiology*, 2008, with permission from Oxford University Press.

# Biochemistry reference intervals



See p662 for the *philosophy of the normal range*; see OHCS p220 for *children*.

Drugs (and other substances) may interfere with any chemical method; as these effects may be method dependent, it is difficult for the clinician to be aware of all the possibilities. If in doubt, discuss with the lab.

Table 17.1

| **Substance** | **Specimen** | **Reference interval** *(labs vary, so a guide only)* | **Your hospital** |
|---|---|---|---|
| Adrenocorticotrophic hormone | P | <80ng/L | |
| Alanine aminotransferase (ALT) | P | 5–35U/L | |
| Albumin[1] | P | 35–50g/L | |
| Aldosterone[2] | P | 100–500pmol/L | |
| Alkaline phosphatase[1] | P | 30–130U/L (adults) | |
| α-amylase | P | 0–180IU/dL | |
| α-fetoprotein | S | <10kU/L | |
| Angiotensin II[2] | P | 5–35pmol/L | |
| Antidiuretic hormone (ADH) | P | 0.9–4.6pmol/L | |
| Aspartate transaminase | P | 5–35U/L | |
| Bicarbonate[1] | P | 24–30mmol/L | |
| Bilirubin | P | 3–17µmol/L | |
| BNP (see p137) | P | <50ng/L | |
| C-reactive protein | P | <10mg/L | |
| Calcitonin | P | <0.1mcg/L | |
| Calcium (ionized) | P | 1.0–1.25mmol/L | |
| Calcium[1] (total)<br>See p676 to correct for albumin | P | 2.12–2.60mmol/L | |
| Chloride | P | 95–105mmol/L | |
| Cholesterol[3] (see p690) | P | <5.0mmol/L | |
| VLDL (see p690) | P | 0.128–0.645mmol/L | |
| LDL | P | <2.0mmol/L | |
| HDL | P | 0.9–1.93mmol/L | |
| Cortisol | P | AM 450–700nmol/L<br>Midnight 80–280nmol/L | |
| Creatine kinase (CK) | P | ♂ 25–195U/L<br>♀ 25–170U/L | |
| Creatinine[1] (proportional to lean body mass) | P | 70–100µmol/L | |
| Ferritin | P | 12–200mcg/L | |
| Folate | S | 2.1mcg/L | |
| Follicle-stimulating hormone (FSH) | P/S | 2–8U/L in ♀ (luteal);<br>>25U/L in menopause | |
| Gamma-glutamyl transpeptidase | P | ♂ 11–51U/L<br>♀ 7–33U/L | |
| Glucose (fasting) | P | 3.5–5.5mmol/L | |
| Growth hormone | P | <20mu/L | |
| HbA1c = glycosylated Hb (DCCT) | B | 4–6%. 7% ≈ good DM control | |
| HbA1c IFCC (more specific than DCCT) | B | 20–42mmol/mol; 53 ≈ good DM control | |
| Iron | S | ♂ 14–31µmol/L<br>♀ 11–30µmol/L | |
| Lactate | P<br>ABG | Venous 0.6–2.4mmol/L<br>Arterial 0.6–1.8mmol/L | |
| Lactate dehydrogenase (LDH) | P | 70–250U/L | |
| Lead | B | <1.8mmol/L | |
| Luteinizing hormone (LH) (premenopausal) | P | 3–16U/L (luteal) | |
| Magnesium | P | 0.75–1.05mmol/L | |
| Osmolality | P | 278–305mosmol/kg | |

| | | |
|---|---|---|
| Parathyroid hormone (PTH) | P | <0.8–8.5pmol/L |
| Potassium | P | 3.5–5.3mmol/L |
| Prolactin | P | ♂ <450U/L<br>♀ <600U/L |
| Prostate-specific antigen (PSA) | P | 0–4mcg/mL, age specific, see p530 |
| Protein (total) | P | 60–80g/L |
| Red cell folate | B | 0.36–1.44μmol/L<br>(160–640mcg/L) |
| Renin[2] (erect/recumbent) | P | 2.8–4.5/<br>1.1–2.7pmol/mL/h |
| Sodium[1] | P | 135–145mmol/L |
| Thyroid-binding globulin (TBG) | P | 7–17mg/L |
| Thyroid-stimulating hormone (TSH) | P | 0.5–4.2mU/L<br>widens with age, p216<br>assays vary; 4–5 is a grey area |
| Thyroxine ($T_4$) | P | 70–140nmol/L |
| Thyroxine (free) | P | 9–22pmol/L |
| Total iron-binding capacity | S | 54–75μmol/L |
| Triglycerides (fasting) | P | 0.50–2.3mmol/L |
| Triiodothyronine ($T_3$) | P | 1.2–3.0nmol/L |
| Troponin T (see p119) | P | <0.1mcg/L |
| Urate[1] | P | ♂ 210–480μmol/L<br>♀ 150–390μmol/L |
| Urea[1] | P | 2.5–6.7mmol/L |
| Vitamin $B_{12}$ | S | 0.13–0.68nmol/L<br>(>150ng/L) |
| Vitamin D | S | 50nmol/L (total) |

P=plasma (eg citrate bottle); S=serum (clotted; no anticoagulant); B=whole blood (EDTA bottle); ABG=arterial blood gas).

**1** See *OHCS* p9 for reference intervals in pregnancy.
**2** The sample requires special handling: contact the laboratory.
**3** Desired upper limit of cholesterol would be <6mmol/L. In some populations, 7.8mmol/L is the top end of the distribution.

## Table 17.2

| Arterial blood gases reference intervals | |
|---|---|
| pH: 7.35–7.45 | $P_aCO_2$: 4.7–6.0kPa |
| $P_aO_2$: >10.6kPa | Base excess: ±2mmol/L |

Note: 7.6mmHg = 1kPa (atmospheric pressure ≈ 100kPa)

## Table 17.3

| Urine reference intervals | Reference interval | Your hospital |
|---|---|---|
| Cortisol (free) | <280nmol/24h | |
| Hydroxyindole acetic acid | 16–73μmol/24h | |
| Hydroxymethylmandelic acid | 16–48μmol/24h | |
| Metanephrines | 0.03–0.69μmol/mmol creatinine (or <5.5μmol/day) | |
| Osmolality† | 350–1000mosmol/kg | |
| 17-oxogenic steroids | ♂ 28–30μmol/24h | |
| | ♀ 21–66μmol/24h | |
| 17-oxosteroids (neutral) | ♂ 17–76μmol/24h | |
| | ♀ 14–59μmol/24h | |
| Phosphate (inorganic) | 15–50mmol/24h | |
| Potassium | 14–120mmol/24h | |
| Protein | <150mg/24h | |
| Protein | creatinine ratio <3mg/mmol | |
| Sodium† | 100–250mmol/24h | |

† Interpret based upon plasma values.

# Haematology reference intervals

Table 17.4 (For $B_{12}$, folate, Fe, and TIBC, see pp752–3.)

| Measurement | | Reference interval | Your hospital |
|---|---|---|---|
| White cell count (WCC) | | $4.0–11.0 \times 10^9$/L | |
| Red cell count | ♂ | $4.5–6.5 \times 10^{12}$/L | |
| | ♀ | $3.9–5.6 \times 10^{12}$/L | |
| Haemoglobin | ♂ | 130–180g/L | |
| | ♀ | 115–160g/L | |
| Packed red cell volume (PCV) or haematocrit | ♂<br>♀ | 0.4–0.54L/L<br>0.37–0.47L/L | |
| Mean cell volume (MCV) | | 76–96fL | |
| Mean cell haemoglobin (MCH) | | 27–32pg | |
| Mean cell haemoglobin concentration (MCHC) | | 300–360g/L | |
| Red cell distribution width (RCDW, RDW) | | 11.6–14.6% (p325) | |
| Neutrophils | | $2.0–7.5 \times 10^9$/L<br>40–75% WCC | |
| Lymphocytes | | $1.0–4.5 \times 10^9$/L<br>20–45% WCC | |
| Eosinophils | | $0.04–0.44 \times 10^9$/L<br>1–6% WCC | |
| Basophils | | $0.0–0.10 \times 10^9$/L<br>0–1% WCC | |
| Monocytes | | $0.2–0.8 \times 10^9$/L<br>2–10% WCC | |
| Platelet count | | $150–400 \times 10^9$/L | |
| Reticulocyte count | | 0.8–2.0%[1] $25–100 \times 10^9$/L | |
| Erythrocyte sedimentation rate | | Depends on age (p372) | |
| Prothrombin time (citrated bottle) (factors I, II, VII, X) | | 10–14s | |
| Activated partial thromboplastin time (VIII, IX, XI, XII) | | 35–45s | |

Therapeutic ranges for INR: see p351.

1 Only use percentages as reference interval if red cell count is normal; otherwise, use the absolute value.

# Drug therapeutic ranges in plasma



►Ranges should only be used as a guide to treatment. A drug in an apparently too low concentration may still be clinically useful, while some patients require (and tolerate) levels in the 'toxic' range.

►The time since the last dose should be specified on the request form.

*Amikacin.*[1] Peak (1h post IV dose): 20–30mg/L. Trough: <10mg/L.

*Carbamazepine.*[1] Optimal concentration: 20–50μmol/L (4–12mg/L).

*Digoxin*[1] (6–12h post dose) 1–2.6nmol/L (0.8–2mcg/L). <1.3nmol/L may be toxic if there is hypokalaemia. *Signs of toxicity*—CVS: arrhythmias, heart block. CNS: confusion, insomnia, agitation, seeing too much yellow (xanthopsia), delirium. GI: nausea.

*Gentamicin*[1,2] (p387) and *tobramycin*.[1,2] The potential for oto- and nephrotoxicity is high if aminoglycosides are used inappropriately, so only prescribe for short therapeutic courses and follow local expert advice/guidelines. *CI* in severe renal or liver failure, ascites, burns, high cardiac output states (eg anaemia, Paget's disease), children, and pregnancy. *Signs of toxicity:* tinnitus, deafness, nystagmus, vertigo, renal failure. *Once-daily dosing* with dose adjustment is prefered as this has fewer SEs and better bactericidal activity (eg gentamicin 5mg/kg/d or tobramycin 4mg/kg/d; check with pharmacist in obese patients or when using tobramycin in cystic fibrosis). An exception to this dosing is endocarditis: split dosing (eg gentamicin 1mg/kg/8h or 12-hourly in renal failure) increases the synergistic bactericidal effect of other agents. Trough (just before dose) levels and renal function should initially be monitored daily (can be twice weekly in stable patients with normal renal function)—aim for *trough:* <1mg/L for both gentamicin and tobramycin. If the trough level is out of range, withold the next dose until level <1mg/L (recheck after 12–24h).

*Lithium*[2] (12h post dose). Guidelines vary: 0.4–0.8mmol/L is reasonable. *Early* signs of toxicity ($Li^+$ >1.5mmol/L): tremor. *Intermediate:* lethargy. *Late:* ($Li^+$ >2mmol/L) spasms, coma, fits, arrhythmias, renal failure (haemodialysis may be needed). See *OHCS* p349.

*Phenytoin.*[1,2] Trough: 40–80μmol/L (10–20mg/L). Beware if ↑↓albumin, as the assay is for bound phenytoin, while it is free phenytoin that is pharmacologically important. *Signs of toxicity:* ataxia, diplopia, nystagmus, sedation, dysarthria.

*Theophylline* 10–20mg/mL (55–110μmol/L). (►See p810.) Take sample 4–6h after starting an infusion (which should be stopped for ~15min just before the specimen is taken). *Signs of toxicity:* arrhythmias, anxiety, tremor, convulsions.

*Vancomycin.*[1,2] Renally excreted; dosing guided by age and renal function but typically 500mg–1g/12h. Check trough levels prior to 3rd dose, aiming for: 5–10 mg/L (10–15mg/L in SBE/IE and less-sensitive MRSA infections). If levels too low check drug being given then cautiously increase dose; if levels high then confirm timing of dose/levels, omit next dose, recheck levels, and consider decreasing dose or frequency.

**1** Trough levels should be taken just before the next dose. If values abnormally high, check that sample was indeed a trough level and not taken post-dose.
**2** Drugs for which *routine* monitoring is indicated.

## Some important drug interactions

►See *BNF*.

►See p689 for a list of cytochrome P450 inducers and inhibitors. Note: '↑' = effect of drug increased; '↓' = effect decreased.

### Drugs

*Adenosine:* ↓ by: aminophylline. ↑ by: dipyridamole.
*Aminoglycosides:* ↑ by: loop diuretics.
*Antidiabetic drugs:* (All) ↑ by: alcohol, β-blockers, bezafibrate, monoamine oxidase inhibitors. ↓ by: contraceptive steroids, corticosteroids, diazoxide, diuretics, (possibly also lithium).
- *Metformin* ↑ by: cimetidine. With alcohol: lactic acidosis risk.
- *Sulfonylureas* ↑ by: azapropazone, chloramphenicol, bezafibrate, co-trimoxazole, miconazole, sulfinpyrazone. ↓ by: rifampicin (nifedipine occasionally).

*Antiretroviral agents (HIV):* See p402.
*Angiotensin-converting enzyme (ACE) inhibitors:* ↓ by: NSAIDs, oestrogens.
*Antihistamines:* Avoid anything that ↑concentrations and risk of arrhythmias, eg anti-arrhythmics, antifungals, antipsychotics, β-blockers, diuretics, halofantrine, macrolide antibiotics (erythromycin, azithromycin, etc), protease inhibitors (p402), SSRIs (p448), tricyclics.
*Azathioprine:* ↑ by: allopurinol.
*β-blockers:* Avoid verapamil. ↓ by: NSAIDs. Lipophilic β-blockers (eg propranolol) are metabolized by the liver, and concentrations are ↑ by cimetidine. This does not happen with hydrophilic β-blockers (eg atenolol).
*Carbamazepine:* ↑ by: erythromycin, isoniazid, verapamil.
*Ciclosporin:* ↑ by: erythromycin, grapefruit juice, nifedipine. ↓ by: phenytoin.
*Cimetidine:* ↑ the effect of: amitriptyline, lidocaine, metronidazole, pethidine, phenytoin, propranolol, quinine, theophylline, warfarin.
*Contraceptive steroids:* ↓ by: antibiotics, barbiturates, carbamazepine, phenytoin, rifampicin.
*Digoxin:* ↑ by: amiodarone, carbenoxolone and diuretics (due to ↓$K^+$), quinine, verapamil.
*Diuretics:* ↓ by: NSAIDs—particularly indometacin.
*Ergotamine:* ↑ by: erythromycin (ergotism may occur).
*Fluconazole:* Avoid concurrent astemizole.
*Lithium:* ↑ by: thiazide diuretics.
*Methotrexate:* ↑ by: aspirin, NSAIDs. Many antibiotics (check *BNF*).
*Phenytoin:* ↑ by: chloramphenicol, cimetidine, disulfiram, isoniazid, sulfonamides. ↓ by: carbamazepine.
*Potassium-sparing diuretics with ACE-inhibitors:* Hyperkalaemia.
*Theophyllines:* ↑ by: cimetidine, ciprofloxacin, erythromycin, contraceptive steroids, propranolol. ↓ by: barbiturates, carbamazepine, phenytoin, rifampicin. See p810.
*Valproate:* ↓ by: carbamazepine, phenobarbital, phenytoin.
*Warfarin and nicoumalone:* (Nicoumalone=acenocoumarol) ↑ by: alcohol, allopurinol, amiodarone, aspirin, chloramphenicol, cimetidine, ciprofloxacin, co-trimoxazole, danazol, dipyridamole, disulfiram, erythromycin (and broad-spectrum antibiotics), gemfibrozil, glucagon, ketoconazole, metronidazole, miconazole, nalidixic acid, neomycin, NSAIDs, phenytoin, quinidine, simvastatin (but not pravastatin), sulfinpyrazone, sulfonamides, tetracyclines, levothyroxine.
*Warfarin and nicoumalone:* ↓ by: aminoglutethimide, barbiturates, carbamazepine, contraceptive steroids, dichloralphenazone, griseofulvin, rifampicin, phenytoin, vitamin K.
*Zidovudine (AZT):* ↑ by: paracetamol (↑marrow toxicity).

### IVI solutions to avoid

*Glucose:* Avoid furosemide, ampicillin, hydralazine, insulin, melphalan, phenytoin, and quinine.
*0.9% saline:* Avoid amphotericin, lidocaine, nitroprusside.

# 18 Practical procedures

**Contents**




**Fig 18.1** NHS 'clean your hands' campaign poster. Contains public sector information licensed under Open Government Licence v3.0. https://www.whatdotheyknow.com/request/21861/response/56086/attach/3/04072%20Hand%20Hygiene%205%201.1.pdf

Hungarian obstetrician Ignaz Semmelweis demonstrated the benefits of handwashing in the 1840s: he observed that maternal mortality was nearly three times as high on a doctor-run maternity ward compared to a midwife-run ward. The explanation remained elusive until Semmelweis' friend Jakob Kolletschka died after receiving an accidental scalpel cut from a student during a post-mortem demonstration. Semmelweis recognized in Kolletschka's death many of the features of the dying mothers. The explanation: the maternity ward doctors' day started with post-mortem examinations, from which they would procede to perform vaginal examinations on the living without washing their hands. Noticing this, Semmelweis introduced the practice of washing hands with chloride of lime and cut death rates to that of the midwives' patients. Despite the evidence he amassed, Semmelweis's theory was rejected by his contemporaries, a rejection which undoubtedly contributed to his psychiatric distress, eventual commitment to an asylum, and ultimate death from the blows of his guards. It would take another 20 years and countless deaths before Lister published his landmark work on the use of carbolic acid in surgery.

Take a minute to wash your hands thoroughly before undertaking any procedure. This prerequisite will not only reduce infection risk for your patients, but give you a moment for mindfulness: focus on the hot water running over your hands, breathe deeply, and for a while forget about your list of jobs. Perhaps spare Dr Kolletschka a thought. You may find that the subsequent procedure goes more smoothly than anticipated.

## Training and the business of medicine

As medical training has evolved in an environment where patient safety is paramount, the old adage of 'see one, do one, teach one' is no longer relevant. 'Just having a go' when you aren't confident can have devastating consequences for the patient, and also for you and your future. This creates tensions for training, but these are not insurmountable. Seek out opportunities to learn practical procedures, ideally in a controlled, elective setting, so that your first attempt isn't a life-or-death emergency attempt—time spent in theatres or ICU will pay dividends in this regard. Many seniors will be happy to make time to teach if you contact them in advance—let them know you are interested and leave your bleep. ►Even in an emergency setting, it is still wiser to seek help rather than attempting an urgent procedure for the first time.


*We thank our Specialist Reader, Dr Andrew Johnston, for his contribution to this chapter.*

## Nasogastric tubes

These tubes are passed into the stomach via the nose. Large (eg 16F) are good for drainage but can be uncomfortable for patients. Small (eg 10F) are more comfortable for feeding but can be difficult to aspirate and are poor for drainage. Used:

- To decompress the stomach/gastrointestinal tract especially when there is obstruction, eg gastric outflow obstruction, ileus, intestinal obstruction.
- For gastric lavage.
- To administer feed/drugs, especially in critically ill patients or those with dysphagia, eg motor neuron disease, post CVA.

**Passing the tube** Nurses are experts and will ask you (who may never have passed one) to do so only when they fail—so the first question to ask is: 'Have you asked the charge-nurse from the ward next door?'

- Wear non-sterile gloves and an apron to protect both you and the patient.
- Explain the procedure. Take a new, cool (hence less flexible) tube. Have a cup of water to hand. Lubricate well with aqueous gel.
- Use the tube, by holding it against the patient's head, to estimate the length required to get from the nostril to the back of the throat.
- Place lubricated tube in nostril with its natural curve promoting passage down, rather than up. The right nostril is often easier than the left but, if feasible, ask the patient for their preference. Advance directly backwards (not upwards).
- When the tip is estimated to be entering the throat, rotate the tube by ~180° to discourage passage into the mouth.
- Ask the patient to swallow a sip of water, and advance as they do, timing each push with a swallow. *If this fails:* Try the other nostril.
- The tube has distance markings along it: the stomach is at ~35–40cm in adults, so advance > this distance, preferably 10–20cm beyond. Tape securely to the nose.

**Confirming position** This is vital prior to commencing any treatment through the tube. Misplaced nasogastric tubes have led to a number of preventable deaths, and feeding via a misplaced tube is considered an NHS Never Event (a serious, largely preventable patient safety incident that should *never* occur if the available preventative measures have been implemented).

- Use pH paper to test that you are in the stomach: aspirated gastric contents are acid (pH ≤5.5) although antacids or PPIs may increase the pH. Small tubes can be difficult to aspirate, try withdrawing or advancing a few cm or turning the patient on the left side to help dip the tube in gastric contents. Aspirates should be >0.5mL and tested directly on unhandled pH paper. Allow 10s for colour change to occur.
- If the pH is >5.5 and the NGT is needed for drug or feed administration then the position must be checked radiologically. Request a CXR/abdo X-ray (tell the radiologist why you need it). Look for the radio-opaque line/tip (this can be hard to see, look below the diaphragm, but if in doubt, ask for help from the radiologist).
- *The 'whoosh' test is NOT an accepted method of testing for tube position.*
- Either spigot the tube, or allow to drain into a dependent catheter bag secured to clothing (zinc oxide tape around tube to form a flap, safety pin through flap).

►Do not pass a tube nasally if there is any suspicion of a facial fracture.

►Get senior help if the patient has recently had upper GI surgery—it is not good practice to push the tube through a fresh anastomosis.

**Complications** •Pain, or, rarely: •Loss of electrolytes •Oesophagitis •Tracheal or duodenal intubation •Necrosis: retro- or nasopharyngeal •Stomach perforation.

**Weaning** When planning removal of an NGT *in situ* for decompression or relief of obstruction, it is wise to wean it so that the patient manages well without it. Drainage should be <750mL/24 hours for successful weaning.

- First it should be on free drainage with, eg, 4hrly aspirations.
- Then spigot with 4hrly aspirations.
- Then spigot only. If this is tolerated along with oral intake then it is probably safe to remove the tube; if not, then take a step backwards.

# Placing IV cannulae



►Much of what we do is not evidence based; however, in more recent years, particularly in intensive care units, the rise of hospital-acquired infections and multidrug-resistant organisms has prompted a review of standard practice and a series of evidence-based interventions put together as a 'care bundle' to reduce hospital-acquired infections. The technique for placing a cannula is best shown at the bedside by an expert, but following these simple rules will significantly reduce the risk of infection from the cannula.

**Preparation is key, remember the following before you start**

1. *Equipment:* Set up a tray with cleaning swabs, gauze, cannulae (swallow your pride and take at least three of different sizes, see table 18.1), dressings, 0.9% saline, 10mL syringe, needle-free adaptor (eg octopus with bionector), blood tubes if required, portable sharps bin → needlestick injuries do happen.
2. *Patient:* Have them lying down, explain procedure, obtain verbal consent, place tourniquet around arm, rest the arm below the heart to aid venous filling.
3. *Site:* Look for the best vein—it should be palpable; some of the best veins are not easily visible, some of the most visible collapse on insertion. Tapping gently helps. ►*Never cannulate:* AV fistulae arms, limbs with lymphoedema. ►*Avoid:* Sites crossing a joint (if possible), the cephalic vein in a renal patient.
4. *Consider:* *EMLA®* cream, cold spray, or 1% lidocaine for children or those with needle phobia. *EMLA®* takes 45min to work, but can save you hassle later.

**Insertion care bundle**

**1** Aseptic technique. **2** Hand hygiene. **3** Apron + non-sterile gloves. **4** Skin preparation—2% chorhexidine in 70% isopropyl alcohol (allow to dry for 30 seconds). Do not repalpate vein after cleaning unless wearing sterile gloves. **5** Dressing—sterile and transparent so that insertion site can be observed.

**After insertion**

**1** Take blood with syringe or adaptor. **2** Remove tourniquet. **3** Attach needle-free device (if appropriate) and flush with 10mL 0.9% saline. **4** Apply dressing. **5** Let nursing staff know that cannula is in place and ready for use. **6** Document insertion according to local policy. **7** Write up appropriate fluids or parenteral medication.

When seeing your patient on the daily ward round (and to avoid being called to review or replace cannulae at 6pm) do a RAID[1] assessment: consider if the drip is:

**R**equired—can the patient manage with oral medication/fluids?

**A**ppropriate—should you consider a PICC, central line, long-term line, etc?

**I**nfected—any signs of inflammation or infection? Remove if yes. Peripheral cannulae should be replaced every 72–96 hours.

**D**ressed properly—many drips are replaced early because they have 'fallen out', or are kinked from poor dressings.

Tissued or infected cannulae need replacing, either with another peripheral cannula, or with a longer-term access device, such as a PICC line.

**If you fail after three attempts** ►Shocked patients need fluid quickly: if you are having trouble putting in a drip, call your senior. The following advice assumes that the drip is not immediately life-saving. ►►If it is, see BOX.

- Ask for help—from colleagues or seniors—do not be ashamed, everyone has to learn and even senior doctors have bad days; a fresh pair of eyes can be all it takes. As a house officer, one of us was asked to place a drip when a very shame-faced consultant had 'had a go' to prove he still could, and found out that he couldn't!
- Help yourself—try putting the hand in warm water, using a small amount of GTN paste over the vein, or using ultrasound if available to help you identify the vein.
- If there is no one else to help, take a break and come back in half an hour. Veins come and go, and coming back with fresh eyes can make all the difference.

**Table 18.1** Intravenous cannulae sizes and UK colour conventions

| Gauge | Colour | Diameter (mm) | Length (mm) | Flow rate (mL/min) |
|---|---|---|---|---|
| 14G | ORANGE/BROWN | 2.0 | 45 | 250 |
| 16G | GREY | 1.7 | 42 | 170 |
| 18G | GREEN | 1.2 | 40 | 90 |
| 20G | PINK | 1.0 | 32 | 55 |
| 22G | BLUE | 0.28 | 25 | 25 |
| 24G | YELLOW | 0.07 | 19 | 24 |

Flow rate is given as maximum flow rate under gravity; faster rates may be achievable with rapid infusion devices. According to Poiseuille's law[1] the flow rate ($Q$) of a fluid through a tubular structure is inversely proportional to viscosity ($\eta$) and length ($l$) and proportional to the pressure difference across it ($P_i - P_o$) and the radius *to the power of* 4($r^4$). Hence:

$$Q \propto \frac{(P_i - P_o)\, r^4}{\eta\, l}$$

## A last throw of the dice

Just once it may come down to you. For some, this is one of the challenges and thrills in medicine. There may be no one else available to help when there is an absolute and urgent indication for IV drugs/fluids/blood—and all of the previously discussed measures have been tried, and have failed. Think of lonesome night shifts, over-run emergency departments, a disaster scene, war, or medicine in the field. The following measures are not recommended for non-life-threatening scenarios:

- ►►Don't worry. Have a good look again. Feet (avoid in diabetics)? Inside of the forearm? Upper arm?
- ►►Have you really exhausted all of your options for help from a colleague? Maybe the anaesthetist or ICU registrar is approachable—they do have remarkable skills.
- ►►Is the patient familiar with his/her own veins (eg previous IV drug abuser)?
- ►►If there is only a small amount of IV medication required and a small, short vein, you may be able to gain access with a carefully placed butterfly needle that is taped down. Some drugs cannot be passed this way (eg amiodarone, $K^+$).
- ►►The external jugular vein may become prominent when the patient is head down (Trendelenburg) by 5–10° (►not in situations of fluid overload, LVF, ↑ICP). Only attempt cannulation of this vein if you are not going to jeopardize future central line insertion, and if you can clearly determine the surrounding anatomy.
- ►►In an arrest situation, the 2015 Advanced Life Support Guidelines recommend the intraosseous route in both adults and children if venous access is not possible; access devices should be available within resuscitation settings (eg emergency department).

*Only do the following if you have had the appropriate training/experience:*

- ►►In children, consider cannulating a scalp vein.
- ►►Central venous catheterization (p775). This may be just as hard in a profoundly hypovolaemic arrest patient, and a good knowledge of local anatomy and of the procedure (± ultrasound guidance) will be invaluable.

If you don't have an intraosseous access device, a cut down to the long saphenous vein may be attempted, *in extremis, even if you have no prior experience* (at this site you won't kill by being ham-fisted). ►►Make a transverse incision 1–2cm anterior and superior to the medial malleolus. ►►Free vein with forceps. ►►Cannulate it under direct vision. ►*Here, 'first do no harm' is trumped by 'nothing ventured, nothing gained'.*

Hopefully, it shouldn't ever have to come to these measures, but one day…

1 Poiseuille's law is a neat piece of physiology and worth remembering—it is applicable in some form to almost every system in the body. Note that it is a 4th-power law: a small change in the radius makes a huge difference to flow.

# Catheterizing bladders



Urinary tract infections are the second commonest health care-associated infection, and urinary catheters are frequently to blame. Think, does the patient really need a catheter? If so, use the smallest you can and take out as soon as possible.

**Size** (in French gauge): 12=small; 16=large; 20=very large (eg 3-way). **Material** Coated latex catheters are soft and a good short-term option ►but unsuitable in true latex allergy. Silastic (silicone) catheters may be used long term, but cost more. Silver alloy coating reduces infections. **Shape** Foley is typical (fig 18.2); coudé (elbow) catheters have an angled tip to ease around prostates but are more risky; 3-way catheters are used in clot or debris retention and have an extra lumen for irrigation fluid, attached to the irrigation set via an extra port on the distal end (fig 18.3). Get urology advice before starting irrigation. Condom catheters are often preferred by patients (less discomfort) even though they may leak and fall off.

**Catheter problems** • *Infection:* ~5% develop bacteraemia (most will have bacterial colonization, antibiotics may not be required unless systemically unwell—discuss with microbiology). A stat dose of, eg gentamicin 80mg is sometimes given pre-insertion despite a lack of evidence for benefit. Check your local policy. • *Bladder spasm:* May be painful—try reducing the water in the balloon or an anticholinergic drug, eg oxybutynin.

**Per urethram** Aseptic technique required.

*Indications* • Relieve urinary retention. • Monitor urine output in critically ill patients. • Collect uncontaminated urine for diagnosis. ► It is contraindicated in urethral injury (eg pelvic fracture) and acute prostatitis.

- Explain the procedure, and obtain verbal consent. Prepare a catheterization trolley: gloves, catheter, lidocaine jelly, cleaning solution, drape, kidney dish, gauze swabs, drainage bag, 10mL water and syringe, specimen container.
- Lie the patient supine: women with knees flexed and hips abducted with heels together. Use a gloved hand to clean urethral meatus in a pubis-to-anus direction, holding the labia apart with the other hand. With uncircumcised men, retract the foreskin to clean the glans; use a gloved hand to hold the penis still. The hand used to hold the penis or labia should not touch the catheter. Place a sterile drape with a hole in the middle to help you maintain asepsis. Remember: left hand dirty, right hand clean.
- Put sterile lidocaine 1–2% gel on the catheter tip and ≤10mL into the urethra (≤5mL if ♀). In men, lift and gently stretch the penis upwards to eliminate any urethral folds that may lead to false passage formation.
- Use steady gentle pressure to advance the catheter, rotating slightly can help it slide in. ►Never force the catheter. Tilting the penis up towards the umbilicus while inserting may help negotiate the prostate. Insert to the hilt; wait until urine emerges before inflating the balloon. Remember to check the balloon's capacity before inflation (written on the outer end). Collect a sterile specimen and attach a drainage bag. Pull the catheter back so that the balloon comes to rest at the bladder neck.
- If you are having trouble getting past the prostate, try: more lubrication, a gentle twisting motion; a larger catheter; or call the urologists, who may use a guidewire.

►Remember to reposition the foreskin in uncircumcised men after the catheter is inserted to prevent oedema of the glans and paraphimos.

*Documentation:* In the notes be sure to document the indication for catheterization, size of catheter, whether insertion was difficult or straightforward, any complications, residual volume and colour of urine. It is good practice to document that the foreskin has been replaced. Sign with your name, date, and designation.

**Suprapubic catheterization:** Sterile technique required ►Absolutely contraindicated unless there is a large bladder palpable or visible on ultrasound, because of the risk of bowel perforation. Be wary, particularly if there is a history of abdominal or pelvic surgery. Suprapubic catheter insertion is high risk and you should be trained before attempting it, speak to the urologists first.

## Self-catheterization

This is a good, safe way of managing chronic retention from a neuropathic bladder (eg in multiple sclerosis, diabetic neuropathy, spinal tumour, or trauma). Never consider a patient in difficulties from a big residual volume to be too old, young, or disabled to learn. 5-yr-old children can learn the technique, and can have their lives transformed—so motivation may be excellent. There may be fewer UTIs as there is no residual urine—and less reflux obstructive uropathy. Assessing suitability entails testing sacral dermatomes: a 'numb bum' implies ↓sensation of a full bladder; higher sensory loss may mean catheterization will be painless. Get help from your continence adviser who will be in a position to teach the patient or carer that catheterizations must be gentle (the catheter is of a much smaller calibre), particularly if sensation is lacking, and must number >4/d ('always keep your catheter with you; don't wait for an urge before catheterizing'). See fig 18.4.

Fig 18.2 A size 14F latex Foley catheter with the balloon inflated via the topmost port of the outer end (green).




Fig 18.3 The external end of a size 20F 3-way catheter. The lowest port is for the bladder irrigation fluid and the uppermost port (yellow) is for balloon inflation.



Fig 18.4 A size 10F catheter for self-catheterization. They are usually smaller than indwelling catheters, eg 10F compared to 14F. Note that this catheter also has no balloon.



## 'The catheter is not draining…'

You will be asked to check catheters that are not draining. Check the fluid chart and the patient:

- *Previously good output, now anuric:* Blocked catheter until proven otherwise. Was the urine clear previously or bloodstained? Consider flushing the catheter: with aseptic technique flush and withdraw 20mL of sterile 0.9% saline with a bladder syringe. This may get the flow going again. A 3-way catheter may be needed if there is clot or debris retention. If it blocks again, replace it. Repeated flushes lead to infection.
- *Slow decline in urine output over several hours:* In a dehydrated/post-op patient a fluid challenge of 500mL STAT (250mL if cardiac comorbidity) may help, come back and check the response in 30min. Check all other parameters (eg pulse, BP, CVP) and increase rate of background IV fluids if appropriate. ►►Acute kidney injury (p298): if urine output has tailed off and now stopped, the cause is often renal hypoperfusion (ie pre-renal failure), but consider other factors, eg nephrotoxic drugs.
- *Catheter is bypassing:* A condom catheter may be more appropriate.
- *Catheter has dislodged into the proximal (prostatic) urethra:* Possible even if the balloon is fully inflated. Consider this if a flush enters but cannot be withdrawn. If the patient still needs a catheter then replace it, consider a larger size.
- *The catheter has perforated the lower urinary tract on insertion and is not lying in the bladder or urethra:* ►►If suspected, call the urologists immediately.

*Remember:* urine output should be >400mL in 24h or >0.5mL/kg/h (see p576).

## Trial without catheter (TWOC)

When it is time to remove a catheter, the possibility of urinary retention must be considered. Remove the catheter first thing one morning. If retention does occur, insert a long-term catheter (eg silicone), consider an α-blocker (p642), and arrange urology TWOC clinic follow-up.

## Draining ascites

For patients with refractory or recurrent ascites that is symptomatic, it is possible to drain the ascites using a long pig-tail catheter. Paracentesis in such patients even in the presence of spontaneous bacterial peritonitis may be safe. Learn at the bedside from an expert.

**Contraindications (these are relative, not absolute)** End-stage cirrhosis; coagulopathy; hyponatraemia (≤126mmol/L); sepsis. The main complication of the procedure is severe hypovolaemia secondary to reaccumulation of the ascites, so intravascular replenishment with a plasma expander is required. For smaller volumes, eg less than 5L, 500mL of 5% human albumin or Gelofusine® would be sufficient. For volumes over 5L, reasonable replacement would be 100mL 20% human albumin IV for each 1–3 litres of ascites drained (check your local policy). You may need to call the haematology lab to request this in advance.

**Procedure** Requires sterile technique.

- Ensure you have good IV access—eg 18G cannula in the antecubital fossa.
- Explain the procedure including the risks of infection, bleeding, hyponatraemia, renal impairment, and damage to surrounding structures (such as liver, spleen, and bowel), and obtain consent from the patient. Serious complications occur in less than 1 in 1000 patients. Ask the patient to empty their bladder.
- Examine the abdomen carefully, evaluating the ascites and checking for organomegaly. Mark where you are going to enter. If in doubt, ask the radiology department to ultrasound the abdomen and mark a spot for drainage. Approach from the left side unless previous local surgery/stoma prevents this—call a senior for support and advice if this is the case.
- Prepare a tray with 2% chlorhexidine solution, sterile drapes, 1% lidocaine, syringes, needles, sample bottles, and your drain. Clean the abdomen thoroughly and place sterile drapes, ensure you maintain sterile technique throughout. Infiltrate the local anaesthetic.
- Perform an ascitic tap (see p765) first so that you know you are in the correct place: remove 20mL fluid for MC&S.
- Away from the patient, carefully thread the catheter over the (large and long) needle using the guide so that the pig-tail has been straightened out. Remove the guide.
- With the left hand hold the needle ~2.5cm (1 inch) from the tip—this will stop it from advancing too far (and from performing an aortic biopsy). With the right hand, hold the other end.
- Gently insert the needle perpendicular to the skin at the site of the ascitic tap up to your hold with your left hand—ascites should now drain easily. If necessary, advance the needle and catheter a short distance until good flow is achieved.
- Advance the catheter over the needle with your left hand, keeping the needle in exactly the same place with your right hand. ►Do not re-advance the needle because it will go through the curled pig-tail and do not withdraw it because you won't be able to thread in the catheter.
- When fully inserted, remove the needle, connect the catheter to a drainage bag (keep it below the level of the abdomen), and tape it down securely to the skin.
- The patient should stay in bed as the ascites drains.
- Document clearly in the notes the indication for the procedure, that consent was obtained, clotting and U&Es checked pre-procedure, how much lidocaine was required, how much fluid was removed for investigations, and whether there were any complications to the procedure.
- Replenish intravascular volume with human albumin (see 'Contraindications' earlier in topic).
- Ask the nursing staff to remove the catheter after 6h or after a pre-determined volume has been drained (up to 20L can come off in 6h) and document this clearly in the medical notes. Drains are removed after 4–6h to prevent infection.
- Check U&Es after the procedure and re-examine the patient.

## Diagnostic taps

If you are unsure whether a drain is needed, a diagnostic tap can be helpful. Whatever fluid you are sampling, a green needle carries far less risk than a formal drain. It also allows you to decide whether a drain is required.

Ascites may be sampled to give a cytological or bacterial diagnosis, eg to exclude spontaneous bacterial peritonitis (SBP; p276). Before starting, know the patient's platelets + clotting times. If they are abnormal, seek help before proceeding.

- Place the patient flat and tap out the ascites, marking a point where fluid has been identified, avoiding vessels, stomas, and scars (adhesions to the anterior abdominal wall). The left side may be safer—less chance of nicking liver (fig 18.5).
- Clean the skin. Infiltrate some local anaesthetic, eg 1% lidocaine (see p573).
- Insert a 21G needle on a 20mL syringe into the skin and advance while aspirating until fluid is withdrawn, try to obtain 60mL of fluid.
- Remove the needle, apply a sterile dressing.
- Send fluid to microbiology (15mL) for microscopy and culture, biochemistry (5mL for protein, see p192), and cytology (40mL). Call microbiology to forewarn them if urgent analysis of the specimen is required.

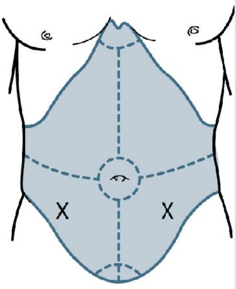



Fig 18.5 Always tap out the ascites, but aim approximately for 5cm medial to and superior to the anterior superior iliac spine. If in doubt, ask for an ultrasound to mark the spot.

## Diagnostic aspiration of a pleural effusion

- If not yet done, a CXR may help evaluate the side and size of the effusion.
- Ideally use US guidance at the bedside (↑ chance of successful aspirate and ↓ chance of organ puncture). If this is unavailable, ask an ultrasonographer to mark a spot, or percuss the upper border of the pleural effusion and choose a site 1 or 2 intercostal spaces below it (usually posteriorly or laterally).
- Clean the area around the marked spot with 2% chlorhexidine solution.
- Infiltrate down to the pleura with 5–10mL of 1% lidocaine.
- Attach a 21G needle to a syringe and insert it just above the upper border of the rib below the mark to avoid the neurovascular bundle (fig 18.6). Aspirate whilst advancing the needle. Draw off 10–30mL of pleural fluid. Send fluid to the lab for chemistry (protein, glucose, pH, LDH); bacteriology (microscopy and culture, auramine stain, TB culture); cytology, and, if indicated, amylase and immunology (rheumatoid factor, antinuclear antibodies, complement).

▸▸If you cannot obtain fluid with a 21G needle, seek help.

- If any cause for concern, arrange a repeat CXR.

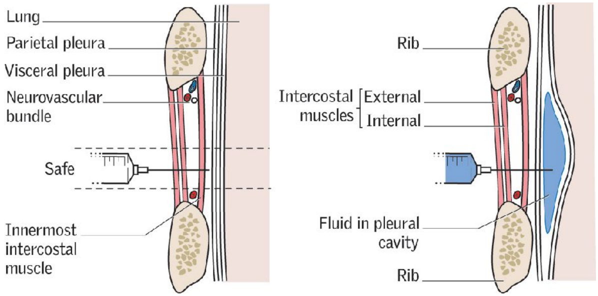



Fig 18.6 Safe approach to entering the pleura by the intercostal route.

# Inserting a chest drain



## Indications

- Pneumothorax (p814): ventilated; tension; persistent/recurrent despite aspiration (eg <24h after 1st aspiration); large 2nd spontaneous pneumothorax if >50yrs old.
- Malignant pleural effusion, empyema, or complicated parapneumonic effusion.
- Pleural effusion compromising ventilation, eg in ICU patients.
- Traumatic haemopneumothorax.
- Post-operatively: eg thoracotomy; oesophagectomy; cardiothoracic surgery.

►Pleural effusions are best drained under US guidance using a Seldinger technique. This technique is also used for pneumathoraces (except in traumatic or post-operative situations) without US guidance; for this reason it is detailed here.

## Sterile procedure

- Identify the point for drainage. In effusions, this should be done with US, ideally under direct guidance or with a marked spot. For pneumothoraces, check the drainage point from CXR/CT/examination.
- Preparation: trolley with dressing pack; 2% chlorhexidine; needles; 10mL syringes; 1% lidocaine; scalpel; suture; Seldinger chest drain kit; underwater drainage bottle; connection tubes; sterile $H_2O$; dressings. Incontinence pad under patient.
- Choose insertion site: 4–6th intercostal space, anterior- to mid-axillary line—the 'safe triangle' (see BOX 'The "safe triangle" for insertion' and fig 18.7). A more posterior approach, eg the 7th space posteriorly, may be required to drain a loculated effusion (under direct US visualization) and occasionally the 2nd intercostal space in the mid-clavicular line may be used for apical pneumothoraces—however, both approaches tend to be less comfortable.
- Maintain sterile technique—clean and place sterile drapes. Scrub for insertion.
- Prepare your underwater drain by filling the bottle to the marked line with sterile water. Ensure this is kept sterile until you need it.
- Infiltrate down to pleura with 10mL of 1% lidocaine and a 21G needle. Check that air/fluid can be aspirated from the proposed insertion site; if not, do not proceed.
- Attach the Seldinger needle to the syringe containing 1–2mL of sterile saline. The needle is bevelled and will direct the guidewire; in general advance bevel up for pneumothoraces, bevel down for effusions.
- Insert the needle gently, aspirating constantly. When fluid/air is obtained in the syringe, stop, note insertion depth from the markings on the Seldinger needle. Remove syringe, thread the guidewire through the needle. Remove the needle and clamp the guidewire to the sterile drapes to ensure it does not move. Using the markings on the Seldinger needle, move the rubber stops on the dilators to the depth noted earlier, to prevent the dilator slipping in further than intended.
- Make a nick in the skin where the wire enters, and slide the dilators over the wire sequentially from smallest to largest to enlarge the hole, keep gauze on hand. Slide the Seldinger drain over the wire into the pleural cavity. Remove the wire and attach a 3-way tap to the drain, then connect to the underwater drainage bottle.
- Suture the drain in place using a drain stitch—make a stitch in the skin close to the drain site, tie this fairly loosely with a double knot. Then tie the suture to the drain. It is usually best to be shown this before attempting it for yourself. Dress the drain, and ensure it is well taped down.
- Check that the drain is swinging (effusion) or bubbling (pneumothorax) and ensure the water bottle remains below the level of the patient at all times. If the drain needs to be lifted above the patient, clamp it briefly. ►You should never clamp chest drains inserted for pneumothoraces. Clamping for pleural effusions can control the rate of drainage and prevent expansion pulmonary oedema.
- Request a CXR to check the position of the drain.

**Removal** *In pneumothorax:* Consider when drain is no longer bubbling and CXR shows re-inflation. Give analgesia beforehand, eg morphine. Smartly withdraw during expiration or Valsalva. There is no need to clamp the drain beforehand as reinsertion is unlikely. *In effusions:* Generally the drain can be removed when drainage is <200mL/24h, but for cirrhotic hydrothoraces the chest drain is treated similarly to the ascitic drain (see p764) with HAS supplementation and removal at 4–6 hours.

## The 'safe triangle' for insertion of a chest drain

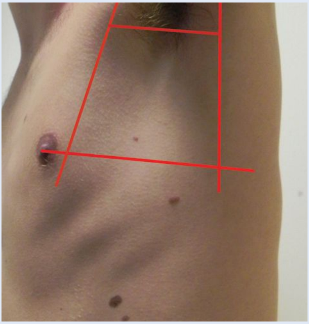


Fig 18.7 The safe 'triangle' is not really a triangle, as the axilla cuts off the point of the triangle. Draw a line along the lateral border of pectoralis major, a line along the anterior border of latissimus dorsi, and a line superior to the horizontal level of the nipple. The apex of the triangle is the axilla. Often chest drains are inserted directly under ultrasound guidance, or with a pre-marked spot; however, in an emergency or for aspiration, the landmarks of the safe triangle are important to know.

### Complications

- Thoracic or abdominal organ injury. • Lymphatic damage ∴ chylothorax.
- Damage to long thoracic nerve of Bell ∴ wing scapula. • Rarely, arrhythmia.

### Watch out for:

- Retrograde flow back into the chest.
- Persistent bubbling—there may be a continual leak from the lung.
- Blockage of the tube from clots or kinking—no swinging or bubbling.
- Malposition—check position with CXR.

## Relieving a tension pneumothorax

**Symptoms** Acute respiratory distress, chest pain, ►►respiratory arrest.

**Signs** Hypotension; distended neck veins; asymmetrical lung expansion; trachea and apex deviated away from side of reduced air entry and hyperresonance to percussion. ►►There is no time for a CXR (but see fig 16.43, p749).

**Aim** To release air from the pleural space. In a tension pneumothorax, air is drawn into the intrapleural space with each breath, but cannot escape due to a valve-like effect of the tiny flap in the parietal pleura. The increasing pressure progressively embarrasses the heart and the other lung.

### ►►100% oxygen.

### Prodedure

- Insert a large-bore IV cannula (eg Venflon®) usually through the 2nd intercostal space in the midclavicular line or the 'safe triangle' for chest drain insertion (see BOX 'The "safe triangle" for insertion'). Remove the stylet, allowing the trapped air to escape, usually with an audible hiss. This converts the tension pneumothorax to an open pneumothorax. Tape securely. • Don't recover the cannula as tensioning will recur.
- Proceed to formal chest drain insertion (see p766).

## Aspiration of a pneumothorax

Identify the 2nd intercostal space in the midclavicular line (or 4–6th intercostal space in the midaxillary line) and infiltrate with 1% lidocaine down to the pleura overlying the pneumothorax.

- Insert a 16G cannula into the pleural space. Remove the needle and connect the cannula to a 3-way tap and a 50mL syringe. Aspirate up to 2.5L of air (50mL×50). Stop if resistance is felt, or if the patient coughs excessively.
- Request a CXR to confirm resolution of the pneumothorax. If successful, consider discharging the patient and repeating the CXR after 24h to exclude recurrence, and again after 7–10d. Advise to avoid air travel for 6 weeks after a normal CXR. Diving should be permanently avoided.
- If aspiration is unsuccessful (in a significant, symptomatic pneumothorax), insert an intercostal drain (see p766).

## Lumbar puncture (LP)

**Contraindications** •Bleeding diathesis. •Cardiorespiratory compromise. •Infection at site of needle insertion. Most importantly: ▸▸ ↑ICP (suspect if very severe headache, ↓level of consciousness with falling pulse, rising BP, vomiting, focal neurology, or papilloedema)—LP in these patients will cause coning, so unless it is a routine procedure, eg for known idiopathic intracranial hypertension, obtain a CT prior to LP. CT is not infallible, so be sure your indication for LP is strong.

**Method** Explain to the patient what sampling CSF entails, why it is needed, that co-operation is vital, and that they can communicate with you at all stages.

- Place the patient on his or her left side, with the back on the edge of the bed, fully flexed (knees to chin). A pillow under the head and another between the knees may keep them more stable.
- Landmarks: plane of iliac crests through the level of L3/4 (see fig 18.8). In adults, the spinal cord ends at the L1/2 disc (fig 18.9). Mark L3/4 intervertebral space (or one space below, L4/5), eg by a gentle indentation of a needle cap on the overlying skin (better than a ballpoint pen mark, which might be erased by the sterilizing fluid).
- Use aseptic technique (hat, mask, gloves, gown) and 2% chlorhexidine in 70% alcohol to clean the skin, allow to dry and then place sterile drapes.
- Open the spinal pack. Assemble the manometer and 3-way tap. Have three plain sterile tubes and one fluoride tube (for glucose) ready.
- Using a 25G (orange) needle, raise a bleb of local anaesthetic, then use a 21G (green) needle to infiltrate deeper.
- Wait 1min, then insert spinal needle (22G, stilette in place) perpendicular to the body, through your mark, aiming slightly up towards the umbilicus. Feel resistance of spinal ligaments, and then the dura, then a 'give' as the needle enters the subarachnoid space. NB: keep the bevel of the needle facing up, parallel with dural fibres.
- Withdraw stilette. Check CSF fills needle and attach manometer (3-way tap turned off towards you) to measure 'opening' pressure.
- Catch fluid in three sequentially numbered bottles (10 drops per tube).
- Reinsert stilette then remove needle and apply dressing. Document the procedure clearly in the notes including CSF appearance and opening pressure.
- Send CSF promptly for *microscopy*, *culture*, *protein*, *lactate*, *and glucose* (do plasma glucose too)—call the lab to let them know. If applicable, also send for: cytology, fungal studies, TB culture, virology (± herpes and other PCR), syphilis serology, oligoclonal bands (+serum sample for comparison) if multiple sclerosis suspected. Is there xanthochromia (p478)?
- If you fail; ask for help—try with the patient sitting or with radiological guidance.

**CSF composition** *Normal values:* Lymphocytes <5/mm$^3$; no polymorphs; protein <0.4g/L; glucose >2.2mmol/L (or ≥50% plasma level); pressure <200mm CSF. *In meningitis:* See p822. *In multiple sclerosis:* See p496.

*Bloody tap:* This is an artefact due to piercing a blood vessel, which is indicated (unreliably) by fewer red cells in successive bottles, and no yellowing of CSF (xanthochromia). To estimate how many white cells ($W$) were in the CSF before the blood was added, use the following:

$$W = \text{CSF WCC} - [(\text{blood WCC} \times \text{CSF RBC}) \div \text{blood RBC}].$$

If the blood count is normal, the rule of thumb is to subtract from the total CSF WCC (per μL) one white cell for every 1000 RBCs. To estimate the true protein level, subtract 10mg/L for every 1000 RBCs/mm$^3$ (be sure to do the count and protein estimation on the same bottle). NB: high protein levels in CSF make it appear yellow. *Subarachnoid haemorrhage:* Xanthochromia (yellow supernatant on spun CSF). Red cells in equal numbers in all bottles (unreliable). RBCs will excite an inflammatory response (eg CSF WCC raised), most marked after 48h. *Raised protein:* Meningitis; MS; Guillain–Barré syndrome. *Very raised CSF protein:* Spinal block; TB; or severe bacterial meningitis.

## Complications

- Post-dural puncture headache. • Infection. • Bleeding. • Cerebral herniation (rare, check for signs of ↑ICP before proceeding). • Minor/transient neurological symptoms, eg paraesthesia, radiculopathy.

Any change in lower body neurology after an LP (pain, weakness, sensory changes, bladder/bowel disturbance) should be treated as cauda equina compression (haematoma/abscess) until proven otherwise. Obtain an urgent MRI spine.

Post-LP brain MRI scans often show diffuse meningeal enhancement with gadolinium. This is thought to be a reflection of increased blood flow secondary to intracranial hypotension. Interpret these scans with caution and in the context of the patient's clinical situation. Ensure the reason for the scan and current neurological examination are discussed with the radiologist pre procedure.

## Post-LP headache

**Risk** 10–30%, typically occurring within 24h of LP, resolution over hours to 2wks (mean: 3–4d). Patients describe a constant, dull ache, more frontal than occipital. The most characteristic symptom is of positional exacerbation—worse when upright. There may be mild meningism or nausea. The pathology is thought to be continued leakage of CSF from the puncture site and intracranial *hypo*tension, though there may be other mechanisms involved.

**Prevention** Use the smallest spinal needle that is practical (22G) and keep the bevel aligned as described on p768. Blunt needles (more expensive) can reduce risk and are recommended (ask an anaesthetist about supply); however, collection of CSF takes too long (>6min) if needles smaller than 22G are used. Before withdrawing the needle, reinsert the stilette.

**Treatment** Despite years of anecdotal advice to the contrary, none of the following has ever been shown to be a risk factor: position during or after the procedure; hydration status before, during, or after; amount of CSF removed; immediate activity or rest post-LP. Time is a consistent healer. For severe or prolonged headaches, ask an anaesthetist about a blood patch. This is a careful injection of 20mL of autologous venous blood into the adjacent epidural space (said to 'clog up the hole'). Immediate relief occurs in 95%.

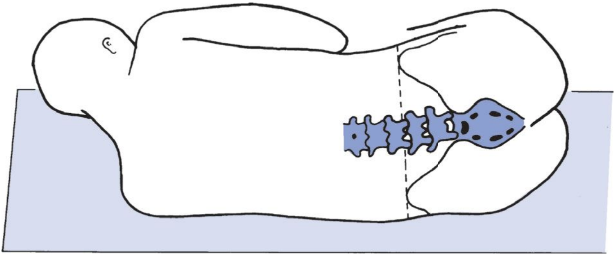


Fig 18.8 Defining the 3rd–4th lumbar vertebral interspace.

Adapted with permission from Vakil *et al.*, *Diagnosis and Management of Medical Emergencies*, 1977 Oxford University Press.

Fig 18.9 Axial T2-weighted MRI of the lumbar spine. The conus ends at the L1/L2 level with continuation of the cauda equina. Lumbar puncture below the L2 level will not damage the cauda equina as the nerve roots will part around an LP needle.

Image courtesy of Norwich Radiology Dept.

# Cardioversion/defibrillation

►Do not wait for a crisis before familiarizing yourself with the defibrillator, as there are several types. All hospitals should include this information in your induction but check how the machine on your ward works.

**Indications** To restore sinus rhythm if VF/VT; AF, flutter, or supraventricular tachycardias if other treatments (p126) have failed, or there is haemodynamic compromise (p130 & p806). This may be done as an emergency, eg VF/VT, or electively, eg AF.

**Aim** To completely depolarize the heart using a direct current.

**Procedure** ►►*For VF/pulseless VT follow the ALS algorithm on p894 and call the arrest team!*

- Unless critically unwell, conscious patients require a general anaesthetic or monitored heavy sedation.
- If elective cardioversion of AF ensure adequate anticoagulation beforehand.
- Almost all defibrillators are now paddle-free and use 'hands-free' pads instead (less chance of skin arc than jelly). Place the pads on chest, one over apex (p39) and one below right clavicle. The positions are often given by a diagram on the reverse of the pad.

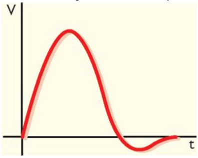



Fig 18.10 The dampened sine monophasic waveform.

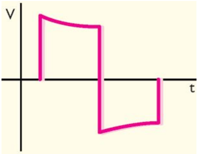



Fig 18.11 Rectilinear biphasic waveform with truncated exponential decay. Most new external defibrillators use this waveform.

*Cardioversion:* Synchronize the shock with the rhythm by pressing the 'SYNC' button on the machine. This ensures the shock does not initiate a ventricular arrhythmia. However, this only works for cardioversion; if the sync mode is engaged in *VF*, the defibrillator will not discharge.

- *Monophasic defibrillators:* (fig 18.10) Set the energy level at 360J for VF/VT (►►arrest situation); 200J for AF; 50J for atrial flutter.
- *Biphasic defibrillators:* (fig 18.11) Impedance is less with a biphasic shock and 120–200J is used for shocks for VF/VT. They use less energy and are just as effective as monophasic defibs in cardioversion. 120–200J will cardiovert most arrhythmias.
- *Automatic external defibrillators:* (AEDs) Can be used by anyone who can turn them on and apply the pads. Follow the instructions given by the AED.

**Shocking**

1. Consider anticoagulation in AF (see p130).
2. Clearly state that you are charging the defibrillator.
3. Make sure no one else is touching the patient, the bed, or anything is in turn touching these.
4. Clearly state that you are about to shock the patient.
5. Give the shock. If there is a change in rhythm before you shock and the shock is no longer required, turn the dial to 'discharge'. Do not allow anyone to approach until the reading has dropped to 0J.
6. After a shock: ►►in resuscitation, resume CPR immediately and do not reassess rhythm until the end of the cycle (see p894, fig A3); in cardioversion, watch ECG; consider need to repeat the shock. Up to three are usual for AF/flutter.
7. Get an up-to-date 12-lead ECG.

►In children, use 4J/kg in VF/VT; see *OHCS* p239.

## Taking arterial blood gas (ABG) samples

Having an artery sampled is more unpleasant for the patient than venepuncture: explain that it is going to feel different and is for a different purpose (p162 for indications and analysis). The usual site is the radial artery at the wrist. ►*Check with the patient that they do not have an arterio-venous fistula for haemodialysis. Never, ever sample from a fistula.*

Fig 18.12 The ideal position for the wrist, slightly hyperextended, resting on an unopened litre bag of fluid or a bandage is ideal. In an unconscious patient or for arterial line insertion, taping the thumb to the bed can hold the wrist in the perfect position if you do not have an assistant.

*Procedure:*

- Get kit ready; include: portable sharps bin; pre-heparinized syringe; needle (blue size (23G) is good, although many syringes now come pre-made with needle); gloves; 2% chlorhexidine/70% alcohol swab; gauze; tape.
- Feel thoroughly for the best site. Look at both sides.
- Wipe with cleaning swab. Let the area dry. Get yourself comfortable.
- If the patient is drowsy or unconscious, ask an assistant to hold the hand and arm with the wrist slightly extended (fig 18.12).
- Before sampling, expel any excess heparin in the syringe. Infiltration over the artery with a small amount of 1% lidocaine (p573) through a 25G (orange) needle makes the procedure painless.
- Hold the syringe like a pen, with the needle bevel up. Let the patient know you are about to take the sample. Feel for the pulse with your other hand and enter at 45°, aiming beneath the finger you are feeling with.
- In most syringes, the plunger will move up on its own in a pulsatile manner if you are in the artery; rarely, entry into a vein next to the artery will give a similar result. Colour of the blood is little guide to its source.
- Allow the syringe to fill with 1–2mL, then remove the needle and apply firm pressure for 5 minutes (10 if anticoagulated).
- Expel any air from the syringe as this will alter the oxygenation of the blood. Cap and label the sample, check the patient's temperature and $FiO_2$ (0.21 if on air). Take the sample to the nearest analysis machine or send it by express delivery to the lab (which may be by your own feet, get someone else to apply pressure) as it should be analysed within 15 minutes of sampling.
- Syringes and analysis machines differ, so get familiar with the local nuances.

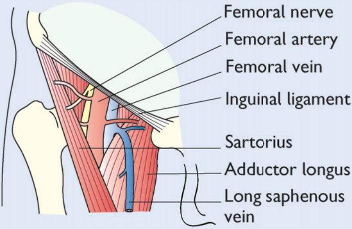



Fig 18.13 The femoral artery is amenable to ABG sampling.

The other site that is amenable to ABG sampling is the femoral artery (fig 18.13). Surprisingly this may be less uncomfortable as it is a relatively less sensitive area and because, when supine, the patient cannot see the needle and thus may feel less apprehensive. The brachial artery can also be used, but be aware that the median nerve sits closely on its medial side and it is an end-artery. Normal values: p753.

# Emergency airway management



**Cricothyroidotomy** Thsi is an emergency procedure to overcome airway obstruction above the level of the larynx. It should only be done in absolute 'can't intubate, can't ventilate' situations, ie where ventilation is impossible with a bag and mask (± airway adjuncts) and where there is an immediate threat to life. If not, call anaesthetics or ENT for immediate help.

**Indications** Upper airway obstruction when endotracheal intubation not possible, eg irretrievable foreign body; facial oedema (burns, angio-oedema); maxillofacial trauma; infection (epiglottitis).

**Procedure** Lie the patient supine with neck extended (eg pillow under shoulders) unless there is suspected cervical-spine instability. Run your index finger down the neck anteriorly in the midline to find the notch in the upper border of the thyroid cartilage (the Adam's apple): just below this, between the thyroid and cricoid cartilages, is a depression—the cricothyroid membrane (see fig 18.14). If you cannot feel the depression and it is an emergency, you can access the trachea directly approximately halfway between the cricoid cartilage and the suprasternal notch.

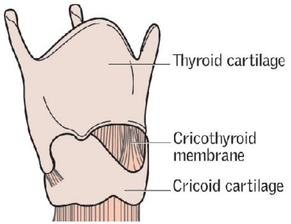



Fig 18.14 The cricothyroid membrane.

Ideally use a purpose-designed kit (eg QuickTrach®, MiniTrach®), all hospitals will stock one version. If no kit is available then a cannula (needle cricothyroidotomy) can buy time, and in out-of-hospital situations a blade and empty biro case have saved lives. ▸▸Needle and kit cricothyroidotomies are temporary measures pending formal tracheostomy.

**1** *Needle cricothyroidotomy:* Pierce the membrane perpendicular to the skin with large-bore cannula (14G) attached to syringe: withdrawal of air confirms position; lidocaine may or may not be required. Slide cannula over needle at 45° to the skin superiorly in the sagittal plane. Use a Y-connector (see fig 18.16) or improvise connection to $O_2$ supply at 15L/min: use thumb on Y-connector to allow $O_2$ in over 1s and $CO_2$ out over 4s ('transtracheal jet insufflation'). This is the preferred method in children <12yrs. This will only sustain life for 30–45min before $CO_2$ builds up. However, if the patient has a completely obstructed airway then they will not be able to exhale through this, and it will lead to cardiovascular compromise and pneumothoraces.

**2** *Cricothyroidotomy kit:* Most contain a guarded blade, and a large (4–6mm) shaped cannula (cuffed or uncuffed depending on brand) over an introducer, plus a connector and binding tape. The patient will have to be ventilated via a bag, as the resistance is too high to breathe spontaneously. This will sustain for 30–45min.

**3** *Surgical cricothyroidotomy:* Smallest tube for prolonged ventilation is 6mm. Introduce high-volume, low-pressure cuff tracheostomy tube through a horizontal incision in membrane. Take care not to cut the thyroid or cricoid cartilages.

**Complications** Local haemorrhage ± aspiration; posterior perforation of trachea ± oesophagus; subglottic stenosis; laryngeal stenosis if membrane over-incised in childhood; tube blockage; subcutaneous tunnelling; vocal cord paralysis or hoarseness (the recurrent laryngeal nerve runs superiorly in the tracheo-oesophageal groove).

## Emergency needle pericardiocentesis

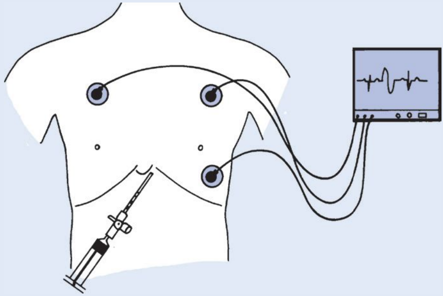


Fig 18.15 Emergency needle pericardiocentesis.

- Get your senior's help (for whom this page may serve as an *aide-memoire*).
- Equipment: 20mL syringe, long 18G cannula, 3-way tap, ECG monitor, skin cleanser. Use ECHO guidance if there is time.
- If time allows, use full aseptic technique, at a minimum clean skin with 2% chlorhexidine in 70% alcohol and wear sterile gloves, and, if conscious, use local anaesthesia and sedation, eg with slow IV midazolam: titrate up to 3.5–5mg—start with 2mg over 1min, 0.5–1mg in elderly (in whom the maximum dose is 3.5mg; inject at the rate of 2mg/min)—antidote: flumazenil 0.2mg IV over 15s, then 0.1mg every 60s, up to 1mg in total.
  ►*Ensure you have IV access and full resuscitation equipment to hand.*
- Introduce needle at 45° to skin just below and to left of xiphisternum, aiming for tip of left scapula (fig 18.15). Aspirate continuously and watch ECG. Frequent ventricular ectopics or an injury pattern (↓ST segment) on ECG imply that the myocardium has been breached—withdraw slightly. As soon as fluid is obtained through the needle, slide the cannula into place.
- Evacuate pericardial contents through the syringe and 3-way tap. Removal of only a small amount of fluid (eg 20mL) can produce marked clinical improvement. If you are not sure whether the fluid you are aspirating is pure blood (eg on entering a ventricle), see if it clots (heavily bloodstained pericardial fluid does not clot), or measure its PCV (though this may be difficult in the acute setting but some blood gas analysers may give this).
- You can leave the cannula *in situ* temporarily, for repeated aspiration. If there is reaccumulation, insert a drain but pericardiectomy may be needed.
- Send fluid for microscopy and culture, as needed, including tests for TB.

*Complications:* Laceration of ventricle or coronary artery (± subsequent haemopericardium); aspiration of ventricular blood; arrhythmias (ventricular fibrillation); pneumothorax; puncture of aorta, oesophagus (± mediastinitis), or peritoneum (± peritonitis).

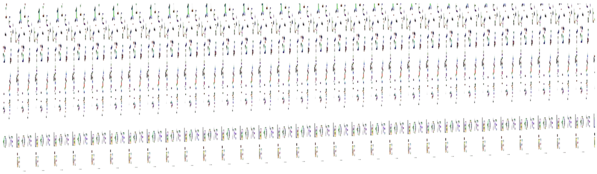


Fig 18.16 Methods of providing oxygen.

# Central venous cannulation



Central venous cannulae may be inserted to measure central venous pressure (CVP), to administer certain drugs (eg amiodarone, chemotherapy), or for intravenous access (fluid, parenteral nutrition). In an emergency, the procedure can be done using the landmark method (see p775), though NICE recommends that all routine internal jugular catheters should be placed with US guidance. Even if the line is not placed under direct US visualization, a look to check vessel size, position in relation to artery, and patency (no thrombus or stenosis) is extremely useful. For contraindications, see table 18.2.

Table 18.2 Contraindications to central venous cannulation

| Absolute | Relative |
|---|---|
| Infection at insertion site | Coagulopathy |
| | Ipsilateral carotid endarterectomy |
| | Newly inserted cardiac pacemaker leads |
| | Thrombus within the vein |
| | Venous stenosis |
| | Ipsilateral abnormal anatomy |

**Sites of insertion** These include the internal jugular vein (see p775 and p43), subclavian vein, and the femoral vein. The choice depends largely on operator experience, but evidence suggests that the femoral approach is associated with a higher rate of line infection and thrombosis. Overall, the internal jugular approach (with ultrasound guidance) is most commonly used and risks fewer complications than the subclavian. If possible, get written consent (p568). Check clotting and platelets. The technique for internal jugular (routine) and femoral (emergency) are given here.

**Complications** (~20%.) Insertion is not without hazard, so decide whether the patient requires a line first, and then ask for help if you are inexperienced.
►Bleeding; arterial puncture/cannulation; AV fistula formation; air embolism; pneumothorax; haemothorax; chylothorax (lymph); phrenic nerve palsy (the right phrenic nerve passes over the brachiocephalic artery, posterior to the subclavian vein—hiccups may be a sign of injury); phlebitis; thrombus formation on tip or in vein (if high risk for thromboembolism, eg malignancy, consider anticoagulation, eg LMWH); bacterial colonization; cellulitis; sepsis (can be reduced by adherence to a strict aseptic technique; if taking blood cultures in a febrile patient with a central venous line, remember to take samples from the central line and from a peripheral vein).

**Peripherally inserted central cannulas (PICC lines)** These are a good alternative to central lines, as they can stay *in situ* for up to 6 months, and provide access for blood sampling, fluids, antibiotics (allowing home IV therapy). They are placed using a Seldinger technique, puncturing the brachial or basilic vein then threading the line into the subclavian or superior vena cava. Because of the insertion site there is a much lower risk of pneumo- or haemothorax, but they are tricky to insert in an emergency.

**Removing central lines** Should be done carefully with aseptic technique. Position the patient slightly head down, remove dressings, clean and drape the area, remove sutures. Ask the patient to inhale and hold their breath, then ►breathe out smoothly while you are pulling the line out. This helps to prevent air emboli. Ask the patient to rehearse this sequence with you to ensure they have understood their role. Apply pressure for 5 minutes (longer if coagulopathic).

## Internal jugular catheterization

**Internal jugular** Should be the approach of choice in a non-emergency situation. Ideally the right side as it offers a direct route to the heart and there is less chance of misplacement of the line compared to the left. The subclavian approach is trickier and best taught by an expert. Use US guidance if at all possible, ideally to insert the line under direct vision, but at least to define the anatomy. If possible, have the patient attached to a cardiac monitor in case of arrhythmias.

- Position the patient slightly head down to avoid air embolism and fill the veins to improve your chances of success. ►This can compromise cardiac function and precipitate acute LVF so check if your patient has a cardiac history. Minimize the time the patient is head down; if they are unable to lie flat, consider a femoral approach. Turn their head slightly to the left.
- This should be a sterile procedure so use full aseptic technique (hat, mask, gloves, gown) and clean with 2% chlorhexidine in 70% isopropyl alcohol before draping. Ensure your equipment is prepared, flush the catheter lumens with saline.
- If US is unavailable, the *landmark procedure* can be used to identify insertion point—approximately at the junction of the two heads of sternocleidomastoid at about the level of the thyroid cartilage (fig 18.17). Feel gently for the carotid pulse, then infiltrate with 1% lidocaine just lateral to this. The vein is usually superficial (fig 18.18).
- Insert the introducer needle with a 5mL syringe attached, advance gently at a 45° angle, aiming for the ipsilateral nipple and aspirating continuously. If you are using US, watch the needle tip enter the vein, if the landmark approach keep your fingers on the carotid pulse.
- As soon as blood is aspirated, lay down the US probe and hold the introducer needle in position, remove the syringe and thread the guidewire through the needle. It should pass easily, if there is resistance try lowering the angle of the needle and gently advancing the wire. If the wire will not pass do not remove it alone, the tip can shear off and embolize; remove the needle with the wire, apply pressure and attempt a second puncture.

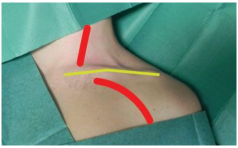


**Fig 18.17** Position of internal jugular and subclavian veins (red) compared to the clavicle (yellow).

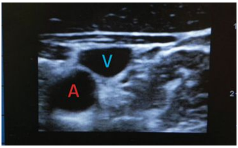



**Fig 18.18** Vessels seen on ultrasound. The compressible vein is above the artery.

- If the wire threads easily, insert to 30cm (see markings on the wire), remove the needle keeping hold of the wire at all times. Make a nick in the skin with a scalpel at the insertion point, and gently thread the dilator over the wire. You do not need to insert the dilator far, only as far as the vein (you often feel a loss of resistance as the dilator enters the vein, so insert gently: a pneumothorax can result from enthusiastic dilating).
- Remove the dilator, keeping hold of the wire, thread the flushed catheter over the wire, then remove the wire. The line should sit at about 13cm on the right side (17cm on the left). Check you can aspirate blood from each lumen, then flush them.
- Suture the catheter in place (many have little 'wings' for suturing) and dress. Request a CXR to confirm position and exclude pneumothorax. The tip of the catheter should sit vertically in the SVC.

**Femoral vein** In an emergency situation where ultrasound is not easily accessible, if the patient is unable to lie flat, or where speed is of the essence, the femoral approach is often the safest, as there is no risk of pneumothorax or haemothorax and a much reduced risk of arrhythmia. The technique is similar to internal jugular, except the insertion point is just medial to the femoral artery at the groin crease.

**Subclavian vein** Should be taught by an expert and should ideally be carried out under US guidance. Some physicians prefer this approach, but even in experienced hands there is a ↑risk of complications compared to US-guided internal jugular lines.

# Inserting a temporary cardiac pacemaker



Often it is wiser to liaise with a specialist pacing centre to arrange prompt, definitive pacing than to try temporary transvenous pacing, which often has complications (see later in topic) and therefore may delay a definitive procedure.

## Possible indications in the acute phase of myocardial infarction

- *Complete AV block:*
  - With inferior MI (right coronary artery occlusion) pacing may only be needed if symptomatic; spontaneous recovery may occur.
  - With anterior MI (representing massive septal infarction).
- *Second-degree block:*
  - Wenckebach (p99; implies decremental AV node conduction; may respond to atropine in an inferior MI; pace if anterior MI).
  - Mobitz type 2 block is usually associated with distal fascicular disease and carries high risk of complete heart block, so pace in both types of MI.
- *First-degree block:* Observe carefully: 40% develop higher degrees of block.
- *Bundle branch block:* Pace prophylactically if evidence of trifascicular disease (p100) or non-adjacent bifascicular disease.
- *Sino-atrial disease + serious symptoms:* Pace unless responds to atropine.

## Other indications where temporary pacing may be needed

- Pre-op: if surgery is required in patients with type 2 or complete heart block (whether or not MI has occurred); do 24h ECG; liaise with the anaesthetist.
- Drug poisoning, eg with β-blockers, digoxin, or verapamil.
- Symptomatic bradycardia, uncontrolled by atropine or isoprenaline.
- Suppression of drug-resistant VT and SVT (overdrive pacing; do on ICU).
- Asystolic cardiac arrest with P-wave activity (ventricular standstill).
- During or after cardiac surgery—eg around the AV node or bundle of His.

**Technique for temporary transvenous pacing** Learn from an expert.

- *Preparation:* Monitor ECG; have a defibrillator to hand, ensure the patient has peripheral access; check that a radiographer with screening equipment is present. If you are screening, wear a protective lead apron.
- *Insertion:* Using an aseptic technique, place the introducer into the (ideally right) internal jugular vein (p775) or subclavian. If this is difficult, access to the right atrium can be achieved via the femoral vein. Pass the pacing wire through the introducer into the right atrium, ideally under radiological screening. It will either pass easily through the tricuspid valve or loop within the atrium. If the latter occurs, it is usually possible to flip the wire across the valve with a combined twisting and withdrawing movement (fig 18.19). Advance the wire slightly. At this stage the wire may try to exit the ventricle through the pulmonary outflow tract. A further withdrawing and rotation of the wire will aim the tip at the apex of the right ventricle. Advance slightly again to place the wire in contact with the endocardium. Remove any slack to ↓risk of subsequent displacement.
- *Checking the threshold:* Connect the wire to the pacing box and set the 'demand' rate slightly higher than the patient's own heart rate and the output to 3V. A paced rhythm should be seen. Find the pacing threshold by slowly reducing the voltage until the pacemaker fails to stimulate the tissue (pacing spikes are no longer followed by paced beats). The threshold should be less than 1V, but a slightly higher value may be acceptable if it is stable—eg after a large infarction.
- *Setting the pacemaker:* Set the output to 3V or over 3 times the threshold value (whichever is higher) in 'demand' mode. Set the rate as required. Suture the wire to the skin, and fix with a sterile dressing.
- Check the position of the wire (and exclude pneumothorax) with a CXR.
- Recurrent checks of the pacing threshold are required over the next few days. The formation of endocardial oedema can raise the threshold by a factor of 2–3.

**Complications** Pneumothorax; sepsis; cardiac perforation; pacing failure: from loss of capture, loss of electrical continuity in pacing circuit, or electrode displacement.

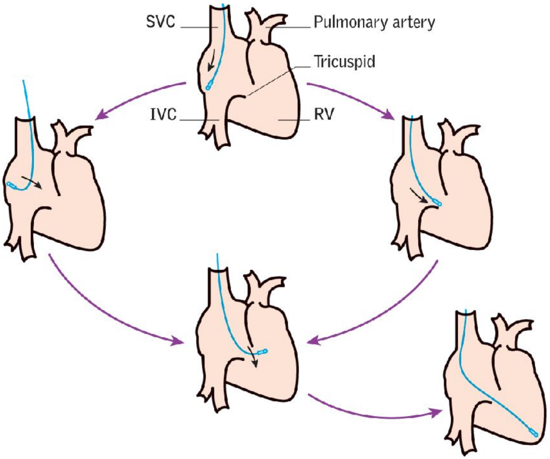



Fig 18.19 Siting a temporary cardiac pacemaker.

## Non-invasive transcutaneous cardiac pacing

This method (performed through a defibrillator with external pacing facility) has the advantages of being quicker, less risky than the transvenous route, and easier to perform. Its main disadvantage is the pain caused by skeletal muscle contraction in the non-sedated patient. Indications for pacing via the transcutaneous route are as p776, *plus* if transvenous pacing (or someone able to perform it) is unavailable or will be delayed in an emergency situation.

- Give sedation and analgesia, eg midazolam + morphine IV titrated to effect.
- Clipping chest hair may help improve electrical contact; ►don't shave the skin, as nicks can predispose to electrical burns. Ensure the skin is dry.
- Almost all modern transcutaneous devices can function through defibrillation 'hands-free' pads, and so these can be applied as for defibrillation (see p770). If necessary, the pads can be placed in an AP position: anteriorly over the $V_2$–$V_3$ electrode position and posteriorly at the same level, just below the scapula.
- Select 'demand' mode, (which synchronizes the stimulus with the R wave, so avoiding pacing on the T wave—which can provoke VF or VT) and adjust the ECG gain so that QRS complexes can be seen.
- Select an appropriate pacing rate: eg 60–90bpm in an adult.
- Set the pacing current at the lowest setting and turn on the pacemaker.
- Increase the pacing current until electrical capture occurs (normally from 50–100mA), which can be confirmed by seeing a wide QRS complex and a T wave on the trace (ventricular electrical capture).
- There will be some interference from skeletal muscle contraction on the ECG trace, as well as possible artefact, which could be mistaken for a QRS complex. The absence of a T wave in the former is an important discriminator between the two.
- CPR can continue with the pads in place, though only when the pacing unit is *off*.
- Once adequate cardiac output has been maintained, seek expert help and arrange transvenous pacing.

# 19 Emergencies

## Contents




**Fig 19.1** Dr Leander Starr Jameson (1853–1917) was a Victorian physician barely known to the history of our profession and yet immortalized by the idealization of his character in Rudyard Kipling's poem '*If*'. Kipling's words contain much that might guide the doctor faced with a sudden and unexpected emergency. Beyond the oft quoted exhortation to '*keep your head when all about you are losing theirs*', Kipling also endorsed an appropriate degree of self-confidence ('*trust yourself when all men doubt you, But make allowance for their doubting too*'), encouraged realism ('*meet with Triumph and Disaster And treat those two impostors just the same*'), and a large measure of courage ('*force your heart and nerve and sinew To serve your turn long after they are gone, And so hold on when there is nothing in you Except the Will which says to them: "Hold on!"*'). But what of Dr James? As a house officer in London, he broke down from overwork and moved to take up what would become a distinguished practice in South Africa. A foray into the military ill-suited him and he ended up in jail for leading a botched raid that was so out-of-character that it seems he allowed himself to become a scapegoat for others higher in government. He died in November 1917, outliving Kipling's son John, who was killed in the trenches in 1915 at the age of 18, and to whom '*If*' was addressed: '*Yours is the Earth and everything that's in it, And—which is more—you'll be a Man, my son.*'




*We thank Dr Andrew Johnston and Bernard Ho, our Junior Reader, for their contribution.*

## Introduction to emergencies

Some doctors enjoy the adrenaline rush of seeing an emergency case and pulling someone back from the edge of death. Some fear the emergency take, worried as to what they will miss, how many will die. There is no right approach; the emergency room needs both thought and practicality to manage patients. A patient who comes in flat and can be resuscitated gives the whole team a boost, as right there in front of us is the proof that we can make a difference. However, a patient who comes in well and collapses, dying before we can even decide what is wrong, can bring the whole team down. There are nights where we save life after life, and nights where we can't; sometimes a death is inevitable, despite our best efforts. You are a doctor, but you are a human being as well, and losing a patient can feel like a personal failure. However, remember that when we lose a patient it is the disease that has killed them, not us. Try to take a few minutes and reflect on what happened; ask if you could have done anything differently. Should you have sought help sooner? Discussing with a senior can be helpful, as can writing down your reflection, not in a portfolio for discussion at your appraisal, but in anonymous format for your own education. Watch the team at an arrest, the best leaders are the ones who have learned to stand back, assess the whole situation, and take enough time to see where the critical intervention is needed. There is no substitute for experience, nobody becomes a consultant overnight, but watch the best clinicians at work and you will learn both practical and life skills.

Most important in an emergency situation is communication. Wherever you can, involve the relatives and the whole team in discussions, but remember that at the heart of this is a patient. What do they want? Never be afraid to ask the patient directly, they may hold very strong views. However, it is up to us as physicians to be honest with them about their prognosis—do not offer a treatment you know is not in the best interests of the patient, and this includes resuscitation. When faced with death, many patients are afraid—our role is to try to relieve that fear, whether by intervening to delay death, or by easing their passing. But we cannot ever prevent death, we simply delay it. As Shakespeare has *Julius Caesar* say:

'*Of all the wonders that I have yet heard, it seems to me most strange that men should fear, seeing that death, a necessary end, will come when it will come.*'

## ABCDE preliminary assessment (primary survey)

| | |
|---|---|
| **A**irway | Protect cervical spine, if injury possible.<br>*Assessment:* any signs of obstruction? Ascertain patency.<br>*Management:* establish a patent airway. |
| **B**reathing | *Assessment:* determine respiratory rate, check bilateral chest movement, percuss, and auscultate.<br>*Management:* if no respiratory effort, treat as arrest (see p894, fig A3), intubate and ventilate. If breathing compromised, give high-concentration $O_2$, manage according to findings, eg relieve tension pneumothorax. |
| **C**irculation | *Assessment:* check pulse and BP; check if peripherally shut down; check capillary refill; look for evidence of haemorrhage.<br>*Management:* if shocked, treat as on p790.<br>If no cardiac output, treat as arrest (see p894, fig A3). |
| **D**isability | Assess 'level of consciousness' with **AVPU** score (**a**lert? responds to **v**oice? to **p**ain? **u**nresponsive?); check pupils: size, equality, reactions. *Glasgow Coma Scale*, if time allows. |
| **E**xposure | Undress patient, but cover to avoid hypothermia. |

Quick history from relatives assists diagnosis: *Events* surrounding onset of illness, evidence of overdose/suicide attempt, any suggestion of trauma? *Past medical history,* especially diabetes, asthma, COPD, alcohol, opiate or street drug abuse, epilepsy or recent head injury, recent travel. *Medication,* current drugs. *Allergies*.

Once ventilation and circulation are adequate, proceed to carry out history, examination, investigations, and management in the usual way.

## Headache: differential diagnosis

The vast majority of headaches are benign, but when taking a history do not forget to ask about the following[1] (early diagnosis can save lives):

### Worrying features or 'red flags'

- First and worst headache—*subarachnoid haemorrhage* (p478).
- Thunderclap headache—*subarachnoid haemorrhage* (p478: p480 for other causes).
- Unilateral headache and eye pain—*cluster headache, acute glaucoma* (p456).
- Unilateral headache and ipsilateral symptoms—*migraine, tumour, vascular* (p458).
- Cough-initiated headache—↑*ICP/venous thrombosis* (p480).
- Worse in the morning or bending forward—↑*ICP/venous thrombosis* (p480).
- Persisting headache ± scalp tenderness in over-50s—*giant cell arteritis* (p556).
- Headache with fever or neck stiffness—*meningitis* (p822).
- Change in the pattern of 'usual headaches' (p456).
- Decreased level of consciousness (p456).

*Two other vital questions:*

- Where have you been? (Malaria, p416).
- Might you be pregnant? (Pre-eclampsia; especially if proteinuria and ↑BP, p458.)

Always examine a patient presenting with a severe headache; if nothing about history or examination is concerning, both you and the patient will be reassured, but subtle abnormalities are important not to miss.

### No signs on examination

- Tension headache (p456).
- Migraine (p458).
- Cluster headache (p457).
- Post-traumatic (p456).
- Drugs (nitrates, calcium-channel antagonists) (p114).
- Carbon monoxide poisoning or anoxia. (p842)
- Subarachnoid haemorrhage (p478).

### Signs of meningism?

- Meningitis (may not have fever or rash—p822).
- Subarachnoid haemorrhage (p478—examination may be normal).

### Decreased conscious level or localizing signs?

- Stroke (p470).
- Encephalitis/meningitis (p822).
- Cerebral abscess (p824).
- Subarachnoid haemorrhage (pp478–9, figs 10.17, 10.18).
- Venous sinus occlusion (p480—focal neurological deficits).
- Tumour (p498).
- Subdural haematoma (p482).
- TB meningitis (p393).

### Papilloedema?

- Tumour (p498).
- Venous sinus occlusion (p480—focal neurological deficits).
- Malignant (accelerated phase) hypertension (p138).
- Idiopathic intracranial hypertension (p498).
- Any CNS infection, if prolonged (eg >2wks)—eg TB meningitis (p393).

### Others

- Giant cell arteritis (p556—↑ESR and tender scalp over temporal arteries).
- Acute glaucoma (p456—painful red eye—get pressures checked urgently).
- Vertebral artery dissection (p470—neck pain and cerebellar/medullary signs).
- Cervical spondylosis (p508).
- Sinusitis.
- Paget's disease (p685—↑ALP).
- Altitude sickness (*OHCS* p770).

# Breathlessness: emergency presentations

There may not be time to ask or the patient may not be able to give you a history in acute breathlessness, this in itself can be a helpful sign (inability to complete sentences in one breath = severe breathlessness, inability to speak/impaired conscious level = life-threatening). Collateral history of known respiratory disease, anaphylaxis, or other history can be extremely helpful but do not delay. Assess the patient for the following:

### Wheezing?
- Asthma (p810).
- COPD (p812).
- Heart failure (p800).
- Anaphylaxis (p794).

**Stridor?** (Upper airway obstruction.)
- Foreign body or tumour.
- Acute epiglottitis (younger patients).
- Anaphylaxis (p794).
- Trauma, eg laryngeal fracture.

### Crepitations?
- Heart failure (p800).
- Pneumonia (p816).
- Bronchiectasis (p172).
- Fibrosis (p198).

### Chest clear?
- Pulmonary embolism (p818).
- Hyperventilation.
- Metabolic acidosis, eg diabetic ketoacidosis (p832).
- Anaemia (p324).
- Drugs, eg salicylates.
- Shock (may cause 'air hunger', p790).
- *Pneumocystis jirovecii* pneumonia (p400).
- CNS causes.

### Others
- Pneumothorax (p814—pain, increased resonance, tracheal deviation if tension pneumothorax).
- Pleural effusion (p192—'stony dullness').

### Key investigations
- Baseline observations—$O_2$ sats, pulse, temperature, peak flow.
- ABG if saturations <94% or concern about acidosis/drugs/sepsis.
- ECG (signs of PE, LVH, MI?).
- CXR.
- Baseline bloods: glucose, FBC, U&E, consider drug screen.

# Chest pain: differential diagnosis



First exclude any potentially life-threatening causes, by virtue of history, brief examination, and limited investigations. Then consider other potential causes. For the full assessment of cardiac pain, see pp94, 118.

## Life-threatening

- Acute myocardial infarction (pp796–9).
- Angina/acute coronary syndrome (pp796–9).
- Aortic dissection (p655).
- Tension pneumothorax (p814).
- Pulmonary embolism (p818).
- Oesophageal rupture (p820).

## Others

- Pneumonia (p816).
- Chest wall pain:
  - Muscular.
  - Rib fractures.
  - Bony metastases.
  - Costochondritis.
- Gastro-oesophageal reflux (p254).
- Pleurisy (p166).
- Empyema (p170).
- Pericarditis (p154).
- Oesophageal spasm (p250).
- Herpes zoster (p404).
- Cervical spondylosis (p508).
- Intra-abdominal:
  - Cholecystitis (p634).
  - Peptic ulceration (p252).
  - Pancreatitis (p270).
- Sickle-cell crisis (p340).

Before discharging patients with undiagnosed chest pain, be sure in your own mind that the pain is not cardiac (this pain is usually dull, may radiate to jaw, arm, or epigastrium, and is usually associated with exertion). Carry out key investigations and discuss options with a colleague, and the patient. ***Safety-net***, telling the patient to return or seek advice if they develop worrying features (specify these) or the pain does not settle.

## Key investigations

- CXR.
- ECG.
- FBC, U&E, and troponin (p118). Consider D-dimer only if low probability of venous thromboembolism. See 'Modified Wells' score for PE, p191.

►Just because the patient's chest wall is tender to palpation, this doesn't mean the cause of the chest pain is musculoskeletal. Even if palpation reproduces the same type of pain, ensure that you exclude all potential life-threatening causes. Although chest wall tenderness has discriminatory value against cardiac pain, it may be a feature of a pulmonary embolism.

# Coma

**Definition** *Unrousable unresponsiveness*. Quantify using *Glasgow Coma Scale* (GCS).

## Causes of impaired conscious level/coma

*Metabolic:*
- Drugs, poisoning, eg carbon monoxide, alcohol, tricyclics.
- Hypoglycaemia, hyperglycaemia (ketoacidotic, or HONK, pp832–4).
- Hypoxia, $CO_2$ narcosis (COPD).
- Septicaemia (p792).
- Hypothermia.
- Myxoedema (p834), Addisonian crisis (p836).
- Hepatic/uraemic encephalopathy (pp275, 298).

*Neurological:*
- Trauma.
- Infection: meningitis (p822); encephalitis (eg herpes simplex—p404), *tropical*: malaria (p416; do thick films), typhoid, typhus, rabies, trypanosomiasis.
- Tumour: 1° or 2° (p528).
- Vascular: stroke (p470), subdural (p482), subarachnoid (p478), hypertensive encephalopathy (p140).
- Epilepsy: non-convulsive status (p484) or post-ictal state.

**Immediate management** See fig 19.2 (and coma CNS exam, p789).
- Assess **A**irway, **B**reathing, and **C**irculation. Consider intubation if GCS <8 (p788). Support the circulation if required (ie IV fluids). Give $O_2$ and treat any seizures. Protect the cervical spine unless trauma is known not to be the cause.
- Check blood glucose; give eg 200mL 10% glucose IV stat if hypoglycaemia possible.
- IV thiamine if any suggestion of Wernicke's encephalopathy; see later in topic.
- IV naloxone (0.4–2mg IV) for opiate intoxication (may also be given IM or via ET tube); IV flumazenil (p842) for benzodiazepine intoxication only if airway compromised as risk of seizures especially if concomitant tricyclic intoxication.

**Examination** ►*Vital signs are vital—obtain full set, including temperature.*
- Signs of trauma—haematoma, laceration, bruising, CSF/blood in nose or ears, fracture 'step' deformity of skull, subcutaneous emphysema, 'panda eyes'.
- Stigmata of other illnesses: liver disease, alcoholism, diabetes, myxoedema.
- Skin for needle marks, cyanosis, pallor, rash (meningitis; typhus), poor turgor.
- Smell the breath (alcohol, hepatic fetor, ketosis, uraemia).
- Opisthotonus (fig 9.45, p436) ≈meningitis or tetanus. Decerebrate/decorticate (p788)?
- Meningism (pp456, 822) ►but do *not* move neck unless cervical spine is cleared.
- Pupils (p789) size, reactivity, gaze.
- Heart/lung exam for BP, murmurs, rubs, wheeze, consolidation, collapse.
- Abdomen/rectal for organomegaly, ascites, bruising, peritonism, melaena.
- Are there any foci of infection (abscesses, bites, middle ear infection)?
- Any features of meningitis: neck stiffness, rash, focal neurology?
- Note the *absence* of signs, eg *no* pin-point pupils in a known heroin addict.

**Quick history** From family, ambulance staff, bystanders: abrupt or gradual onset? How found—suicide note, seizure? If injured, suspect cervical spinal injury and do not move spine (*OHCS* p782). Recent complaints—headache, fever, vertigo, depression? Recent medical history—sinusitis, otitis, neurosurgery, ENT procedure? Past medical history—diabetes, asthma, ↑BP, cancer, epilepsy, psychiatric illness? Drug or toxin exposure (especially alcohol or other recreational drugs)? Any travel?

*If the diagnosis is unclear:*
- Treat the treatable: $O_2$; naloxone as above; glucose (eg 200mL of 10% IV); Pabrinex® IV for Wernicke's encephalopathy, p714; septic specifics: cefotaxime 2g/12h IV (meningitis, p822), artemether/quinine (malaria, p418), aciclovir (encephalitis, p824).
- Do routine biochemistry, haematology, thick films, blood cultures, blood ethanol, drug screen, etc.
- Arrange urgent CT head, if normal, and no CI, proceed to LP.

The diagnosis should now be clear, eg hypo/hyperglycaemia; alcohol excess; poisoning; uraemia; pneumonia; subarachnoid; hypertensive/hepatic encephalopathy.

## ⚠ Managing coma

ABC of life support

↓

IV access

↓

Stabilize the cervical spine (vital if trauma is a possibility)

↓

Blood glucose (fingerprick & lab)

↓

Control seizures

↓

Treat potential causes, eg IV glucose, thiamine, naloxone (if pupils small or if possible narcotic use). Other antidotes: see p842

↓

Brief collateral history & examination. Get details later

↓

**Investigations**

- ABG, FBC, U&E, LFT, CRP, ethanol, toxin screen, drug levels
- Blood cultures, urine culture, consider malaria
- CXR, CT head

↓

Reassess the situation and plan further investigations

Fig 19.2 Managing coma. NB: check pupils every few minutes during the early stages, particularly if trauma is the likely cause. Doing so is the quickest way to find a localizing sign (so helpful in diagnosis, but remember that false localizing signs do occur)—and observing changes in pupil behaviour (eg becoming fixed and dilated) is the quickest way of finding out just how bad things are.

## The Glasgow Coma Scale (GCS)

This gives a reliable, objective way of recording the conscious state of a person. It can be used by medical and nursing staff for initial and continuing assessment. It has value in predicting ultimate outcome. Three types of response are assessed, note in each case the best response (or best of any limb) which should be recorded (table 19.1).

Table 19.1 The Glasgow Coma Scale

| | ⚠ Best motor response | | Best verbal response | | Eye opening |
|---|---|---|---|---|---|
| **6** | Obeying commands | **5** | Oriented (time, place, person) | **4** | Spontaneous |
| **5** | Localizing to pain | **4** | Confused conversation | **3** | In response to speech |
| **4** | Withdrawing to pain | **3** | Inappropriate speech | **2** | In response to pain |
| **3** | Flexor response to pain | **2** | Incomprehensible sounds | **1** | None |
| **2** | Extensor response to pain | **1** | None | | |
| **1** | No response to pain | | | | |

Adapted from 'Assessment of coma and impaired consciousness: a practical scale', Graham Teasdale and Bryan Jennett, *The Lancet*, Vol 304, No. 7872, 81–84 (1974), Elsevier.

An overall score is made by summing the score in the three areas assessed.
- No response to pain + no verbalization + no eye opening = 3.
- Severe injury, GCS ≤8—consider airway protection.
- Moderate injury, GCS 9–12.
- Minor injury, GCS 13–15.

Causing pain is not a pleasant thing, there are acceptable and unacceptable methods. Try fingernail bed pressure with a pen/pencil, sternal pressure (not a rub), or suprascapular squeeze. Abnormal responses to pain can help to localize the damage:
- *Flexion* = decorticate posture (arms bent inwards on chest, thumbs tucked in a clenched fist, legs extended) implies damage above the level of the red nucleus in the midbrain.[1]
- *Extension* = decerebrate posture (adduction and internal rotation of shoulder, pronation of forearm) indicates midbrain damage below the level of the red nucleus.

NB: an abbreviated coma scale, **AVPU**, is sometimes used in the initial assessment ('primary survey') of the critically ill:

**A** = **a**lert

**V** = responds to **v**ocal stimuli

**P** = responds to **p**ain

**U** = **u**nresponsive

NB: GCS scoring is different in young children; see *OHCS* p201.

Emergencies

1 Red nucleus output reinforces upper limb antigravity flexion. When its output is damaged, the unregulated reticulospinal and vestibulospinal tracts reinforce extension tone of upper and lower limbs.

## The neurological examination in coma

This is aimed at locating the pathology in one of two places. Altered level of consciousness implies either:

1. A diffuse, bilateral, cortical dysfunction (usually producing loss of awareness with normal arousal), or
2. Damage to the ascending reticular activating system (ARAS) located throughout the brainstem from the medulla to the thalami (usually producing loss of arousal with unassessable awareness). The brainstem can be affected directly (eg pontine haemorrhage) or indirectly (eg compression from transtentorial or cerebellar herniation secondary to a mass or oedema).

*Systematic examination:*

- Level of consciousness; describe using *objective* words/AVPU.
- Respiratory pattern (p53)—Cheyne-Stokes (brainstem lesions or compression) hyperventilation (acidosis, hypoxia, or, rarely, neurogenic), ataxic or apneustic (breath-holding) breathing (brainstem damage with grave prognosis).
- Eyes—almost all patients with ARAS pathology will have eye findings:
  1. *Visual fields*—in light coma, test fields with visual threat. No blink in one field suggests hemianopia and contralateral hemisphere lesion.
  2. *Pupils—normal direct and consensual reflexes present* = intact midbrain. *Midposition (3-5mm) non-reactive ± irregular* = midbrain lesion. *Unilateral dilated and unreactive* ('fixed') = 3rd nerve compression. *Small, reactive* = pontine lesion ('pin-point pontine pupils') or drugs. *Horner's syndrome* (p702, fig 15.4) = ipsilateral lateral medulla or hypothalamus lesion, may precede uncal herniation. Beware patients with false eyes or who use eye drops for glaucoma.
  3. *Extraocular movements (EOMS)*—observe resting position and spontaneous movement; then test the vestibulo-ocular reflex (VOR) with either the *doll's-head manoeuvre* (normal if the eyes keep looking at the same point in space when the head is quickly moved laterally or vertically) or *ice water calorics* (normal if eyes deviate towards the cold ear with nystagmus to the other side). If present, the VOR exonerates most of the brainstem from the VIIth nerve nucleus (medulla) to the IIIrd (midbrain). *Don't move the head unless the cervical spine is cleared*.
  4. *Fundi*—papilloedema, subhyaloid haemorrhage, hypertensive retinopathy, signs of other disease (eg diabetic retinopathy).
- Examine for CNS asymmetry (tone, spontaneous movements, reflexes). One way to test for hemiplegia in coma is to raise both arms together and compare how they fall under gravity. If one descends fast, like a lead weight, but the other descends more gracefully, you have found a valuable focal sign of cortical dysfunction. The same applies to the legs.

# ►►Shock

Circulatory failure resulting in inadequate organ perfusion. Often defined by ↓BP—systolic <90mmHg—or mean arterial pressure (MAP) <65mmHg—with evidence of tissue hypoperfusion, eg mottled skin, urine output (UO) of <0.5mL/kg/h, serum lactate >2mmol/L. *Signs:* ↓GCS/agitation, pallor, cool peripheries, tachycardia, slow capillary refill, tachypnoea, oliguria.

MAP = cardiac output (CO) × systemic vascular resistance (SVR).
CO = stroke volume × heart rate.
► ∴ shock can result from inadequate CO or a loss of SVR, or both.

## Inadequate cardiac output

- *Hypovolaemia:*
  - Bleeding: trauma, ruptured aortic aneurysm, GI bleed.
  - Fluid loss: vomiting, burns, 'third-space' losses, eg pancreatitis, heat exhaustion.
- *Pump failure:*
  - Cardiogenic shock, eg ACS, arrhythmias, aortic dissection, acute valve failure.
  - Secondary causes, eg PE, tension pneumothorax, cardiac tamponade.

## Peripheral circulatory failure (loss of SVR)

- *Sepsis:* (p792) Infection with any organism can cause acute vasodilation from inflammatory cytokines. Gram -ves can produce endotoxin, causing sudden and severe shock but without signs of infection (fever, ↑WCC). Classically patients with sepsis are warm & vasodilated, but may be cold & shut down. Other diseases, eg pancreatitis, can give a similar picture associated with the inflammatory cascade.
- *Anaphylaxis:* p794.
- *Neurogenic:* Eg spinal cord injury, epidural or spinal anaesthesia.
- *Endocrine failure:* Addison's disease, p836 or hypothyroidism; see p834.
- *Other:* Drugs, eg anaesthetics, antihypertensives, cyanide poisoning.

**Assessment** **►►ABCDE** (p779). With shock we are dealing primarily with '**C**' so get large-bore IV access ×2 and check ECG for rate, rhythm (very fast or very slow will compromise cardiac output), and signs of ischaemia.

- *General review:* Cold and clammy suggests cardiogenic shock or fluid loss. Look for signs of anaemia or dehydration, eg skin turgor, postural hypotension? Warm and well perfused, with bounding pulse points to septic shock. Any features suggestive of anaphylaxis—history, urticaria, angioedema, wheeze?
- *CVS:* Usually tachycardic (unless on β-blocker, or in spinal shock—*OHCS* p757) and hypotensive. But in the young and fit, or pregnant women, the systolic BP may remain normal, although the pulse pressure will narrow, with up to 30% blood volume depletion. Difference between arms (>20mmHg)—aortic dissection (p655)?
- *JVP or central venous pressure:* If raised, cardiogenic shock likely.
- *Check abdomen:* Any signs of trauma, or aneurysm? Any evidence of GI bleed?

**Management** ►If BP unrecordable, call the cardiac arrest team.

- *Septic shock:* See p792.
- *Anaphylaxis:* See p794.
- *Cardiogenic shock:* See p802.
- *Hypovolaemic shock:* Identify and treat underlying cause. Raise the legs.
  - Give fluid bolus 10–15mL/kg crystalloid via large peripheral line, if shock improves, repeat, titrate to HR (aim <100), BP (aim SBP >90) and UO (aim >0.5mL/kg/h).
  - If no improvement after 2 boluses, consider referral to ICU.
- *Haemorrhagic shock:* Stop bleeding if possible. See table 19.2 for grading.
  - If still shocked despite 2L crystalloid or present with class III/IV shock then crossmatch blood (request O Rh-ve in an emergency, see p348).
  - Give FFP with red cells (1:1 ratio); aim for platelets >100 and fibrinogen >1 (guided by results, but eg 1 pool of platelets and 2 pools of cryoprecipitate per 6–8 units of red cells). Consider tranexamic acid 2g IV. Discuss with haematology early.
- *Heat exposure (heat exhaustion):*
  - Tepid sponging + fanning; avoid ice and immersion. Resuscitate with IVI, eg 0.9% saline ± hydrocortisone 100mg IV. Lorazepam 1–2mg IV or chlorpromazine 25mg IM/IV may be used to stop shivering. Stop cooling when core temperature <39°C.

Table 19.2 Categorizing shock

| ⚠ Class of shock | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| Blood loss (estimated mL or % of circulating vol) | <750mL or <15% | 750–1500mL 15–30% | 1500–2000mL 30–40% | >2000mL >40% |
| Heart rate | <100bpm | >100bpm | 120–140bpm | >140bpm |
| Systolic BP | Normal | Normal | Low | Unrecordable |
| Pulse pressure | Normal | Narrow | Narrow | V narrow/ absent |
| Capillary refill | Normal | >2 seconds | >2 seconds | Absent |
| Respiratory rate | 14–20/min | 20–30/min | >30/min | >35/min |
| Urine output | >30mL/h | 20–30mL/h | 5–20mL/h | Negligible |
| Cerebral function | Normal/ anxious | Anxious/ hostile | Anxious/ confused | Confused/unre-sponsive |

# Sepsis



Sepsis is a major killer. There are >150 000 cases of sepsis in the UK each year resulting in >44 000 deaths, and much morbidity.

**Sepsis** Life-threatening organ dysfunction caused by a dysregulated host response to infection.

**Septic shock** Sepsis in combination with:
- EITHER lactate >2mmol/L despite adequate fluid resuscitation
- OR the patient is requiring vasopressors to maintain MAP ≥65mmHg.

## ⚠ Sepsis recognition

Many sepsis-related deaths could be prevented with earlier treatment. Often, the key failure in sepsis management is not recognizing sepsis in time. Early warning scores (p892, fig A1) help identify inpatients who are becoming septic.

Have a low threshold for assessing for sepsis if:
- the patient has communication difficulties: limited English; limited verbal communication; cognitive impairment
- the patient is immunosuppressed, on chemotherapy, or an IV drug user
- the patient recently had surgery or is pregnant/recently gave birth
- the patient has indwelling lines/other foreign material.

## ⚠ Assessing risk in sepsis

Table 19.3 Risk criteria in sepsis

| ⚠ Moderate- to high-risk criteria | High-risk criteria |
|---|---|
| Reports of altered mental state or acute deterioration in functional status | Objective evidence of altered mental state |
| Respiratory rate (RR) 21–24 | RR>24; new requirement for $FiO_2$ >40% to keep sats >92% (>88% in COPD) |
| Systolic blood pressure (SBP) 91–100mmHg | SBP <90 or >40mmHg less than baseline |
| Heart rate 91–130bpm or new arrhythmia | Heart rate >130bpm |
| Urine output: nil for 12–18h; 0.5–1.0mL/kg/h if catheterized | Urine output: nil for 18h; <0.5mL/kg/h if catheterized |
| Local signs of infection, incl. redness, swelling, or discharge around wound | Mottled, ashen, or cyanotic skin. Non-blanching rash (p822) |
| Rigors, or temperature <36°C | |
| Impaired immunity (illness or drugs) | |
| Recent surgery/trauma/invasive procedure | |

When there is a suspicion of sepsis, the criteria in table 19.3 are used to assess the patient's risk of death or serious illness from sepsis as follows:
*High risk:* At least one high-risk criterion OR at least two moderate- to high-risk criteria with AKI or LACTATE >2.
*Moderate to high risk:* At least one moderate- to high-risk criterion.
*Low risk:* No moderate- or high-risk criteria.

**Acute management in sepsis** ►Early recognition and treatment is key. See fig 19.3.[2]
*Antibiotics:* These should be broad spectrum and start within 1h. Consider covering for non-bacterial microbes, eg give aciclovir if HSV encephalitis is suspected.
*Fluids:* Give within 1h if high risk with SBP <90, AKI, or lactate >2 (consider if <2).
- Give 500mL boluses of crystalloids with 130–154mmol/L sodium (eg 0.9% saline) over 15mins. Caution in heart failure.
- If no improvement after two boluses, speak with a senior.

*Oxygen:* Give oxygen as required for target saturations. These will be 94–98% (or 88–92% if the patient is at risk of $CO_2$ retention, eg in severe COPD).
*Critical care review:* Speak with critical care early if intensive care support (eg inotropes, ventilation, haemofiltration, intensive monitoring) may be required.
*Surgical involvement:* Eg emergency wound debridement.
*Manage acute complications:* Shock (p790), AKI (p298), DIC (p352), ARDS (p186), arrhythmias (may spontaneously resolve when sepsis improves).

## ⚠ Acute management of sepsis in adults

**Recognize the need to assess for sepsis**

- Consider possibility of sepsis in any patient with signs or symptoms that are suggestive of infection (eg productive cough or offensive oozing wound)
- Consider non-specific signs (patient/carer concern; altered mental state; feeling generally unwell), particularly when there is reason to have low threshold for sepsis assessment (see BOX)

↓

**Gather information**

*History:* Full medical history: take time to clarify the time course of symptoms and any travel history

*Examination:*

- General: capillary refill time, mottled/ashen skin, conscious level
- Localize source of infection: don't forget to assess wounds/ulcers which may be hidden by dressings

*Bedside tests:* Blood gas for lactate, observations incl. HR, BP, RR, sats, temperature; ECG; urine dip and urine output monitoring

↓

**Assess risk (see BOX)**

**High risk**
Request immediate senior review

**Moderate to high risk**
Clinician review within 1h. Senior review within 3h if cause not identified

**Low risk**
Manage according to clinical judgement

↓

**Investigations**

*Blood tests:* Serial ABGs, or VBGs for lactate; blood cultures; U&E, CRP, FBC, LFT, clotting screen

*Micro samples:* Sputum and urine for MC&S; swab any wounds; consider LP; send fluid from drains and lines; joint aspirates; ascitic tap

*Imaging:* CXR, consider CT/USS/MRI/echo of suspected source

↓

**Treatment**

*Antibiotics ± other antimicrobials*
*Fluids*
*Oxygen*
*Liaise with other teams*

- Critical care, surgeons, medical specialties

*Manage acute complications*

↓

**Review**

*Immediately alert a consultant if, after 1h of antibiotics and fluids:*

- SBP <90 • RR >30 • Reduced GCS
- Raised lactate not reduced by >20%

►►Consider critical care referral.

Fig 19.3 Management of sepsis in adults.

# ►►Anaphylactic shock

Type I IgE-mediated hypersensitivity reaction. Release of histamine and other agents causes: capillary leak; wheeze; cyanosis; oedema (larynx, lids, tongue, lips); urticaria. More common in atopic individuals. An *anaphylactoid reaction* results from direct release of mediators from inflammatory cells, without involving antibodies, usually in response to a drug, eg acetylcysteine.

## Examples of precipitants

- Drugs, eg penicillin, and contrast media in radiology.
- Latex.
- Stings, eggs, fish, peanuts, strawberries, semen (rare).

## Signs and symptoms

- Itching, sweating, diarrhoea and vomiting, erythema, urticaria, oedema.
- Wheeze, laryngeal obstruction, cyanosis.
- Tachycardia, hypotension.

## Mimics of anaphylaxis

- Carcinoid (p271).
- Phaeochromocytoma (p228, p837).
- Systemic mastocytosis.
- Hereditary angioedema.

**Management** ►See fig 19.4.

## ⚠ Management of anaphylaxis

Secure the airway—give 100% $O_2$
Intubate if respiratory obstruction imminent

↓

Remove the cause; raising the feet may help restore the circulation

↓

Give adrenaline IM *0.5mg (ie 0.5mL of 1:1000)*.
Repeat every 5min, if needed as guided by BP, pulse, and respiratory function, until better

↓

Secure IV access

↓

Chlorphenamine 10mg IV and hydrocortisone 200mg IV

↓

IVI (0.9% saline, eg 500mL over ¼h; up to 2L may be needed)
Titrate against blood pressure

↓

If wheeze, treat for asthma (p810)
May require ventilatory support

↓

If still hypotensive, admission to ICU and an IVI of adrenaline may be needed ± aminophylline (p811) and nebulized salbutamol (p811): get expert help

↓

*Further management:*
- Admit to ward. Monitor ECG
- Measure serum tryptase 1–6h after suspected anaphylaxis
- Continue chlorphenamine 4mg/6h PO if itching
- Suggest a 'MedicAlert' bracelet naming the culprit allergen
- Teach about self-injected adrenaline (eg 0.3mg, Epipen®) to prevent a fatal attack
- Skin-prick tests showing specific IgE help identify allergens to avoid

**Fig 19.4** Management of anaphylaxis.

▸▸Adrenaline (=epinephrine) is given IM and NOT IV unless the patient is severely ill, or has no pulse. The IV dose is *different*: 100mcg/min—titrating with the response. This is 0.5mL of *1:10 000 solution* IV per minute. Stop as soon as a response has been obtained.

If on a β-blocker, consider salbutamol IV in place of adrenaline.

# Acute coronary syndrome—with ST-elevation

Acute coronary syndrome (ACS) includes unstable angina, STEMI, and NSTEMI (p798). STEMI is a common medical emergency; prompt appropriate treatment saves lives.[3]

**Initial treatment** ►See fig 19.5. Take brief history, do a quick physical examination and a 12-lead ECG. Observe on cardiac monitor or telemetry in case of dysrhythmia. Other tests on admission: U&E, troponin, glucose, cholesterol, FBC, CXR.

- *Aspirin:* 300mg PO (if not already given); consider ticagrelor (180mg PO) or prasugrel (60mg PO if no history of stroke/TIA and <75yrs) as newer alternatives to clopidogrel (300mg PO) as they have been shown to be superior in outcome studies.
- *Morphine:* 5–10mg IV (repeat after 5min if necessary). Give anti-emetic with the 1st dose of morphine: metoclopramide 10mg IV (1st line), or cyclizine 50mg IV (2nd line).
- *GTN:* routine use now not recommended in the acute setting unless patient is hypertensive or in acute LVF. Useful as anti-anginal in chronic/stable patients.
- *Oxygen* is recommended if patients have $SaO_2$ <95%, are breathless or in acute LVF.
- *Restore coronary perfusion* in those presenting <12h after symptom onset (see BOX).
- *Anticoagulation:* An injectable anticoagulant must be used in primary PCI. Bivalirudin is preferred, if not available use enoxaparin ± a GP IIb/IIIa blocker.
- *β-blockers* provide additional benefit when started early, eg bisoprolol 2.5mg PO OD. Ensure no evidence of cardiogenic shock, heart failure, asthma/COPD, or heart block.

**Right ventricular infarction** Confirm by demonstrating ST elevation in $rV_{3/4}$ and/or echo. NB: $rV_4$ means that $V_4$ is placed in the right 5th intercostal space in the midclavicular line. Treat hypotension and oliguria with fluids (avoid nitrates and diuretics). Monitor BP carefully, and assess early signs of pulmonary oedema. Intensive monitoring and inotropes may be useful in some patients.

## ⚠ Reperfusion therapy

Early coronary reperfusion saves lives; decisions must be taken quickly so seek senior advice early. Look for typical clinical symptoms of MI plus ECG criteria:

- ST elevation >1mm in ≥2 adjacent limb leads or >2mm in ≥2 adjacent chest leads.
- LBBB (unless known to have LBBB previously).
- Posterior changes: deep ST depression and tall R waves in leads $V_1$ to $V_3$.

Therapy may be percutaneous intervention (PCI—with angiographic identification of the culprit blockage(s) and revascularization via deployment of an expandable metal stent) or thrombolysis (with systemically administered clot-dissolving enzymes):

- *Primary PCI:* ►►Should be offered to all patients presenting within 12h of symptom onset with a STEMI who either are at or can be transferred to a primary PCI centre within 120min of first medical contact. If this is not possible, patients should receive thrombolysis and be transferred to a primary PCI centre after the infusion for either rescue PCI (if residual ST elevation) or angiography (if successful). Use beyond 12h if evidence of ongoing ischaemia or in stable patients presenting after 12–24h may be appropriate—seek specialist advice.
- *Thrombolysis:* ►►Benefit reduces steadily from onset of pain, target time is <30min from admission; use >12h from symptom onset requires specialist advice. ►Do not thrombolyse ST depression alone, T-wave inversion alone, or normal ECG. Thrombolysis is best achieved with tissue plasminogen activators (eg tenecteplase as a single IV bolus). *CI:* •Previous intracranial haemorrhage. •Ischaemic stroke <6months. •Cerebral malignancy or AVM. •Recent major trauma/surgery/head injury (<3wks). •GI bleeding (<1 month). •Known bleeding disorder. •Aortic dissection. •Non-compressible punctures <24h, eg liver biopsy, lumbar puncture. *Relative CI:* •TIA <6 months. •Anticoagulant therapy. •Pregnancy/<1wk post partum. •Refractory hypertension (>180mmHg/110mmHg). •Advanced liver disease. •Infective endocarditis. •Active peptic ulcer. •Prolonged/traumatic resuscitation.

►Patients with STEMI who do not receive reperfusion (eg presenting >12h after symptom onset) should be treated with fondaparinux, or enoxaparin/unfractionated heparin if not available.

**Management of an acute STEMI**

Attach ECG monitor and record a 12-lead ECG

IV access.
*Bloods for FBC, U&E, glucose, lipids, troponin* (p119, fig 3.22)

*Brief assessment:*
- History of cardiovascular disease; risk factors for IHD
- Examination: pulse, BP (both arms), JVP, murmurs, signs of CCF, upper limb pulses, scars from previous cardiac surgery, CXR if will not delay Rx
- Contraindications to PCI or fibrinolysis?

*Aspirin:* 300mg (unless already given by GP/paramedics)
*Ticagrelor:* 180mg (or alternative antiplatelet—see 'Aspirin' in text)

*Morphine:* 5–10mg IV + anti-emetic, eg metoclopramide 10mg IV

STEMI on ECG and PCI available within 120min?

Yes → *Primary PCI*

No → *Fibrinolysis* → Transfer to primary PCI centre for either rescue PCI if fibrinolysis unsuccessful or for angiography

*For further management* see p120

Fig 19.5 Management of an acute STEMI.

# Acute coronary syndrome—without ST-elevation

First stabilize with medical therapy; early risk stratification will identify those in need of further treatment and prompt angiography (involve cardiologists).[4]

## Assessment

*Brief history:* (See p36.) Previous angina, relief with rest/nitrates, history of cardiovascular disease, risk factors for IHD.

*Examination:* (See p38.) Pulse, BP, JVP, cardiac murmurs, signs of heart failure, peripheral pulses, scars from previous cardiac surgery.

**Investigations** ECG: ST depression; flat or inverted T-waves; or normal; FBC, U&E, troponin, glucose, random cholesterol; CXR.

**Management** ►See fig 19.6 for acute management, but p796 if ST elevation. The aim of therapy is to control pain then initiate anti-ischaemic and antiplatelet therapy.

*Oral antiplatelet therapy:* Aspirin 300mg PO, followed by 75mg OD. For those with confirmed ACS give a second antiplatelet agent, eg clopidogrel (300mg PO then 75mg OD PO). Ticagrelor (180mg then 90mg/12h PO) is a prefered alternative, particularly in higher risk groups[5] (eg • ≥60yrs age •previous stroke, TIA, MI, or CABG •known coronary artery stenosis ≥50% in ≥2 vessels or carotid stenosis ≥50% •DM •peripheral arterial disease •chronic kidney disease). Prasugrel (60mg then 10mg/d PO) is an alternative to clopidogrel for those undergoing PCI. In practice, if the history is typical but ECG changes are non-diagnostic and troponin results are awaited, treatment with eg clopridogrel is often given on clinical suspicion; if troponin testing then confirms ACS, it is still appropriate to give either ticagrelor or prasugrel to those patients in whom it is indicated, even if clopidogrel has already been given.

*Anticoagulation:* Ideally fondaparinux (factor Xa inhibitor) 2.5mg OD; if not available, use low-molecular-weight heparin (LMWH, eg enoxaparin 1mg/kg/12h) or unfractionated heparin (aim APTT 50–70s) until discharge.

*β-blockers:* In higher-risk patients with no contraindications (consider diltiazem as alternative). ►Do not use β-blockers with verapamil—can precipitate asystole.

*Nitrates (PO or IV):* For recurrent chest pain.

*ACE-i:* Should be given to all patients unless there are CI (monitor renal function).

*Lipid management:* Start early, eg atorvastatin 80mg OD (see p120).

**Prognosis** Overall risk of death ~1–2%, but ~15% for refractory angina despite medical therapy. Risk stratification can help predict those most at risk and allow intervention to be targeted at those individuals: calculate using GRACE score.[2] The following are associated with an increased risk:

- History of unstable angina.
- ST depression or widespread T-wave inversion.
- Raised troponin (except patients with ST elevation MI).
- Age >70 years.
- General comorbidity, previous MI, poor LV function or DM.

►High-risk patients should be considered for inpatient coronary angiography. Symptomatic lesions may be addressed by coronary stenting or CABG (see p123).

## Further measures

- Wean off *glyceryl trinitrate* (GTN) infusion when stabilized on oral drugs.
- Continue fondaparinux (or LMWH or heparin) until discharge.
- Observe on cardiac monitor or telemetry in case of dysrhythmia. Check serial ECGs, and troponin >12h after pain.
- Address modifiable risk factors: smoking, hypertension, hyperlipidaemia, diabetes.
- Gentle mobilization.
- Ensure patient on dual antiplatelet therapy, β-blocker, ACE inhibitor, and statin.

►*If symptoms recur, refer to cardiologist for urgent angiography & PCI or CABG.*

**2** GRACE = Global Registry of Acute Coronary Events. Risk is scored based on age, heart rate, BP, renal function, Killip class of heart failure, and other events, eg raised troponin. Very complicated to calculate so recommendation by European Society of Cardiology is to use an online calculator, eg http://www.outcomes-umassmed.org/grace/.

## ⚠ Acute management of cardiac chest pain

Monitor closely; record ECG while in pain

↓

If $SaO_2$ <90% or breathless, low-flow $O_2$

↓

*Analgesia:* Eg morphine 5–10mg IV + metoclopramide 10mg IV

↓

*Nitrates:* GTN spray or sublingual tablets as required

↓

*Aspirin:* 300mg PO.
Consider need for second antiplatelet agent (see 'Oral antiplatelet therapy' in text)

↓

Measure troponin and clinical parameters to risk assess, eg GRACE score[2]

↓

*Invasive strategy (high-risk pt):*
- *Rise in troponin* OR:
- *Dynamic ST or T-wave changes*
- *Secondary criteria—diabetes, CKD, LVEF <40%, early angina post MI, recent PCI, prior CABG, intermediate to high-risk GRACE score*

*Conservative strategy (low-risk pt):*
- *No recurrence of chest pain*
- *No signs of heart failure*
- *Normal ECG*
- *—ve baseline (± repeat) troponin*

May be discharged (check troponin interval required with your laboratory and retest after delay if necessary). Arrange further outpatient investigation, eg stress test.

↓ (from invasive strategy)

*Fondaparinux:* 2.5mg OD SC or LMWH 1mg/kg/12h SC

↓

Second antiplatelet agent (see text), eg *ticagrelor* 180mg PO (or *clopidogrel* 300mg PO in lower risk, or *prasugrel* 60mg OD if proceeding to PCI)

↓

IV *nitrate* if pain continues
(eg GTN 50mg in 50mL 0.9% saline at 2–10mL/h)
titrate to pain, and maintain systolic BP >100mmHg

↓

Oral *β-blocker*, eg bisoprolol 2.5mg OD
CI: cardiogenic shock, heart failure, asthma/COPD or heart block; consider rate-limiting calcium antagonist (eg verapamil 80–120mg/8h PO, or diltiazem 60–120mg/8h PO)

↓

Prompt cardiologist review for angiography
1. Urgent (<120min after presentation) if ongoing angina and evolving ST changes, signs of cardiogenic shock or life-threatening arrhythmias
2. Early (<24h) if GRACE score >140 and high-risk patient
3. Within 72h if lower-risk patient

Fig 19.6 Acute management of chest pain and ACS without ST-segment elevation.

# ►►Severe pulmonary oedema



**Causes**

- Cardiovascular, usually left ventricular failure (post-MI or ischaemic heart disease). Also valvular heart disease, arrhythmias, and malignant hypertension.
- ARDS (p186) from any cause, eg trauma, malaria, drugs. Look for predisposing factors, eg trauma, post-op, sepsis. *Is aspirin overdose or glue-sniffing/drug abuse likely*? Ask friends/relatives.
- Fluid overload.
- Neurogenic, eg head injury.

**Differential diagnosis** Asthma/COPD, pneumonia, and pulmonary oedema are often hard to distinguish, especially in the elderly, where they may coexist. If the patient is extremely unwell and you are not sure, consider treating all three (eg with salbutamol nebulizer, furosemide IV, diamorphine, amoxicillin—p386).

**Symptoms** Dyspnoea, orthopnoea (eg paroxysmal), pink frothy sputum. NB: note drugs recently given and other illnesses (recent MI/COPD or pneumonia).

**Signs** Distressed, pale, sweaty, ↑pulse, tachypnoea, pink frothy sputum, pulsus alternans, ↑JVP, fine lung crackles, triple/gallop rhythm (p44), wheeze (cardiac asthma). Usually sitting up and leaning forward. Quickly examine for possible causes.

**Investigations**

- CXR (p135, pp722–4): cardiomegaly, signs of pulmonary oedema: look for shadowing (usually bilateral), small effusions at costophrenic angles, fluid in the lung fissures, and Kerley B lines (septal linear opacities).
- ECG: signs of MI, dysrhythmias.
- U&E, troponin, ABG.
- Consider echo.
- BNP (p137) may be helpful if diagnosis in question (high negative predictive value).

**Management** ►See fig 19.7. ►►Begin treatment before investigations.

*Monitoring progress:* BP; pulse; cyanosis; respiratory rate; JVP; urine output; ABG. Observe on cardiac monitor or telemetry in case of dysrhythmia.

*Once stable and improving:*

- Daily weights, aim reduction of 0.5kg/day, check obs at least QDS.
- Repeat CXR.
- Change to oral furosemide or bumetanide.
- If on large doses of loop diuretic, consider the addition of a thiazide (eg bendroflumethiazide or metolazone 2.5–5mg daily PO).
- ACE-i if LVEF <40%. If ACE-i contraindicated, consider hydralazine and nitrate (may also be more effective in African-Caribbeans).
- Also consider β-blocker and spironolactone (if LVEF <35%).
- Is the patient suitable for biventricular pacing or cardiac transplantation?
- Optimize management of AF if present (p130); consider anticoagulation.

## ⚠ Management of acute heart failure

Sit the patient upright

↓

High-flow oxygen if ↓$SpO_2$*

↓

IV access and monitor ECG
Treat any arrhythmias, eg AF (pp124–31)

↓

Investigations whilst continuing treatment
See p800

↓

Diamorphine 1.25–5mg IV *slowly*
*Caution in liver failure* and *COPD*

↓

Furosemide 40–80mg IV slowly
*Larger doses required in renal failure*

↓

GTN spray 2 puffs SL or 2 × 0.3mg tablets SL
*Don't give if systolic* BP *<90mmHg*

↓

Necessary investigations, examination, and history

↓

If systolic BP ≥100mmHg, start a nitrate infusion,
eg *isosorbide dinitrate 2–10mg/h* IVI; *keep systolic* BP *≥90mmHg*

↓

*If the patient is worsening:*
- Further dose of furosemide 40–80mg
- Consider CPAP—improves ventilation by recruiting more alveoli, driving fluid out of alveolar spaces and into vasculature (get help before initiating!)
- Increase nitrate infusion if able to do so without dropping systolic BP <100mmHg
- Consider alternative diagnoses, eg hypertensive heart failure, aortic dissection, pulmonary embolism, pneumonia

↓

If systolic BP <100mmHg, treat as cardiogenic shock (p802) and refer to ICU

*Avoid supplemental oxygen if not hypoxaemic since may cause vasoconstriction and reduce cardiac output. If known COPD, hypoxaemia still needs correcting; give high-flow oxygen but monitor closely for $CO_2$ retention (check serial ABG if needed) and reduce flow as soon as possible.

**Fig 19.7** Management of heart failure.

# ►►Cardiogenic shock

This has a high mortality and is very difficult to treat. ►Ask a senior physician's help both in formulating an *exact* diagnosis and in guiding treatment.

Cardiogenic shock is a state of inadequate tissue perfusion primarily due to cardiac dysfunction. It may occur suddenly, or after progressively worsening heart failure.



### Causes

- Myocardial infarction (pp796–9).
- Arrhythmias (pp124–31).
- Pulmonary embolus (p818).
- Tension pneumothorax (p814).
- Cardiac tamponade (p154 and later in topic).
- Myocarditis; myocardial depression (drugs, hypoxia, acidosis, sepsis) (p152).
- Valve destruction (endocarditis—p150).
- Aortic dissection (p655).

### Management ►See fig 19.8.

►►If the cause is myocardial infarction prompt reperfusion therapy is vital (see p796).

- Manage in Coronary Care Unit, or ICU.
- Investigation and treatment may need to be done concurrently.
- *Investigations:* ECG, U&E, troponin, ABG, CXR, echocardiogram. If indicated, CT thorax (speak with radiologists, this can be protocolled for both aortic dissection and PE).
- *Monitor:* CVP, BP, ABG, ECG, urine output. Keep on cardiac monitor/telemetry. Record a 12-lead ECG every hour until the diagnosis is made. Consider a CVP line and an arterial line to monitor pressure, if these are *in situ* consider measuring cardiac output and volume status.[3] Catheterize for accurate urine output.

**Cardiac tamponade** Pericardial fluid collects → intrapericardial pressure rises → heart cannot fill → pumping stops.

*Causes:* Trauma, lung/breast cancer, pericarditis, myocardial infarct, bacteria, eg TB. *Rarely:* ↑urea, radiation, myxoedema, dissecting aorta, SLE. Also coronary artery dissection (secondary to PCI) and/or ruptured ventricle.

*Signs:* ↓BP, ↑JVP, and muffled heart sounds (Beck's triad); ↑JVP on inspiration (Kussmaul's sign); pulsus paradoxus (pulse fades on inspiration). Echocardiography may be diagnostic. CXR: globular heart; left heart border convex or straight; right cardiophrenic angle <90°. ECG: electrical alternans (p154).

*Management:* This can be very difficult. Everything is against you: time, physiology, and your own confidence, as the patient may be too ill to give a history, and signs may be equivocal—but bitter experience has taught us not to equivocate for long.

►►Request the presence of your senior at the bedside (do not make do with telephone advice). With luck, prompt pericardiocentesis (p773) brings swift relief. While awaiting this, give $O_2$, monitor ECG, and set up IVI. Take blood for group & save. NB: there may be a role for cardiothoracic surgery (eg CABG, ventricular repair, or pericardial window) as a definitive solution to some causes.

3 Eg Pulse contour cardiac output (PICCO) or lithium dilution cardiac output (LIDCO). Both use injection (PICCO = cold water, LIDCO = lithium) to estimate filling pressure, extravascular water (ie pulmonary oedema) and cardiac output. The time from injection via a central vein to detection via an arterial line, plus dilution, gives estimates of cardiac output and volume status and can guide fluid and inotrope therapy.

## ⚠ Management of cardiogenic shock

Oxygen
*Titrate to maintain arterial saturations of 94–98% (88–92% if* COPD*)*

↓

Diamorphine 1.25–5mg IV for pain and anxiety

↓

*Investigations* and *close monitoring*
(see p802)

↓

Correct arrhythmias (pp124–31), U&E abnormalities, or acid-base disturbance

↓

*Optimize filling pressure* with clinical assessment of pulse, BP, JVP/CVP (if in ICU consider using PICCO, LIDCO, transoesophageal doppler or Swan-Gantz catheter to estimate cardiac output and fluid balance)

↓

*Underfilled?*
Give a plasma expander
100mL every 15min IV
Aim MAP 70mmHg, CVP 8–10mmHg

*Well/over-filled?*
Inotropic support, eg dobutamine 2.5–10mcg/kg/min IVI. Aim MAP 70mmHg

↓

*Look for and treat any reversible cause:*
MI or PE—consider thrombolysis;
surgery for: acute VSD, mitral, or aortic incompetence

**Fig 19.8** Management of cardiogenic shock. MAP = mean arterial pressure.

# ►►Broad complex tachycardia

ECG shows rate of >100bpm and QRS complexes >120ms (>3 small squares on ECGs done at the standard UK rate of 25mm/s). Identify the underlying rhythm and treat accordingly.[6]

**Differential diagnosis** (See p128.)

- *Ventricular tachycardia (VT) including torsade de pointes.* Single ventricular ectopics should not cause confusion; if >3 together at a rate >100, this is VT.
- SVT (p806) with aberrant conduction, eg AF or atrial flutter, with bundle branch block.
- Pre-excited tachycardias, eg AF, atrial flutter, or AV re-entry tachycardia, with underlying WPW (p133).

**Identifying the underlying rhythm** (See p128.) ►►If in doubt, treat as VT.

**Management** ►See fig 19.9.[6]

- Connect patient to a cardiac monitor and have a defibrillator to hand.
- Monitor $O_2$ sats and if <90% give supplemental oxygen.
- Correct electrolyte abnormalities, esp $K^+$ and $Mg^{2+}$.
- Check for adverse signs. Low cardiac output (clammy, ↓consciousness, BP <90); oliguria; angina; pulmonary oedema.
- Obtain 12-lead ECG (request CXR) and obtain IV access.

*If haemodynamically unstable VT:* ►►Synchronized DC shock (see p894, fig A3).

- Correct any hypokalaemia and hypomagnesaemia: up to 60mmol KCl at 30mmol/h, and 4mL 50% magnesium sulfate over 30min both via central line.
- Follow with amiodarone 300mg IV over 10–20min (peripherally only in emergency).
- For refractory cases consider procainamide or sotalol.

*If haemodynamically stable VT:* Correct hypokalaemia and hypomagnesaemia: as above.

- Amiodarone 300mg IV over 20–60min (avoid if long QT) via central line.
- If this fails, use synchronized DC shock.

*After correction of VT:* Establish the cause (via the history and tests described above).

- Maintenance anti-arrhythmic therapy may be required. If VT occurs after MI, give IV amiodarone infusion for 12–24h; if 24h after MI, also start oral anti-arrhythmic: sotalol (if good LV function) or amiodarone (if poor LV function).
- Prevention of recurrent VT: surgical isolation of the arrhythmogenic area or an implantable cardioverter defibrillator (ICD) may help.

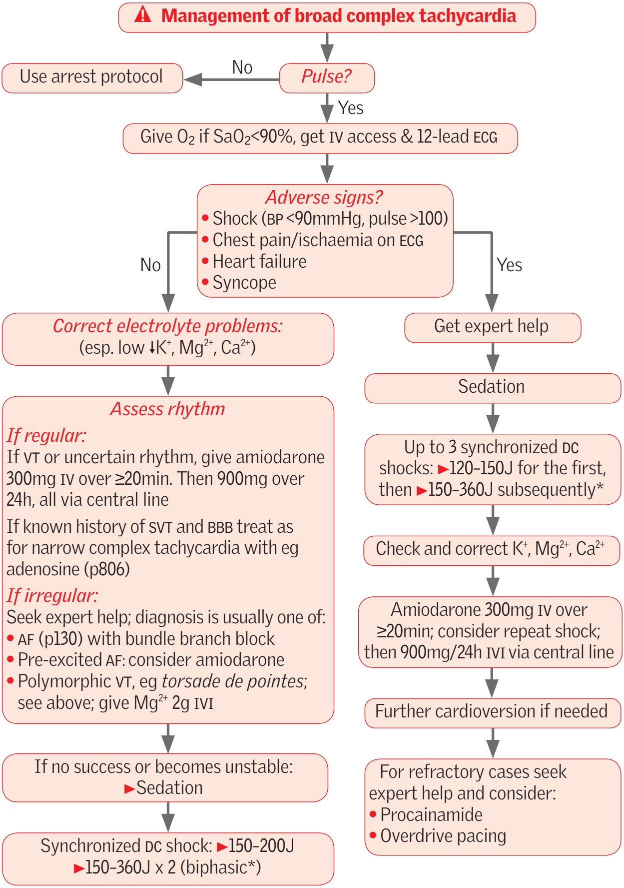
*Ventricular fibrillation:* (ECG p129, fig 3.29) Use non-synchronized DC shock (there is no R wave to trigger defibrillation, p770): see p894, fig A3.

*If SVT with aberrant conduction:* Manage as SVT with eg adenosine (see p806).

*Ventricular extrasystoles (ectopics):* These are the commonest post-MI arrhythmia but they are also seen in healthy people (often >10/h). Patients with frequent ectopics post-MI have a worse prognosis, but there is no evidence that antidysrhythmic drugs improve outcome, indeed they may increase mortality.

*Torsade de pointes:* A form of VT, with a constantly varying axis, often in the setting of long QT syndromes (ECG p129, fig 3.31). Causes (p711): congenital or from drugs (eg some antidysrhythmics, tricyclics, antimalarials, antipsychotics). Torsade in the setting of congenital long-QT syndromes can be treated with high doses of β-blockers.

In acquired long-QT syndromes (p711), stop all predisposing drugs, correct hypokalaemia, and give magnesium sulfate (2g IV over 10min). Alternatives include: overdrive pacing (pace at a faster rate, then slow reduce) or isoprenaline IVI to increase heart rate.




*Check your defibrillator: energies given are for a typical biphasic defibrillator (preferred); if a monophasic shock used, higher energies will be required.

**Fig 19.9** Management of broad complex tachycardia.

# ▸▸Narrow complex tachycardia

ECG shows rate of >100bpm and QRS complex duration of <120ms (<3 small squares on ECGs done at the standard UK rate of 25mm/s).

**Differential diagnosis** (See p126.)

- *Sinus tachycardia:* Normal P wave followed by normal QRS—not an arrhythmia! Do not attempt to cardiovert; if necessary (ie not a physiological response to fever/hypovolaemia) rate control with β-blockers.
- *Atrial tachyarrhythmias:* Rhythm arises in atria, AV node is a bystander.
  - Atrial fibrillation (AF): absent P wave, irregular QRS complexes.
  - Atrial flutter: atrial rate ~260–340bpm. Sawtooth baseline, due to a re-entrant circuit usually in the right atrium. Ventricular rate often 150bpm (2:1 block).
  - Atrial tachycardia: abnormally shaped P waves, may outnumber QRS.
  - Multifocal atrial tachycardia: ≥3 P-wave morphologies, irregular QRS complexes.
- *Junctional tachycardia:* AV node is part of the pathway. P wave either buried in QRS complex or occurring after QRS complex.
  - AV nodal re-entry tachycardia.
  - AV re-entry tachycardia, includes an accessory pathway, eg WPW (p133).

**Management** ▸Be guided by patient status,[6] see fig 19.10.[6]

▸▸If the patient is compromised, use DC cardioversion.

- Otherwise, identify the underlying rhythm and treat accordingly. The most important thing is to decide whether the rhythm is regular or not (irregular is likely AF).
- Vagal manoeuvres (carotid sinus massage, Valsalva manoeuvre) transiently increase AV block, and may unmask an underlying atrial rhythm.
- If unsuccessful, give adenosine, which causes transient AV block. It has a short half-life (10–15s) and works by: **1** transiently slowing ventricles to show the underlying atrial rhythm; **2** cardioverting a junctional tachycardia to sinus rhythm.

*Adenosine:* Consult *BNF* if on dipyridamole or has had a heart transplant. Give 6mg IV bolus into a large vein, followed by 0.9% saline flush, while recording a rhythm strip. If unsuccessful, after 2min give 12mg, then one further 12mg bolus. Warn about SE: transient chest tightness, dyspnoea, headache, flushing. *Relative CI:* Asthma, 2nd/3rd-degree AV block or sinoatrial disease (unless pacemaker). *Interactions:* Potentiated by dipyridamole; antagonized by theophylline.

**Specifics** *Sinus tachycardia:* Identify and treat underlying cause.

*Supraventricular tachycardia:* If adenosine fails, use verapamil 2.5–5mg IV over 2min. NB: NOT if on a β-blocker. If no response, a further 5mg IV over 3min (if age <60yrs). Alternatives: atenolol 2.5mg IV repeated at 5min intervals until 10mg given; or amiodarone. If unsuccessful, use DC cardioversion.

*Atrial fibrillation/flutter:* Manage with rate control; seek help if resistant (p130).

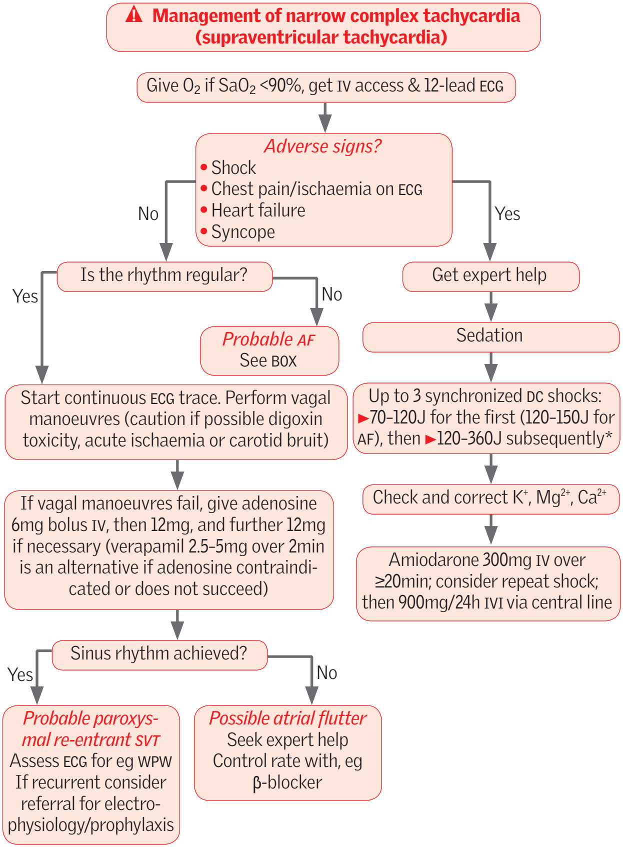
*Atrial tachycardia:* Rare; may be due to digoxin toxicity: withdraw digoxin, consider digoxin-specific antibody fragments. Maintain $K^+$ at 4–5mmol/L.

*Multifocal atrial tachycardia:* Most commonly occurs in COPD. Correct hypoxia and hypercapnia. Consider verapamil if rate remains >110bpm.

*Junctional tachycardia:* Where anterograde conduction through the AV node occurs, vagal manoeuvres are worth trying. Adenosine will usually cardiovert a junctional rhythm to sinus rhythm. If it fails or recurs, β-blockers (or verapamil—*not* with β-blockers, digoxin, or class I agents such as quinidine). If this does not control symptoms, consider radiofrequency ablation.

▸Seek specialist advice if resistant junction tachycardia, or accessory pathway.

**Wolff-Parkinson-White (WPW) syndrome** (*ECG* p133, fig 3.37.) Caused by congenital accessory conduction pathway between atria and ventricles. Resting ECG shows short PR interval and widened QRS complex due to slurred upstroke or 'delta wave'. Two types: WPW type A (+ve δ wave in $V_1$), WPW type B (–ve δ wave in $V_1$). Present with SVT which may be due to an AVRT (p126), pre-excited AF, or pre-excited atrial flutter. Risk of degeneration to VF and sudden death. *℞:* Flecainide, propafenone, sotalol, or amiodarone. Refer to cardiologist for electrophysiology and ablation of the accessory pathway.




*Check your defibrillator: energies given are for a typical biphasic defibrillator (preferred); if a monophasic shock used, higher energies will be required.

**Fig 19.10** Management of narrow complex tachycardia (supraventricular tachycardia).

## ⚠ Irregular narrow complex tachycardia

- Treat as AF—by far the most likely diagnosis.
- Control rate with:
  - β-blocker: eg metoprolol 1–10mg IV, give small increments to slow rate
  - rate-limiting $Ca^{2+}$-channel blocker eg verapamil 5–10mg IV
  - digoxin is an alternative in heart failure (load with eg 500mcg PO then 500mcg PO after 8h and further 250mcg PO after 8h)
  - amiodarine (may also control rhythm—see last bullet point in this box).
- Consider anticoagulation with warfarin or NOAC to ↓ risk of stroke.
- If onset definitely <48h, or if effectively anticoagulated for >3wk, consider cardioversion with synchronized DC cardioversion under sedation (see p770). Chemical cardioversion may be achieved with flecanide 300mg PO (only if definitely no structural heart damage) or amiodarone, 300mg IVI over 20–60min, then 900mg over 24h.

# Bradycardia

Bradycardia is defined as a heart rate <60bpm. This may be normal and asymptomatic in very fit, young individuals whose high stroke volumes will maintain adequate cardiac output at low heart rates.

**Symptoms** Often asymptomatic. Fatigue, nausea, dizziness. The presence of syncope, chest pain, or breathlessness is concerning and suggests the presence of adverse signs; sudden cardiac death can occur.

**Rhythm** The immediate management tends to relate more to cause and adverse signs than to the underlying rhythm, which may be •sinus bradycardia •heart block (see p98) •AF with a slow ventricular response •atrial flutter with a high-degree block •junctional bradycardia.

**Causes**

- *Physiological:* Heart rates as low as 40bpm at rest and 30bpm in sleep can be accepted in asymptomatic trained athletes.
- *Cardiac:*
  - Degenerative changes causing fibrosis of conduction pathways (risk in elderly patients; may have previous ECGs showing bundle branch block or 1st- or 2nd-degree heart block).
  - Post-MI—particularly after an inferior MI (the right coronary artery supplies the sinoatrial node and atrioventricular node in most people).
  - Sick sinus syndrome (p125).
  - Iatrogenic—ablation, surgery.
  - Aortic valve disease, eg infective endocarditis (p150; do daily ECGs looking for heart block).
  - Myocarditis, cardiomyopathy, amyloid, sarcoid, SLE.
- *Non-cardiac origin:*
  - Vasovagal—very common (p460).
  - Endocrine—hypothyroidism, adrenal insufficiency.
  - Metabolic—hyperkalaemia, hypoxia.
  - Other—hypothermia, ↑ICP (Cushing's triad: bradycardia, hypertension, and irregular breathing: ►► urgent senior input needed).
- *Drug-induced:*
  - β-blockers, amiodarone, verapamil, diltiazem, digoxin.

**Management** ►Follow a logical approach,[6] see fig 19.11.[6]

►►Think ahead. If you may need an anaesthetist to sedate the patient for transcutaneous pacing, or a cardiologist for transvenous pacing, call them now.

- Perform a 12-lead ECG, check electrolytes (including $K^+$, $Ca^{2+}$, $Mg^{2+}$), do digoxin levels.
- Connect patient to cardiac monitor/telemetry.
- Address the cause: correct metabolic defects; if the patient has adverse signs or is deteriorating, give antidotes to medicines likely to have caused the bradycardia (eg glucagon if β-blocker overdose).
- If the patient has adverse signs or risk of asystole, give atropine (not to be given if patient has a transplanted heart).
- If atropine is insufficient and adverse signs persist, transcutaneous pacing should be considered (p777). If this cannot be initiated immediately (eg waiting for an anaesthetist), consider other medications such as isoprenaline infusion.
- Remember electrical 'capture' with transcutaneous pacing does not guarantee mechanical 'capture'. Once pacing is established, check the patient's pulse.

►It is possible to have two patients sat next to each other with identical bradycardic ECG tracings, one of whom is peri-arrest, the other is sat comfortably and cannot understand your concern. The clinical state is more important than the numbers on the screen.

## ⚠ Management of bradycardia

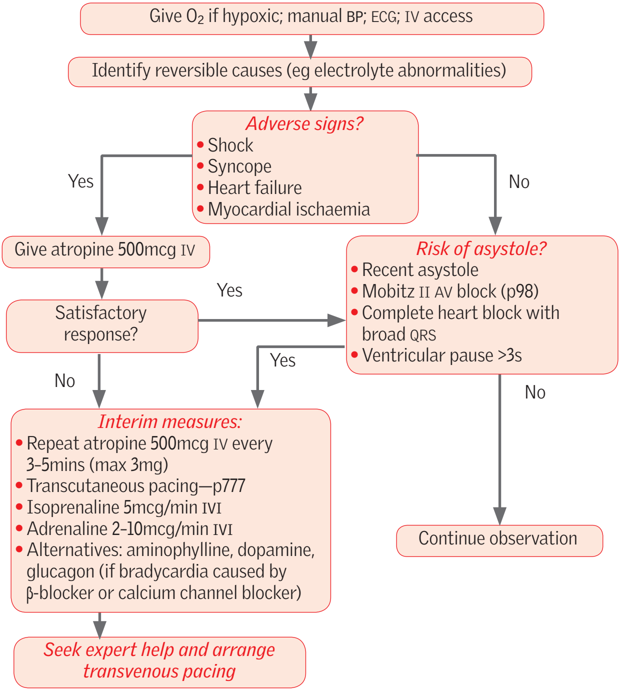



**Fig 19.11** Management of bradycardia.

# ►►Acute severe asthma

►The severity of an attack is easily underestimated.

►An atmosphere of calm helps.

**Presentation** Acute breathlessness and wheeze.

**History** (See p48.) Ask about usual and recent treatment; previous acute episodes and their severity and best peak expiratory flow rate (PEF). Have they been admitted to ICU?

**Differential diagnosis** Acute infective exacerbation of COPD, pulmonary oedema, upper respiratory tract obstruction, pulmonary embolus, anaphylaxis.

**Investigations** PEF—but may be too ill; ABG if saturations <92% or life-threatening features; CXR (if suspicion of pneumothorax, infection or life-threatening attack); FBC; U&E.

## Assessing the severity of an acute asthma attack

*Severe attack:*
- Unable to complete sentences in one breath.
- Respiratory rate ≥ 25/min.
- Pulse rate ≥110 beats/min.
- PEF 33–50% of predicted or best.

*Life-threatening attack:*
- PEF <33% of predicted or best.
- Silent chest, cyanosis, feeble respiratory effort.
- Arrhythmia or hypotension.
- Exhaustion, confusion, or coma.
- Arterial blood gases:
  - Normal/high $P_aCO_2$ >4.6kPa.
  - $P_aO_2$ <8kPa, or $S_aO_2$ <92%.

**Management** ►Rapid treatment and reassessment is key,[7] see fig 19.12.
- Salbutamol 5mg nebulized with oxygen and give prednisolone 30mg PO.
- If PEF remains <75%, repeat salbutamol; add ipratropium.
- Monitor oxygen saturation, heart rate, and respiratory rate.
- Admit all with severe features not responding to initial treatment or with life-threatening features.

NB: the routine use of antibiotics is not recommended in exacerbations of asthma.

**Discharge** Patients with PEF >75% within 1h of initial treatment can be discharged if no other reason to admit. Otherwise, before discharge patients must have:
- been stable on discharge medication for 24h
- had inhaler technique checked
- peak flow rate >75% predicted or best with diurnal variability <25%
- steroid (inhaled *and* oral) and bronchodilator therapy
- their own PEF meter and have written management plan
- GP appointment within 2d
- respiratory clinic appointment within 4wks.

## Drugs used in acute asthma

*Salbutamol* ($\beta_2$-agonist). SE: tachycardia, arrhythmias, tremor, ↓$K^+$.

*Hydrocortisone* and *prednisolone* (steroid; reduces inflammation).

*Aminophylline* is used much less frequently and is not routinely recommended in current *BTS* guidelines, but may be initiated by respiratory team or ICU. It inhibits phosphodiesterase; ↑[CAMP]. *SE:* ↑pulse, arrhythmias, nausea, seizures. The amount of IVI aminophylline may need altering according to the individual patient: always check the *BNF*. Monitor ECG. ►Aim for plasma concentration of 10–20mcg/mL (55–110μmol/L). Serious toxicity (↓BP arrhythmias, cardiac arrest) can occur at concentrations ≥25mcg/mL. Measure plasma $K^+$: theophyllines may cause ↓$K^+$. Don't load patients already on oral preparations. Stick with one brand (bioavailability varies).

## ⚠ Management of acute asthma

*Assess severity of attack:*
PEF, ability to speak, RR, pulse rate, $O_2$ sats
Warn ICU if severe or life-threatening attack

↓

*Immediate treatment:*
Supplemental $O_2$ to maintain sats 94–98%
Salbutamol 5mg (or terbutaline 10mg) nebulized with $O_2$
If severe/life-threatening add in ipratropium 0.5mg/6h to nebulizers
Hydrocortisone 100mg IV or prednisolone 40–50mg PO

↓

*Reassess every 15min:*
- If PEF <75% repeat salbutamol nebulizers every 15–30min, or 10mg/h continuously. Add ipratropium if not already given
- Monitor ECG; watch for arrhythmias
- Consider single dose of magnesium sulfate ($MgSO_4$) 1.2–2g IV over 20min in those with severe/life-threatening features without good initial response to therapy

↓

*►►If not improving:*
Refer to ICU for consideration of ventilatory support and intensification of medical therapy, eg aminophylline, IV salbutamol if any of the following signs are present:
- Deteriorating PEF
- Persistent/worsening hypoxia
- Hypercapnia
- ABG showing low pH or high $H^+$
- Exhaustion, feeble respiration
- Drowsiness, confusion, altered conscious level
- Respiratory arrest

↓

*If improving within 15–30min:*
- Continue nebulized salbutamol every 4–6h (+ ipratropium if started in previous step)
- Prednisolone 40–50mg PO OD for 5–7 days
- Monitor peak flow and $O_2$ sats, aim 94–98% with supplemental if needed
- If PEF >75% 1h after initial treatment, consider discharge with outpatient follow-up

**Fig 19.12** Management of acute asthma.

# Acute exacerbations of COPD



A common medical emergency especially in winter. May be triggered by viral or bacterial infections.

**Presentation** Increasing cough, breathlessness, or wheeze. Decreased exercise capacity.

**History** (See p48.) Ask about usual/recent treatments (especially home oxygen), smoking status, and exercise capacity (may influence a decision to ventilate the patient).

**Differential diagnosis** Asthma, pulmonary oedema, upper respiratory tract obstruction, pulmonary embolus, anaphylaxis.

**Investigations**
- ABG (p771).
- CXR to exclude pneumothorax and infection.
- FBC; U&E; CRP. Theophylline level if patient on therapy at home.
- ECG.
- Send sputum for culture if purulent.
- Blood cultures if pyrexial.

**Management** ►Ensure oxygenation then treat the reversible,[8] see fig 19.13.
- Look for a cause, eg infection, pneumothorax.
- Prior to discharge, liaise with GP regarding steroid reduction, domiciliary oxygen (p184), smoking cessation, and pneumococcal and flu vaccinations (p166).

*Treatment of stable COPD and more advanced disease:* See pp184–5.

*Consider the ceiling of care:* What is in the best interests of the patient? Invasive ventilation for exacerbations of COPD may not be appropriate: it can be difficult to wean patients off ventilatory support, and brings with it the risk of ventilator-associated pneumonias and pneumothoraces from ruptured bullae. If possible, speak to the patient early, before deterioration, try to ascertain their wishes. Patients who have previously been ventilated may not wish to repeat the experience. Consider comorbidities, $FEV_1$, functional status, whether the patient requires home oxygen, and whether the patient has previously been admitted to ICU (and if so, whether they were easily weaned from invasive ventilation). Involve the patient, the family, your seniors, and ICU early in making a decision.

### Oxygen therapy

- The greatest danger is hypoxia, which probably accounts for more deaths than hypercapnia. *Don't leave patients severely hypoxic.*
- However, in some patients, who rely on their hypoxic drive to breathe, too much oxygen may lead to a reduced respiratory rate and hypercapnia, with a consequent fall in conscious level. Always prescribe $O_2$ as if it were a drug.
- Care is always required with $O_2$, especially if there is evidence of $CO_2$ retention. Start with 24–28% $O_2$ in such patients.

►► Whenever you initiate or change oxygen therapy, do consider an ABG within 1h.
- Monitor the patient carefully. Aim to raise the $P_aO_2$ above 8.0kPa with a rise in $P_aCO_2$ <1.5kPa.
- In patients without evidence of retention at baseline use 28–40% $O_2$, but still monitor and repeat ABG.

**⚠ Management of acute COPD**

*Nebulized bronchodilators:*
Salbutamol 5mg/4h and ipratropium 500mcg/6h
*Investigate:* CXR, ABG

↓

*Controlled oxygen therapy if $S_aO_2$ <88% or $P_aO_2$ <7 kPa:*
Start at 24–28%, aim sats 88–92%
Adjust according to ABG, aim $P_aO_2$ >8.0kPa with a rise in $P_aCO_2$ <1.5kPa

↓

*Steroids:*
IV hydrocortisone 200mg and oral prednisolone 30mg OD (continue for 7–14d)

↓

*Antibiotics:*
Use if evidence of infection, eg amoxicillin 500mg/8h PO, alternatively clarithromycin or doxycycline (p387)

↓

Physiotherapy to aid sputum expectoration

↓

*If no response to nebulizers and steroids:*
Consider IV aminophylline*

↓

*If no response:*

**1** Consider non-invasive positive pressure ventilation† (NIPPV) if respiratory rate >30 or pH <7.35, or $P_aCO_2$ rising despite best medical treatment. *OR*:

**2** Consider a respiratory stimulant drug, eg doxapram 1.5–4mg/min IV in patients who are not suitable for mechanical ventilation. *SE*: *agitation*, *confusion*, *tachycardia*, *nausea*. It is a short-term measure, used only if NIV is not available

↓

Consider intubation and ventilation if pH <7.26 and $P_aCO_2$ is rising despite non-invasive ventilation only where appropriate (see 'Consider the ceiling of care' in text)

*Load with 250mg over 20min, then infuse at a rate of ~500mcg/kg/h (300mcg/kg/h if elderly), where kg is ideal body weight. Do not give a loading dose to patients on maintenance methylxanthines (theophyllines/aminophylline; see p811). Check plasma levels if given for >24h. ECG monitoring is required.

†This may alone serve as a rescue therapy, be an intermittent step before ventilation, or be considered as a 'ceiling of therapy' for those deemed not suitable for mechanical ventilation.

**Fig 19.13** Management of acute COPD.

## ▶▶Pneumothorax

### Causes

- *Spontaneous:* (Especially in young thin men) due to rupture of a subpleural bulla.
- *Chronic lung disease:* Asthma; COPD; cystic fibrosis; lung fibrosis; sarcoidosis.
- *Infection:* TB; pneumonia; lung abscess.
- *Traumatic:* Including iatrogenic (CVP line insertion, pleural aspiration or biopsy, percutaneous liver biopsy, positive pressure ventilation).
- *Carcinoma.*
- *Connective tissue disorders:* Marfan's syndrome, Ehlers–Danlos syndrome.

### Clinical features

*Symptoms:* Can be asymptomatic (especially in fit young people with small pneumothoraces) or sudden onset of dyspnoea and/or pleuritic chest pain. Patients with asthma or COPD may present with a sudden deterioration. Mechanically ventilated patients can suddenly develop hypoxia or an increase in ventilation pressures.

*Signs:* Reduced expansion, hyper-resonance to percussion, and diminished breath sounds on the affected side. With a *tension pneumothorax*, the trachea will be deviated away from the affected side and the patient will be very unwell.

**Tests** ▶*A CXR should not be performed if a tension pneumothorax is suspected, as it will delay immediate necessary treatment.* Otherwise, request an expiratory film, and look for an area devoid of lung markings, peripheral to the edge of the collapsed lung (see p725). Ensure the suspected *pneumothorax is not a large emphysematous bulla*. Check *ABG* in dyspnoeic/hypoxic patients and those with chronic lung disease.

**Management** ▶See fig 19.14. Depends on whether it is a primary pneumothorax or secondary (=underlying lung disease or smoker >50yrs old), size, and symptoms.[9]

- Size is measured from the visible lung margin to chest wall *at level of the hilum*.
- Pneumothorax due to trauma or mechanical ventilation requires a chest drain.
- Aspiration of a pneumothorax, see p767.
- Insertion and management of a chest drain, see p766. Use a small tube (10–14F) unless blood/pus is also present. Tubes may be removed 24h after the lung has re-expanded and air leak has stopped (ie the tube stops bubbling). This is done during expiration or a Valsalva manoeuvre.

*Surgical advice:* Arrange if: bilateral pneumothoraces; lung fails to expand within 48h of intercostal drain insertion; persistent air leak; two or more previous pneumothoraces on the same side; or history of pneumothorax on the opposite side.

## ▶▶Tension pneumothorax

▶▶This is a medical emergency. (See fig 16.43, p749.)

**Essence** Air drawn into the pleural space with each inspiration has no route of escape during expiration. The mediastinum is pushed over into the contralateral hemithorax, kinking and compressing the great veins. Unless the air is rapidly removed, cardiorespiratory arrest will occur.

**Signs** Respiratory distress, tachycardia, hypotension, distended neck veins, trachea deviated away from side of pneumothorax. Increased percussion note, reduced air entry/breath sounds on the affected side.

### Treatment

To remove the air, insert a large-bore (14–16G) needle with a syringe, partially filled with 0.9% saline, into the 2nd intercostal interspace in the midclavicular line on the side of the suspected pneumothorax. Remove plunger to allow the trapped air to bubble through the syringe (with saline as a water seal) until a chest tube can be placed. Alternatively, insert a large-bore Venflon in the same location.

▶▶Do this *before* requesting a CXR.

▶▶Then insert a chest drain. See p766.

## Acute management of pneumothorax

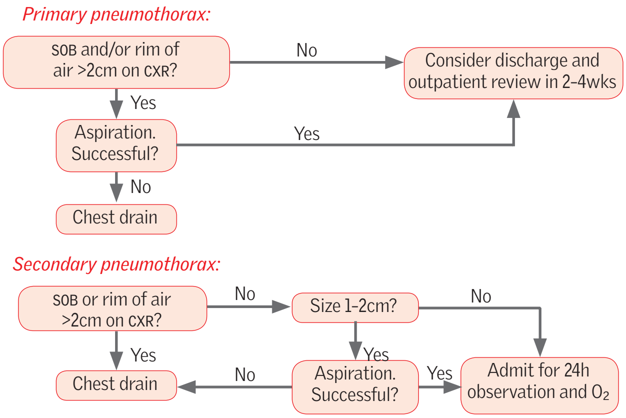



**Fig 19.14** Acute management of pneumothorax.

# Pneumonia

An infection of the lung parenchyma. Incidence of community-acquired pneumonia is 5–11 per 1000 adults. Of these, 1–3 per 1000 will require hospitalization, and mortality in those hospitalized is up to 14%.



## Common organisms

- *Streptococcus pneumoniae* is the commonest cause (60–75%).
- *Haemophilus influenzae*.
- *Mycoplasma pneumoniae*.
- *Staphylococcus aureus* found more commonly in ICU patients.
- *Legionella* species and *Chlamydia psittaci*.
- Gram-negative bacilli, often hospital-acquired or immunocompromised, eg *Pseudomonas*, especially in those with COPD.
- Viruses including influenza account for up to 15%.

**Symptoms** Fever, rigors, malaise, anorexia, dyspnoea, cough, purulent sputum (classically 'rusty' with pneumococcus), haemoptysis, and pleuritic chest pain.

**Signs** Fever, cyanosis, herpes labialis (pneumococcus), confusion, tachypnoea, tachycardia, hypotension, signs of consolidation (diminished expansion, dull percussion note, ↑ tactile vocal fremitus/vocal resonance, bronchial breathing), pleural rub.

**Management** ►Ensure oxygenation then identify and treat reversible pathology,[10] see fig 19.15.

*Investigations:* Assess severity—this will guide both investigation and treatment.
- CXR (X-ray images, fig 16.4 on p725).
- Oxygen saturation and ABG if $S_aO_2$ <92% or severe pneumonia.
- FBC, U&E, LFT, CRP.
- Blood cultures (if CURB-65 ≥2).
- Sputum cultures (if CURB-65 ≥3 or if CURB-65 =2 and not had antibiotics yet).
- Urine pneumococcal antigen (if CURB-65 ≥2); *Legionella* antigen (if CURB-65 ≥3 or if clinical suspicion).
- Consider need for viral throat swabs and mycoplasma PCR/serology.
- Pleural fluid may be aspirated for culture (if CURB-65 ≥2).
- Consider bronchoscopy and bronchoalveolar lavage if the patient is immunocompromised or on ICU.

*Severity***:** Calculate the core adverse features '**CURB-65**' score: •**C**onfusion (abbreviated mental test ≤8). •**U**rea >7mmol/L. •**R**espiratory rate ≥30/min. •**B**P <90/60mmHg. •Age ≥**65**. *Score:* 0–1: home treatment if possible; ≥2: hospital therapy; ≥3: indicates severe pneumonia and should consider ICU referral.

Other features increasing the risk of death are: coexisting disease; bilateral/multilobar involvement; $P_aO_2$ <8kPa or $S_aO_2$ <92%.

*Treatment:* See antibiotic guidance (table 4.2 on p167). Most patients who require IV antibiotics can safely be switched to PO therapy by day 3.

**Complications** (Of infection or treatment.) Pleural effusion, empyema, lung abscess, respiratory failure, septicaemia, pericarditis, myocarditis, cholestatic jaundice, acute kidney injury.

**⚠ Management of pneumonia**

*Assess using ABC:*
Treat hypoxia (sats <88%) with oxygen,
start at 24–28% if history COPD/hypercapnia

↓

Treat hypotension/shock from infection: see p790
Assess for dehydration (common if acutely unwell and fever), consider IV fluid support

↓

*Investigations:*
See p816

↓

*Antibiotics:*
See p167

↓

Analgesia for pleuritic chest pain, eg paracetamol 1g/6h or NSAID

↓

*No improvement?*
If hypoxic despite oxygen, consider CPAP to recruit lung parenchyma and improve oxygenation. But if patient is hypercapnic, they will require non-invasive or invasive (ie intubation) ventilation
►►Discuss with ICU early if patient has rising $P_aCO_2$ or remains hypoxic despite best medical therapy

**Fig 19.15** Management of pneumonia.

# Pulmonary embolism (PE)

►Always suspect pulmonary embolism (PE) in sudden collapse 1–2wks after surgery.

**Mechanism** Venous thrombi, usually from DVT, pass into the pulmonary circulation and block blood flow to lungs. The source is often occult.

## Risk factors

- Malignancy; myeloproliferative disorder; antiphospholipid syndrome.
- Surgery—especially pelvic and lower limb (much lower if prophylaxis used).
- Immobility; active inflammation (eg infection, IBD).
- Pregnancy; combined OCP; HRT.
- Previous thromboembolism and inherited thrombophilia, see p374.



## Signs and symptoms

- Acute dyspnoea, pleuritic chest pain, haemoptysis, and syncope.
- Hypotension, tachycardia, gallop rhythm, ↑JVP, loud $P_2$, right ventricular heave, pleural rub, tachypnoea, cyanosis, AF.

With thromboprophylaxis, PE following surgery is far less common, but PE may occur after any period of immobility, or with no predisposing factors. Breathlessness may be the only sign. Multiple small emboli may present less dramatically with pleuritic pain, haemoptysis, and gradually increasing breathlessness. ►Look for a source of emboli—especially DVT (is a leg swollen?).

**Investigations** ►Risk stratify based upon clinical features (use 2-level Wells' score for PE—p191). A −ve D-dimer in a low-probability patient effective excludes PE.[11]

- U&E, FBC, baseline clotting.
- *ECG:* commonly normal or sinus tachycardia; right ventricular strain pattern $V_1$–$V_3$ (p98), right axis deviation, RBBB, AF, may be deep S waves in I, Q waves in III, inverted T waves in III ('$S_I\ Q_{III}\ T_{III}$').
- *CXR:* often normal; decreased vascular markings, small pleural effusion. Wedge-shaped area of infarction. Atelectasis.
- *ABG:* hyperventilation + poor gas exchange: ↓$P_aO_2$, ↓$P_aCO_2$, ↑pH.
- *Serum D-dimer:* low specificity (↑ if thrombosis, inflammation, post-op, infection, malignancy) ∴ check only in patients with low pre-test probability (p191).
- *CT pulmonary angiography* (CTPA) is sensitive and specific and is the test of choice for high-risk patients or low-risk patients with a +ve D-dimer. If unavailable, a *ventilation-perfusion (V/Q) scan* can aid diagnosis but frequently produces equivocal results

**Management** ►See fig 19.16 for immediate management.[11]

►If good story and signs, make the diagnosis. Start treatment (fig 19.16) before definitive investigations: most PE deaths occur within 1h.

- Commence LMWH or fondaparinux.
- If there is haemodynamic instability, consider thrombolysis.
- Long-term anticoagulation: either DOAC (p350—switch directly from LMWH) or warfarin (continue LMWH until INR >2).
- Is there an underlying cause, eg thrombophilia (p374), SLE, or polycythaemia? Consider malignancy (take a careful history and perform a full physical examination; check CXR, FBC, LFT, $Ca^{2+}$; urinalysis; consider CT abdomen/pelvis and mammogram).
- If obvious remedial cause, 3 months of anticoagulation (p351) may be enough; otherwise, continue for ≥3–6 months (long term if recurrent emboli, or underlying malignancy).

**Prevention** See p190.

## ⚠ Management of large pulmonary embolism

Oxygen if hypoxic, 10–15L/min

↓

Morphine 5–10mg IV with anti-emetic
if the patient is in pain or very distressed

↓

IV access and start LMWH/fondaparinux

↓

If ↓BP give 500mL IV fluid bolus
Get ICU input

↓

►►Haemodynamically unstable?

No ↓

If persistent ↓BP consider vasopressors,
eg dobutamine 2.5–10mcg/kg/min IV or
noradrenaline; aim for systolic BP >90mmHg

Yes ↓

Consider thrombolysis
(eg alteplase 10mg IV bolus
then IVI 90mg/2h)

↓

Initiate long-term anticoagulation

**Fig 19.16** Management of large pulmonary embolism.

# ►►Acute upper gastrointestinal bleeding



**Causes**

- Peptic ulcer disease (PUD) 35–50%.
- Gastroduodenal erosions 8–15%.
- Oesophagitis 5–15%.
- Mallory-Weiss tear 15%.
- Varices 5–10%.
- Other: upper GI malignancy, vascular malformations. Consider also facial trauma, nose bleed, or haemoptysis as causes of *swallowed* blood.

**Signs and symptoms** (See pp56–63.) Haematemesis, or melaena, dizziness (especially postural), fainting, abdominal pain, dysphagia? Hypotension (in young may be postural only), tachycardia (not if on β-blocker), ↓JVP, ↓urine output, cool and clammy, signs of chronic liver disease (p276), eg telangiectasia, purpura, jaundice. NB: ask about previous GI problems, drug use, alcohol.

**Management** ►Focus on circulation,[12] see fig 19.17. Risk stratify based upon, eg Rockall score (see table 6.6, p257).

*Is the patient shocked?*

- Peripherally cool/clammy; capillary refill time >2s; urine output <0.5mL/kg/h.
- ↓GCS or encephalopathy (p275).
- Tachycardic (pulse >100bpm).
- Systolic BP <100mmHg; postural drop >20mmHg.

*If shocked:* See fig 19.17 for management.

*If haemodynamically stable:*

- Insert two large-bore (14–16G) IV cannulae and take blood for FBC, U&E, LFT, clotting, and group & save.
- Give IV fluids (p821) to restore intravascular volume; avoid saline if cirrhotic/varices; consider a CVP line to monitor and guide fluid replacement.
- Organize a CXR, ECG, and check ABG.
- Consider a urinary catheter and monitor hourly urine output.
- Transfuse if significant Hb drop (<70g/L).
- Correct clotting abnormalities (vitamin K (p274), FFP, platelets).
- If suspicion of varices (eg known history of liver disease or alcohol excess) then give terlipressin IV (1–2mg/6h for ≤3d) and initiate broad-spectrum IV antibiotics (eg piperacillin/tazobactam IV 4.5g/8h).
- Monitor pulse, BP, and CVP (keep >5cm$H_2O$) at least hourly until stable.
- Arrange an urgent *endoscopy* (p248).
- If endoscopic control fails, surgery or emergency mesenteric angiography/embolization may be needed. For uncontrolled oesophageal variceal bleeding, a Sengstaken-Blakemore tube may compress the varices, but should only be placed by someone with experience.

*Acute drug therapy:* There is no role for routine administration of PPI pre-endoscopy (provided endoscopy can be arranged in a timely manner). In patients undergoing successful endoscopic haemostasis, give PPI (eg omeprazole 40mg/12h IV/PO). Treat if positive for *H. pylori* (p253).

**Rebleeds** Serious event: 40% of patients who rebleed will die. If 'at risk' maintain a high index of suspicion. If a rebleed occurs, check vital signs every 15min and call senior cover for repeat endoscopy and/or surgical intervention.

*Signs of a rebleed:*

- Rising pulse rate.
- Falling JVP ± decreasing hourly urine output.
- Haematemesis or 'fresh' melaena (NB: it is normal to pass decreasing amounts of melaena for 24h post-haemostasis, as blood makes its way through the GI tract).
- Fall in BP (a late and sinister finding) and decreased conscious level.

**Immediate management if shocked**

Protect airway and keep NBM
Insert two large-bore cannulae (14–16G)

↓

Urgent bloods: FBC, U&E, LFT, glucose, clotting screen, crossmatch 4–6 units

↓

Rapid IV crystalloid infusion up to 1L

↓

If signs of grade III or IV shock (p790) give blood
Group specific or O Rh-ve until crossmatch done

↓

Otherwise continue IV fluids to maintain BP and transfuse if eg Hb <7

↓

Correct clotting abnormalities
Vitamin K, FFP, platelet concentrate

↓

If risk of varices (eg known liver disease or alcohol excess), give terlipressin IV 1–2mg/6h and broad-spectrum IV antibiotics

↓

Consider referral to ICU or HDU, and consider CVP line to guide fluid replacement. Aim for >5cm$H_2O$
CVP may mislead if there is ascites or CCF

↓

Catheterize and monitor urine output. Aim for >30mL/h

↓

Monitor vital signs every 15min until stable, then hourly

↓

Notify surgeons of all severe bleeds

↓

Urgent endoscopy for diagnosis ± control of bleeding at the earliest possible point after adequate resuscitation

**Fig 19.17** Immediate management of suspected upper GI bleed with shock.

# ▸▸Meningitis



▸**Primary care** Prompt actions save lives. ▸▸*If suspect meningitis arrange urgent transfer to secondary care. If a non-blanching rash is present,* give benzylpenicillin 1.2g IM/IV before admitting.

**Organisms** Meningococcus or pneumococcus. Less commonly *Haemophilus influenzae*; *Listeria monocytogenes*. HSV, VZV, enteroviruses. CMV, cryptococcus (p400), or TB (p393) if immunocompromised, eg HIV +ve, organ transplant, malignancy.

**Differential** Malaria, encephalitis, septicaemia, subarachnoid, dengue, tetanus.

Fig 19.18 Glass test for petechiae.
Courtesy of Meningitis Research Foundation.

**Features**

*Early:* Headache, fever, leg pains, cold hands and feet, abnormal skin colour.

*Later:*

- Meningism: neck stiffness, photophobia, Kernig's sign (pain + resistance on passive knee extension with hip fully flexed).
- ↓GCS, coma.
- Seizures (~20%) ± focal CNS signs (~20%) ± opisthotonus (p436, fig 9.46).
- Petechial rash (non-blanching—fig 19.18; may only be 1 or 2 spots, or none).
- Shock: prolonged capillary refill time; DIC; ↓BP.

*Signs of disease causing meningitis:* Zoster; cold sore/genital vesicles (HSV); HIV signs (lymphadenopathy, dermatitis, candidiasis, uveitis); bleeding ± red eye (leptospirosis); parotid swelling (mumps); sore throat ± jaundice ± nodes (glandular fever, p405); splenectomy scar (∴ immunodeficient).

**Management** ▸See fig 19.19; investigations and treatment proceed in parallel.[13]

- *If ↑ICP*, summon help immediately and inform ICU.
- *Initiate early antibiotics.* Take blood cultures first. Then perform LP prior to antibiotics only in patients where no evidence of shock, petechial rash or ↑ICP and where able to obtain LP within 1h (table 19.4). Consult local policies and seek advice. Empirical options include ceftriaxone 2g/12h IV; add eg amoxicillin 2g/4h IV if >60yrs age or immunocompromised. If suspect viral encephalitis see p824.
- *If features of menigism* give dexamethasone 10mg/6h IV
- *Other investigations,* U&E, FBC (↓WBC≈immunocompromise: get help), LFT, glucose, coagulation. Throat swabs (1 for bacteria, 1 for virology). CXR. Consider HIV, TB tests.
- *Prophylaxis* (discuss with public health/ID): •Household contacts in droplet range. •Those who have kissed the patient's mouth. Give ciprofloxacin (500mg PO, 1 dose; child 5–12yrs: 250mg; child <5yrs: 30mg/kg to max 125mg).

## ⚠ Lumbar puncture in meningitis

Table 19.4 CSF analysis in meningitis

| CSF in meningitis | Bacterial | Tuberculous (p393) | Viral ('aseptic') |
|---|---|---|---|
| *Appearance* | Often turbid | Often fibrin web | Usually clear |
| *Predominant cell* | Polymorphs* | Mononuclear* | Mononuclear |
| *Cell count/mm$^3$* | Eg 90–1000 or more | 10–1000 | 50–1000 |
| *Glucose* | <½ plasma | <½ plasma | >½ plasma |
| *Protein (g/L)* | >1.5 | 1–5 | <1 |
| *Bacteria* | In smear & culture | Often none in smear | None seen or cultured |

*Predominant cell type may also be lymphocytes in TB, listerial, and cryptococcal meningitis.

Perform LP (p768) without waiting for CT (not if GCS ≤12 or focal neurology). Wait for clotting screen only if suspect coagulopathy. Record opening pressure—7–18cm CSF normal but ↑ in meningitis. Send CSF for MC&S, protein, lactate, glucose, virology/PCR. *Normal values:* ≤5 lymphocytes/mm$^3$ with no neutrophils is normal. Protein: 0.15–0.45g/L. CSF glucose: 2.8–4.2mmol/L.

## ⚠ Management of suspected bacterial meningitis and meningococcal sepsis

*ABCs*: IVI + fluid resus. Check and correct blood glucose

*Meningitic:* (eg neck stiffness; photophobia) without shock

Take blood cultures

Signs of ↑ICP/shift of brain (papilloedema, uncontrolled seizures, focal neurology, GCS ≤12)

Yes

*Get ICU help:*
- IV antibiotics (see text)
- Dexamethasone 10mg IV
- Airway support
- Fluid resuscitation
- Delay LP until stable
- Nurse at 30°

No

*Get senior help:*
- Perform LP ≤1h
- IV antibiotics (see text)
  ►pre-LP, if LP delayed >1h
- Dexamethasone 10mg IV

*Septicaemic:* eg shock (prolonged capillary refill time; cold hands + feet; ↓BP), evolving rash

*Get ICU help:*
- Take blood cultures
- IV antibiotics (see text)
- Airway support /pre-emptive intubation
- Fluid resuscitation/ionotropes/vasopressors (aim for: MAP >70mmHg; urine output >30mL/h)
- Delay LP until stable

►►*Careful monitoring*

*Subsequent therapy: Discuss antibiotic therapy* with microbiology and adjust based on organism and local sensitivities. *Maintain normovolaemia* with IVI if needed. Isolate for 1st 24h. Inform Public Health, p381.

**Fig 19.19** Management of suspected bacterial meningitis and meningococcal sepsis in immunocompetent adults.

# Encephalitis



►Suspect encephalitis whenever odd behaviour, ↓consciousness, focal neurology or seizure is preceded by an infectious prodrome (↑T°, rash, lymphadenopathy, cold sores, conjunctivitis, meningeal signs). It is often wise to treat (see below) before the exact cause is known—usually viral, and often never identified. Without the infectious prodrome consider *encephalopathy:* hypoglycaemia, hepatic encephalopathy, diabetic ketoacidosis, drugs, hypoxic brain injury, uraemia, SLE, Wernicke's (give vit $B_1$ if in doubt p714).

### Signs and symptoms

- Bizarre encephalopathic behaviour or confusion.
- ↓GCS or coma.
- Fever.
- Headache.
- Focal neurological signs.
- Seizures.
- History of travel or animal bite.

### Causes

- *Viral:* HSV-1 & 2, arboviruses, CMV, EBV, VZV (varicella-zoster virus), HIV (seroconversion), measles, mumps, rabies, Japanese B encephalitis, West Nile virus, tick-borne encephalitis.
- *Non-viral:* Any bacterial meningitis, TB, malaria, listeria, Lyme disease, legionella, leptospirosis, aspergillosis, cryptococcus, schistosomiasis, typhus.

### Investigations

- *Bloods:* Blood cultures; serum for viral PCR (also throat swab and MSU); toxoplasma IgM titre; malaria film.
- *Contrast-enhanced CT:* Focal bilateral temporal lobe involvement is suggestive of HSV encephalitis. Meningeal enhancement suggests meningoencephalitis. Do before LP. MRI is alternative if allergic to contrast.
- *LP:* Typically moderately ↑CSF protein and lymphocytes, and ↓glucose (p822). Send CSF for viral PCR including HSV.
- *EEG:* Urgent EEG showing diffuse abnormalities may help confirm a diagnosis of encephalitis, but does not indicate a cause.

### Management

- ►Mortality in untreated viral encephalitis is ~70%. ►►Aim to start aciclovir within 30min of the patient arriving (10mg/kg/8h IV over 1h) for 14d as empirical treatment for HSV (21d if immunosuppressed). Specific therapies also exist for CMV and toxoplasmosis (p425).
- Supportive therapy, in high-dependency unit or ICU environment if necessary.
- Symptomatic treatment: eg phenytoin for seizures (p826).

# Cerebral abscess

Suspect this in any patient with ↑ICP, especially if there is fever or ↑WCC. It may follow ear, sinus, dental, or periodontal infection; skull fracture; congenital heart disease; endocarditis; bronchiectasis. It may also occur in the absence of systemic signs of inflammation.

**Signs** Seizures, fever, localizing signs, or signs of ↑ICP. Coma. Signs of sepsis elsewhere (eg teeth, ears, lungs, endocarditis).

**Investigations** CT/MRI p746 (eg 'ring-enhancing' lesion); ↑WCC, ↑ESR; biopsy.

**Treatment** Urgent neurosurgical referral; treat ↑ICP (p830). If frontal sinuses or teeth are the source, the likely organism will be *Strep. milleri* (microaerophilic), or oropharyngeal anaerobes. In ear abscesses, *B. fragilis* or other anaerobes are most common. Bacterial abscesses are often peripheral; toxoplasma lesions (p425) are deeper (eg basal ganglia). NB: ask yourself: is the patient immunocompromised? Discuss with infectious diseases/microbiology.

# ►►Status epilepticus

This means seizures lasting for >30min, or repeated seizures without intervening consciousness. Mortality and the risk of permanent brain damage increase with the length of attack. Aim to terminate seizures lasting more than a few minutes as soon as possible (<20min).

Status usually occurs in patients with known epilepsy. If it is the 1st presentation, the chance of a structural brain lesion is high (>50%). Diagnosis of tonic-clonic status is usually clear. Non-convulsive status (eg absence status or continuous partial seizures with preservation of consciousness) may be more difficult: look for subtle eye or lid movement. For other signs, see pp484, 490–3. An EEG can be very helpful. ►*Could the patient be pregnant* (any pelvic mass)? If so, eclampsia (*OHCS* p48) is the likely diagnosis, check the urine and BP: call a senior obstetrician—immediate delivery may be needed.



## Investigations

- Bedside glucose, the following tests can be done once ℞ has started: lab glucose, ABG, U&E, $Ca^{2+}$, FBC, ECG.
- Consider anticonvulsant levels, toxicology screen, LP, culture blood and urine, EEG, CT, carbon monoxide level.
- Pulse oximetry, cardiac monitor.

**Management** ►See fig 19.20. Basic life support—and these agents:[14]

1. *Lorazepam:* 0.1mg/kg (usually 4mg) as a slow bolus into a large vein. If no response after 10–20min give a second dose. Beware respiratory arrest during the last part of the injection. Have full resuscitation facilities to hand for all IV benzodiazepine use. The rectal route is an alternative for diazepam if IV access is difficult. *Buccal midazolam* is an easier to use oral alternative where no IV access (eg in a community setting); dose for those 10yrs old and older 10mg; if 1–5yrs 5mg, if 5–10yrs 7.5mg; squirt half the volume between the lower gum and the cheek on each side. While waiting for this to work, prepare other drugs. If fits continue ...
2. *Phenytoin infusion:* 15–18mg/kg IVI (roughly 1g if 60kg, and 1.5g if 80kg; max 2g), at a maximum rate of 50mg/min (don't put diazepam in same line: they don't mix). Beware ↓BP and do not use if bradycardic or heart block. Requires BP and ECG monitoring. 100mg/6–8h is a maintenance dose (check levels). If fits continue ...
3. *Seek ICU help:* Paralysis and anaesthesia with eg propofol is required. Close monitoring, especially respiratory function, is vital. Consider whether this could be pseudoseizures (p490), particularly if there are odd features (pelvic thrusts; resisting attempts to open lids and your attempts to do passive movements; arms and legs flailing around).
4. *Dexamethasone:* 10mg IV if vasculitis/cerebral oedema (tumour) possible.

As soon as seizures are controlled, start oral drugs (p492). Ask what the cause was, eg hypoglycaemia, pregnancy, alcohol, drugs, CNS lesion or infection, hypertensive encephalopathy, inadequate anticonvulsant dose/compliance (p490).

## ⚠ Management of convulsive status epilepticus

Open and secure the airway (adjuncts as necessary)
Remove false teeth if poorly fitting

↓

Oxygen, 100% + suction (as required)

↓

*IV access and take blood:*
U&E, LFT, FBC, glucose, $Ca^{2+}$
Toxicology screen if indicated
Anticonvulsant levels

↓

*IV bolus—to stop seizures:* eg lorazepam 4mg
Give 2nd dose of lorazepam if no response after 10–20min

↓

Thiamine 250mg IV over 30min if alcoholism or malnourishment suspected
Glucose 50mL 50% IV, unless glucose known to be normal
Treat acidosis if severe (contact ICU)

↓

Correct hypotension with fluids

↓

*IV infusion:* If seizures continue, start phenytoin, 15–18mg/kg IVI, at a rate of 50mg/min. Monitor ECG and BP. 100mg/6–8h is a maintenance dose (check levels)

↓

*General anaesthesia:* Continuing seizures after 60–90mins of above therapies require expert help with paralysis (eg propofol infusion) and ventilation with continuous EEG monitoring in ICU. NB: ►*never* spend longer than 20min on someone with status epilepticus without having help at the bedside from an anaesthetist

**Fig 19.20** Management of status epilepticus.

# ▸▸Head injury

▸If the pupils are unequal, diagnose rising intracranial pressure (ICP), eg from extradural haemorrhage, and summon urgent neurosurgical help (p482). Retinal vein pulsation at fundoscopy helps exclude ↑ICP.

**Initial management** ▸See fig 19.21. Write full notes. Record times.

▸▸Stabilization of airway, breathing, and circulation (ABC) remains the 1st priority. If GCS ≤8 then seek urgent anaesthetic and ICU help to protect airway.[15]

- Involve neurosurgeons early, especially if ↓GCS, or if ↑ICP suspected.
- Examine the CNS. Chart pulse, BP, T°, respirations + pupils every 15min.
- Assess anterograde amnesia (loss from the time of injury, ie post-traumatic) and retrograde amnesia (for events prior to injury)—extent of retrograde loss correlates with severity of injury, and never occurs without anterograde amnesia.
- Nurse semi-prone if no spinal injury; meticulous care to airway and bladder.

*Perform a CT head <1h if:*

- GCS <13 on initial assessment, or GCS <15 at 2h following injury.
- Focal neurological deficit.
- Suspected open or depressed skull fracture, or signs of basal skull fracture: periobital ecchymoses ('panda' eyes/racoon sign), postauricular ecchymosis (Battle's sign), CSF leak through nose/ears, haemotympanum.
- Post-traumatic seizure.
- Vomiting more than once.

*Perform a CT head <8h if:* Any loss of consciousness or amnesia ***AND*** any of: •age ≥65 •coagulopathy •high-impact injury, eg struck by or ejected from motor vehicle; fall >1m or >5 stairs •retrograde amnesia of >30min.

*Suspected cervical spine injuries:* Perform a CT cervical spine <1h if:

- GCS <13 on initial assessment.
- The patient has been intubated.
- Definitive diagnosis of cervical spine injury is needed urgently (eg before surgery).
- The patient is having other body areas scanned, eg multi-region trauma.
- Clinical suspicion of cervical spine injury ***AND ANY OF:*** •age 65 years or older •high-impact injury •focal neurological deficit •paraesthesia in the upper or lower limbs.

▸If above-listed criteria are ***NOT*** met ***AND IF ANY OF*** the following low-risk features are present, then assess neck movement: •simple rear-end motor vehicle collision •comfortable in a sitting position •ambulatory since injury •no midline cervical spine tenderness •delayed onset of neck pain. If patient unable actively to rotate neck 45° to left and right or if a low-risk feature not present, then obtain plain X-rays of cervical spine <1h. If X-rays technically inadequate, suspicious, or definitely abnormal, proceed to CT.

*Admit if:* •new, clinically significant abnormalities on CT •GCS<15 after CT, regardless of result or continuing worrying signs (eg vomiting) •when CT indications met but CT unavailable •other concerns (eg drugs or alcohol, other injuries, CSF leak, shock, suspected non-accidental injury, meningism).

▸Do not attribute ↓GCS to alcohol until a significant head injury has been excluded. Alcohol is an unlikely cause of coma if plasma alcohol <44mmol/L. If unavailable, estimate blood alcohol level from the osmolar gap (p668). If blood alcohol ≈ 40mmol/L, osmolar gap ≈ 40mmol/L.

*Discuss with neurosurgical unit* all with significant abnormalities on CT or with eg persistent GCS ≤8 or deteriorating GCS (especially motor component), persistent confusion, progressive focal neurology, seizure without full recovery, penetrating injuries, or CSF leak. If transfer is required, ensure skilled medical escort and consider need for intubation prior to transfer.

**Complications** *Early:* Extradural/subdural haemorrhage, seizures. *Late:* Subdural (p482), seizures, diabetes insipidus, parkinsonism, dementia.

**Indicators of a bad prognosis** Old age, decerebrate rigidity, extensor spasms, prolonged coma, ↑BP, ↓$P_aO_2$ (on blood gases), T° >39°C. 60% of those with loss of consciousness of >1 month will survive 3–25yrs, but may need daily nursing care.

## Immediate management plan for head injury

ABC

↓

Oxygen if saturations <92% or hypoxic on ABG
Intubate and hyperventilate if necessary
Immobilize neck until injury to cervical spine excluded

↓

Stop blood loss and support circulation
Treat for shock if required (p790)

↓

Treat seizures with lorazepam ± phenytoin (p826)

↓

*Assess level of consciousness (GCS):*
►► If GCS ≤8 ICU involvement to manage airway
Assess for anterograde and retrograde amnesia

↓

Rapid examination survey

↓

*Investigations:*
U&Es, glucose, FBC, blood alcohol,
toxicology screen, ABGs, and clotting

↓

Neurological examination

↓

*Brief history:*
When? Where? How? Had a fit? Lucid interval? Alcohol?

↓

*Evaluate lacerations of face or scalp:*
Palpate deep wounds with sterile glove to check for
step deformity. Note obvious skull/facial fractures

↓

*Check for CSF leak, from nose (rhinorrhoea) or ear (otorrhoea)*
*Any blood behind the ear drum?*
If either is present, suspect basilar skull fracture: do CT
Give tetanus toxoid, and refer at once to neurosurgeons

↓

*Palpate the neck posteriorly for tenderness and deformity:*
If detected, or if the patient has other indicators for neck imaging, immobilize the neck and get cervical spine X-ray or CT neck (see text)

↓

*Radiology:*
As indicated: CT of head/neck (see text)
Consider need for trauma series (eg CT chest/abdo/pelvis)

**Fig 19.21** Immediate management plan for head injury.

# ▶▶Raised intracranial pressure (ICP)

The volume inside the cranium is fixed, so any increase in the contents can lead to raised ICP. This can be mass effect, oedema, or obstruction to fluid outflow. Normal ICP in adults is <15mmHg.



### Causes

- Primary or metastatic tumours.
- Head injury.
- Haemorrhage (subdural, extradural, subarachnoid, intracerebral, intraventricular).
- Infection: meningitis, encephalitis, brain abscess.
- Hydrocephalus.
- Cerebral oedema.
- Status epilepticus.

### Signs and symptoms

- Headache (worse on coughing, leaning forwards), vomiting.
- Altered GCS: drowsiness, listlessness, irritability, coma.
- History of trauma.
- ↓HR and ↑BP (Cushing's response); Cheyne-Stokes respiration.
- Pupil changes (constriction at first, later dilation—do not mask these signs by using agents such as tropicamide to dilate the pupil to aid fundoscopy).
- ↓Visual acuity; peripheral visual field loss.
- Papilloedema is an unreliable sign, but venous pulsation at the disc may be absent (absent in ~50% of normal people, but *loss* of it is a useful sign).

### Investigations

- U&E, FBC, LFT, glucose, serum osmolality, clotting, blood culture.
- Consider toxicology screen.
- CXR—any source of infection that might indicate abscess?
- CT head.
- Then consider LP if safe. Measure the opening pressure!

**Management** ►See fig 19.22. The goal is to ↓ICP and avert secondary injury. Urgent neurosurgery is required for the definitive treatment of ↑ICP from focal causes (eg haematomas). This is achieved via a craniotomy or burr hole. Also, an ICP monitor (or bolt) may be placed to monitor pressure. *Holding measures* are listed in fig 19.22.

### Herniation syndromes

*Uncal herniation* is caused by a lateral supratentorial mass, which pushes the ipsilateral inferomedial temporal lobe (uncus) through the temporal incisura and against the midbrain. The IIIrd nerve, travelling in this space, gets compressed, causing a dilated ipsilateral pupil, then ophthalmoplegia (a fixed pupil localizes a lesion poorly but is 'ipsilateralizing'). This may be followed (quickly) by contralateral hemiparesis (pressure on the cerebral peduncle) and coma from pressure on the ascending reticular activating system (ARAS) in the midbrain.

*Cerebellar tonsil herniation* is caused by ↑pressure in the posterior fossa forcing the cerebellar tonsils through the foramen magnum. Ataxia, VIth nerve palsies, and upgoing plantar reflexes occur first, then loss of consciousness, irregular breathing, and apnoea. This syndrome may proceed very rapidly given the small size of, and poor compliance in, the posterior fossa.

*Subfalcian (cingulate) herniation* is caused by a frontal mass. The cingulate gyrus (medial frontal lobe) is forced under the rigid falx cerebri. It may be silent unless the anterior cerebral artery is compressed and causes a stroke—eg contralateral leg weakness ± abulia (lack of decision-making).

## ⚠ Immediate management plan for raised intracranial pressure

ABC

↓

Correct hypotension, maintain MAP >90mmHg and treat seizures

↓

*Brief examination; history if available:*
Any clues, eg meningococcal rash, previous carcinoma

↓

Elevate the head of the bed to 30–40°

↓

*If intubated, hyperventilate to ↓$P_aCO_2$ (aim 3.5–4kPa):*
This causes cerebral vasoconstriction and reduces ICP almost immediately. Maintain $P_aO_2$ >12kPa

↓

*Osmotic agents (eg mannitol) can be useful but may lead to rebound ICP after prolonged use (~12–24h):*
Give 20% solution 0.25–0.5g/kg IV over 10–20min (eg 5mL/kg). Effect is seen after ~20min and lasts for 2–6h. Follow serum osmolality—aim for about 300mosmol/kg but don't exceed 310

↓

*Corticosteroids are not effective in reducing ICP except for oedema surrounding tumours:*
Eg dexamethasone 10mg IV and follow with 4mg/6h IV/PO

↓

Consider other measures,
eg sedation, anti-epileptics, therapeutic hypothermia

↓

Restrict fluid to <1.5L/d

↓

Monitor the patient closely; consider monitoring ICP

↓

Aim to make a diagnosis

↓

Treat cause or exacerbating factors,
eg hyperglycaemia, hyponatraemia

↓

Definitive treatment if possible

**Fig 19.22** Immediate management plan for raised intracranial pressure.

# Diabetic ketoacidosis (DKA)

**Mechanism** Normally the body metabolizes carbohydrates, leading to efficient energy production. Ketoacidosis is an alternative metabolic pathway used in starvation states; it is far less efficient, and produces acetone as a byproduct (hence the fruity breath of patients in ketosis). In acute diabetic ketoacidosis, there is excessive glucose, but because of a lack of insulin, this cannot be taken up into cells to be metabolized, so pushing the body into a starvation-like state where ketoacidosis is the only mechanism of energy production. The combination of severe acidosis and hyperglycaemia can be deadly, so early recognition and treatment is important.

**Typical picture** Gradual drowsiness, vomiting, and dehydration in type 1 diabetic (very rarely type 2). ►Do glucose in *all* those with unexplained vomiting, abdo pain, polyuria, polydipsia, lethargy, anorexia, ketotic breath, dehydration, coma, or deep breathing (sighing 'Kussmaul' hyperventilation). *Triggers:* Infection, eg UTI; surgery; MI; pancreatitis; chemotherapy; antipsychotics; wrong insulin dose/non-compliance.

**Diagnosis**

1. Acidaemia (venous blood pH <7.3 or $HCO_3^-$ < 15.0mmol/L).
2. Hyperglycaemia (blood glucose >11.0mmol/L) or known DM.
3. Ketonaemia (≥3.0mmol/L) or significant ketonuria (more than 2+ on dipstick).

*Tests:* ECG, CXR. *Urine:* Dipstick and MSU. *Blood:* Capillary and lab glucose, ketones, pH (use venous blood; ABG only if ↓GCS or hypoxia), U&E, $HCO_3^-$, osmolality, FBC, blood culture.

**Severe DKA** If one or more of the following features is present on admission, consider transfer to HDU/ICU for monitoring and central venous access. Get senior help!

- Blood ketones >6mmol/L.
- Venous bicarbonate <5mmol/L.
- Venous/arterial pH <7.0.
- K <3.5mmol/L on admission.
- GCS <12.
- $O_2$ sats <92% on air (assuming no respiratory disease).
- Systolic BP <90mmHg.
- Pulse >100 or <60 bpm.
- Anion gap above 16 (p670).

**Management** ►Replace volume then correct metabolic defects,[16] see fig 19.23.

**Pitfalls in diabetic ketoacidosis**

- *Plasma glucose* is usually high, but not always, especially if insulin continued.
- *High WCC* may be seen in the absence of infection.
- *Infection.* Often there is no fever. Do MSU, blood cultures, and CXR. Start broad-spectrum antibiotics (eg co-amoxiclav, p386) early if infection is suspected.
- *Creatinine.* Some assays for creatinine cross-react with ketone bodies, so plasma creatinine may not reflect true renal function.
- *Hyponatraemia* is common, due to osmolar compensation for the hyperglycaemia. ↑ or ↔ [$Na^+$] indicates severe water loss. As treatment commences $Na^+$ rises as water enters cells. $Na^+$ is also low due to an artefact; corrected plasma [$Na^+$] = $Na^+$ + 2.4[(glucose −5.5)/5.5].
- *Ketonuria* does not equate with ketoacidosis. Anyone may have up to ++ketonuria after an overnight fast. Not all ketones are due to diabetes—consider alcohol if glucose normal. Always check venous blood ketones.
- *Recurrent ketoacidosis.* Blood glucose may return to normal long before ketones are removed from the blood, and premature termination of insulin infusion may lead to lack of clearance and return to DKA. This may be avoided by maintaining a constant rate of insulin infusion (with co-infusion of glucose 10% to maintain plasma glucose at 6–10mmol/L) until blood ketones <0.6mmol/L and pH >7.3.
- *Acidosis* but without gross elevation of glucose may occur, but consider overdose (eg aspirin) and lactic acidosis (in elderly diabetics).
- *Serum amylase* is often raised (up to ×10) and non-specific abdominal pain is common, even in the absence of pancreatitis.

**Complications** ►Cerebral oedema (get help if sudden CNS decline), aspiration pneumonia, hypokalaemia, hypomagnesaemia, hypophosphataemia, thromboembolism.

**Prevention** ►Talk to the patient: evaluate compliance and educate about triggers.

## ⚠ Management plan for diabetic ketoacidosis

*ABC approach, 2 large-bore cannulae:*
Start fluid: 1L 0.9% saline over 1h (if systolic BP <90mmHg then give 500mL bolus over 15mins and reassess—if systolic BP still <90mmHg then seek senior review and give further 500mL bolus; if BP remains <90mmHg then involve ICU)

↓

*Tests:* Venous blood gas for pH, bicarbonate; bedside and lab glucose and ketones; U&Es, FBC, CRP; CXR, ECG

↓

*Insulin:* Add 50 units human soluble insulin to 50mL 0.9% saline. Infuse continuously at 0.1 unit/kg/h. Continue patient's regular long-acting insulin at usual doses and times; consider initiating long-acting insulin in newly diagnosed T1DM. ►Aim for a fall in blood ketones of 0.5mmol/L/h, or a rise in venous bicarbonate of 3mmol/L/h with a fall in glucose of 3mmol/L/h. If not achieving this, increase insulin infusion by 1 unit/h until target rates achieved

↓

Check capillary blood glucose and ketones hourly
Check VBG (pH, $HCO_3^-$, $K^+$) at 2h, 4h, 8h, 12h, and 24h (or more frequent)

↓

Continue fluids and assess need for $K^+$ (see below)

↓

Consider catheter if not passed urine by 1h, aim for urine output 0.5mL/kg/h. Consider NG tube if vomiting or drowsy
Start all patients on LMWH

↓

Avoid hypoglycaemia! When glucose <14mmol/L start 10% glucose at 125mL/h to run alongside saline and prevent hypoglycaemia

↓

Continue fixed-rate insulin until ketones <0.6mmol/L, venous pH >7.3, and venous bicarb >15mmol/L. Do not rely on urinary ketones to indicate resolution—they stay positive after DKA resolved
Find and treat infection/cause for DKA

Fig 19.23 Management plan for diabetic ketoacidosis.

### ⚠ Fluid replacement

- 0.9% saline is the replacement fluid of choice.
- Typical fluid deficit is 100mL/kg, so for an average 70kg man = 7 litres.
- Give eg 1L in 1h (faster if systolic BP <90mmHg) then: 1L over 2h, 1L over 2h, 1L over 4h, 1L over 4 h, 1L over 8h. This regimen may not be appropriate for all: reassess frequently, especially if young, elderly, pregnant, or comorbidities.
- Bicarbonate may increase risk of cerebral oedema and is not recommended.

### ⚠ Potassium replacement

- Typical deficit = 3-5mmol/kg, plasma $K^+$ falls with treatment as $K^+$ enters cells.
- Don't add $K^+$ to the 1st bag. Thereafter add $K^+$ according to most recent VBG result (table 19.5).

Table 19.5 Potassium replacement in DKA

| Serum $K^+$ (mmol/L) | Amount of KCl to add per litre of IV fluid |
|---|---|
| >5.5 | Nil |
| 3.5–5.5 | 40mmol |
| <3.5 | Seek help from HDU/ICU for higher doses |

## Other diabetic emergencies

**Hypoglycaemia** Usually *rapid* onset; may be accompanied by odd behaviour (eg aggression), sweating, ↑pulse, seizures. *Management:* ►►If conscious, orientated, and able to swallow, give 15–20g of quick-acting carbohydrate snack (eg 200mL orange juice) and recheck blood glucose after 10–15mins (repeat snack up to 3 times). If conscious but uncooperative, squirt glucose gel between teeth and gums. In unconscious patients, or those not responding to these measures, start glucose IVI (eg 10% at 200mL/h if conscious; 10% at 200mL/15mins if unconscious), or give glucagon 1mg IV/IM (will not work in malnourished patients). Expect prompt recovery. Once blood glucose >4.0mmol/L and patient has recovered, give long-acting carbohydrate (eg slice of toast).

**Hyperglycaemic hyperosmolar state (HHS)** Seen in unwell patients with type 2 DM.[17] The history is longer (eg 1wk), with marked dehydration and glucose >30mmol/L. There is no switch to ketone metabolism, so ketonaemia stays <3mmol/L and pH >7.3. Osmolality is typically >320mosmol/kg: ►*Occlusive events are a danger* (focal CNS signs, chorea, DIC, leg ischaemia/rhabdomyolysis)—give LMWH prophylaxis to all unless contraindication (p350). Rehydrate slowly with 0.9% saline IVI over 48h, typical deficits are 110–220mL/kg, ie 8–15L for a 70kg adult. Replace $K^+$ when urine starts to flow (see DKA BOX, p833). ►Only use insulin if blood glucose not falling by 5mmol/L/h with rehydration or if ketonaemia—start slowly 0.05u/kg/h. Keep blood glucose at least 10–15mmol/L for first 24 hours to avoid cerebral oedema. Look for the cause, eg MI, drugs, sepsis, or bowel infarct.

**Lactic acidosis** A rare but serious complication of DM with metformin use or septicaemia. Blood lactate: >5mmol/L. Seek expert help. Treat sepsis vigorously, maintain blood pressure and hence tissue perfusion. Stop metformin.

## Thyroid emergencies

**Myxoedema coma** The ultimate hypothyroid state before death.

*Signs and symptoms:* Looks hypothyroid (p220, p221, fig 5.16); often >65yrs; hypothermia; hyporeflexia; ↓glucose; bradycardia; coma; seizures. May have had radioiodine, thyroidectomy, or pituitary surgery (signs of hypopituitarism, p232). Psychosis (myxoedema madness) may precede coma. *Precipitants:* infection, MI, stroke, or trauma.

*Examination:* Goitre; cyanosis; ↓BP (cardiogenic); heart failure; signs of precipitants.

*Treatment:* Preferably on ICU.

- Blood for: $T_3$, $T_4$, TSH, FBC, U&E (often ↓$Na^+$), cultures, cortisol, glucose.
- ABG for $P_aO_2$. High-flow $O_2$ if cyanosed. Ventilation may be needed.
- Correct any hypoglycaemia.
- Give $T_3$ (liothyronine) 5–20mcg/12h IV slowly. Be cautious: you may precipitate manifestations of ischaemic heart disease. Alternative regimens involve levothyroxine.
- Give hydrocortisone 100mg/8h IV—vital if pituitary hypothyroidism is suspected (ie no goitre, no previous radioiodine, and no previous thyroid surgery).
- If infection suspected, give antibiotic, eg co-amoxiclav 1.2g/8h IV.
- Caution with fluid, rehydrate as needed but watch for cardiac dysfunction; BP may not respond to fluid and inotropes may be needed.
- Active warming (blankets, fluids) may be needed for *hypothermia*. Beware complications (hypoglycaemia, pancreatitis, arrhythmias). See p848.

*Further ℞:* $T_3$ 5–20mcg/4–12h IV until sustained improvement (~2–3d) then levothyroxine 50mcg/24h PO. Hydrocortisone + IV fluids as needed (hyponatraemia is dilutional).

**Hyperthyroid crisis (thyrotoxic storm)** *Signs and symptoms:* ♀:♂≈4:1. Severe hyperthyroidism (see p218): ↑T°, agitation, confusion, coma, tachycardia, AF, D&V, goitre, thyroid bruit, acute abdomen (exclude surgical causes), heart failure.

*Precipitants:* Recent thyroid surgery or radioiodine; infection; MI; trauma.

*Diagnosis:* Do not wait for test results if urgent treatment is needed. Do TSH, free $T_4$, and free $T_3$. Confirm with technetium uptake if possible.

*Treatment:* ►See fig 19.24. Seek endocrinology advice and aim to: **1** Counteract peripheral effects of thyroid hormones. **2** Inhibit thyroid hormone synthesis. **3** Treat systemic complications. If you are not making headway in 24h, thyroidectomy may be an option.

**Management plan for thyrotoxic storm**

IV access, fluids if dehydrated. NG tube if vomiting

↓

Take blood for: $T_3$, $T_4$, TSH, cultures (if infection suspected)

↓

Sedate if necessary (eg chlorpromazine 50mg PO/IM). Monitor BP

↓

If no contraindication, and cardiac output OK, give propranolol 60mg/4–6h PO; max IV dose: 1mg over 10min; may need repeating every few hours. In asthma/poor cardiac output, propranolol has caused cardiac arrest in thyroid storm, so ultra-short-acting β-blockers have a role, eg IV esmolol. Consider diltiazem if β-blockers contraindicated. ►*Get help*

↓

High-dose digoxin may be needed to slow the heart, but ensure adequately β-blocked, give with cardiac monitoring

↓

Antithyroid drugs: carbimazole 15–25mg/6h PO (or via NGT); after 4h give Lugol's solution (aqueous iodine oral solution) 0.3mL/8h PO well diluted in water for 7–10d to block thyroid

↓

Hydrocortisone 100mg/6h IV or dexamethasone 2mg/6h PO to prevent peripheral conversion $T_4$ to $T_3$

↓

Treat suspected infection, eg with co-amoxiclav 1.2g/8h IV

↓

Adjust IV fluids as necessary; cool with tepid sponging ± paracetamol

↓

*Continuing treatment:* After 5d reduce carbimazole to 15mg/8h PO. After 10d stop propranolol and iodine. Adjust carbimazole (p218)

**Fig 19.24** Management plan for thyrotoxic storm.

## Addisonian crisis

**Signs and symptoms** Patients may present in shock (↑HR; vasoconstriction; postural hypotension; oliguria; weak; confused; comatose)—often (but not always) in a patient with known Addison's (eg when oral steroid has not been increased to cover stress such as pneumonia), or someone on long-term steroids who has forgotten their tablets. Remember bilateral adrenal haemorrhage (eg meningococcaemia) as a cause. An alternative presentation is with hypoglycaemia.

**Precipitating factors** Infection, trauma, surgery, missed medication.

**Management** If suspected, treat before biochemical results.

- Bloods for cortisol and ACTH (this needs to go straight to laboratory, call ahead!), U&Es—can have ↑$K^+$ (check ECG and give calcium gluconate if needed, see p301) and ↓$Na^+$ (salt depletion, should resolve with rehydration and steroids).
- Hydrocortisone 100mg IV stat.
- IV fluid bolus eg 500mL 0.9% saline to support BP, repeated as necessary.
- Monitor blood glucose: the danger is hypoglycaemia.
- Blood, urine, sputum for culture, then antibiotics if concern about infection.

**Continuing treatment**

- Glucose IV may be needed if hypoglycaemic.
- Give IV fluids as guided by clinical state and to correct U&E imbalance.
- Continue hydrocortisone, eg 100mg/8h IV or IM.
- Change to oral steroids after 72h if patient's condition good.
- Fludrocortisone may well be needed if the cause is adrenal disease: ask an expert.
- Search for (and vigorously treat) the underlying cause. Get endocrinological help.

## Hypopituitary coma

Think of decompensated chronic hypophyseal failure whenever hypothermia, refractory hypotension ± septic signs *without* fever occur with short stature or loss of axillary/pubic hair ± gonadal atrophy. ►Waiting for lab confirmation may be fatal. It *usually* develops gradually in a person with known hypopituitarism. If rapid onset due to pituitary infarction (eg postpartum, Sheehan's, p232), subarachnoid haemorrhage is often misdiagnosed as symptoms include headache and meningism.

**Presentation** Headache; ophthalmoplegia; ↓consciousness; hypotension; hypothermia; hypoglycaemia; signs of hypopituitarism (p232).

**Tests** Cortisol; $T_4$; TSH; ACTH; glucose. Pituitary fossa CT/MRI.

**Treatment** ►Hydrocortisone, eg 100mg IV/6h.

- Only after hydrocortisone begun: liothyronine (L-tri-iodothyronine sodium), eg 10mcg/12h PO or by slow IV: 5–20mcg/12h (4-hourly may be needed).
- Prompt surgery is needed if the cause is pituitary apoplexy (p234).

## Phaeochromocytoma emergencies

Patients with phaeochromocytoma may have had undiagnosed symptoms for some time, but stress, abdominal palpation, parturition, general anaesthetic, or contrast media used in imaging can cause acute *hypertensive crises*.

**Signs and symptoms** Pallor, pulsating headache, hypertension, feels 'about to die', pyrexial. ***ECG***: signs of LVF, ↑ST segment, VT, and cardiogenic shock.

**Treatment** ►Get help. Take to ICU.

Principle is combined α- and β-adrenoreceptor blockade, but α must be started first, as unopposed β-blockade can worsen hypertension.

- Start with short-acting, IV α-blocker, eg phentolamine 2–5mg IV. Repeat to maintain safe BP.
- When BP controlled, give long-acting α-blocker, eg phenoxybenzamine 10mg/24h PO (increase by 10mg/d as needed, up to 30mg/12h PO); SE: postural hypotension; dizziness; tachycardia; nasal congestion; miosis; idiosyncratic marked BP drop after 1st dose. The idea is to ↑ the dose until BP is controlled and there is no significant postural hypotension. Alternative $\alpha_1$-selective blockers, eg doxazosin, are preferred in some centres, particularly if surgery is not an option, eg metastatic tumour.
- A $\beta_1$-blocker may also be given at this stage to control any tachycardia or myocardial ischaemia/dysrhythmias (p114).
- Surgery is usually done electively after 4–6wks to allow full α-blockade and volume expansion. When admitted for surgery the phenoxybenzamine dose is increased until significant postural hypotension occurs.

# Acute poisoning—general measures



**Diagnosis** Mainly from the history. The patient may not tell the truth about what has been taken. If there are any tablets with the patient, use *MIMS Colour Index*, *EMIMS* images, *BNF* descriptions, or the computerized system 'TICTAC' (ask pharmacy) to identify tablets and plan specific treatment.

**TOXBASE** The best resource for managing acute poisoning: www.toxbase.org—check with your Emergency Department about log-in details for your hospital.

**Clues** May become apparent from examination:
- *Fast or irregular pulse:* Salbutamol, antimuscarinics, tricyclics, quinine, or phenothiazine poisoning.
- *Respiratory depression:* Opiate (p842) or benzodiazepine (p842) toxicity.
- *Hypothermia:* Phenothiazines (p843), barbiturates.
- *Hyperthermia:* Amphetamines, MAOIs, cocaine, or ecstasy (p843).
- *Coma:* Benzodiazepines, alcohol, opiates, tricyclics, or barbiturates.
- *Seizures:* Recreational drugs, hypoglycaemic agents, tricyclics, phenothiazines, or theophyllines.
- *Constricted pupils:* Opiates (p842) or insecticides (organophosphates, p843).
- *Dilated pupils:* Amphetamines, cocaine, quinine, or tricyclics.
- *Hyperglycaemia:* Organophosphates, theophyllines, or MAOIs.
- *Hypoglycaemia:* (p834) Insulin, oral hypoglycaemics, alcohol, or salicylates.
- *Renal impairment:* Salicylate (p844), paracetamol (p844), or ethylene glycol.
- *Metabolic acidosis:* Alcohol, ethylene glycol, methanol, paracetamol, or carbon monoxide poisoning (p842).
- *↑Osmolality:* Alcohols (ethyl or methyl); ethylene glycol. See p668.

**Management** ►See fig 19.25.
- *Take blood* as appropriate (p840). Always check paracetamol and salicylate levels.
- *Empty stomach* if appropriate (p840).
- *Consider specific antidote* (p842) or oral activated charcoal (p840).
- *If you are not familiar with the poison* get more information. *Toxbase* (www.toxbase.org) should be your first thought. If no information here or in doubt how best to act, phone the Poisons Information Service: in the UK phone 0844 892 0111.

**Continuing care** Measure temperature, pulse, BP, and blood glucose regularly. Keep on cardiac monitor. If unconscious, nurse semi-prone, turn regularly. Catherize if the bladder is distended, or acute kidney injury (p298) is suspected, or forced diuresis undertaken. Consider ICU, eg if ↓respiration.

**Psychiatric assessment** Be sympathetic despite the hour! Interview relatives and friends if possible. Aim to establish:
- *Intentions at time.* Was this a suicide attempt, if so was the act planned? What precautions against being found? Did the patient seek help afterwards? Does the patient think the method was dangerous? Was there a final act (eg suicide note)?
- *Present intentions.* Do they still feel suicidal? Do they wish it had worked?
- *What problems* led to the act: do they still exist?
- Is there a *psychiatric disorder* (depression, alcoholism, personality disorder, schizophrenia, dementia)?
- What are the patient's *resources* (friends, family, work, personality)?

*The assessment of suicide risk:* The following ↑chance of future suicide: original intention was to die; present intention is to die; presence of psychiatric disorder; poor resources; previous suicide attempts; socially isolated; unemployed; male; >50yrs old. See *OHCS* p358. There is ↑risk of death in the first year following initial presentation.

⚠ *Referral to psychiatrist:* This depends partly on local resources. Refer all with presence of psychiatric disorder or high suicide risk. Consider discussing all presentations with deliberate self-poisoning.

*Mental Capacity Act or the Mental Health Act:* (In England and Wales) may provide for the detention of the patient against his or her will: see *OHCS* p406.

## Emergency care in acute poisoning

ABC, clear airway

↓

Consider ventilation (if the respiratory rate is <8/min, or $P_aO_2$ <8kPa, when breathing 60% $O_2$, or the airway is at risk, eg GCS ≤ 8)

↓

Treat shock (p790)

↓

If unconscious, nurse semi-prone

## *Further management*

Assess the patient

↓

History from patient, friends, or family is vital

↓

Features from the examination may help (see p838)

↓

*Investigations:*
- Glucose, U&E, FBC, LFT, INR, ABG, ECG, paracetamol, and salicylate levels
- Urine/serum toxicology, specific assays as appropriate

↓

*Monitor:*
- T°, pulse, and respiratory rate, BP, $O_2$ saturations, urine output ± ECG

↓

*Treatment:*
- Supportive measures: may need catheterization
- ↓Absorption: consider gastric lavage ± activated charcoal (see p840)

↓

*Specific measures:* See p840; for *antidotes*, see p842
*Consider naloxone* if ↓conscious level and pin-point pupils
Consider Pabrinex® and glucose if drowsy/confused

**Fig 19.25** Emergency care in acute poisoning.

# Acute poisoning—specific points



**Plasma toxicology** For all unconscious patients, paracetamol and aspirin levels and blood glucose are required. The necessity of other assays depends on the drug taken and the index of suspicion. Be guided by the Poisons Information Service. More common assays include: digoxin; methanol; lithium; iron; theophylline. Toxicological screening of urine, especially for recreational drugs, may be of use in some cases (although not always, see BOX).

**GI decontamination** Recommended for many drugs. The treatment of choice is now activated charcoal rather than gastric lavage. If in doubt, consult Toxbase or Poisons Information Service.

**Activated charcoal** Reduces the absorption of many drugs from the gut when given as a single dose of 50g with water, eg salicylates, paracetamol. It is given in repeated doses (50g/4h) to increase elimination of some drugs from the blood, eg carbamazepine, dapsone, theophyllines, quinine, phenobarbital, and paraquat. Lower doses are used in children. Do not use with petroleum products, corrosives, alcohols, clofenotane, malathion, or metal salts (eg iron, lithium).

**Gastric lavage** Rarely used. Lavage after 30–60min may make matters worse. ►*Do not empty stomach* if petroleum products or corrosives such as acids, alkalis, bleach, descalers have been ingested (*exception:* paraquat), or if the patient is unconscious or unable to protect their airway (unless intubated). ►Never induce vomiting.

**Gastric emptying and lavage** NB: we do not recommend gastric lavage is attempted unless specifically suggested by Toxbase or Poisons Information Service. If comatose, or no gag reflex, ask for an anaesthetist to protect airway with cuffed endotracheal tube. If conscious, get verbal consent.

- Monitor $O_2$ by pulse oximetry. See p162.
- Have suction apparatus to hand and working.
- Position the patient in left lateral position.
- Raise the foot of the bed by 20cm.
- Pass a lubricated tube (14mm external diameter) via the mouth, asking the patient to swallow.
- Confirm position in stomach (see p759).
- Siphon the gastric contents. Check pH with litmus paper.
- Perform gastric lavage using 300–600mL tepid water at a time. Massage the left hypochondrium then siphon fluid.
- Repeat until no tablets in siphoned fluid.
- Leave activated charcoal (50g in 200mL water) in the stomach unless alcohol, iron, $Li^+$, or ethylene glycol ingested.
- When pulling out tube, occlude its end (prevents aspiration of fluid remaining in the tube).

**Haemodialysis** This may be needed for poisoning from ethylene glycol, lithium, methanol, phenobarbital, salicylates, and sodium valproate.

**Help on the web** Healthcare providers in the UK can register for free toxological advice at www.toxbase.org or call 0344 892 0111 for Poisons Information Service.

## Legal highs

Increasingly, 'designer' drugs, with chemical properties similar to illegal drugs (but yet sufficiently distinct to fall outside current legislation), are leading to acute poisoning and complications requiring admission. These drugs pose a difficult problem for the admitting physician, as the precise chemistry and mechanisms of action of both the active compound as well as any impurities are often unclear. Although many are legal, they can be as deadly as many well-known recreational drugs, leading to death or life-threatening complications such as rhabdomyolysis. Be aware that these drugs are out there, and ask specifically about legal highs when taking a history, as there are often no screening tools for these drugs.

## Some specific poisons and their antidotes



**Benzodiazepines** Flumazenil (for respiratory arrest) 200mcg over 15s; then 100mcg at 60s intervals if needed. Usual dose range: 300–600mcg IV over 3–6min (up to 1mg; 2mg if on ICU). May provoke fits. Use only after expert advice.

**β-blockers** Severe bradycardia or hypotension. Try atropine up to 3mg IV. Give glucagon 2–10mg IV bolus + 5% glucose if atropine fails then infusion of 50mcg/kg/h. Also consider including phosphodiesterase inhibitor infusions (eg enoximone 5–20mcg/kg/min). If unresponsive, consider pacing.

**Cyanide** This fast-killing poison has affinity for $Fe^{3+}$, and inhibits the cytochrome system, ↓aerobic respiration (therefore patients are acidotic with raised lactate). Depending on degree of poisoning presentation can be:

- *Mild:* Dizziness, anxiety, tachycardia, nausea, drowsiness/confusion.
- *Moderate:* Vomiting, reduced consciousness, convulsions, cyanosis.
- *Severe:* Deep coma, fixed unreactive pupils, cardiorespiratory failure, arrhythmias, pulmonary oedema.

*Treatment:* ►►100% $O_2$, GI decontamination. If mild, supportive care is usually sufficient. If moderate/severe then specific treatment to bind cyanide is required. Give sodium nitrite/sodium thiosulfate, or dicobalt edetate 300mg IV over 1min, then 50mL 50% glucose IV (repeat once if no response after a minute); or hydroxocobalamin (Cyanokit®) 5g over 15min repeated once if required. *Get expert help.* See p847.

**Carbon monoxide** Despite hypoxaemia skin is pink (or pale), not blue, as carboxyhaemoglobin (COHb) displaces $O_2$ from Hb binding sites. For the same reasons $SpO_2$ from a pulse oximeter may be normal. Check ABG in a co-oximeter (ie ensure it measures haemoglobin, $SaO_2$, Meth-Hb and COHb) which will show low $SaO_2$ and high COHb (normal <5%). *Symptoms:* Headache, vomiting, ↑pulse, tachypnoea, and, if COHb >50%, fits, coma, and cardiac arrest.

*Treatment:* ►►Remove the source. Give 100% $O_2$ until COHb <10%. Metabolic acidosis usually responds to correction of hypoxia. If severe, anticipate cerebral oedema and give mannitol IVI (p831). Confirm diagnosis with an ABG quickly as levels may soon return to normal. Monitor ECG. If COHb >20%, patient has neurological or psychological features, or cardiovascular impairment, fails to respond to treatment, or is pregnant, consider hyperbaric $O_2$: discuss with the poisons service.

**Digoxin** *Symptoms:* ↓Cognition, yellow-green visual halos, arrhythmias, nausea, and anorexia. If serious arrhythmias are present, correct hypokalaemia, and inactivate with digoxin-specific antibody fragments (DigiFab®). If load or level is unknown, give 20 vials (800mg)—adult or child >20kg.

**Heavy metals** Enlist expert help.

**Iron** Desferrioxamine 15mg/kg/h IVI; max 80mg/kg/d. NB: gastric lavage if iron ingestion in last hour; consider whole-bowel irrigation.

**Oral anticoagulants** See p351. If major bleed, give vitamin K 5mg slow IV and prothrombin complex concentrate 50U/kg IV (or if unavailable, fresh frozen plasma 15mL/kg IVI). For abnormal INR with no (or minimal) bleeding, see *BNF*. If it is vital that anticoagulation continues, enlist expert help. Discuss with haematology. NB: coagulation defects may be delayed for 2–3d following ingestion.

**Opiates** (Many analgesics contain opiates.) Give naloxone, eg 0.4–2mg IV; repeat every 2min until breathing is adequate (it has a short *t*½, so it may need to be given often or IM; max ~10mg). Naloxone may precipitate features of opiate withdrawal—diarrhoea and cramps, which will normally respond to diphenoxylate and atropine (eg co-phenotrope). Sedate as needed (see p15). High-dose opiate misusers may need methadone (eg 10–30mg/24h PO) to combat withdrawal. Refer for help (*OHCS* p374).

**Phenothiazine poisoning** (Eg chlorpromazine.) No specific antidote. *Dystonia* (torticollis, retrocollis, glossopharyngeal dystonia, opisthotonus): try procyclidine, eg 5–10mg IM or IV. Treat shock by raising the legs (± plasma expander IVI, or inotropes if desperate). Restore body temperature. *Monitor ECG*. Use lorazepam IV for prolonged fits in the usual way (p826). *Neuroleptic malignant syndrome* consists of: hyperthermia, rigidity, extrapyramidal signs, autonomic dysfunction (labile BP, ↑HR, sweating, urinary incontinence), mutism, confusion, coma, ↑WCC, ↑CK; it may be treated with cooling. Dantrolene 1–2.5mg/kg IV (p572) (max 10mg/kg/day) can help, bromocriptine and amantadine are alternatives.

**Carbon tetrachloride poisoning** This solvent, used in many industrial processes, causes vomiting, abdominal pain, diarrhoea, seizures, coma, renal failure, and tender hepatomegaly with jaundice and liver failure. IV acetylcysteine may improve prognosis. Seek expert help.

**Organophosphate insecticides** Inactivate cholinesterase—the resulting increase in acetylcholine causes the **SLUD** response: **s**alivation, **l**acrimation, **u**rination, and **d**iarrhoea. Also look for sweating, small pupils, muscle fasciculation, coma, respiratory distress, and bradycardia. *Treatment:* Wear gloves; remove soiled clothes. Wash skin. Take blood (FBC and serum cholinesterase activity). Give atropine IV 2mg every 10min till full atropinization (skin dry, pulse >70, pupils dilated). Up to 3d treatment may be needed. Also give pralidoxime 30mg/kg IVI over 20min, then 8mg/kg/h, max 12g in 24h. Even if fits are not occurring, diazepam 5–10mg IV slowly seems to help.

**Paraquat poisoning** (Found in weed-killers.) This causes D&V, painful oral ulcers, alveolitis, and renal failure. Diagnose by urine test. Give activated charcoal at once (100g followed by a laxative, then 50g/3–4h). ►*Get expert help.* Avoid $O_2$ early on (promotes lung damage).

**Ecstasy poisoning** Ecstasy is a semi-synthetic, hallucinogenic substance (MDMA, 3,4-methylenedioxymethamphetamine). Its effects range from nausea, muscle pain, blurred vision, amnesia, fever, confusion, and ataxia to tachyarrhythmias, hyperthermia, hyper/hypotension, water intoxication, DIC, ↑$K^+$, acute kidney injury (AKI), hepatocellular and muscle necrosis, cardiovascular collapse, and ARDS. There is no antidote and treatment is supportive. Management depends on clinical and lab findings, but may include:

- Administration of activated charcoal and monitoring of BP, ECG, and temperature for at least 12h (rapid cooling may be needed).
- Monitor urine output and U&E (AKI pp298–9), LFT, CK, FBC, and coagulation (DIC p352). Metabolic acidosis may benefit from treatment with bicarbonate.
- Anxiety: lorazepam 1–2mg IV as a slow bolus into a large vein. Repeat doses may be administered until agitation is controlled (see p826).
- Narrow complex tachycardias (p806) in adults: consider metoprolol 5mg IV.
- Hypertension can be treated with nifedipine 5–10mg PO or phentolamine 2–5mg IV. Treat hypotension conventionally (p790).
- Hyperthermia: attempt to cool, if rectal T° >39°C consider dantrolene 1mg/kg IV (may need repeating: discuss with your senior and a poisons unit). Hyperthermia with ecstasy is akin to serotonin syndrome, and propranolol, muscle relaxation, and ventilation may be needed.

**Snakes (adders)** *Anaphylaxis:* p794. *Signs of envenoming:* ↓BP (vasodilation, viper cardiotoxicity); D&V; swelling spreading proximally within 4h of bite; bleeding gums or venepuncture sites; anaphylaxis; ptosis; trismus; rhabdomyolysis; pulmonary oedema. *Tests:* ↑WCC; abnormal clotting; ↓platelets; U&E; ↑urine RBC; ↑CK; ↓$P_aO_2$, ECG. *Management:* Avoid active movement of affected limb (so use splints/slings). *Avoid incisions and tourniquets.* ►Get help from local/national poisons service. Is antivenom indicated (IgG from venom-immunized sheep)?—eg 10mL IV over 15min (adults *and* children) of *European Viper Antiserum* (from Monviato) for adder bites (see *BNF*);—20mL if severe envenoming have adrenaline to hand—p794. Monitor ECG. For non-UK endemic snakes, see *BNF*.

## Paracetamol poisoning

12g (=24 tablets) or 150mg/kg in adults may be fatal. ►If the patient weighs >110kg, calculate ingested dose using a body weight of 110kg to avoid underestimating toxicity. If the patient is malnourished then 75mg/kg can kill.

**Signs and symptoms** None initially, or vomiting ± RUQ pain. Later: jaundice and encephalopathy from liver damage (the main danger) ± acute kidney injury (AKI).

**Management** *General measures:* See p838. GI decontamination is recommended in those presenting <4h after overdose: give activated charcoal 1g/kg (max 50g).

- Glucose, U&E, LFT, INR, ABG, FBC, $HCO_3^-$; blood paracetamol level at 4h post-ingestion.
- If <10–12h since overdose, not vomiting, and plasma paracetamol is above the line on the graph (see fig 19.26), start acetylcysteine.
- If >8–24h and suspicion of large overdose (>7.5g) err on the side of caution and start acetylcysteine, stopping it if level below treatment line and INR/ALT normal.
- If ingestion time is unknown, or it is staggered, or presentation is >15h from ingestion, treatment *may* still help. ►Get advice.

*Acetylcysteine* is given by IVI: 150mg/kg in 5% glucose over 15–60min; then 50mg/kg in 500mL of 5% glucose over 4h; then 100mg/kg/16h in 1L of 5% glucose. Rash is a common SE: treat with chlorphenamine + observe; do not stop unless anaphylatoid reaction with shock, vomiting, and wheeze (occur <10%). An alternative (if acetylcysteine unavailable) is methionine 2.5g/4h PO for 16h (total: 10g), but absorption is unreliable if vomiting.

**Ongoing management**
- Next day do INR, U&E, LFT. If INR rising, continue acetylcysteine until <1.4.
- If continued deterioration, discuss with the liver team. Don't hesitate to get help.
- Consider referral to specialist liver unit guided by eg King's College criteria (BOX, p275).

## Salicylate poisoning

Aspirin is a weak acid with poor water solubility. It is present in many over-the-counter preparations. Uncoupling of oxidative phosphorylation leads to anaerobic metabolism and the production of lactate and heat. Effects are dose related and potentially fatal: •150mg/kg: mild toxicity. •250mg/kg: moderate •>500mg/kg: severe toxicity. Levels over 700mg/L are potentially fatal.

**Signs and symptoms** Unlike paracetamol, there are many early features:
- Vomiting, dehydration, hyperventilation, tinnitus, vertigo, sweating.
- Rarely ↓GCS, seizures, ↓BP and heart block, pulmonary oedema, hyperthermia.

Patients present initially with respiratory alkalosis due to a direct stimulation of the central respiratory centres and then develop a metabolic acidosis. Hyper- or hypoglycaemia may occur.

**Management** *General:* (pp838–9.) Correct dehydration. Keep patient on ECG monitor. Give activated charcoal to all presenting ≲1h—consider even if delayed presentation, slow-release formations, or bezoar formation (can delay absorption): at least one dose of 1g/kg (max 50g). Consider repeat doses (two further doses of 50g, 4h apart).

1. *Bloods.* Paracetamol and salicylate level, glucose, U&E, LFT, INR, ABG, $HCO_3^-$, FBC. Salicylate level may need to be repeated after 2h, due to continuing absorption if a potentially toxic dose has been taken. Monitor blood glucose 1–2hrly, beware hypoglycaemia, if severe poisoning, monitor salicylate levels, serum pH, and U&E.
2. *Urine.* Check pH, consider catheterization to monitor output and pH.
3. *Correct acidosis.* If plasma salicylate level >500mg/L (3.6mmol/L) or severe metabolic acidosis, consider alkalinization of the urine, eg with 1.5L 1.26% sodium bicarbonate IV over 3h. Aim for urine pH 7.5–8. NB: monitor serum $K^+$ as hypokalaemia may occur, and should be treated (caution if AKI).
4. *Dialysis* may well be needed if salicylate level >700mg/L, and if AKI or heart failure, pulmonary or cerebral oedema, confusion or seizures, severe acidosis despite best medical therapy, or persistently ↑plasma salicylate. Contact nephrology early.

Discuss any serious cases with the local toxicological service or national poisons information service.

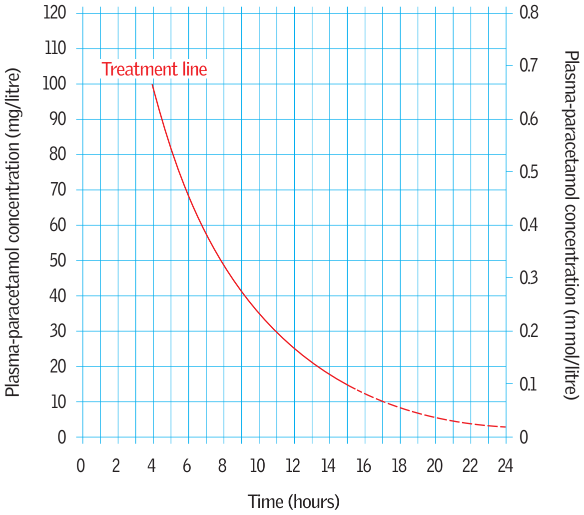



**Fig 19.26** Plasma concentration of paracetamol vs time, see p844 for interpretation. The graph may mislead if HIV +ve (↓hepatic glutathione), or if long-acting paracetamol has been taken, or if pre-existing liver disease or induction of liver enzymes has occurred.

# ▸▸Burns



**Assessment** *Burn size* is important to assess as it influences the size of the inflammatory response (vasodilation, increased vascular permeability) and thus fluid shift from the intravascular volume. Use Lund and Browder charts (see fig 19.27) or the '*rule of nines*' (arm: 9%; front of trunk 18%; head and neck 9%; leg 18%; back of trunk 18%; perineum 1%). Ignore erythema. *Burn depth* determines healing time/scarring; assessing this can be hard, even for the experienced. The big distinction is whether the burn is partial thickness (painful, red, and blistered) or full thickness (insensate/painless; grey-white). NB: burns can evolve, particularly over the first 48h.

**Resuscitation** Resuscitate and arrange transfer to specialist burns unit for all major burns (>25% partial thickness in adults and >20% in children). Assess site, size, and depth of burn (fig 19.27, to help calculate fluid requirements). Referral is still warranted in cases of full thickness burns >5%, partial thickness burns >10% in adults or >5% in children or the elderly, burns of special sites, chemical and electrical burns, and burns with inhalational injury.

- *Airway:* Beware of upper airway obstruction developing if hot gases inhaled. Suspect if history of fire in enclosed space, soot in oral/nasal cavity, singed nasal hairs or hoarse voice. A flexible laryngo/bronchoscopy is useful. Involve anaesthetists early and consider early intubation. Obstruction can develop in the first 24h.
- *Breathing:* Exclude life-threatening chest injuries (eg tension pneumothorax) and constricting burns—consider escharotomy if chest burns are impairing thorax excursion (*OHCS* p766). Give 100% $O_2$. Suspect carbon monoxide poisoning (p842) from history, cherry-red skin, and carboxyhaemoglobin level (COHb). With 100% $O_2$ *t½* of COHb falls from 250min to 40min (consider hyperbaric $O_2$ if: pH<7.1; CNS signs; >25% COHb or >20% if pregnant). $SpO_2$ (oximetry) is unreliable in CO poisoning.
- *Circulation:* Partial thickness burns >10% in a child and >15% in adults require IV fluid resuscitation. Put up 2 large-bore (14G or 16G) IV lines. Do not worry if you have to put these through burned skin; intraosseous access is valuable in infants and can be used in adults (see *OHCS* p236). Secure them well: they are literally lifelines.

Use a *burns calculator* flow chart or a formula, eg: *Parkland formula* (popular): 4 × weight (kg) × % burn = mL Hartmann's solution in 24h, half given in 1st 8h. Replace fluid from the time of burn, not from the time first seen in hospital. *Formulae are only guides:* adjust IVI according to clinical response and urine output; aim for 0.5mL/kg/h (1mL/kg/h in children), ~50% more in electrical burns and inhalation injury. Monitor T° (core and surface); catheterize the bladder. Beware of over-resuscitation ('fluid creep') which can lead to complications such as abdominal compartment syndrome.

**Treatment** 'Cool the burn, warm the patient.' Do *not* apply cold water to extensive burns for long periods: this may intensify shock. Take care with circumferential full-thickness burns of the limbs as compartment syndrome may develop rapidly particularly after fluid resuscitation. Decompress (escharotomy and fasciotomy) as needed. If transferring to a burns unit, do not burst blisters or apply any special creams as this can hinder assessment. Simple saline gauze or paraffin gauze is suitable; cling film is useful as a temporary measure and relieves pain. Titrate morphine IV for good analgesia. Ensure tetanus immunity. Antibiotic prophylaxis is not routinely used.

**Definitive dressings** There are many dressings for partial thickness burns, eg biological (pigskin, cadaveric skin), synthetic, and silver sulfadiazine cream alone or with cerium nitrate as Flammacerium®; it forms a leathery eschar which resists infection. Major full-thickness burns benefit from early tangential excision and split-skin grafts as the burn is a major source of inflammatory cytokines and forms a rich medium for bacterial growth.

## Smoke inhalation

Consider if:

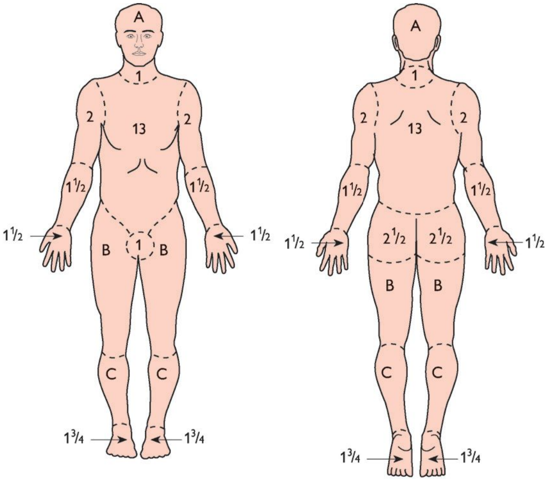
- History of exposure to fire and smoke in an enclosed space
- Hoarseness or change in voice
- Harsh cough
- Stridor.
- Burns to face
- Singed nasal hairs
- Soot in saliva or sputum
- Inflamed oropharynx

Initially laryngospasm leads to hypoxia and straining (leading to petechiae), then hypoxic cord relaxation leads to true inhalation injury. Free radicals, cyanide compounds, and carbon monoxide (CO) accompany thermal injury. Cyanide (p842) compounds (generated, eg from burning plastics) stop oxidative phosphorylation, causing dizziness, headaches, and seizures. Tachycardia + dyspnoea soon give way to bradycardia + apnoea. CO is generated later in the fire as oxygen is depleted. NB: COHb levels do not correlate well with the severity of poisoning and partly reflect smoking status and urban life. Use nomograms to extrapolate peak levels.

▸▸100% $O_2$ is given to elute both cyanide and CO.

▸▸Involve ICU/anaesthetists early if any signs of airway obstruction or respiratory failure: early intubation and ventilation may be useful.

▸▸Enlist expert help in cyanide poisoning—see p842.




Relative percentage of body surface area affected by growth

| Area | Age 0 | 1 | 5 | 10 | 15 | Adult |
|---|---|---|---|---|---|---|
| A: half of head | 9½ | 8½ | 6½ | 5½ | 4½ | 3½ |
| B: half of thigh | 2¾ | 3¼ | 4 | 4¼ | 4½ | 4¾ |
| C: half of leg | 2½ | 2½ | 2¾ | 3 | 3¼ | 3½ |

**Fig 19.27** Lund and Browder charts.

**Acknowledgement**

We thank Professor Tor Chiu for help in preparing this topic.

# Hypothermia

►*Have a high index of suspicion and a low-reading thermometer.* Most patients are elderly and do not complain of, or feel, cold—so they have not tried to warm up. In the young, hypothermia is usually from cold exposure (eg near-drowning), or is secondary to impaired consciousness (eg following excess alcohol or drug overdose).

**Definition** Hypothermia implies a core (rectal) temperature <35°C.

**Causes** In the elderly, hypothermia is often caused by a combination of factors:
- Impaired homeostatic mechanisms: usually age-related.
- Low room temperature: poverty, poor housing.
- Impaired thermoregulation: pneumonia, MI, heart failure.
- Reduced metabolism: immobility, hypothyroidism, diabetes mellitus.
- Autonomic neuropathy (p505); eg diabetes mellitus, Parkinson's.
- Excess heat loss: psoriasis and any other widespread dermatological diseases (ie TEN/ erythrodermic eczema).
- ↓Cold awareness: dementia, confusion.
- Increased exposure to cold: falls, especially at night when cold.
- Drugs: major tranquillizers, antidepressants, diuretics, alcohol.

**The patient**
- If the patient is shivering then the hypothermia is mild, if they are not shivering despite temp <35°C then the hypothermia is severe.
- Symptoms and signs include confusion, agitation, ↓GCS, coma, bradycardia, hypotension and arrhythmias (AF, VT, VF), especially if temp <30°C.

There are many stories of people 'returning to life' when warmed despite absence of vital signs, see BOX. It is essential to rewarm (see 'Treatment' later in topic) and re-examine.

**Diagnosis** Check oral or axillary T°. If ordinary thermometer shows <36.5°C, use a low-reading one PR. Is the rectal temperature <35°C? Infra-red ear thermometers can accurately reflect core temperature.

**Tests** Urgent U&E, plasma glucose, and amylase. Thyroid function tests; FBC; blood cultures. Consider blood gases. The ECG may show J-waves (fig 19.28).

**Treatment** Use ABCDE approach (p779)—but don't expose to cold.
- All patients should receive warm, humidified $O_2$; ventilate if comatose or respiratory insufficiency.
- Remove wet clothing, *slowly rewarm*, aiming for rise of ½°C/h (check temperature, BP, HR, and respiratory rate every 30min) using blankets or active external warming (hot air duvets). If T° rising too quickly stop and allow to cool slightly. Rapid rewarming causes peripheral vasodilation and shock. A falling BP can be a sign of too rapid warming.
- Warm IVI.
- Cardiac monitor is essential (AF, VF, and VT can occur at any time during rewarming or on stimulation).
- Consider antibiotics for the prevention of pneumonia (p166). Give these routinely in patients over 65yrs with T° <32°C.
- Consider urinary catheter (to monitor renal function).

NB: in sudden hypothermia from immersion or profound hypothermia with cardiovascular instability/cardiac arrest, the temperature needs to be raised rapidly. Options include warmed fluid lavage (intravesical, nasogastric, intrapleural, intraperitoneal) and intravascular warming (cardiopulmonary bypass, dialysis). In the event of cardiac arrest, defibrillation is usually unsuccessful if T° <30°C (consider amiodarone, bretylium). Resuscitation must continue until core T° >33°C (*OHCS* p786).

**Complications** Arrhythmias (if there is a cardiac arrest continue resuscitating until T° >33°C, as cold brains are less damaged by hypoxia); pneumonia; pancreatitis; AKI; DIC. *Prognosis:* Depends on age and degree of hypothermia. If age >70yrs and T° <32°C then mortality >50%.

**Before hospital discharge** Anticipate problems. Will it happen again? What is their network of support? Review medication (could you stop tranquillizers)? How is progress to be monitored? Liaise with GP/social worker.

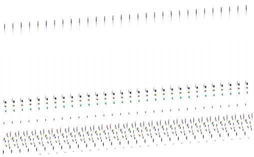


Fig 19.28 J-wave in hypothermia.

Courtesy of Dr R Luke and Dr E McLachlan.

**'I did not die but nothing of life remained...'**

Remember that death is a process not an event, and that in hypothermia, *all* processes are suspended: metabolism may slow to as much as 10% of baseline, drastically diminishing the oxygen requirements of all tissues. Perhaps this is what Dante had in mind for the last round of the 9th circle of *Hell*, in which those betraying their benefactors are encased in ice (*canto XXXIV*) 'Com'io divenni allor gelato e fioco...Io non mori e non rimasi vivo—How frozen I then became: I did not die but nothing of life remained'.

*Human records:* 13 month old Canadian Erica Nordby came to life 2 hours after her heart stopped (core T°: 16°C). Anna Bågenholm, a Swedish trainee orthopaedic surgeon, became trapped under freezing water covered by a layer of ice for 80 minutes following a skiing accident, suffering a cardiac arrest (core T°: 13.7°C). After resuscitation and 20 days in intensive care, she regained consciousness, suffering no permanent brain damage. She is now a radiologist. ►Do not declare anybody dead until they are warm and dead.

# Major disasters

**Planning** All hospitals have a detailed *Major Incident Plan*, but additionally the tasks of key personnel can be distributed on individual *Action Cards*.

**At the scene** Call the police to notify them of the Major Incident and ask them to take command. They will set up a central command centre to assess and manage the incident, depending on casualty numbers they will inform multiple hospitals of the need to prepare for the imminent arrival of casualties.

*Safety:* Paramount—your own and others. Be visible (luminous monogrammed jacket) and wear protective clothing where appropriate (safety helmet; waterproofs; boots; respirator in chemical environment).

*Triage:* See OHCS p800. There are several commercial systems available to label patients so emergency personnel can see at a glance the scale of the incident. The key is to divide patients by the urgency of care/transfer to hospital:

1. Emergency (label **RED** = will die in a few minutes if no treatment)
2. Urgent (label **YELLOW** =will die in ~2h if no treatment)
3. Non-urgent (label **GREEN** = walking wounded/those who are stable and can wait)
4. Deceased (label **BLUE/WHITE**).

*Communications:* Essential; each emergency service will dispatch a control vehicle and will have a designated incident officer for liaison. Support medical staff from hospital report to the medical incident officer (MIO)—he or she is usually the first doctor on the scene. Their job is to assess then communicate to the receiving hospital the number + severity of casualties, to organize resupply of equipment and to replace fatigued staff. The MIO must resist temptation to treat casualties as this compromises their role.

*Equipment:* Must be portable and include: intubation and cricothyrotomy set; intravenous fluids (colloid); bandages and dressings; chest drain (+flutter valve); amputation kit (when used, ideally 2 doctors should concur); drugs—*analgesic:* morphine; *anaesthetic:* ketamine 2mg/kg IV over >60s (0.5mg/kg is a powerful analgesic without respiratory depression); limb splints (may be inflatable); defibrillator/monitor ± pulse oximeter.

*Evacuation:* Remember that with immediate treatment on scene, the priority for evacuation may be reduced (eg a tension pneumothorax—**RED**—once relieved can wait for evacuation and becomes **YELLOW**), but those who may suffer by delay at the scene must go first. Send any severed limbs to the same hospital as the patient, ideally chilled—but not frozen.

**At the hospital** A 'major incident' is declared. The *first receiving* hospital will take most of the casualties; the *support* hospital(s) will cope with overflow and may provide mobile teams so that staff are not depleted from the first hospital. A control room is established and the medical coordinator ensures staff have been summoned and informed of their roles, nominates a triage officer, and supervises the best use of inpatient beds and ICU/theatre resources.

## Blast injuries

These may be caused by domestic (eg gas explosion) or industrial (eg mining) accidents, or by terrorist bombs. Death may occur without any obvious external injury. Injury occurs in a number of ways:

1 *Blast wave:* A transient (milliseconds) wave of overpressure expands rapidly producing cellular disruption, shearing forces along tissue planes (submucosal/subserosal haemorrhage) and re-expansion of compressed trapped gas—bowel perforation, fatal air embolism.
2 *Blast wind:* This can totally disrupt a body or cause avulsive amputations. Bodies can be thrown and sustain injuries on landing.
3 *Missiles:* Penetration or laceration from missiles are by far the commonest injuries. Missiles arise from the bomb or are secondary, eg glass.
4 *Flash burns:* These are usually superficial and occur on exposed skin.
5 *Crush injuries:* Beware sudden death or acute kidney injury from rhabdomyolysis after release.
6 *Contamination:* There is increasing concern about the use of biological or radioactive material in terrorist bombs. Even domestic or industrial blasts can scatter chemicals widely and cause both superficial and penetrating contamination. Consider the location and mechanism of the blast, and seek advice.
7 *Psychological injury:* Eg post-traumatic stress disorder (*OHCS* p353).

*Treatment:* Approach the same as any major trauma (*OHCS* p778). Rest and observe any suspected of exposure to significant blast but without other injury. Gun-shot injury: see *OHCS* p789. Major blast injuries, whatever the cause, should be reported to the police for investigation.

# 20 References

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*Abbreviations:* **F** indexes a notable figure or image; dis = disease; syn = syndrome.

## A

## B

## C

## D

## *I*

## M

## N

## O

## P

## Q



## R

## S

## U

## V



## W



## X



## Y



## Z

# Early warning score

Early warning scores are scoring systems based on physiological parameters. The magnitude of the given score reflects how far the parameter varies from normal. The collated score from different parameters is used in:

- the assesssment of acute illness
- the detection of a clinical deterioration
- the initiation of a timely and competent clinical response.

A standardized National Early Warning Score (NEWS) is recommended for use across the NHS.[1] The components of the NEWS are detailed in fig A1. An appropriate clinical response to the aggregate score from fig A1 is outlined in fig A2.

| PHYSIOLOGICAL PARAMETERS | 3 | 2 | 1 | 0 | 1 | 2 | 3 |
|---|---|---|---|---|---|---|---|
| Respiration Rate | ≤8 | | 9 - 11 | 12 - 20 | | 21 - 24 | ≥25 |
| Oxygen Saturations | ≤91 | 92 - 93 | 94 - 95 | ≥96 | | | |
| Any Supplemental Oxygen | | Yes | | No | | | |
| Temperature | ≤35.0 | | 35.1 - 36.0 | 36.1 - 38.0 | 38.1 - 39.0 | ≥39.1 | |
| Systolic BP | ≤90 | 91 - 100 | 101 - 110 | 111 - 219 | | | ≥220 |
| Heart Rate | ≤40 | | 41 - 50 | 51 - 90 | 91 - 110 | 111 - 130 | ≥131 |
| Level of Consciousness | | | | A | | | V, P, or U |

*The NEWS initiative flowed from the Royal College of Physicians' NEWSDIG, and was jointly developed and funded in collaboration with the Royal College of Physicians, Royal College of Nursing, National Outreach Forum and NHS Training for Innovation.

**Fig A1** National Early Warning Score for adult patients. © RCP 2012.

1 Royal College of Physicians. *National Early Warning Scores (NEWS): standardising the assessment of acute illness severity in the NHS*. London: RCP, 2012.

| NEWS SCORE | FREQUENCY OF MONITORING | CLINICAL RESPONSE |
|---|---|---|
| **0** | Minimum 12 hourly | • Continue routine NEWS monitoring with every set of observations |
| **Total: 1-4** | Minimum 4-6 hourly | • Inform registered nurse who must assess the patient;<br>• Registered nurse to decide if increased frequency of monitoring and / or escalation of clinical care is required; |
| **Total: 5 or more or 3 in one parameter** | Increased frequency to a minimum of 1 hourly | • Registered nurse to urgently inform the medical team caring for the patient;<br>• Urgent assessment by a clinician with core competencies to assess acutely ill patients;<br>• Clinical care in an environment with monitoring facilities; |
| **Total: 7 or more** | Continuous monitoring of vital signs | • Registered nurse to **immediately** inform the medical team caring for the patient – this should be at least at Specialist Registrar level;<br>• Emergency assessment by a clinical team with critical care competencies, which also includes a practitioner/s with advanced airway skills;<br>• Consider transfer of Clinical care to a level 2 or 3 care facility, i.e. higher dependency or ITU; |

**Fig A2** Clinical response to NEWS triggers. © RCP 2012.

►Early warning scores are tools to aid assessment. They do not replace clinical judgement: use yours and respect the clinical opinion of others.

►Refer to local early warning scores where available.

# Cardiac arrest: advanced life support algorithm 2015

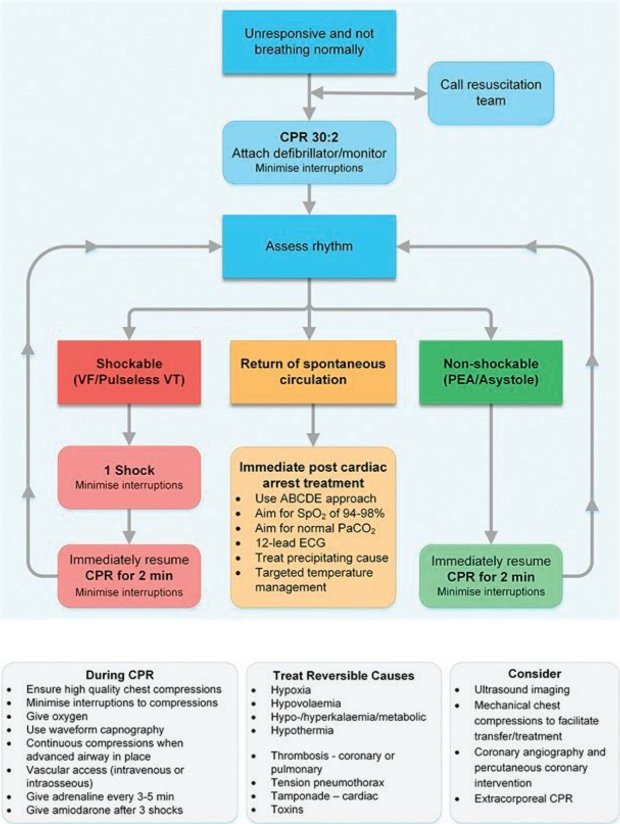


**Fig A3** Cardiac arrest: advanced life support algorithm 2015.

## ▸▸Cardiorespiratory arrest

▸▸Ensure the safety of the patient and yourself.

▸▸Confirm diagnosis: a patient who is unresponsive and not breathing properly is in cardiac arrest (a manual pulse check is inaccurate and not recommended).

**Basic life support** Shout for help. Ask someone to call the arrest team and bring the defibrillator. Note the time. **ABC**:

▸▸*Airway:* Head tilt (if no spine injury) and chin lift/jaw thrust.

▸▸*Breathing:* Look, listen, and feel for breathing for no more than 10 seconds. If there is any doubt whether breathing is normal, proceed to chest compressions.

▸▸*Chest compressions:* Place the heel of one hand on the centre of the chest (lower half of the stenum). Place your second hand on top and interlock fingers. Use straight arms. Give compressions at a rate of 100–120/min. Aim to compress the sternum 5-6cm. After 30 compressions give 2 rescue breaths. Do not interrupt compressions >10s. Continue with a ratio of 30:2 until defibrillator is available.

**Advanced life support** See algorithm fig A3.

- Continue chest compressions while adhesive defibrillation/monitoring pads are put in place. Plan all actions before pausing chest compressions.
- Stop chest compression for <5s to assess rhythm. Determine whether the rhythm is shockable (VF/pulseless VT) or non-shockable (asystole, pulseless electrical activity).

### Shockable rhythm: VF/pulseless VT

▸▸A single person performs uninterrupted chest compressions while everyone else prepares for defibrillation: stand clear, move oxygen delivery device 1m away.

▸▸Select the appropriate energy on the defibrillator (150J or manufacturer's guidelines). When defibrillator is charged and safety check complete, the rescuer performing chest compressions stands clear and the shock is delivered.

▸▸CPR is resumed immediately (30:2). Reassess pulse/rhythm only after 2 minutes of CPR.

▸▸Repeat if shockable rhythm remains. Give drugs after 3 shocks (see Drugs, this topic).

### Non-shockable rhythm: asystole/pulseless electrical activity (PEA)

▸▸Continue CPR 30:2. Obtain IV access and secure airway. Once airway secure switch to continuous compressions and ventilation. Give adrenaline 1mg IV.

▸▸Check rhythm every 2 minutes.

▸▸Consider reversible causes (4Hs and 4Ts: hypoxia, hypovolaemia, hyper/hypokalaemia/other metabolic derangement, hypothermia, thrombosis, tension pneumothorax, tamponade, toxins).

### Drugs

- Give adrenaline 1mg IV every 3–5 mins for both shockable (from 3rd shock) and non-shockable rhythms. In practice this means at every other rhythm check or shock.[1]
- In shockable rhythms give amiodarone 300mg IV after 3 defibrillation attempts. Consider a further 150mg IV after 5 shocks. Lidocaine is an alternative.

**Discontinuing resuscitation** Needs clinical judgement: what is the likelihood of achieving a successful return of spontaneous circulation? If there is a shockable rhythm or a reversible cause then attempts are usually continued. It is reasonable to discontinue if asystole >20mins without a reversible cause. Ask for the opinion of others in the resuscitation team.

**Resuscitation decisions** Consider, discuss, and record CPR decisions:

- at the request of a patient with capacity
- as part of end-of-life care (p12, p536)
- in deteriorating, severe illness.

▸Your patient should be involved in decisions about CPR (unless it would cause physical or psychological harm). Explain your clinical decision to them, including futility.

▸Do not make judgements about the quality of life of others based on your own perception.

1 Meta-analysis fails to show that adrenaline increases survival to hospital discharge (http://www.ncbi.nlm.nih.gov/pubmed/24193240). RCT results are awaited (Paramedic 2: The Adrenaline Trial ISRCTN 73485024).

## Useful doses for the new doctor

►These pages outline the typical adult doses of drugs that a foundation doctor will be called upon to prescribe. Refer to local guidelines first. If in any doubt, consult a drug formulary (eg British National Formulary www.bnf.org) especially if ↓eGFR or weight <50kg. Always check allergies before prescribing.

*Analgesics*

| Drug | Dose and frequency | Notes |
|---|---|---|
| Paracetamol | 1g/6h PO/PR/IV, max. 4g/24h | Avoid if hepatic impairment. |
| Ibuprofen | 400mg/8h PO, max 2.4g/24h | SE: gastritis; bronchospasm; AKI; fluid retention; hypersensitivity. CI: peptic ulcer; NSAID-induced asthma; coagulopathy; advanced CKD; heart failure. |
| Diclofenac sodium | 50mg/8h PO/PR | |
| Codeine phosphate | 30–60mg/4h PO/IM, max 240mg/24h | Chronic pain, eg malignancy, may require higher doses (see p536). Reduce dose if ↓eGFR. Care in head injury, as may hinder neurological assessment. SE: N&V; constipation; drowsiness; hypotension; respiratory depression, dependence. CI: respiratory depression. |
| Dihydrocodeine tartrate | 30mg/4–6h PO, or 50mg/4–6h IM/SC | |
| Morphine | 5–10mg/4h PO/IM | |
| Oxycodone | 2.5–5mg/4h PO | |
| Tramadol | 50–100mg/4h PO/IM/IV | |

*Antibiotics (refer to local guidelines)*

| Drug | Dose and frequency | Notes |
|---|---|---|
| Phenoxymethylpenicillin | 500mg/6h PO (max 4g/24h) | SE: rash; hypersensitivity and anaphylaxis; diarrhoea. CI: history of allergy. |
| Benzylpenicillin | 0.6–1.2g/6h IV/IM | |
| Flucloxacillin | 250–500mg/6h PO/IM<br>1g/6h IV | |
| Erythromycin | 250–500mg/6h PO | IV only if oral treatment not possible. Beware of cytochrome P450 interactions (not azithromycin). SE: N&V; diarrhoea; cholestasis; QT prolongation; pancreatitis |
| Clarithromycin | 250–500mg/6h PO | |
| Azithromycin | 500mg/24h PO | |
| Doxycycline | 200mg/24h PO as a single dose then 100mg/24h | SE: hypersensitivity; hepatotoxicity; may exacerbate myasthenia gravis and SLE. CI: pregnancy; age <12y. |
| Metronidazole | 400mg/8h PO, or<br>500mg/8h IV, or<br>1g/8h PR | IV only if oral treatment not possible. |
| Gentamicin | 5mg/kg/24h IV adjusted to serum concentration | Adjust dose for renal function. SE: nephrotoxicity (correct volume depletion); electrolyte disturbance; ototoxicity. |
| Trimethoprim | 200mg/12h PO | CI: 1st trimester (folate antagonist). |

*Anti-emetics*

| Drug | Dose and frequency | Notes |
|---|---|---|
| Cyclizine | 50mg/8h PO/IM/IV | SE: drowsiness. |
| Metoclopramide | 10mg/8h PO/IM/IV | SE: extrapyramidal SE, especially in young adults. |
| Ondansetron | 4–8mg/8–12h PO/IV | SE: constipation; headache<br>CI: long QT syndrome. |

| Drug | Dose and frequency | Notes |
|---|---|---|
| ***Acute asthma/COPD*** | | |
| Salbutamol | 5mg via nebulizer as required according to clinical response | Oxygen-driven nebulizer in asthma, air-driven in COPD with appropriate concentration of oxygen provided in addition. SE: tachycardia; hypokalaemia. |
| Ipratropium bromide | 500mcg/4–6h via nebulizer | SE: GI disturbance, cough. |
| Prednisolone | 30–50mg/24h (refer to local guidelines). | Oral steroids are as effective as IV. Specify course length. SE: DM; peptic ulceration; psychosis; ↑BP; fluid retention. |
| ***Antihistamines*** | | |
| Chlorphenamine | 4mg/4–6h PO<br>In anaphylaxis: 10mg IM | SE: drowsiness; urinary retention; dry mouth; blurred vision; GI disturbance. |
| Cetirizine | 10mg/24h PO | |
| Loratidine | 10mg/24h PO | |
| ***Gastric acid-reducing drugs*** | | |
| Ranitidine | 150mg/12h PO or 50mg/8h IV | SE: diarrhoea; dizziness; cholestasis. |
| Omeprazole | 20–40mg/24h PO or 40mg/24h IV | SE: GI disturbance; hypersensitivity.<br>► May mask symptoms of gastric cancer. |
| Lansoprazole | 15–30mg/24h PO | |
| Pantoprazole | 20–40mg/24h PO or 40mg/24h IV | |
| ***Heparins (refer to local guidelines)*** | | |
| Unfractionated heparin | *DVT prophylaxis*:<br>5000U/8–12h SC | Dose needs correcting for renal function and pregnancy (see local guidelines). SE: bleeding; thrombocytopenia (watch for ↓ by 30% or thrombosis 5–10 days into treatment); hyperkalaemia; osteoporosis after prolonged use (↓risk with LMWH). CI: heparin-induced thrombocytopenia; bleeding disorders; epidural anaesthesia, recent cerebral bleed; recent trauma or surgery; active bleeding. |
| Enoxaparin | *DVT prophylaxis*:<br>20–40mg/24h SC<br>*DVT/PE treatment*:<br>1.5mg/kg/24h SC | |
| Tinzaparin | *DVT prophylaxis:*<br>3500U/24h SC<br>*DVT/PE treatment:*<br>175U/kg per 24h SC | |
| Dalteparin | *DVT prophylaxis:*<br>2500–5000U/24h SC<br>*DVT/PE treatment:*<br>200U/kg/d SC | |
| ***Tranquillizers for non-psychotic behavioural disturbance*** | | |
| Lorazepam | 1–2mg PO/IM | De-escalation techniques first. Medication only to ↓risk of harm to self/others. ↑Risk of medication in frail, elderly, comorbid conditions. Wait 60 min for response. Seek expert help if needs repeat dose. SE: respiratory depression; drowsiness; ataxia; confusion; GI disturbance; urinary retention. CI: respiratory disease |

**See also:** prescribing in palliative care (pp532–537), laxatives (p260), inhalers (p183), digoxin (p115), insulin (p589), fluid (p666), oxygen prescribing (p189), naloxone (p842).